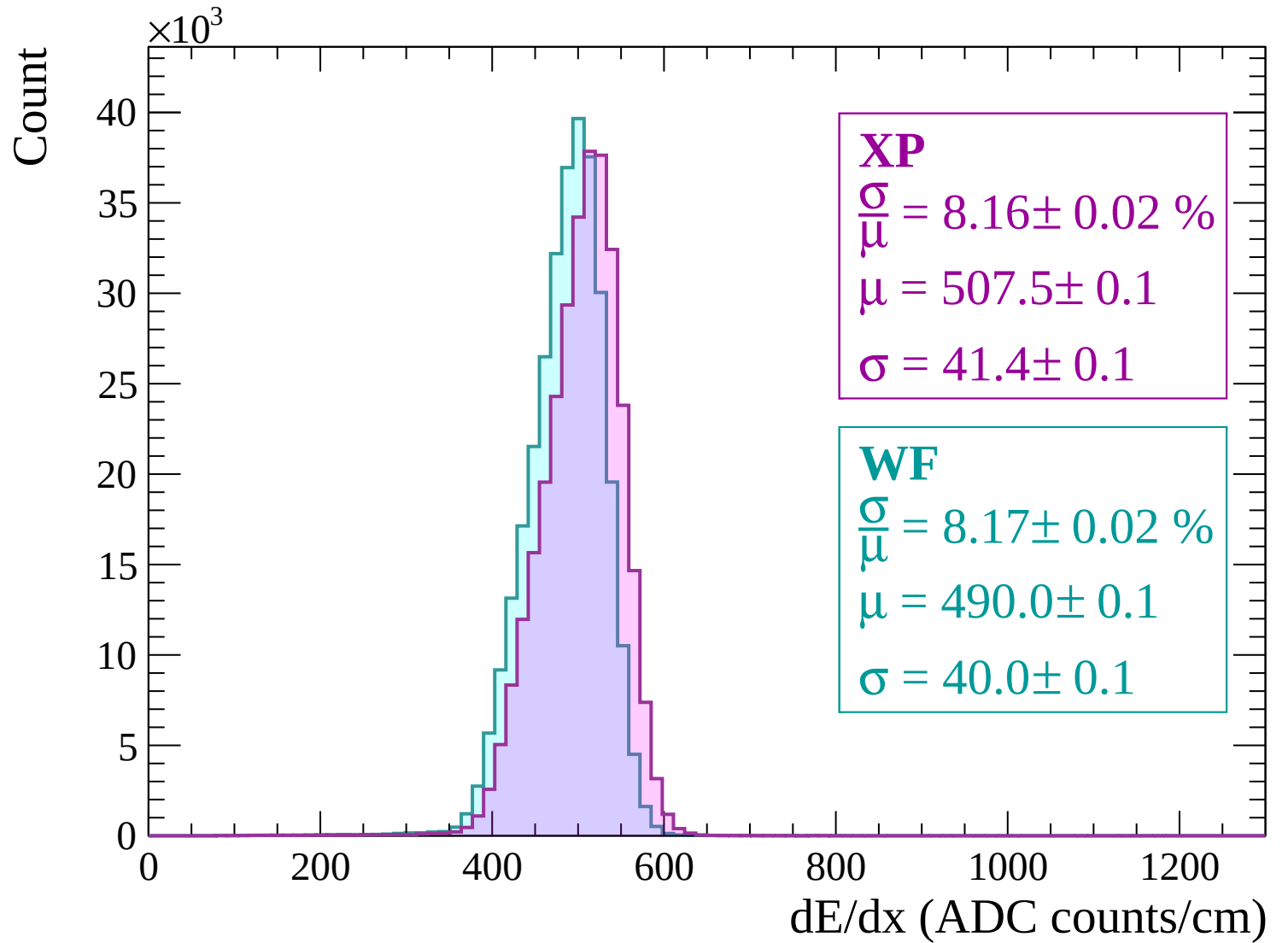
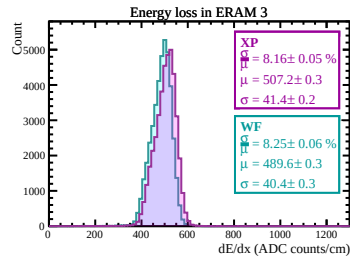
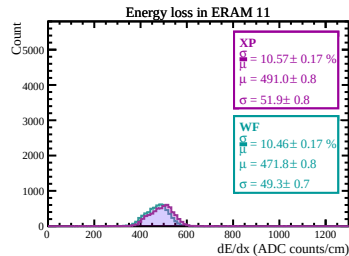
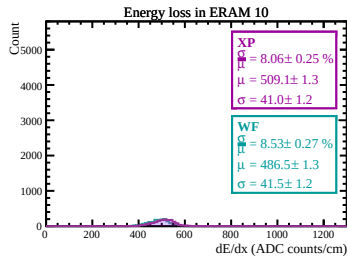
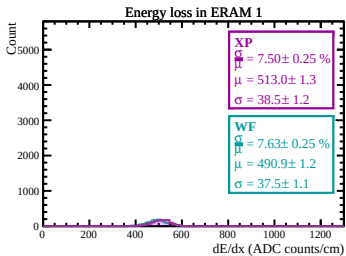
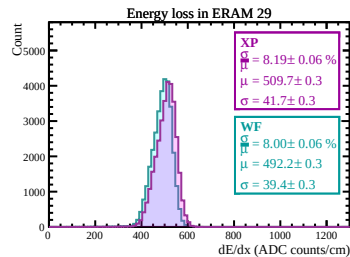
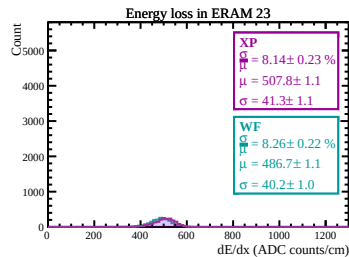
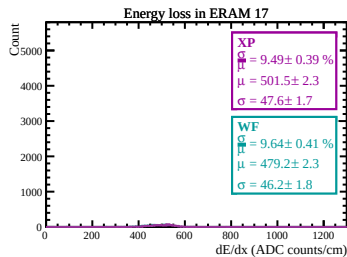
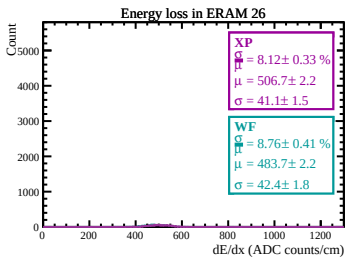
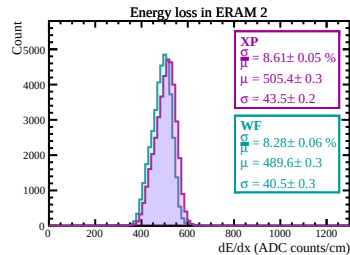
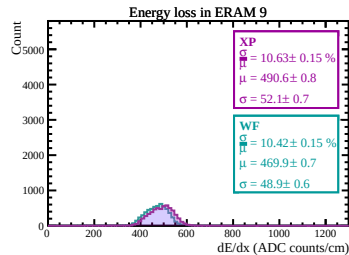
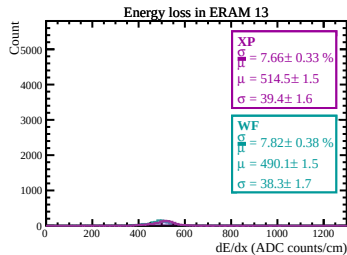
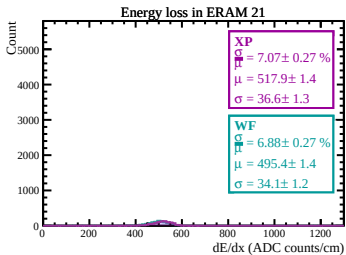
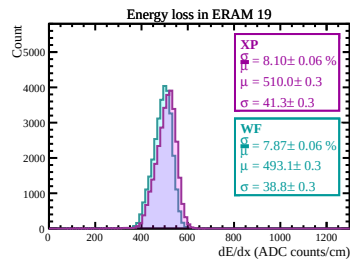
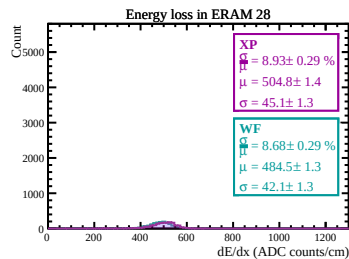
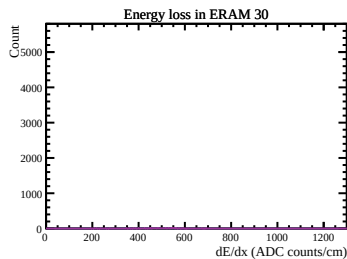
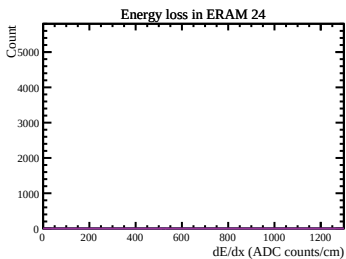
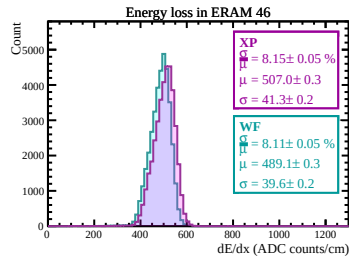
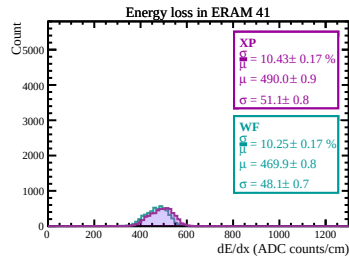
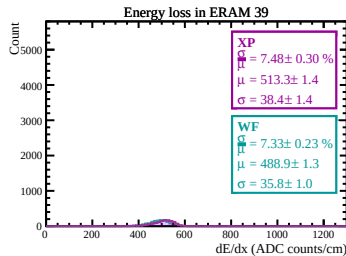
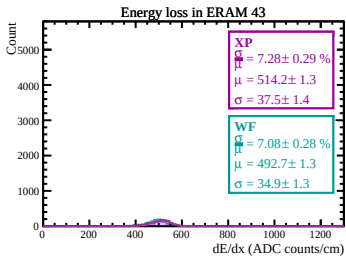
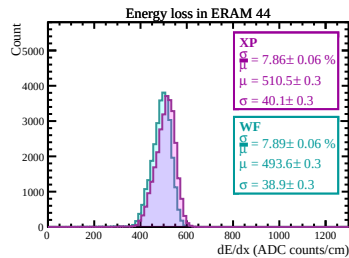
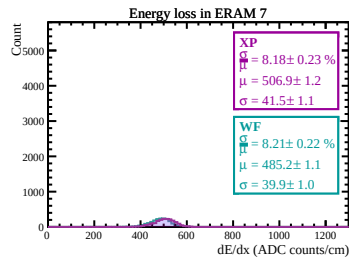
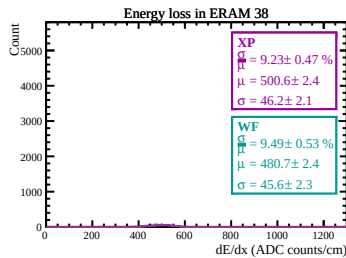
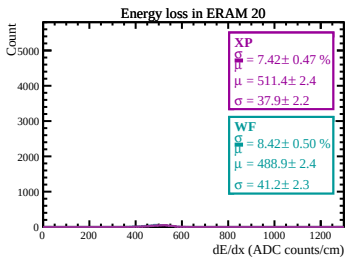
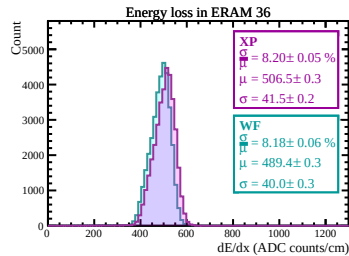
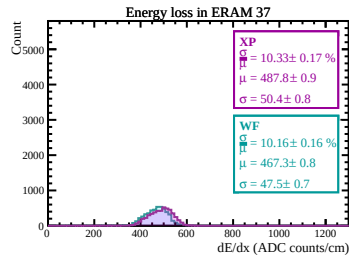
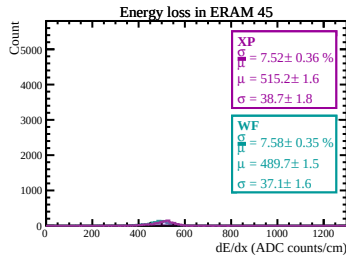
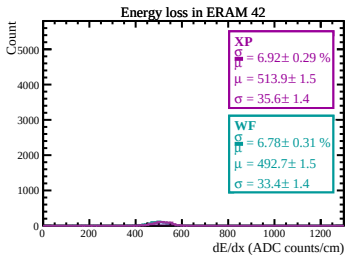
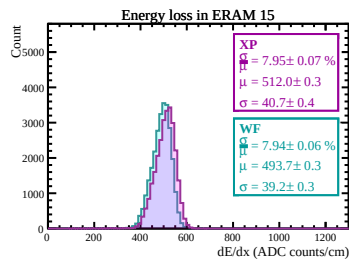
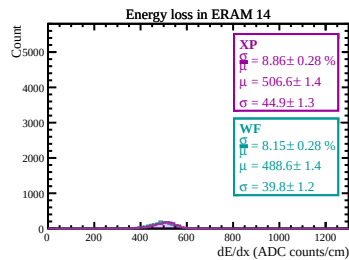
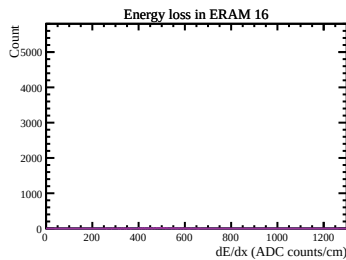
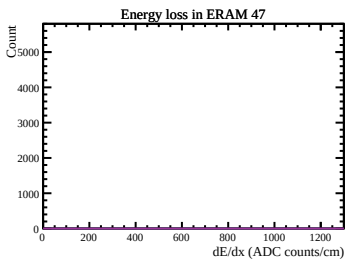
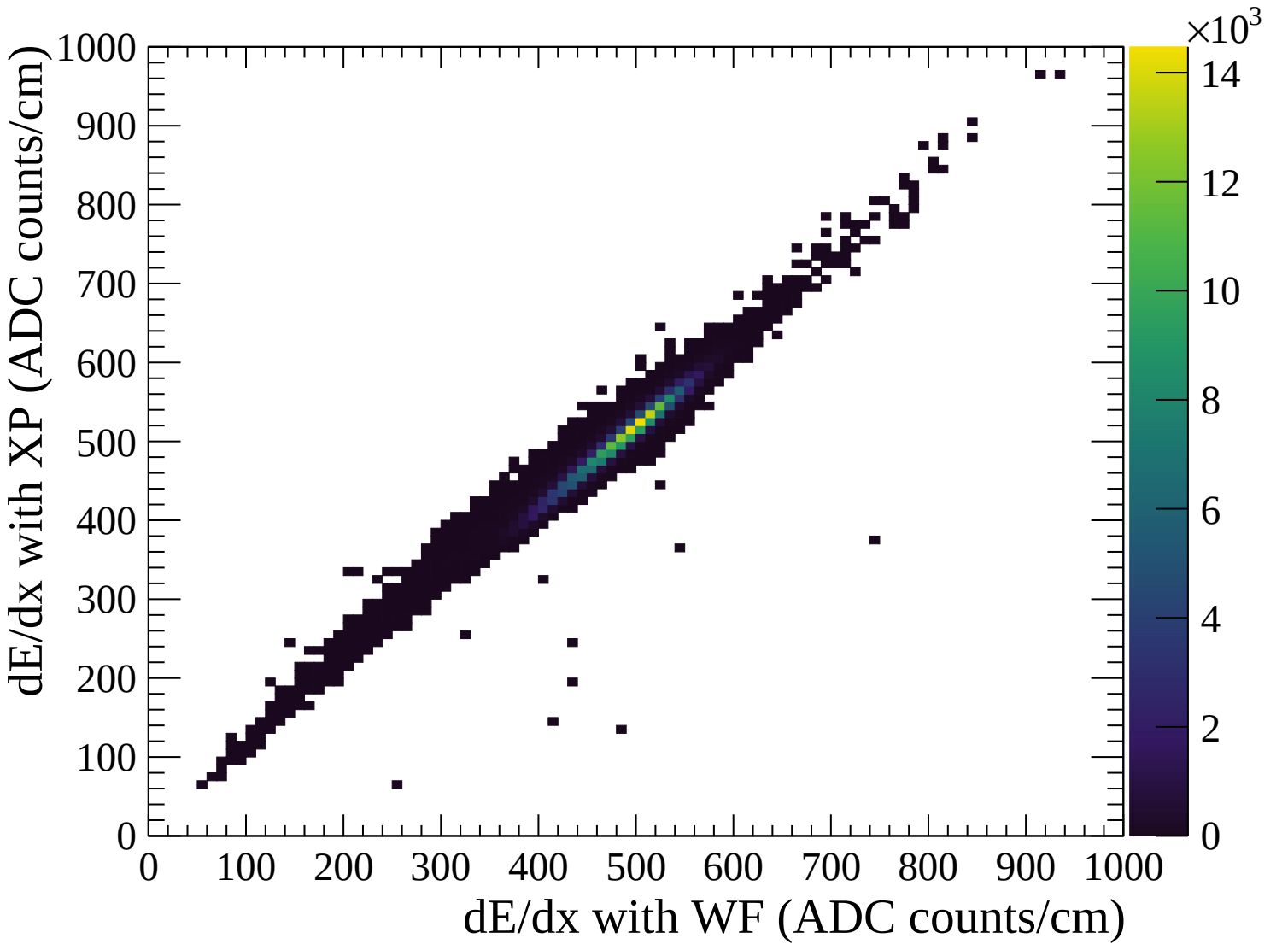
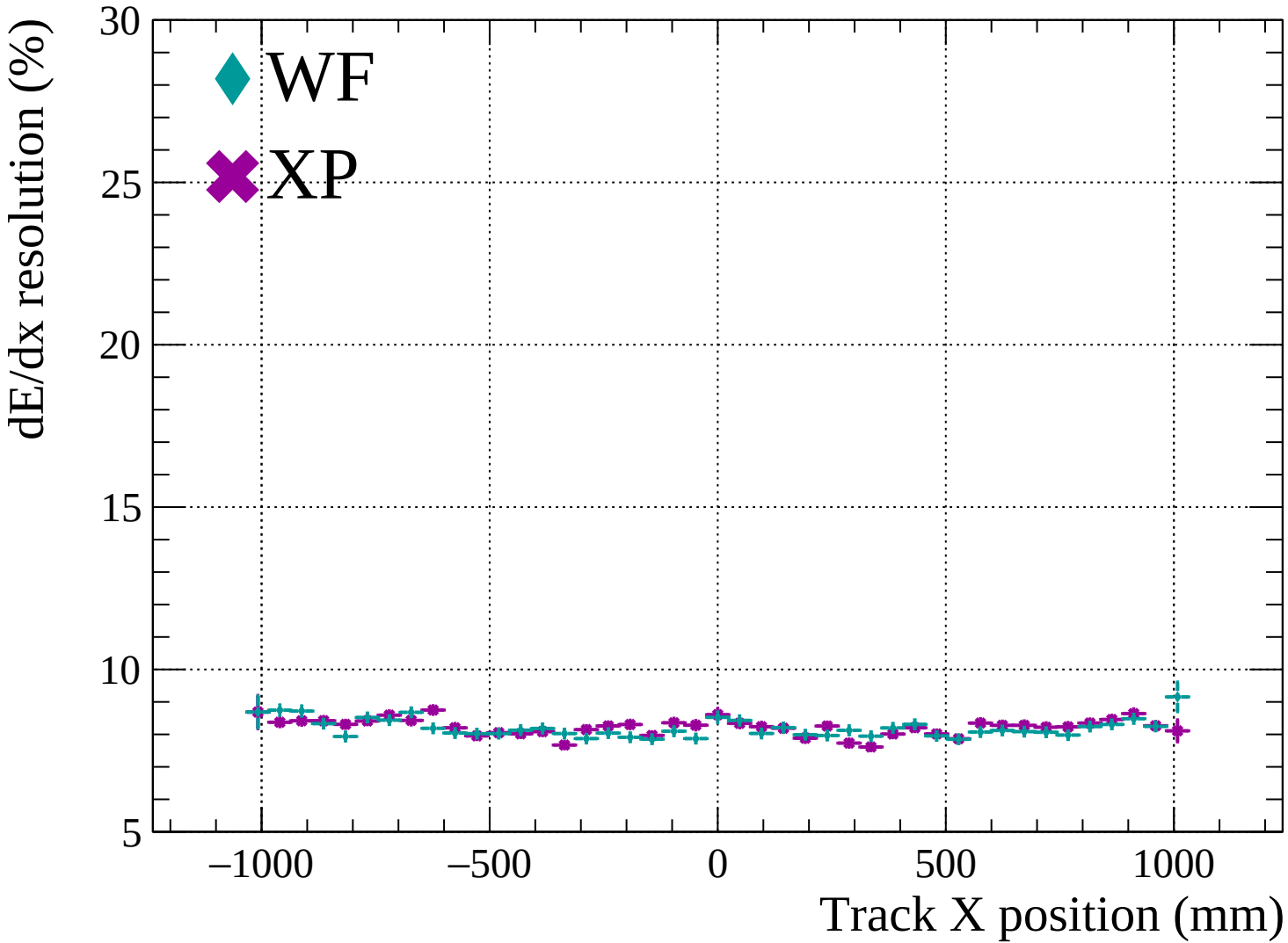


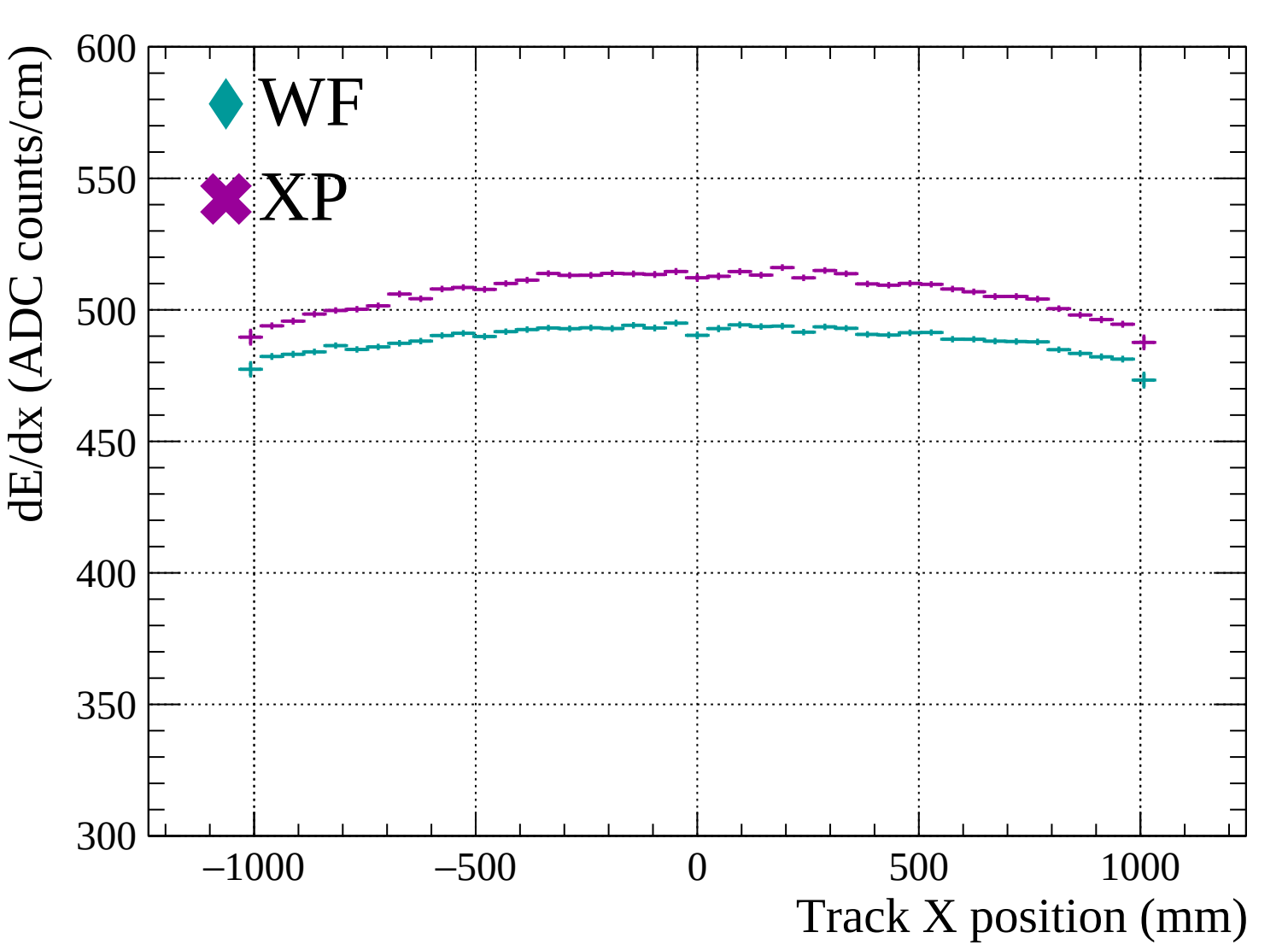


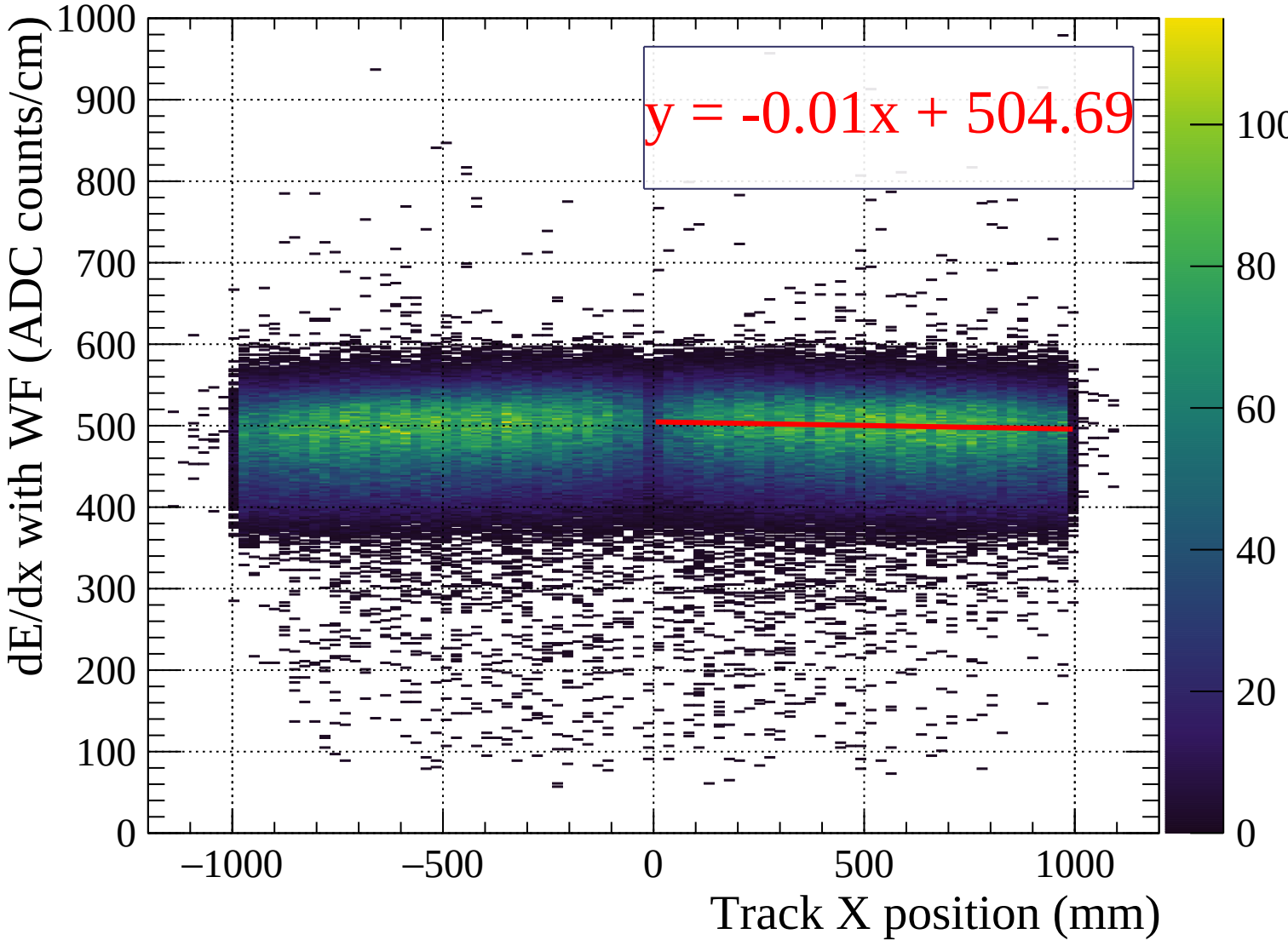


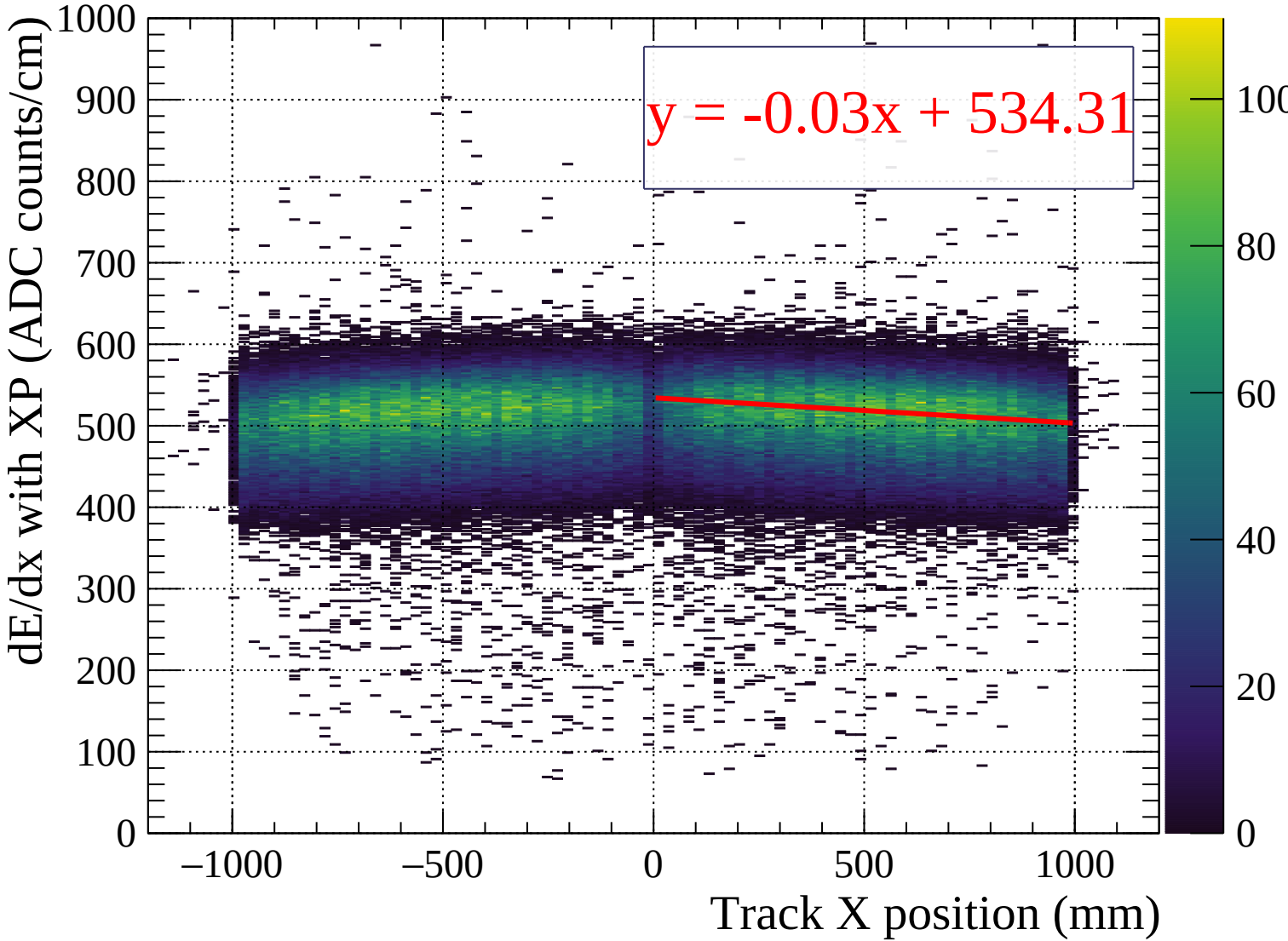


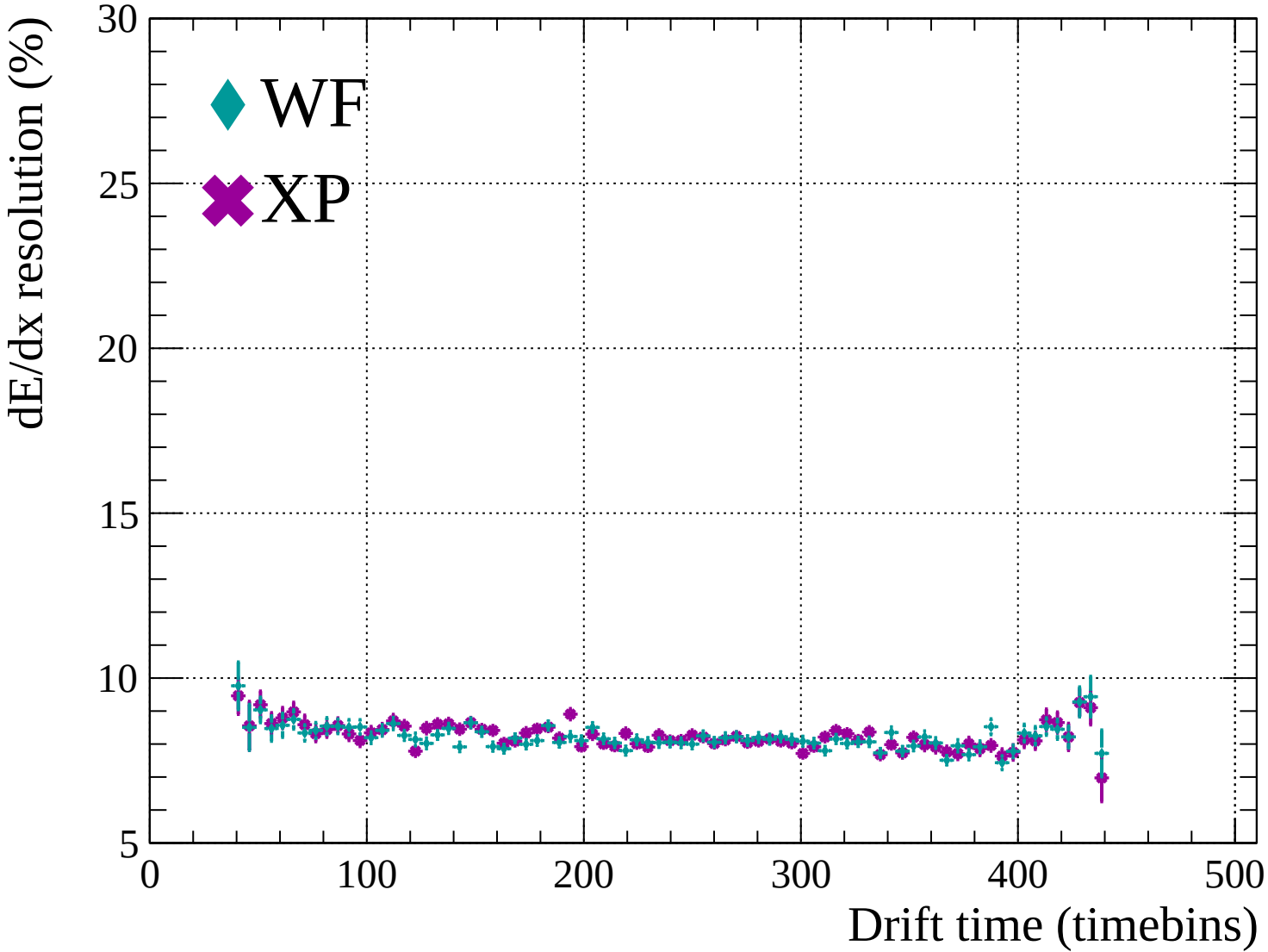


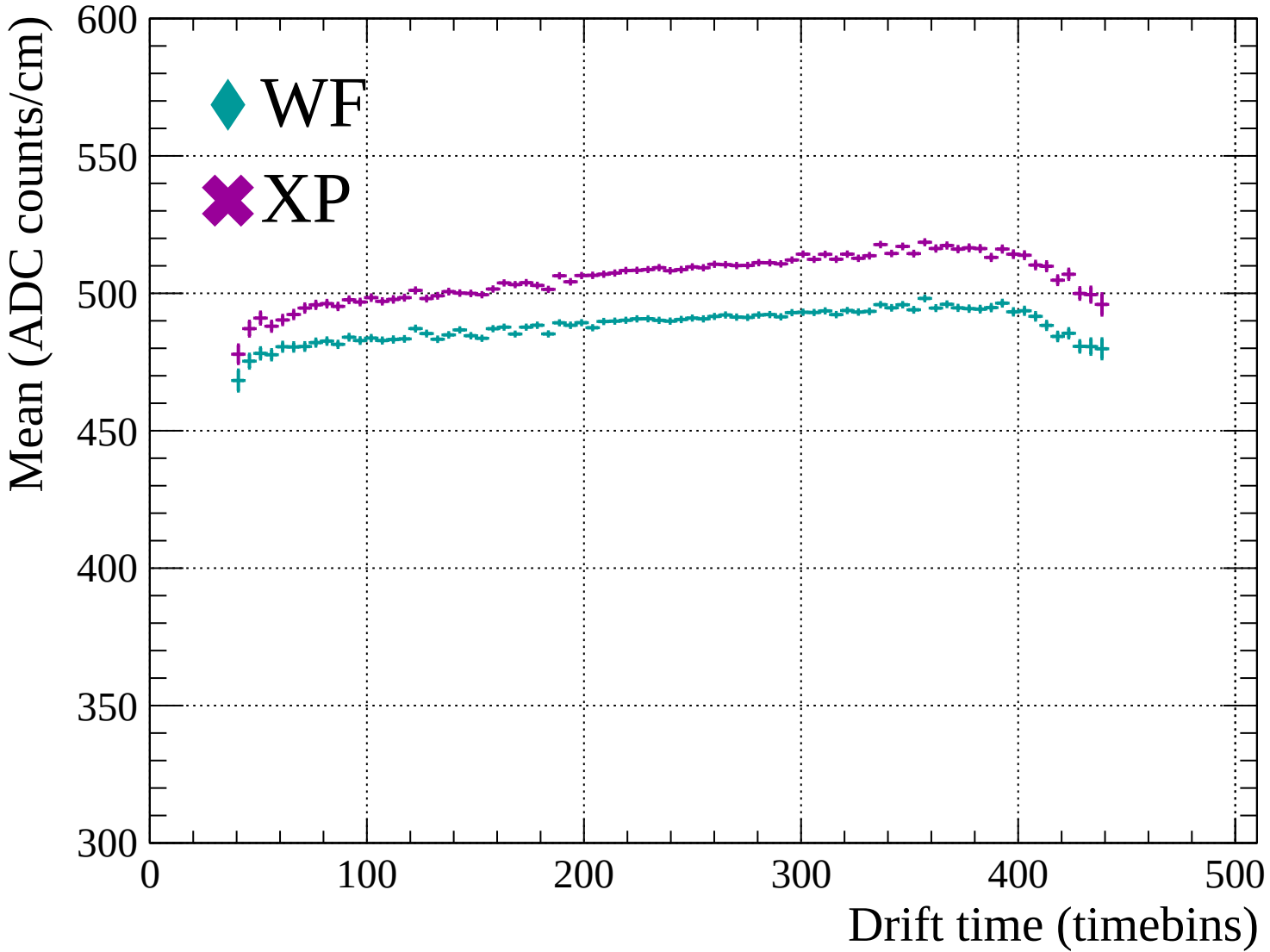


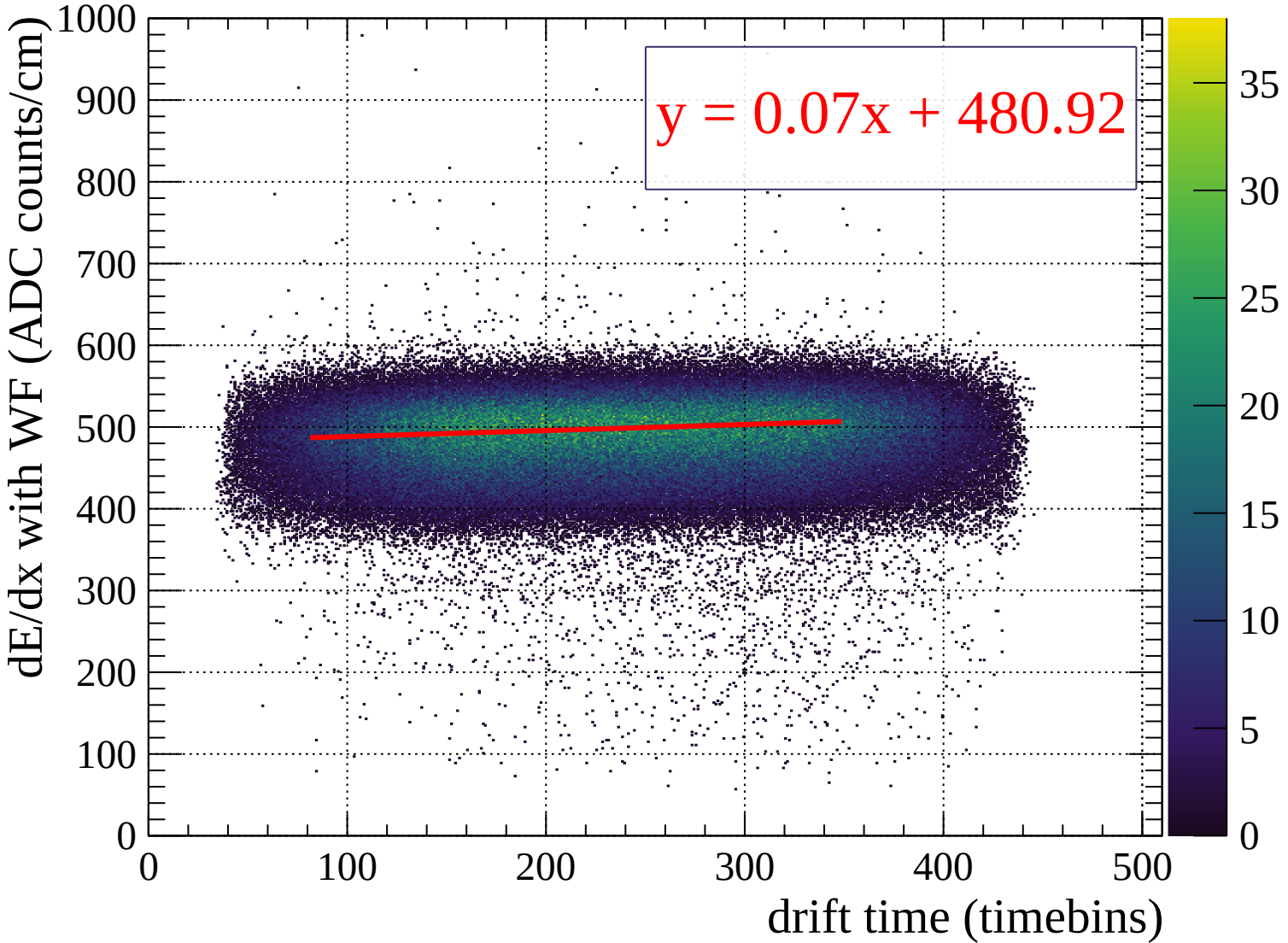


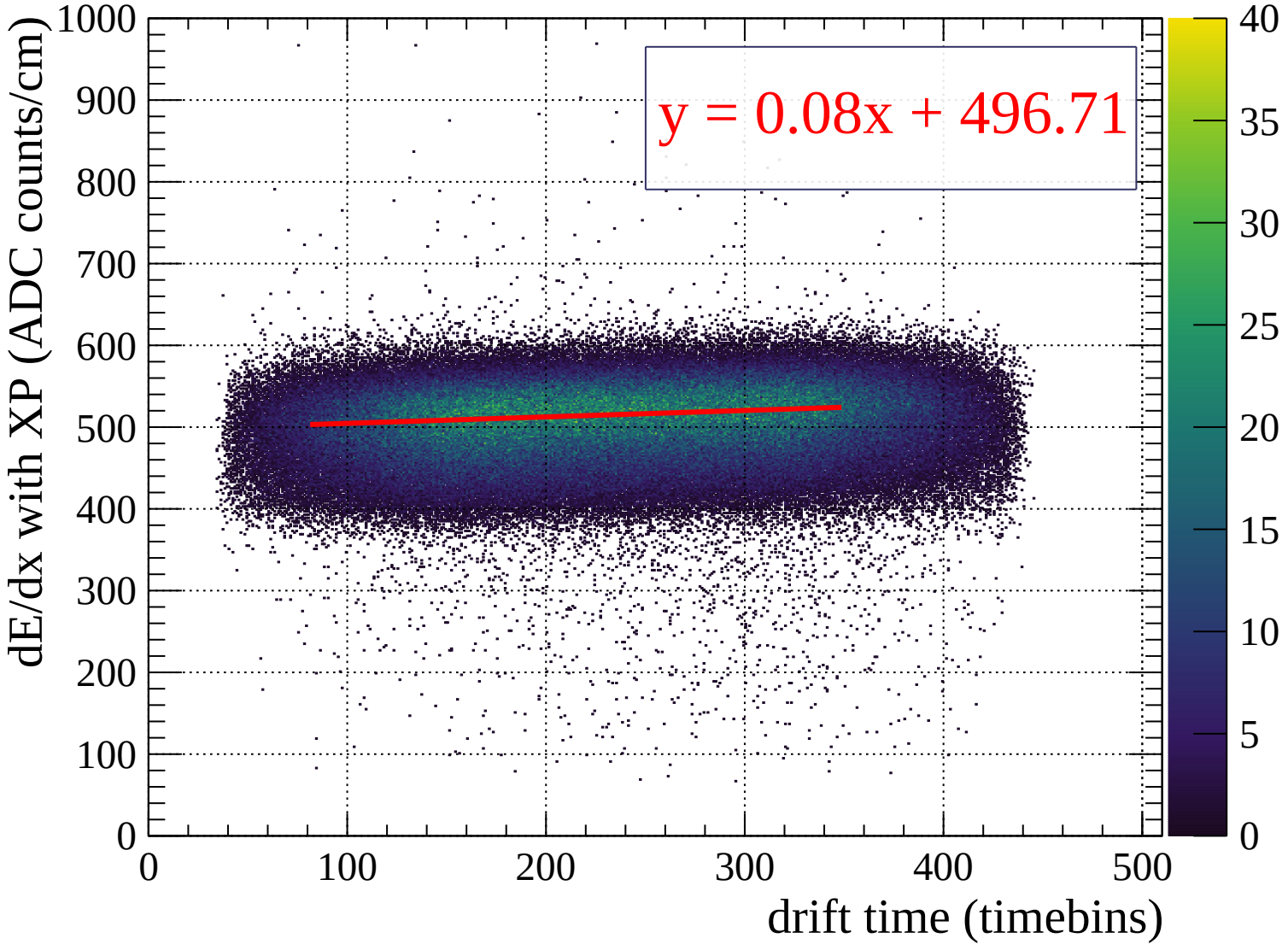


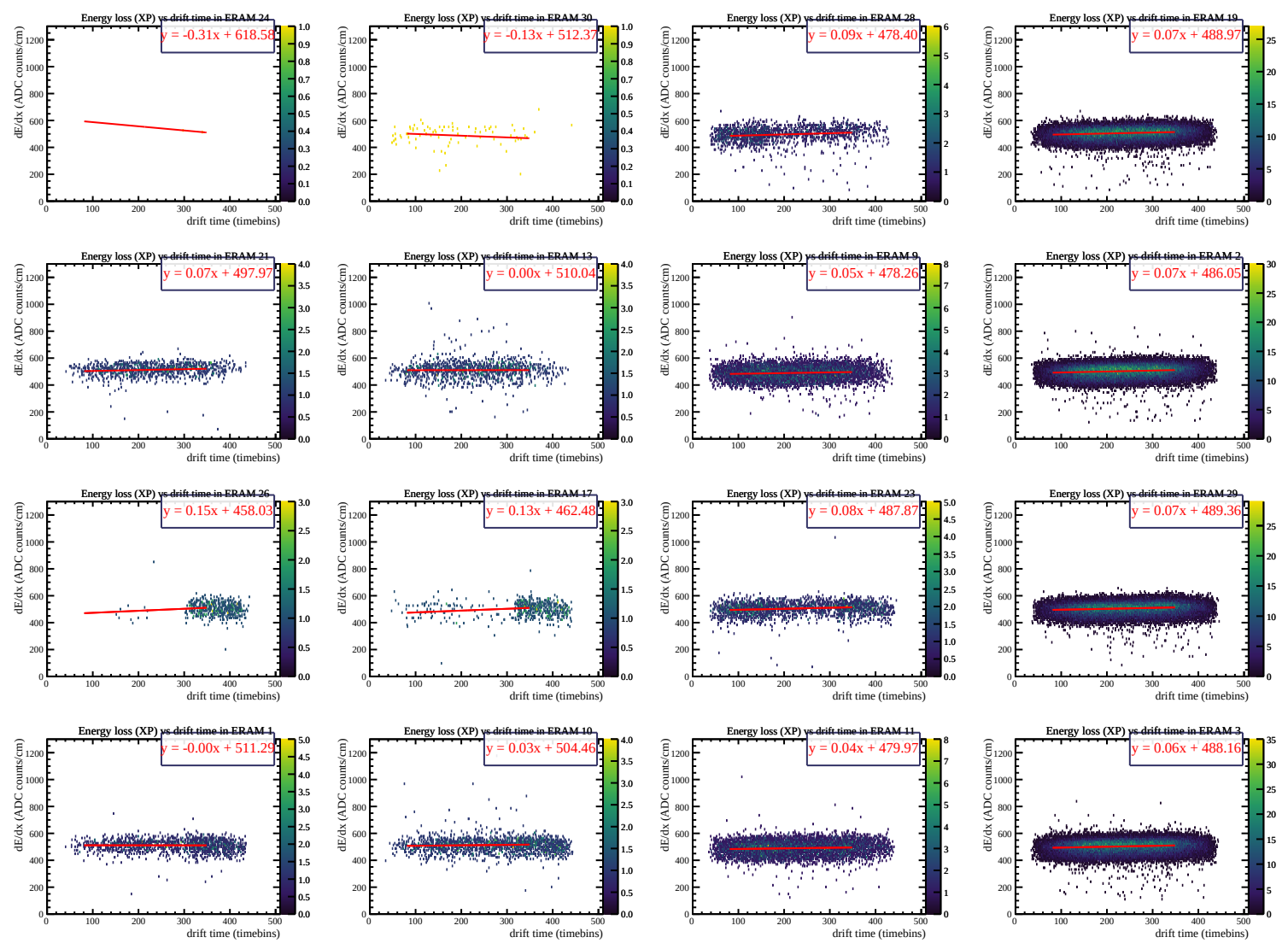


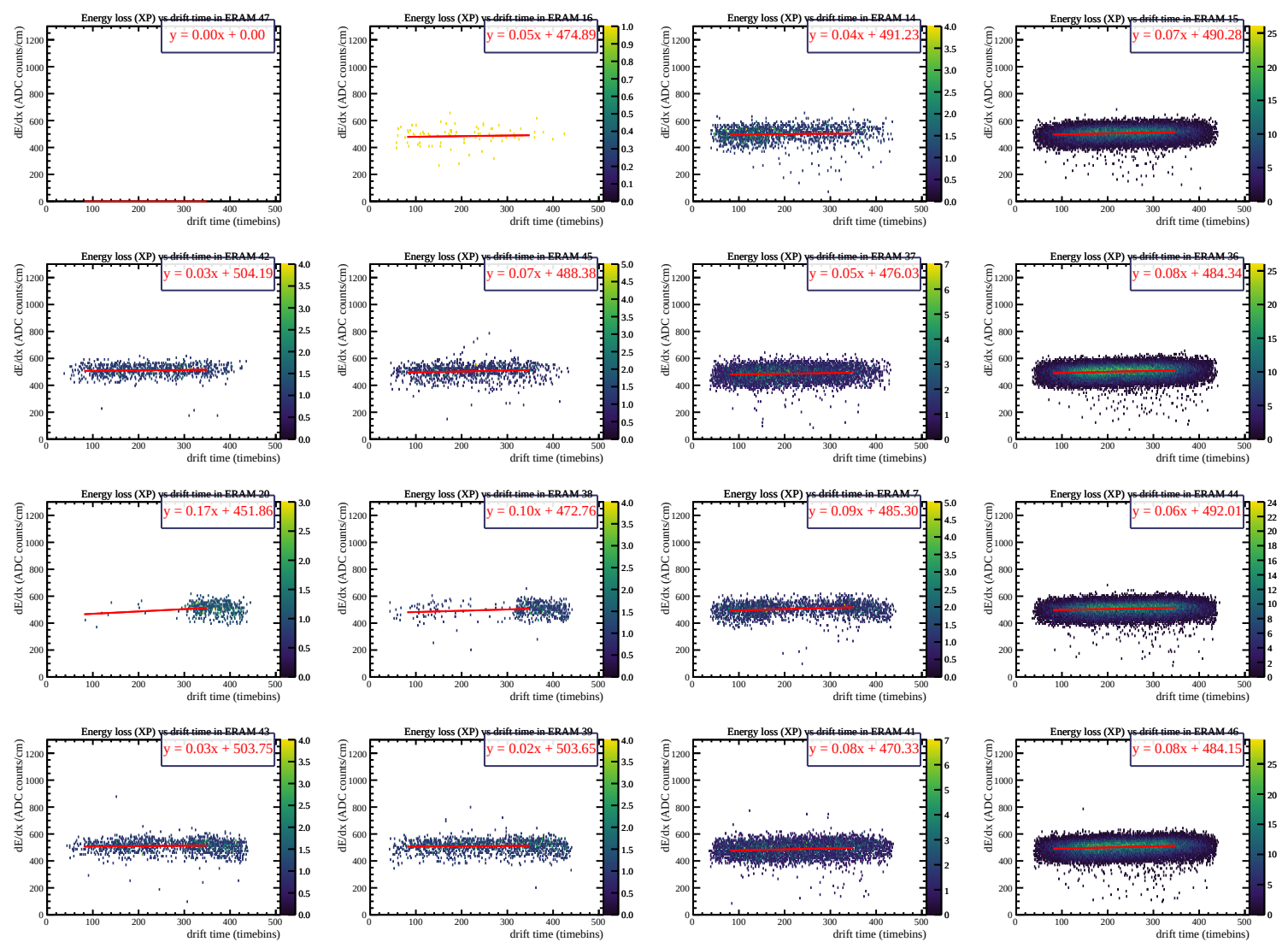


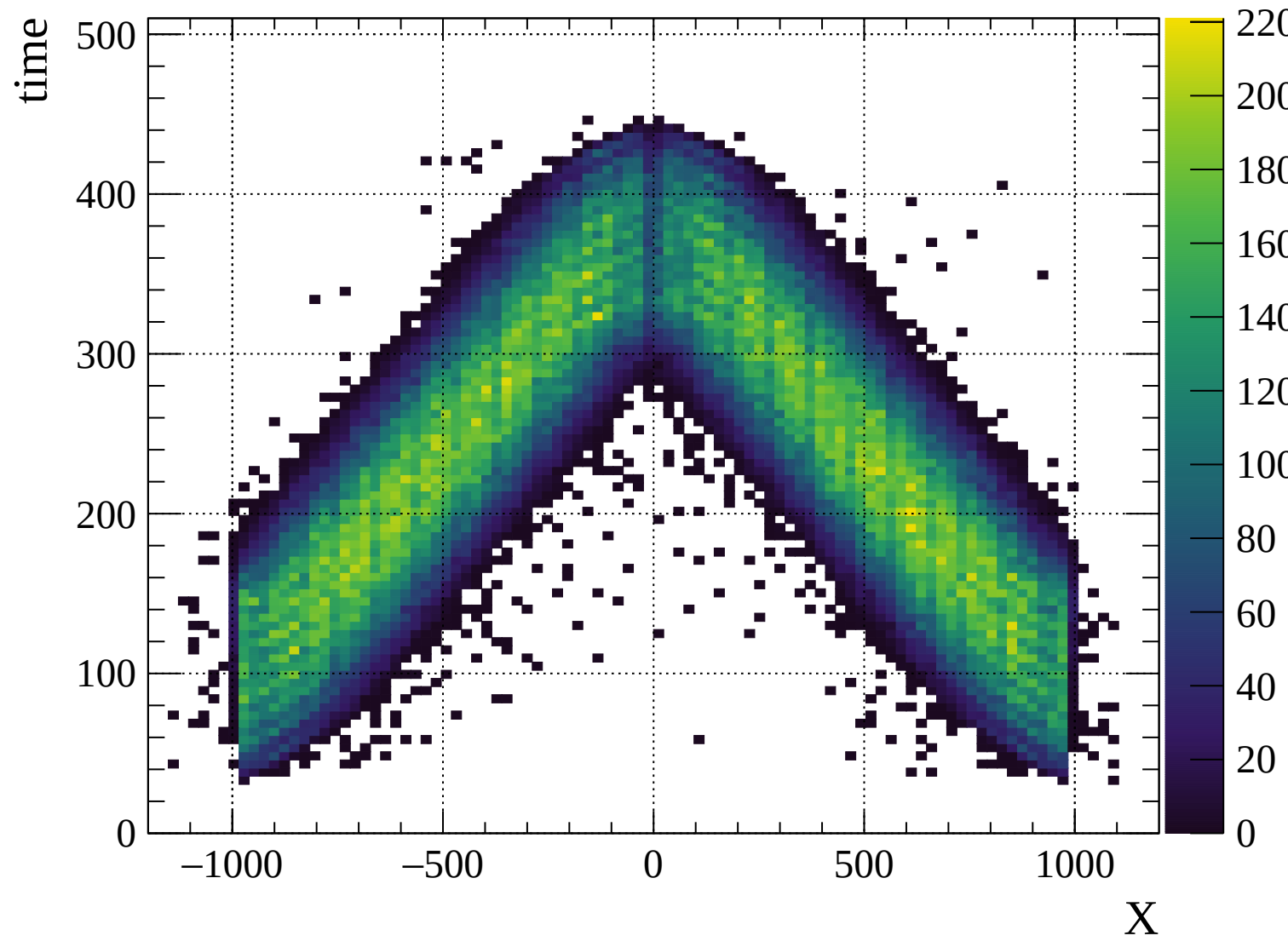


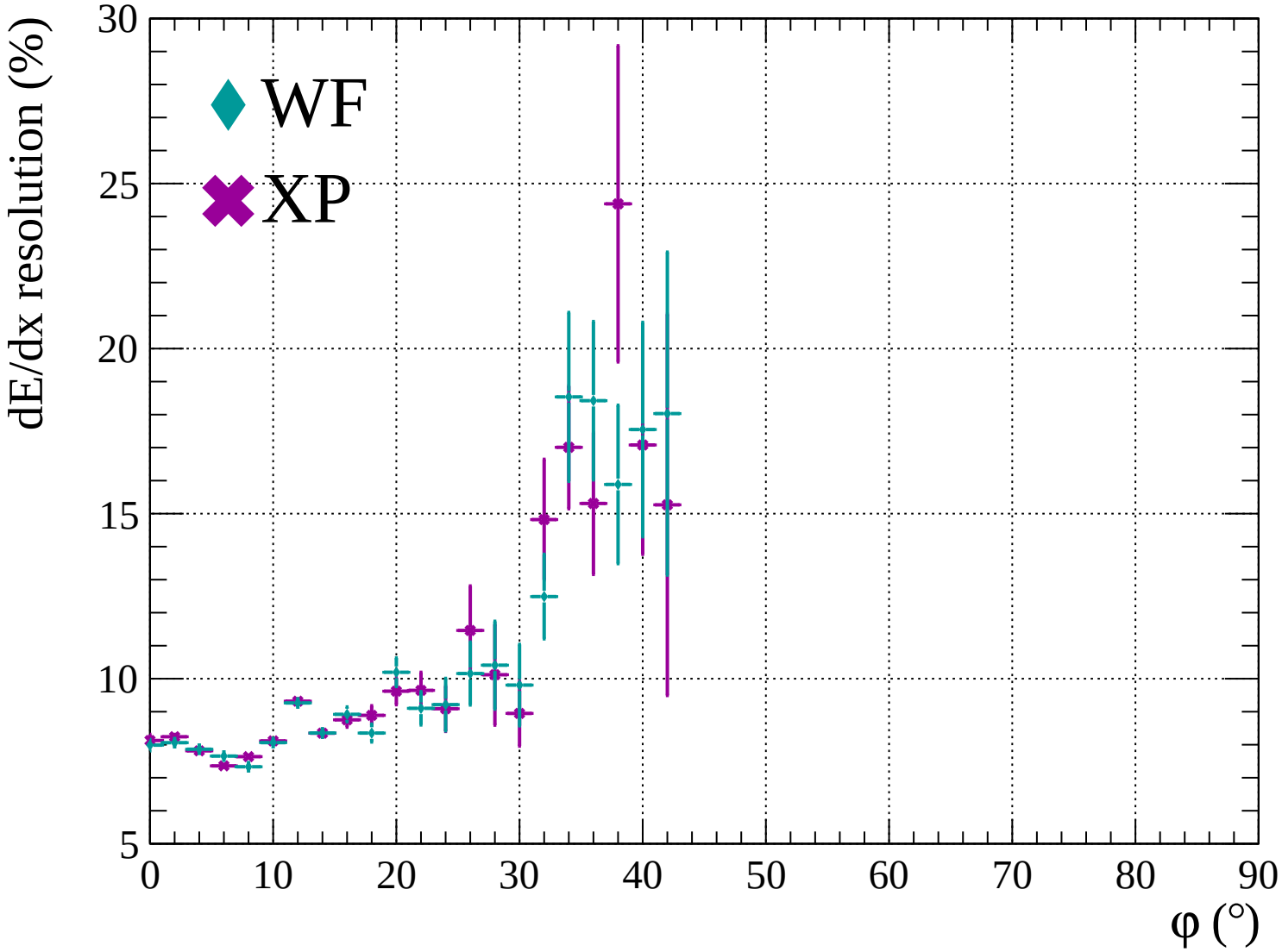


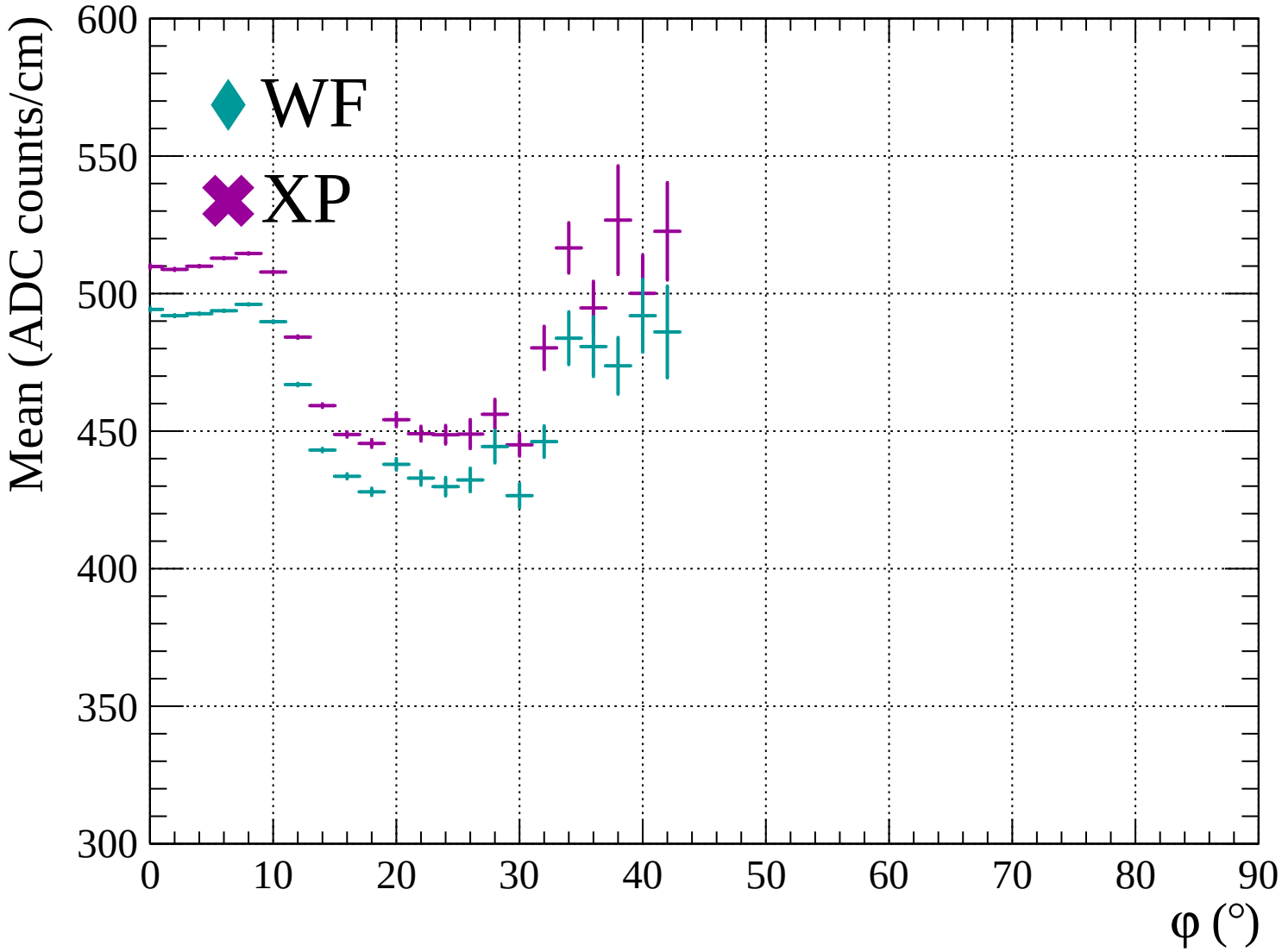


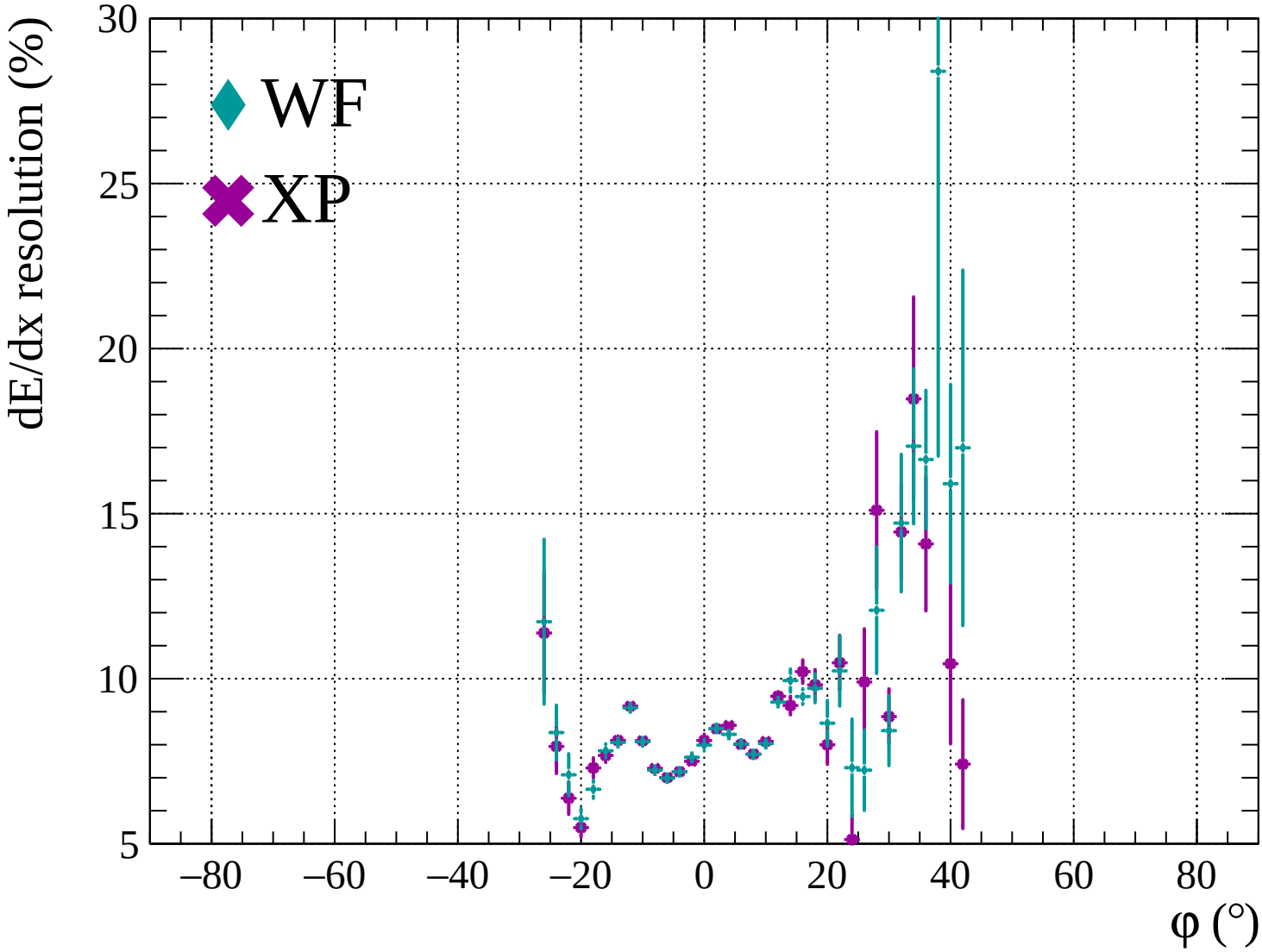


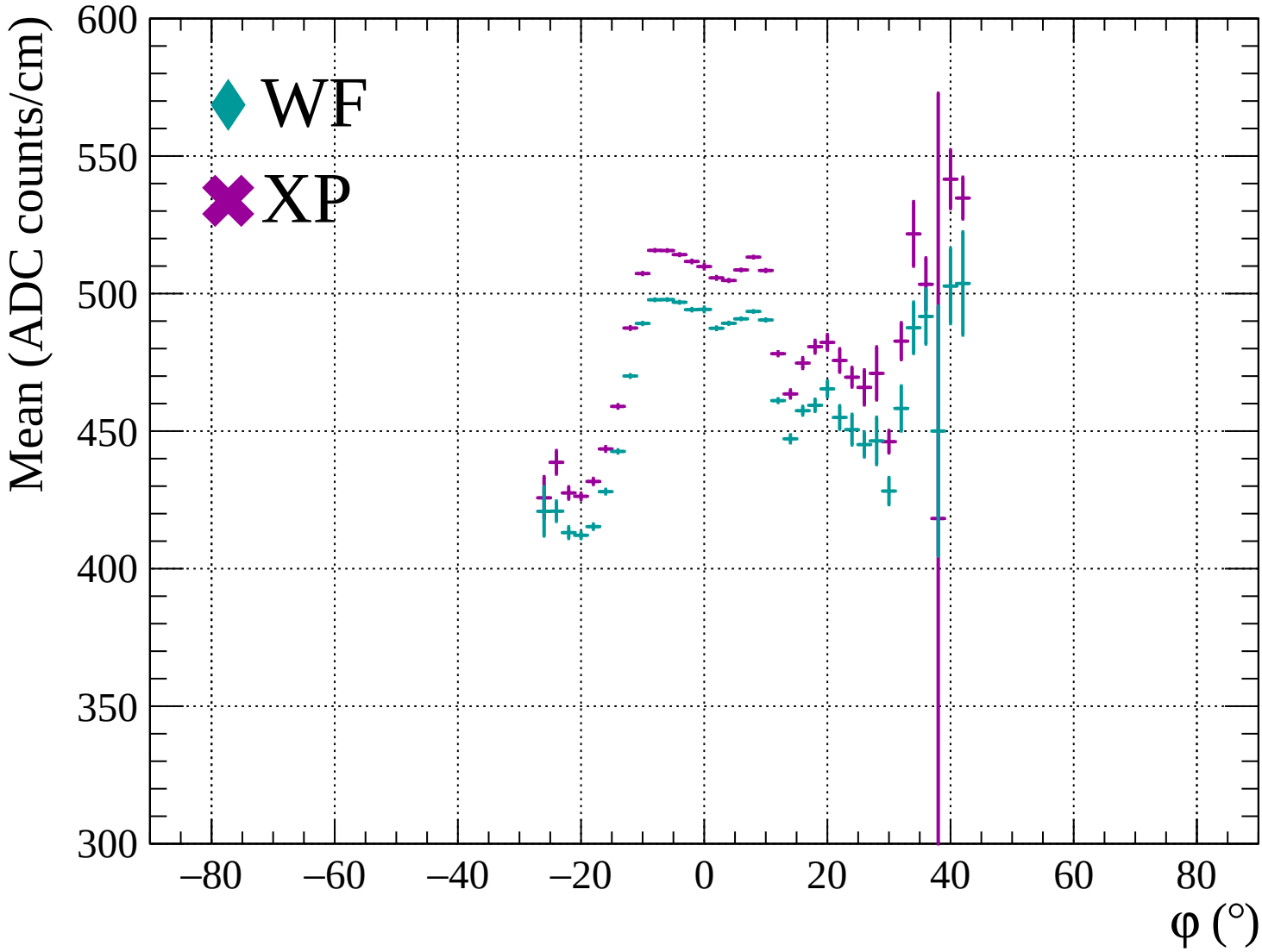


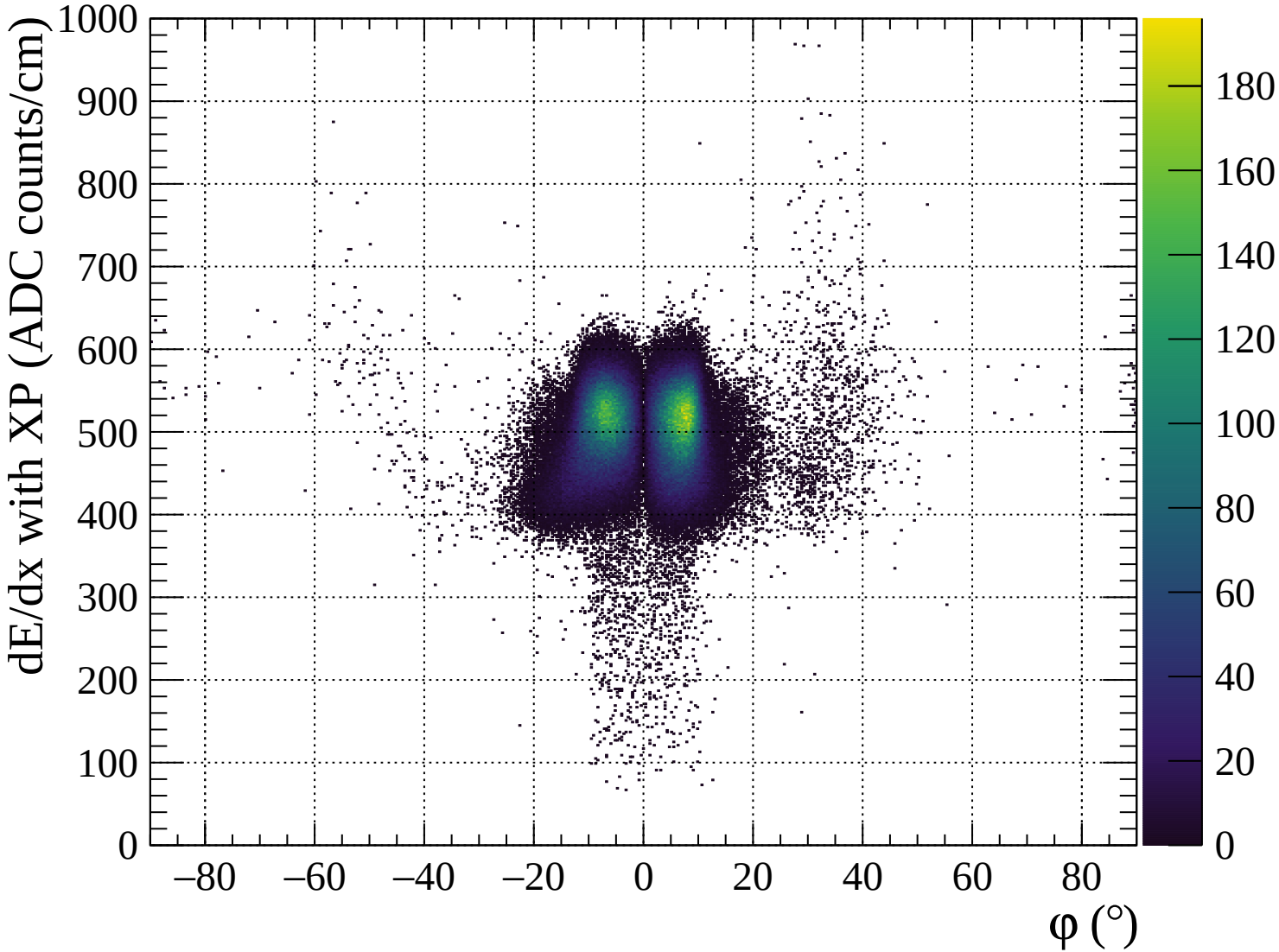


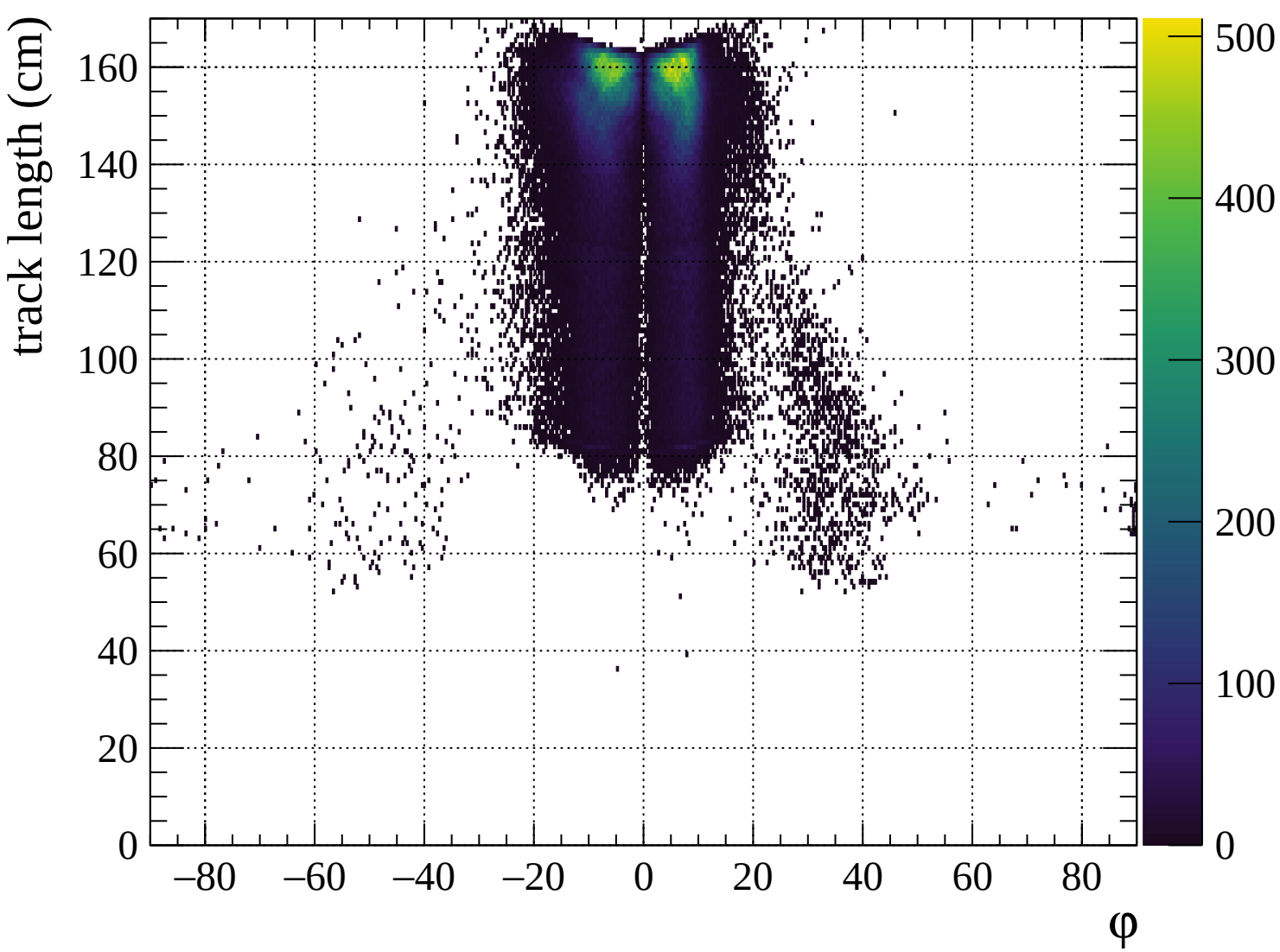


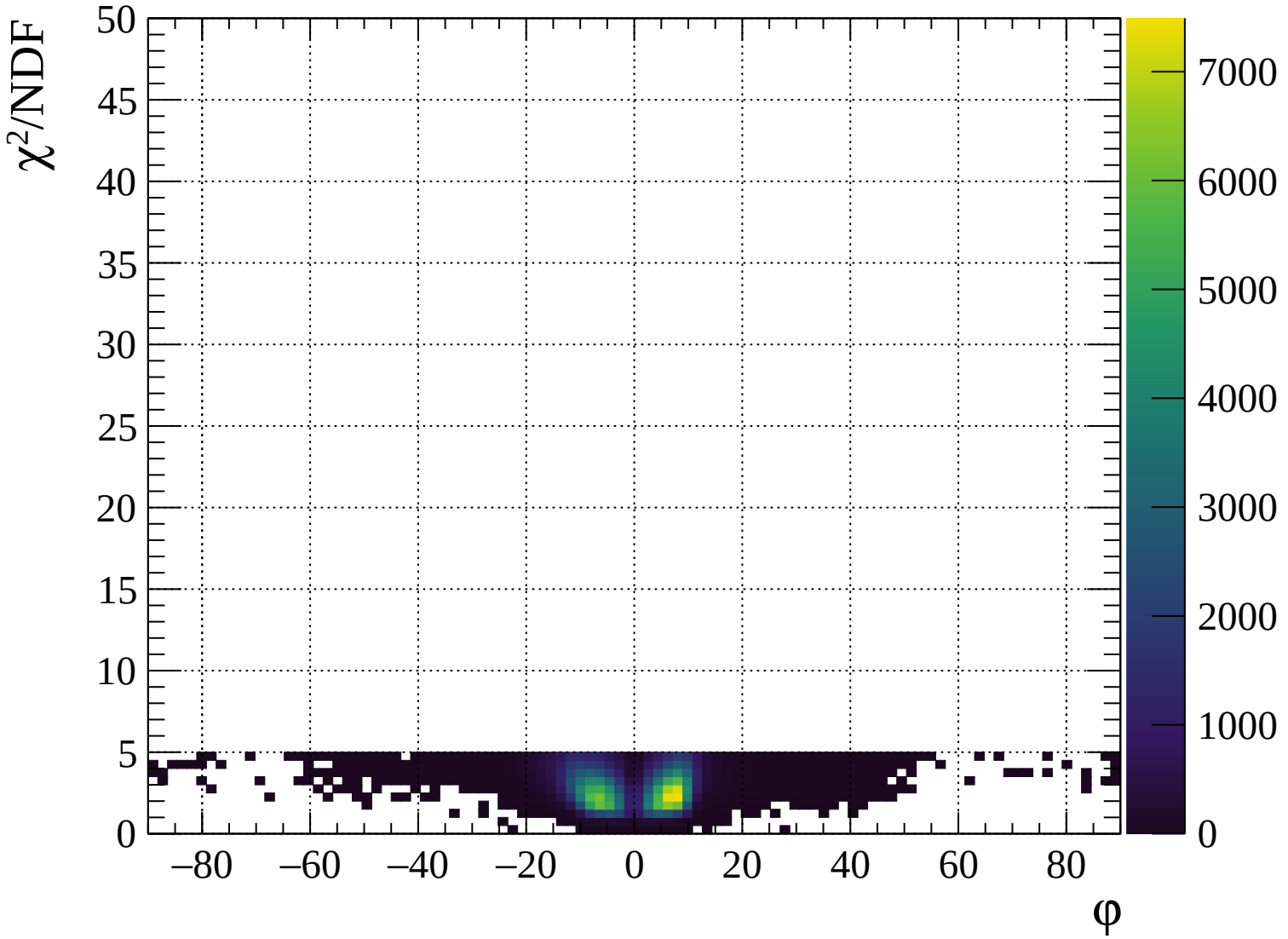


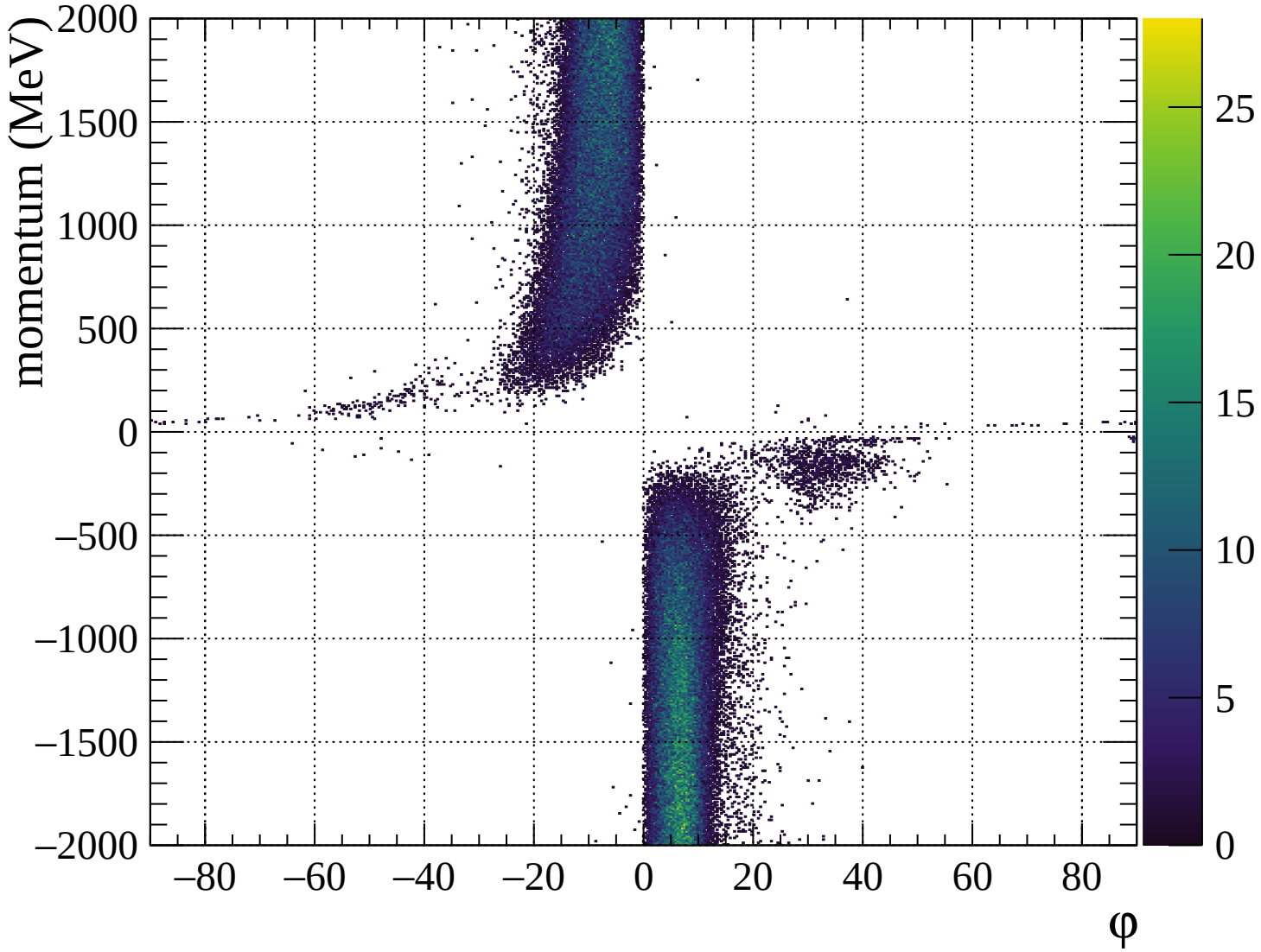


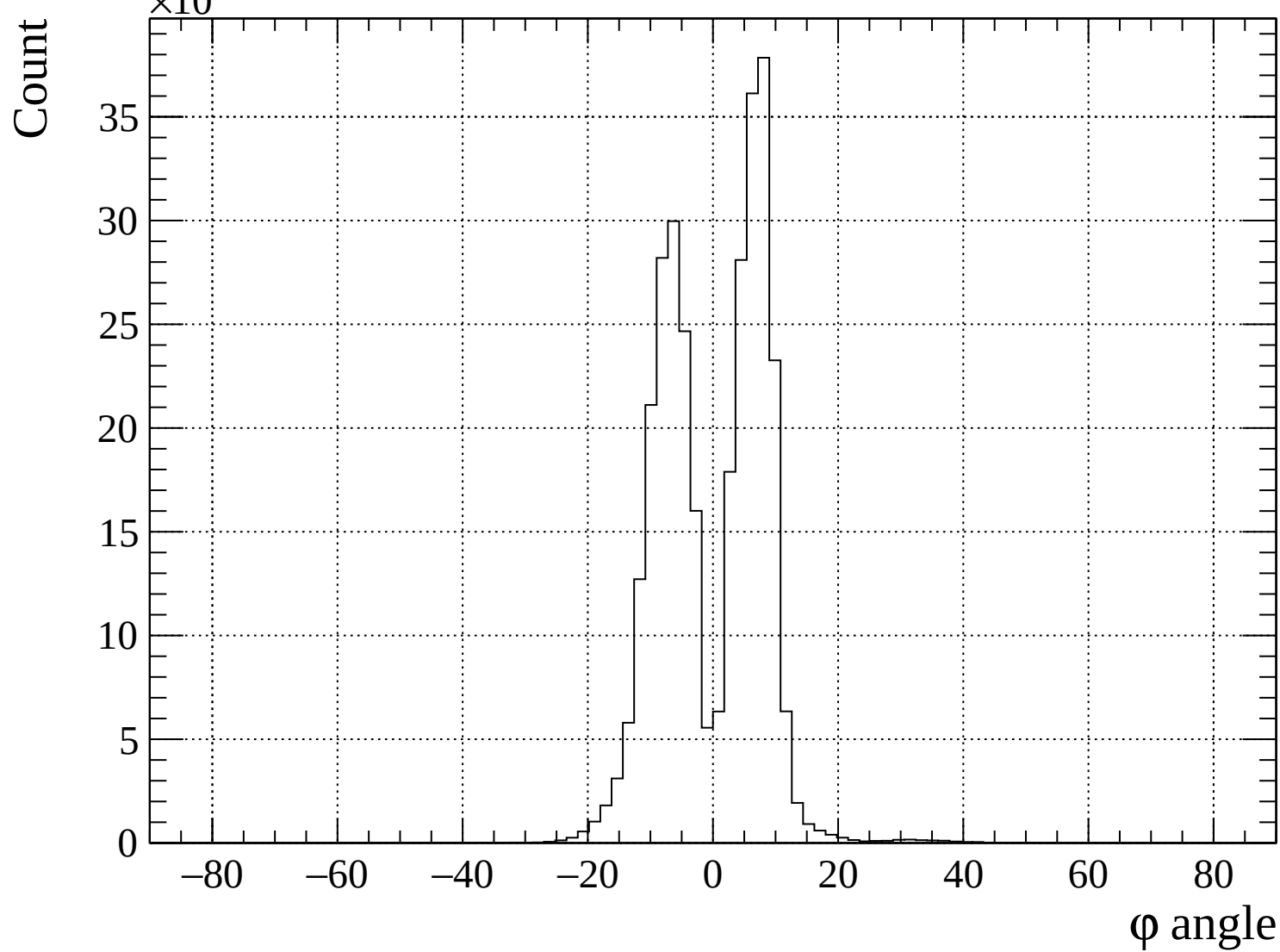


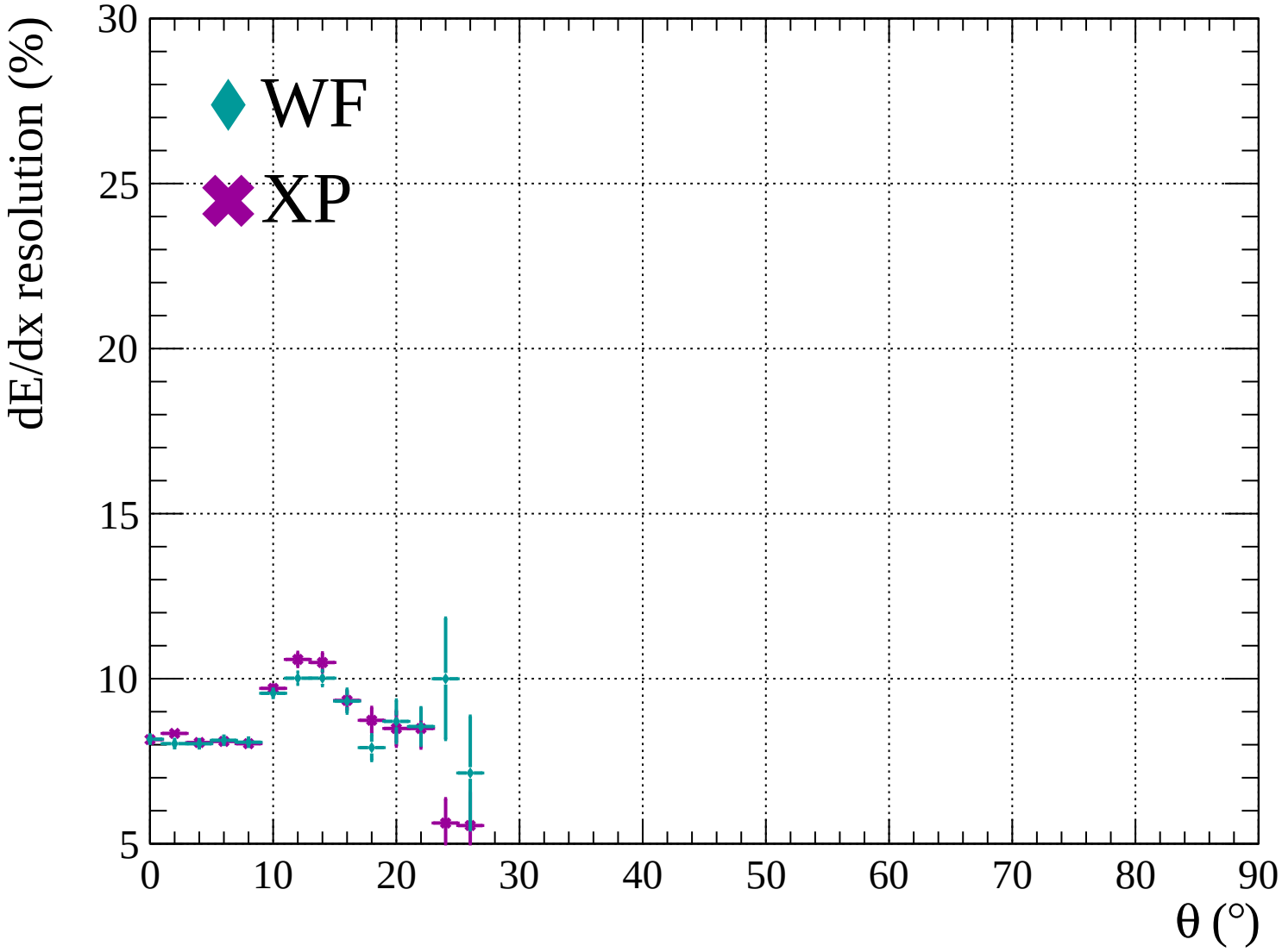


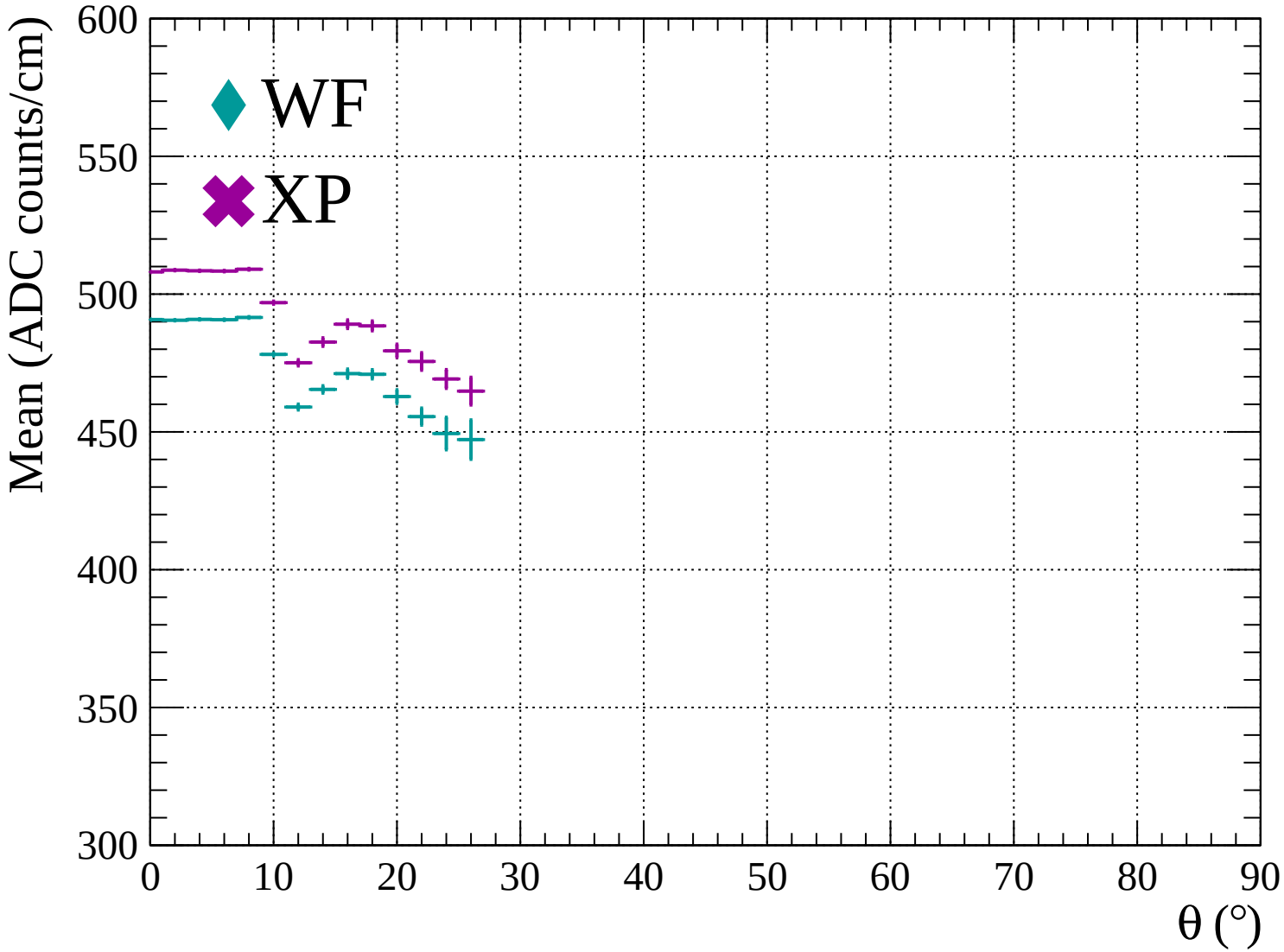


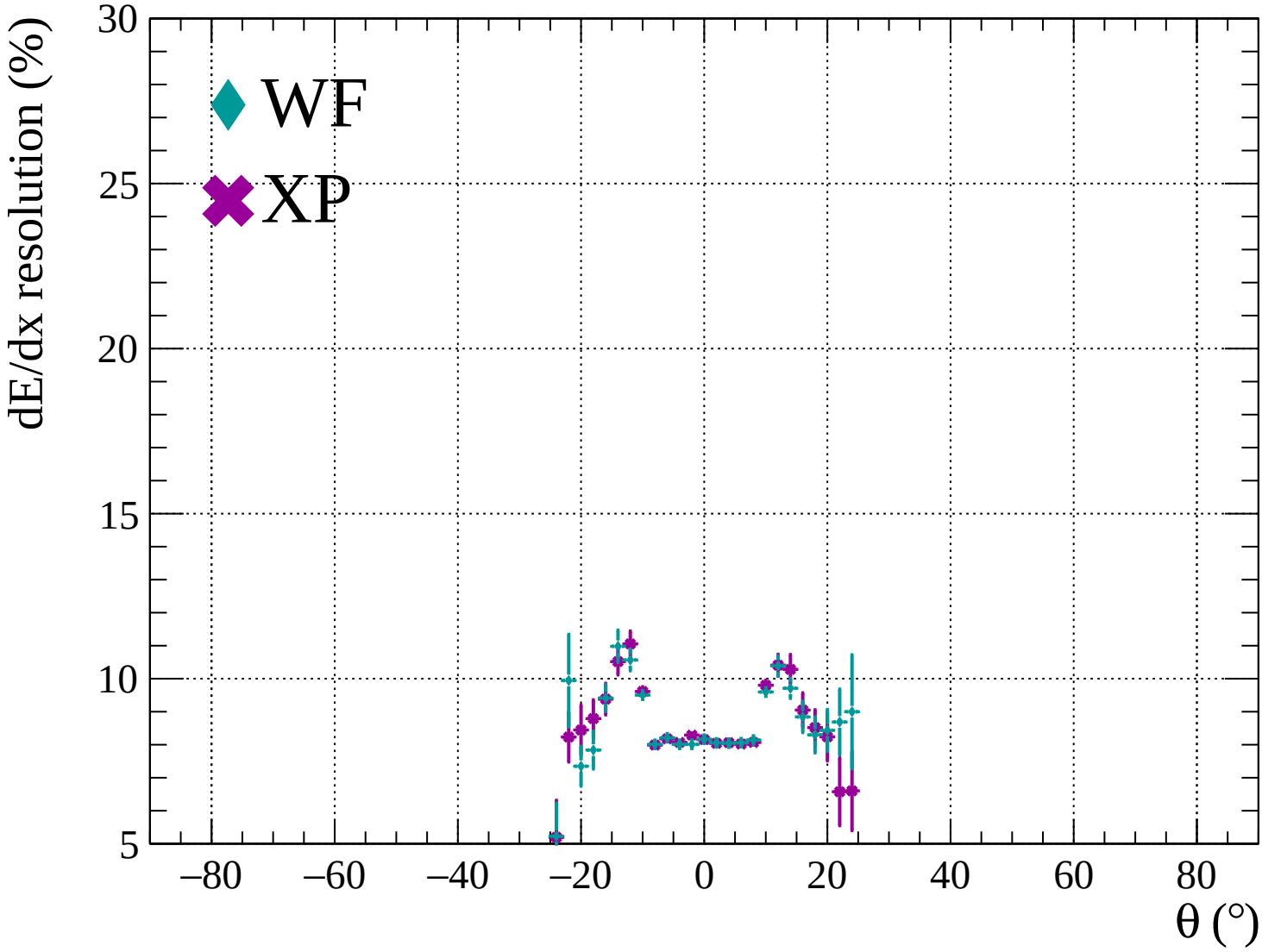


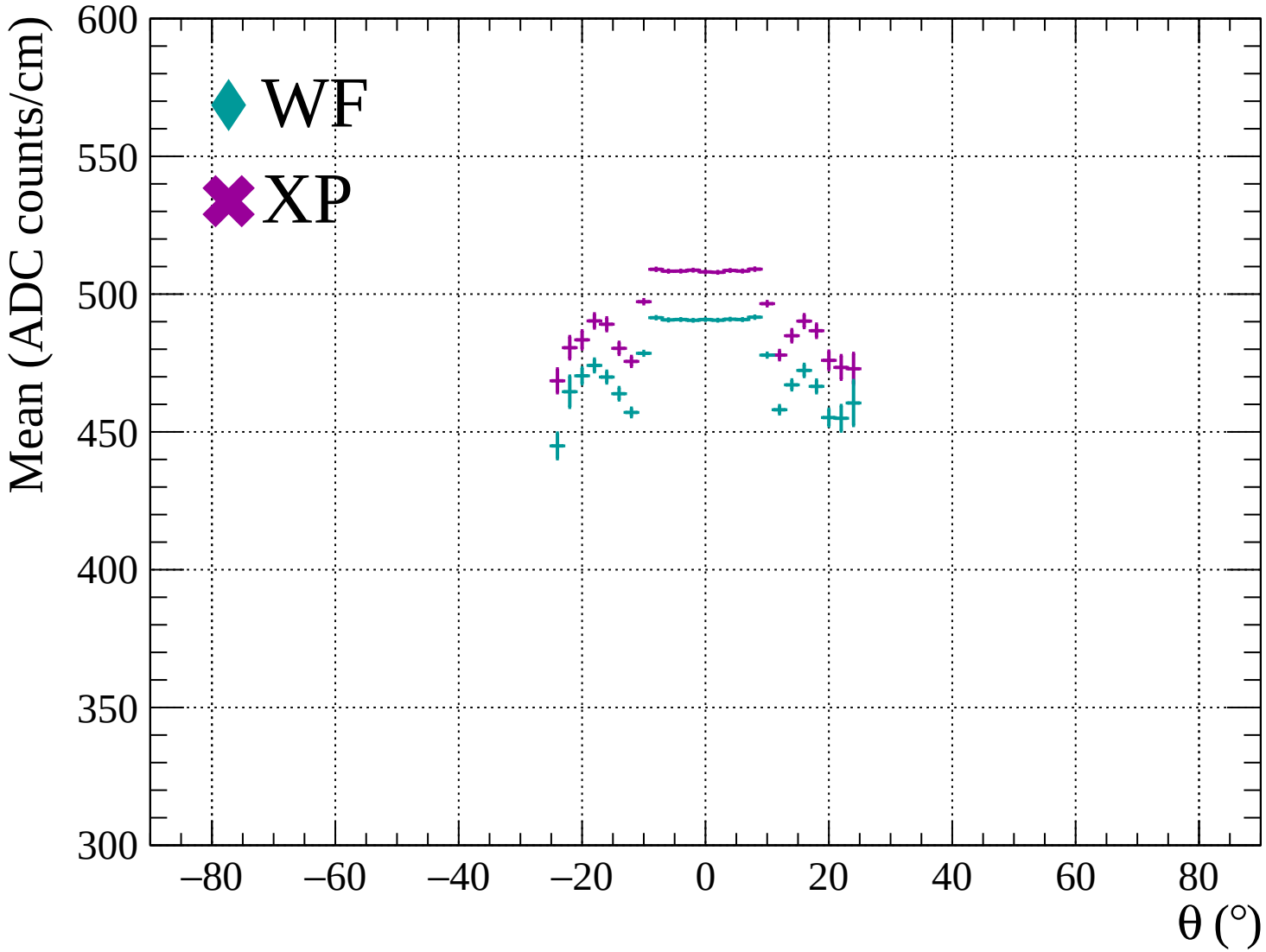


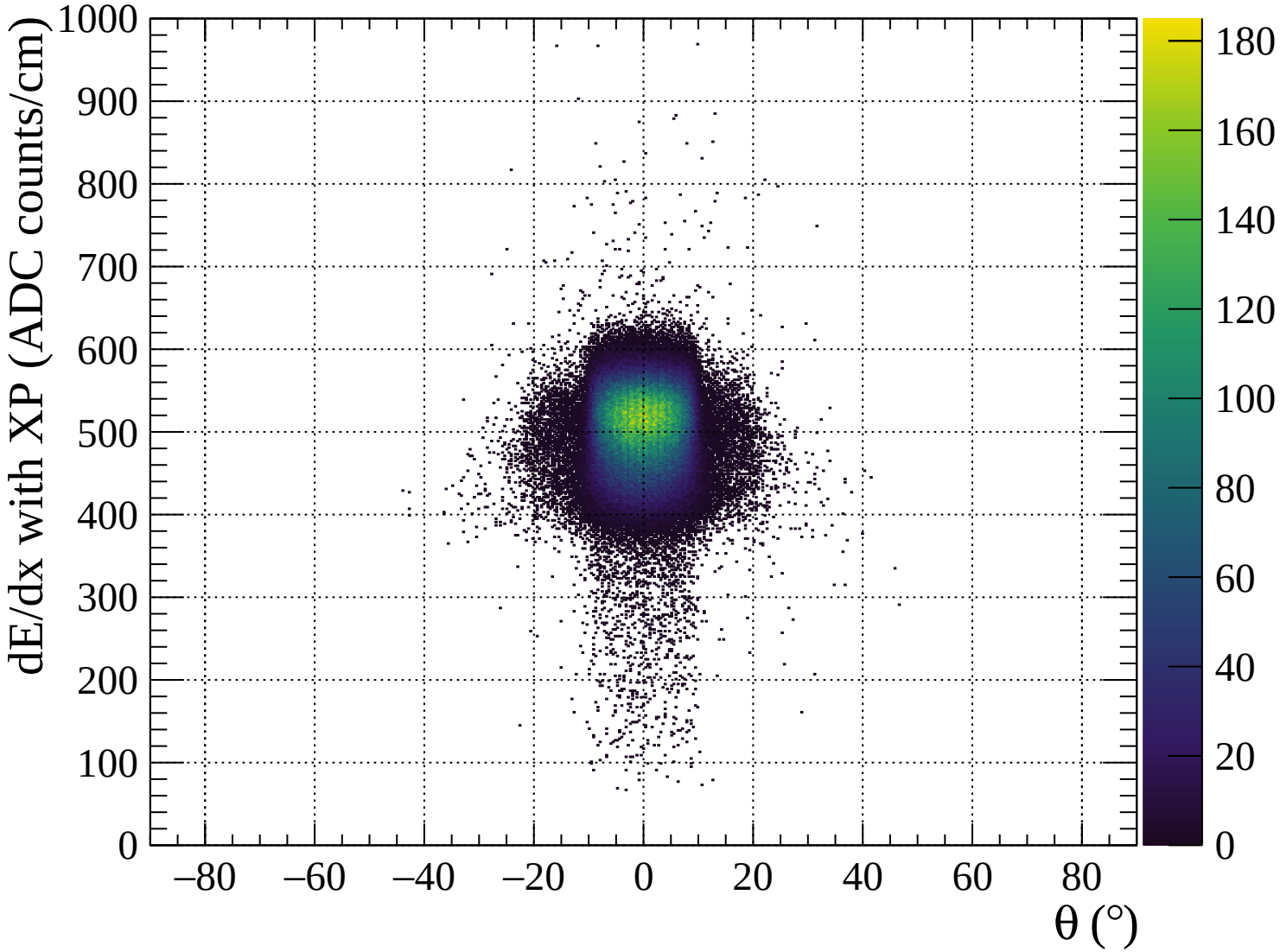


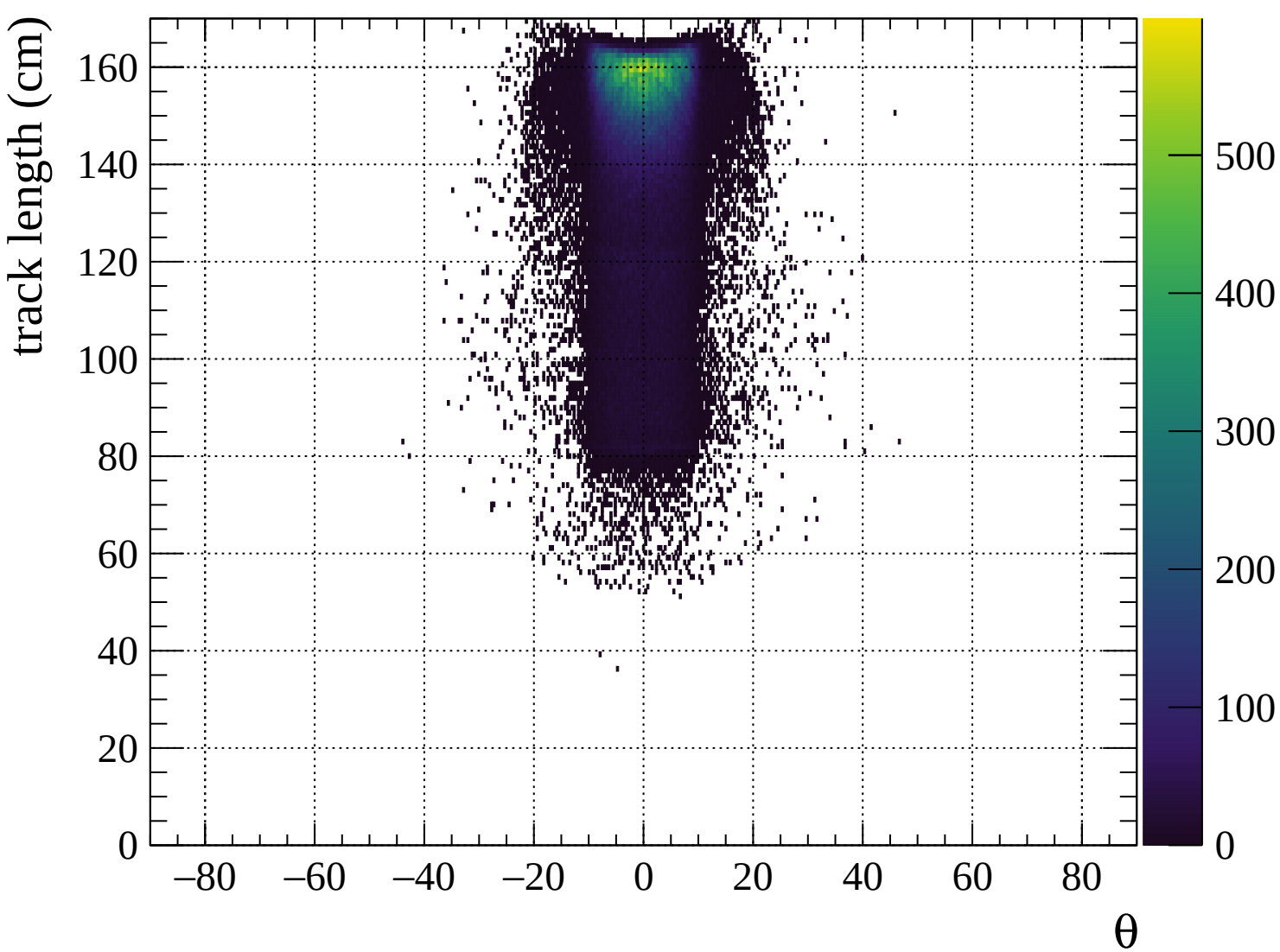


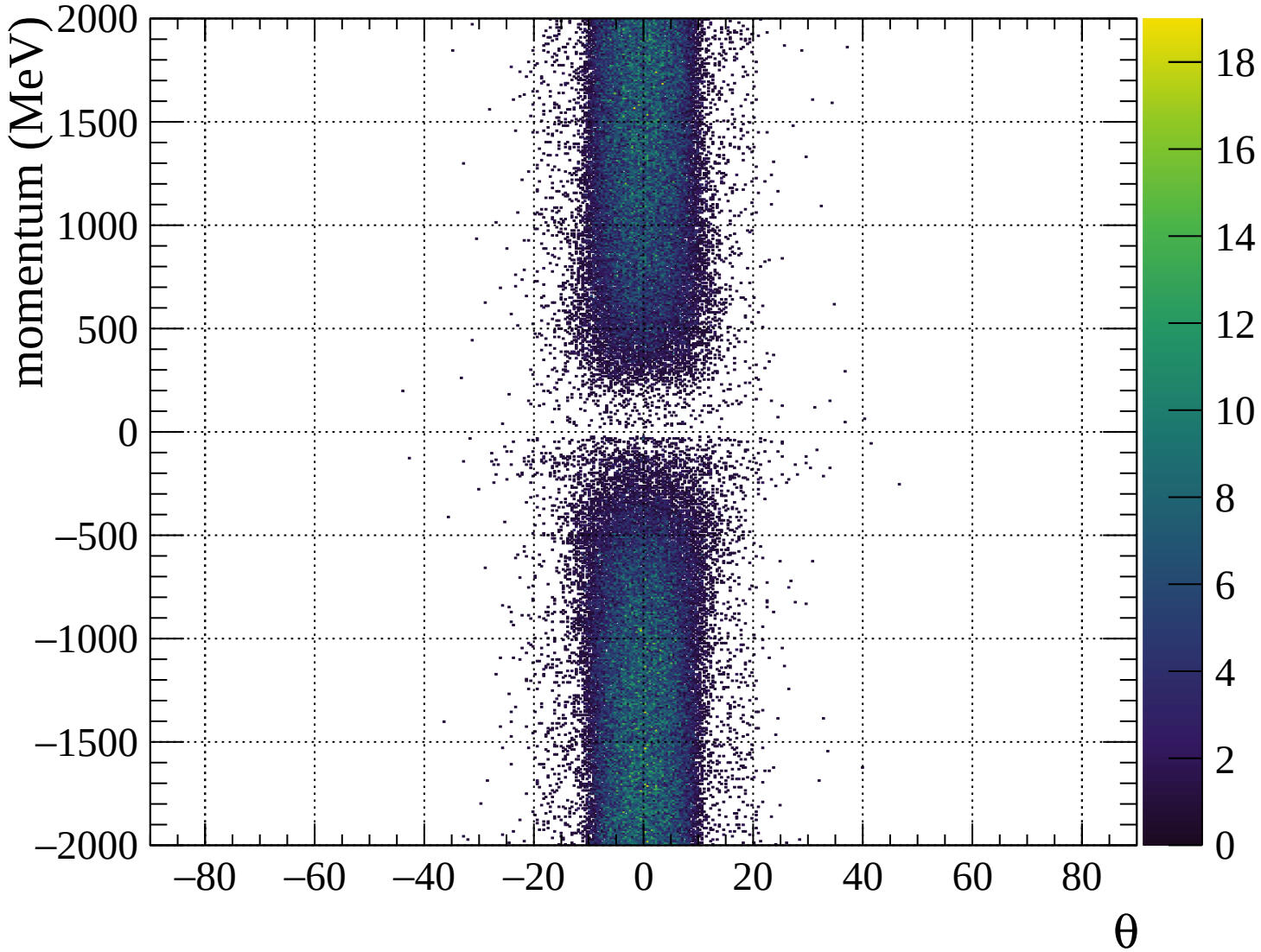


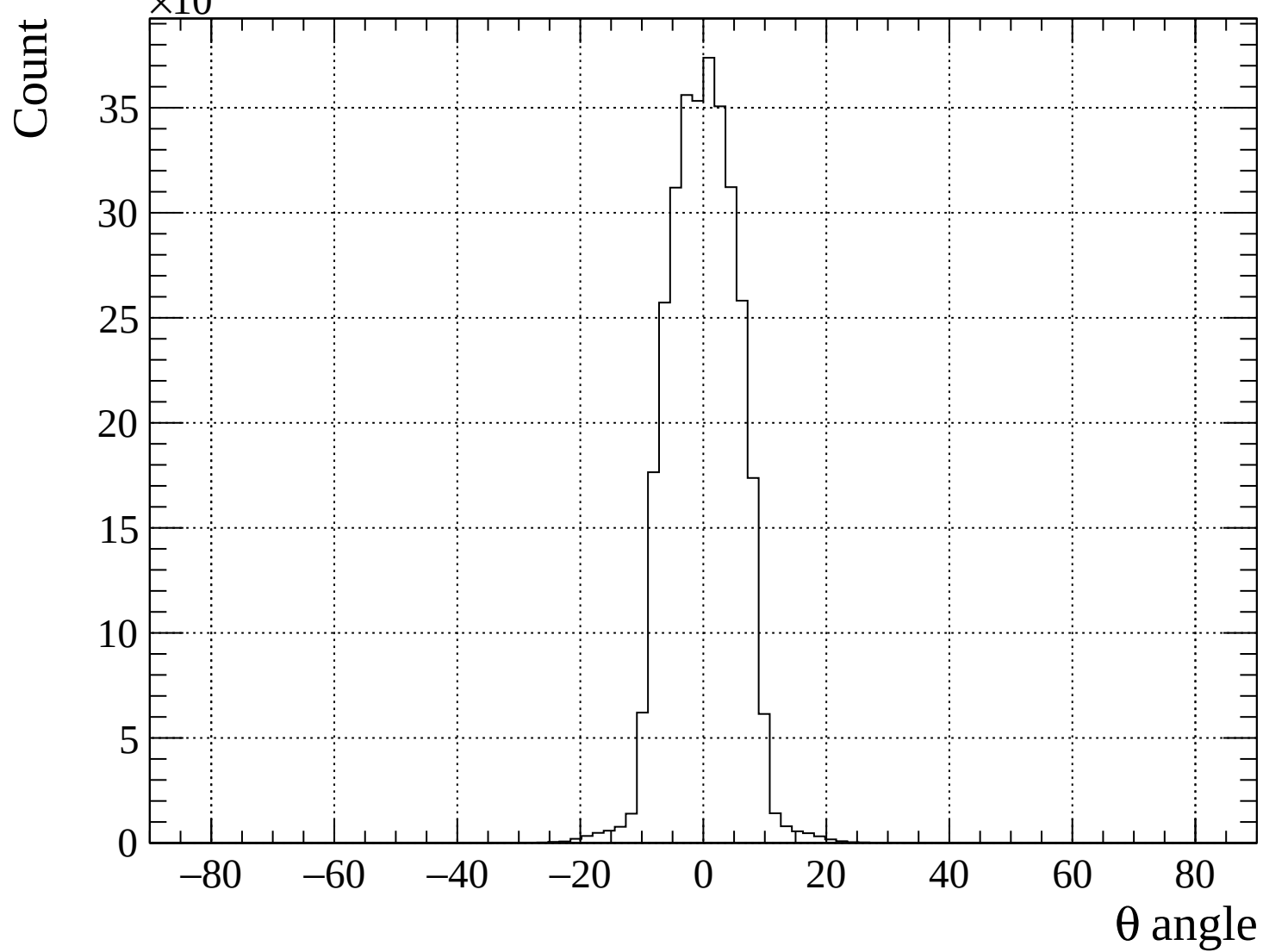


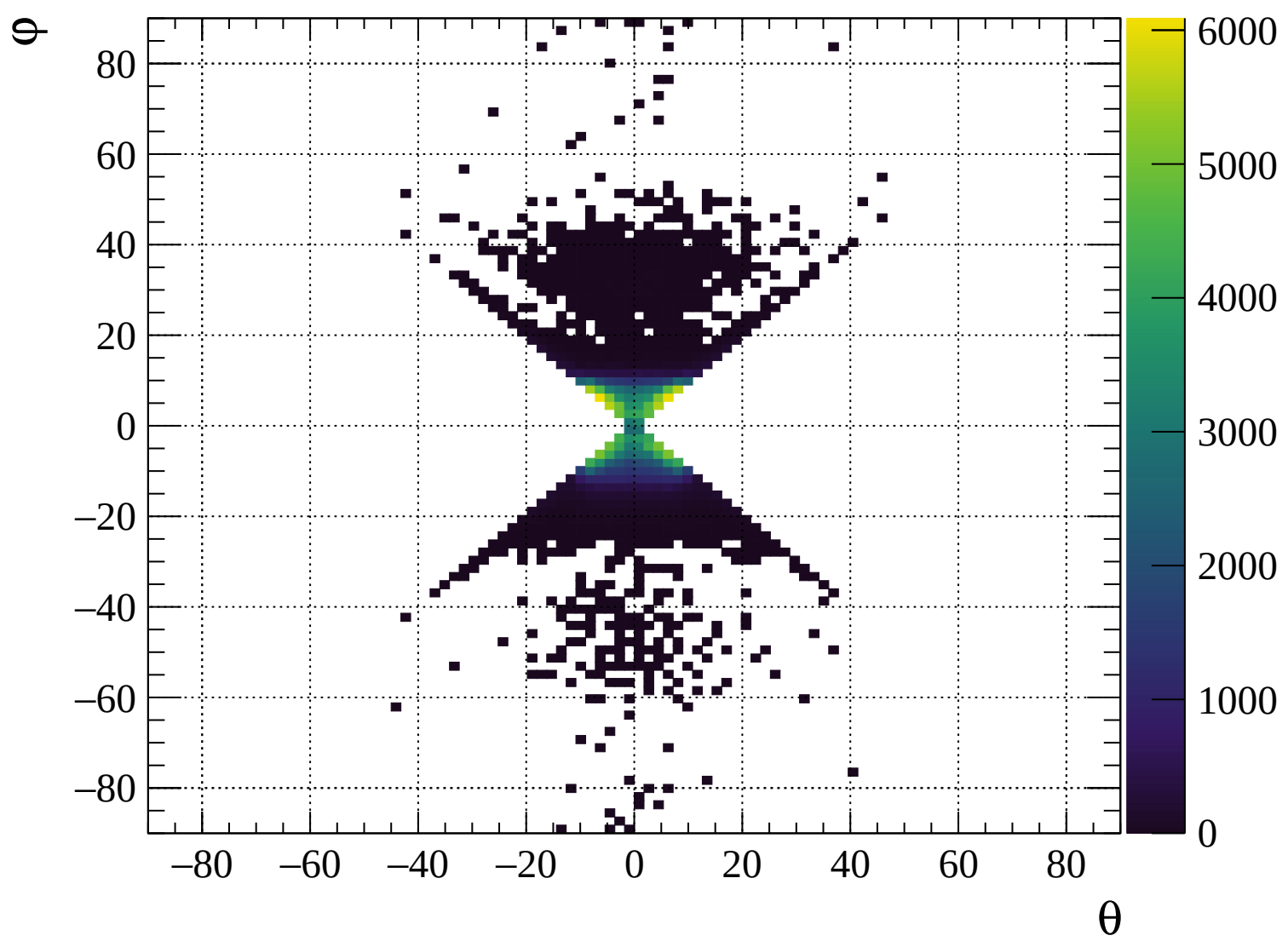


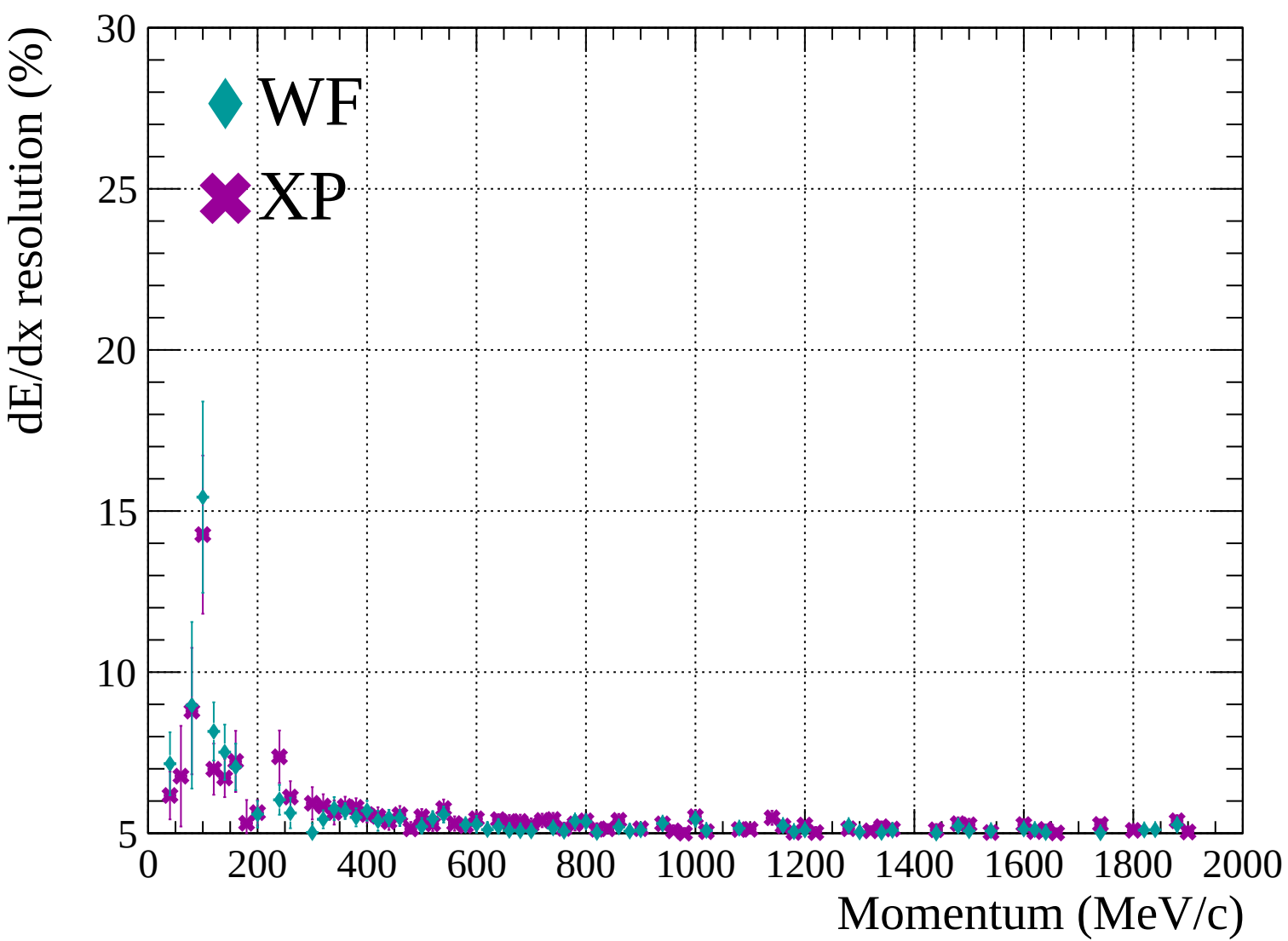


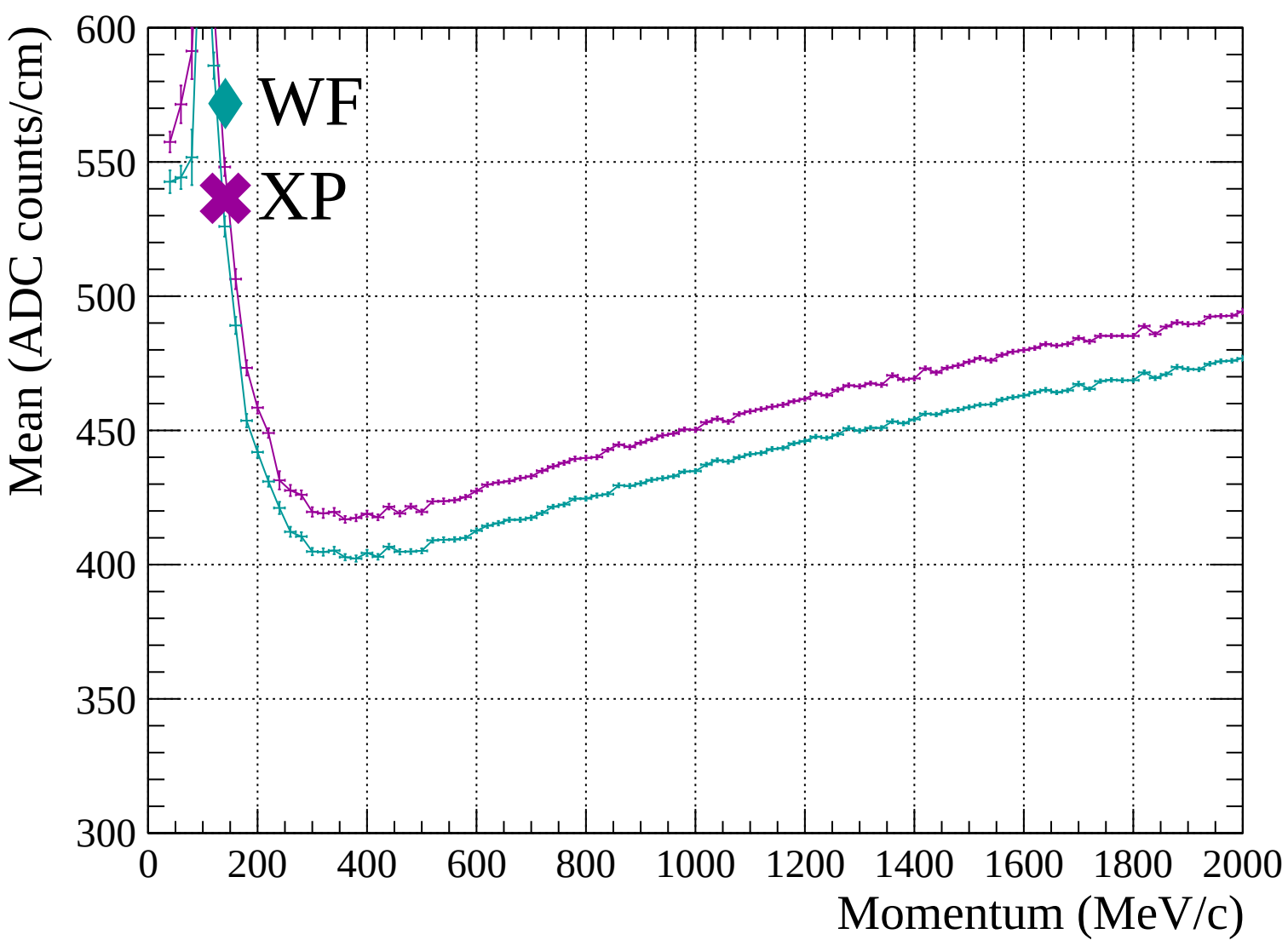


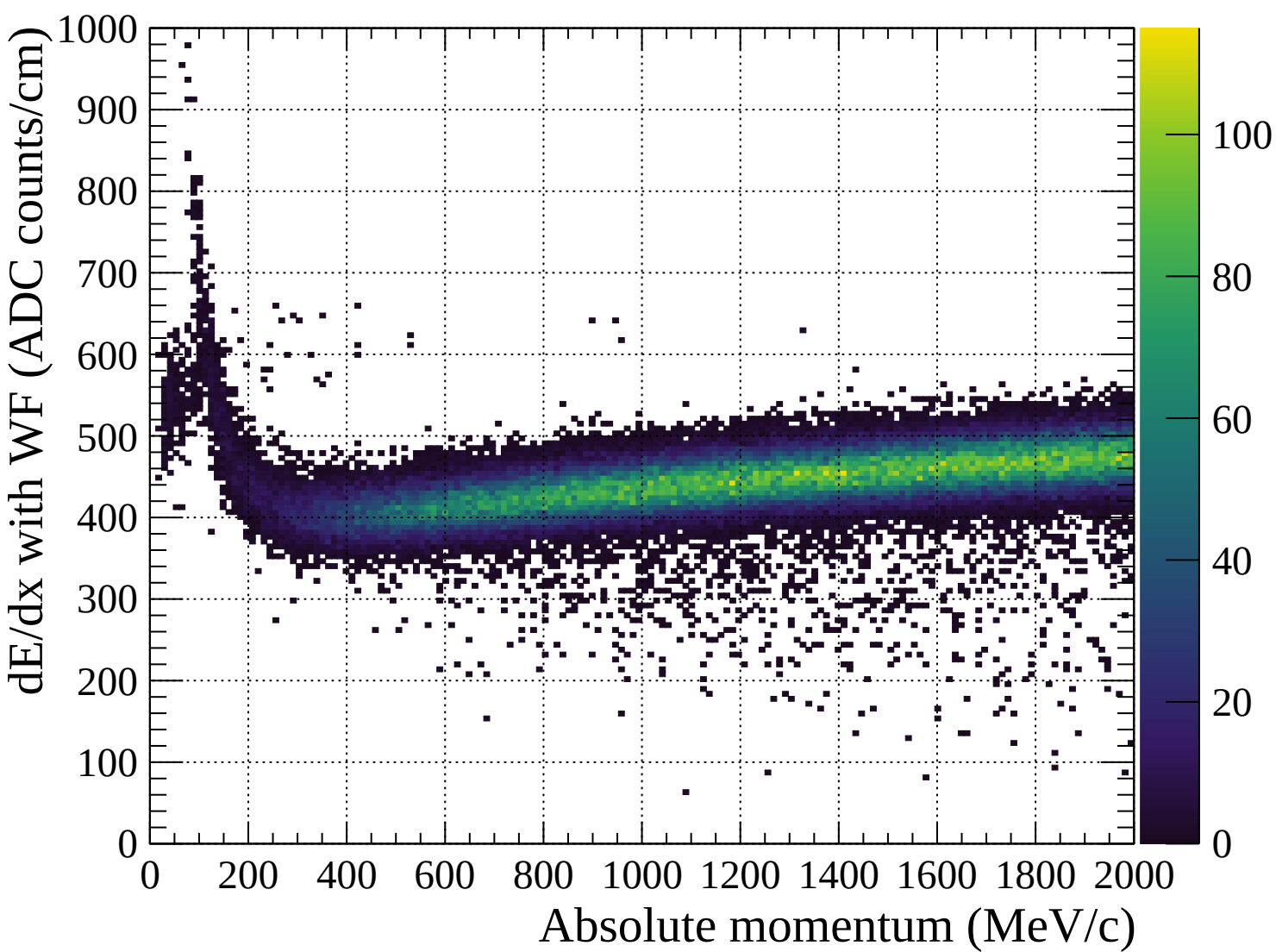


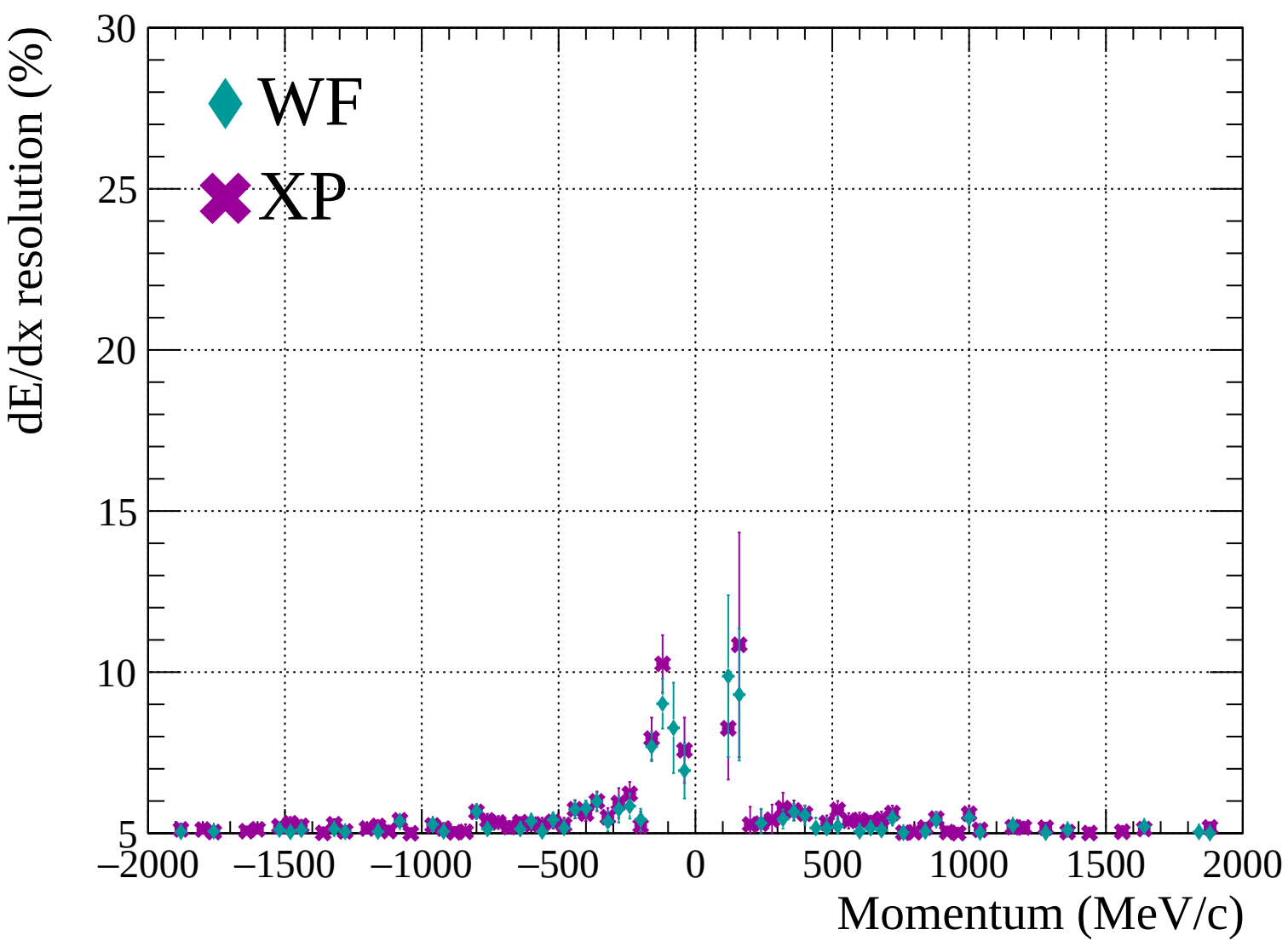


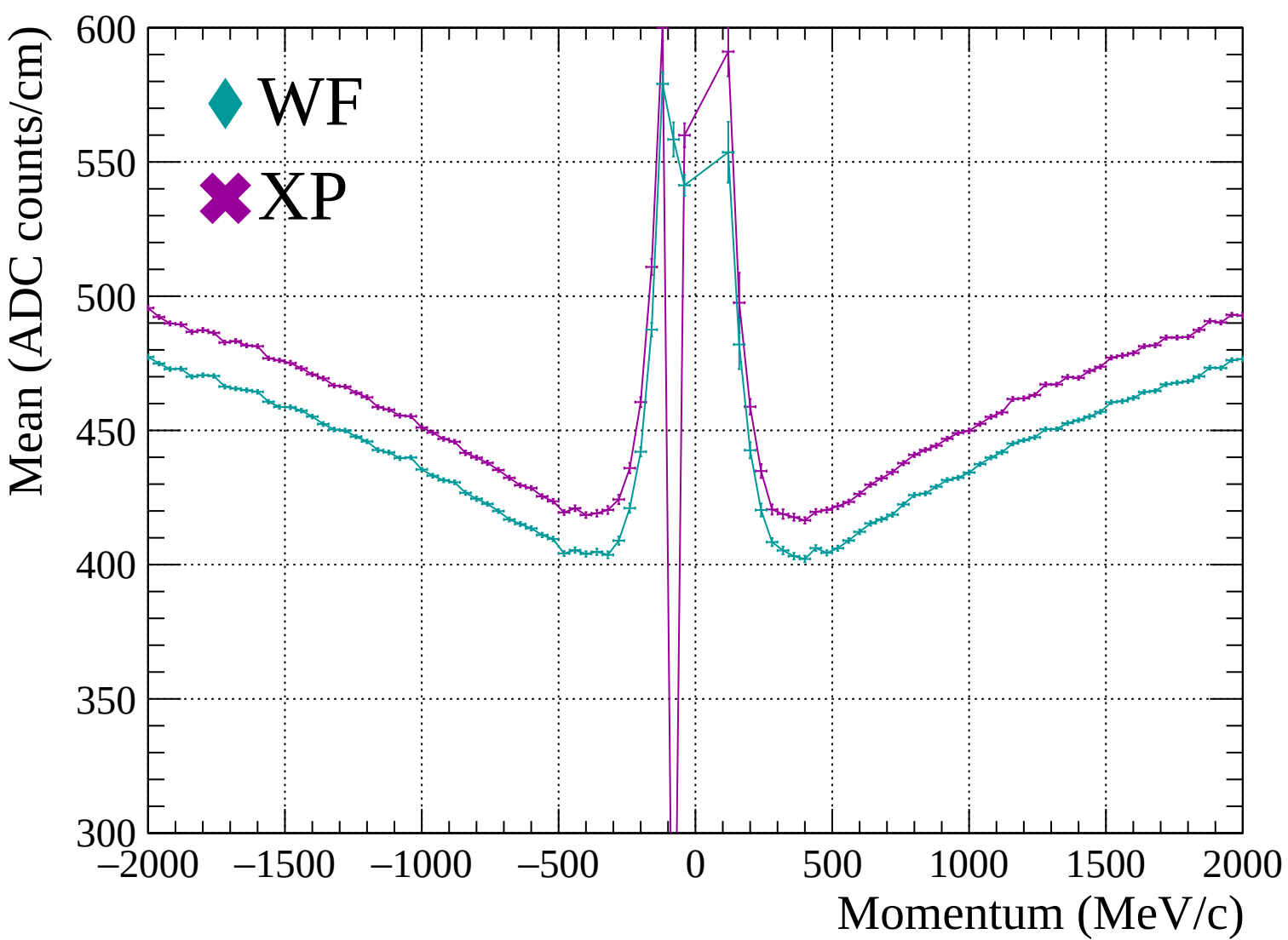


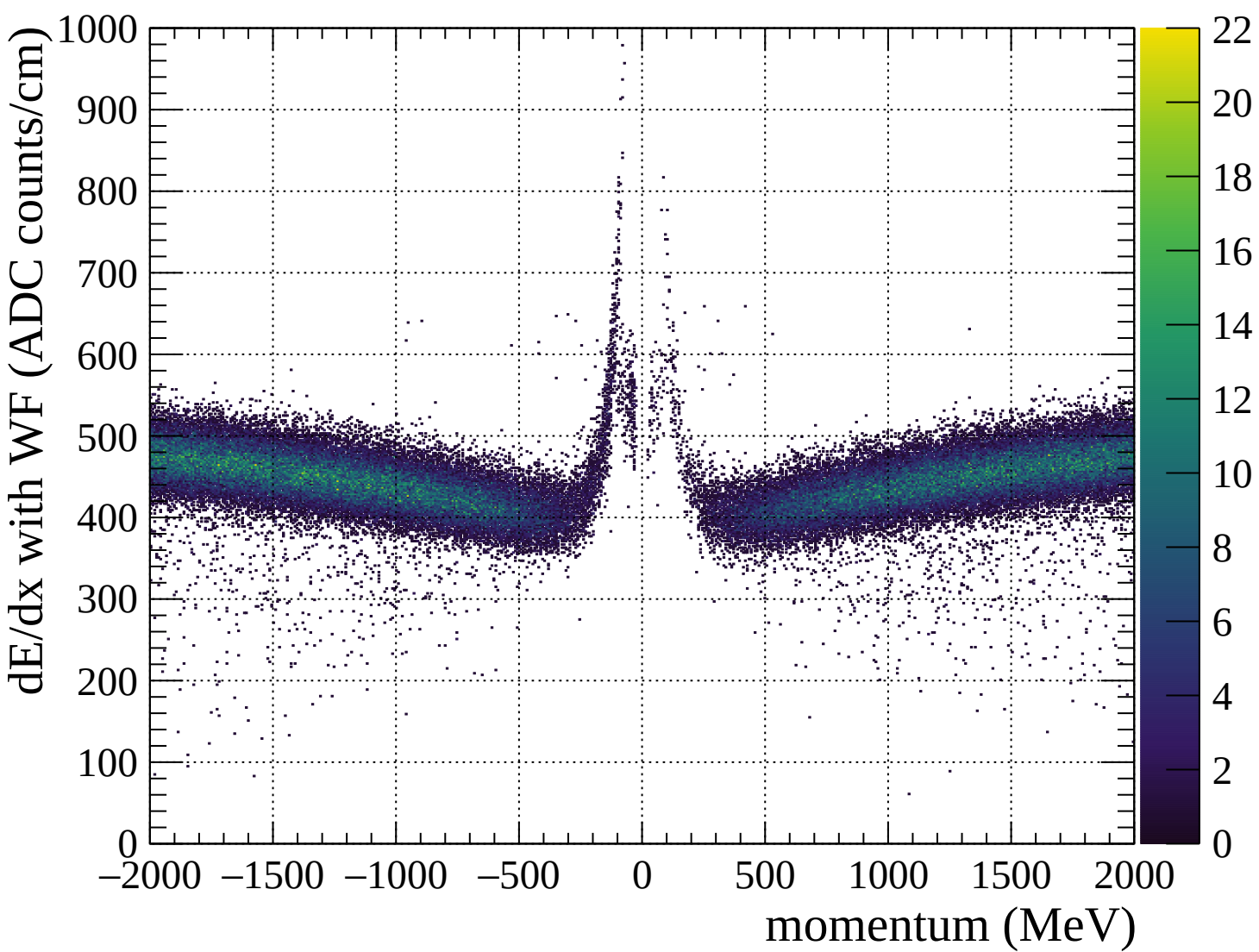


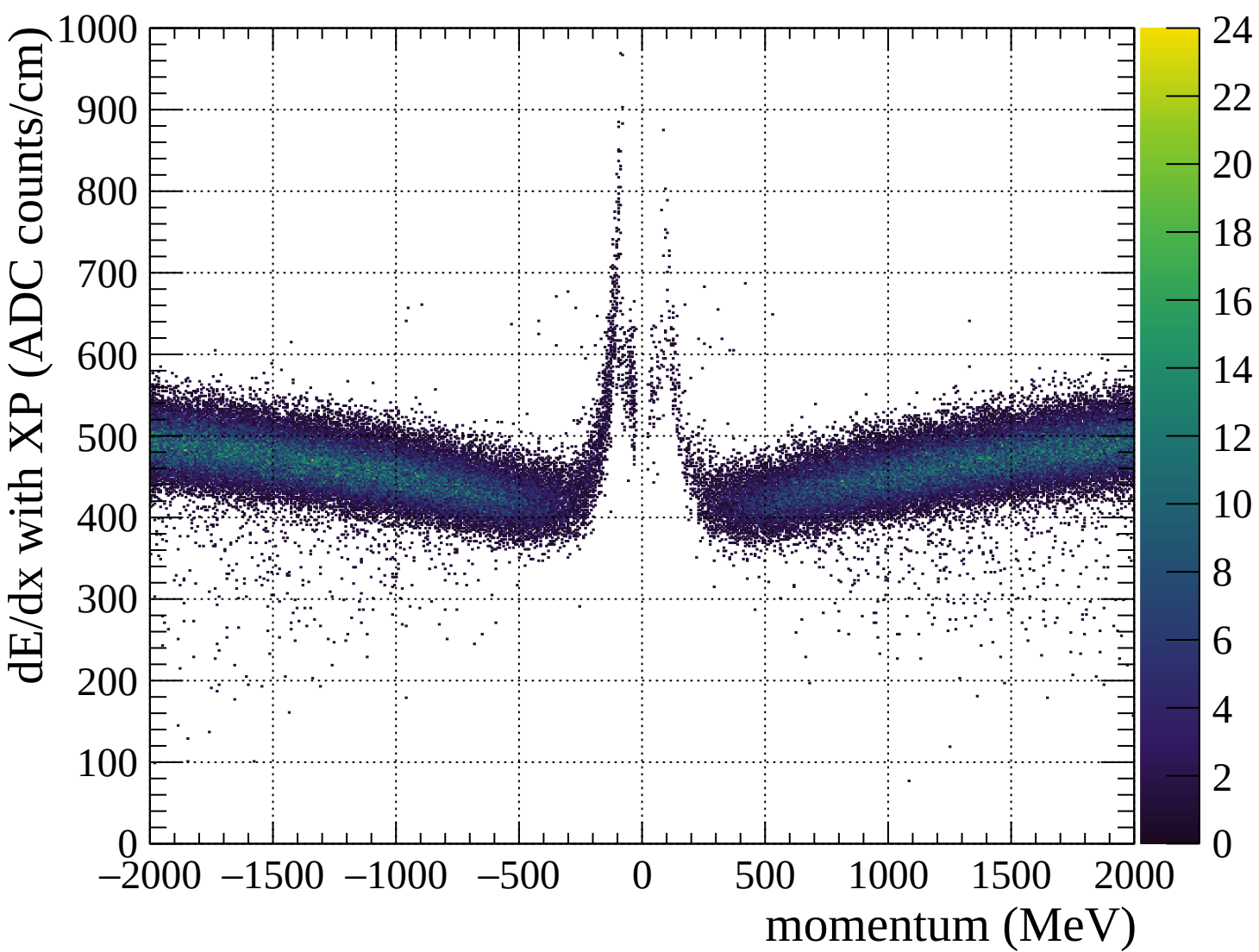


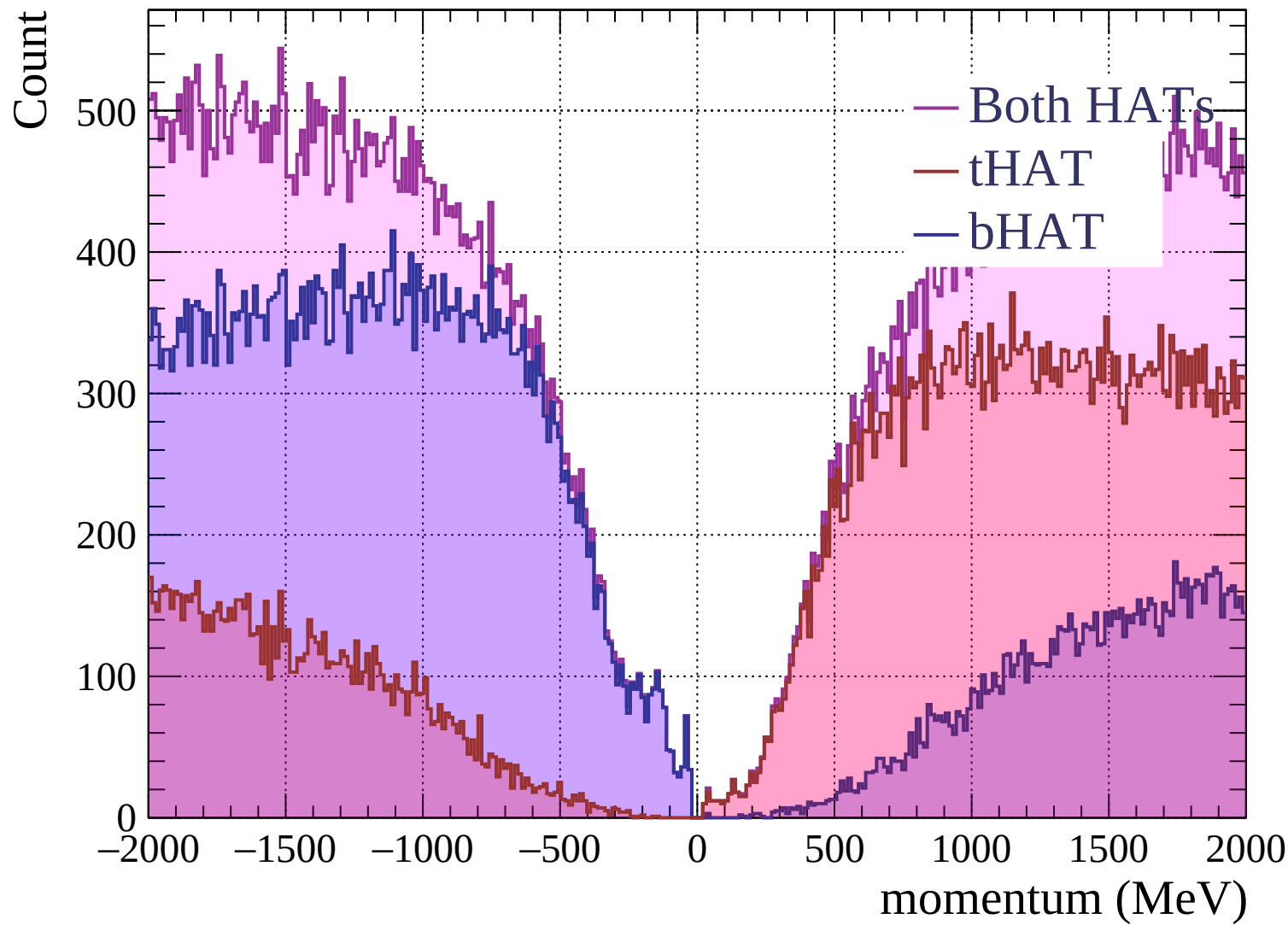


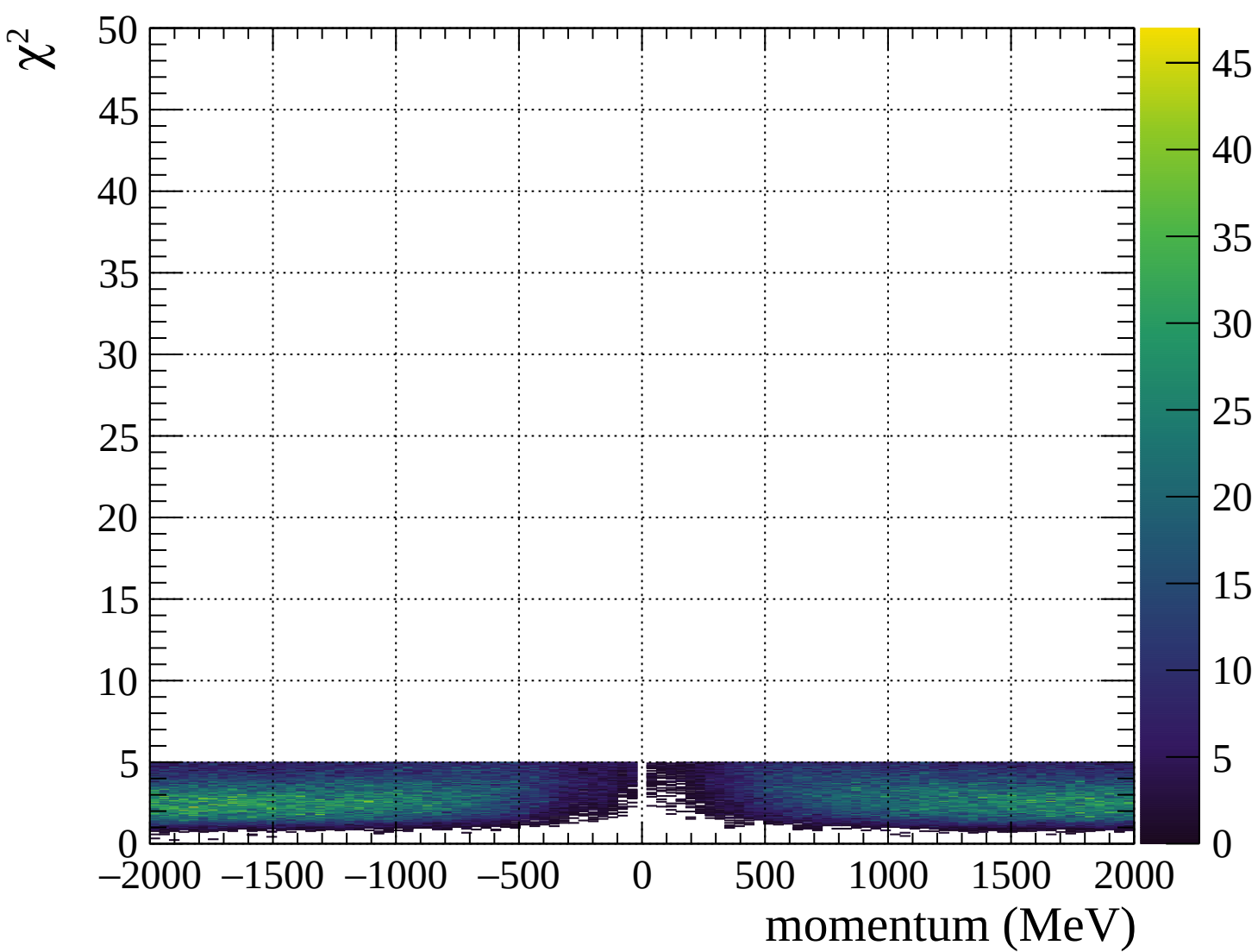




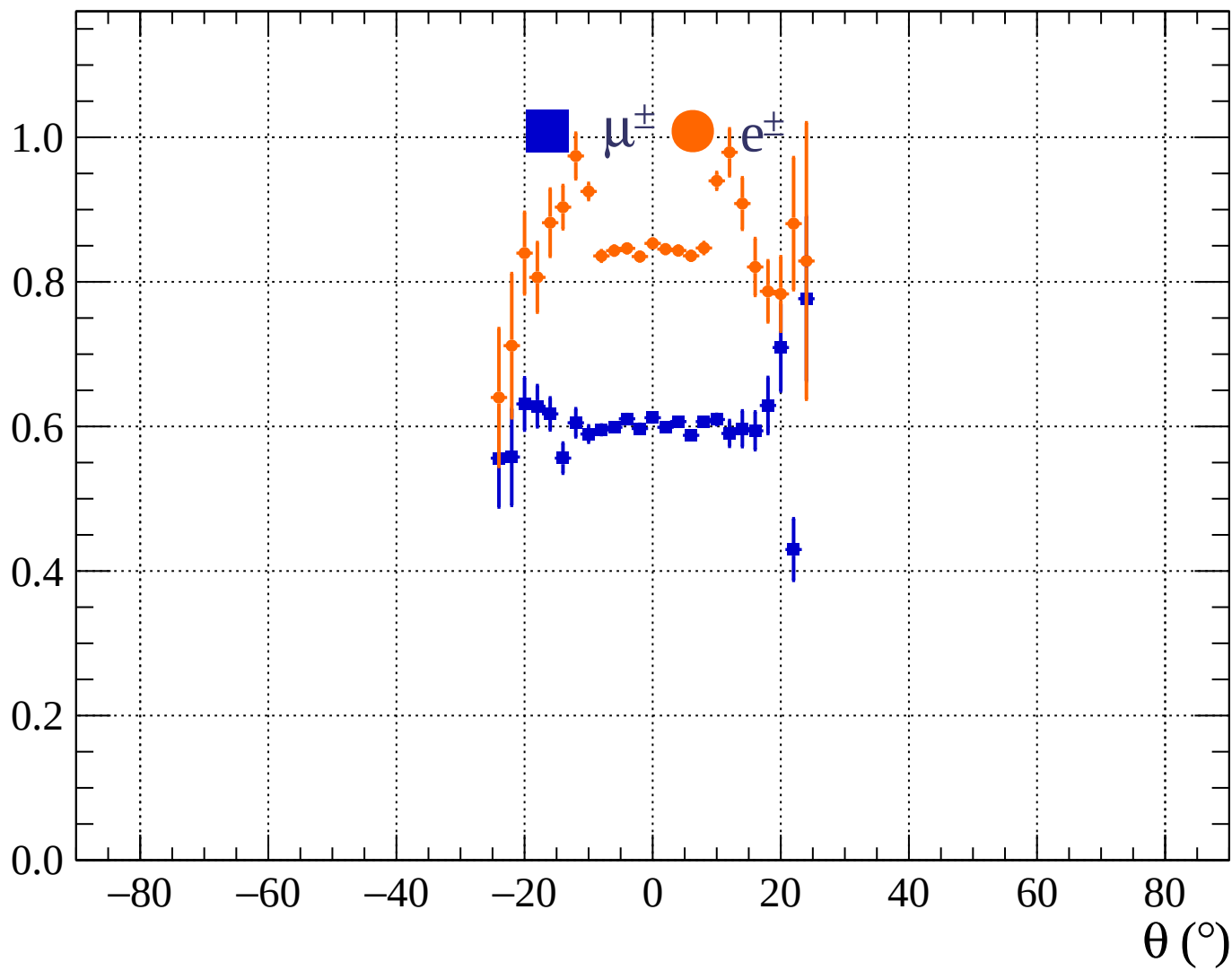


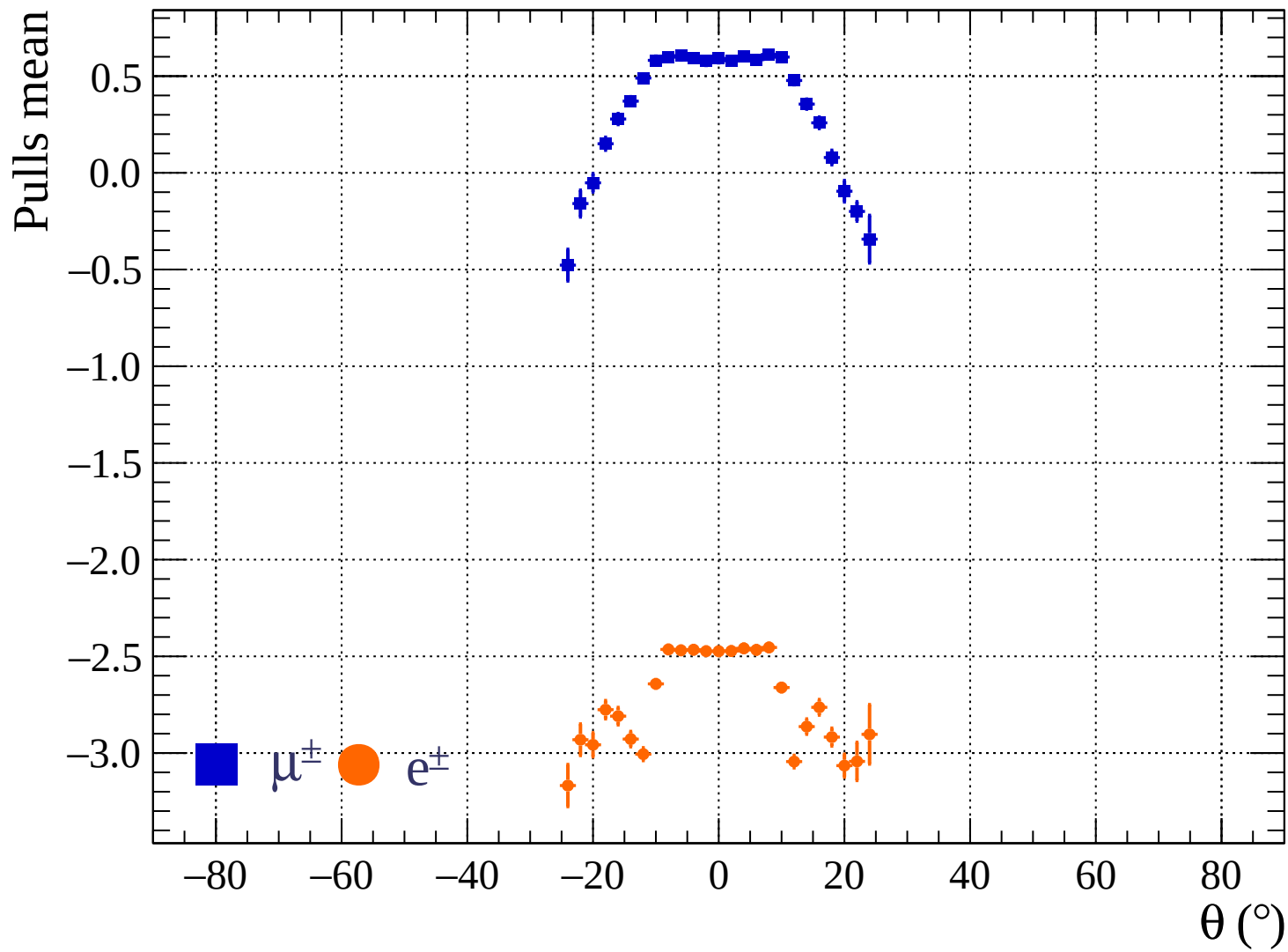




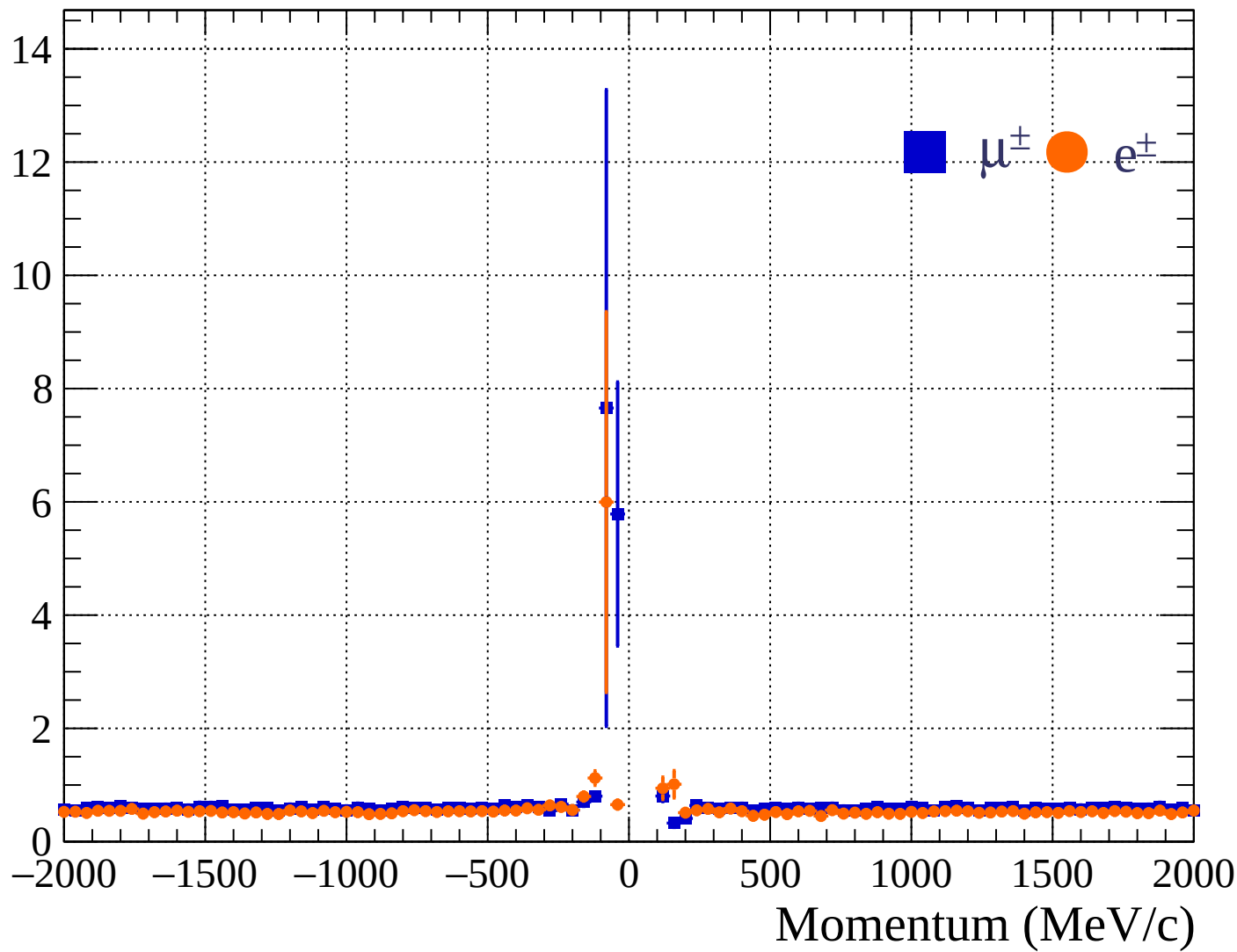


Pulls standard deviation

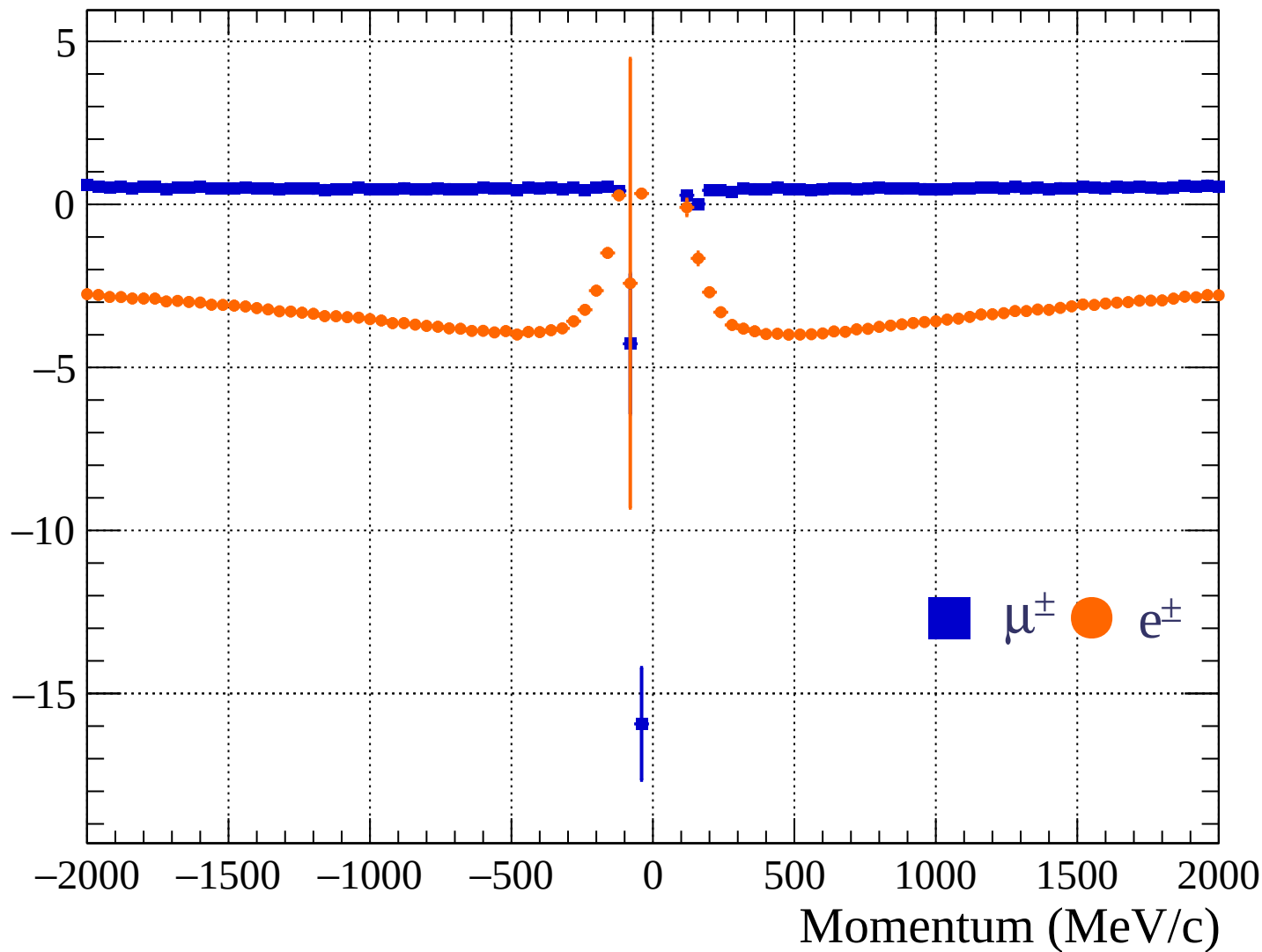


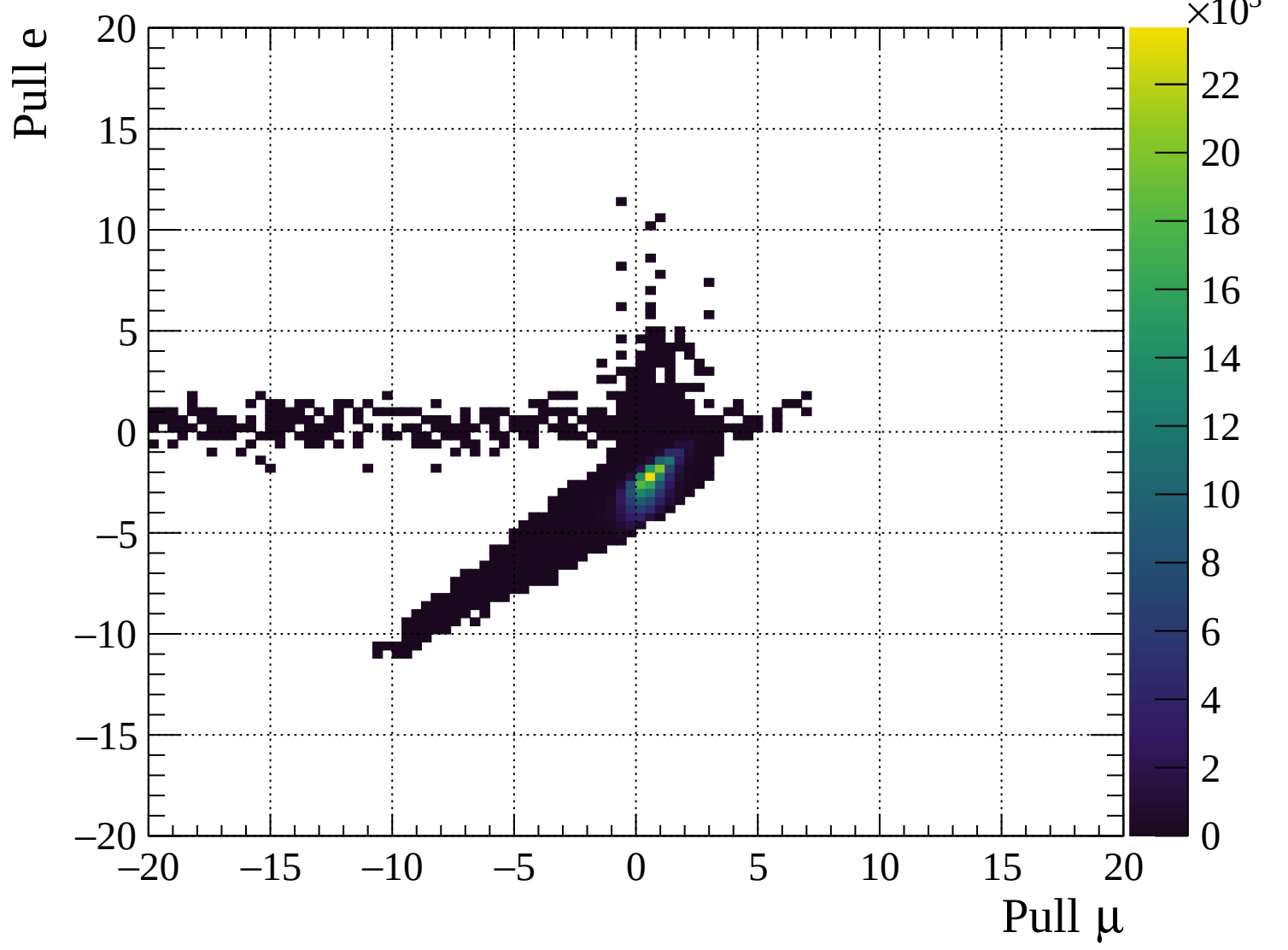


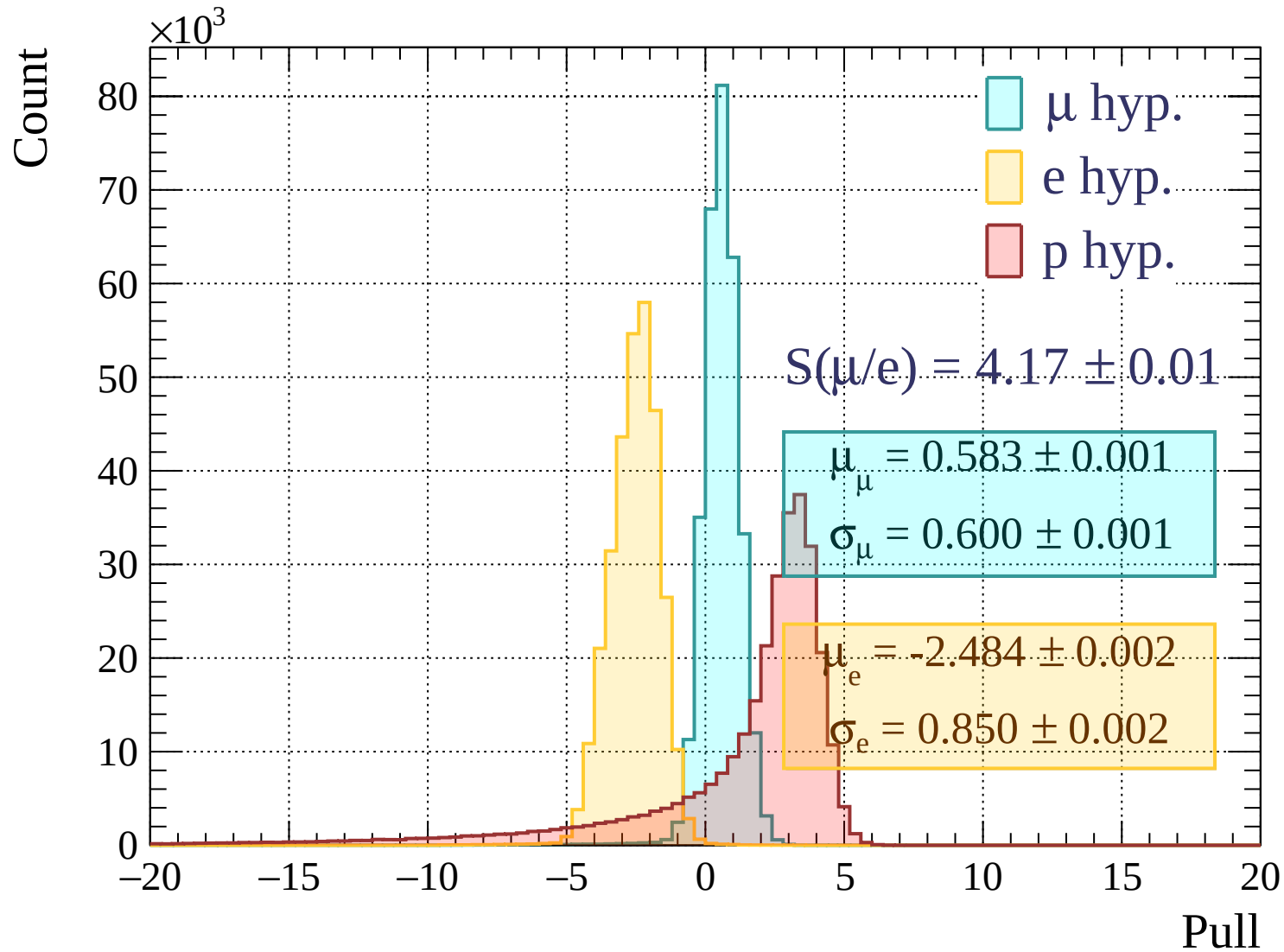
Pulls standard deviation

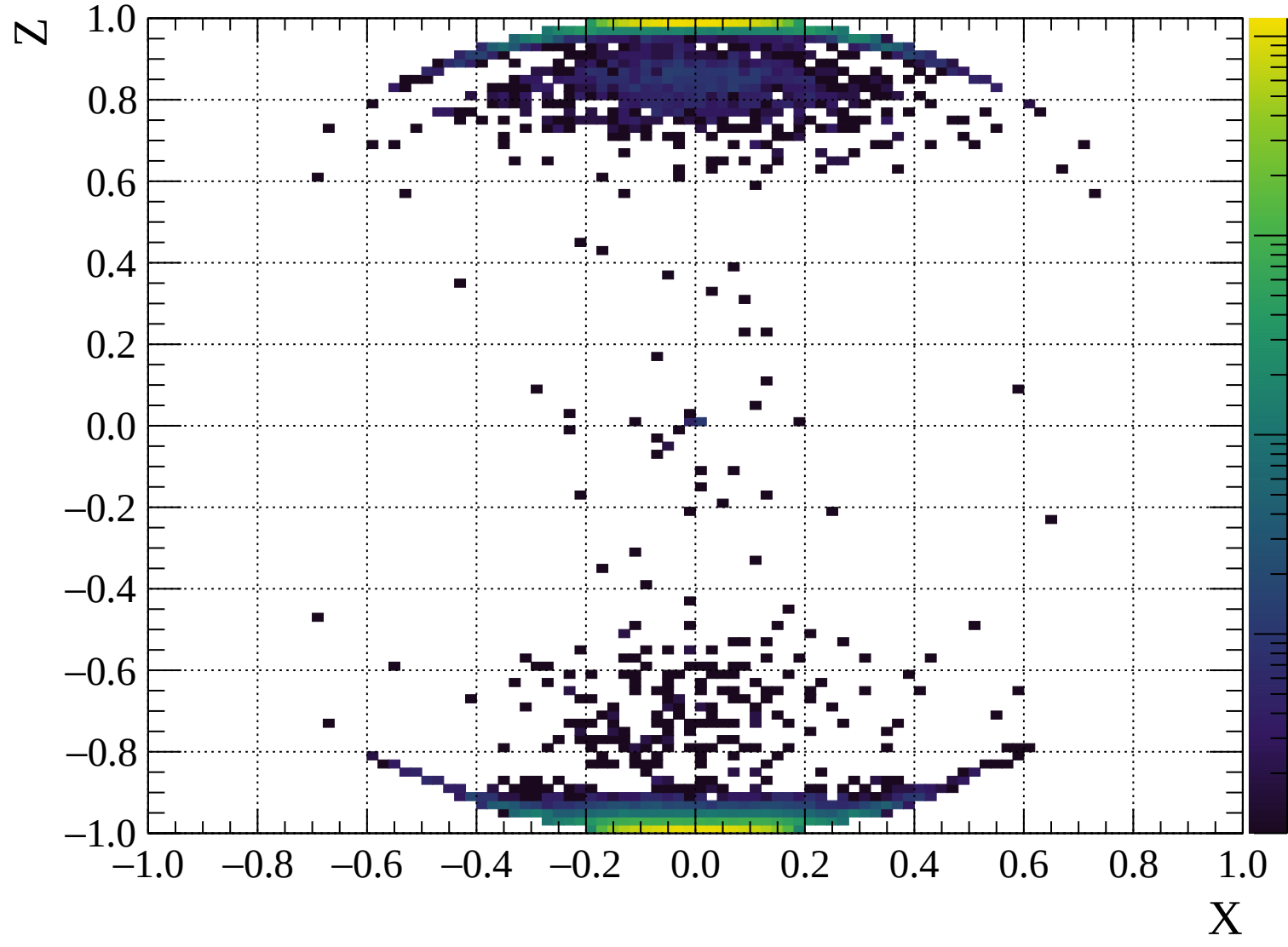


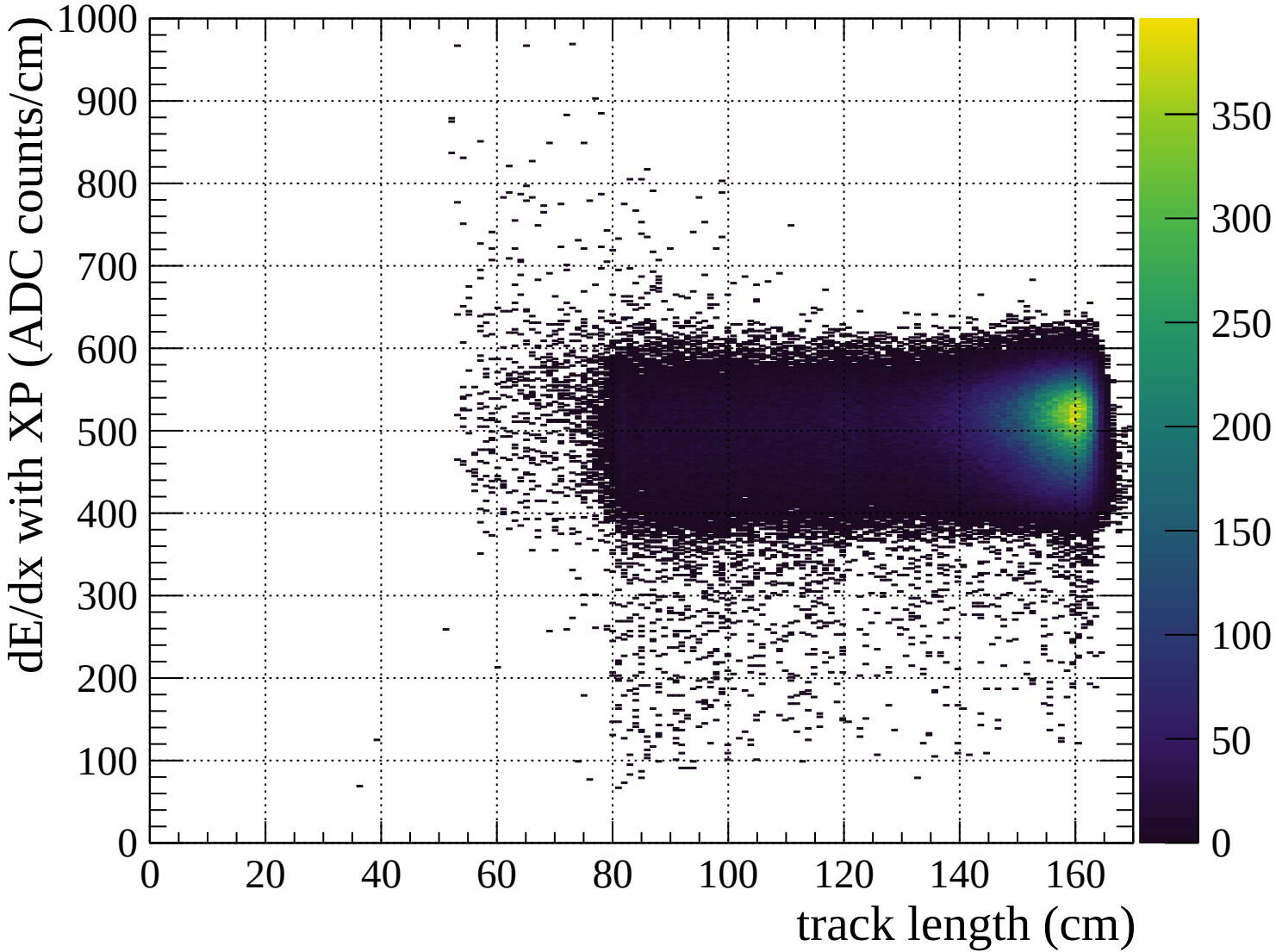
Pulls mean

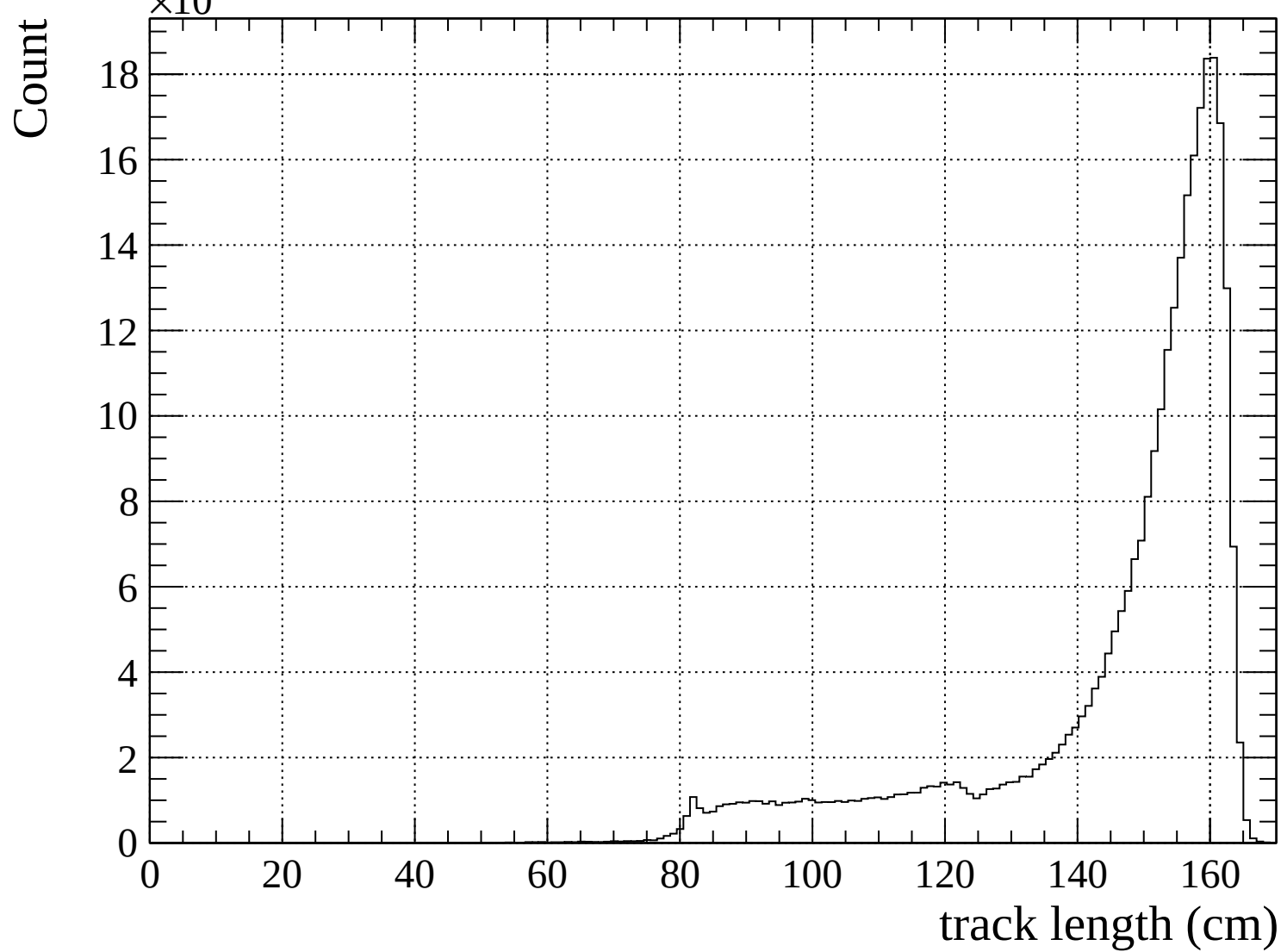


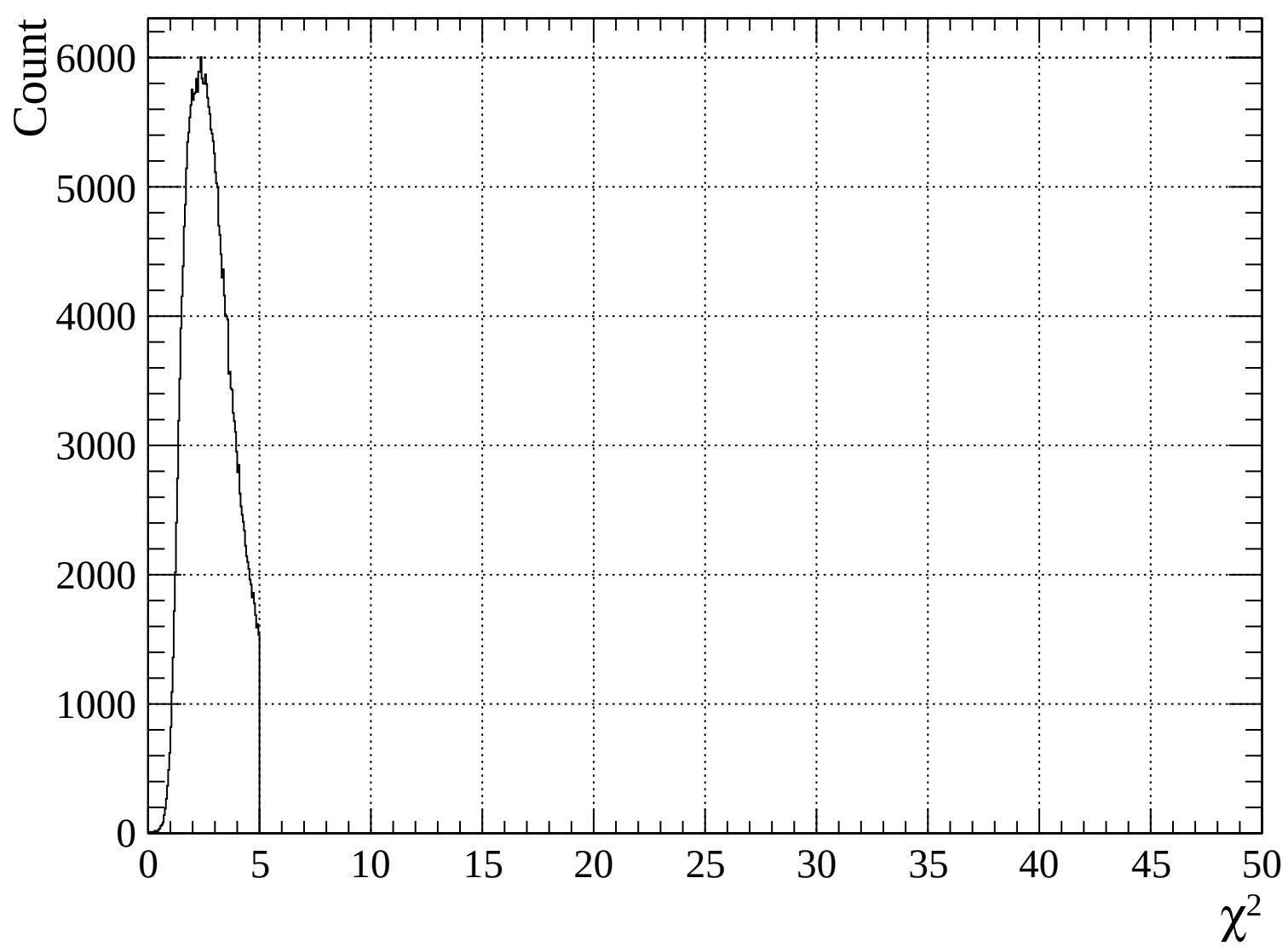


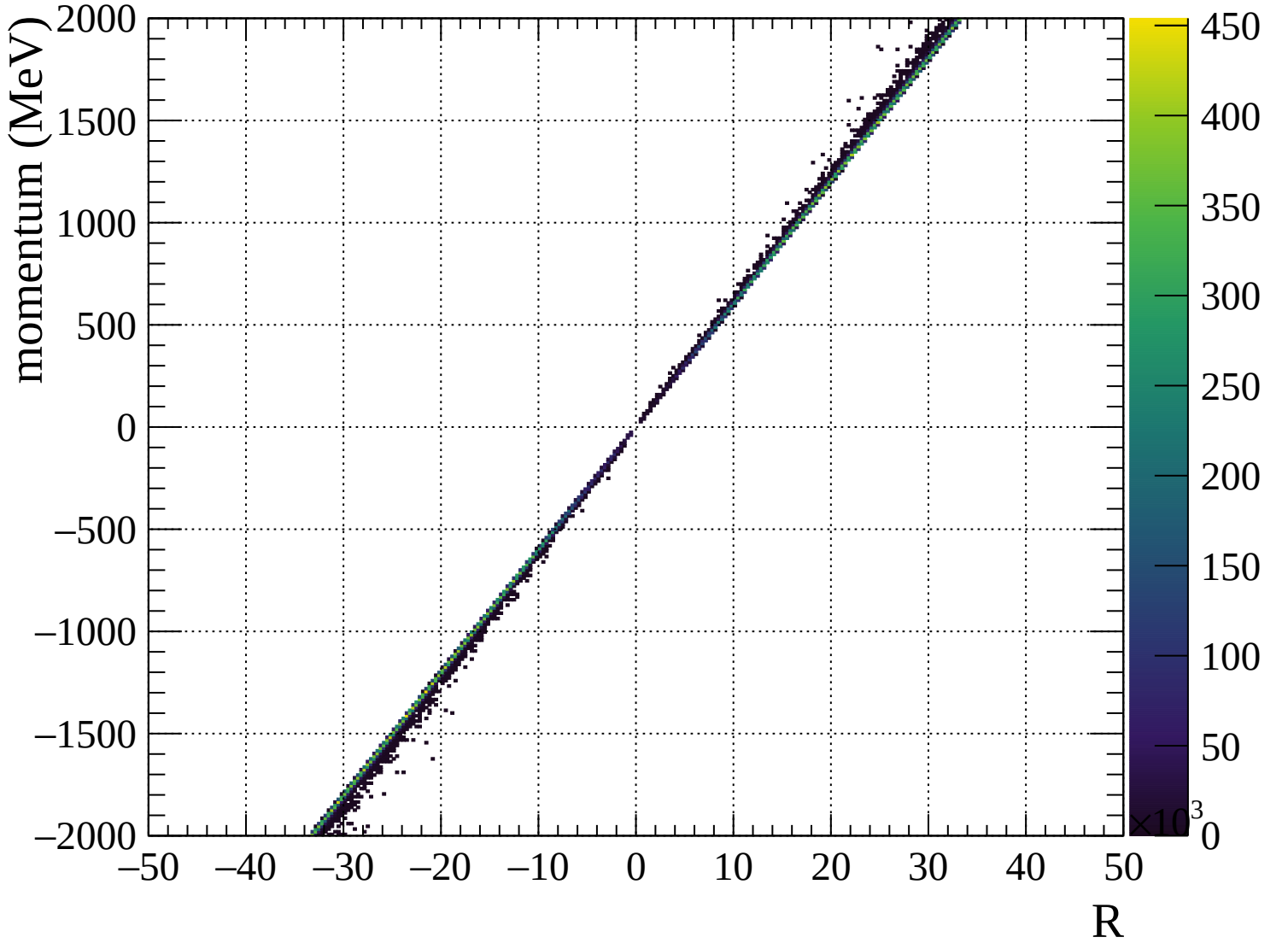


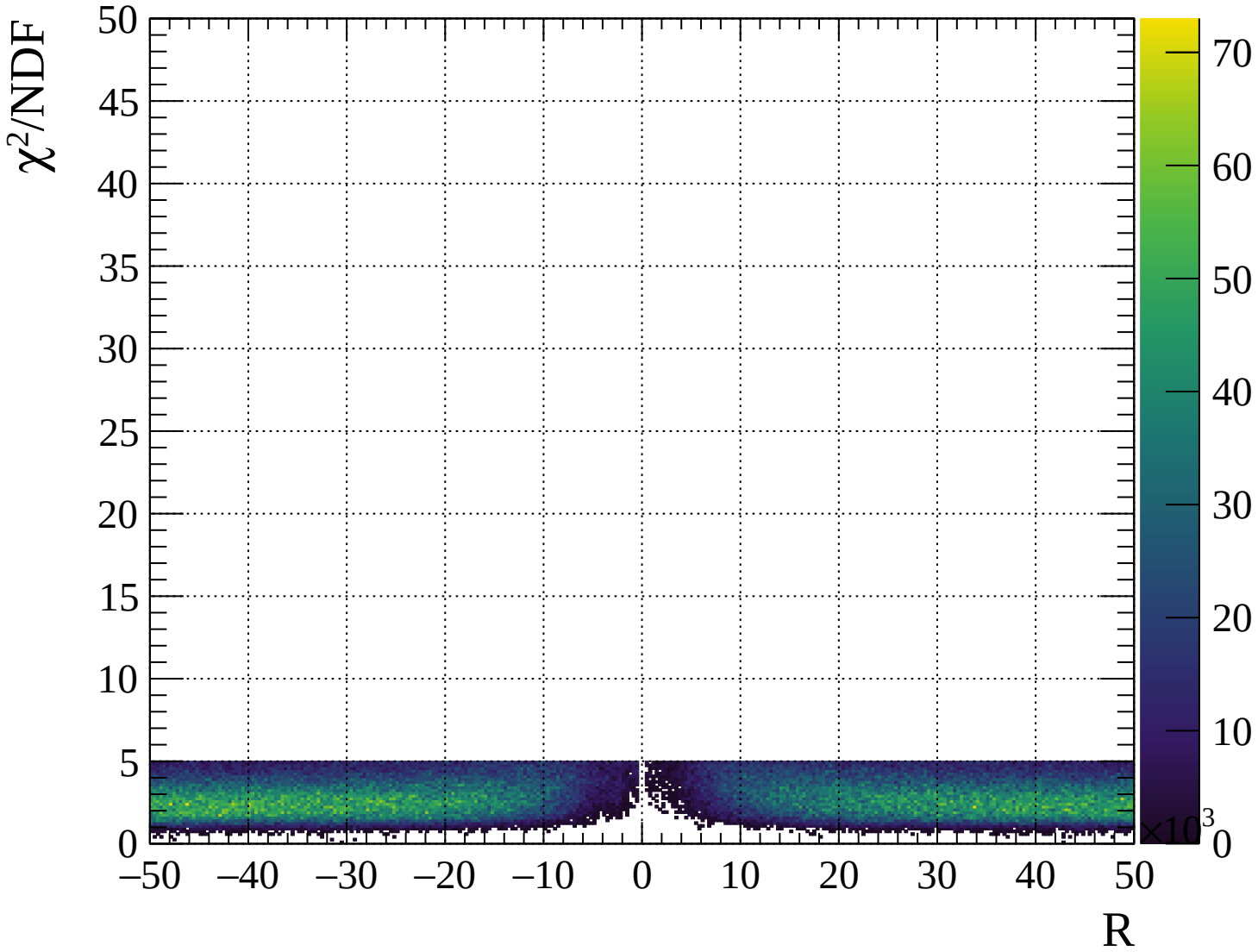






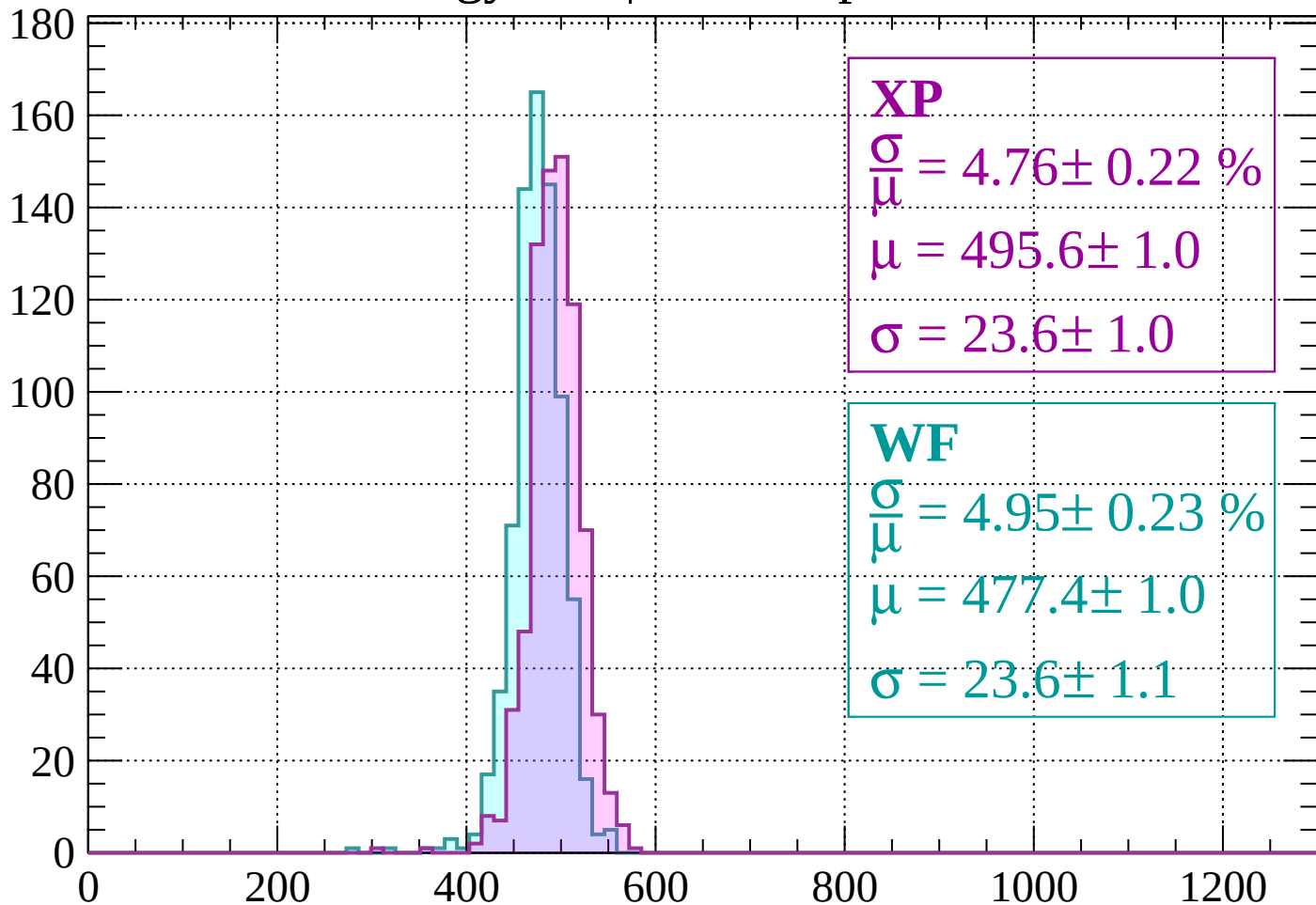






Energy loss | $-2000 < p < -1960$

Count



XP

$$\frac{\sigma}{\mu} = 4.76 \pm 0.22 \%$$

$$\mu = 495.6 \pm 1.0$$

$$\sigma = 23.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.95 \pm 0.23 \%$$

$$\mu = 477.4 \pm 1.0$$

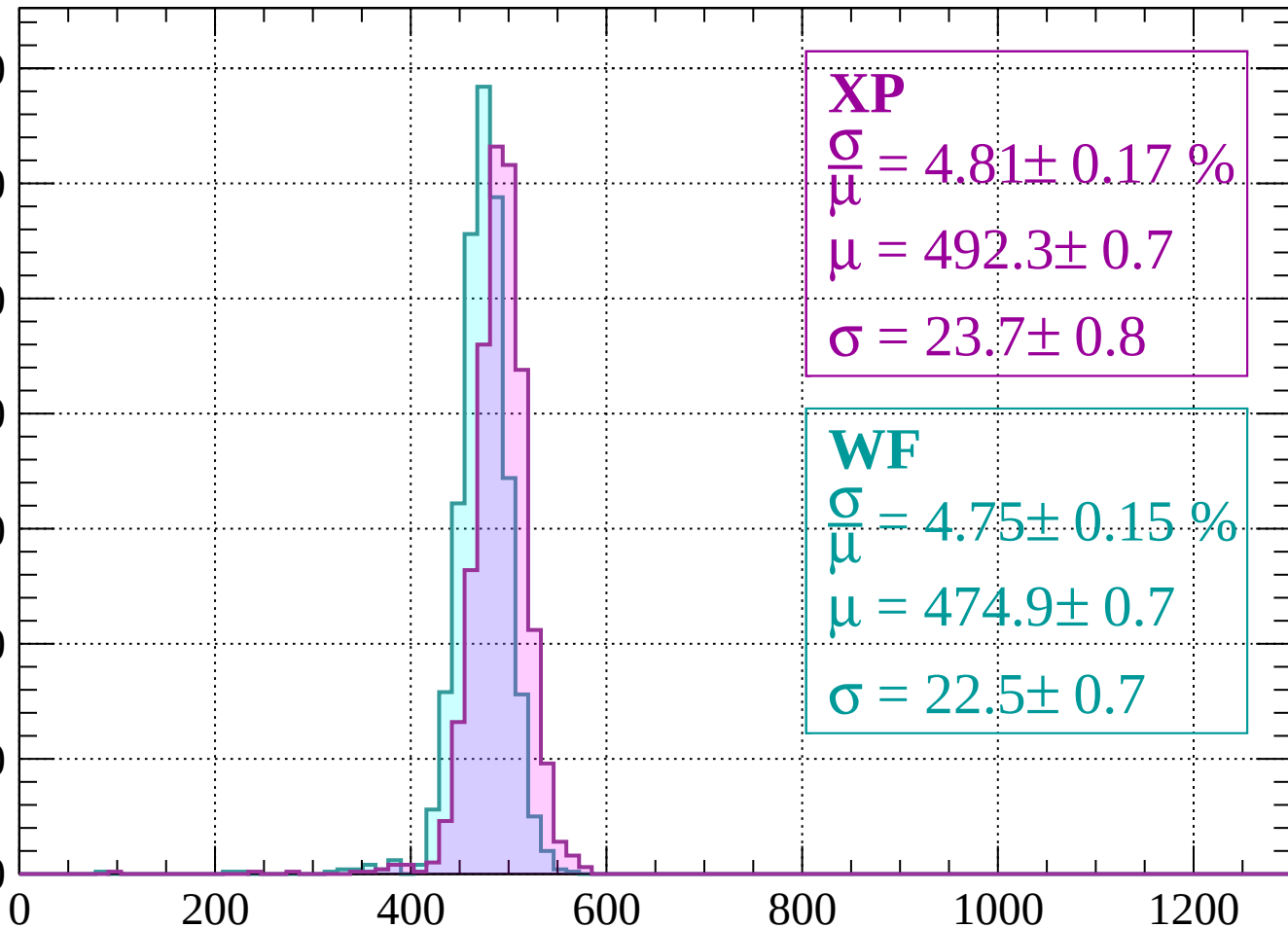
$$\sigma = 23.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.17 \%$$

$$\mu = 492.3 \pm 0.7$$

$$\sigma = 23.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.75 \pm 0.15 \%$$

$$\mu = 474.9 \pm 0.7$$

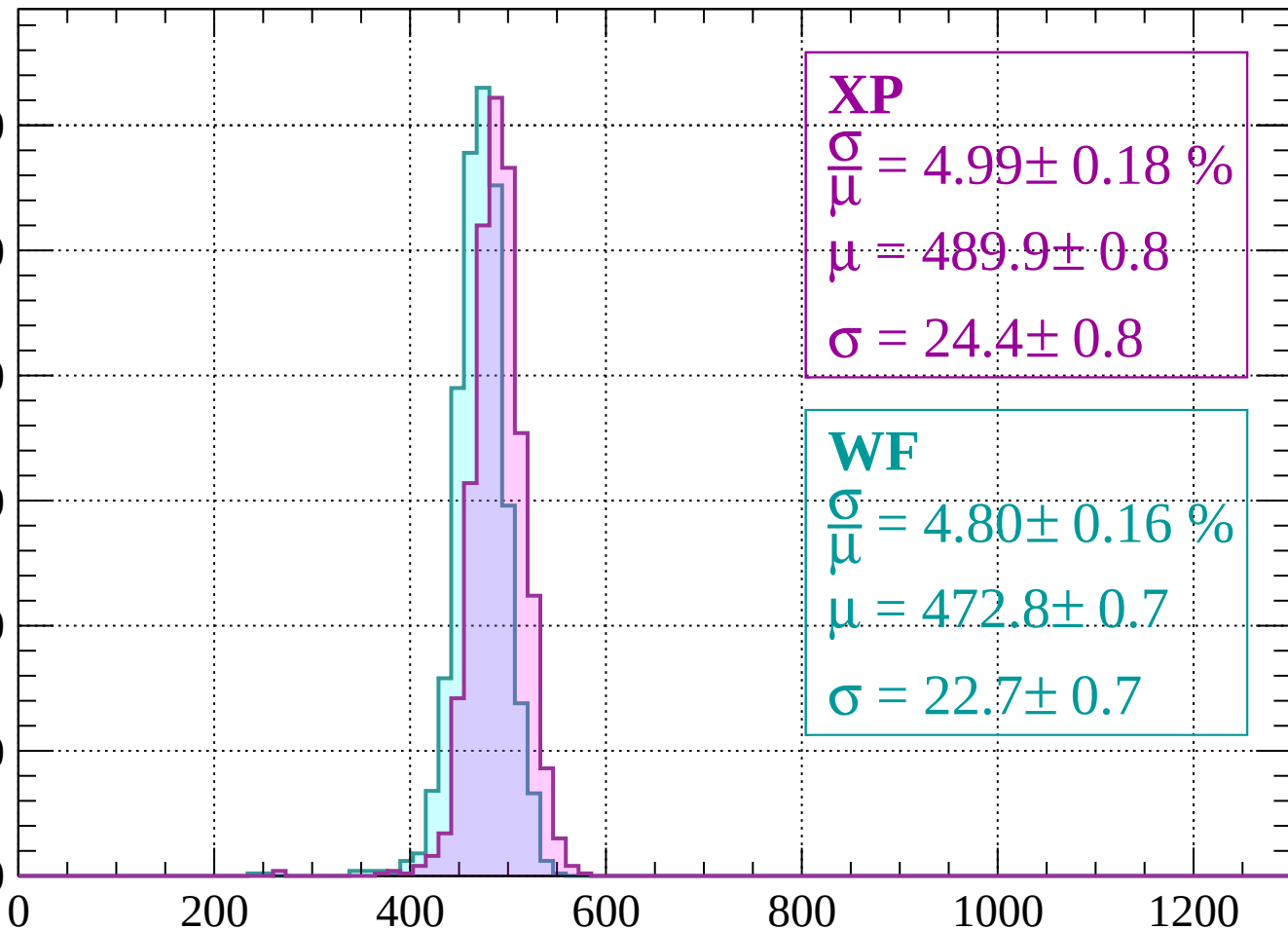
$$\sigma = 22.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.99 \pm 0.18 \%$$

$$\mu = 489.9 \pm 0.8$$

$$\sigma = 24.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.80 \pm 0.16 \%$$

$$\mu = 472.8 \pm 0.7$$

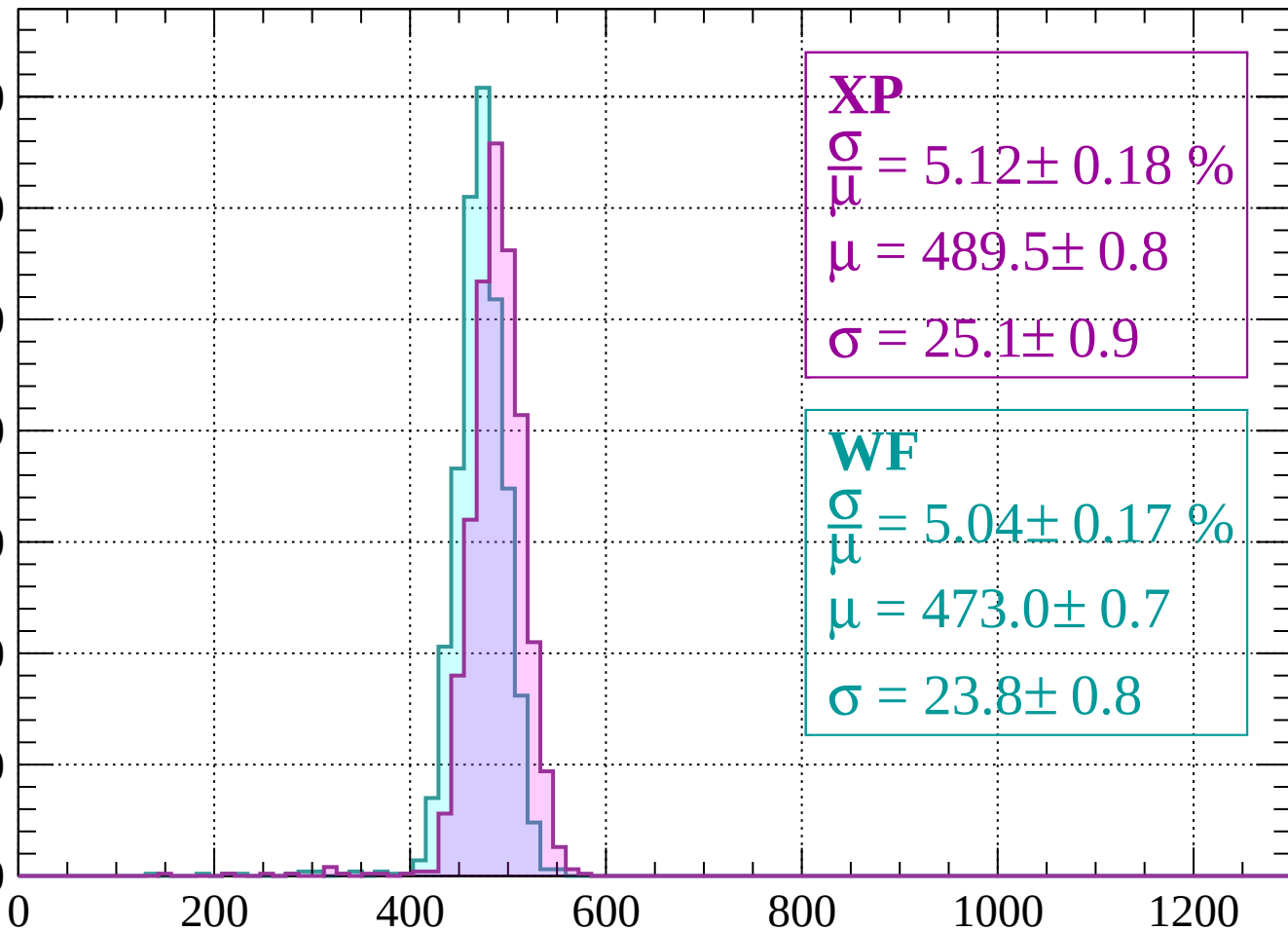
$$\sigma = 22.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1880 < p < -1840$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.12 \pm 0.18 \%$$

$$\mu = 489.5 \pm 0.8$$

$$\sigma = 25.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.04 \pm 0.17 \%$$

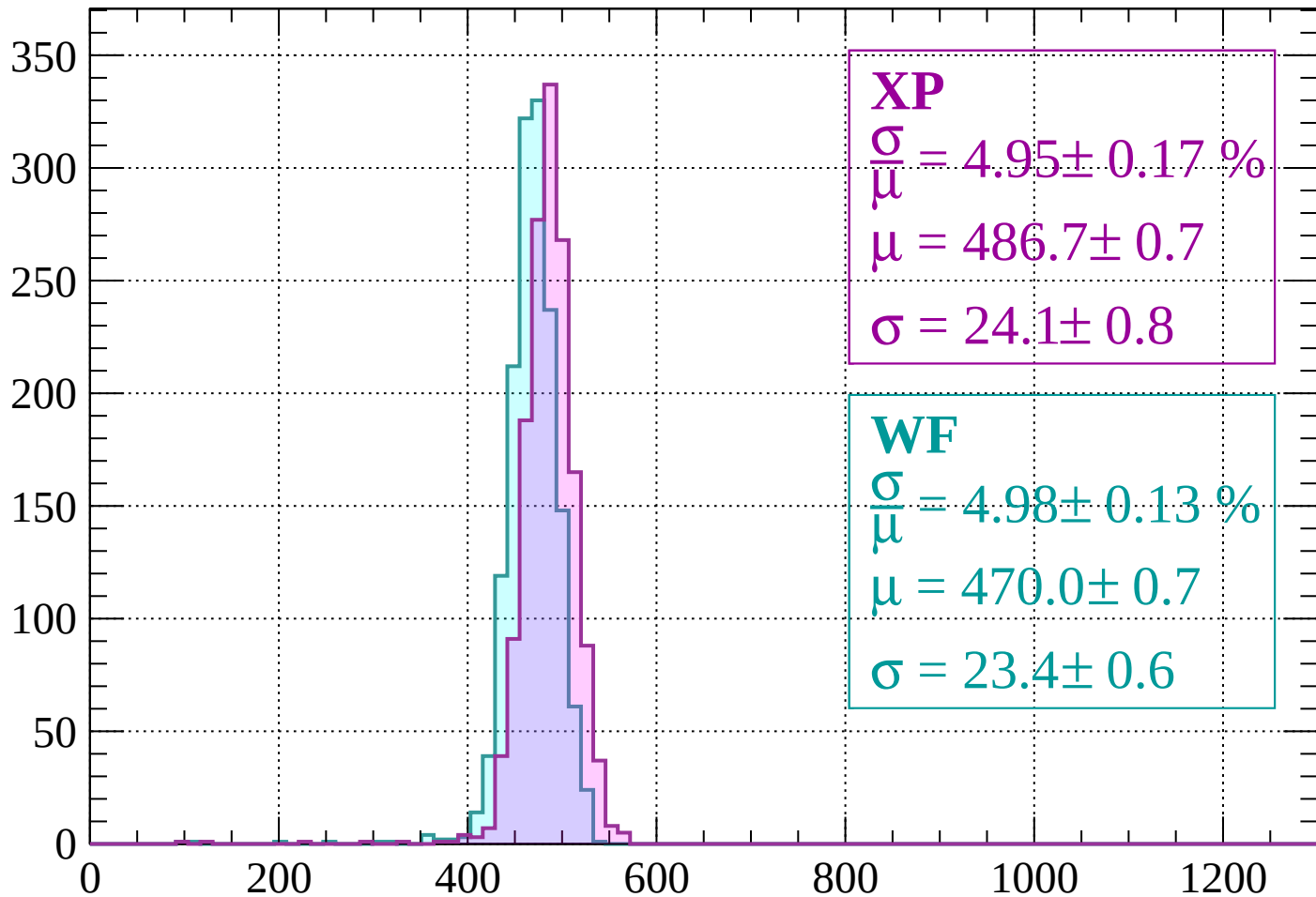
$$\mu = 473.0 \pm 0.7$$

$$\sigma = 23.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 4.95 \pm 0.17 \%$$

$$\mu = 486.7 \pm 0.7$$

$$\sigma = 24.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.98 \pm 0.13 \%$$

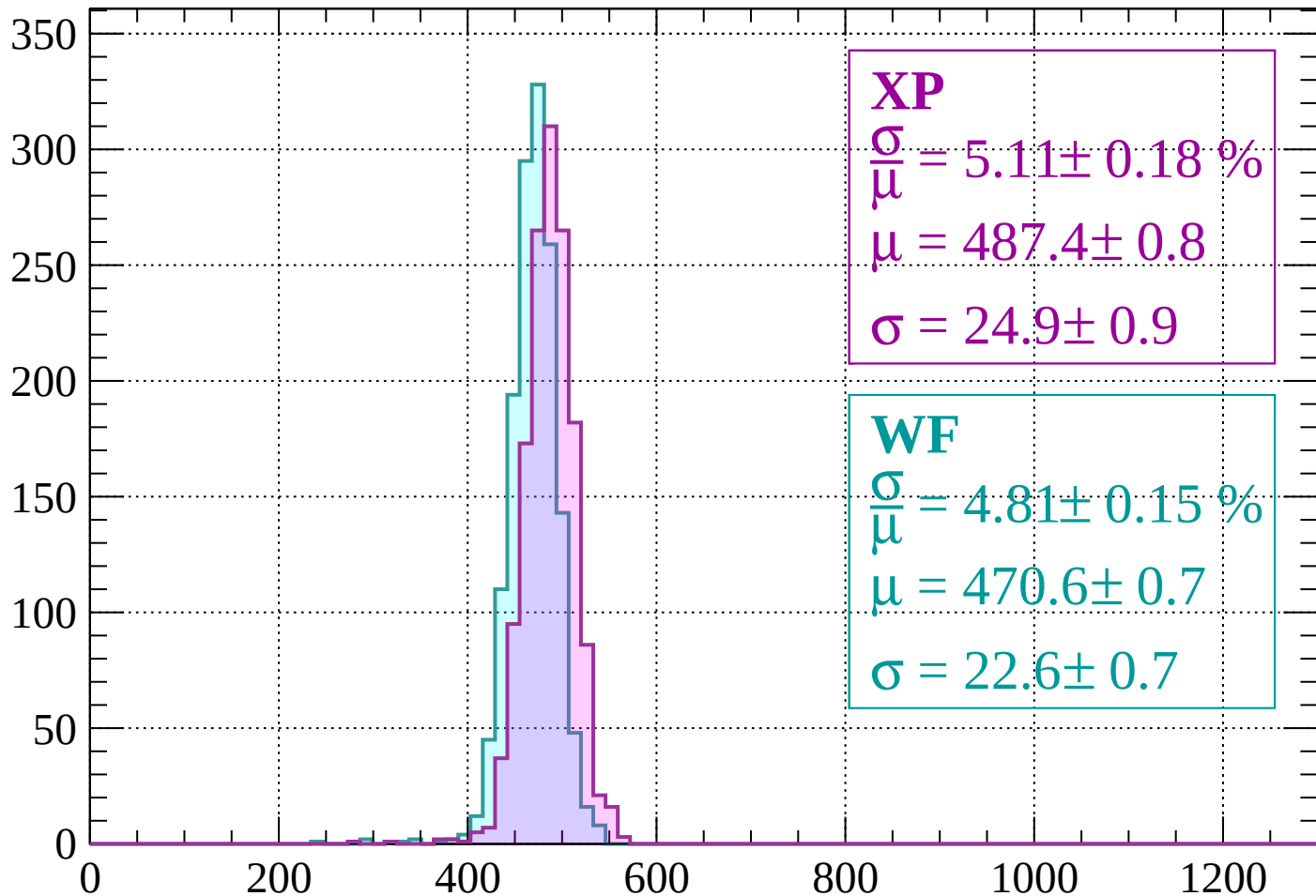
$$\mu = 470.0 \pm 0.7$$

$$\sigma = 23.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\mu} = 5.11 \pm 0.18 \%$$

$$\mu = 487.4 \pm 0.8$$

$$\sigma = 24.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.15 \%$$

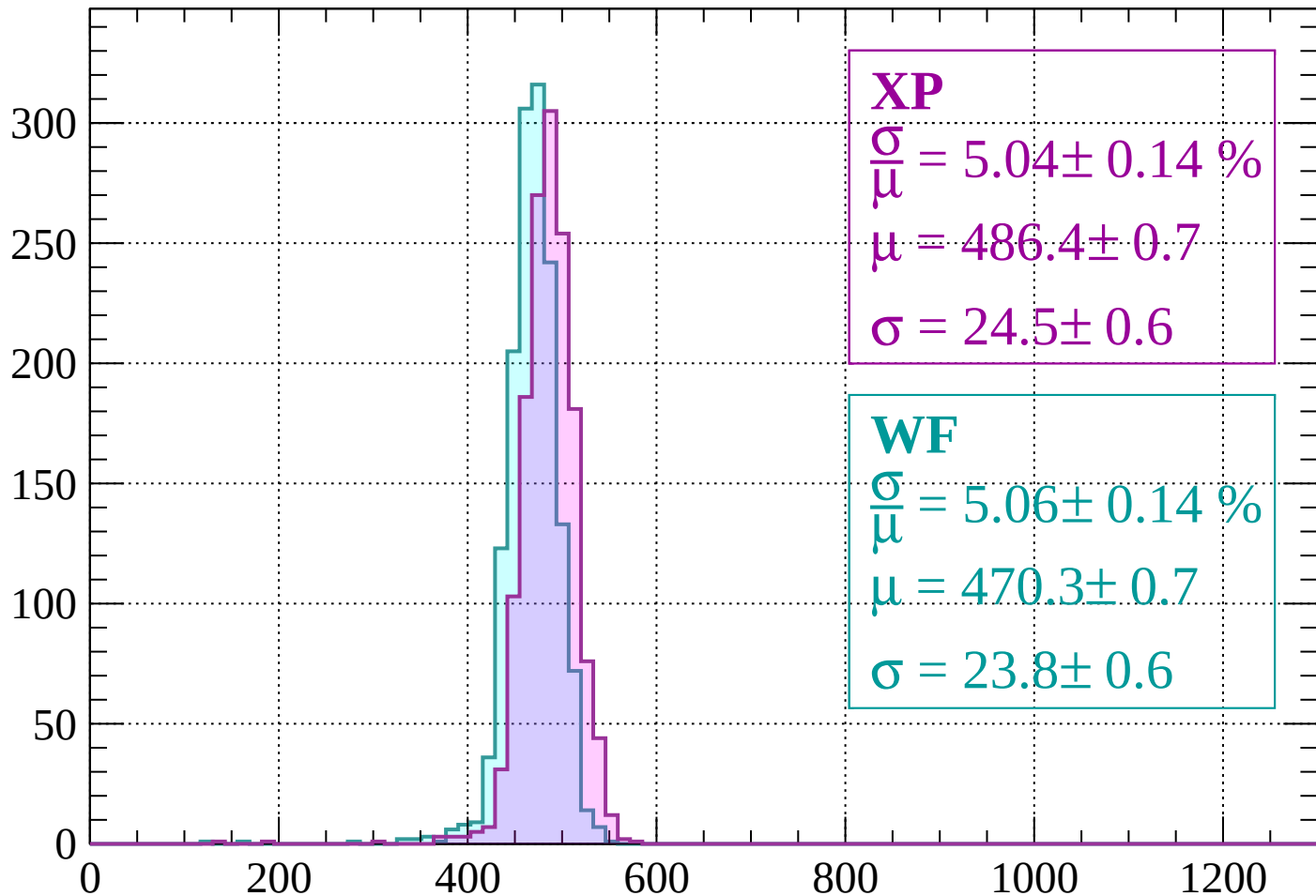
$$\mu = 470.6 \pm 0.7$$

$$\sigma = 22.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.14 \%$$

$$\mu = 486.4 \pm 0.7$$

$$\sigma = 24.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.06 \pm 0.14 \%$$

$$\mu = 470.3 \pm 0.7$$

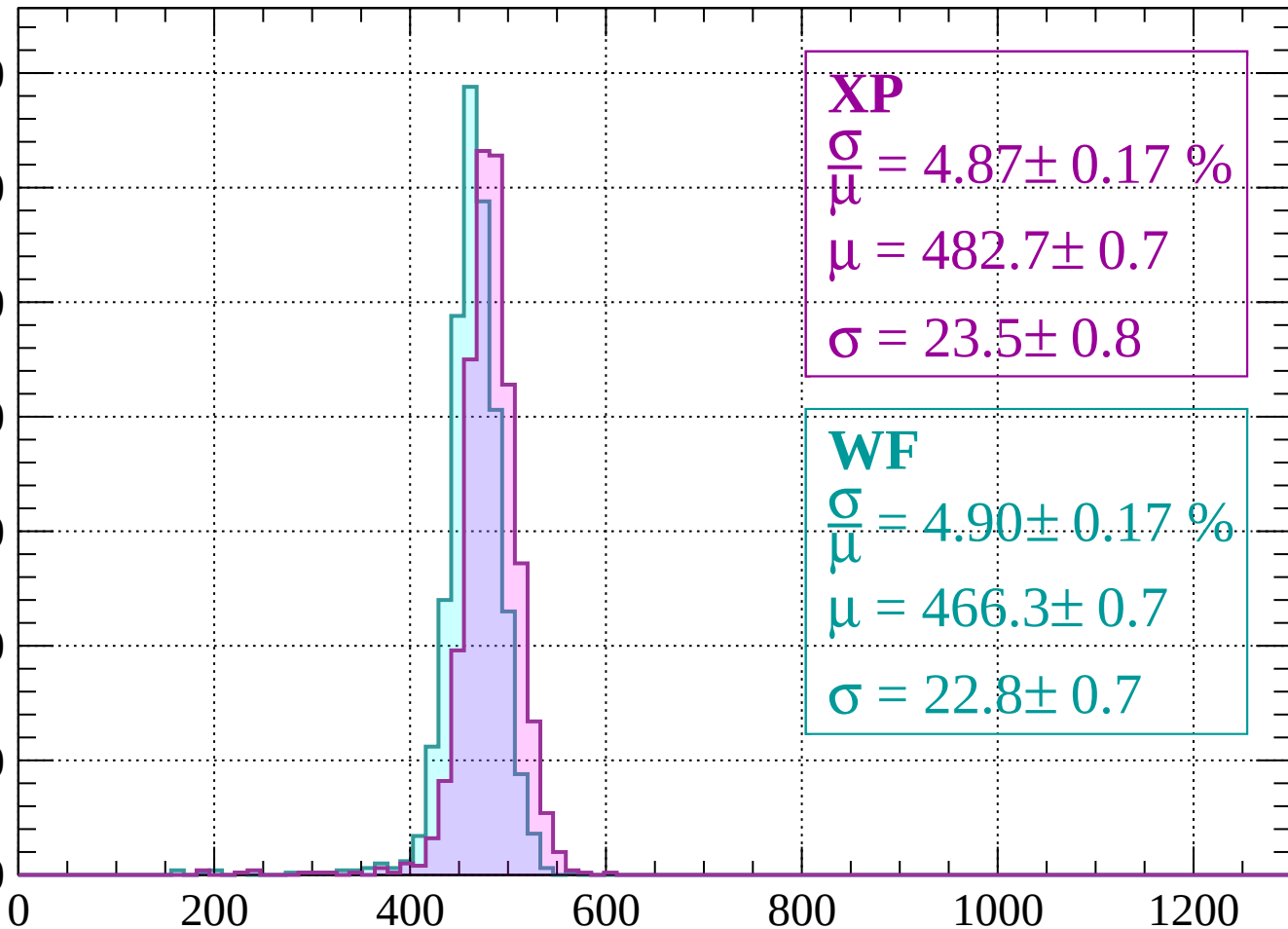
$$\sigma = 23.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.87 \pm 0.17 \%$$

$$\mu = 482.7 \pm 0.7$$

$$\sigma = 23.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.90 \pm 0.17 \%$$

$$\mu = 466.3 \pm 0.7$$

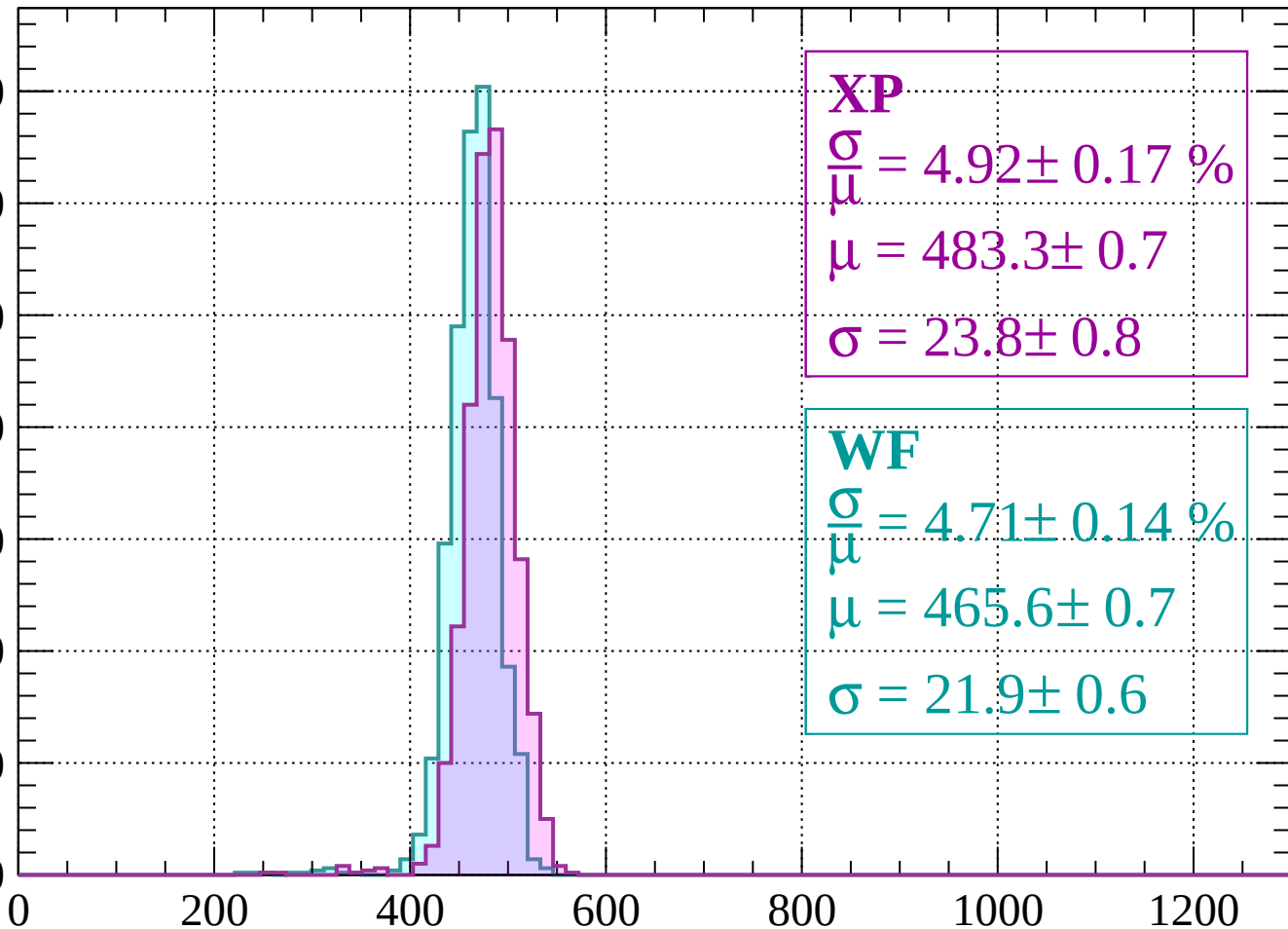
$$\sigma = 22.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.92 \pm 0.17 \%$$

$$\mu = 483.3 \pm 0.7$$

$$\sigma = 23.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.71 \pm 0.14 \%$$

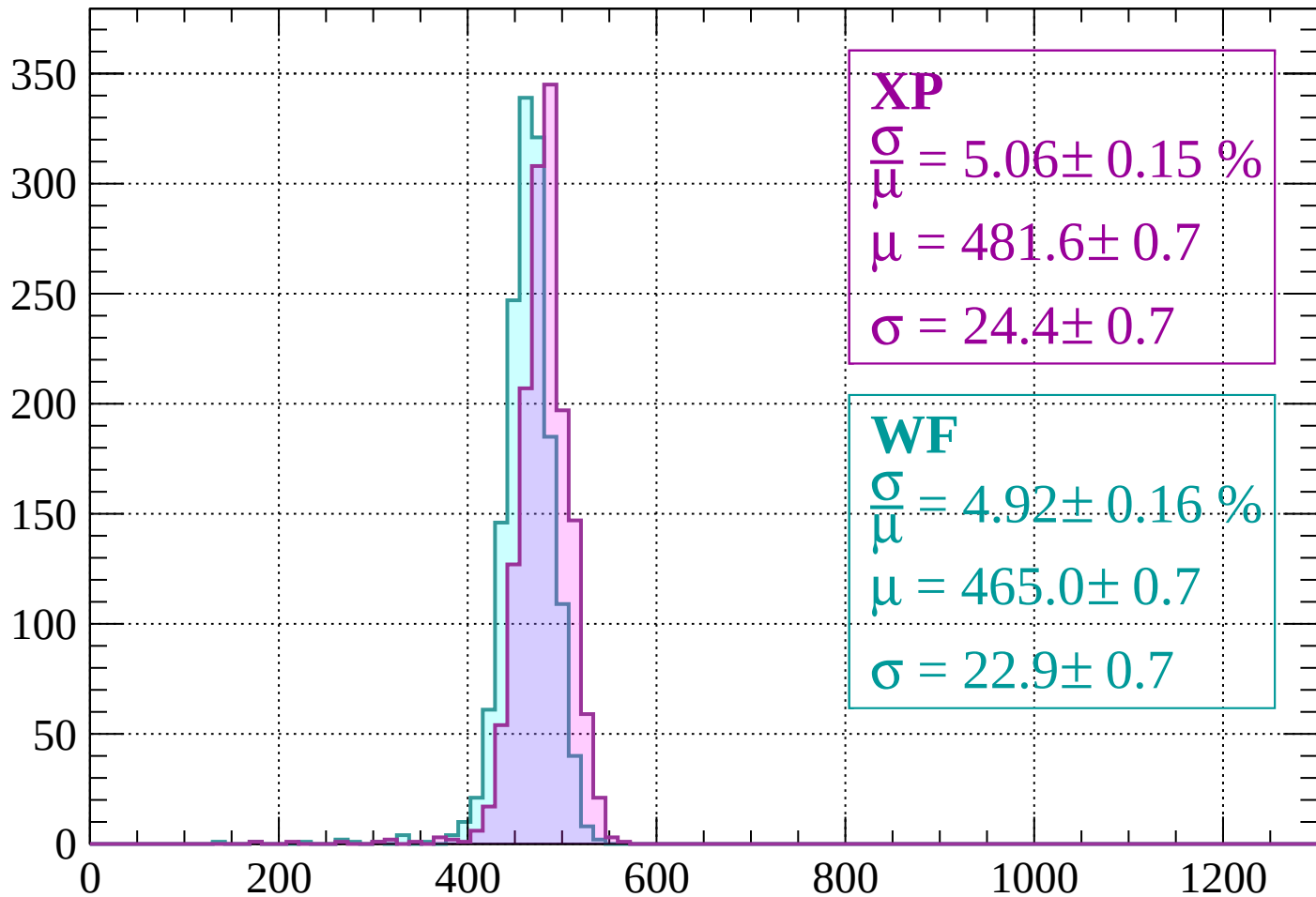
$$\mu = 465.6 \pm 0.7$$

$$\sigma = 21.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 5.06 \pm 0.15 \%$$

$$\mu = 481.6 \pm 0.7$$

$$\sigma = 24.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.92 \pm 0.16 \%$$

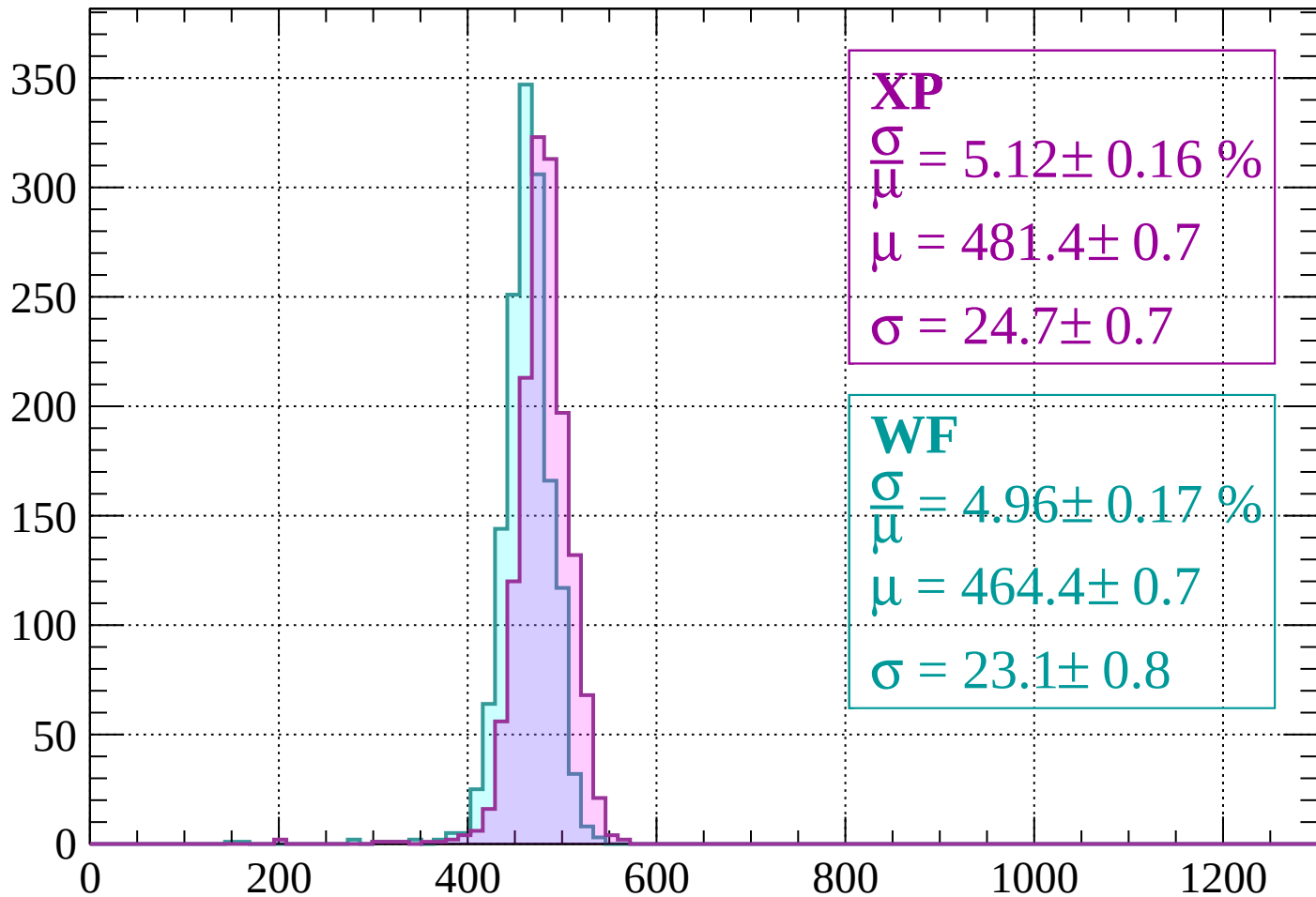
$$\mu = 465.0 \pm 0.7$$

$$\sigma = 22.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 5.12 \pm 0.16 \%$$

$$\mu = 481.4 \pm 0.7$$

$$\sigma = 24.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.96 \pm 0.17 \%$$

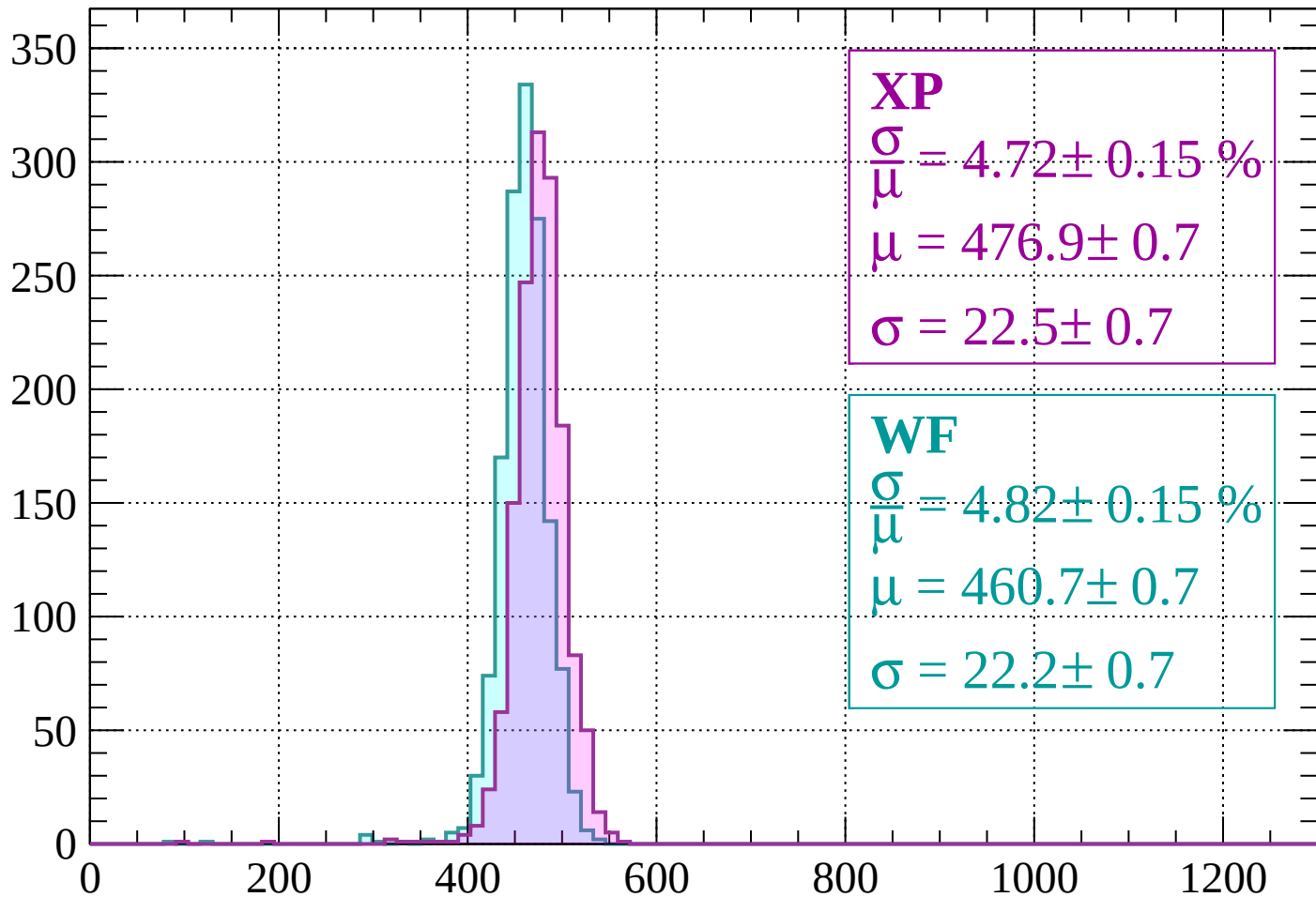
$$\mu = 464.4 \pm 0.7$$

$$\sigma = 23.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 4.72 \pm 0.15 \%$$

$$\mu = 476.9 \pm 0.7$$

$$\sigma = 22.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.15 \%$$

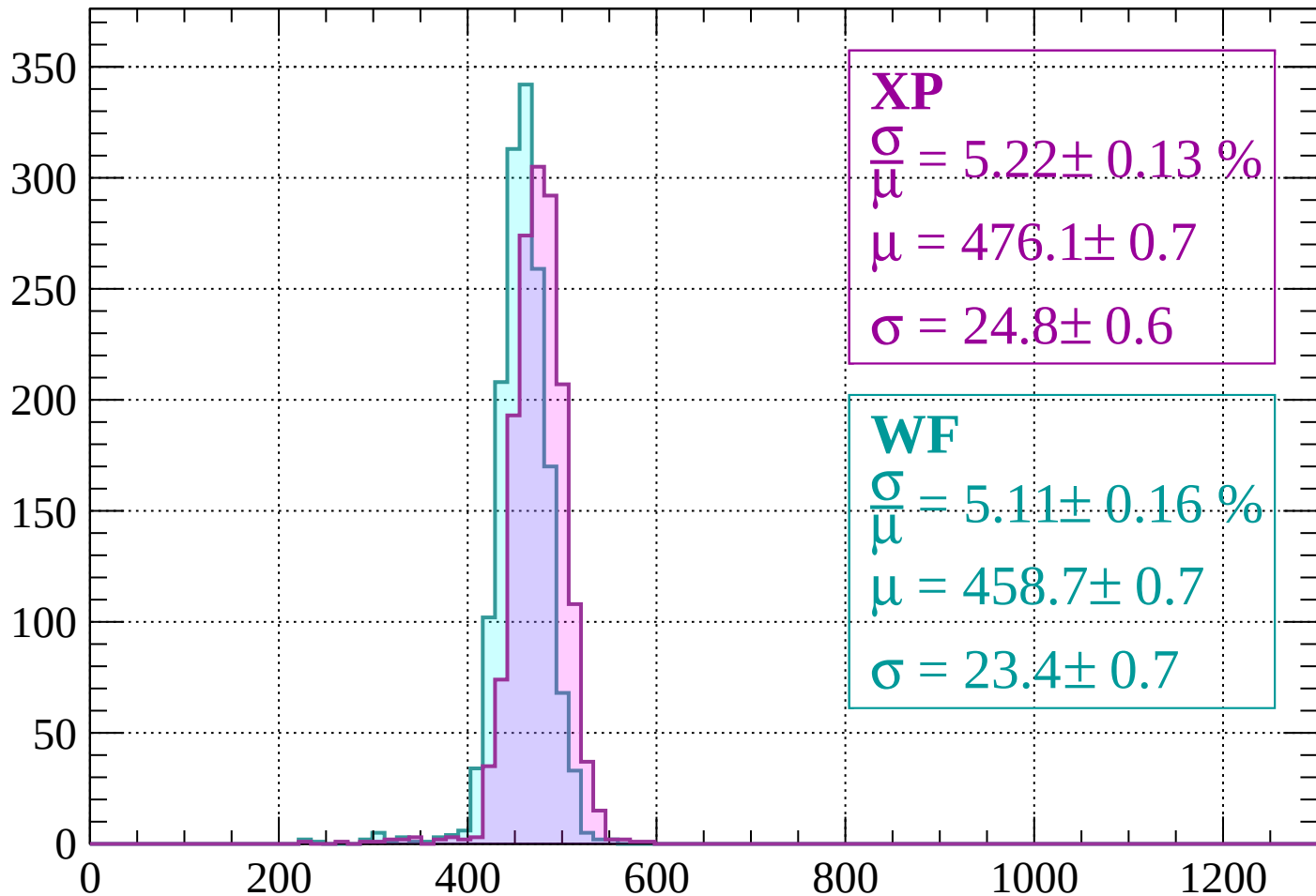
$$\mu = 460.7 \pm 0.7$$

$$\sigma = 22.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 5.22 \pm 0.13 \%$$

$$\mu = 476.1 \pm 0.7$$

$$\sigma = 24.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.11 \pm 0.16 \%$$

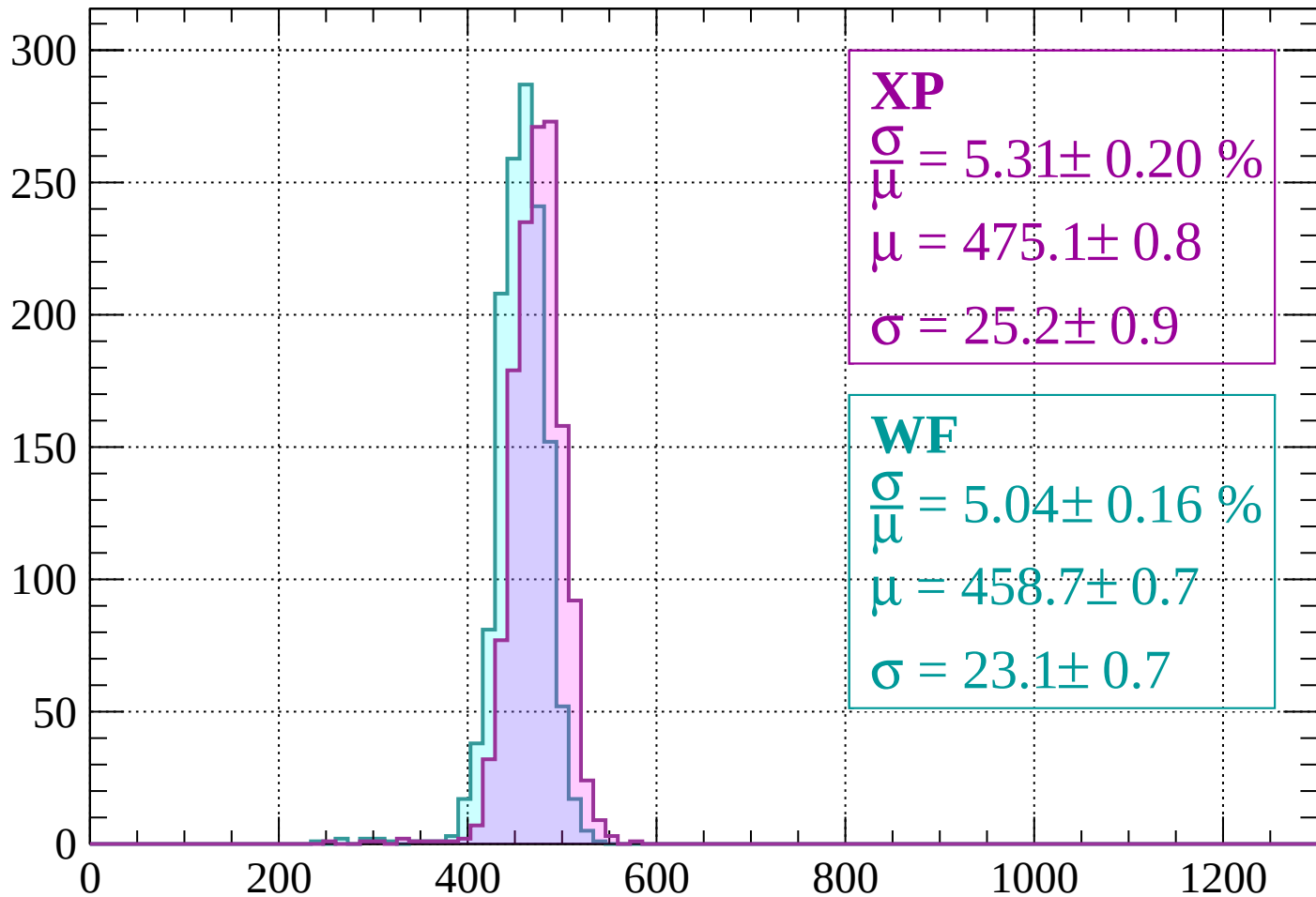
$$\mu = 458.7 \pm 0.7$$

$$\sigma = 23.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 5.31 \pm 0.20 \%$$

$$\mu = 475.1 \pm 0.8$$

$$\sigma = 25.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.04 \pm 0.16 \%$$

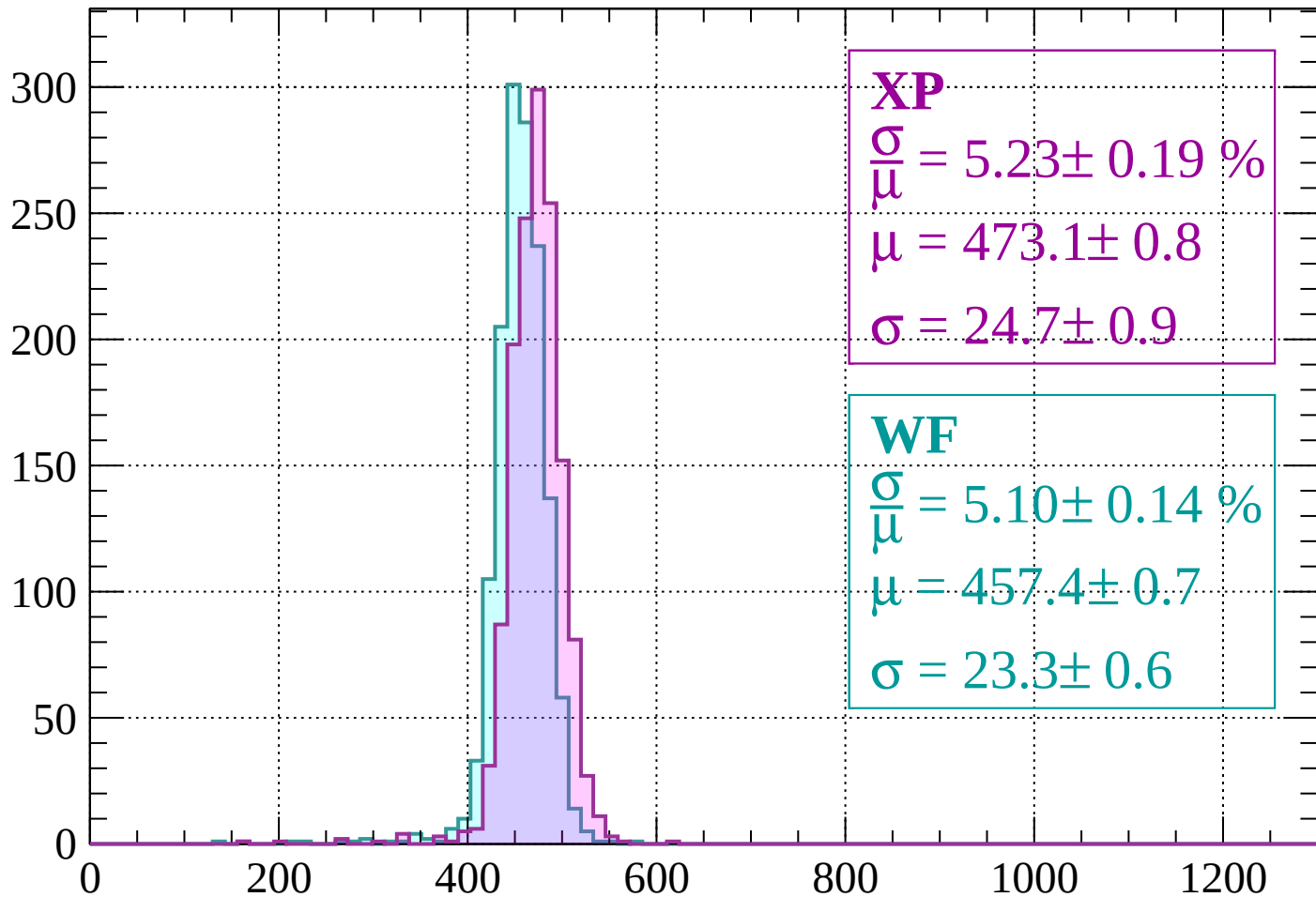
$$\mu = 458.7 \pm 0.7$$

$$\sigma = 23.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.19 \%$$

$$\mu = 473.1 \pm 0.8$$

$$\sigma = 24.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.10 \pm 0.14 \%$$

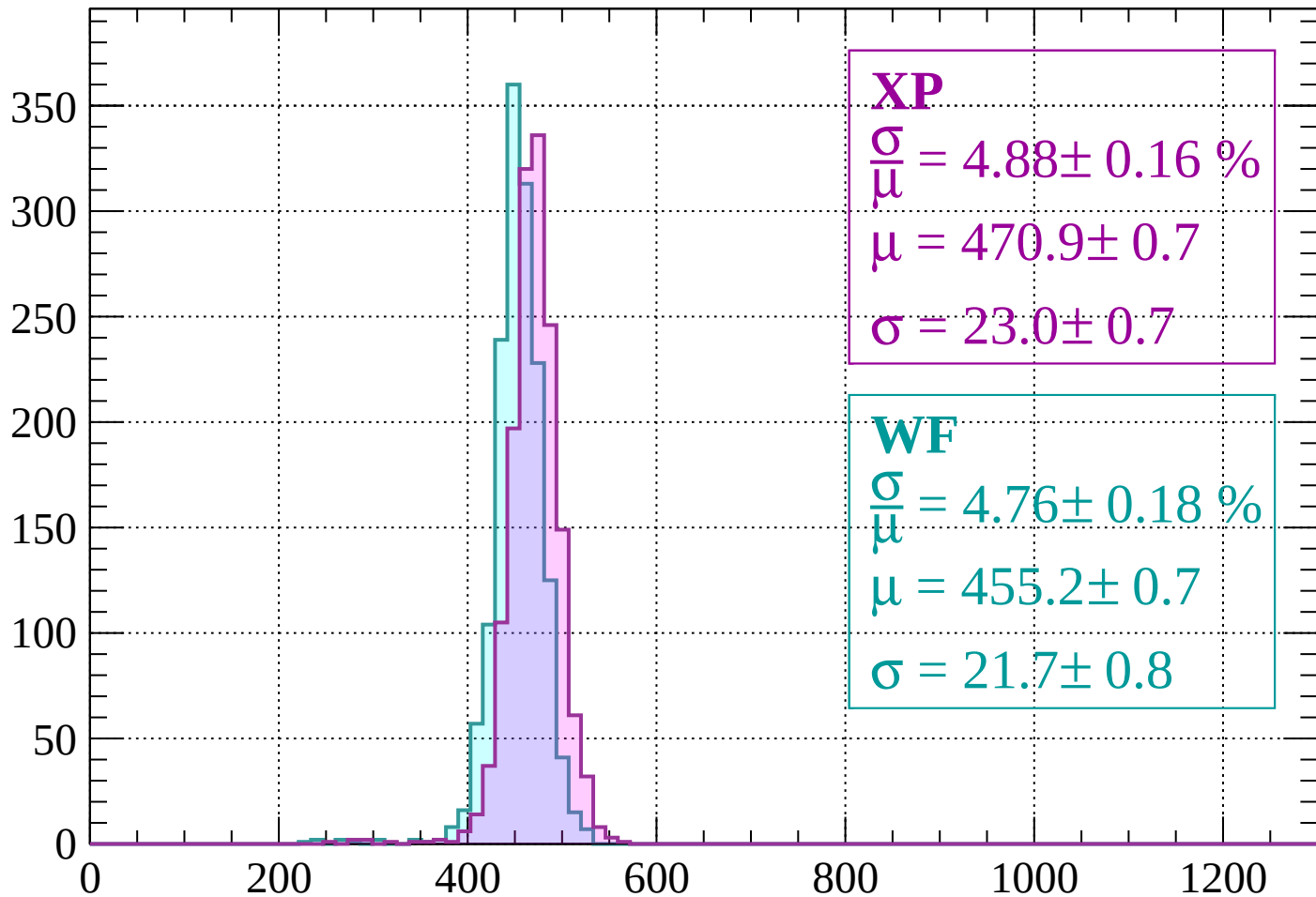
$$\mu = 457.4 \pm 0.7$$

$$\sigma = 23.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 4.88 \pm 0.16 \%$$

$$\mu = 470.9 \pm 0.7$$

$$\sigma = 23.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.76 \pm 0.18 \%$$

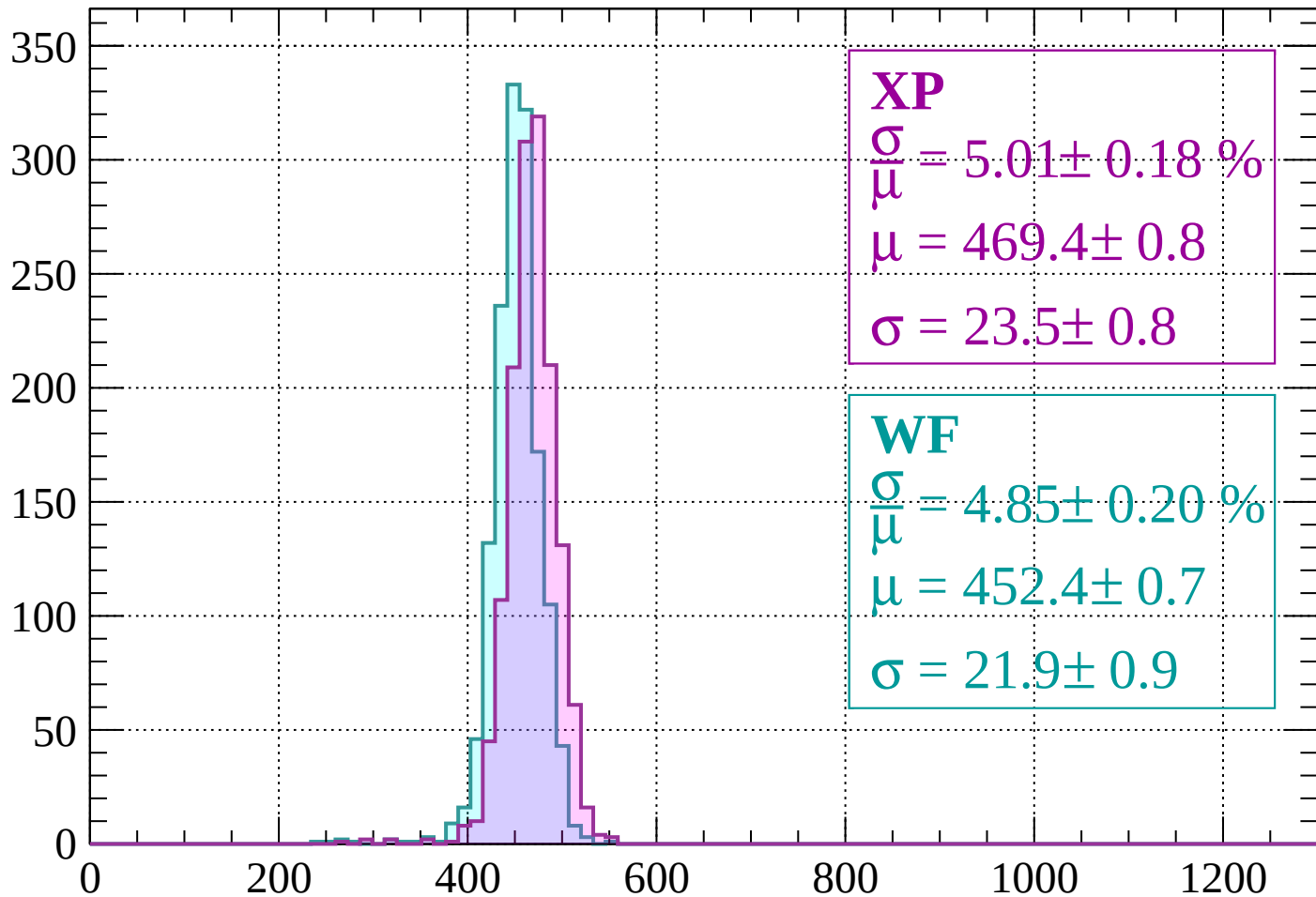
$$\mu = 455.2 \pm 0.7$$

$$\sigma = 21.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 5.01 \pm 0.18 \%$$

$$\mu = 469.4 \pm 0.8$$

$$\sigma = 23.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.85 \pm 0.20 \%$$

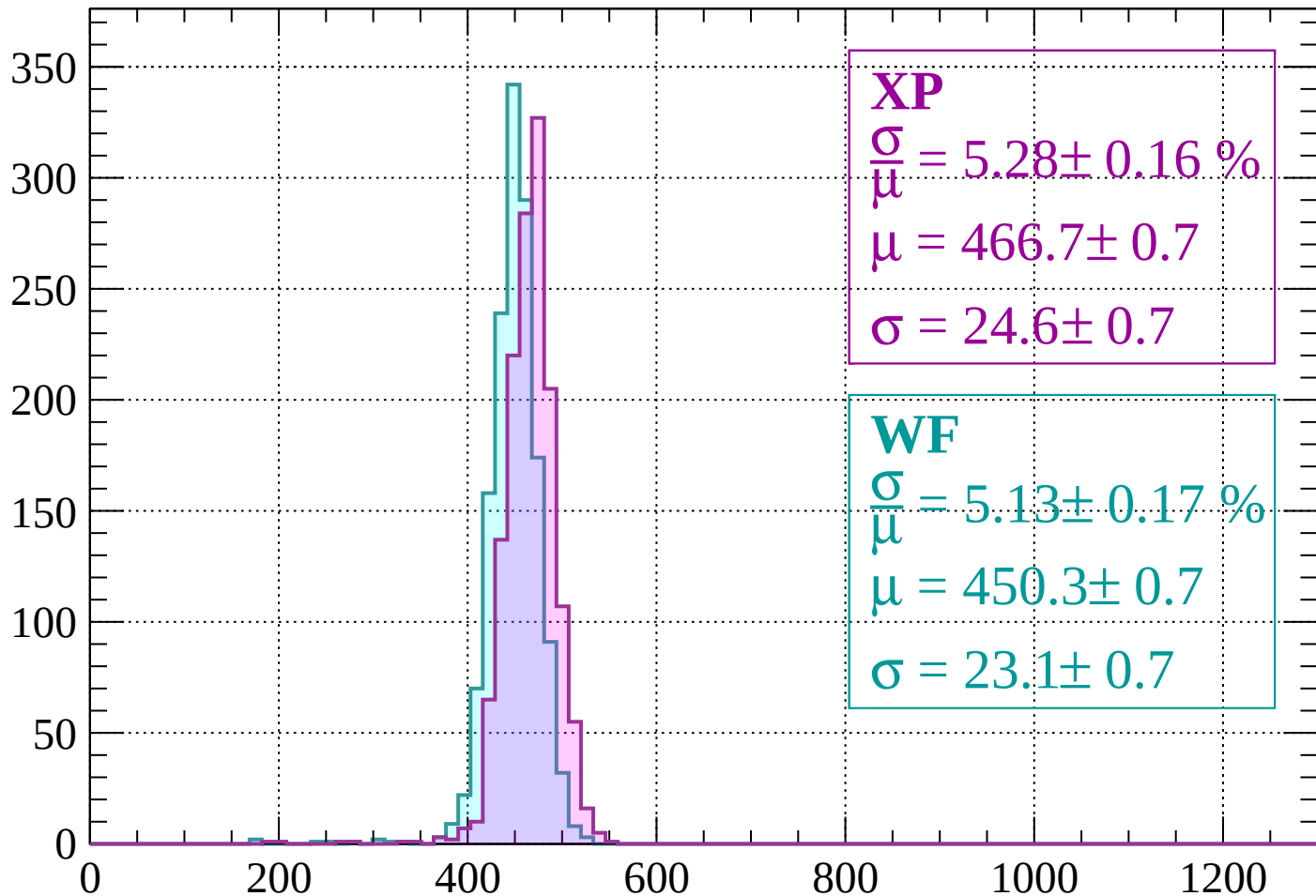
$$\mu = 452.4 \pm 0.7$$

$$\sigma = 21.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.16 \%$$

$$\mu = 466.7 \pm 0.7$$

$$\sigma = 24.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.13 \pm 0.17 \%$$

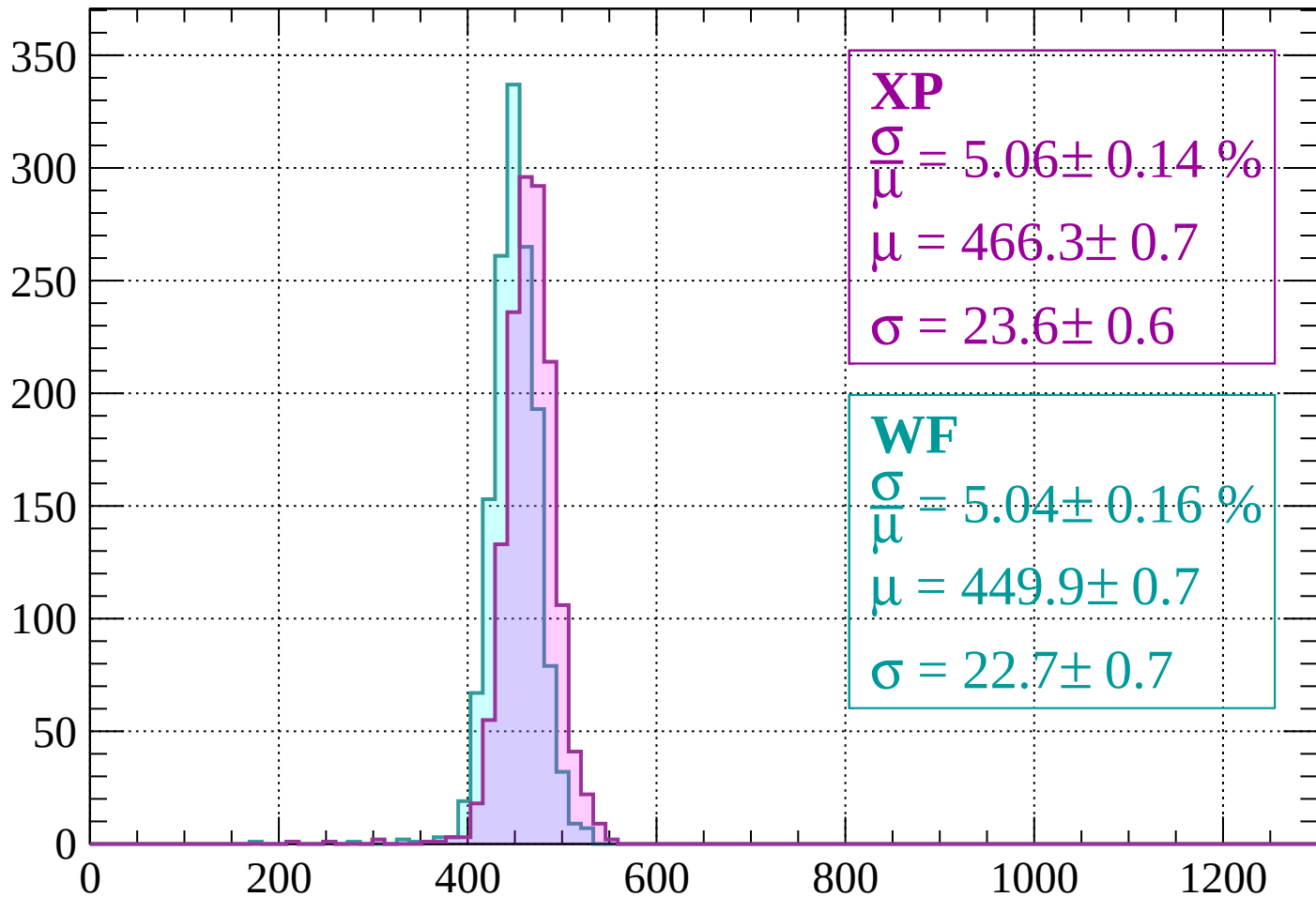
$$\mu = 450.3 \pm 0.7$$

$$\sigma = 23.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 5.06 \pm 0.14 \%$$

$$\mu = 466.3 \pm 0.7$$

$$\sigma = 23.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.04 \pm 0.16 \%$$

$$\mu = 449.9 \pm 0.7$$

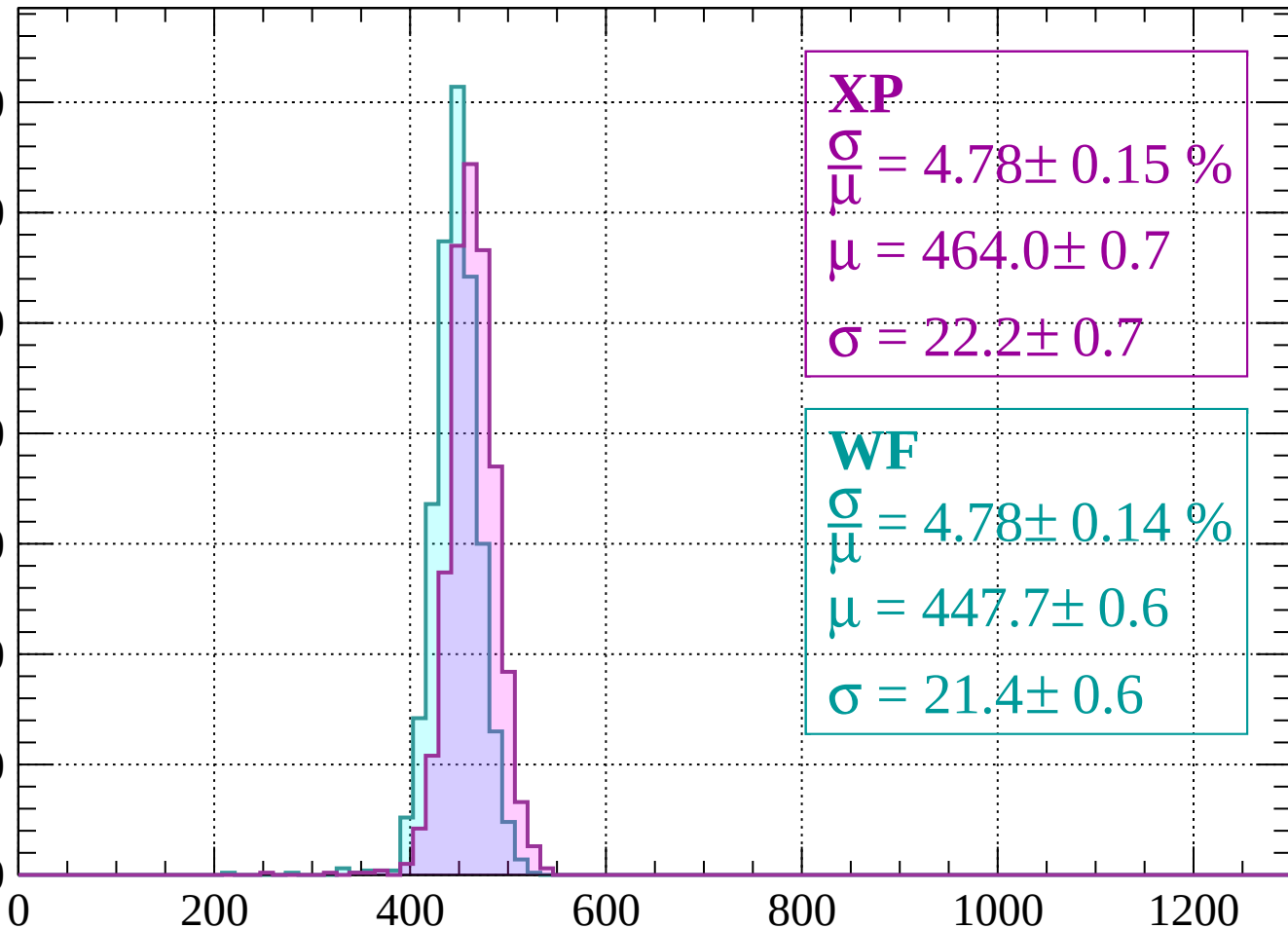
$$\sigma = 22.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.15 \%$$

$$\mu = 464.0 \pm 0.7$$

$$\sigma = 22.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.78 \pm 0.14 \%$$

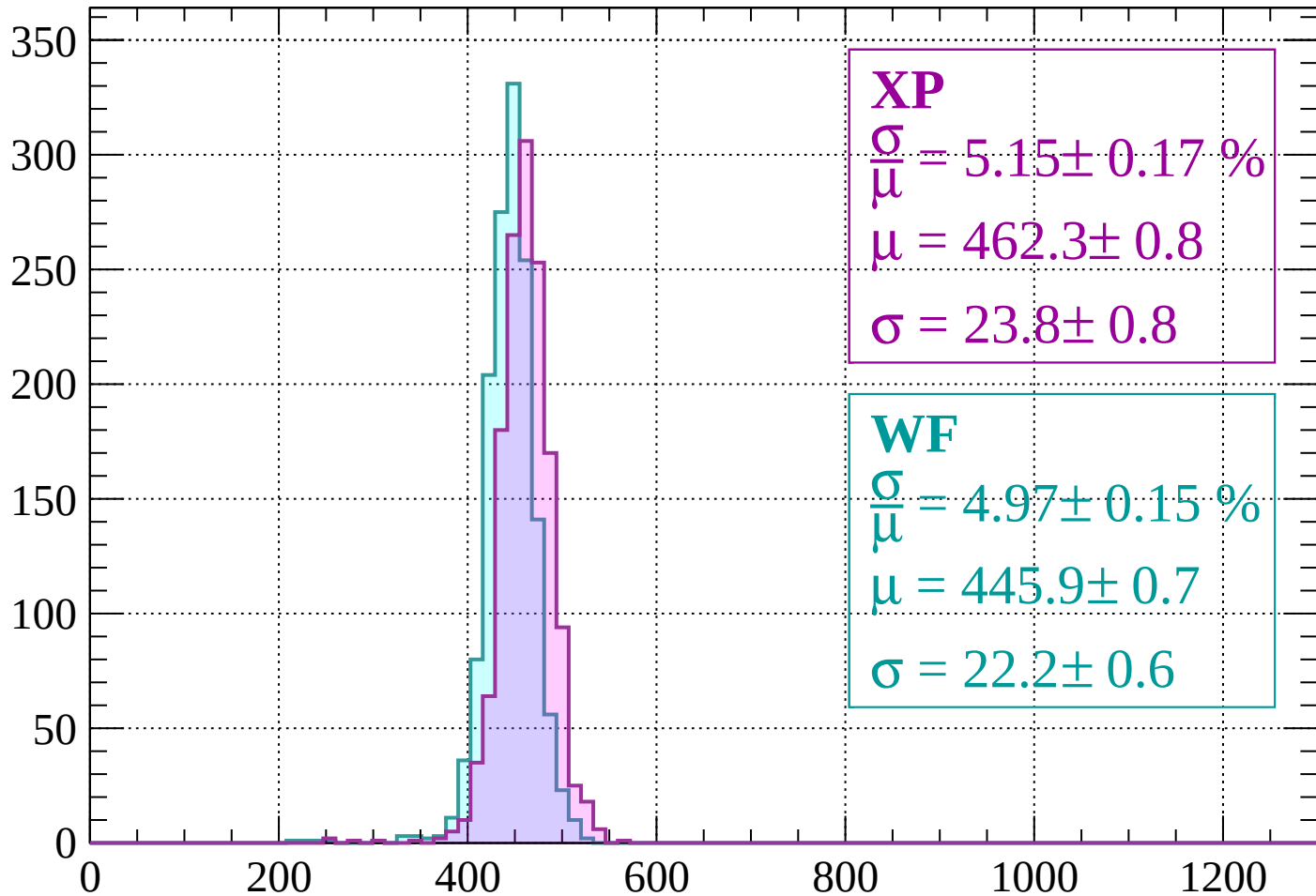
$$\mu = 447.7 \pm 0.6$$

$$\sigma = 21.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 5.15 \pm 0.17 \%$$

$$\mu = 462.3 \pm 0.8$$

$$\sigma = 23.8 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 4.97 \pm 0.15 \%$$

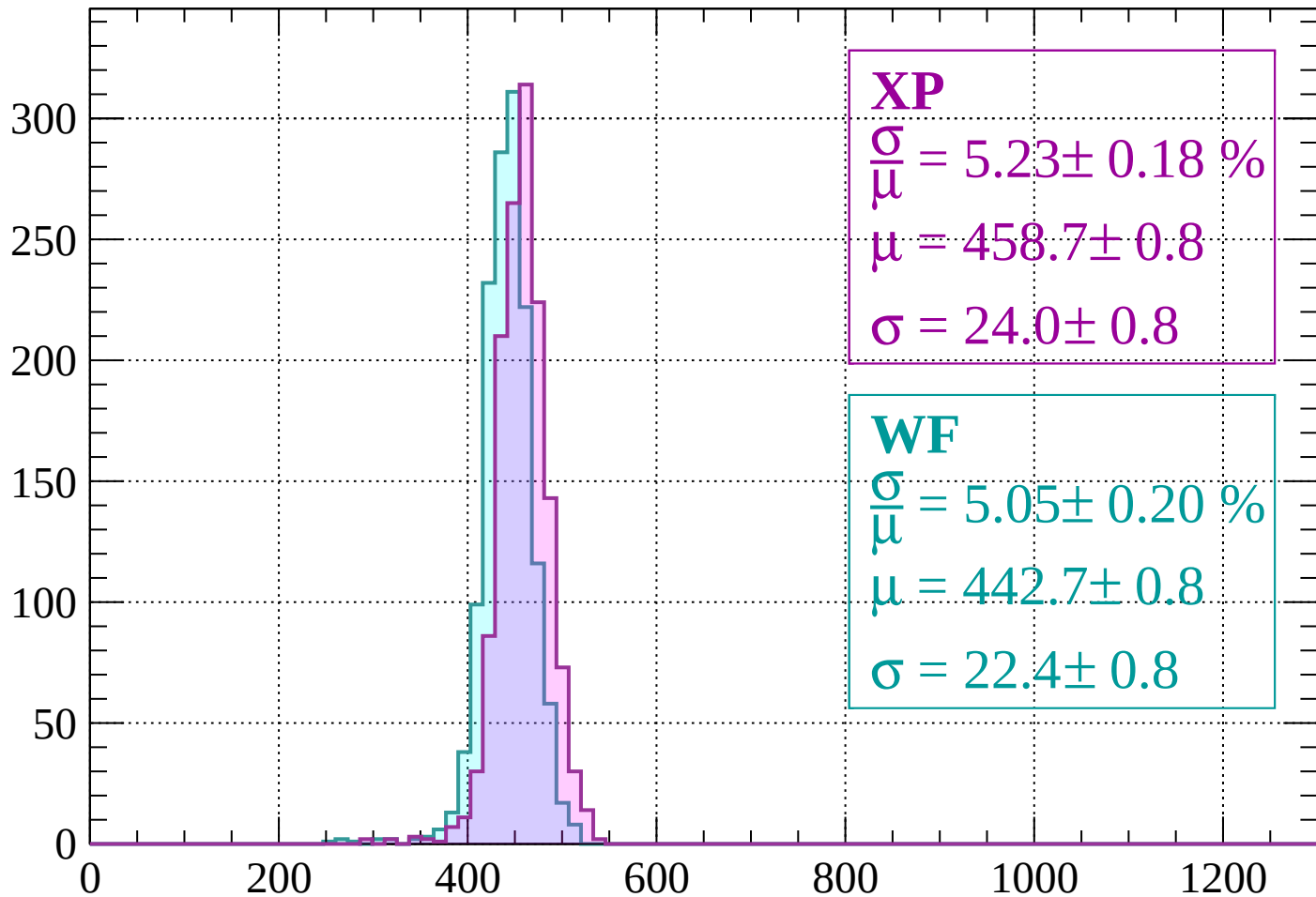
$$\mu = 445.9 \pm 0.7$$

$$\sigma = 22.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.18 \%$$

$$\mu = 458.7 \pm 0.8$$

$$\sigma = 24.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.05 \pm 0.20 \%$$

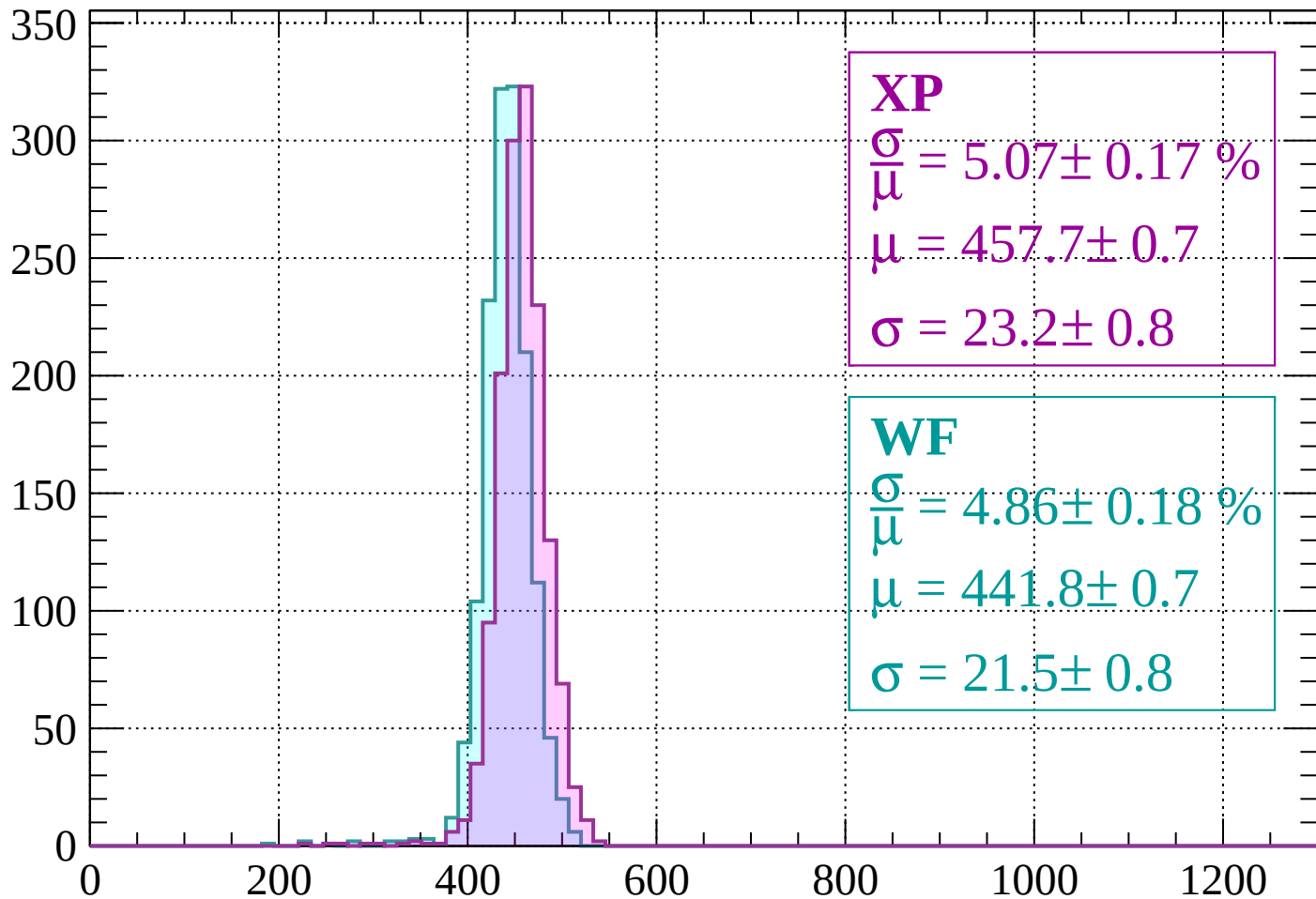
$$\mu = 442.7 \pm 0.8$$

$$\sigma = 22.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 5.07 \pm 0.17 \%$$

$$\mu = 457.7 \pm 0.7$$

$$\sigma = 23.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.86 \pm 0.18 \%$$

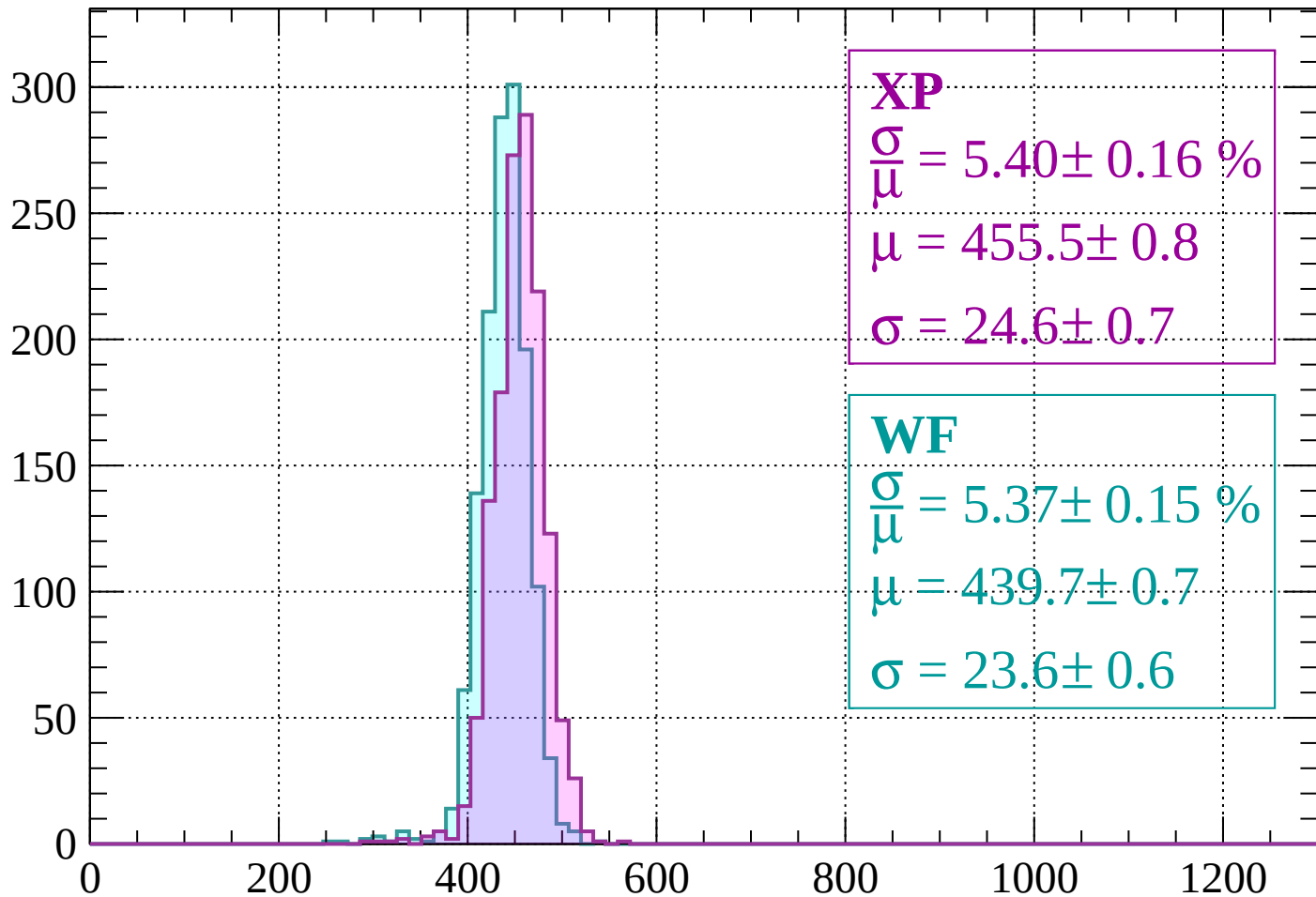
$$\mu = 441.8 \pm 0.7$$

$$\sigma = 21.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 5.40 \pm 0.16 \%$$

$$\mu = 455.5 \pm 0.8$$

$$\sigma = 24.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.37 \pm 0.15 \%$$

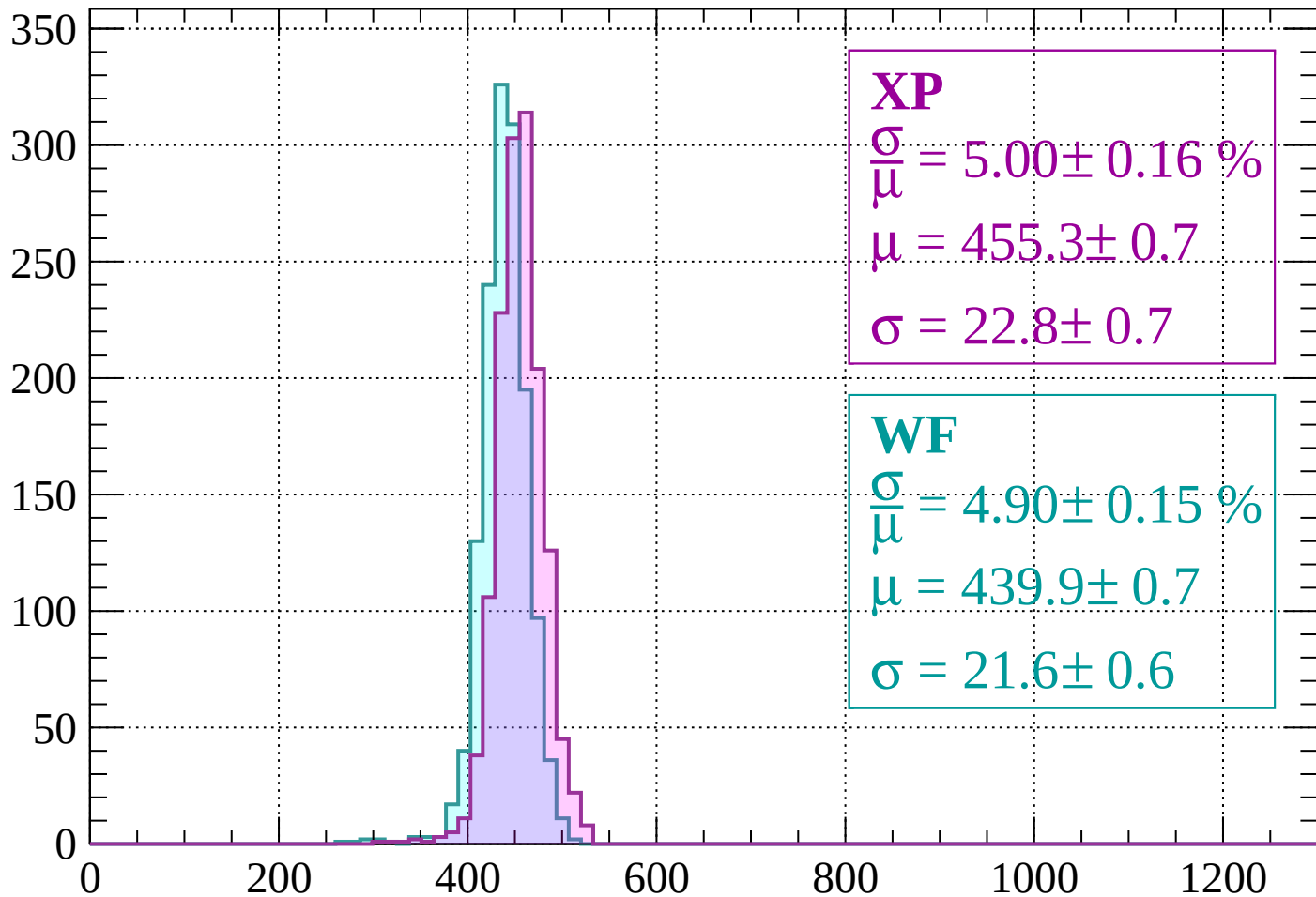
$$\mu = 439.7 \pm 0.7$$

$$\sigma = 23.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 5.00 \pm 0.16 \%$$

$$\mu = 455.3 \pm 0.7$$

$$\sigma = 22.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.90 \pm 0.15 \%$$

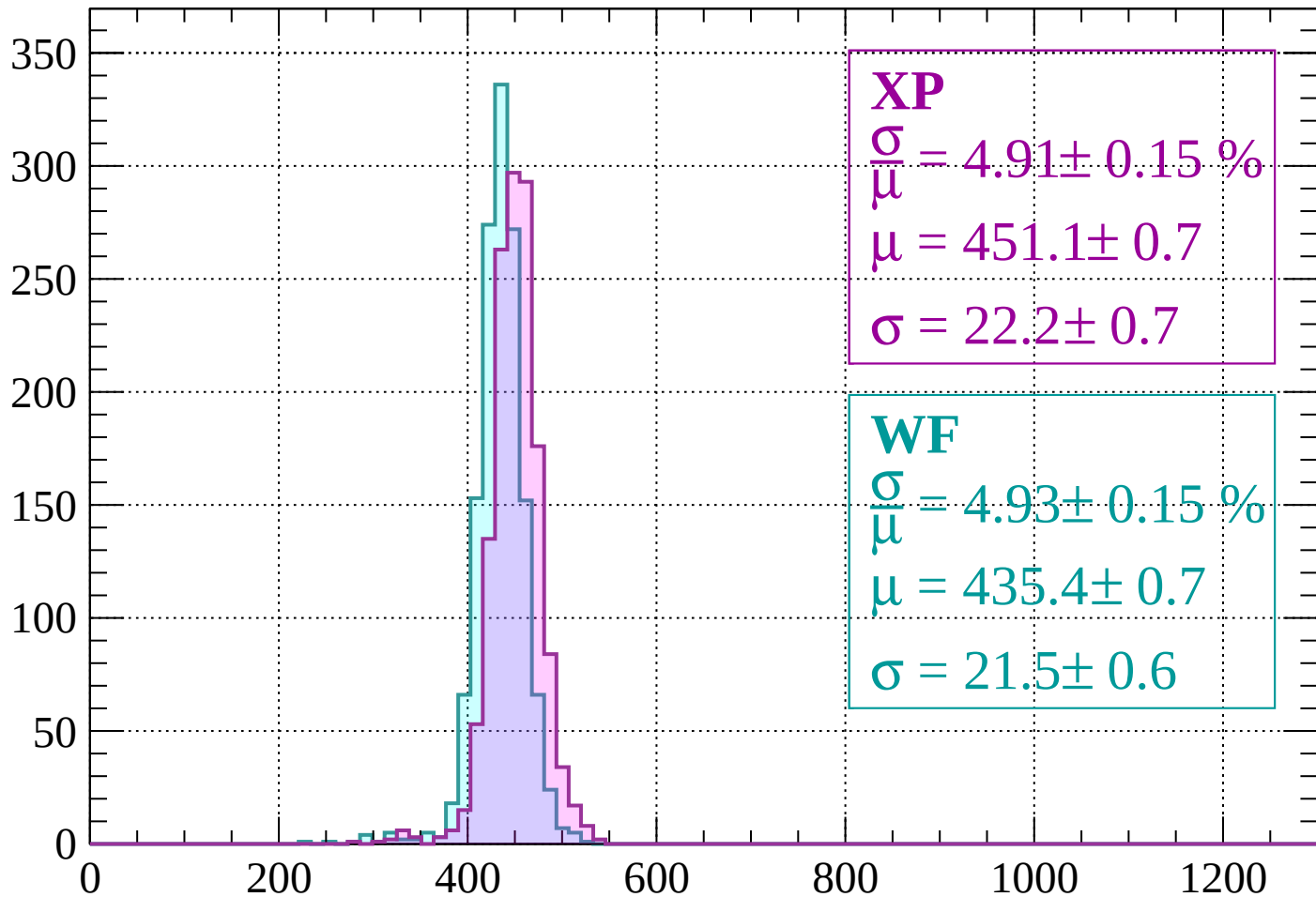
$$\mu = 439.9 \pm 0.7$$

$$\sigma = 21.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 4.91 \pm 0.15 \%$$

$$\mu = 451.1 \pm 0.7$$

$$\sigma = 22.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.93 \pm 0.15 \%$$

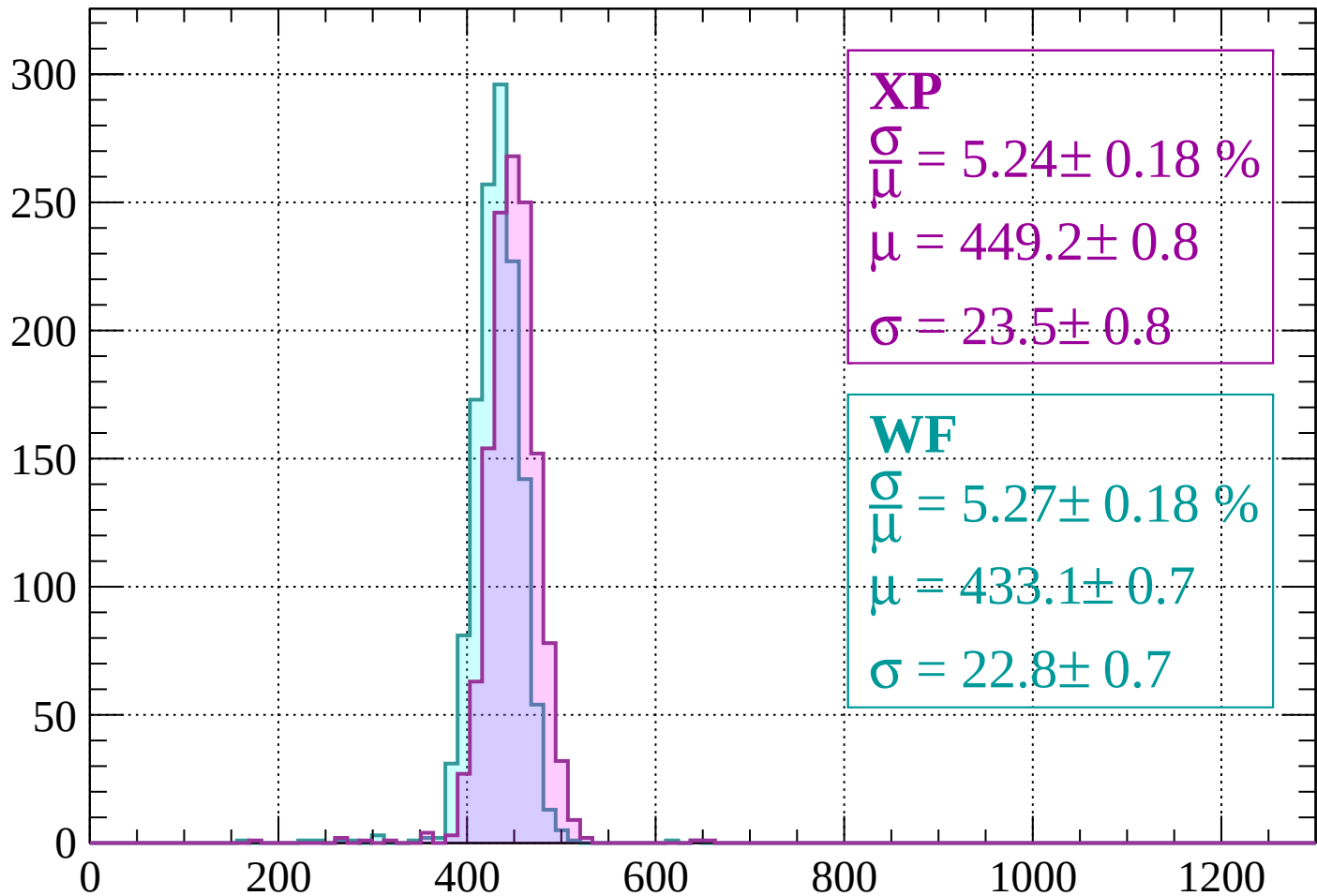
$$\mu = 435.4 \pm 0.7$$

$$\sigma = 21.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 5.24 \pm 0.18 \%$$

$$\mu = 449.2 \pm 0.8$$

$$\sigma = 23.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.27 \pm 0.18 \%$$

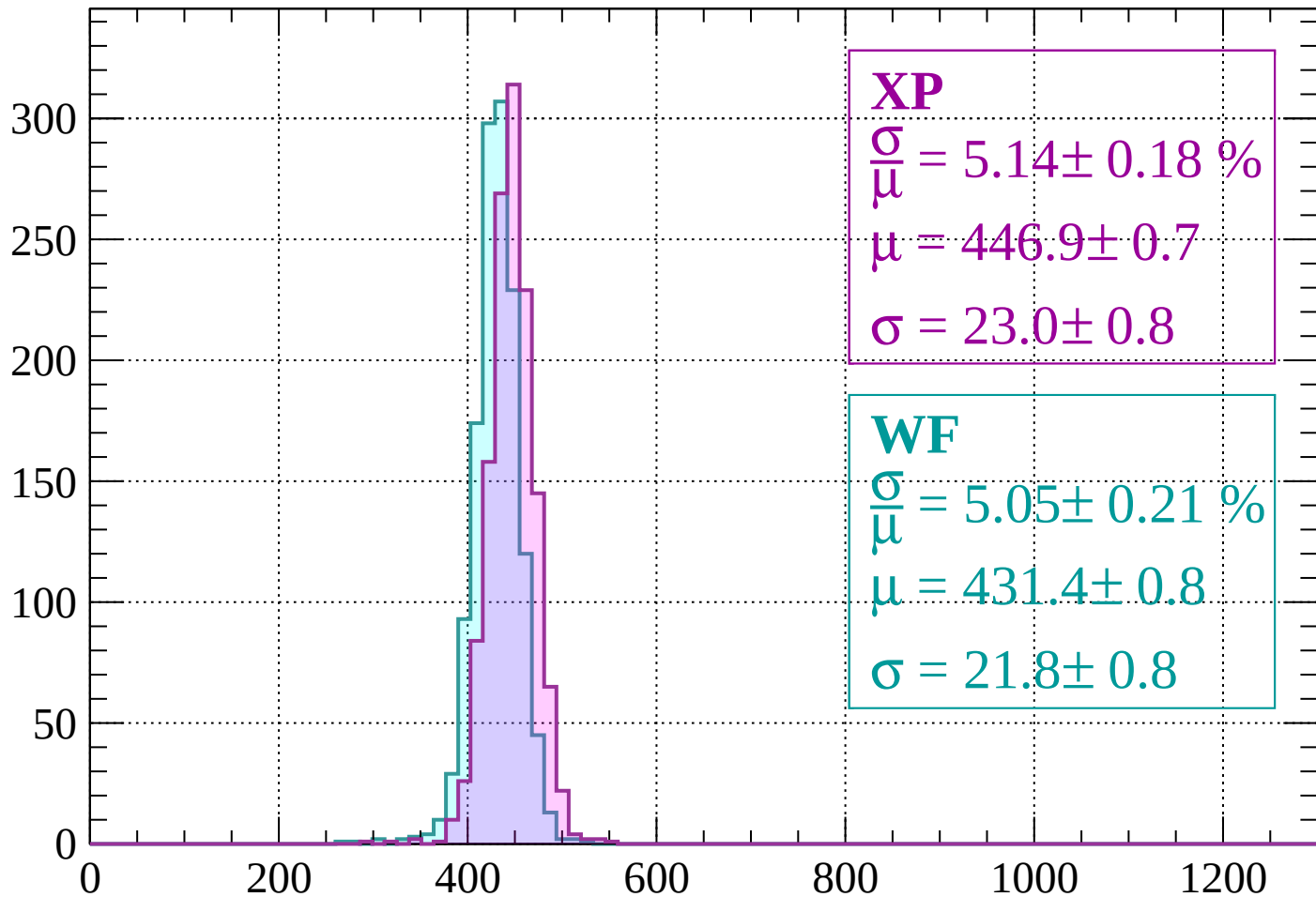
$$\mu = 433.1 \pm 0.7$$

$$\sigma = 22.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.18 \%$$

$$\mu = 446.9 \pm 0.7$$

$$\sigma = 23.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.05 \pm 0.21 \%$$

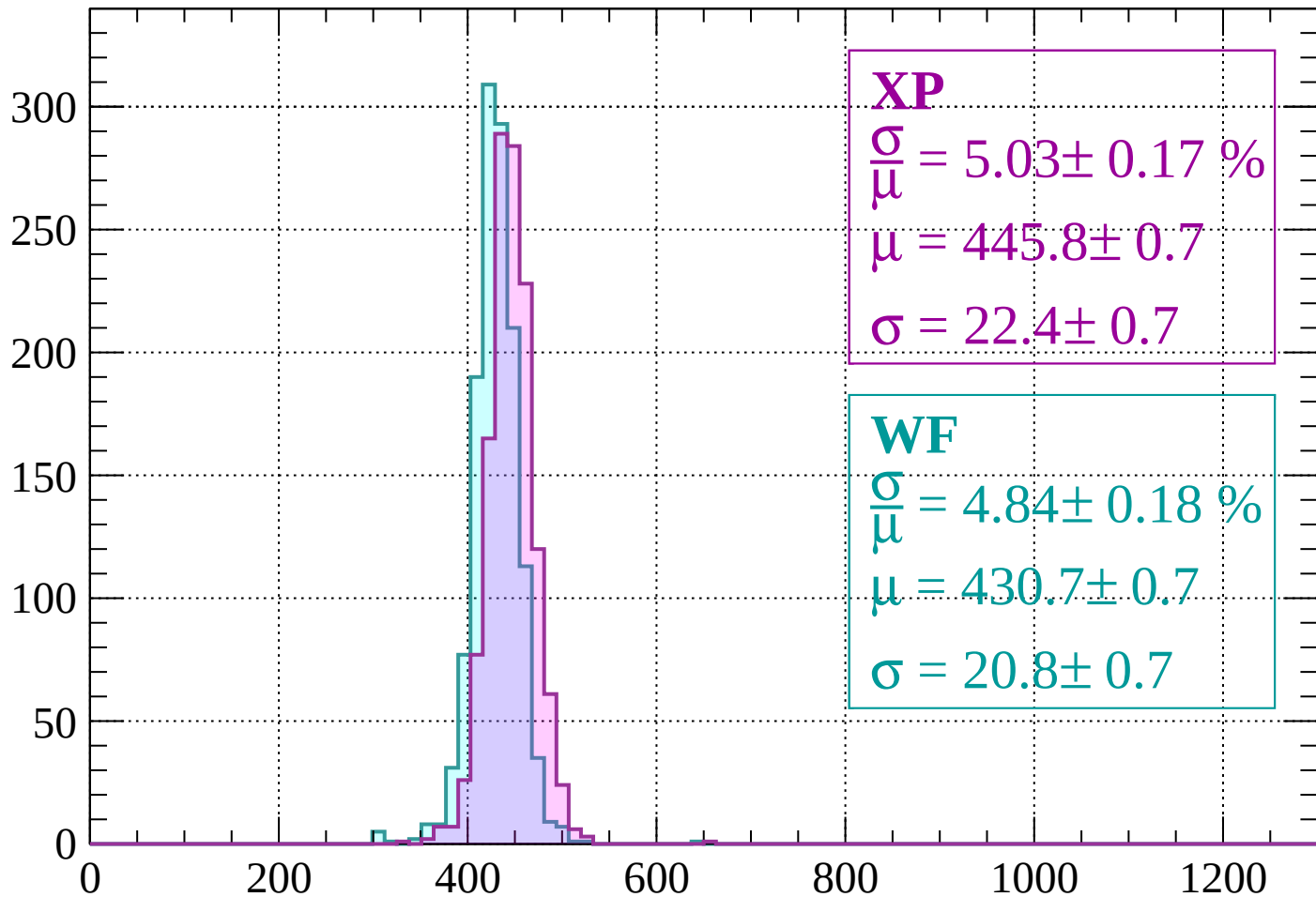
$$\mu = 431.4 \pm 0.8$$

$$\sigma = 21.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 5.03 \pm 0.17 \%$$

$$\mu = 445.8 \pm 0.7$$

$$\sigma = 22.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.84 \pm 0.18 \%$$

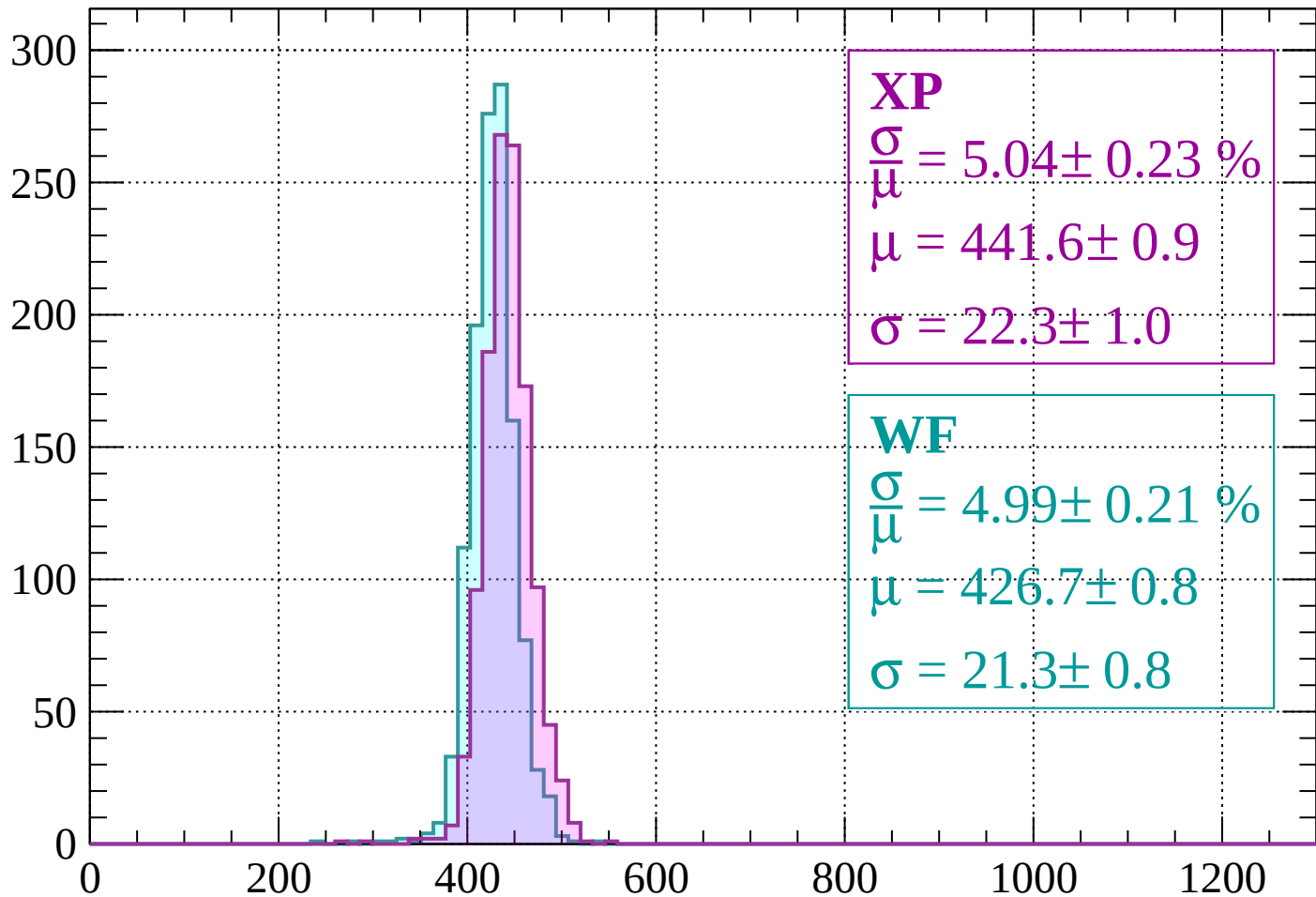
$$\mu = 430.7 \pm 0.7$$

$$\sigma = 20.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.23 \%$$

$$\mu = 441.6 \pm 0.9$$

$$\sigma = 22.3 \pm 1.0$$

WF

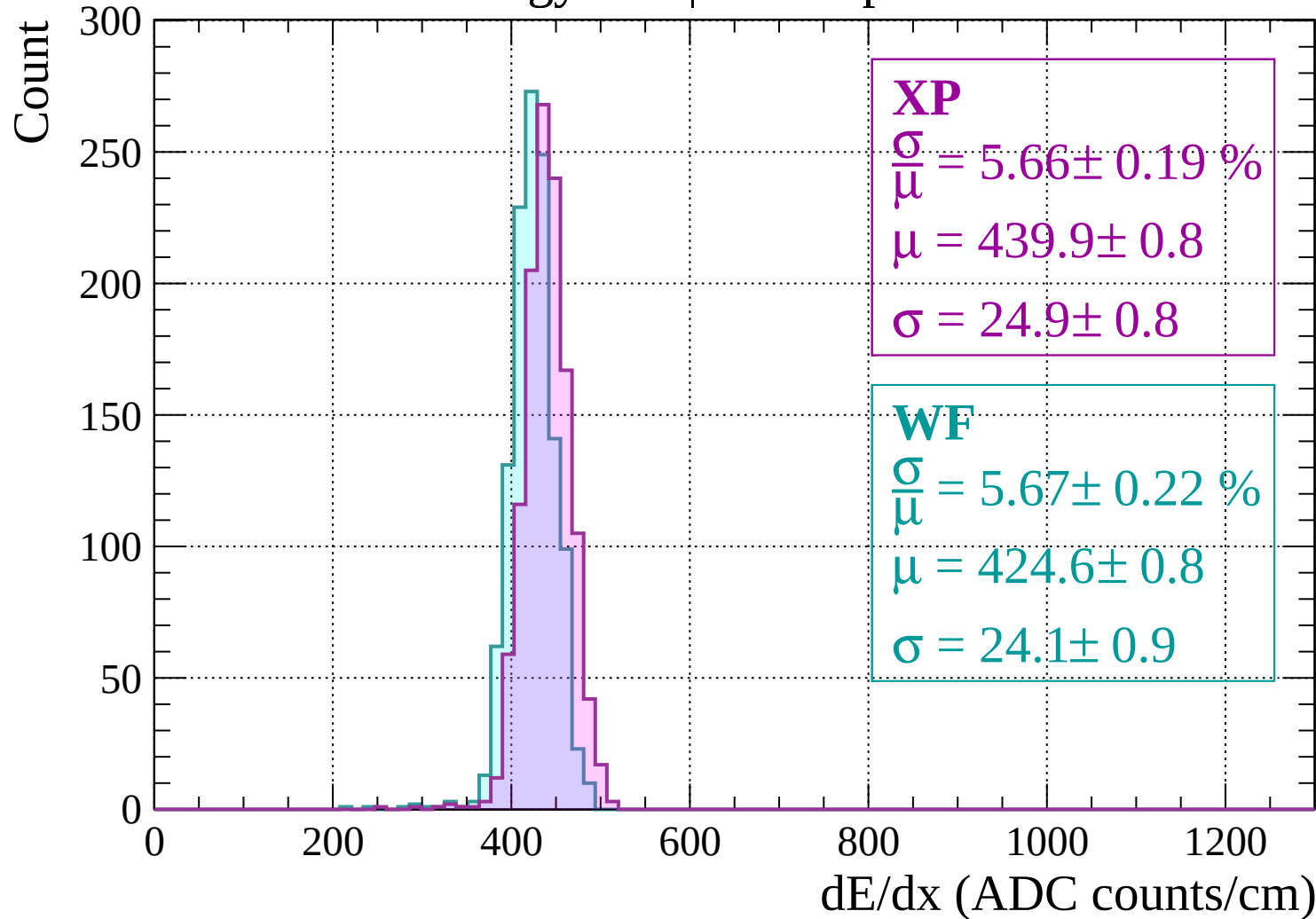
$$\frac{\sigma}{\mu} = 4.99 \pm 0.21 \%$$

$$\mu = 426.7 \pm 0.8$$

$$\sigma = 21.3 \pm 0.8$$

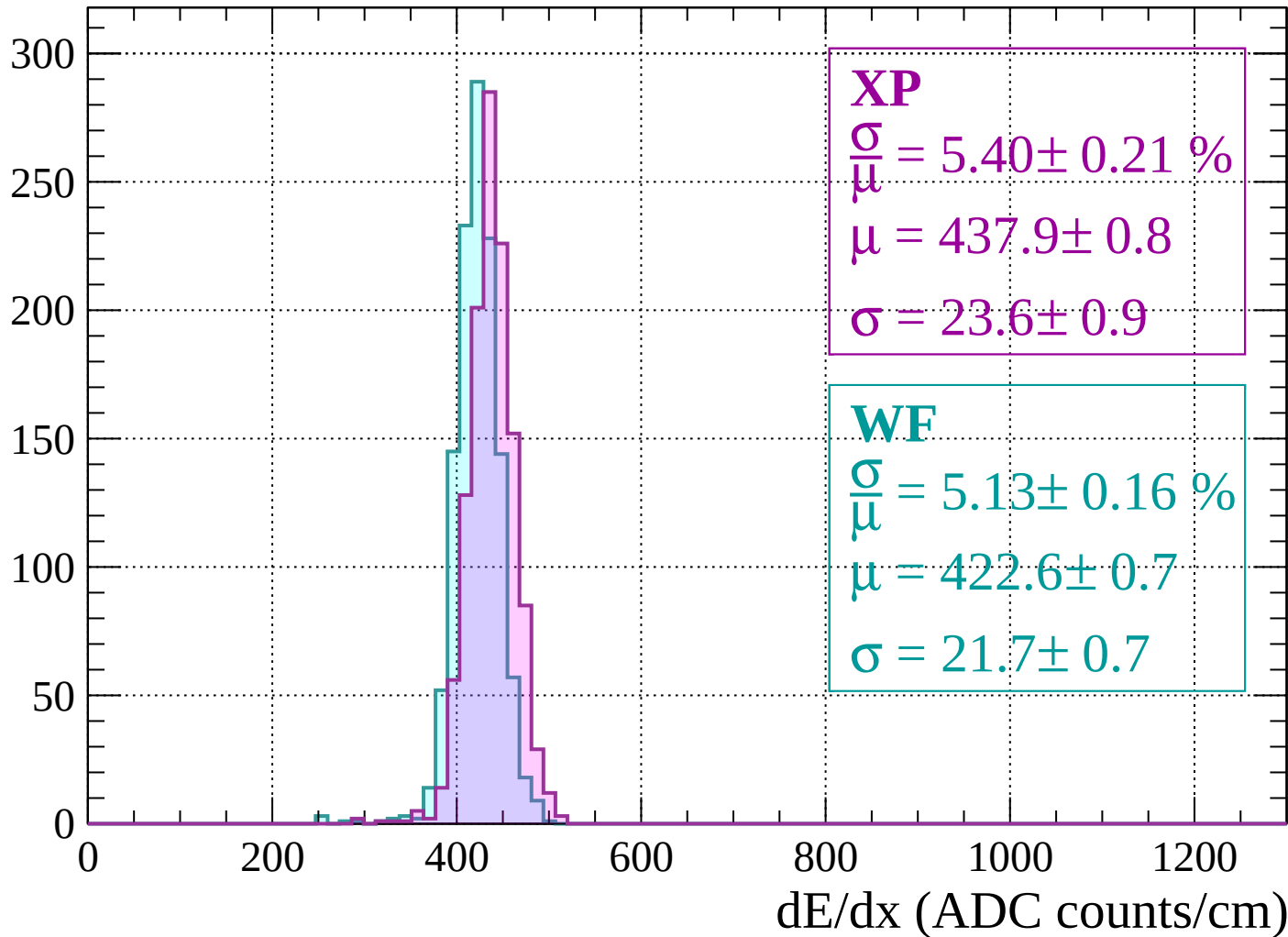
dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$



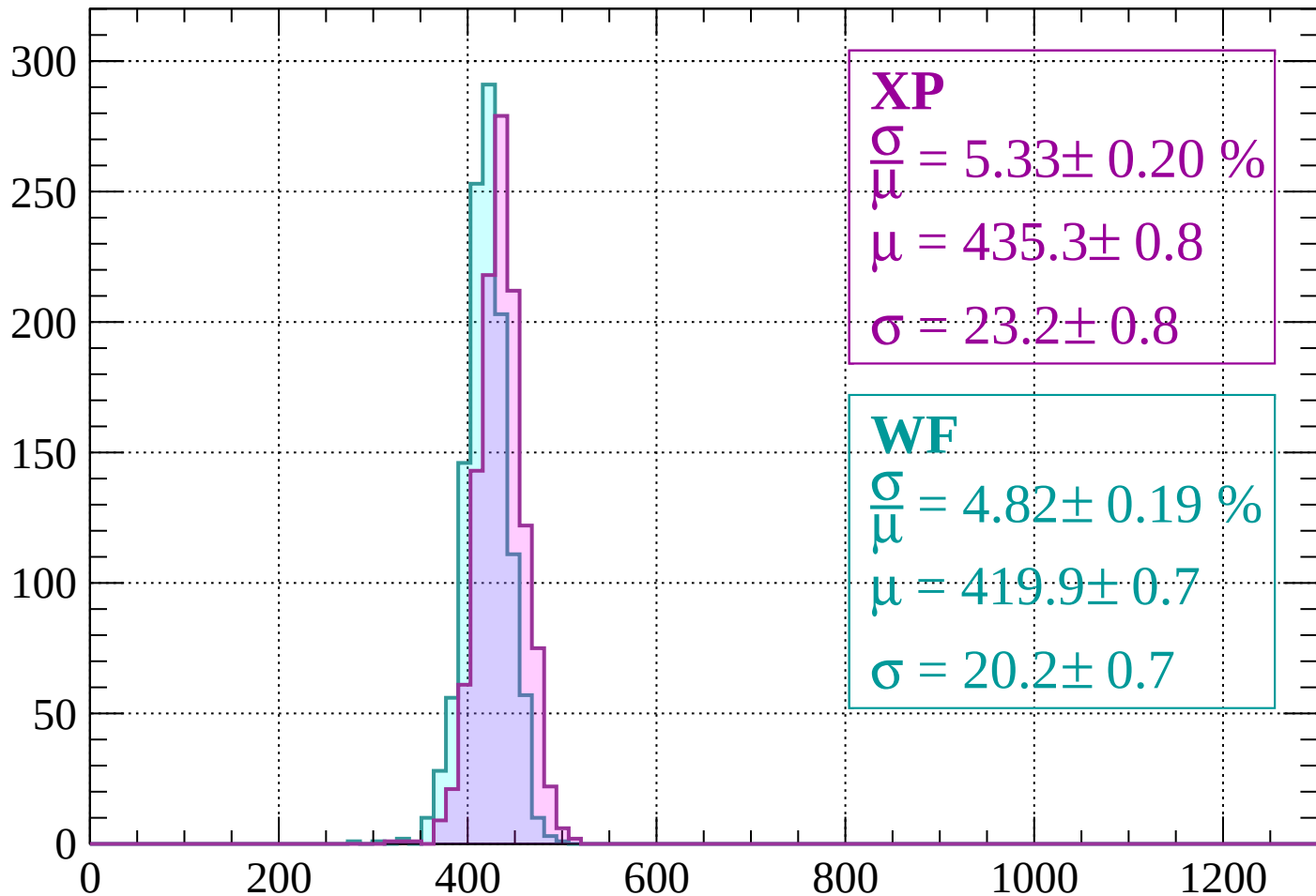
Energy loss | $-760 < p < -720$

Count



Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 5.33 \pm 0.20 \%$$

$$\mu = 435.3 \pm 0.8$$

$$\sigma = 23.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.19 \%$$

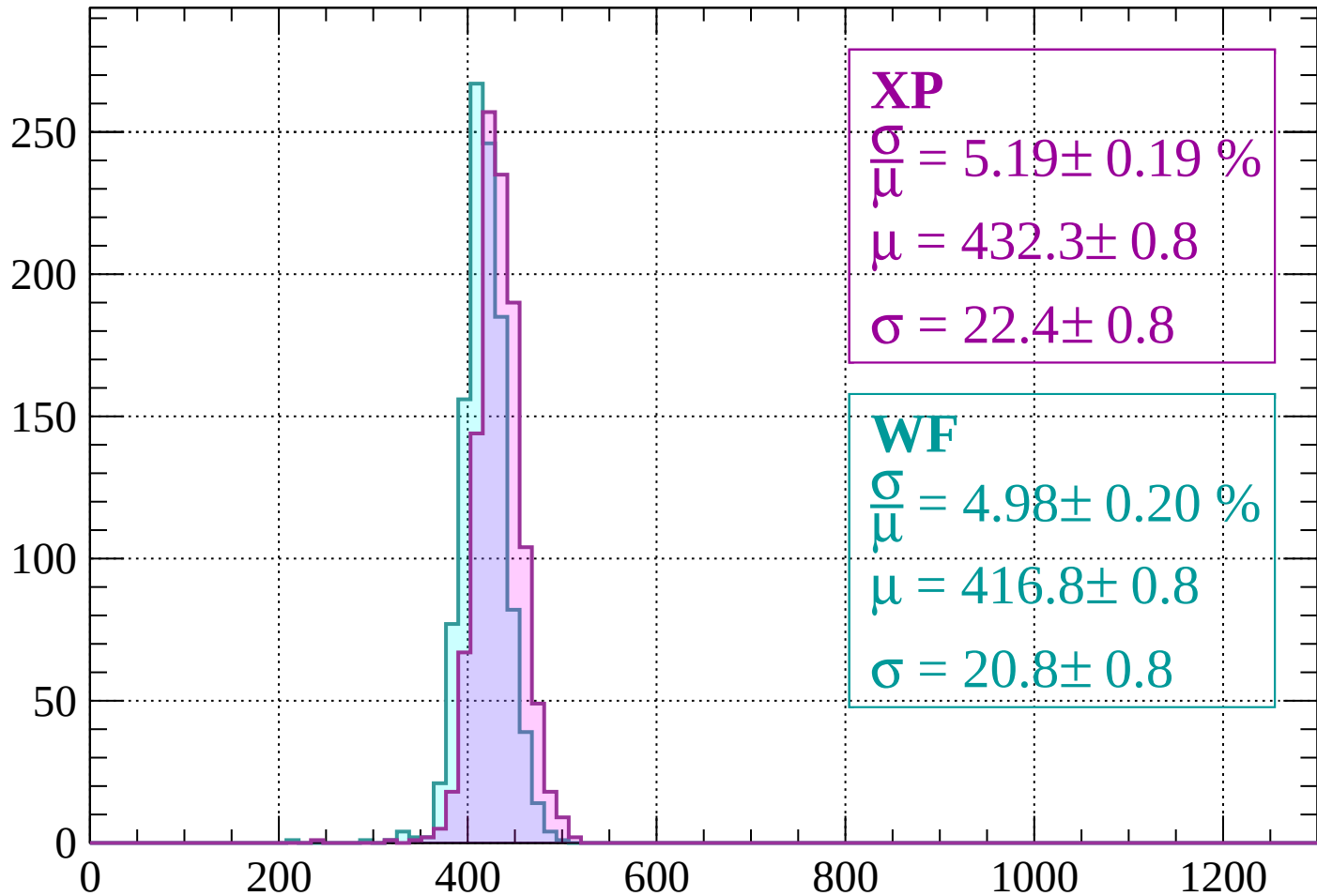
$$\mu = 419.9 \pm 0.7$$

$$\sigma = 20.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 5.19 \pm 0.19 \%$$

$$\mu = 432.3 \pm 0.8$$

$$\sigma = 22.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.98 \pm 0.20 \%$$

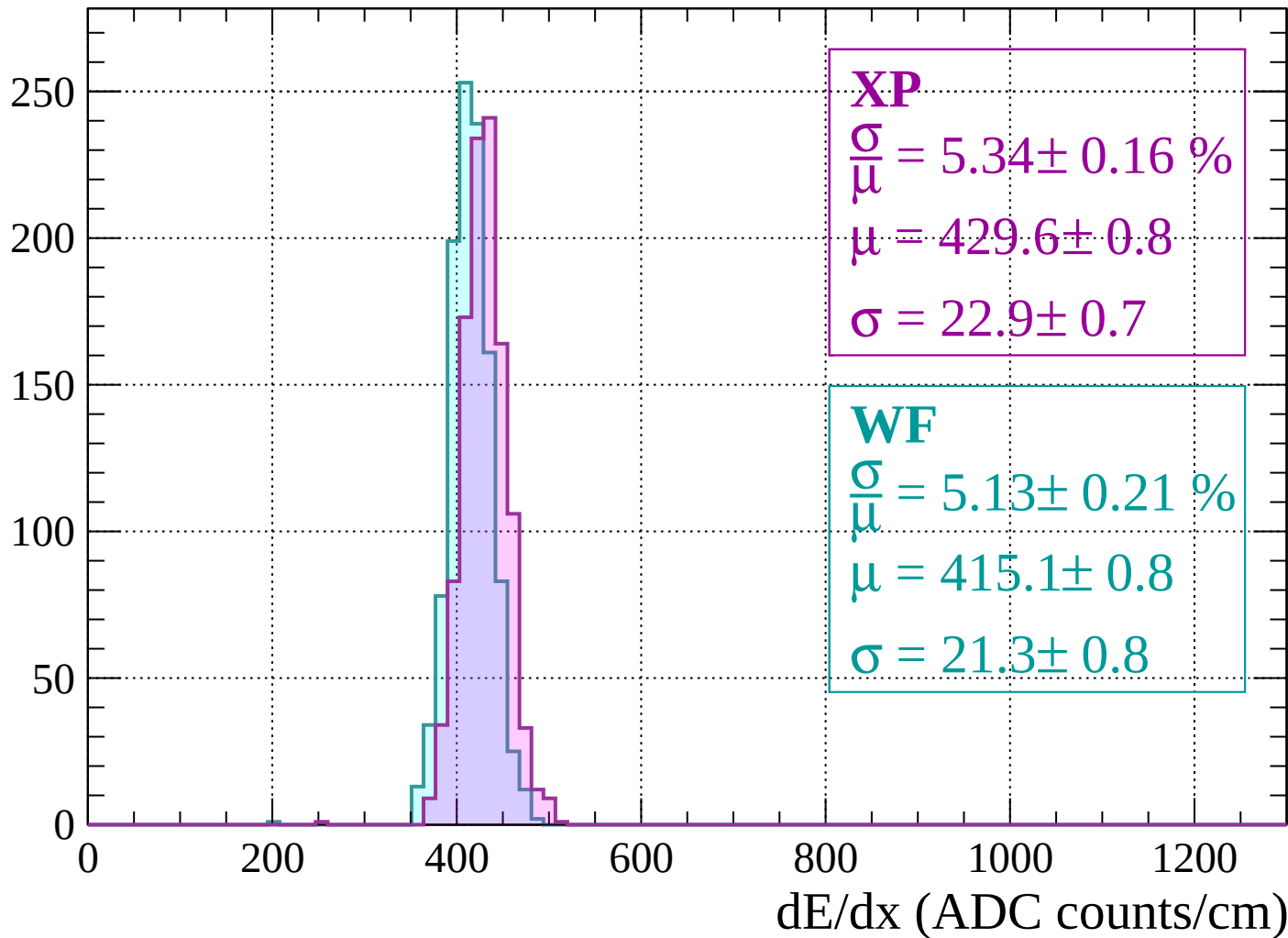
$$\mu = 416.8 \pm 0.8$$

$$\sigma = 20.8 \pm 0.8$$

dE/dx (ADC counts/cm)

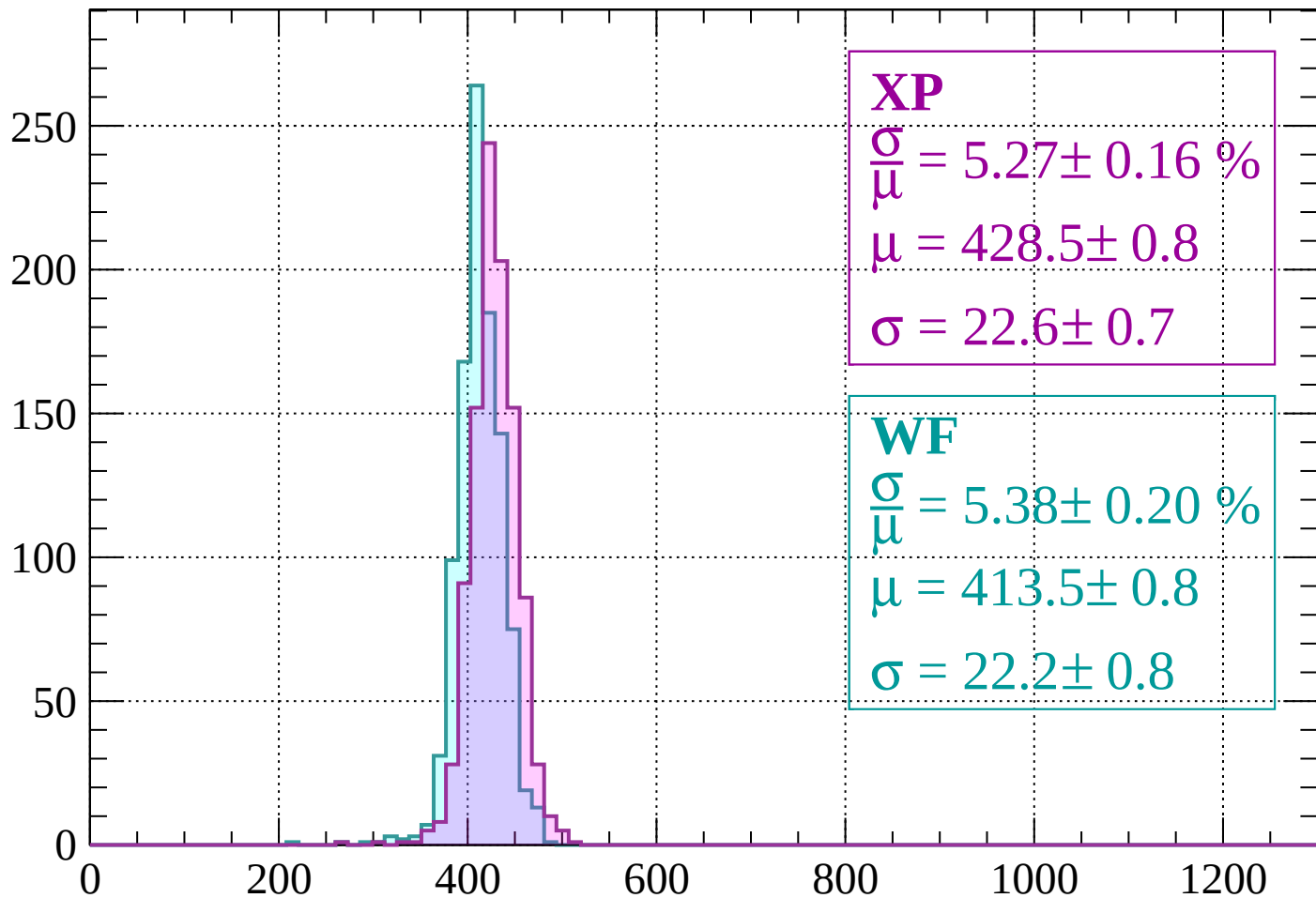
Energy loss | $-640 < p < -600$

Count



Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 5.27 \pm 0.16 \%$$

$$\mu = 428.5 \pm 0.8$$

$$\sigma = 22.6 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 5.38 \pm 0.20 \%$$

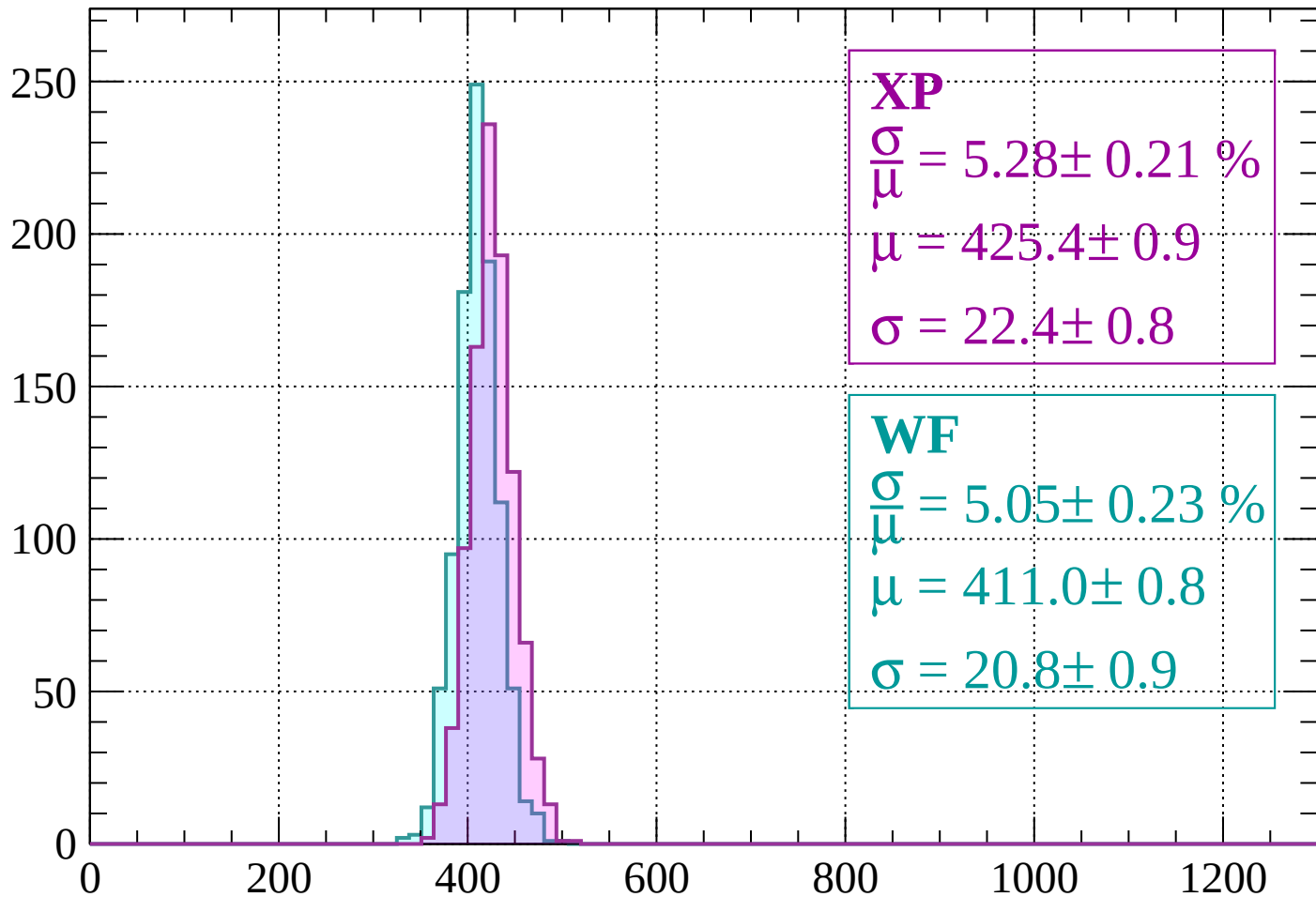
$$\mu = 413.5 \pm 0.8$$

$$\sigma = 22.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.21 \%$$

$$\mu = 425.4 \pm 0.9$$

$$\sigma = 22.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.05 \pm 0.23 \%$$

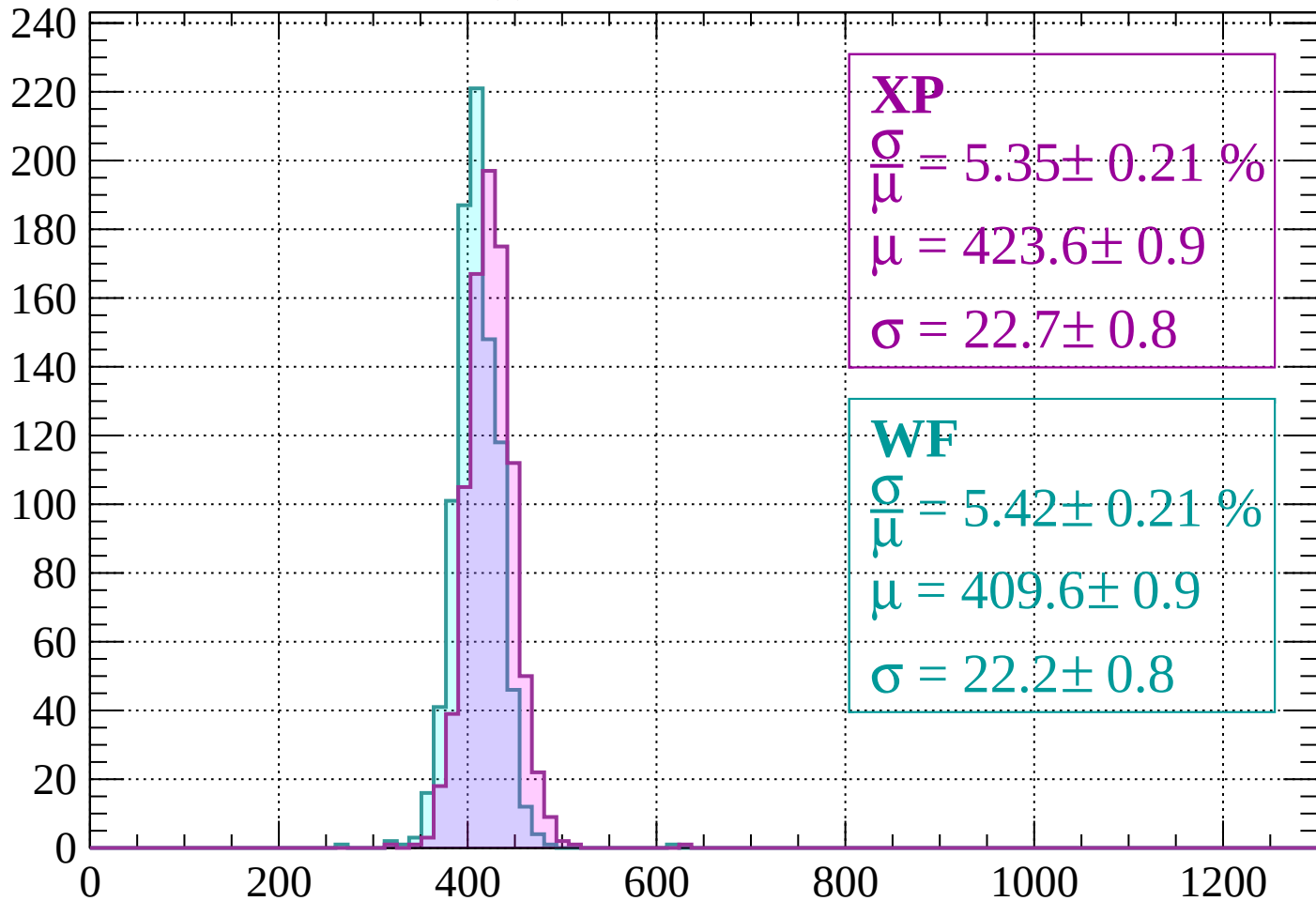
$$\mu = 411.0 \pm 0.8$$

$$\sigma = 20.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 5.35 \pm 0.21 \%$$

$$\mu = 423.6 \pm 0.9$$

$$\sigma = 22.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.21 \%$$

$$\mu = 409.6 \pm 0.9$$

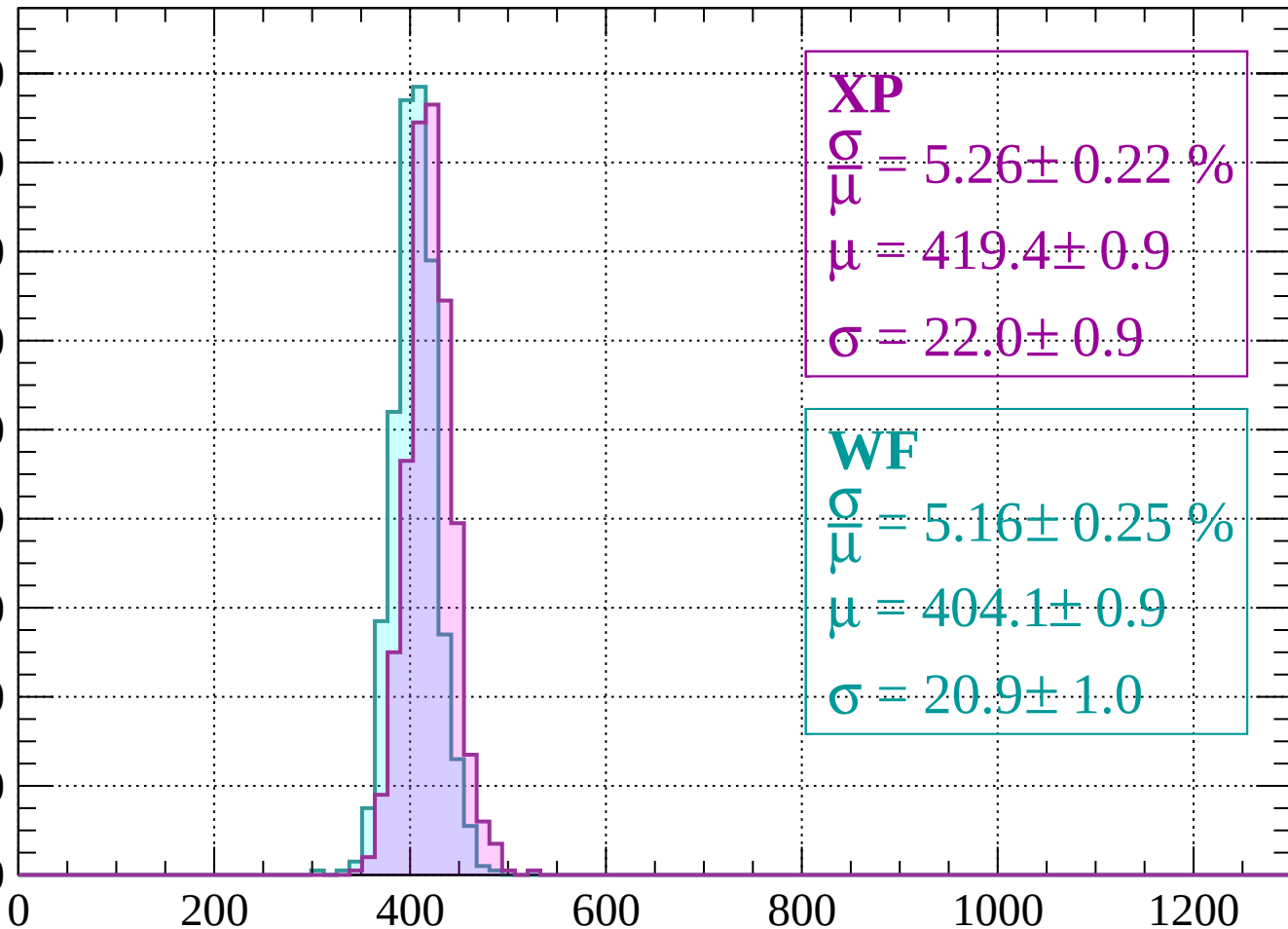
$$\sigma = 22.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 5.26 \pm 0.22 \%$$

$$\mu = 419.4 \pm 0.9$$

$$\sigma = 22.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.25 \%$$

$$\mu = 404.1 \pm 0.9$$

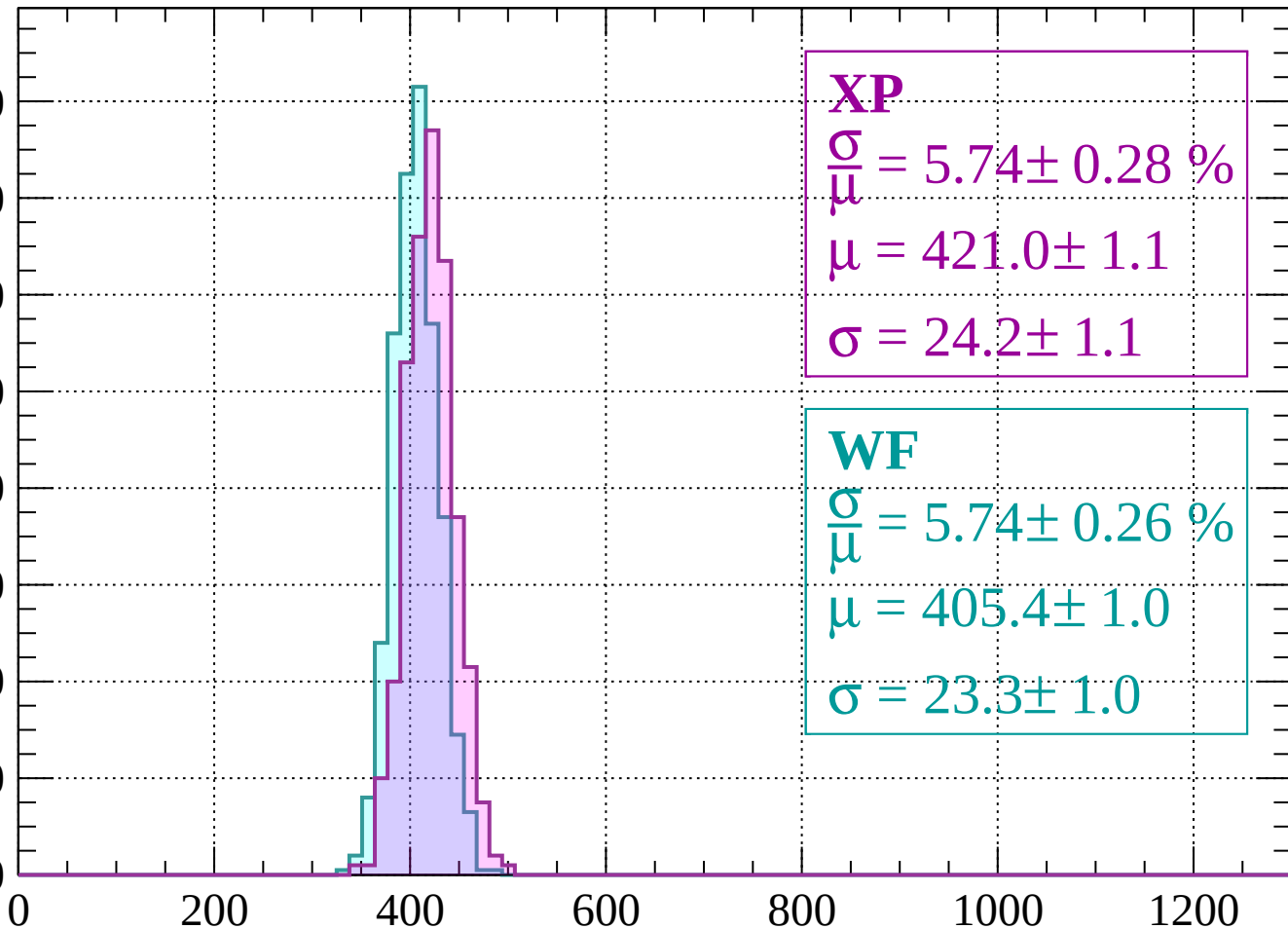
$$\sigma = 20.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 5.74 \pm 0.28 \%$$

$$\mu = 421.0 \pm 1.1$$

$$\sigma = 24.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.74 \pm 0.26 \%$$

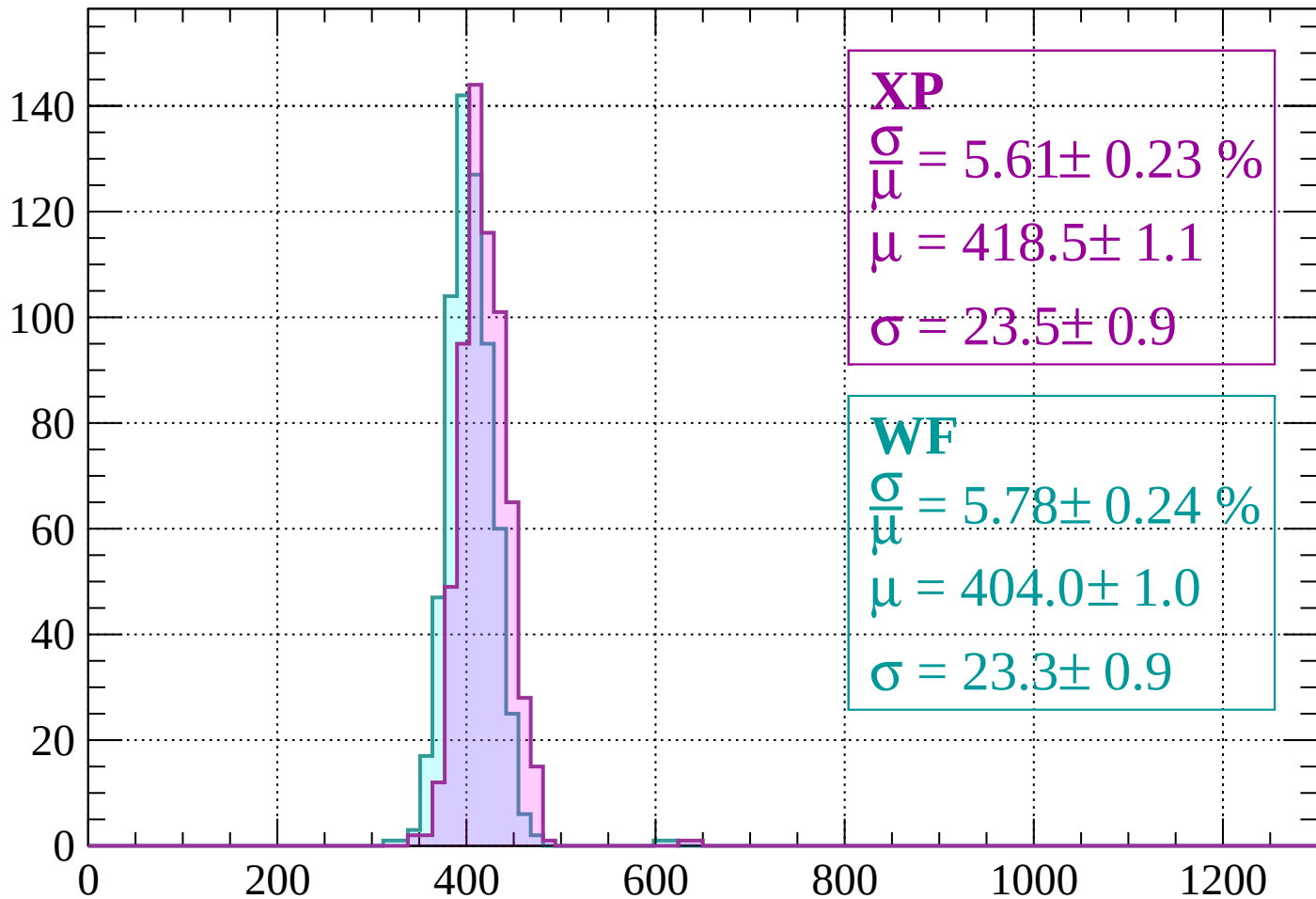
$$\mu = 405.4 \pm 1.0$$

$$\sigma = 23.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 5.61 \pm 0.23 \%$$

$$\mu = 418.5 \pm 1.1$$

$$\sigma = 23.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.78 \pm 0.24 \%$$

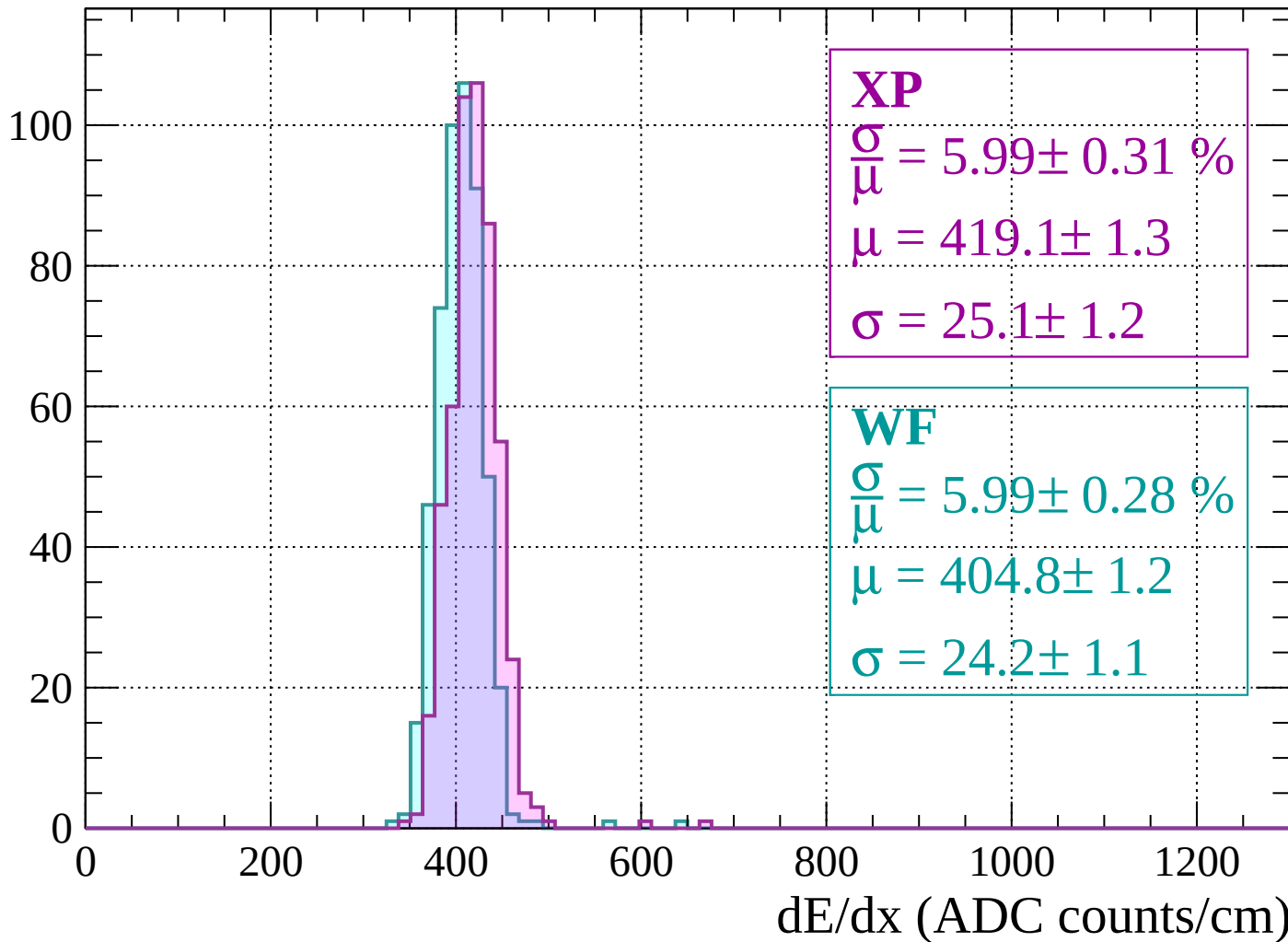
$$\mu = 404.0 \pm 1.0$$

$$\sigma = 23.3 \pm 0.9$$

dE/dx (ADC counts/cm)

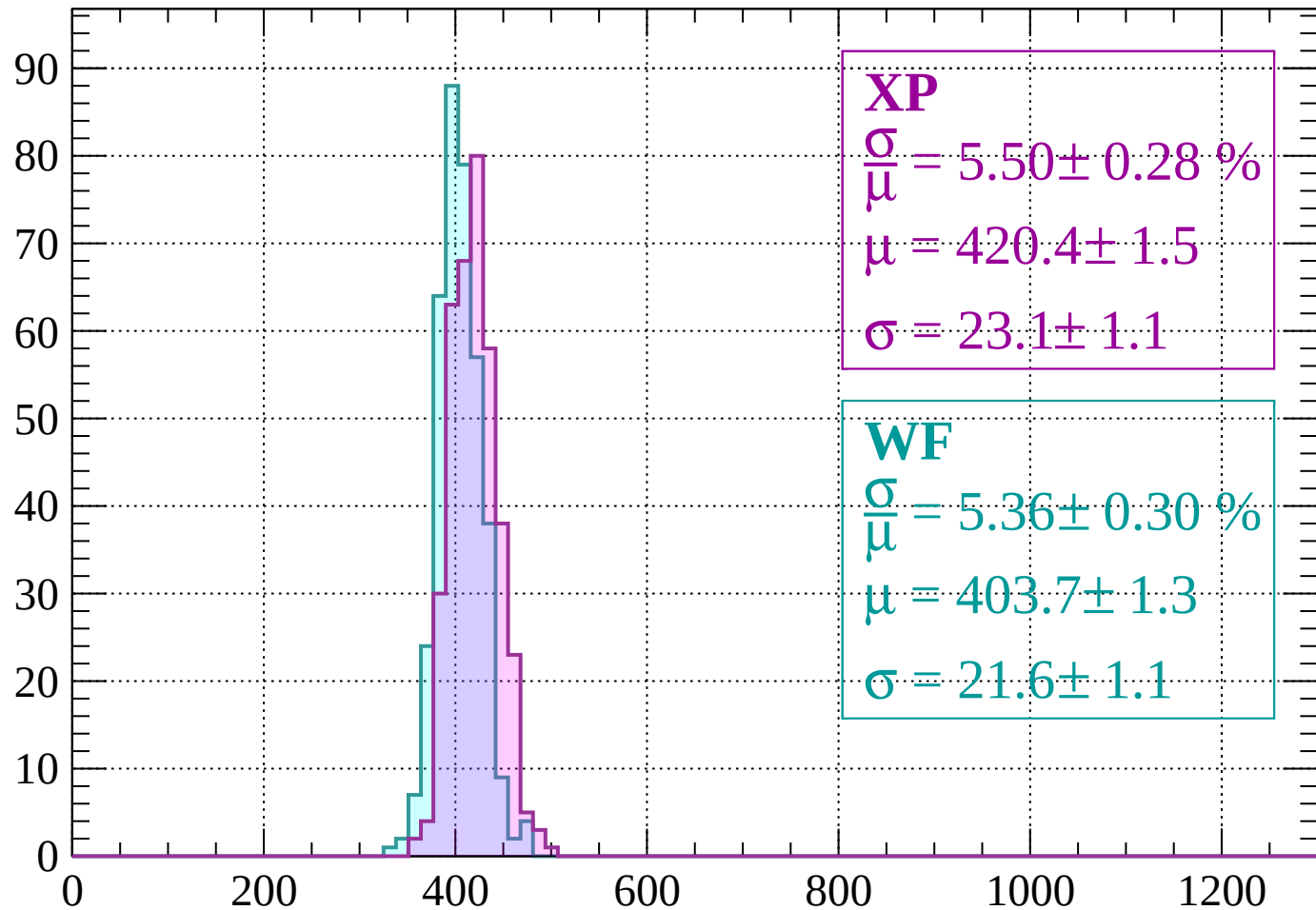
Energy loss | $-360 < p < -320$

Count



Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 5.50 \pm 0.28 \%$$

$$\mu = 420.4 \pm 1.5$$

$$\sigma = 23.1 \pm 1.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 5.36 \pm 0.30 \%$$

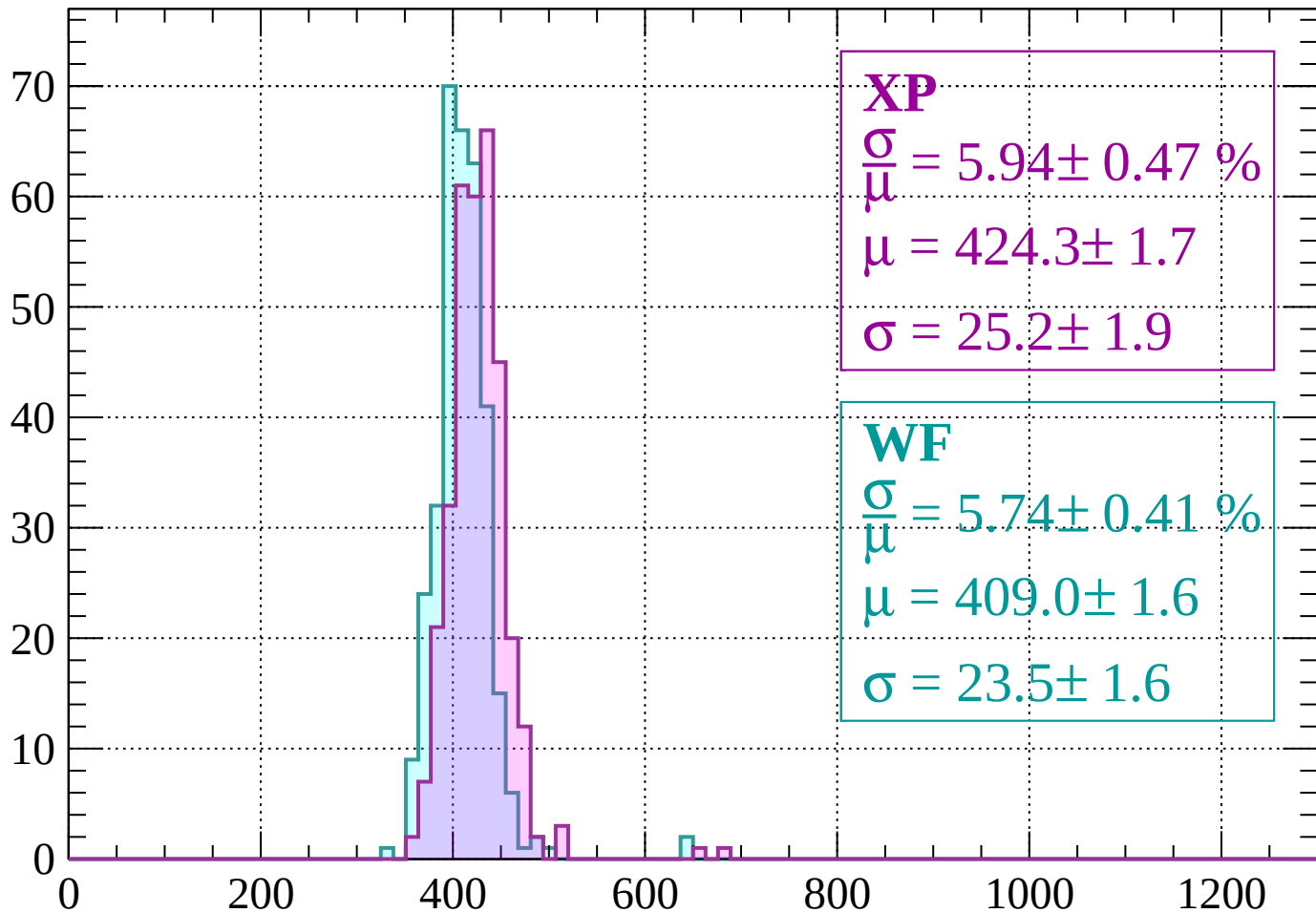
$$\mu = 403.7 \pm 1.3$$

$$\sigma = 21.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 5.94 \pm 0.47 \%$$

$$\mu = 424.3 \pm 1.7$$

$$\sigma = 25.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 5.74 \pm 0.41 \%$$

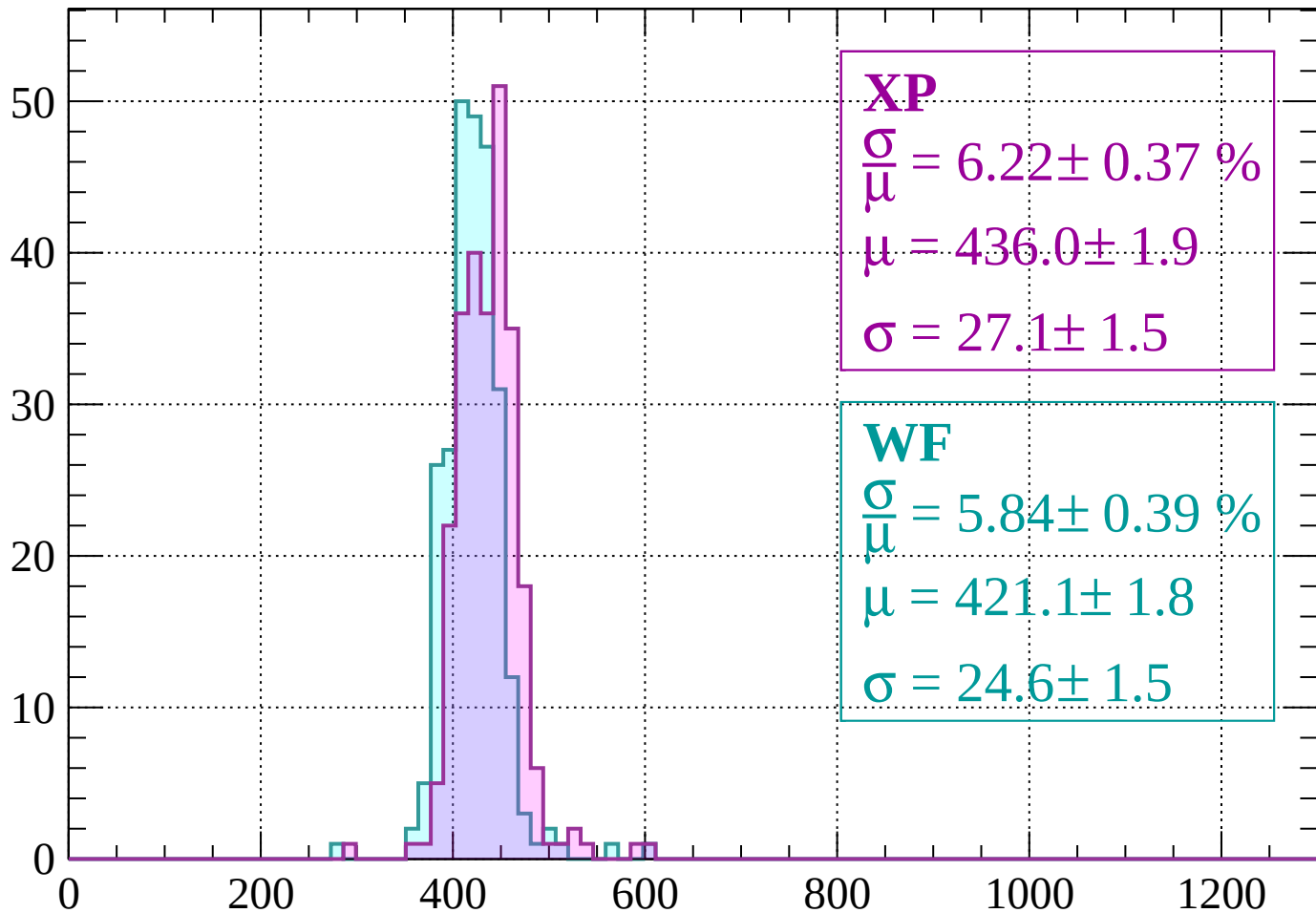
$$\mu = 409.0 \pm 1.6$$

$$\sigma = 23.5 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.37 \%$$

$$\mu = 436.0 \pm 1.9$$

$$\sigma = 27.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 5.84 \pm 0.39 \%$$

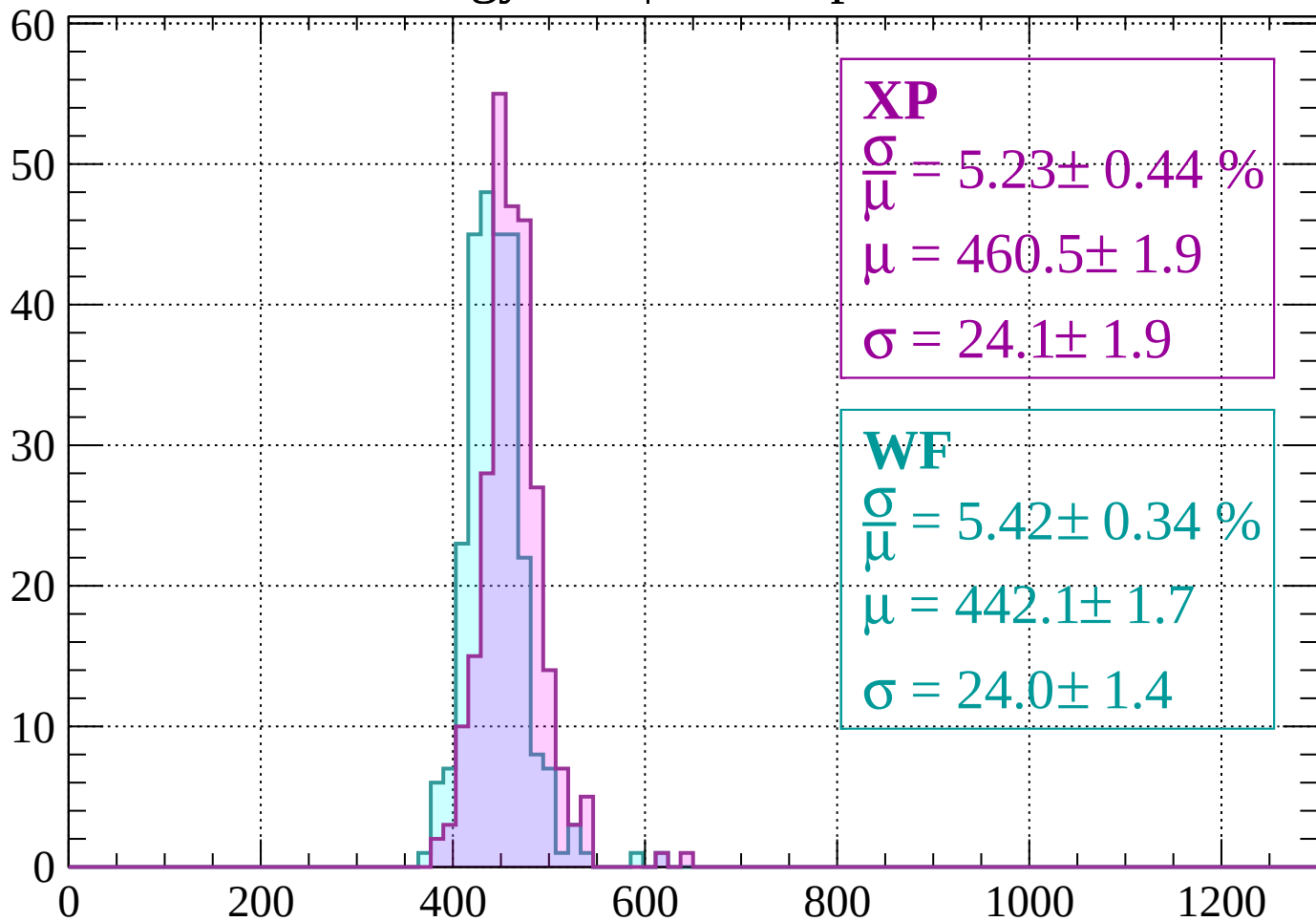
$$\mu = 421.1 \pm 1.8$$

$$\sigma = 24.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.44 \%$$

$$\mu = 460.5 \pm 1.9$$

$$\sigma = 24.1 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.34 \%$$

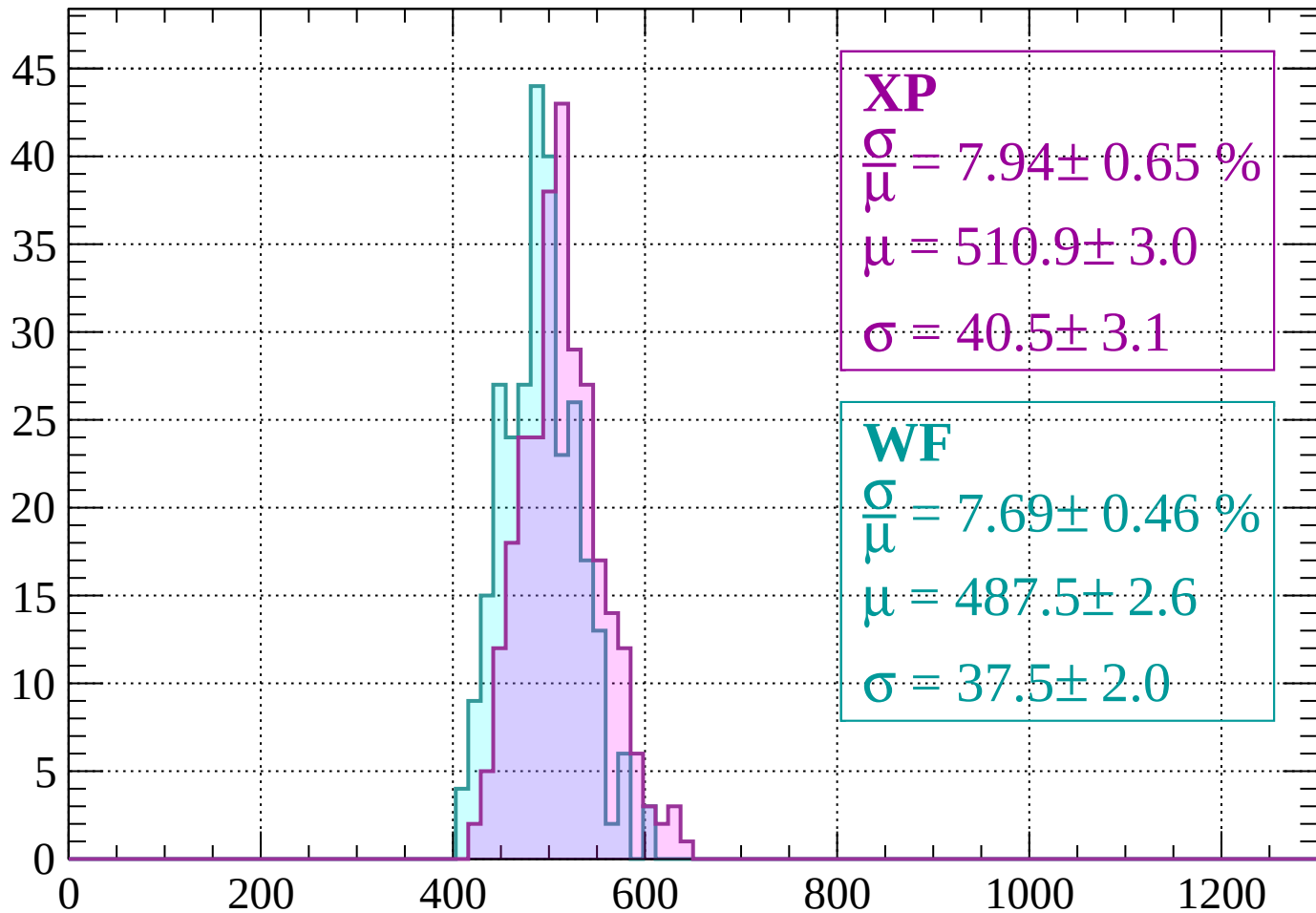
$$\mu = 442.1 \pm 1.7$$

$$\sigma = 24.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.94 \pm 0.65 \%$$

$$\mu = 510.9 \pm 3.0$$

$$\sigma = 40.5 \pm 3.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.69 \pm 0.46 \%$$

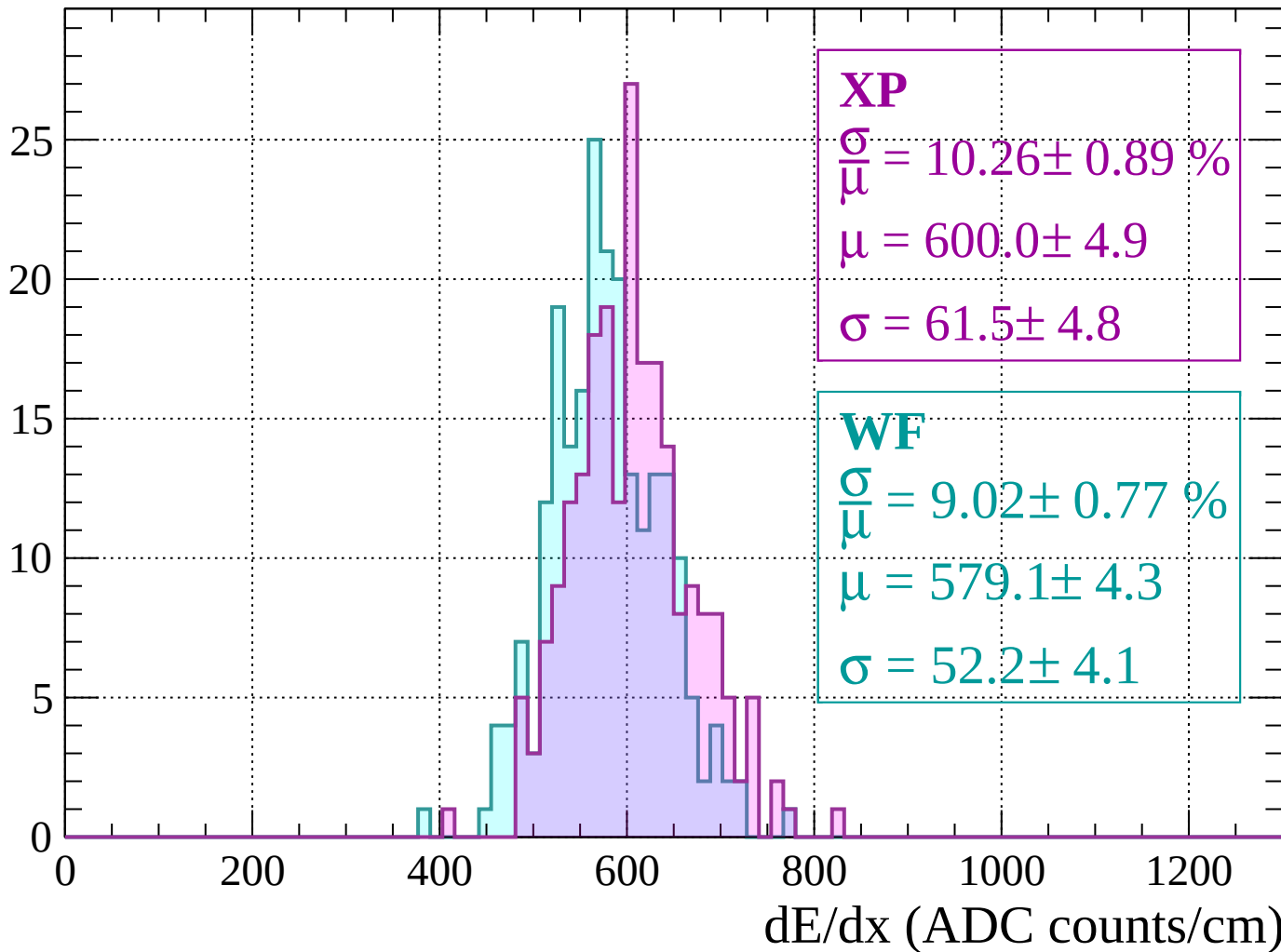
$$\mu = 487.5 \pm 2.6$$

$$\sigma = 37.5 \pm 2.0$$

dE/dx (ADC counts/cm)

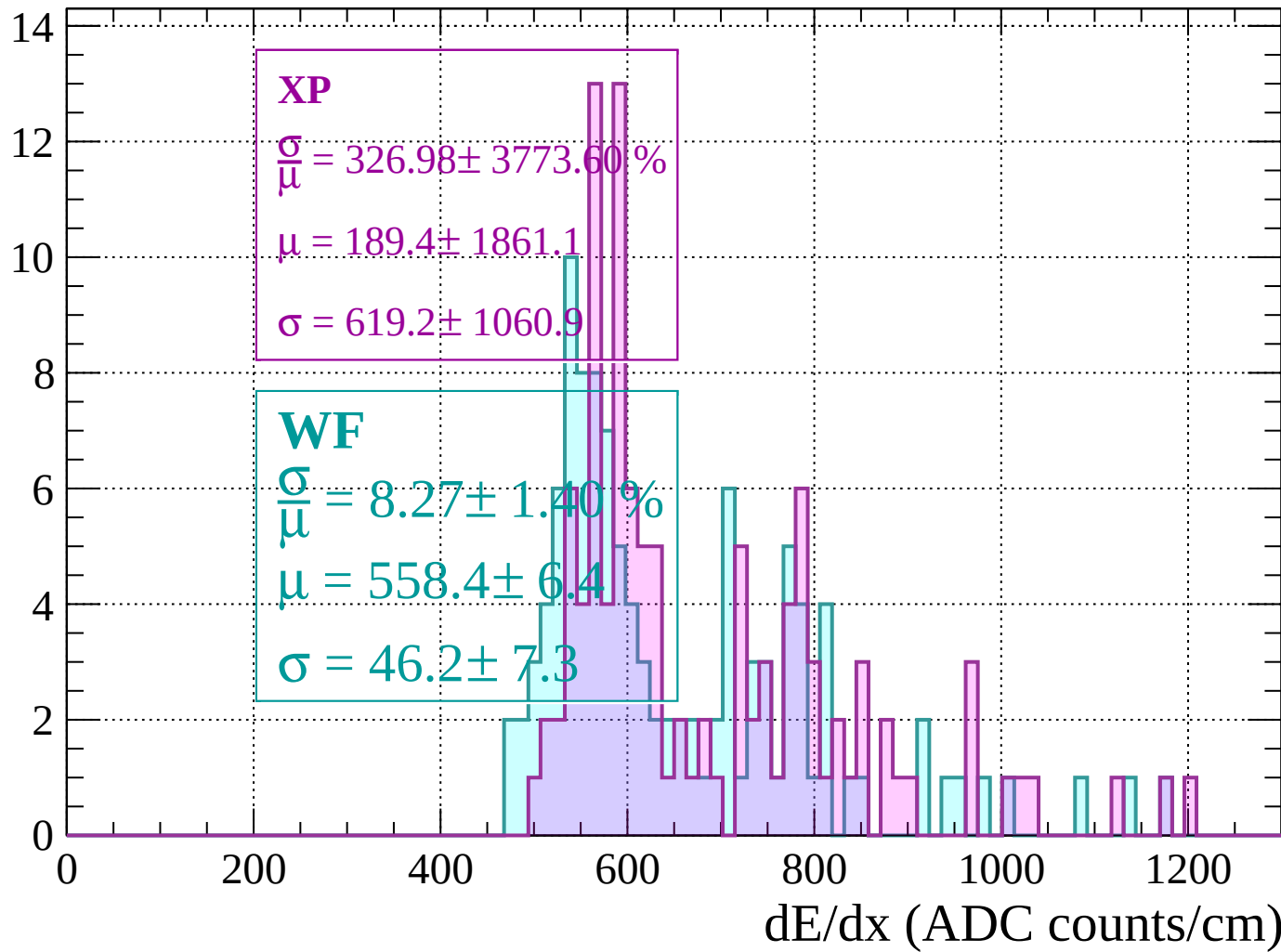
Energy loss | $-120 < p < -80$

Count



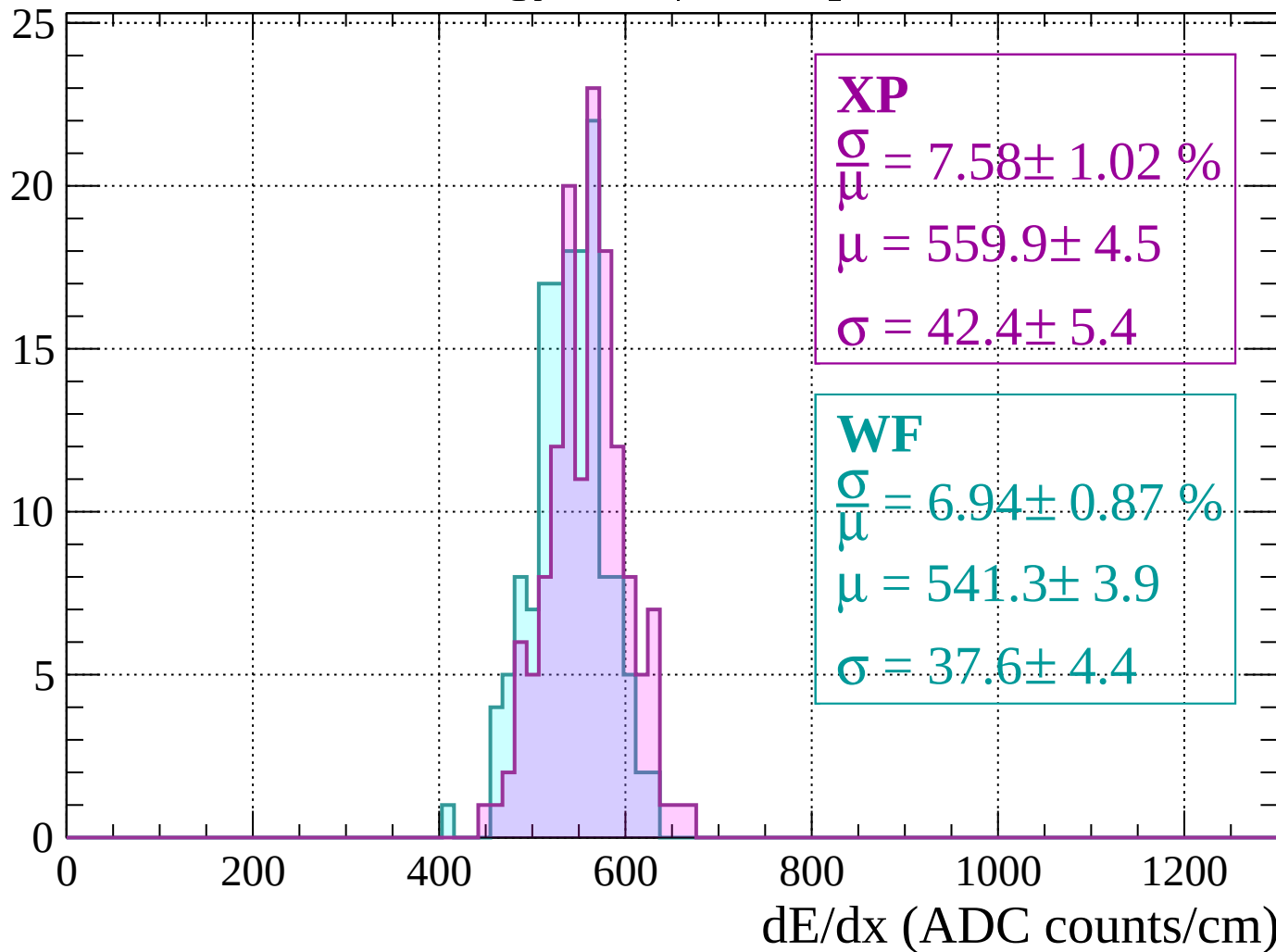
Energy loss | -80 < p < -40

Count



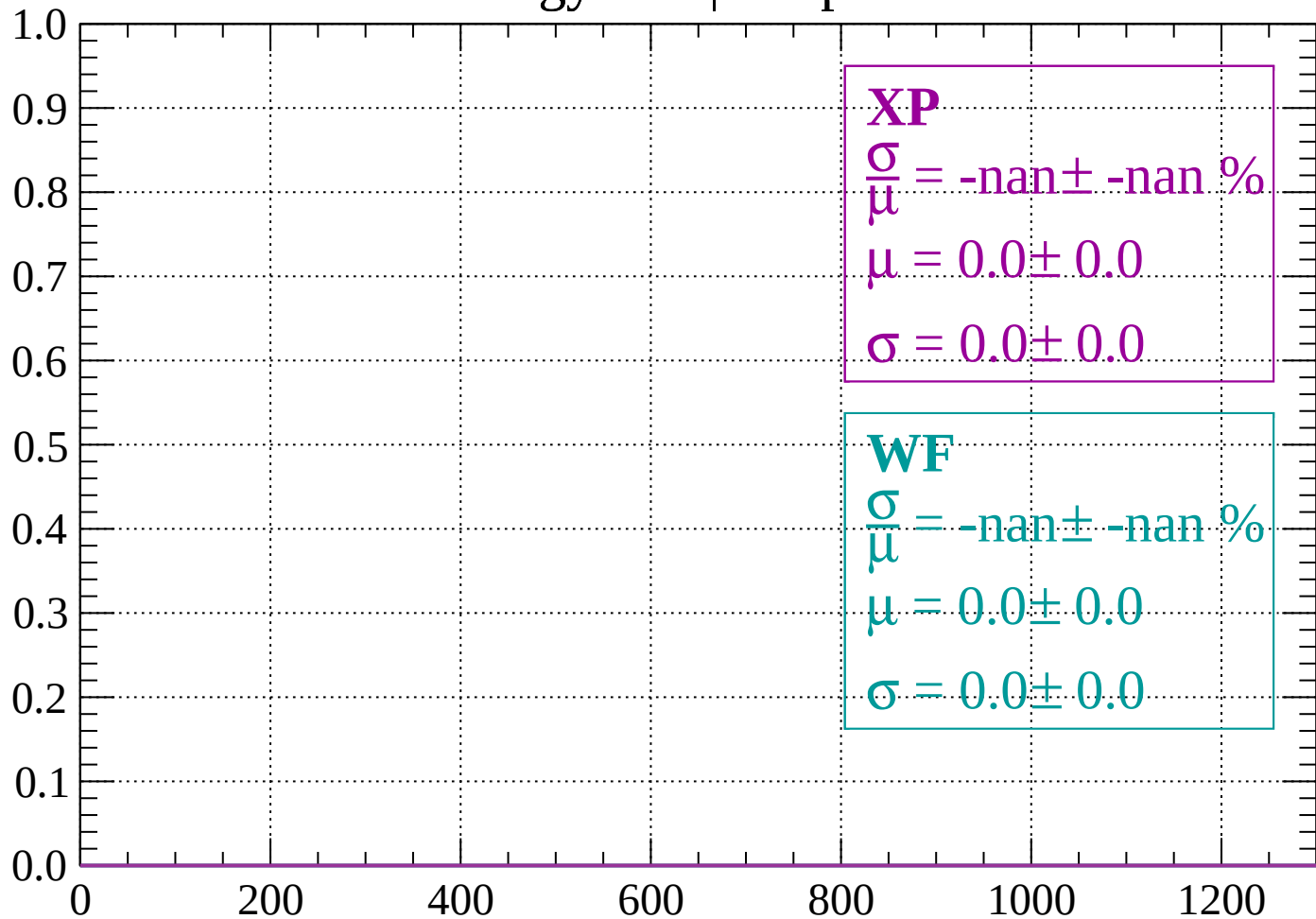
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

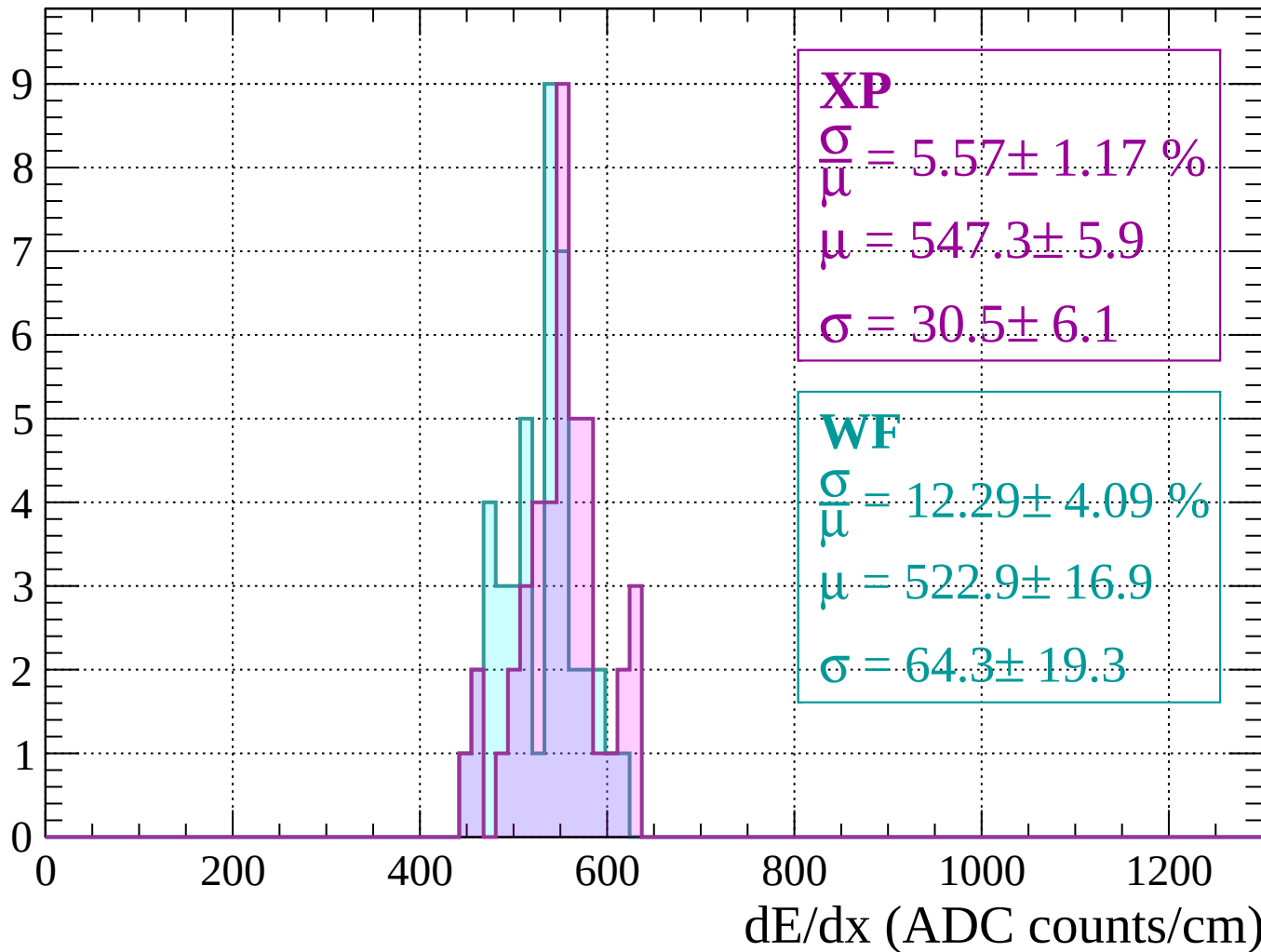
Count



dE/dx (ADC counts/cm)

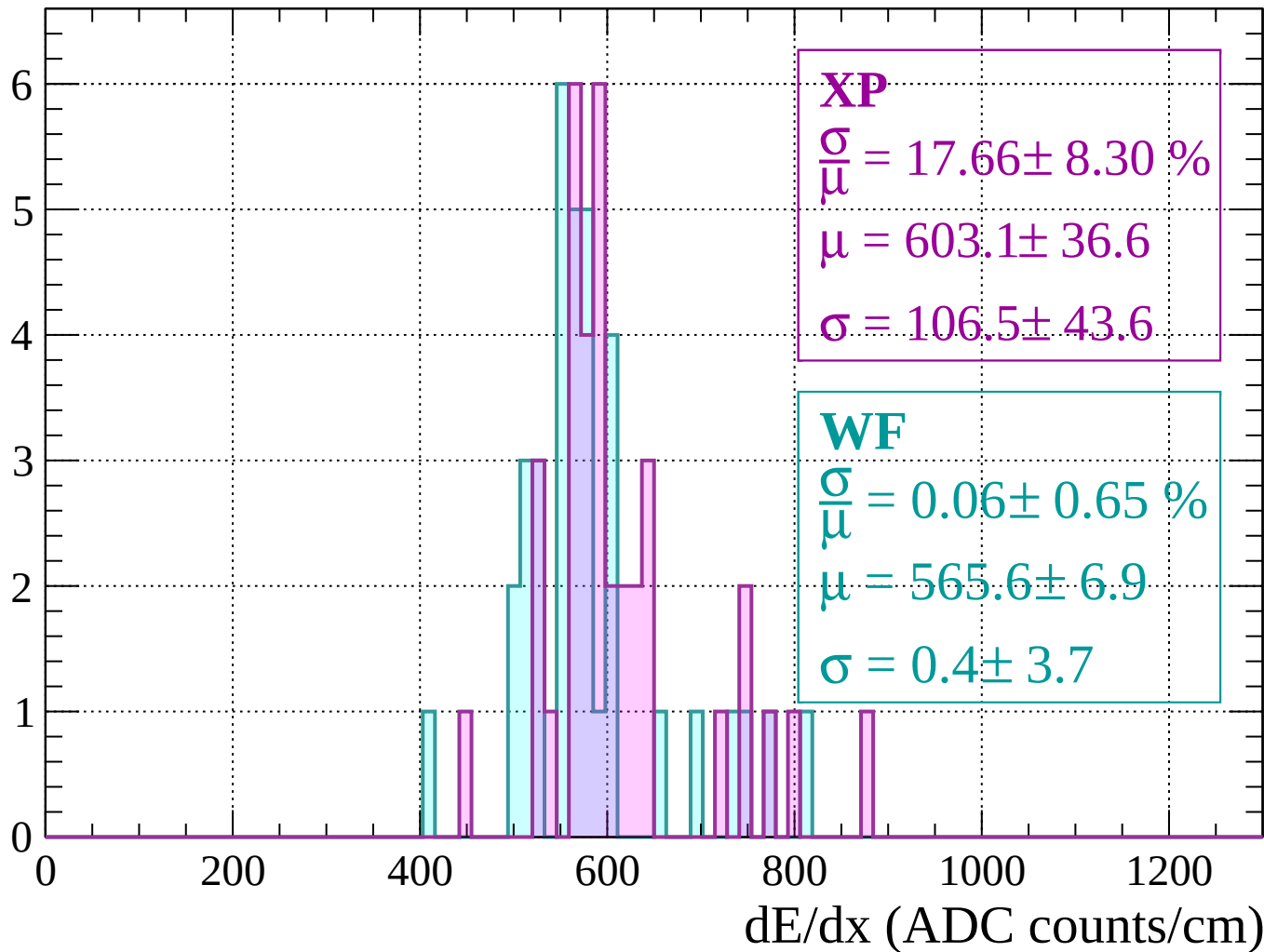
Energy loss | $40 < p < 80$

Count



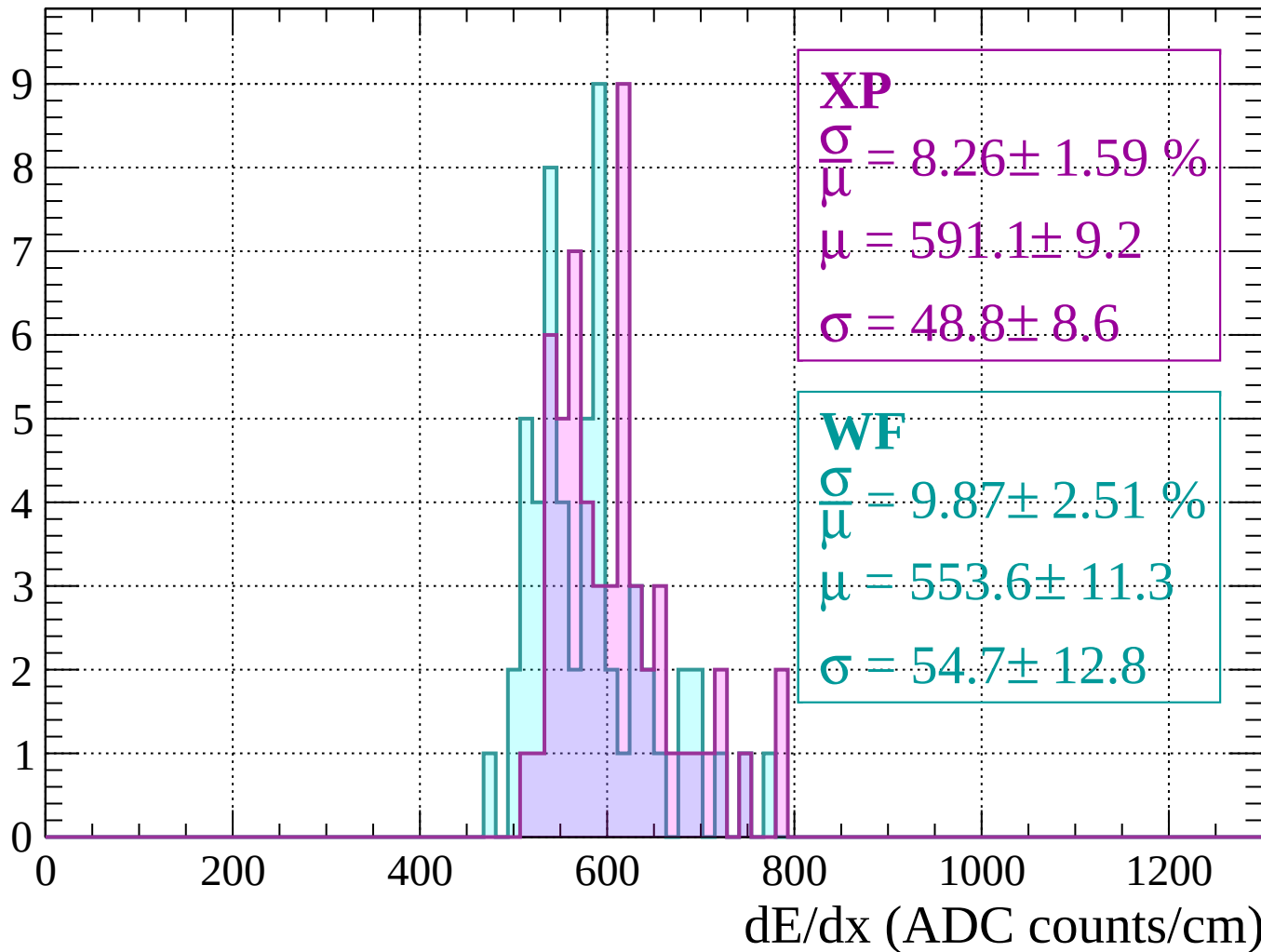
Energy loss | $80 < p < 120$

Count



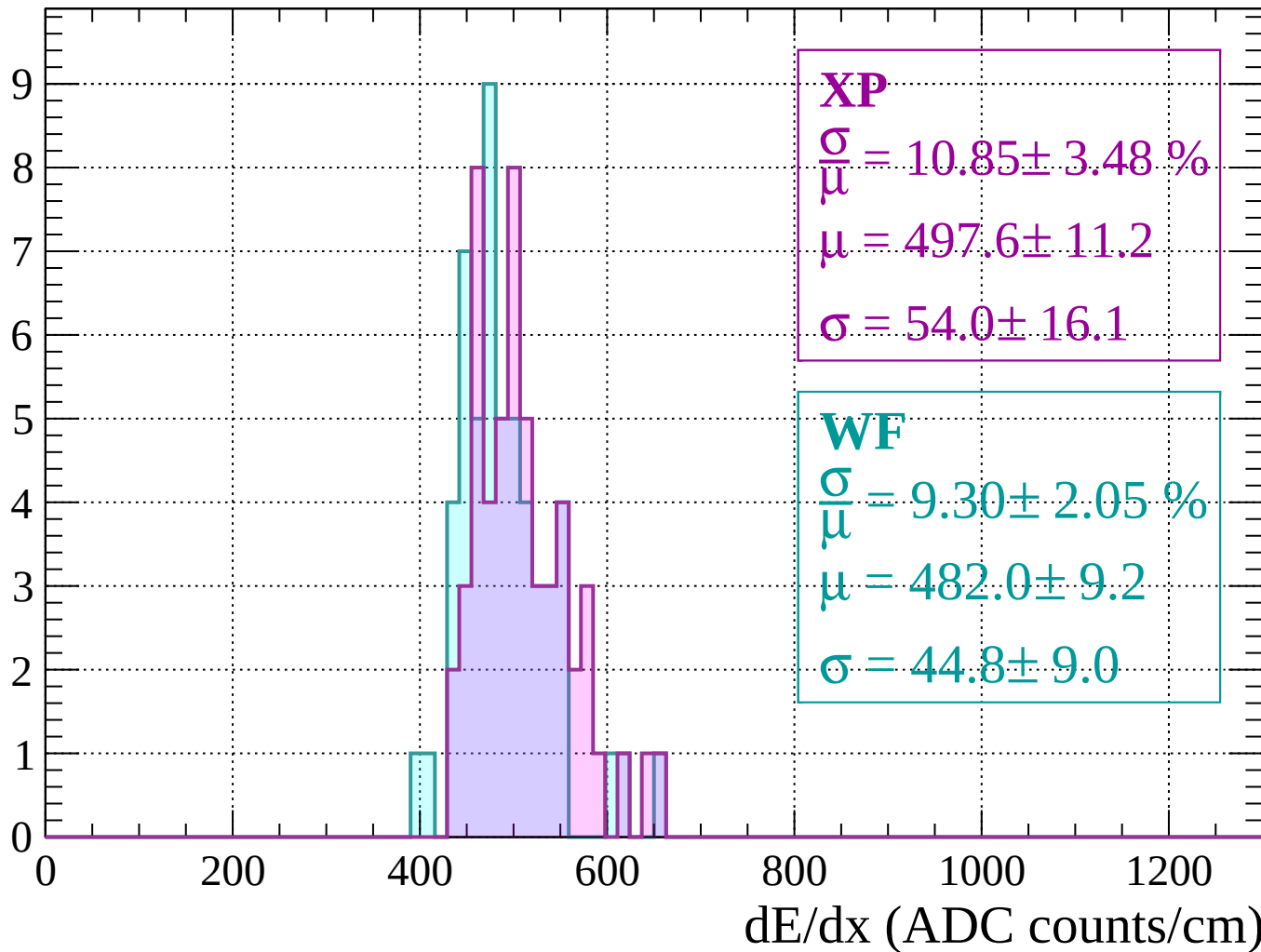
Energy loss | $120 < p < 160$

Count



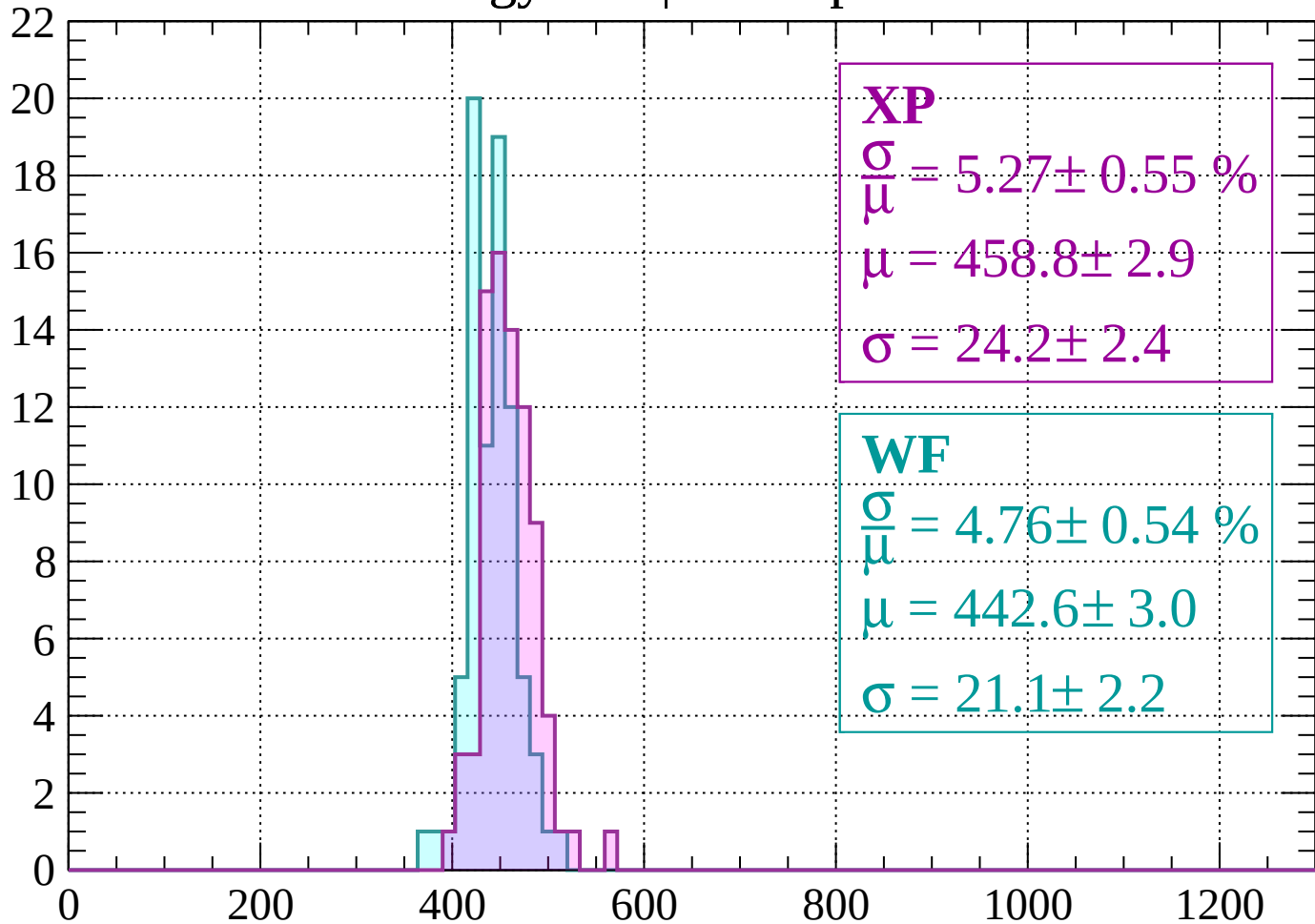
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

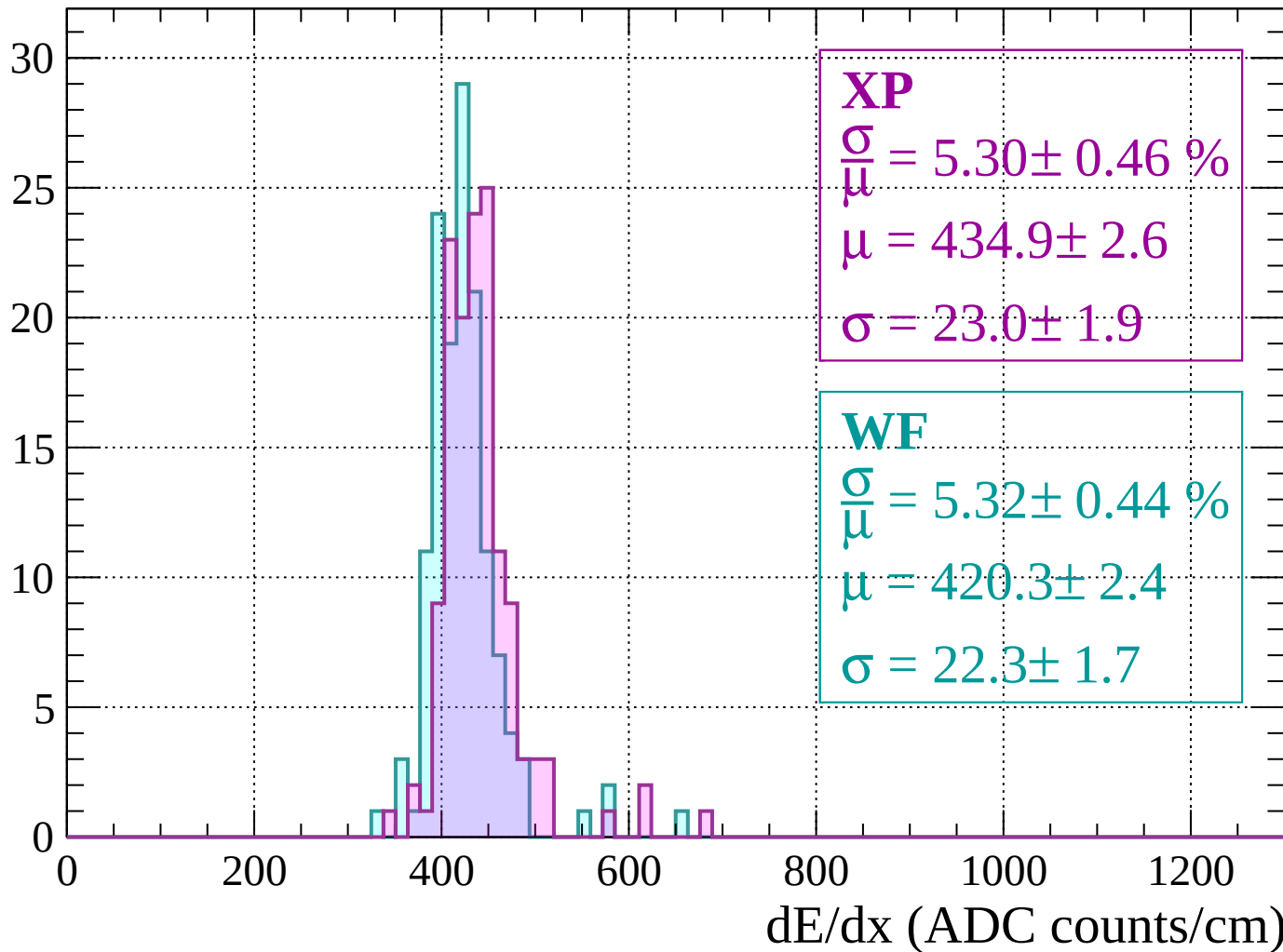
Count



dE/dx (ADC counts/cm)

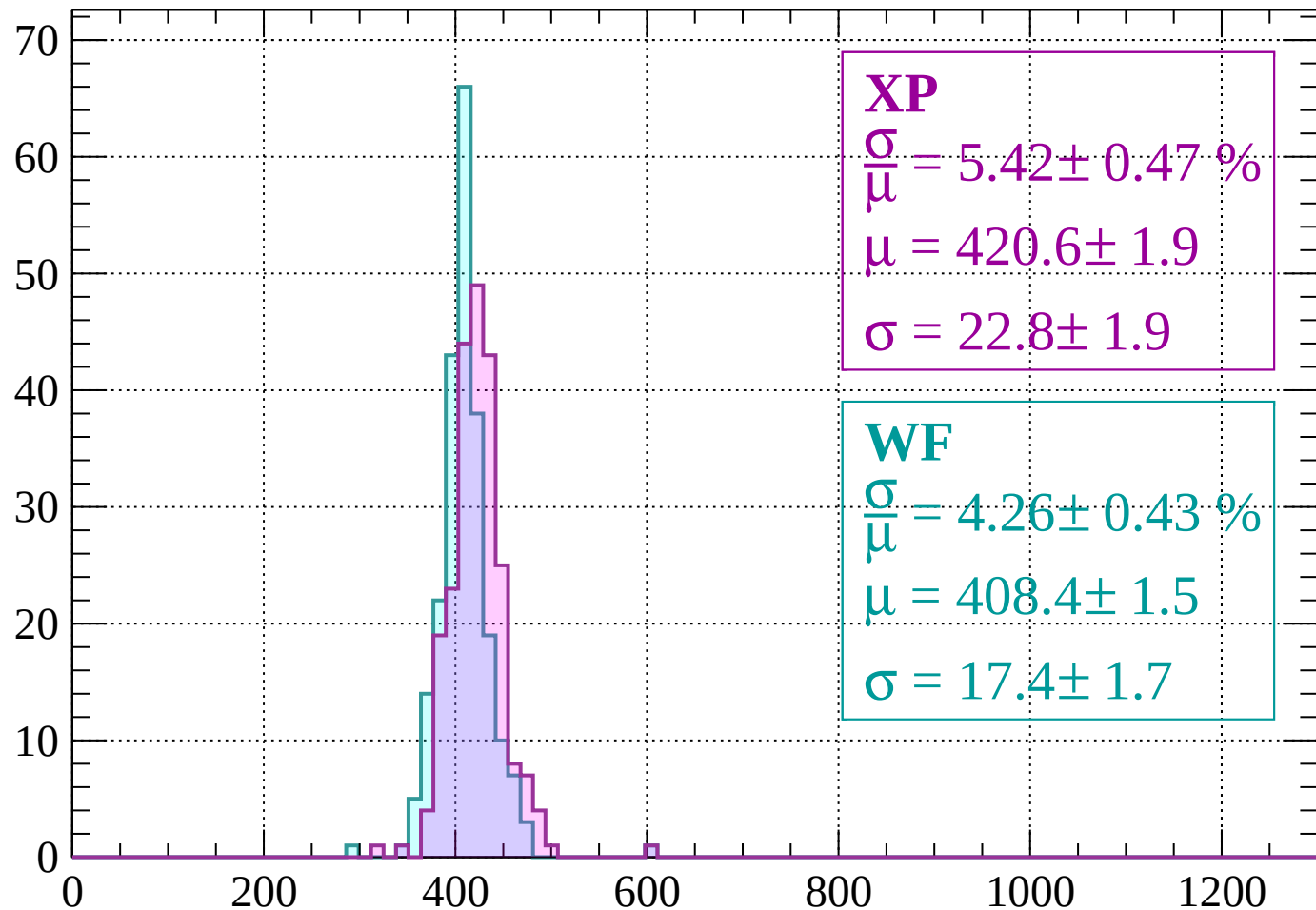
Energy loss | $240 < p < 280$

Count



Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 5.42 \pm 0.47 \%$$

$$\mu = 420.6 \pm 1.9$$

$$\sigma = 22.8 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 4.26 \pm 0.43 \%$$

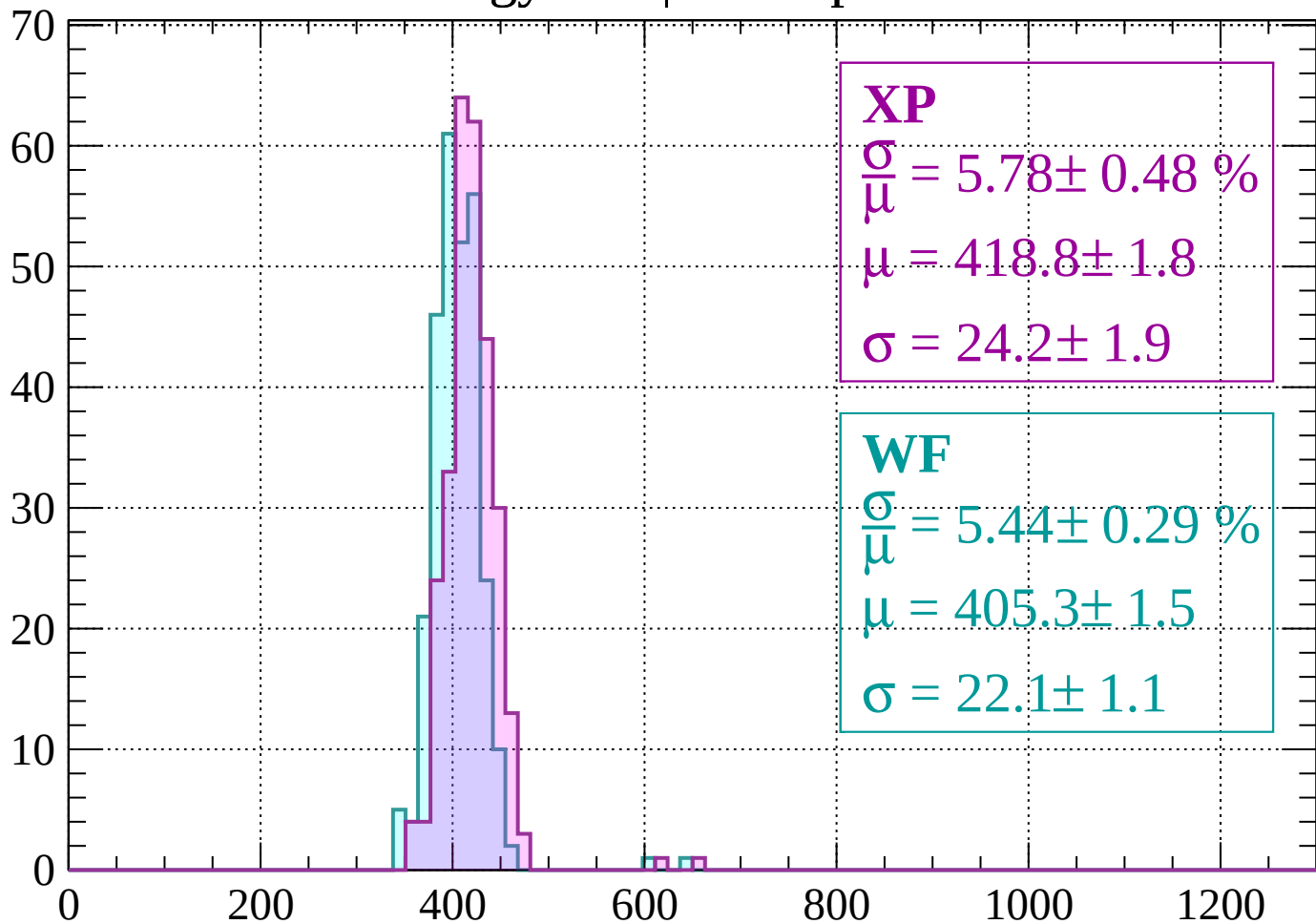
$$\mu = 408.4 \pm 1.5$$

$$\sigma = 17.4 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 5.78 \pm 0.48 \%$$

$$\mu = 418.8 \pm 1.8$$

$$\sigma = 24.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 5.44 \pm 0.29 \%$$

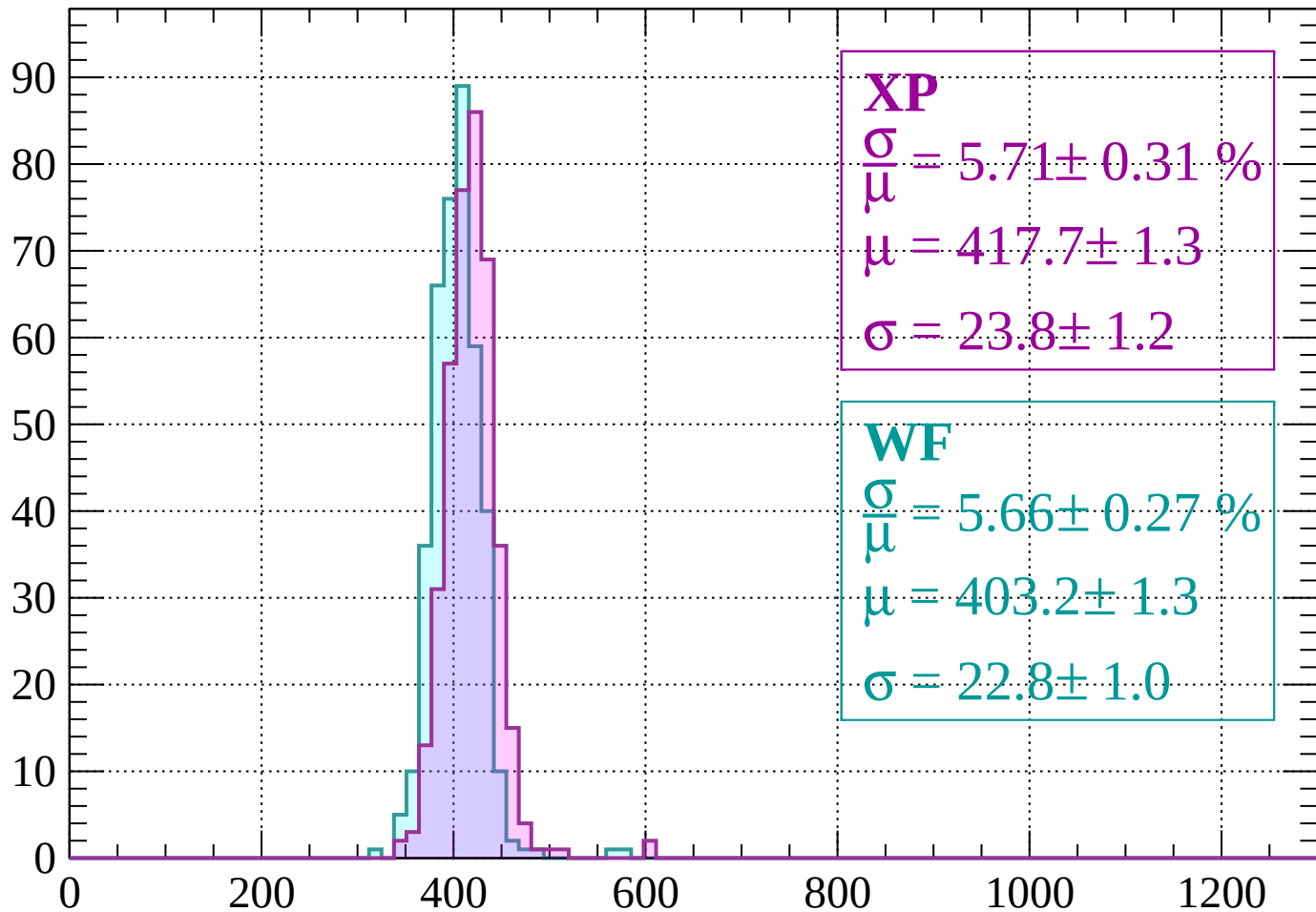
$$\mu = 405.3 \pm 1.5$$

$$\sigma = 22.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

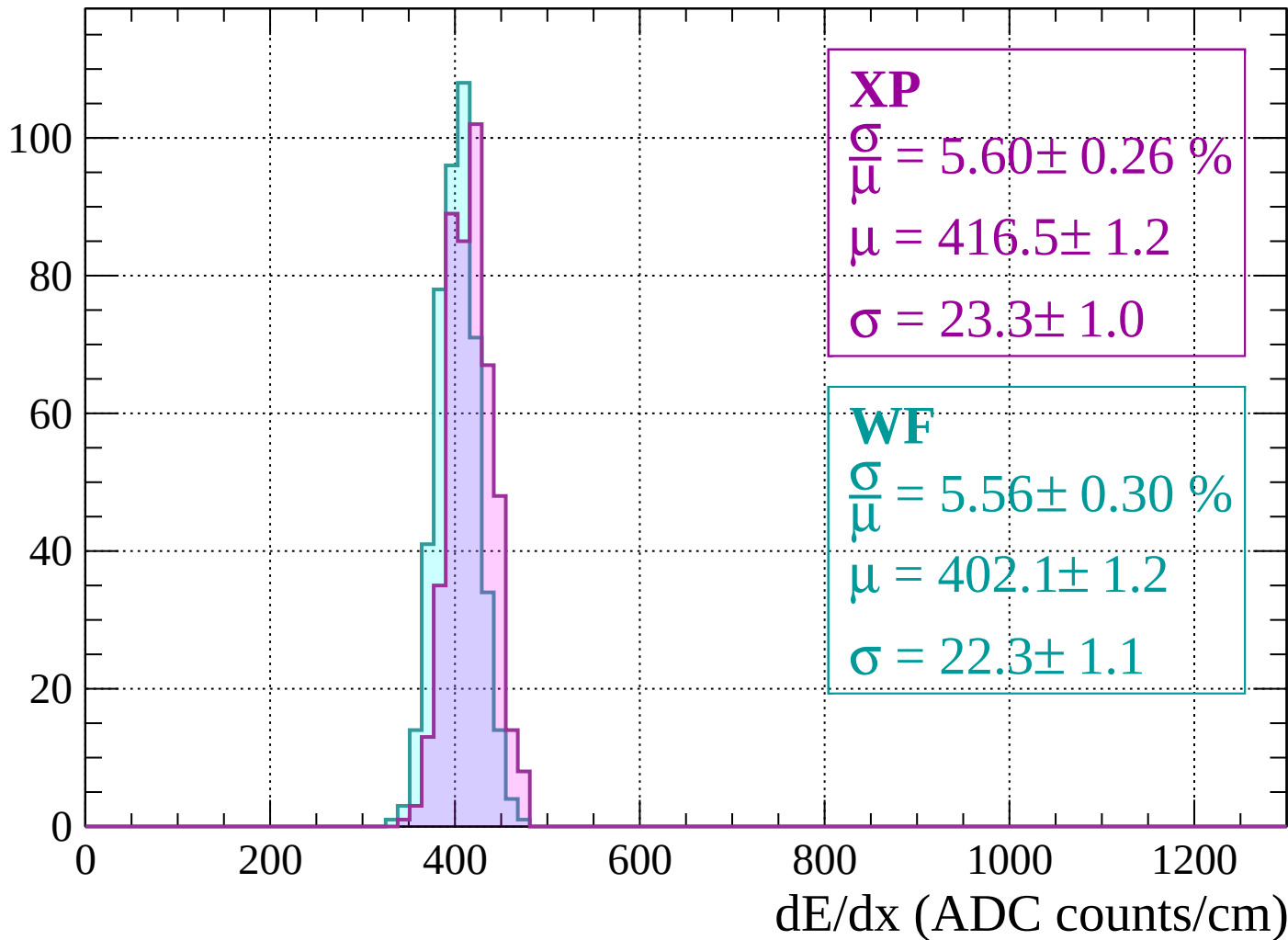
Count



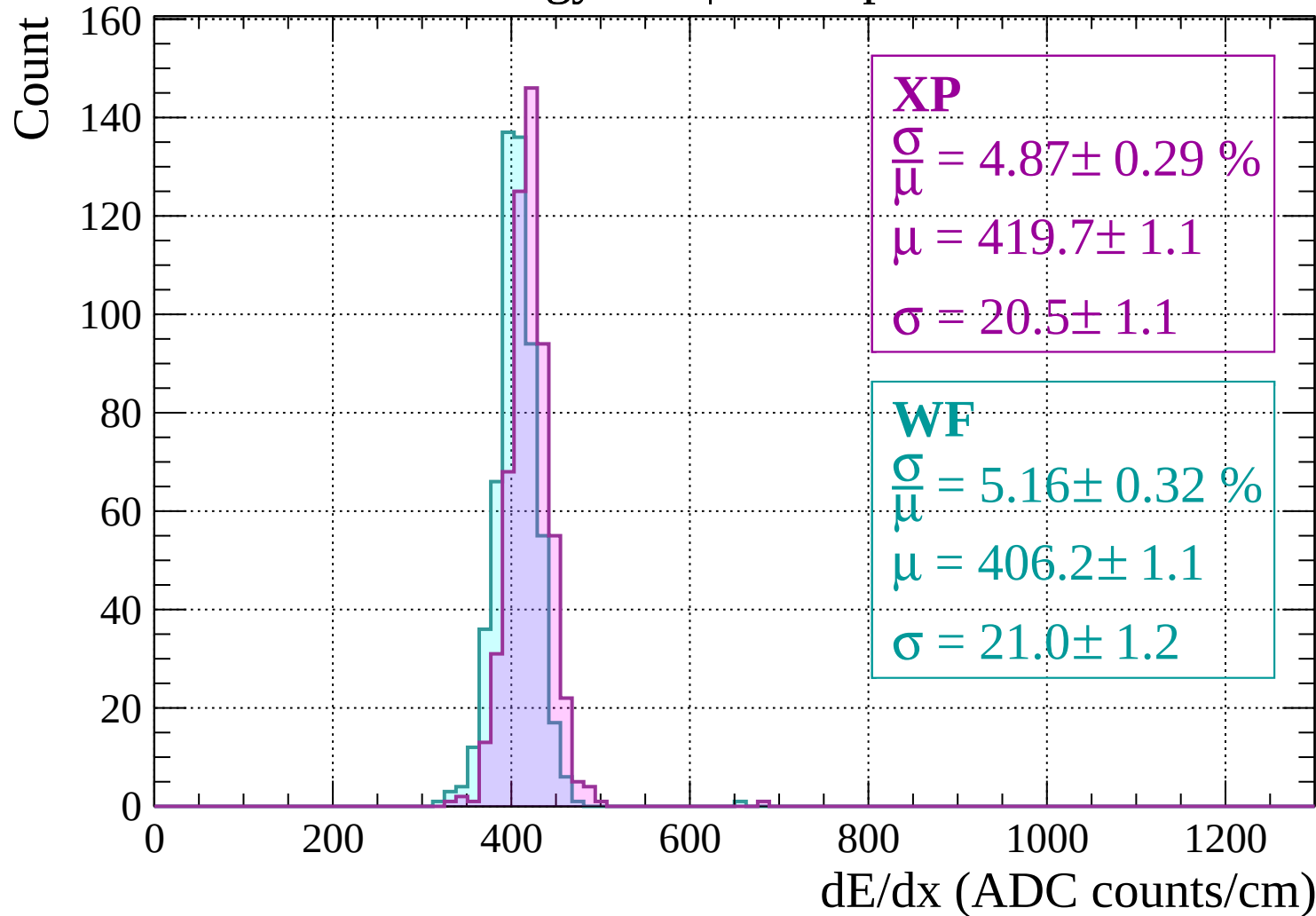
dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



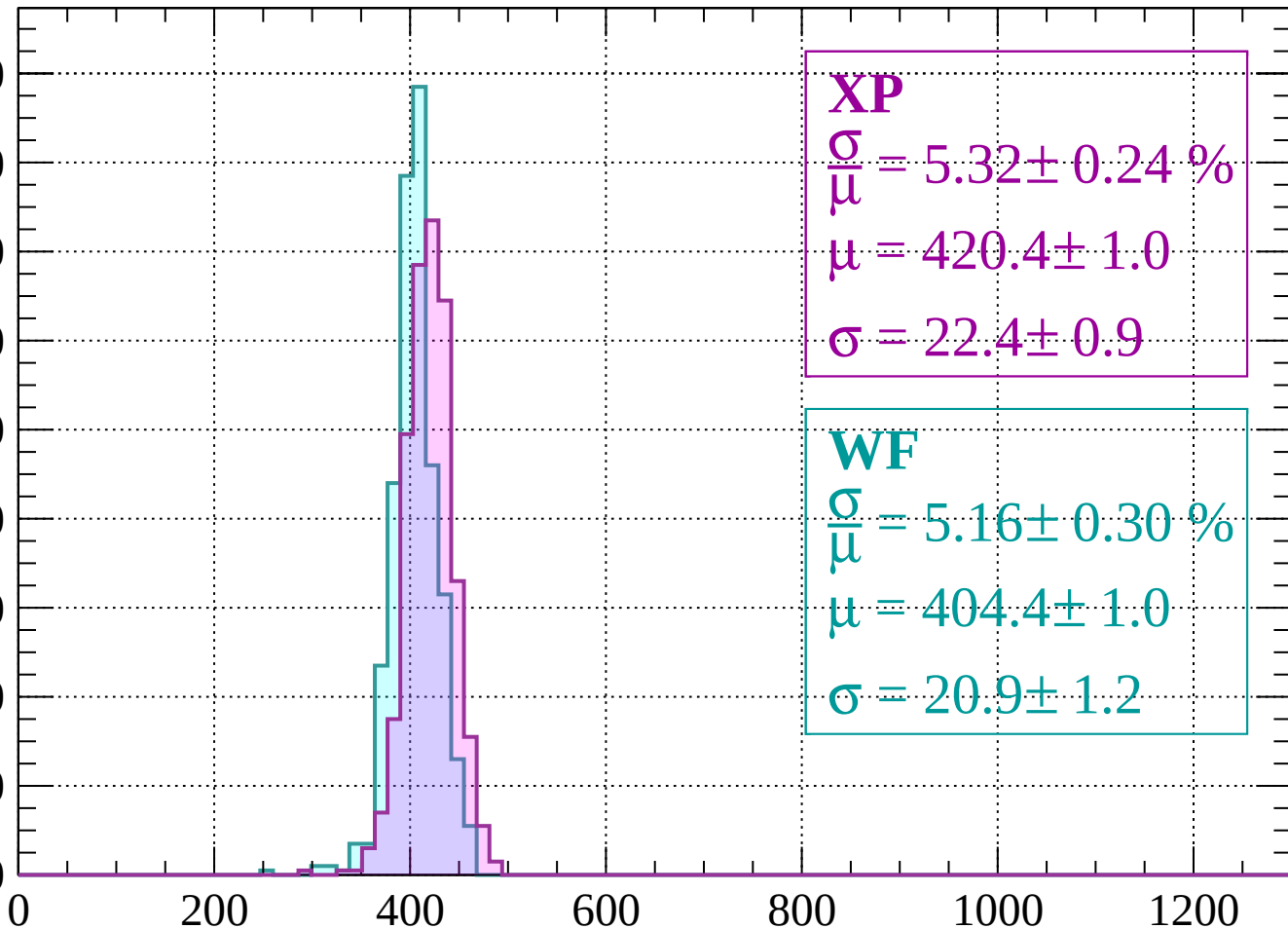
Energy loss | $440 < p < 480$



Energy loss | $480 < p < 520$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 5.32 \pm 0.24 \%$$

$$\mu = 420.4 \pm 1.0$$

$$\sigma = 22.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.30 \%$$

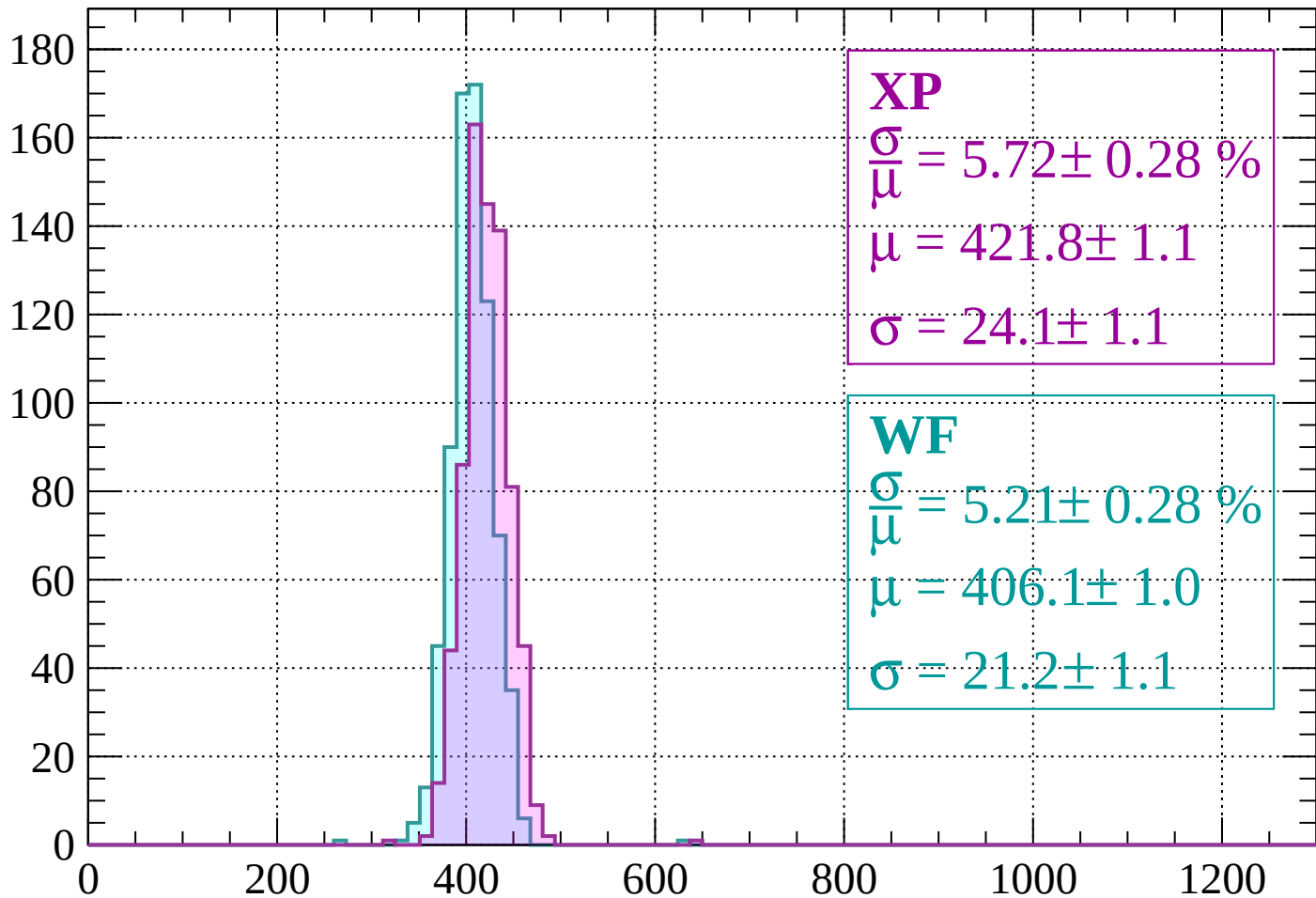
$$\mu = 404.4 \pm 1.0$$

$$\sigma = 20.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 5.72 \pm 0.28 \%$$

$$\mu = 421.8 \pm 1.1$$

$$\sigma = 24.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.21 \pm 0.28 \%$$

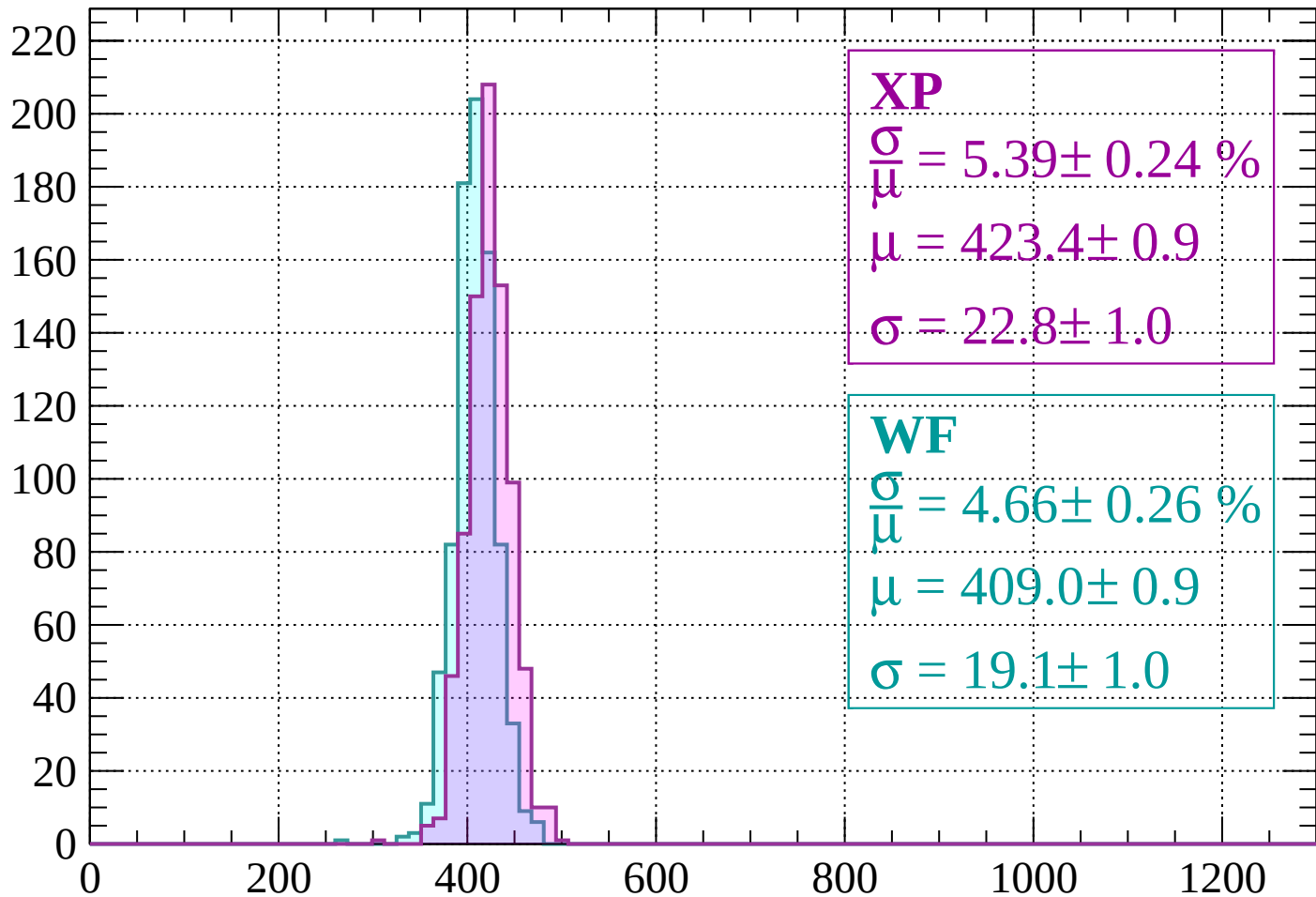
$$\mu = 406.1 \pm 1.0$$

$$\sigma = 21.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 5.39 \pm 0.24 \%$$

$$\mu = 423.4 \pm 0.9$$

$$\sigma = 22.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.66 \pm 0.26 \%$$

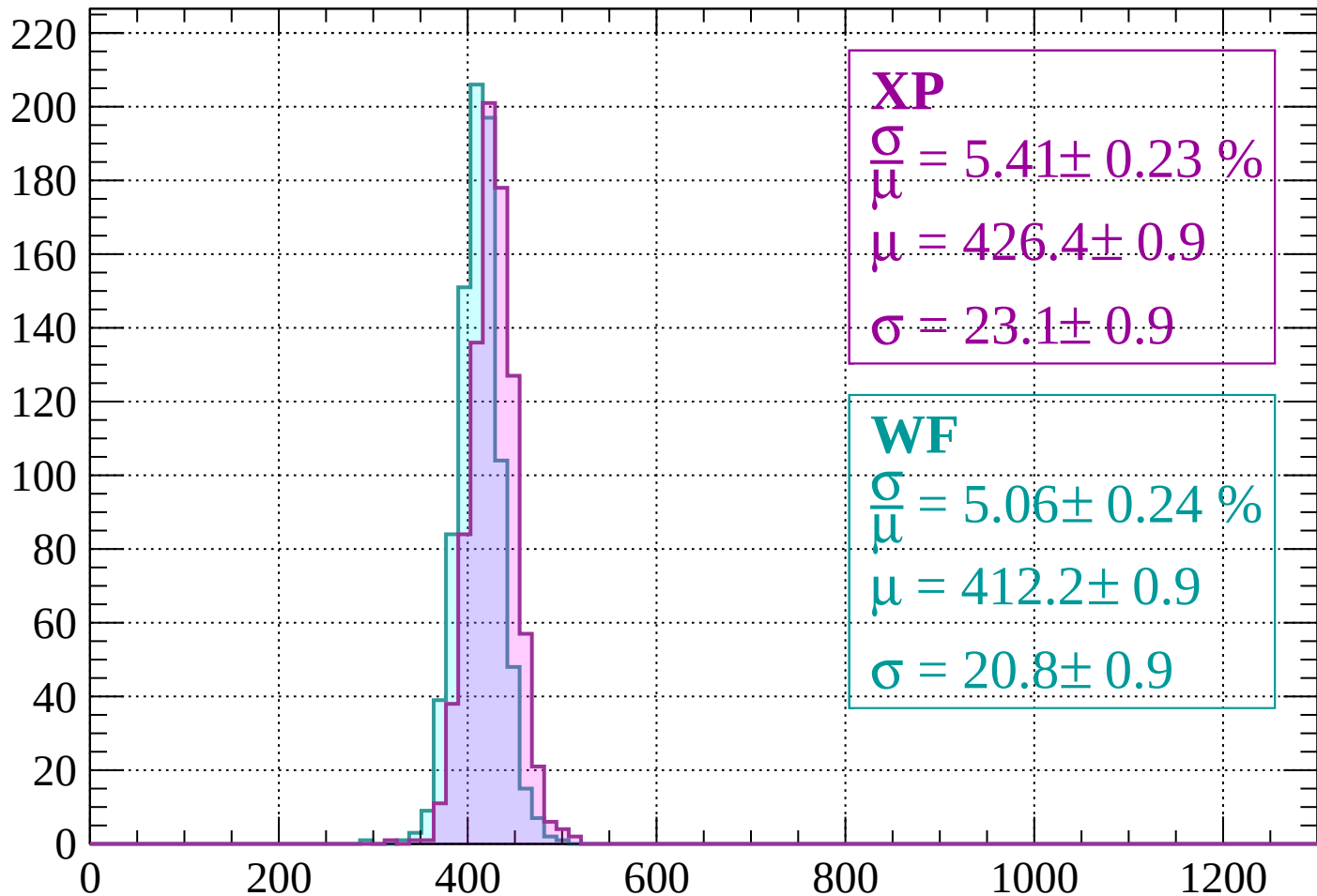
$$\mu = 409.0 \pm 0.9$$

$$\sigma = 19.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 5.41 \pm 0.23 \%$$

$$\mu = 426.4 \pm 0.9$$

$$\sigma = 23.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.06 \pm 0.24 \%$$

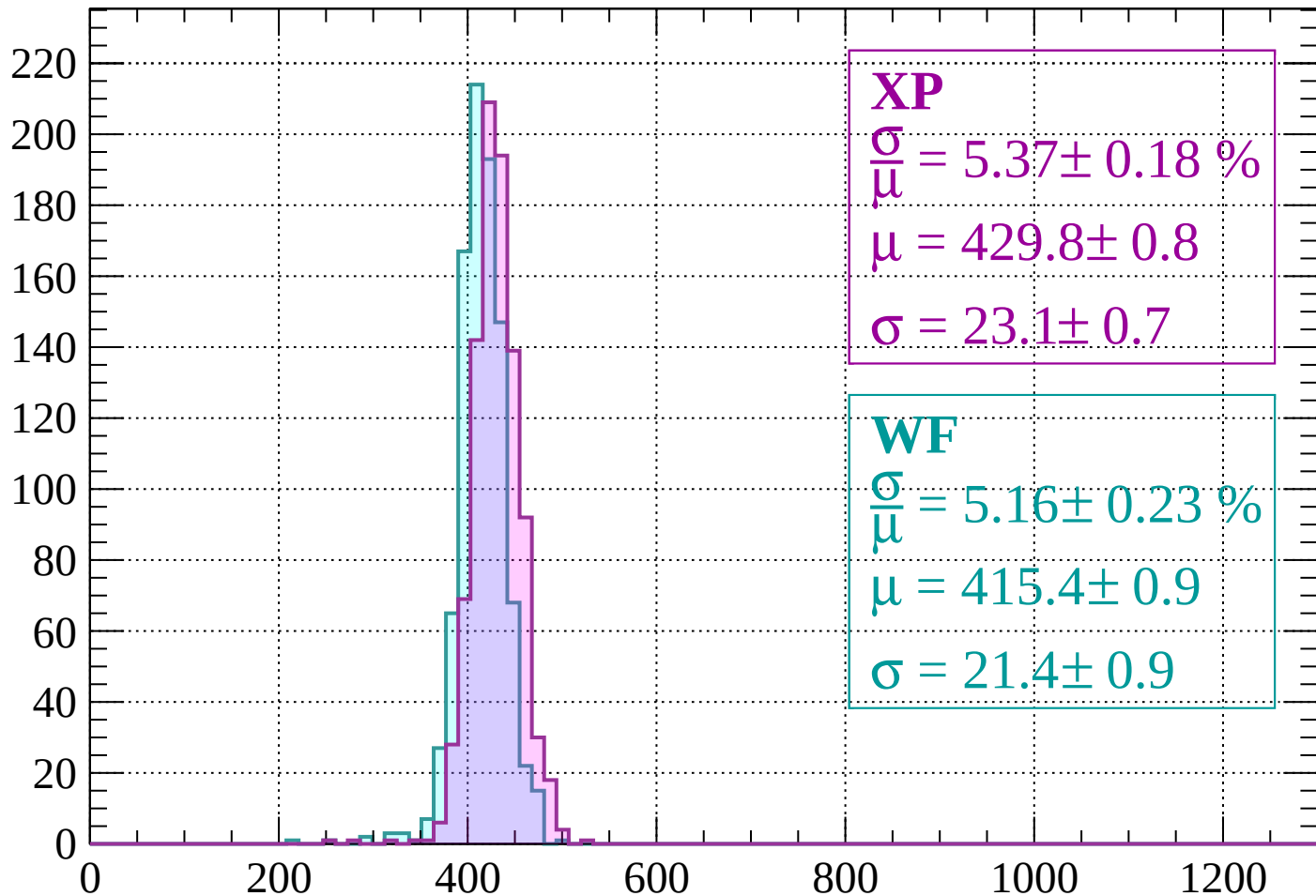
$$\mu = 412.2 \pm 0.9$$

$$\sigma = 20.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 5.37 \pm 0.18 \%$$

$$\mu = 429.8 \pm 0.8$$

$$\sigma = 23.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.23 \%$$

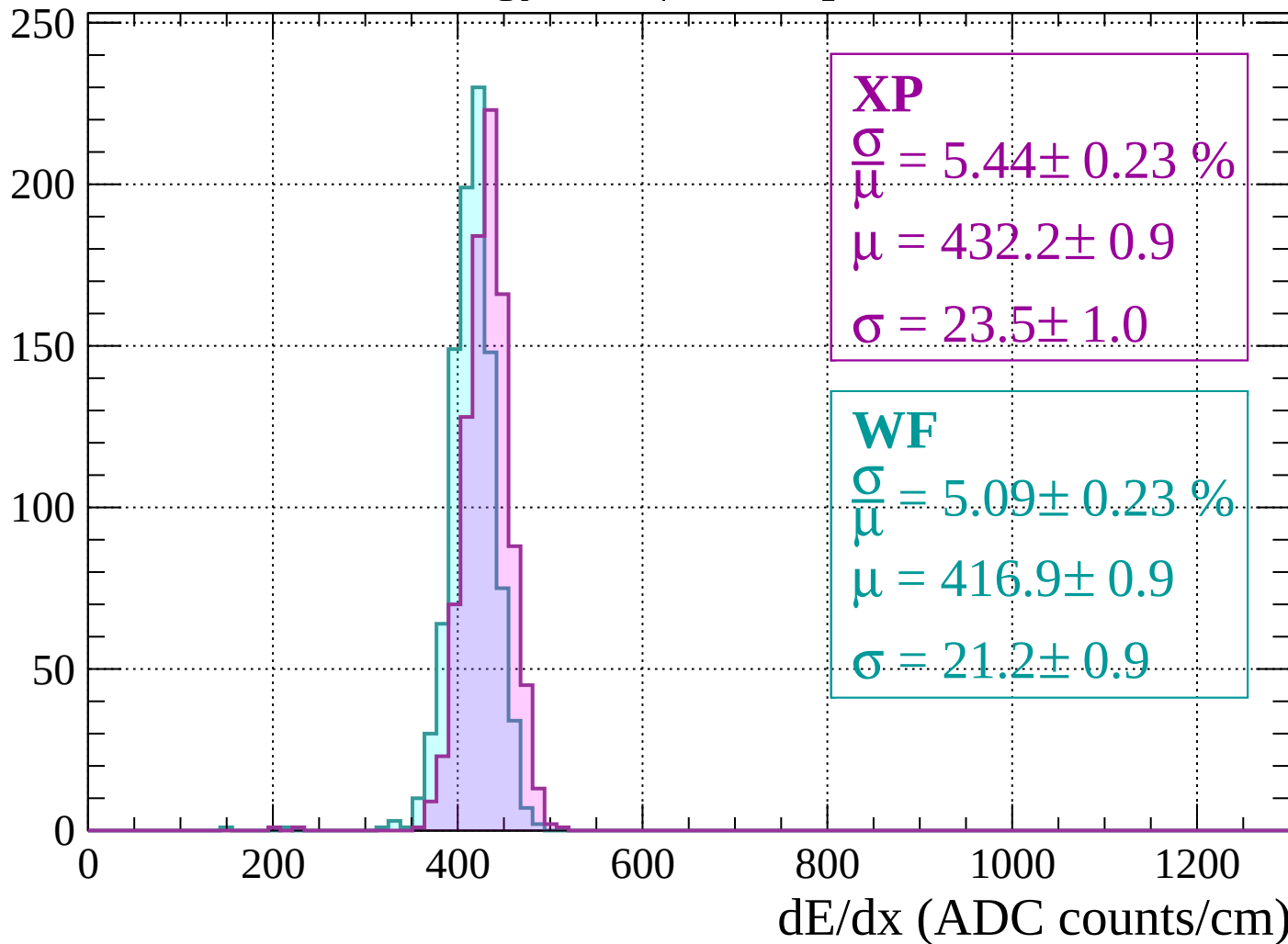
$$\mu = 415.4 \pm 0.9$$

$$\sigma = 21.4 \pm 0.9$$

dE/dx (ADC counts/cm)

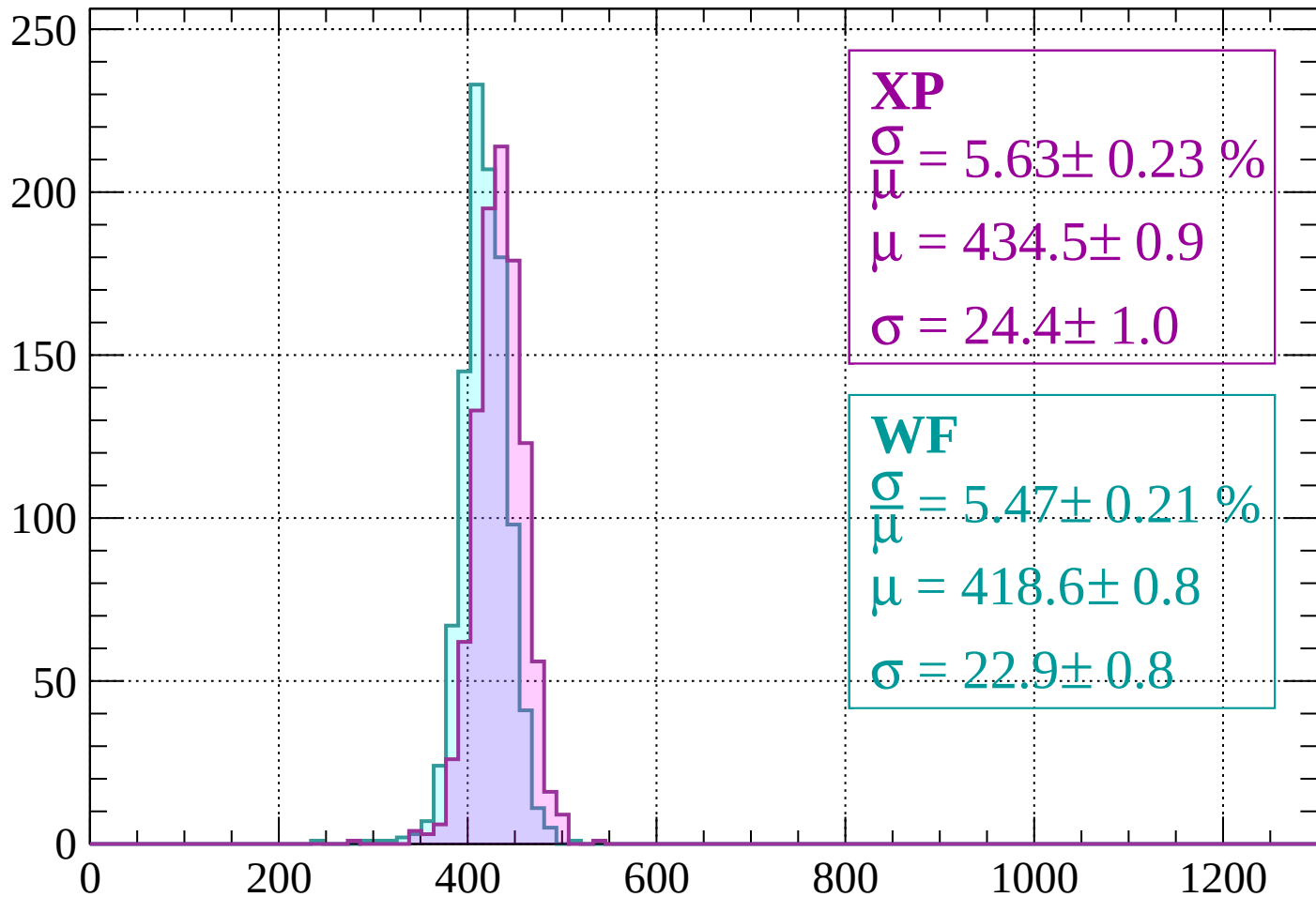
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 5.63 \pm 0.23 \%$$

$$\mu = 434.5 \pm 0.9$$

$$\sigma = 24.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.47 \pm 0.21 \%$$

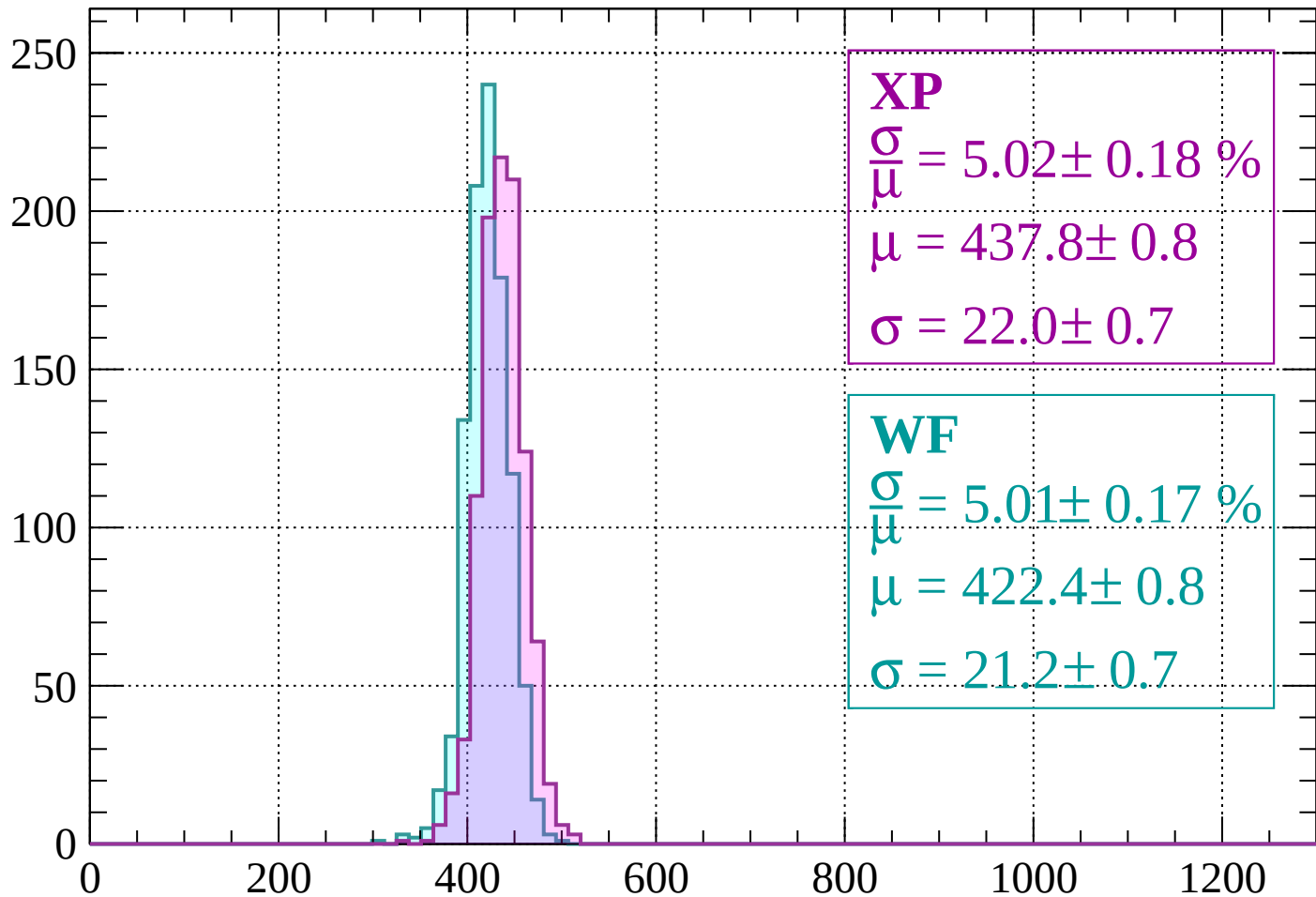
$$\mu = 418.6 \pm 0.8$$

$$\sigma = 22.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 5.02 \pm 0.18 \%$$

$$\mu = 437.8 \pm 0.8$$

$$\sigma = 22.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.01 \pm 0.17 \%$$

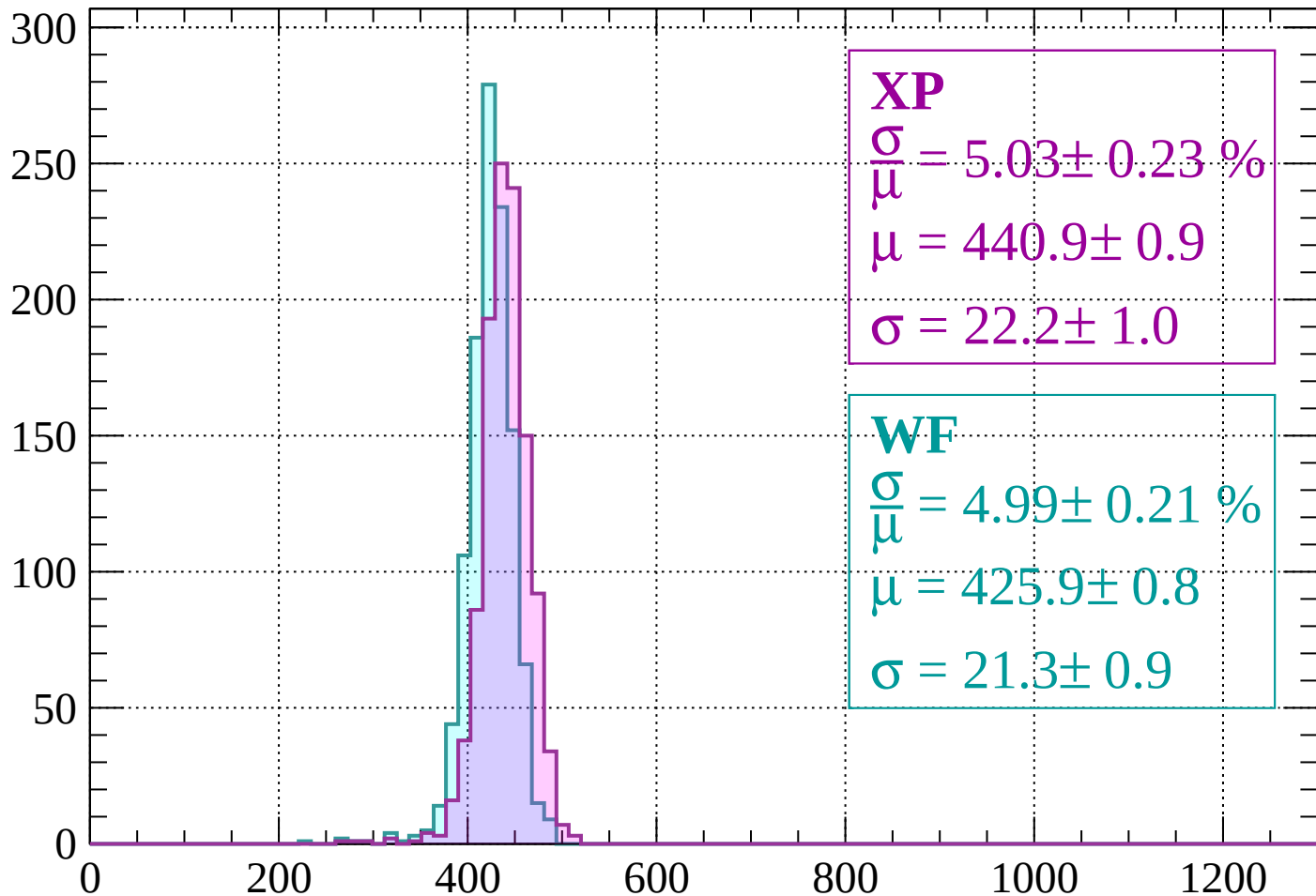
$$\mu = 422.4 \pm 0.8$$

$$\sigma = 21.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 5.03 \pm 0.23 \%$$

$$\mu = 440.9 \pm 0.9$$

$$\sigma = 22.2 \pm 1.0$$

WF

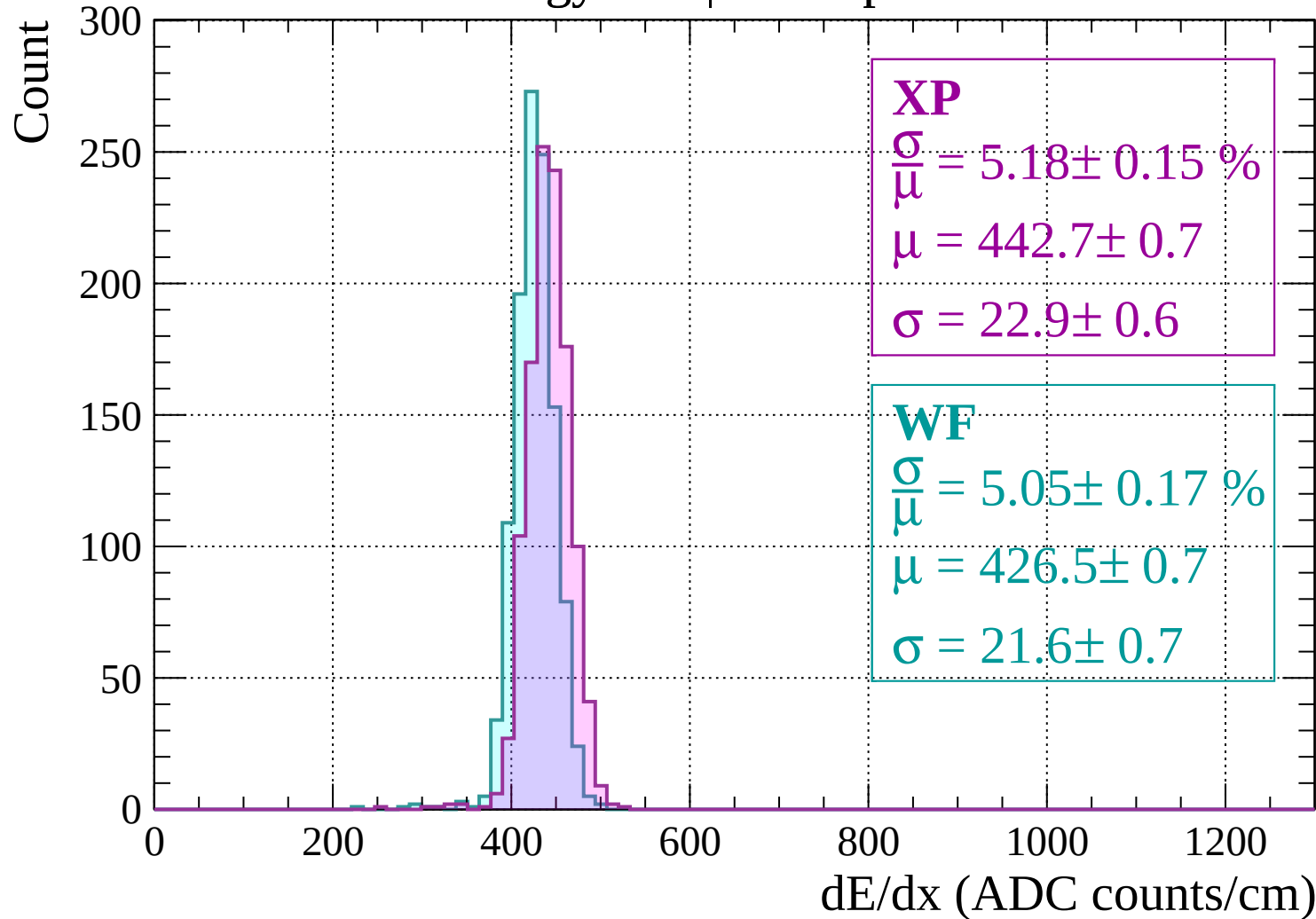
$$\frac{\sigma}{\mu} = 4.99 \pm 0.21 \%$$

$$\mu = 425.9 \pm 0.8$$

$$\sigma = 21.3 \pm 0.9$$

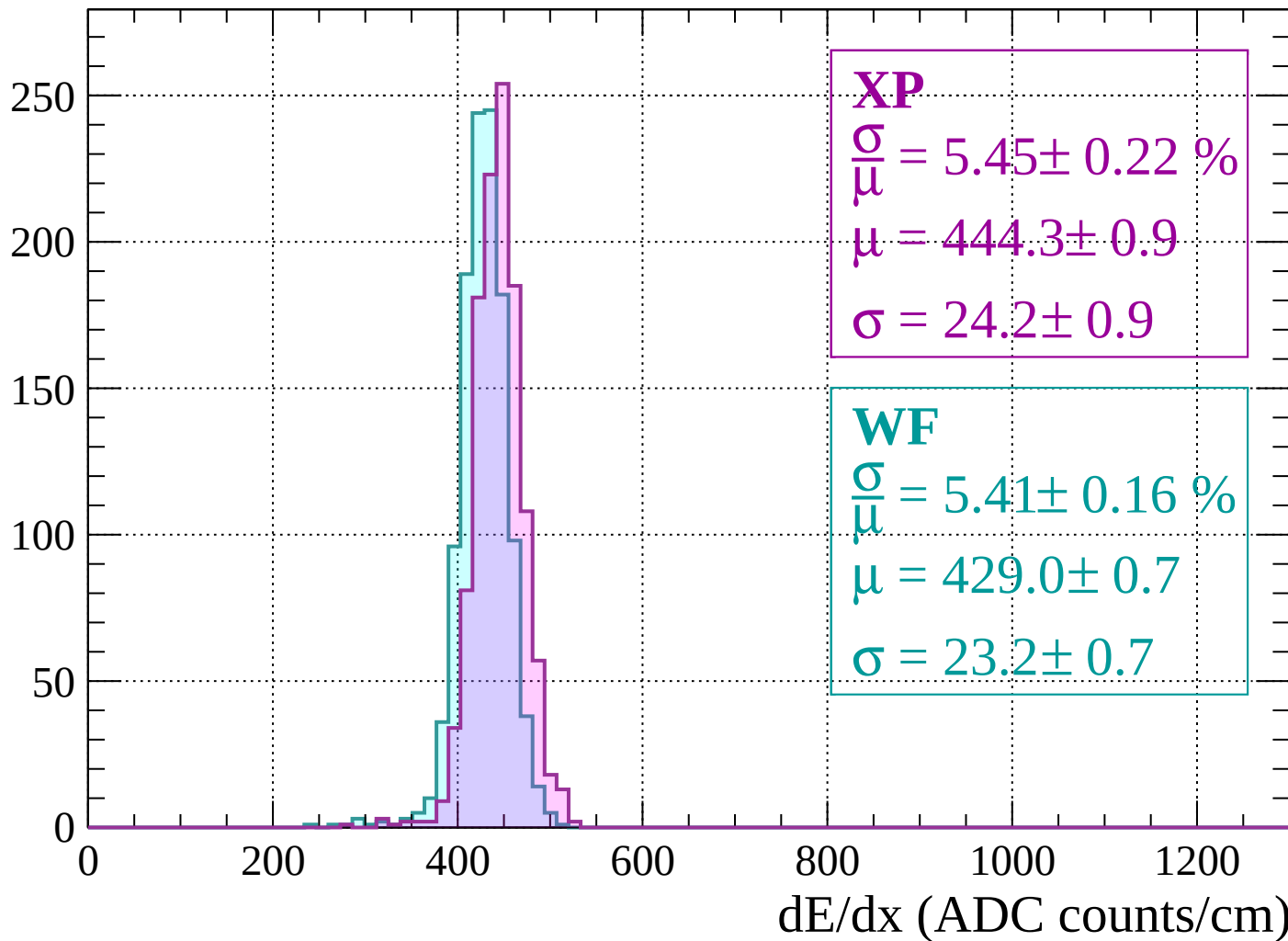
dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$



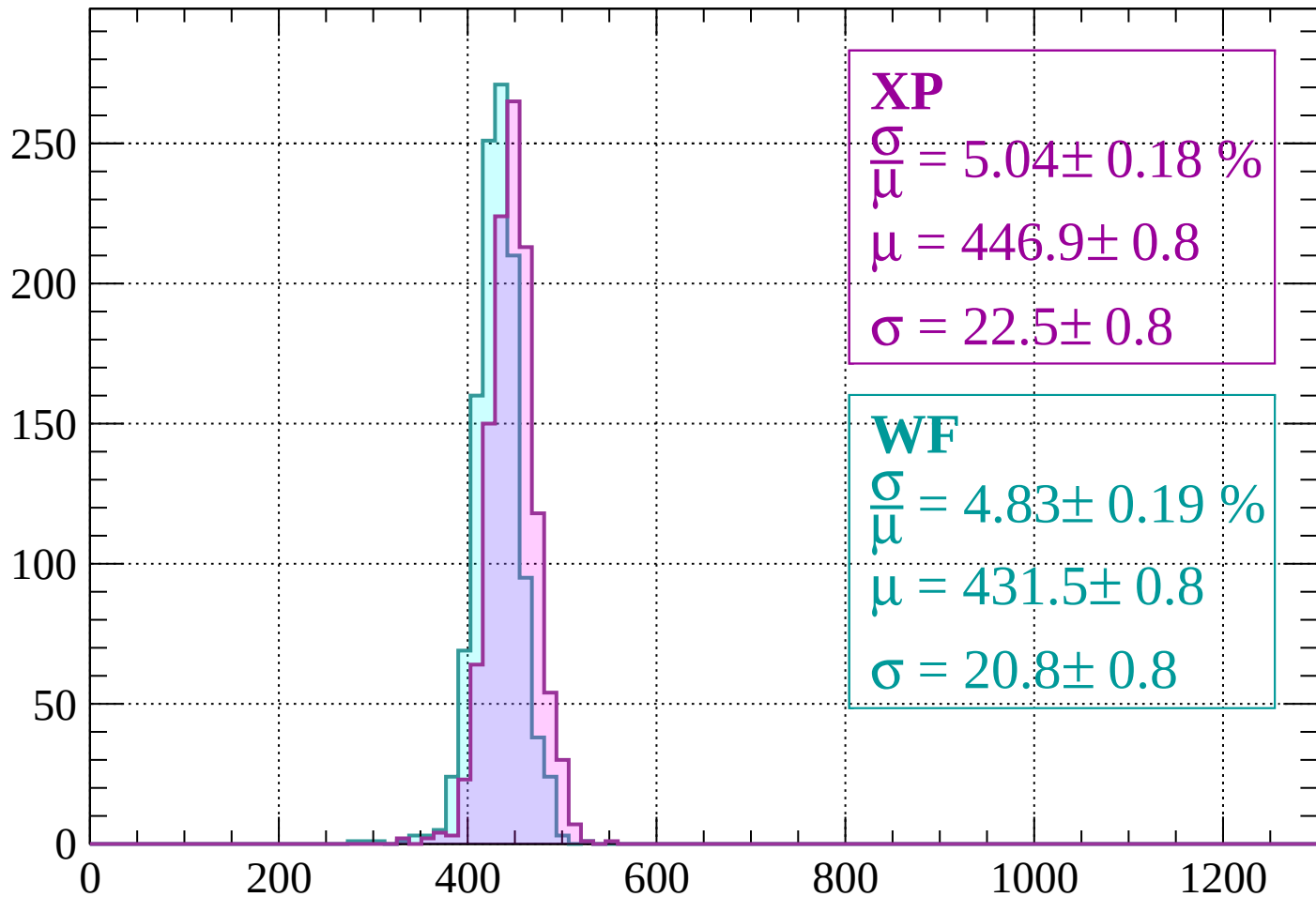
Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.18 \%$$

$$\mu = 446.9 \pm 0.8$$

$$\sigma = 22.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.83 \pm 0.19 \%$$

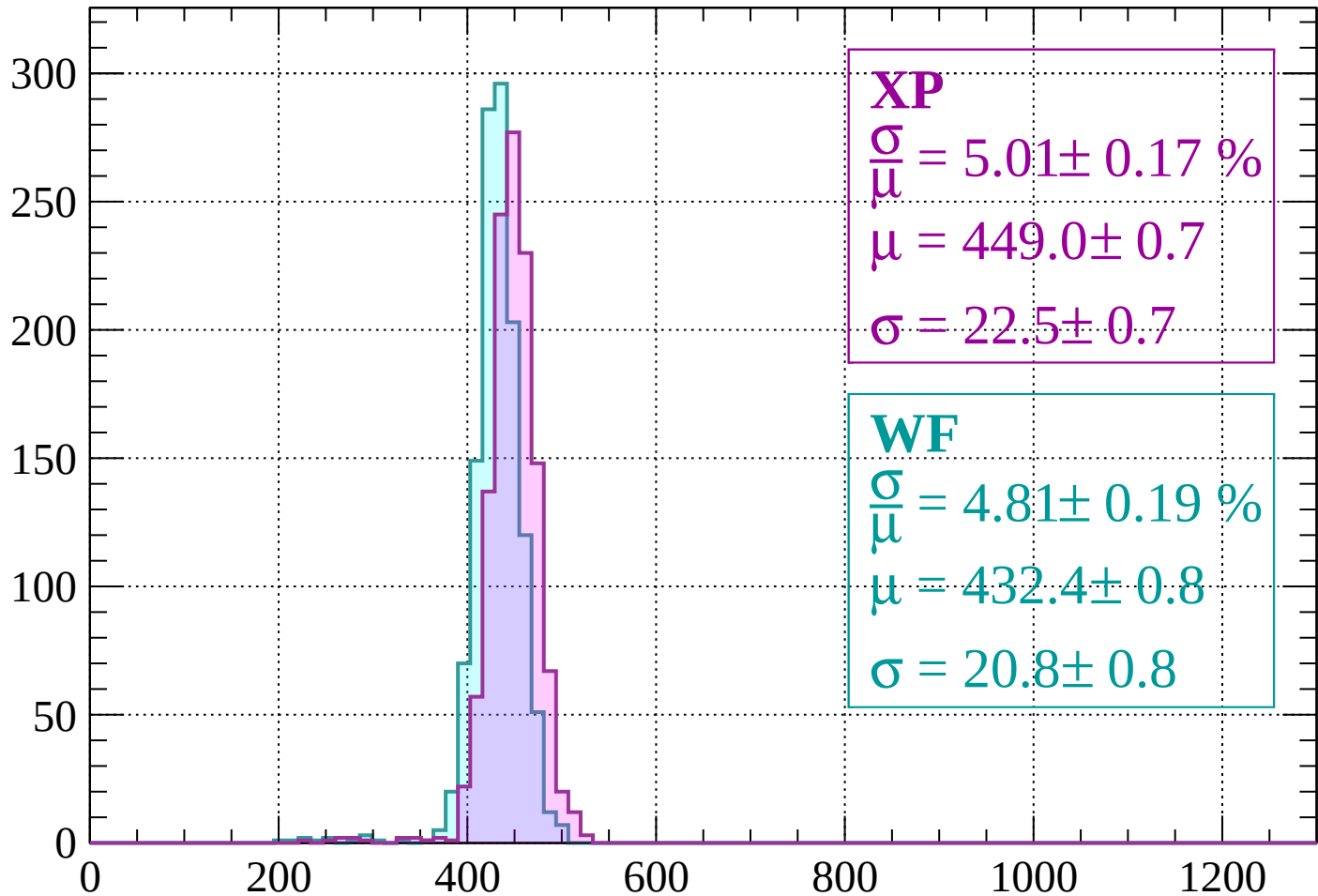
$$\mu = 431.5 \pm 0.8$$

$$\sigma = 20.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 5.01 \pm 0.17 \%$$

$$\mu = 449.0 \pm 0.7$$

$$\sigma = 22.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.19 \%$$

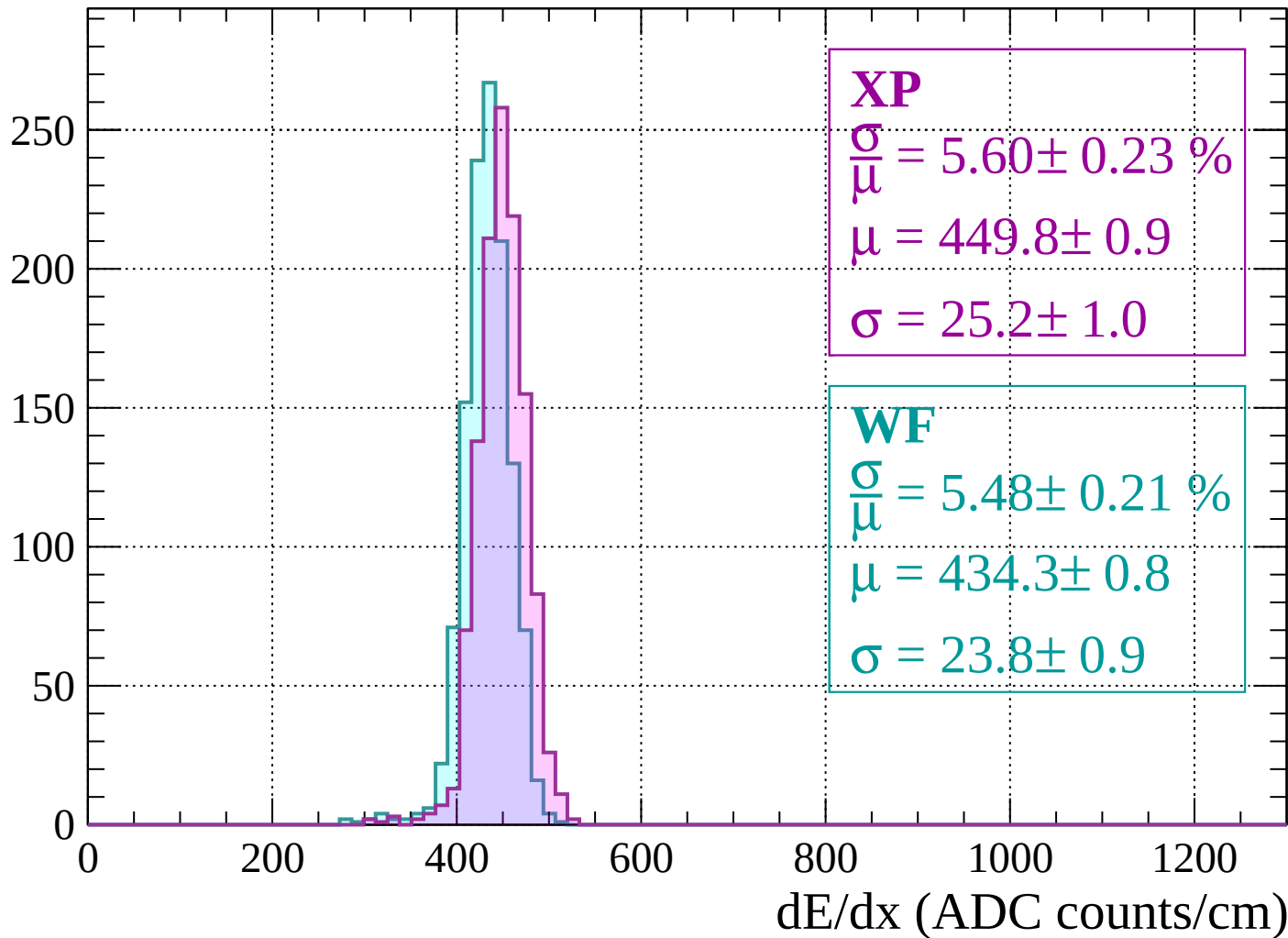
$$\mu = 432.4 \pm 0.8$$

$$\sigma = 20.8 \pm 0.8$$

dE/dx (ADC counts/cm)

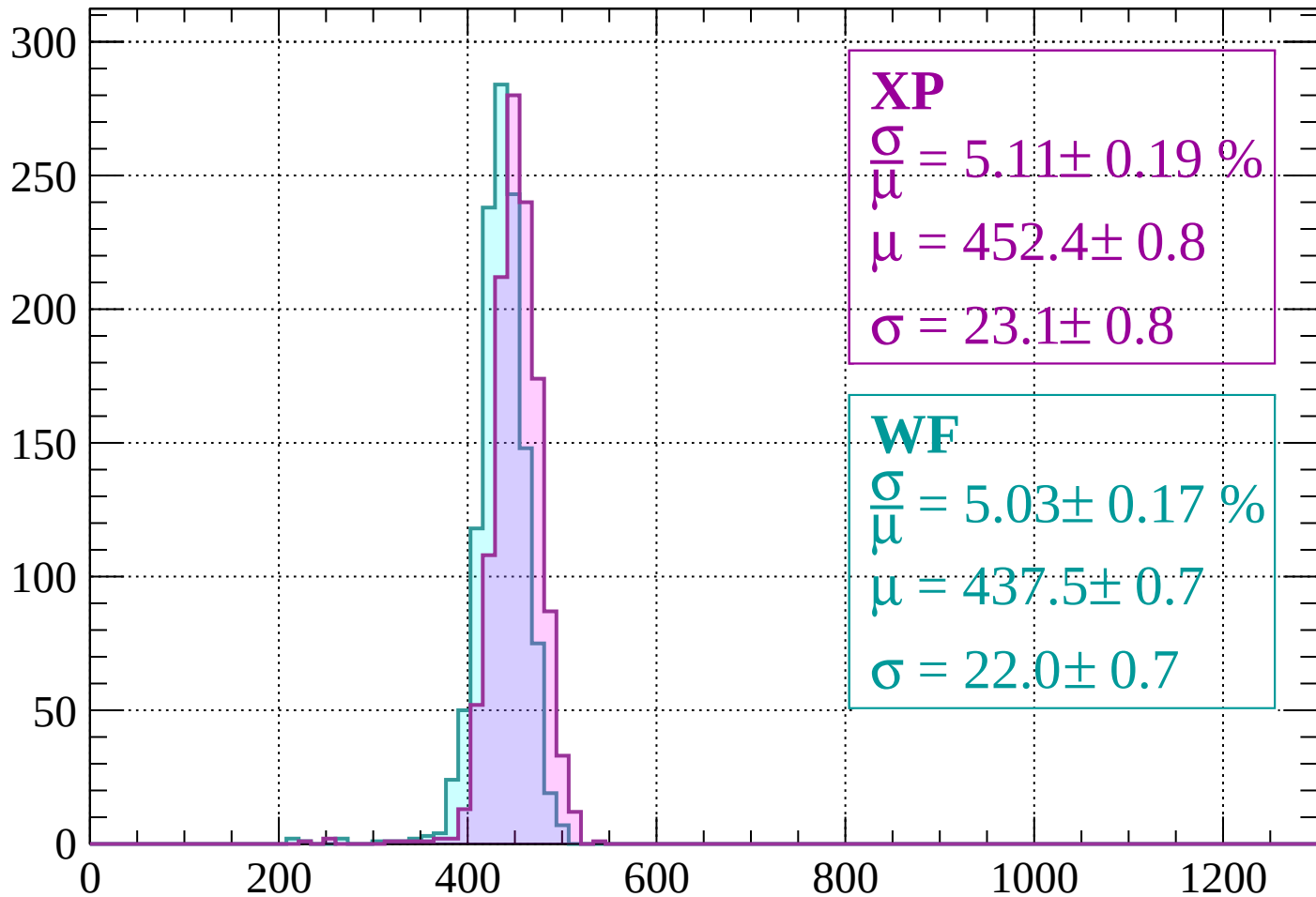
Energy loss | $1000 < p < 1040$

Count



Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 5.11 \pm 0.19 \%$$

$$\mu = 452.4 \pm 0.8$$

$$\sigma = 23.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.03 \pm 0.17 \%$$

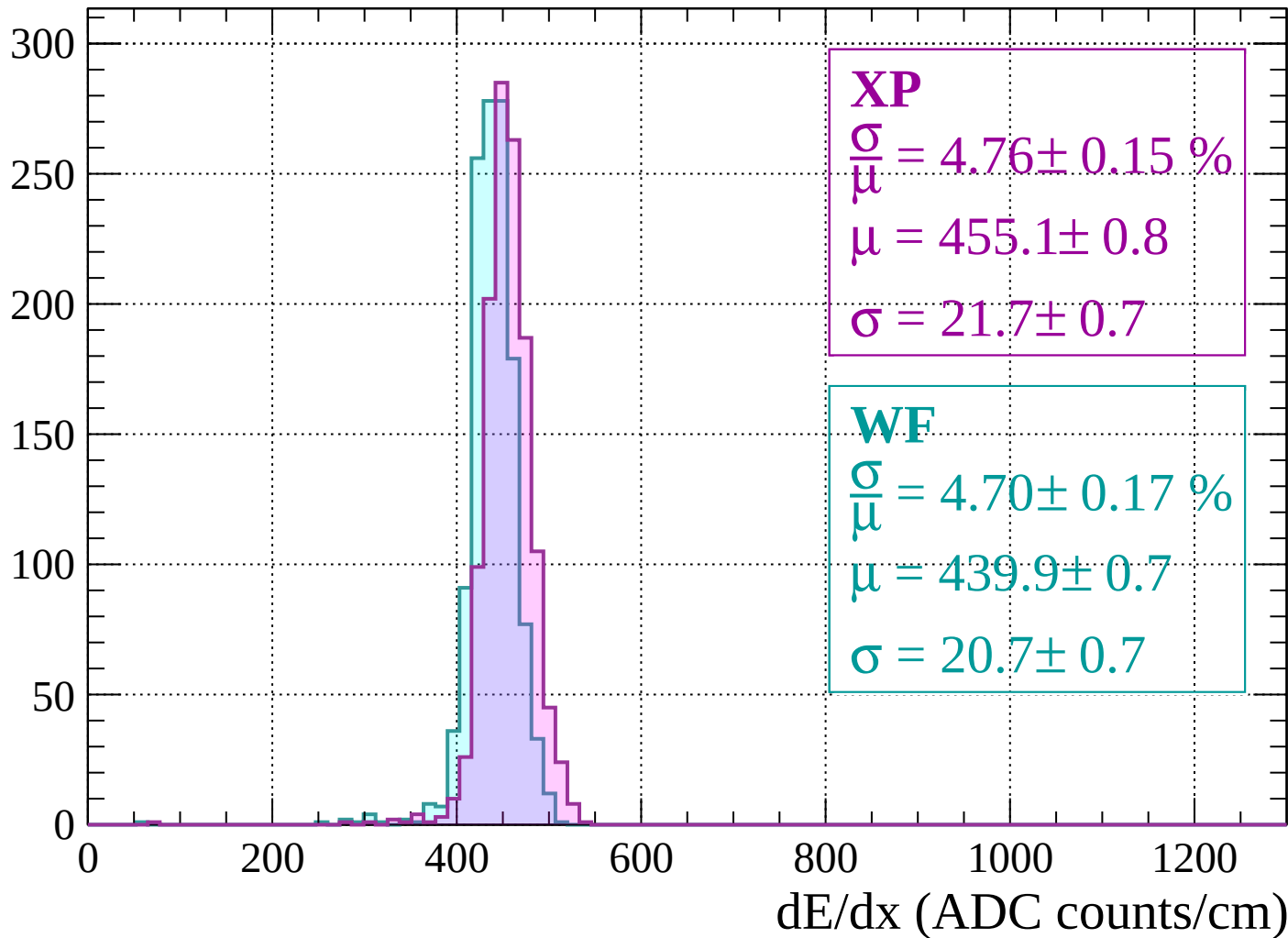
$$\mu = 437.5 \pm 0.7$$

$$\sigma = 22.0 \pm 0.7$$

dE/dx (ADC counts/cm)

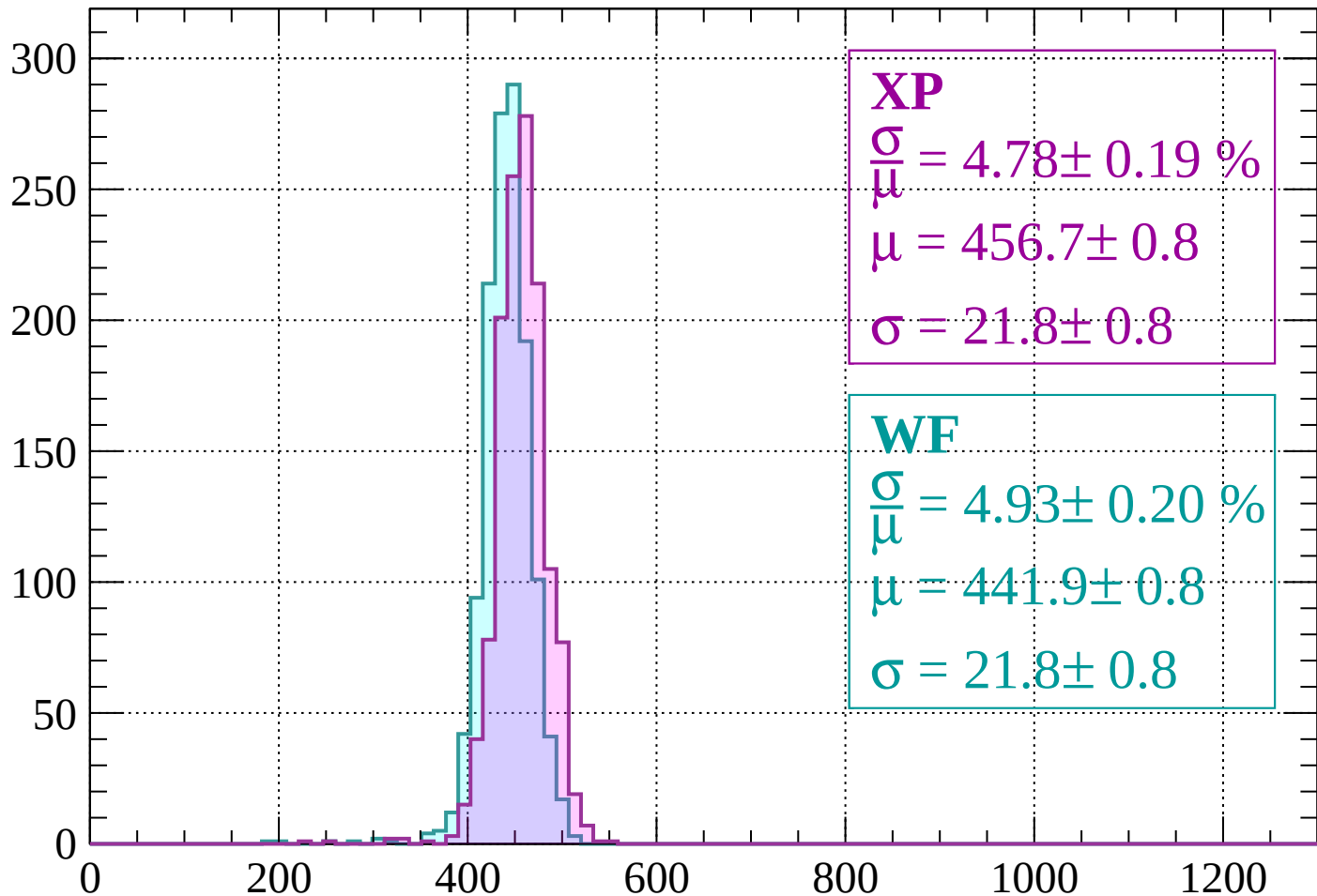
Energy loss | $1080 < p < 1120$

Count



Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.19 \%$$

$$\mu = 456.7 \pm 0.8$$

$$\sigma = 21.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.93 \pm 0.20 \%$$

$$\mu = 441.9 \pm 0.8$$

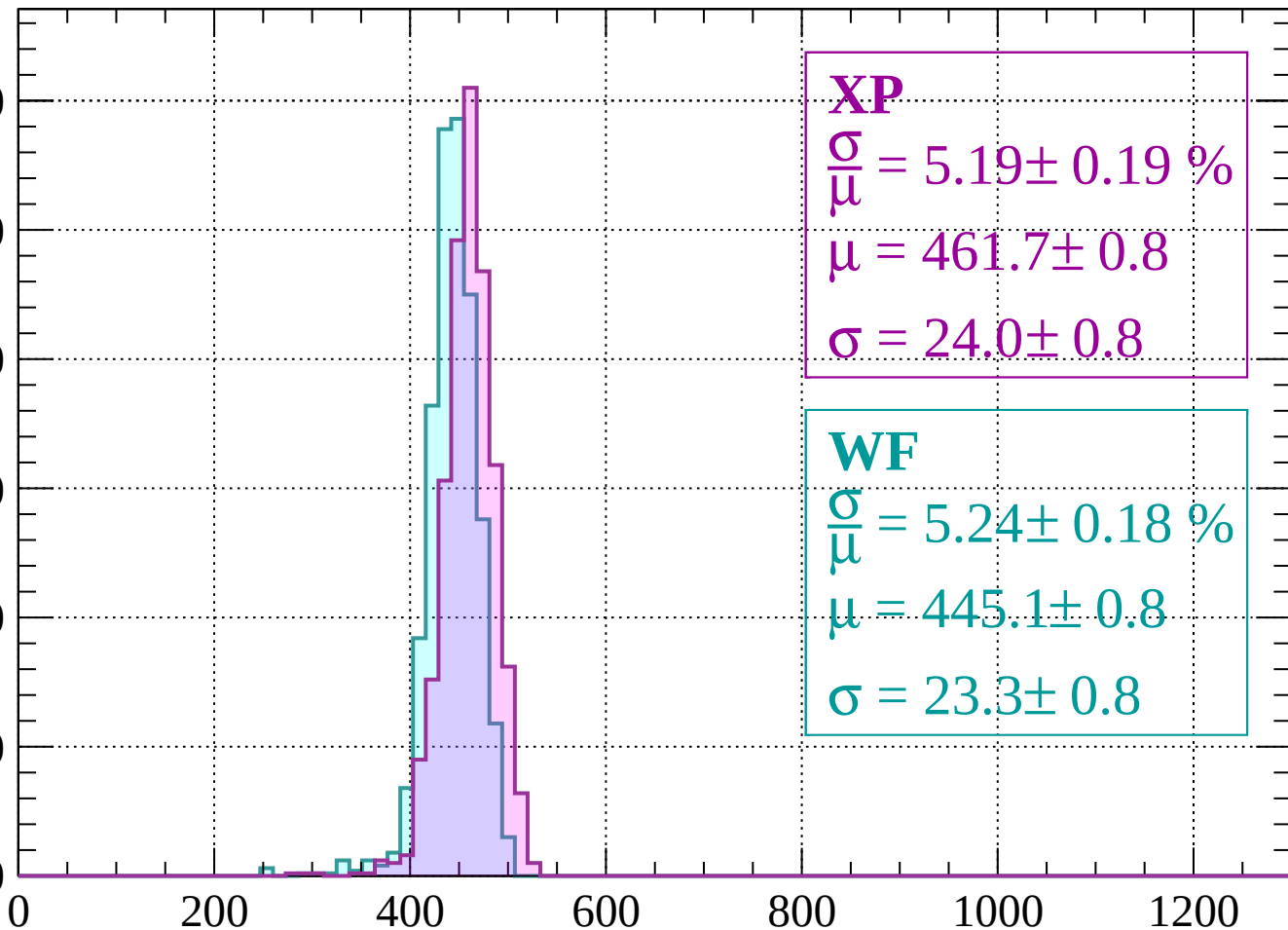
$$\sigma = 21.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.19 \pm 0.19 \%$$

$$\mu = 461.7 \pm 0.8$$

$$\sigma = 24.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.24 \pm 0.18 \%$$

$$\mu = 445.1 \pm 0.8$$

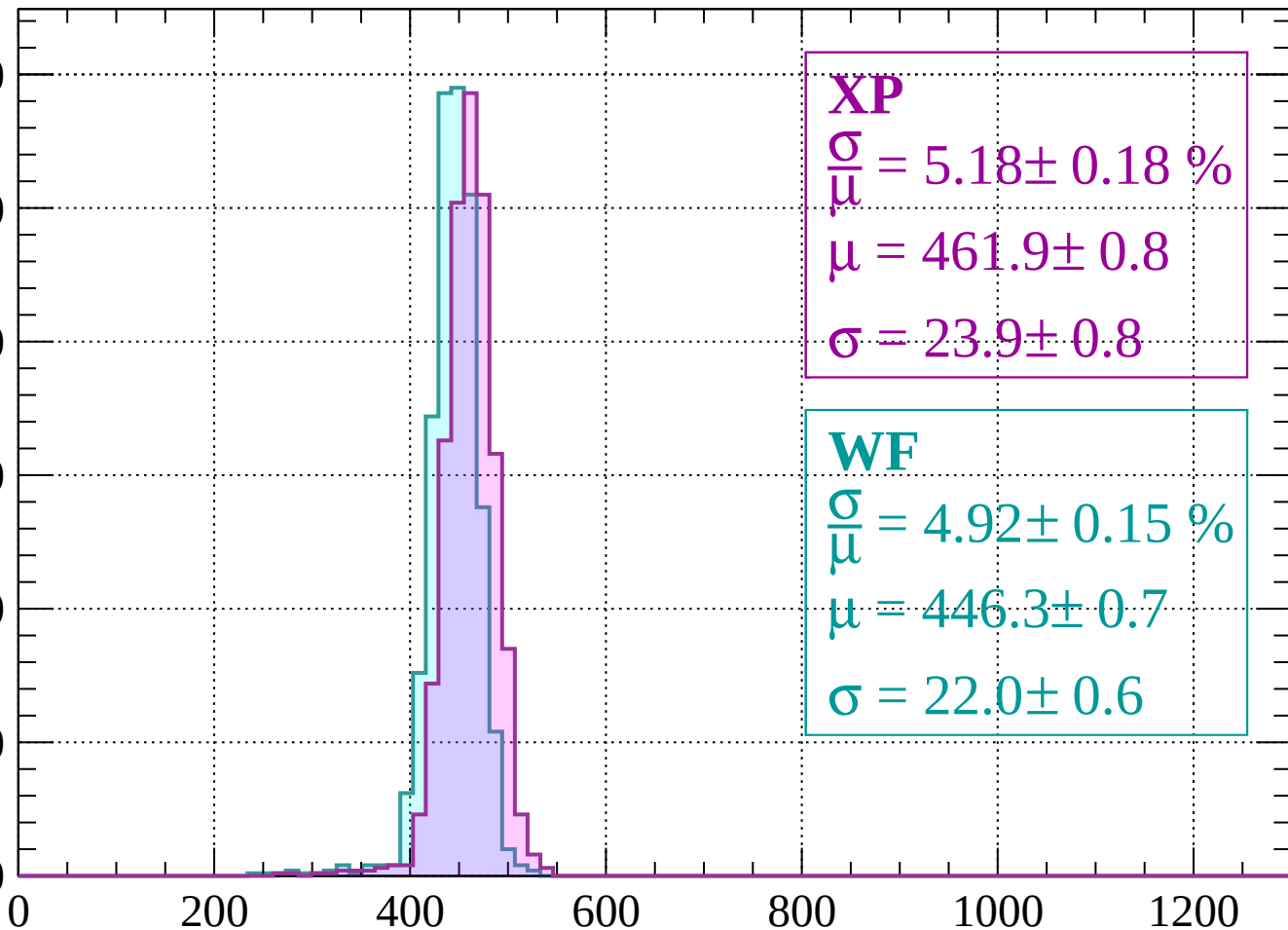
$$\sigma = 23.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.18 \pm 0.18 \%$$

$$\mu = 461.9 \pm 0.8$$

$$\sigma = 23.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.92 \pm 0.15 \%$$

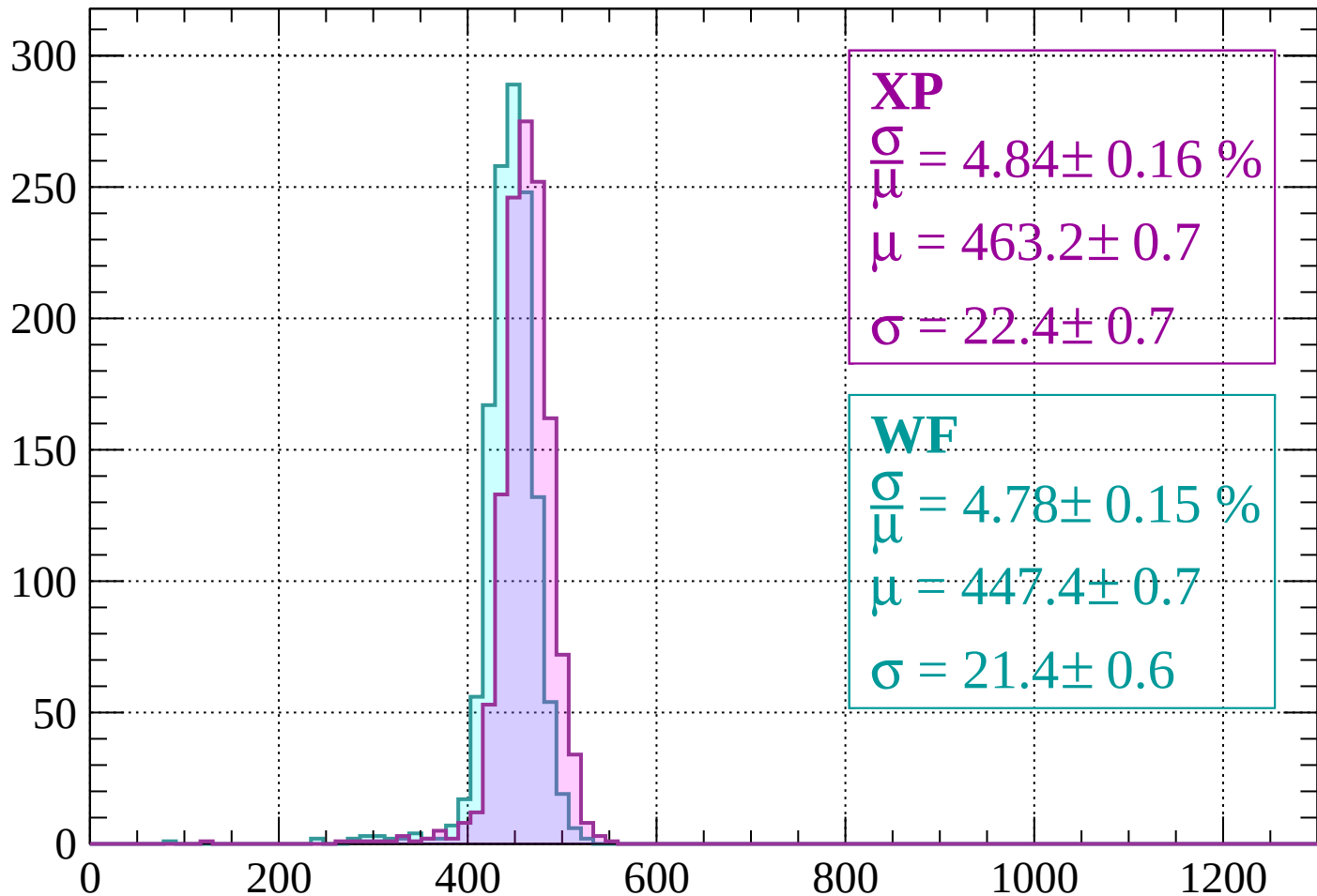
$$\mu = 446.3 \pm 0.7$$

$$\sigma = 22.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 4.84 \pm 0.16 \%$$

$$\mu = 463.2 \pm 0.7$$

$$\sigma = 22.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.78 \pm 0.15 \%$$

$$\mu = 447.4 \pm 0.7$$

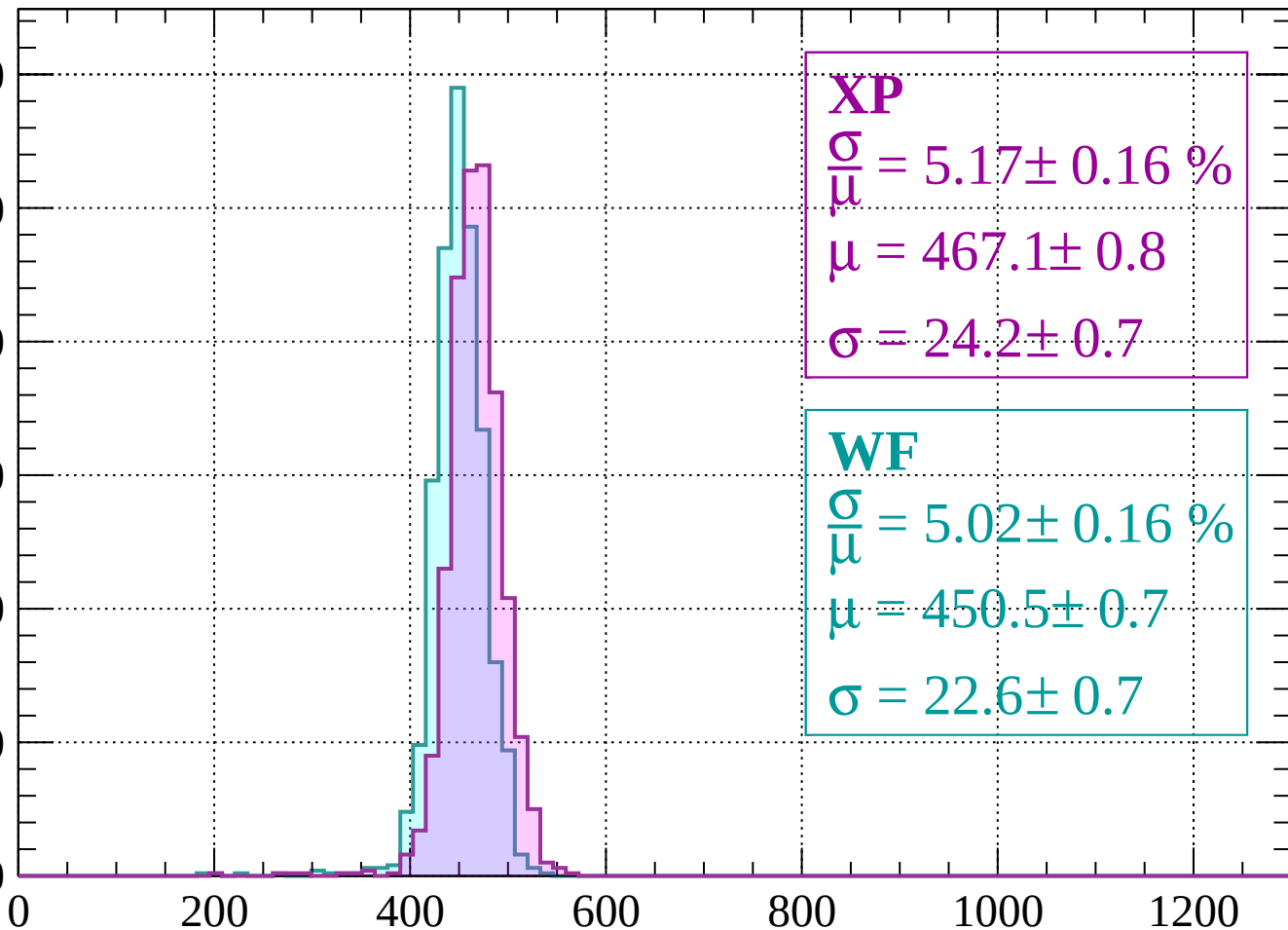
$$\sigma = 21.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.17 \pm 0.16 \%$$

$$\mu = 467.1 \pm 0.8$$

$$\sigma = 24.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.02 \pm 0.16 \%$$

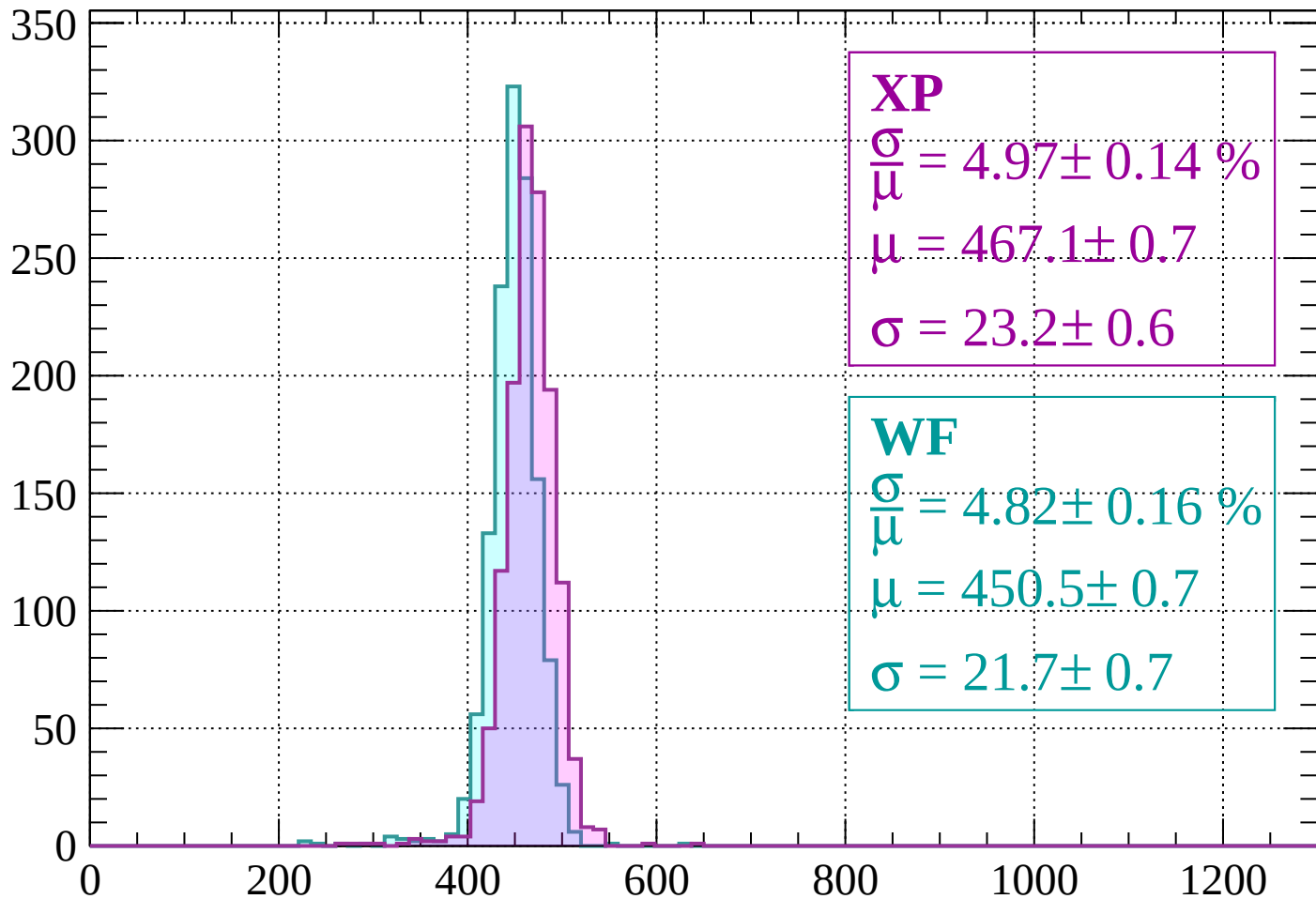
$$\mu = 450.5 \pm 0.7$$

$$\sigma = 22.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 4.97 \pm 0.14 \%$$

$$\mu = 467.1 \pm 0.7$$

$$\sigma = 23.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.16 \%$$

$$\mu = 450.5 \pm 0.7$$

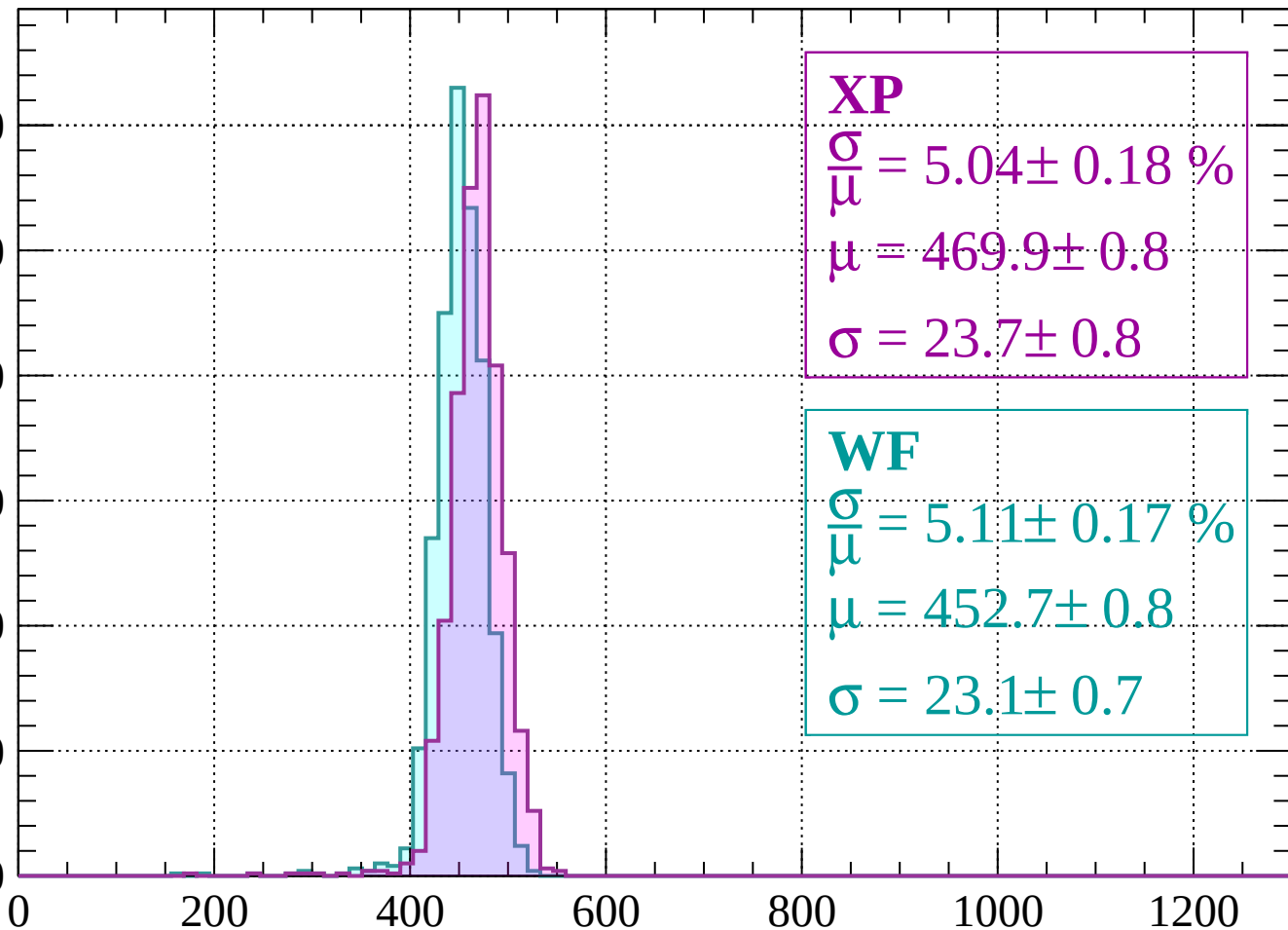
$$\sigma = 21.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.18 \%$$

$$\mu = 469.9 \pm 0.8$$

$$\sigma = 23.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.11 \pm 0.17 \%$$

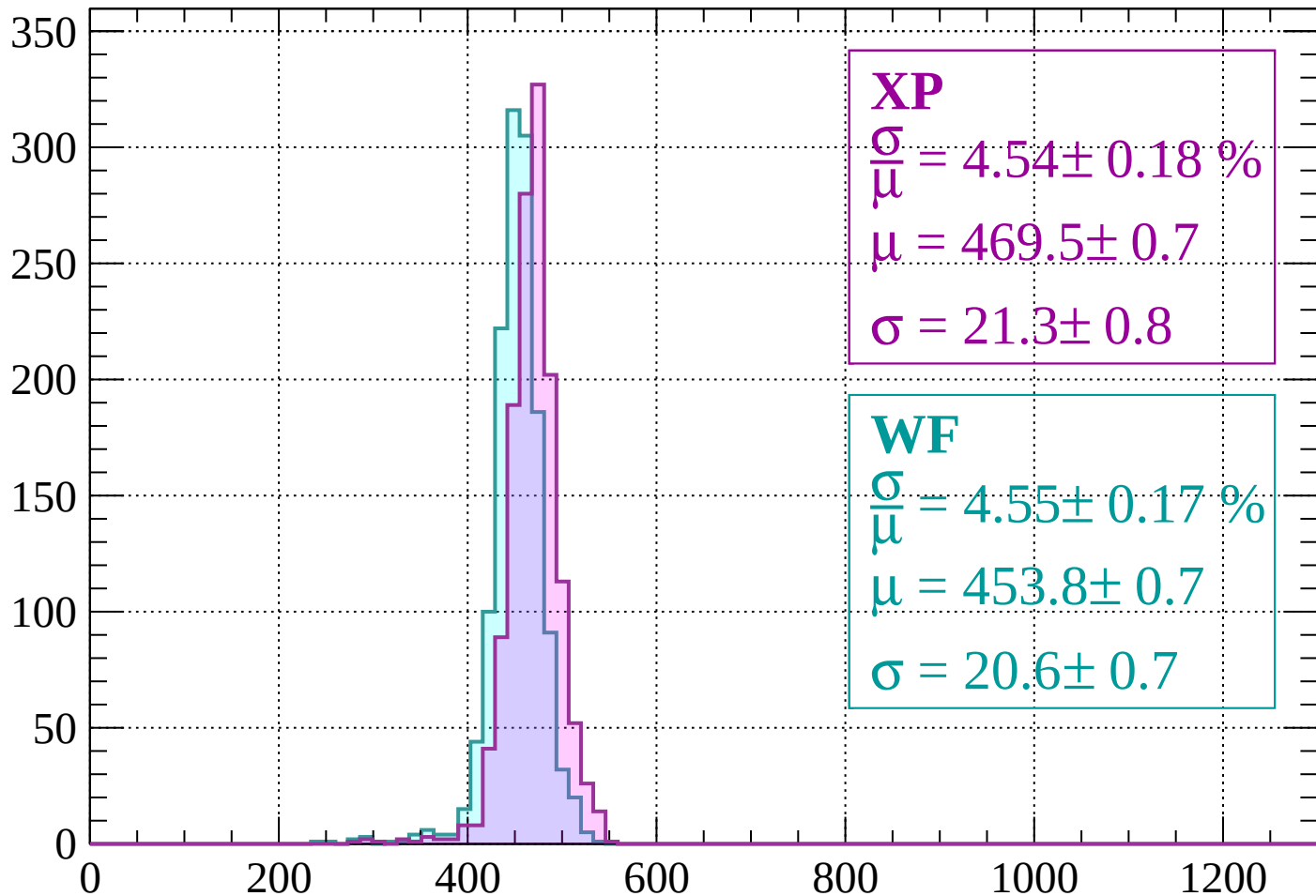
$$\mu = 452.7 \pm 0.8$$

$$\sigma = 23.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 4.54 \pm 0.18 \%$$

$$\mu = 469.5 \pm 0.7$$

$$\sigma = 21.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.55 \pm 0.17 \%$$

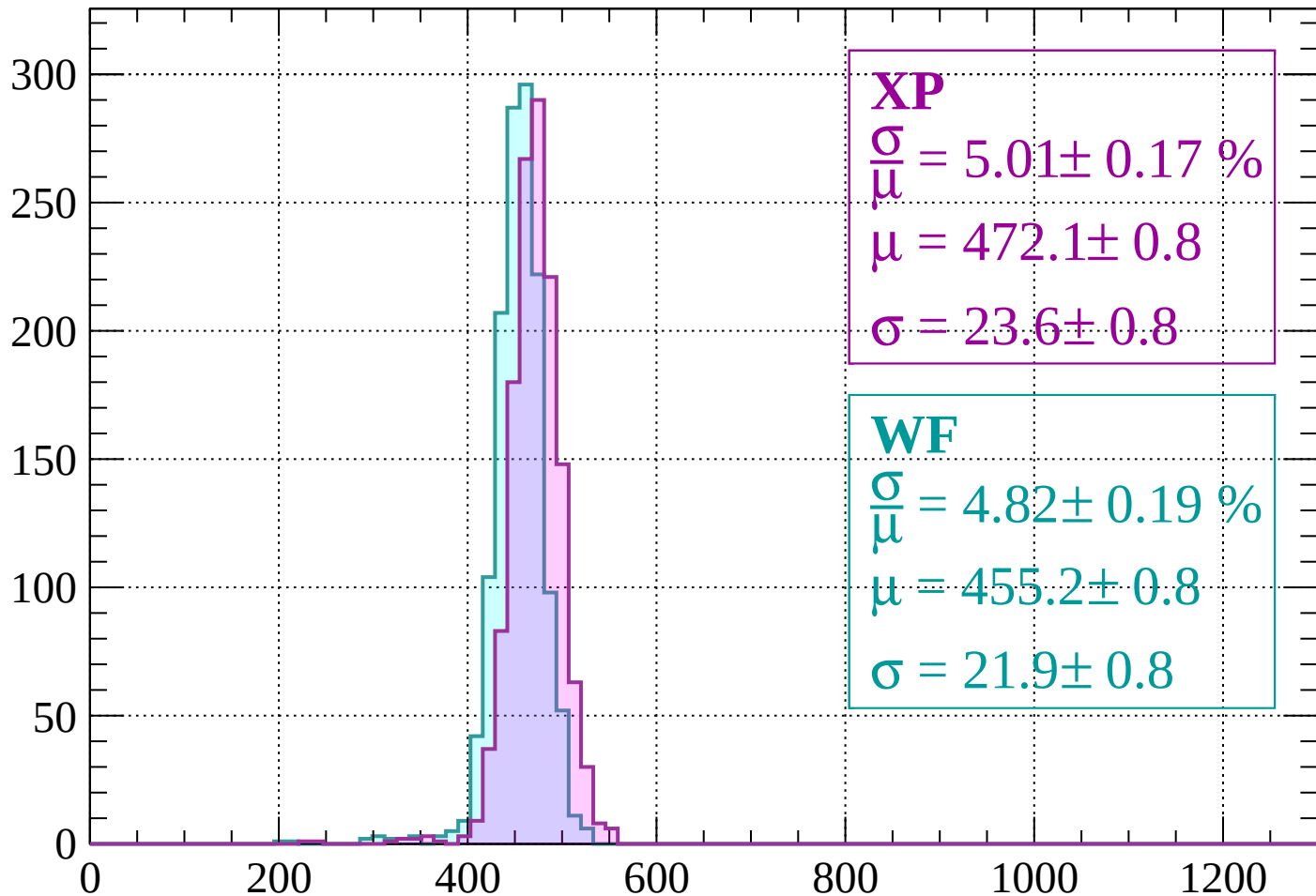
$$\mu = 453.8 \pm 0.7$$

$$\sigma = 20.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 5.01 \pm 0.17 \%$$

$$\mu = 472.1 \pm 0.8$$

$$\sigma = 23.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.19 \%$$

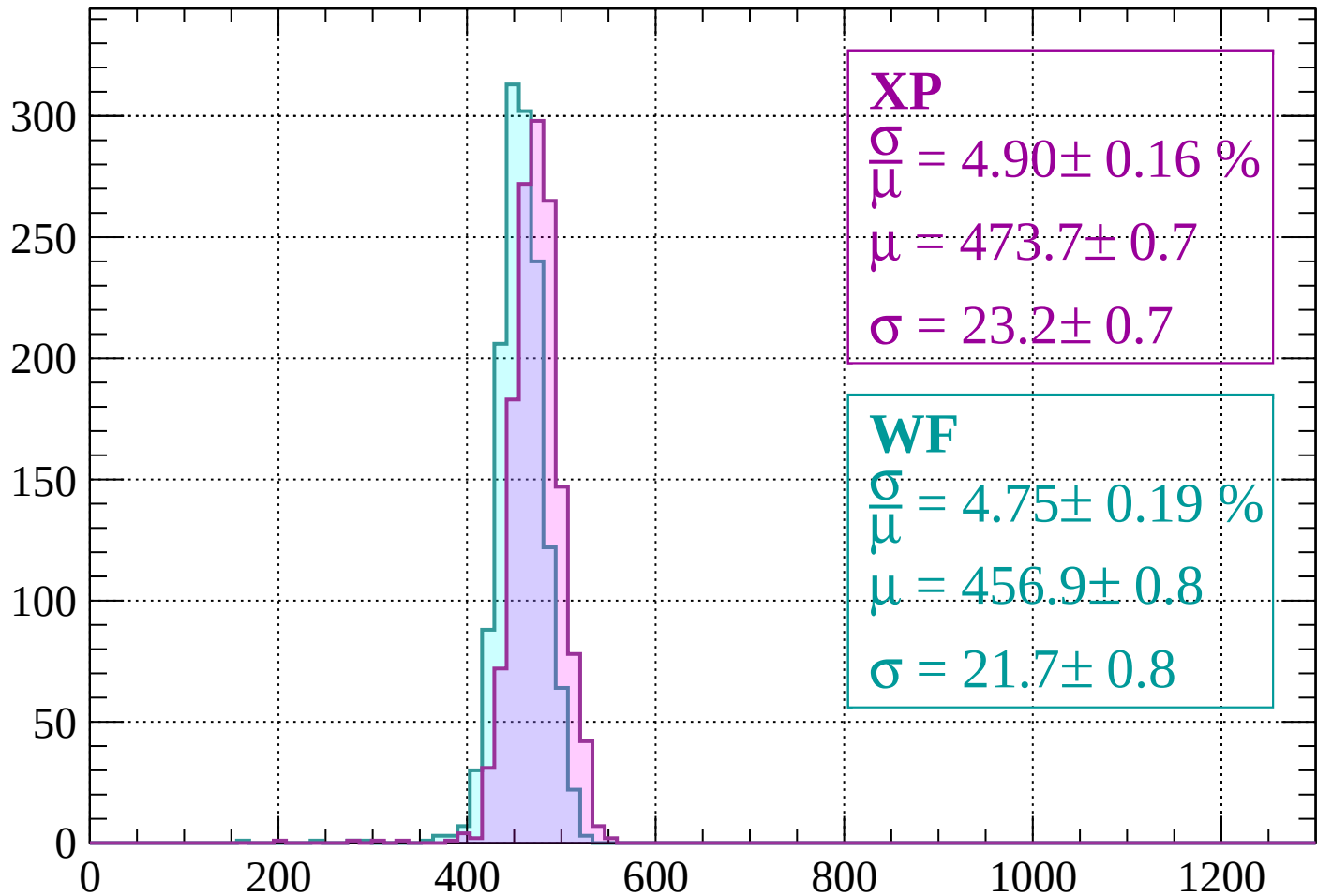
$$\mu = 455.2 \pm 0.8$$

$$\sigma = 21.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 4.90 \pm 0.16 \%$$

$$\mu = 473.7 \pm 0.7$$

$$\sigma = 23.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.75 \pm 0.19 \%$$

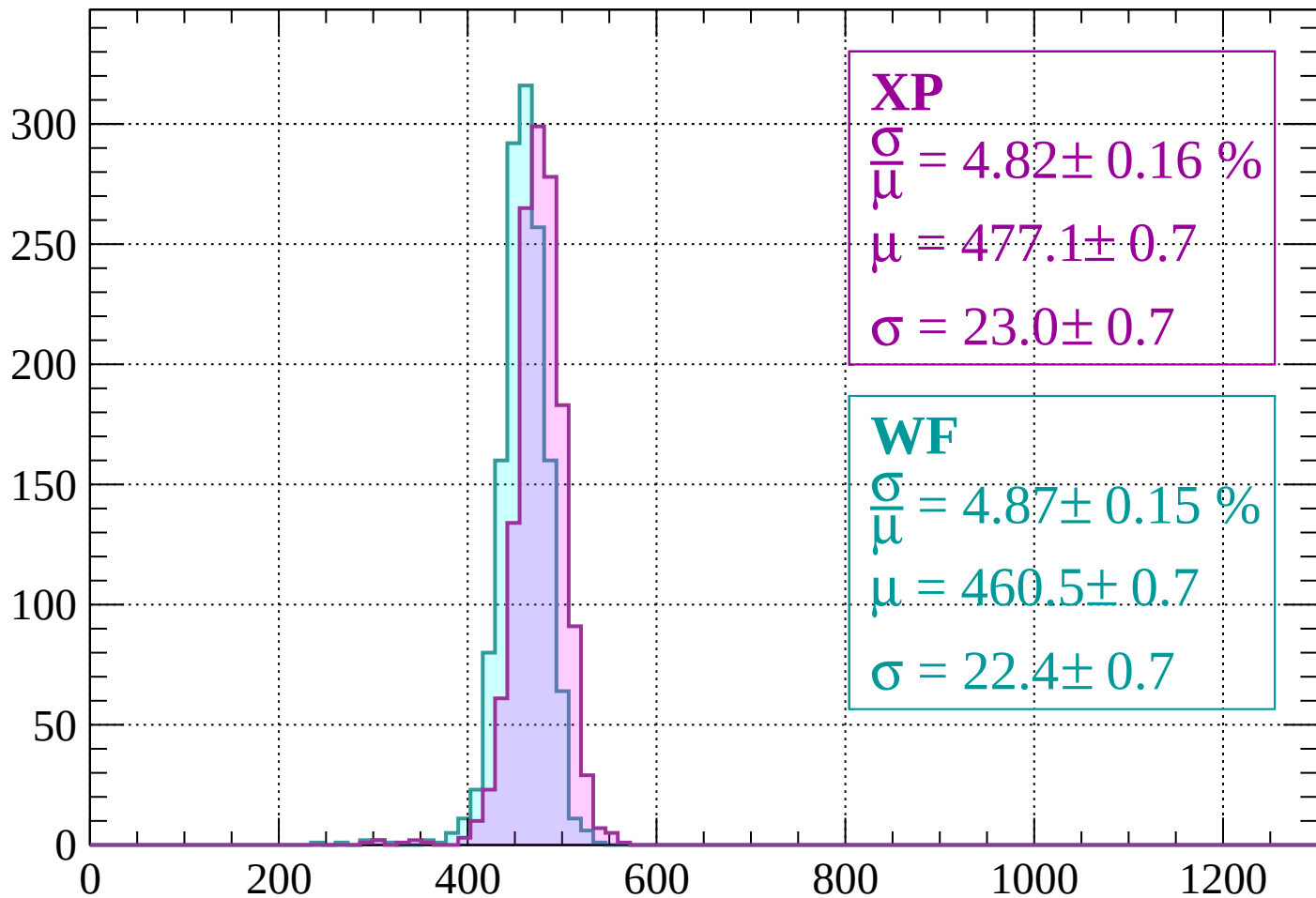
$$\mu = 456.9 \pm 0.8$$

$$\sigma = 21.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 4.82 \pm 0.16 \%$$

$$\mu = 477.1 \pm 0.7$$

$$\sigma = 23.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.87 \pm 0.15 \%$$

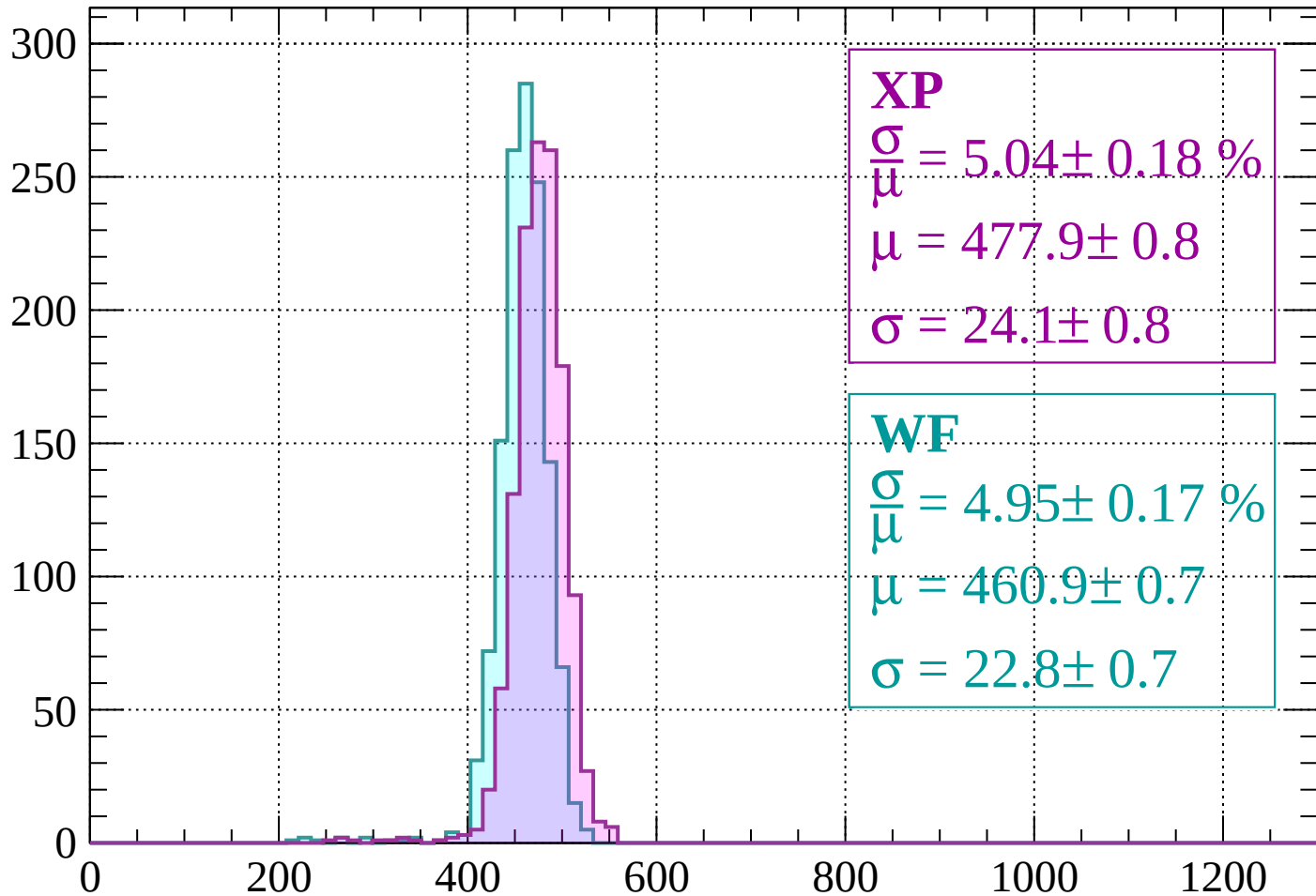
$$\mu = 460.5 \pm 0.7$$

$$\sigma = 22.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.18 \%$$

$$\mu = 477.9 \pm 0.8$$

$$\sigma = 24.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.95 \pm 0.17 \%$$

$$\mu = 460.9 \pm 0.7$$

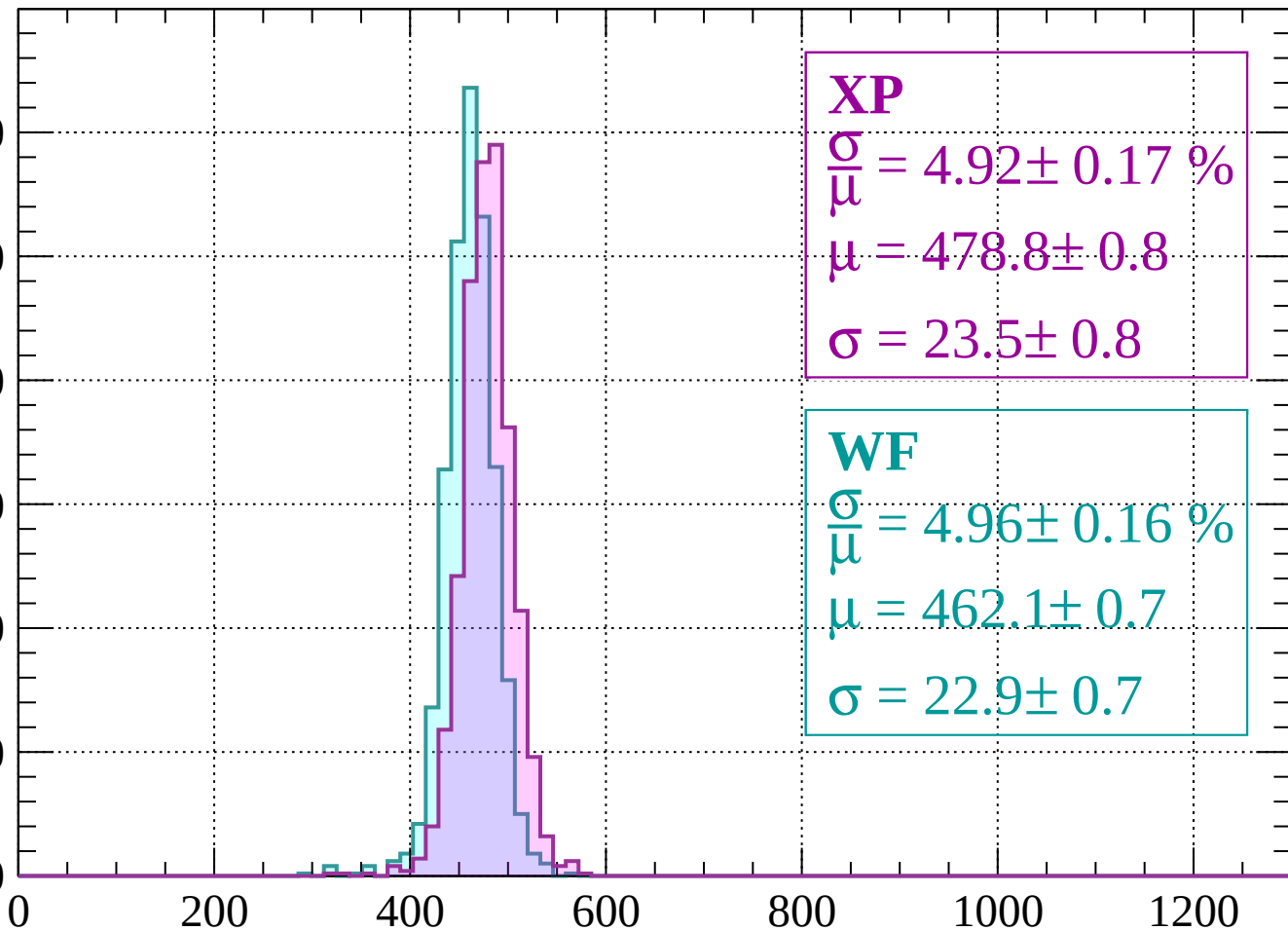
$$\sigma = 22.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.92 \pm 0.17 \%$$

$$\mu = 478.8 \pm 0.8$$

$$\sigma = 23.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.96 \pm 0.16 \%$$

$$\mu = 462.1 \pm 0.7$$

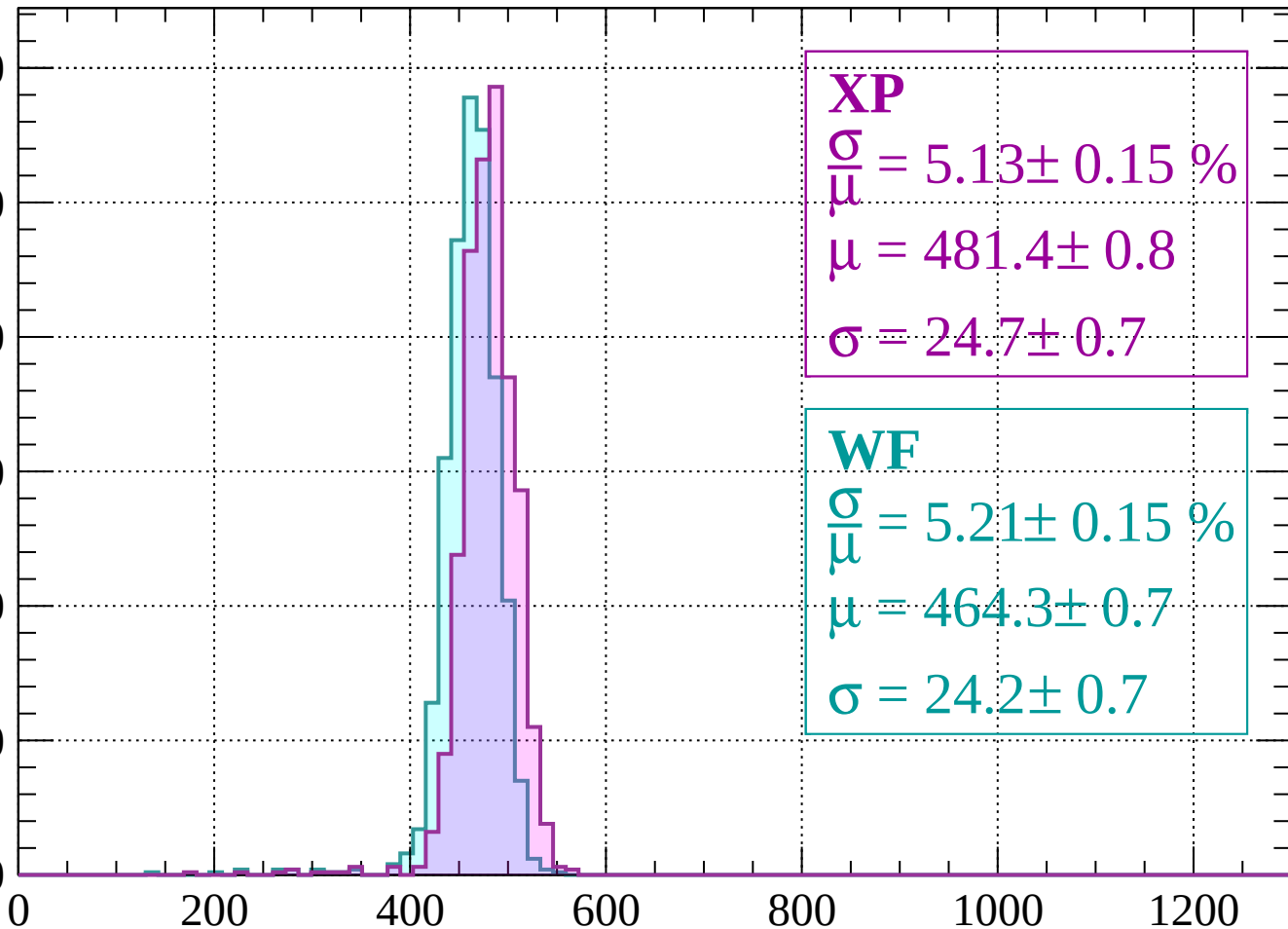
$$\sigma = 22.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.13 \pm 0.15 \%$$

$$\mu = 481.4 \pm 0.8$$

$$\sigma = 24.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.21 \pm 0.15 \%$$

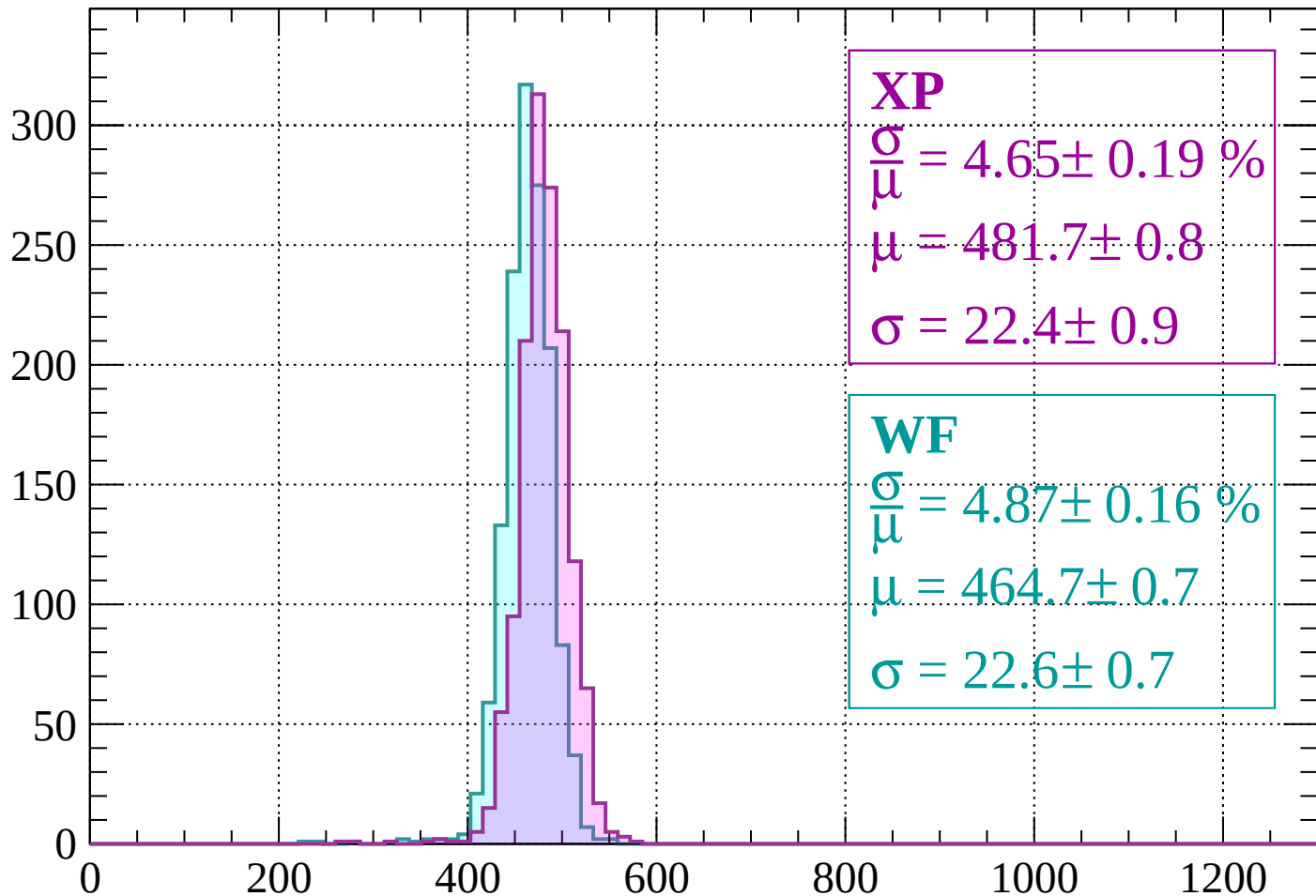
$$\mu = 464.3 \pm 0.7$$

$$\sigma = 24.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 4.65 \pm 0.19 \%$$

$$\mu = 481.7 \pm 0.8$$

$$\sigma = 22.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.87 \pm 0.16 \%$$

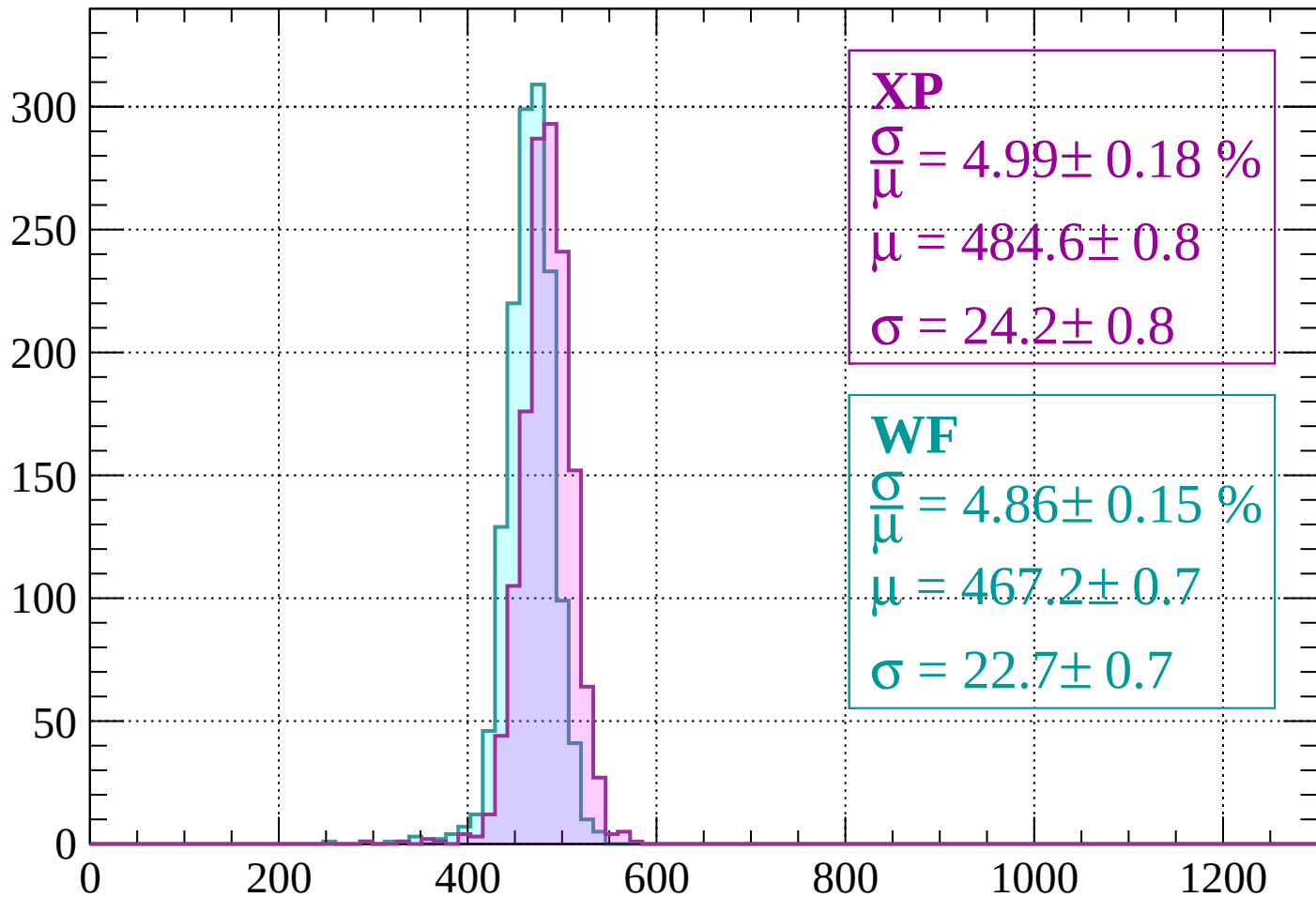
$$\mu = 464.7 \pm 0.7$$

$$\sigma = 22.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 4.99 \pm 0.18 \%$$

$$\mu = 484.6 \pm 0.8$$

$$\sigma = 24.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.86 \pm 0.15 \%$$

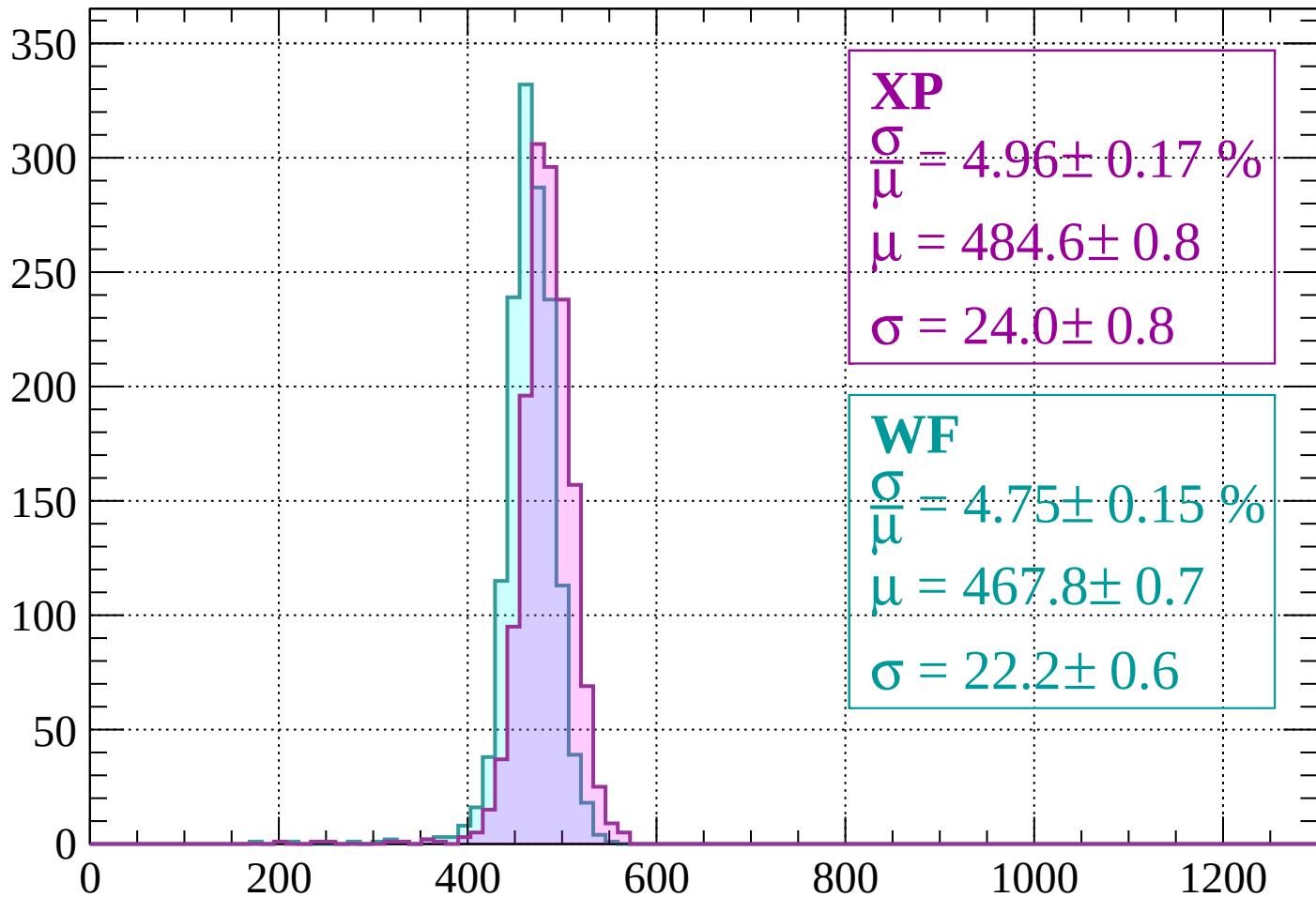
$$\mu = 467.2 \pm 0.7$$

$$\sigma = 22.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.17 \%$$

$$\mu = 484.6 \pm 0.8$$

$$\sigma = 24.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.75 \pm 0.15 \%$$

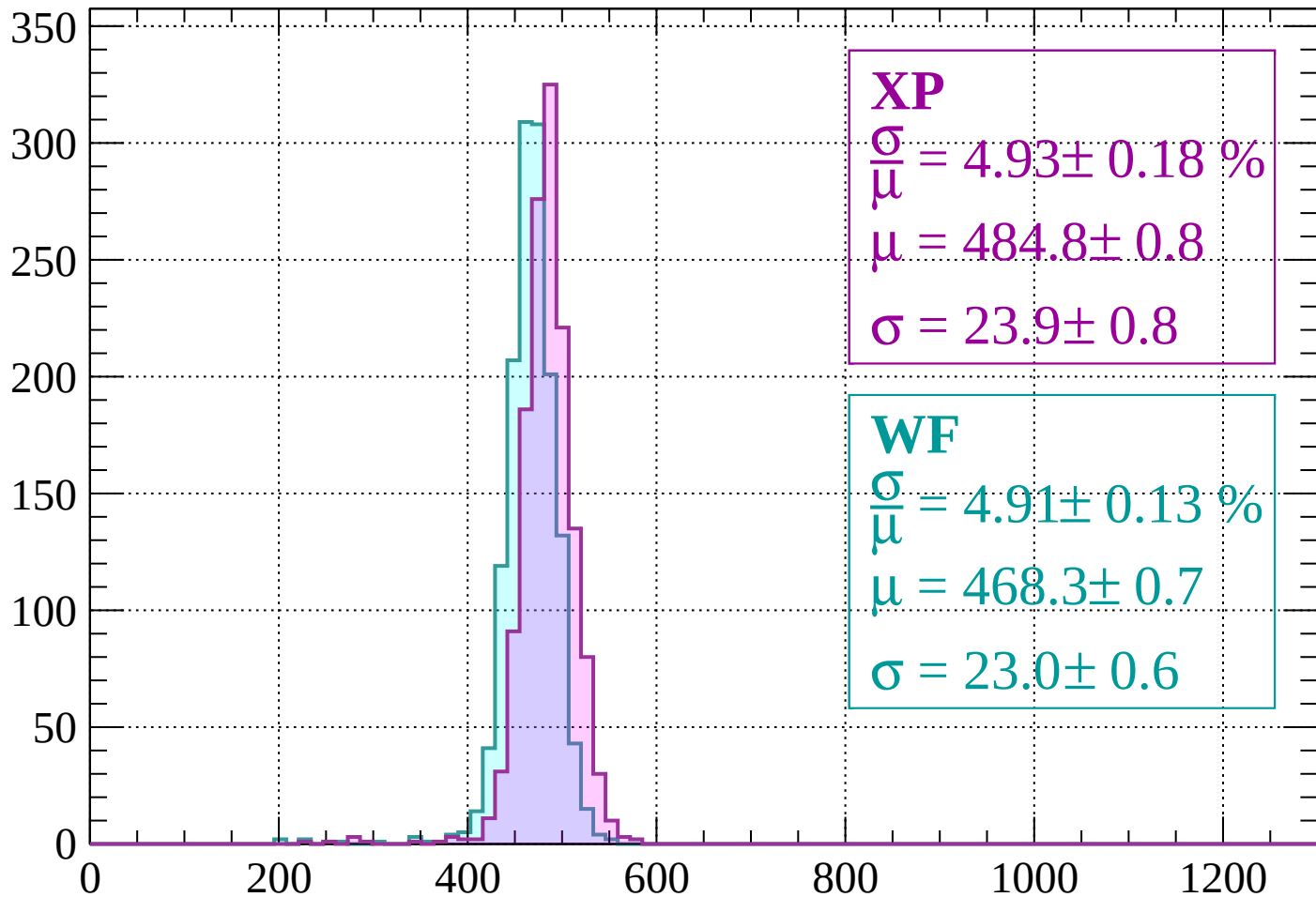
$$\mu = 467.8 \pm 0.7$$

$$\sigma = 22.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 4.93 \pm 0.18 \%$$

$$\mu = 484.8 \pm 0.8$$

$$\sigma = 23.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.91 \pm 0.13 \%$$

$$\mu = 468.3 \pm 0.7$$

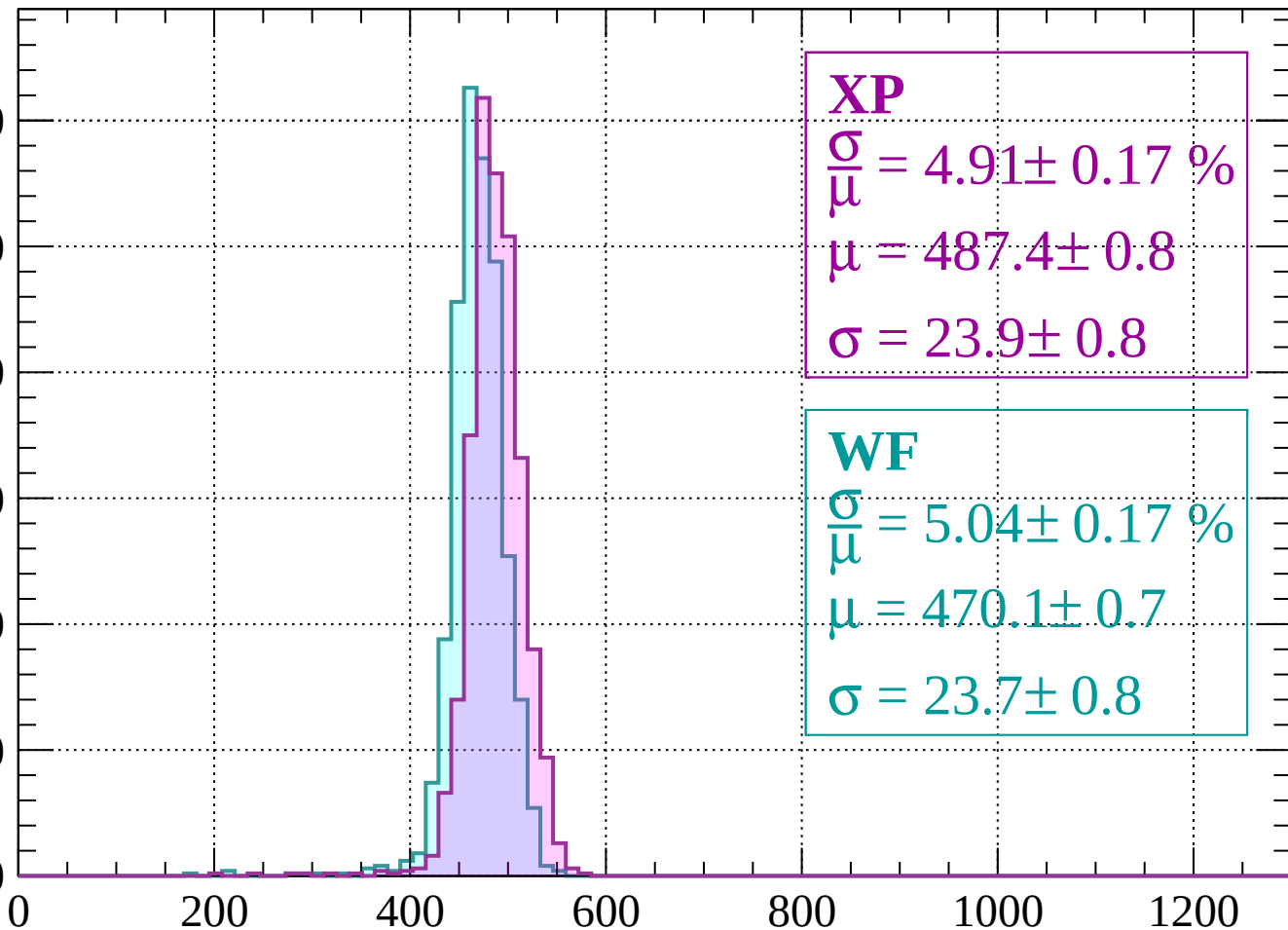
$$\sigma = 23.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.91 \pm 0.17 \%$$

$$\mu = 487.4 \pm 0.8$$

$$\sigma = 23.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.04 \pm 0.17 \%$$

$$\mu = 470.1 \pm 0.7$$

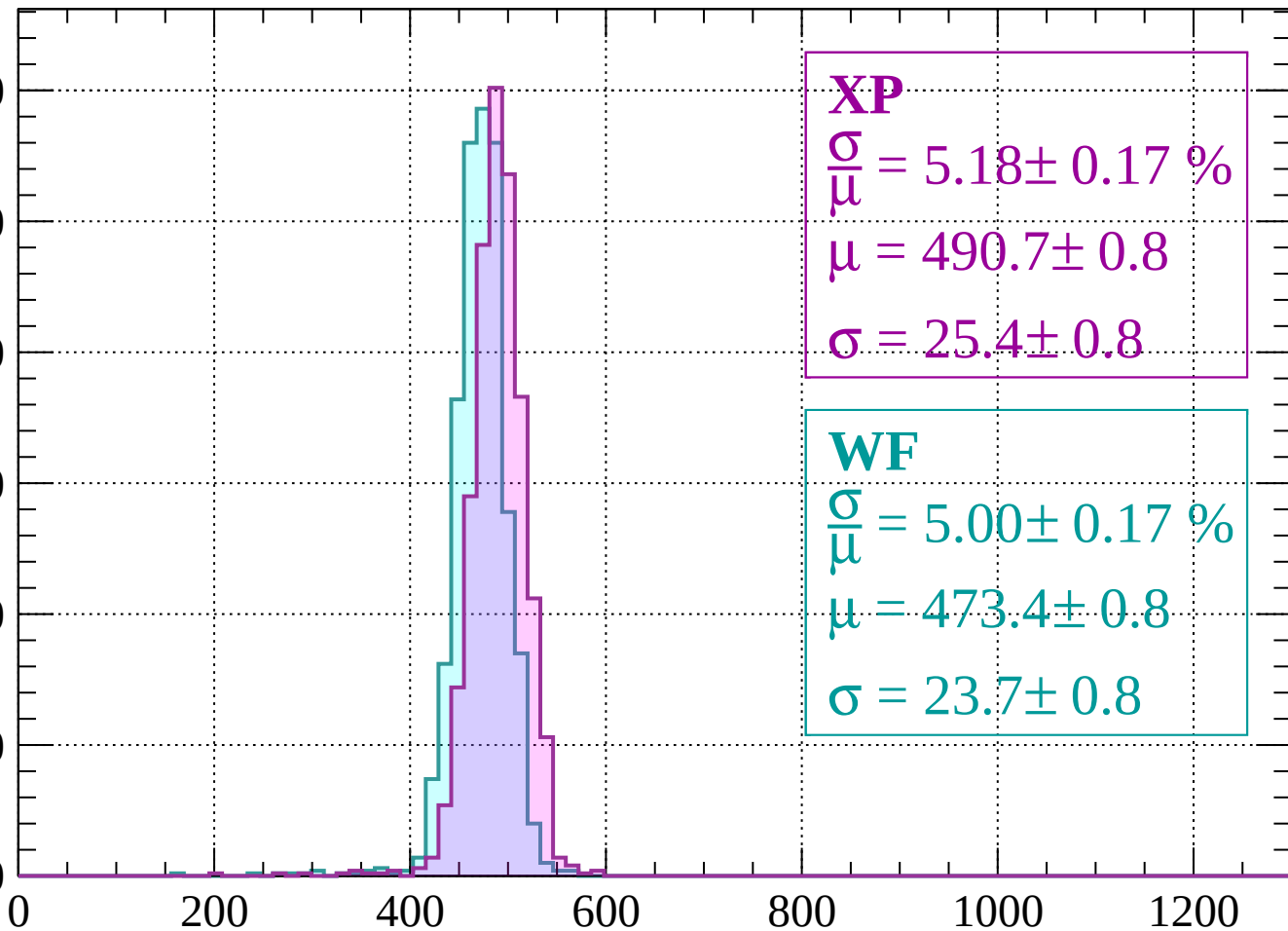
$$\sigma = 23.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.18 \pm 0.17 \%$$

$$\mu = 490.7 \pm 0.8$$

$$\sigma = 25.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.00 \pm 0.17 \%$$

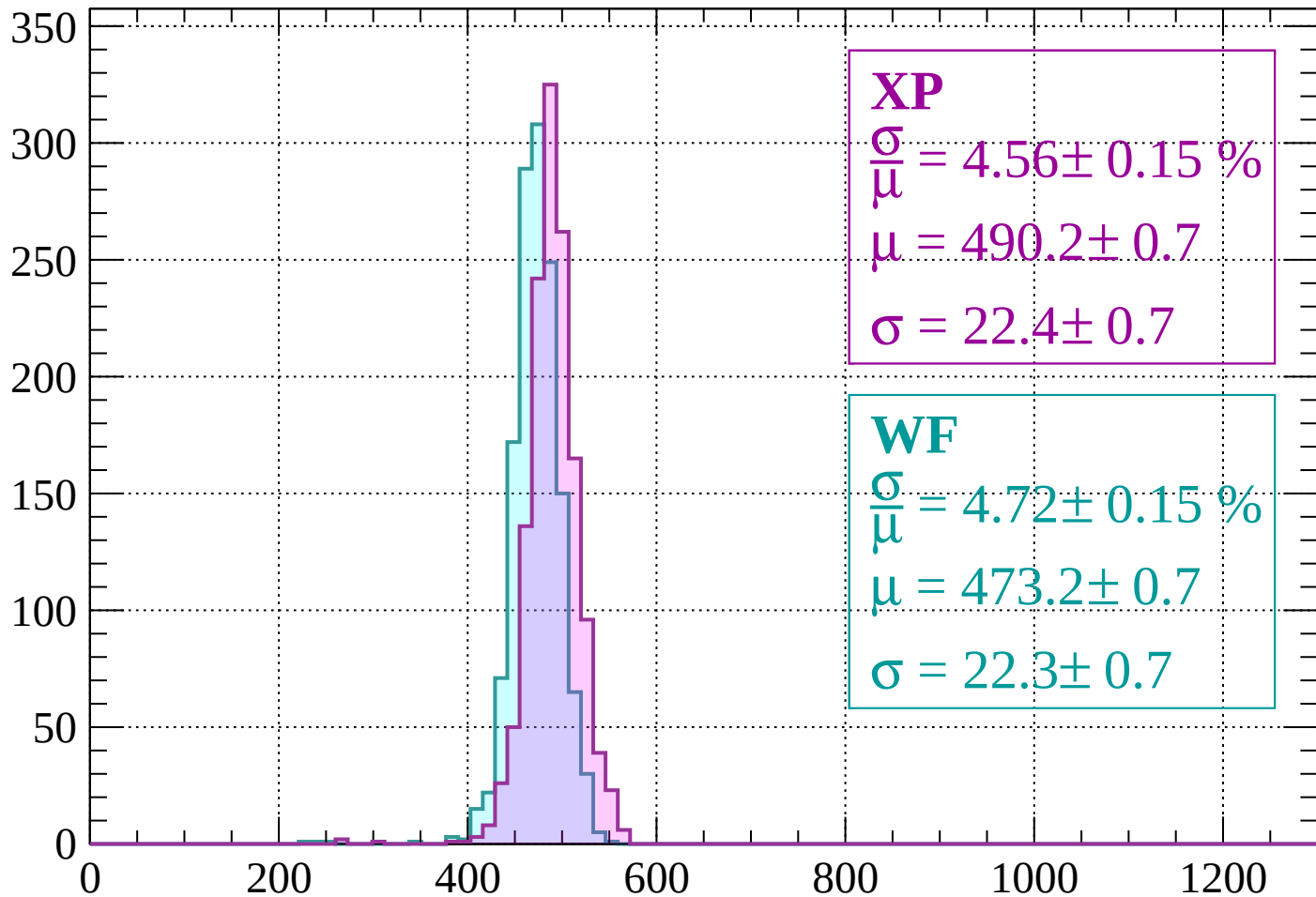
$$\mu = 473.4 \pm 0.8$$

$$\sigma = 23.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 4.56 \pm 0.15 \%$$

$$\mu = 490.2 \pm 0.7$$

$$\sigma = 22.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.72 \pm 0.15 \%$$

$$\mu = 473.2 \pm 0.7$$

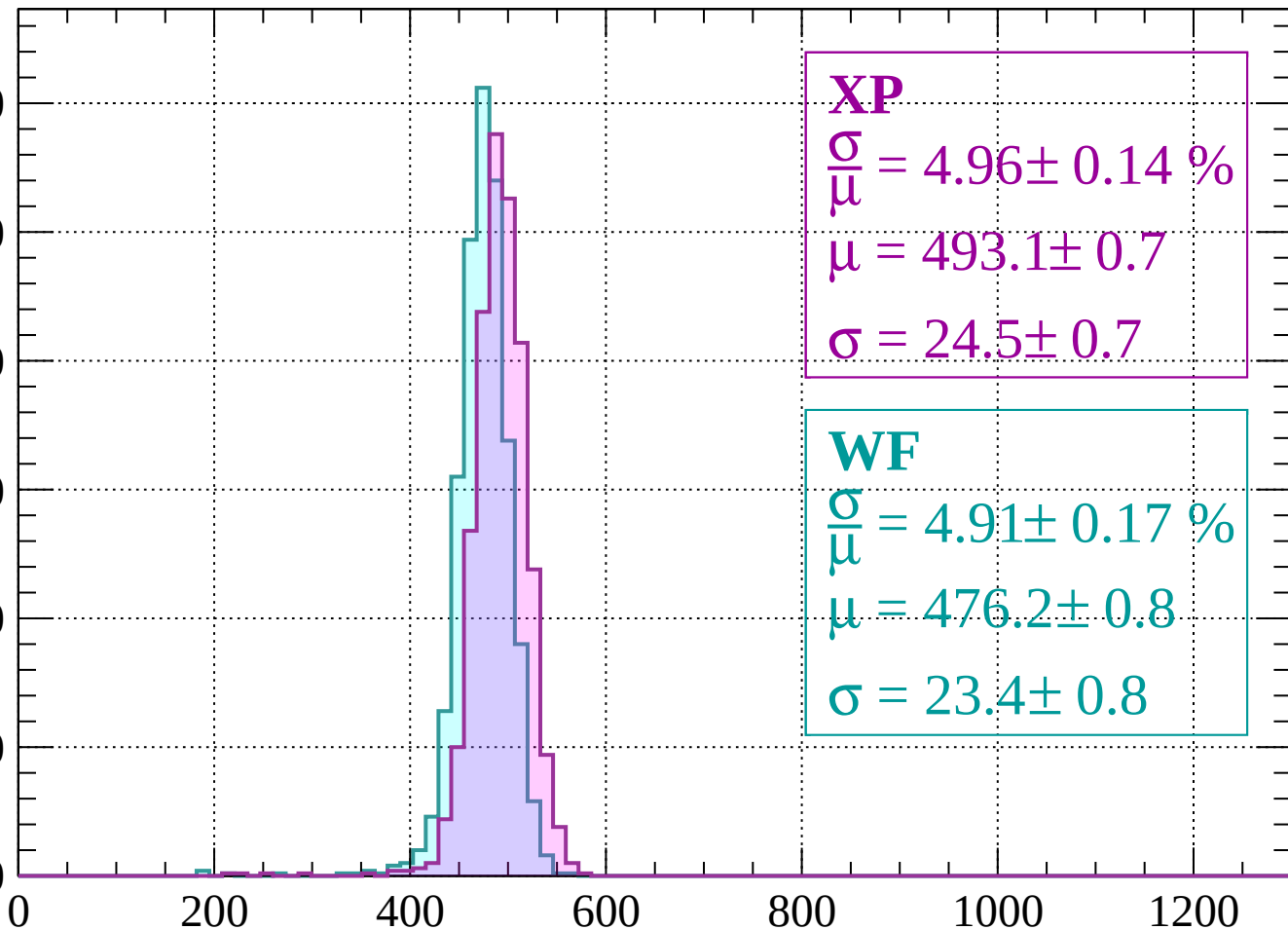
$$\sigma = 22.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1960 < p < 2000$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.14 \%$$

$$\mu = 493.1 \pm 0.7$$

$$\sigma = 24.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.91 \pm 0.17 \%$$

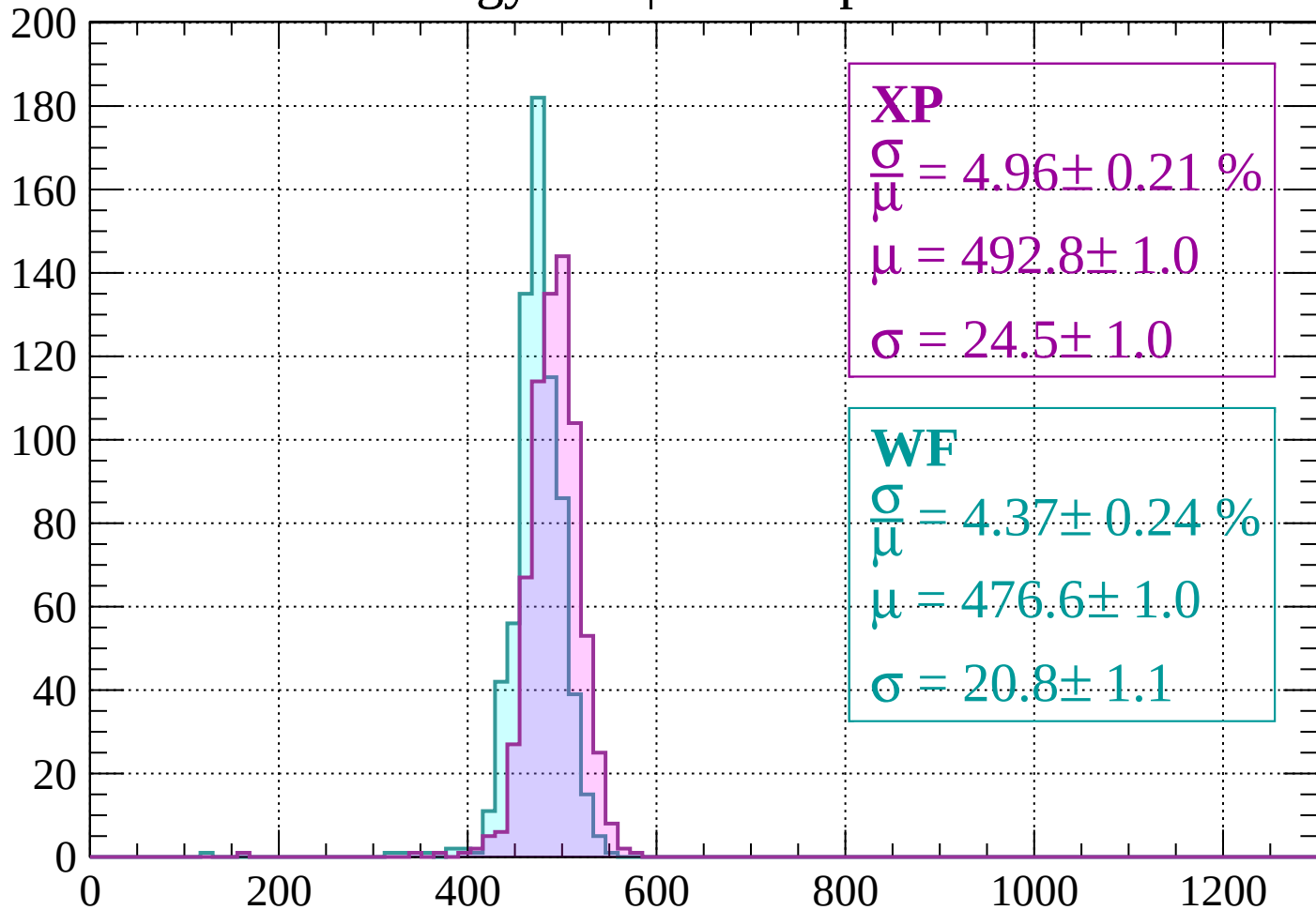
$$\mu = 476.2 \pm 0.8$$

$$\sigma = 23.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$

Count



XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.21 \%$$

$$\mu = 492.8 \pm 1.0$$

$$\sigma = 24.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.37 \pm 0.24 \%$$

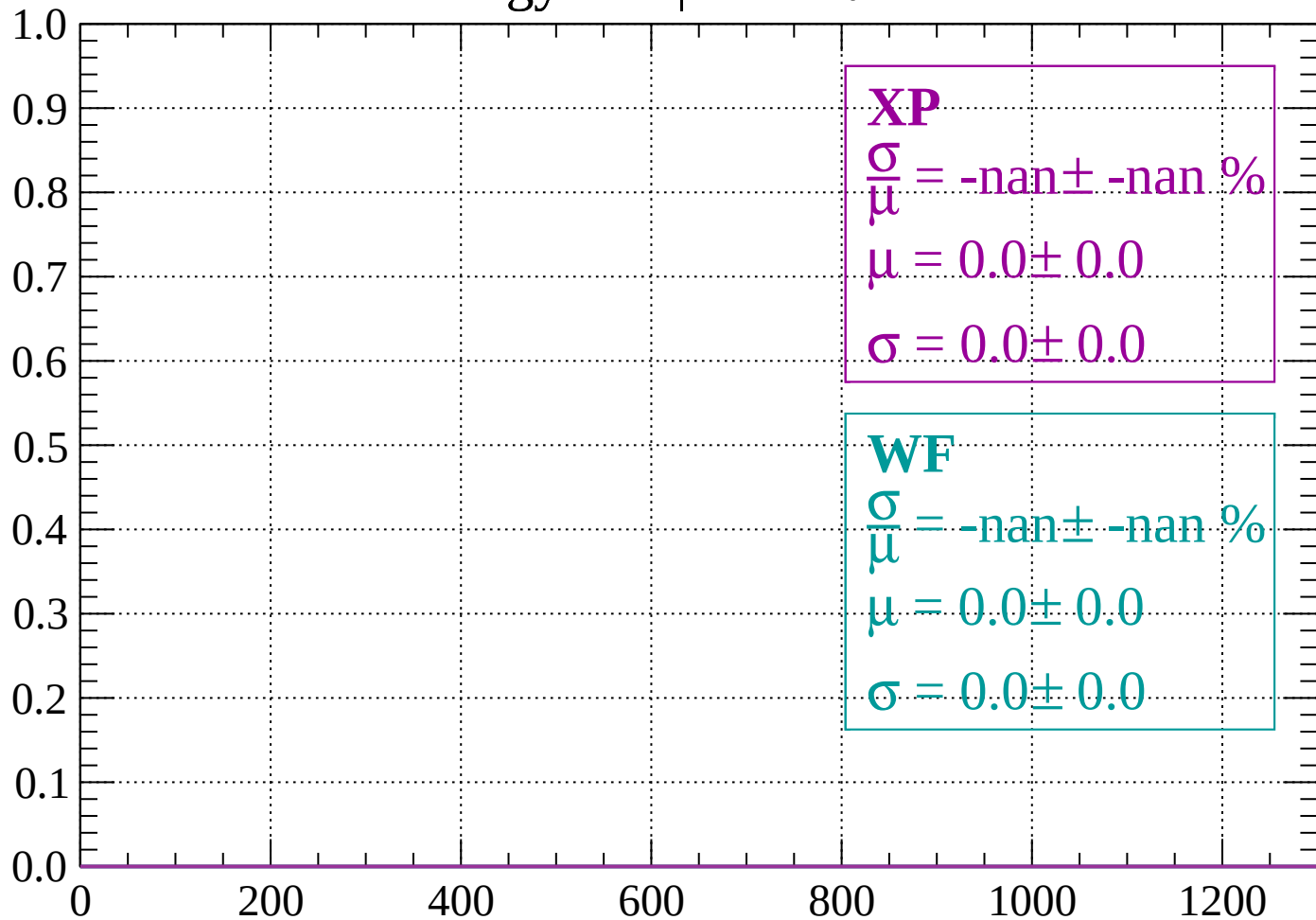
$$\mu = 476.6 \pm 1.0$$

$$\sigma = 20.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

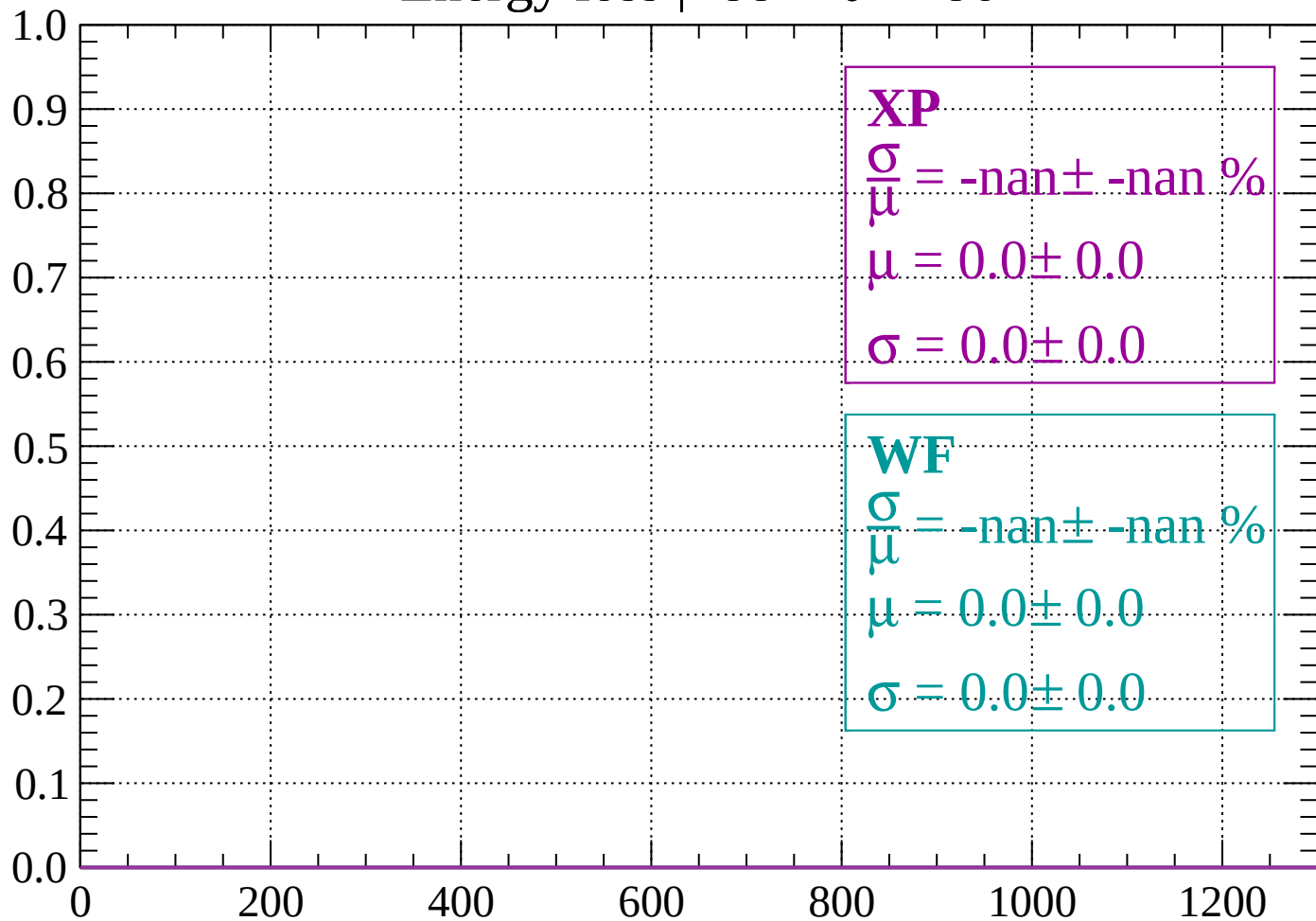
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

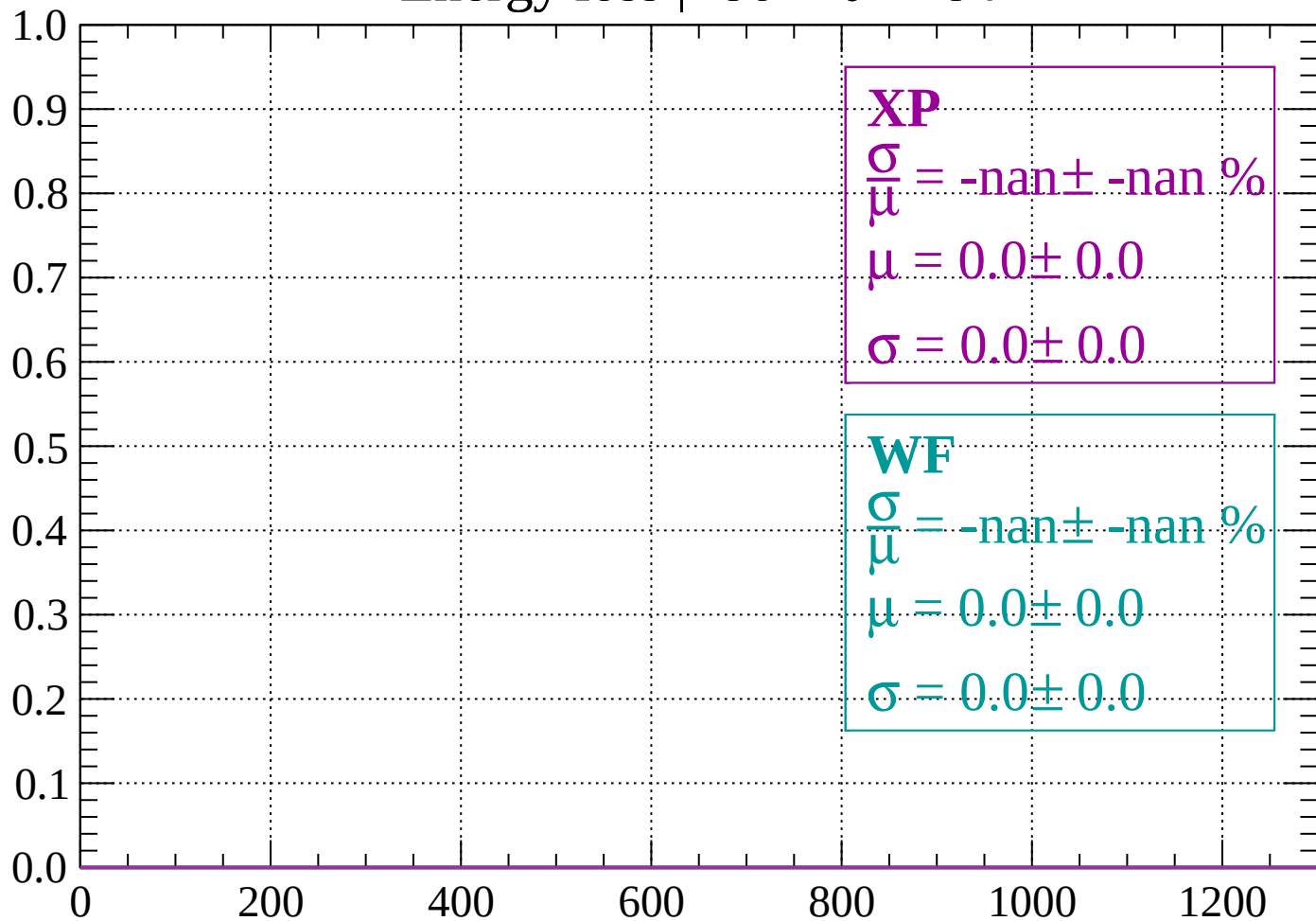
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

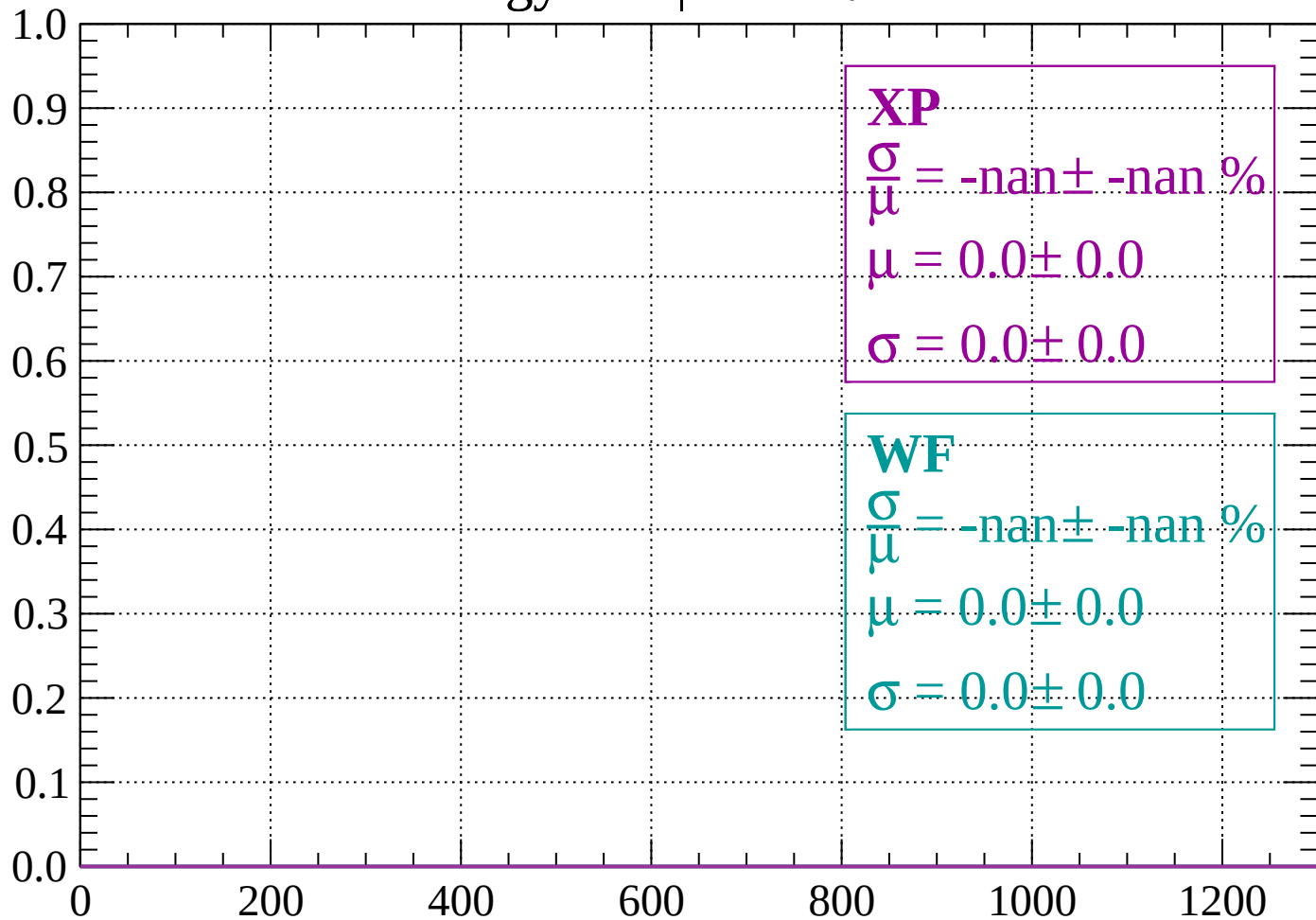
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

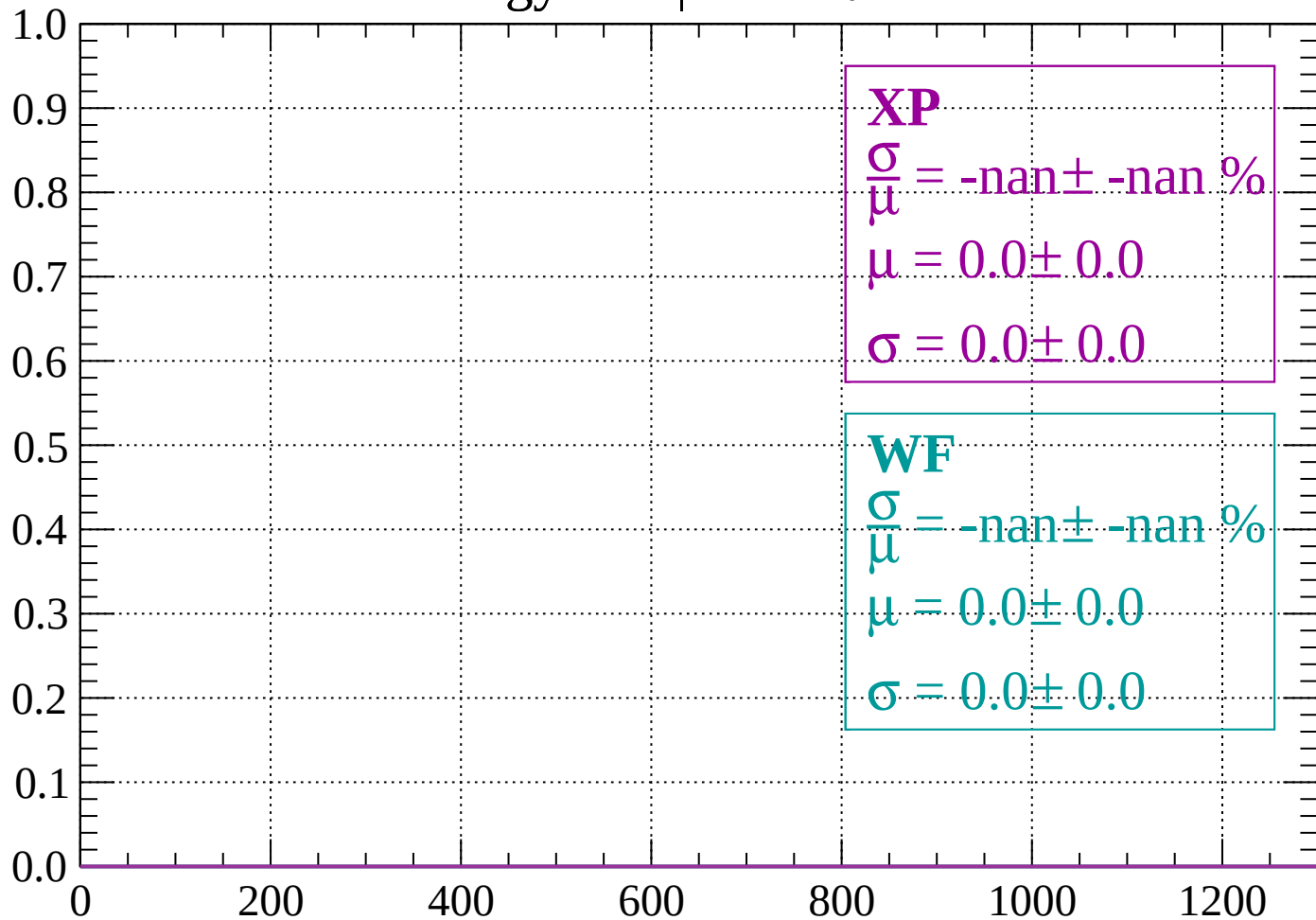
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

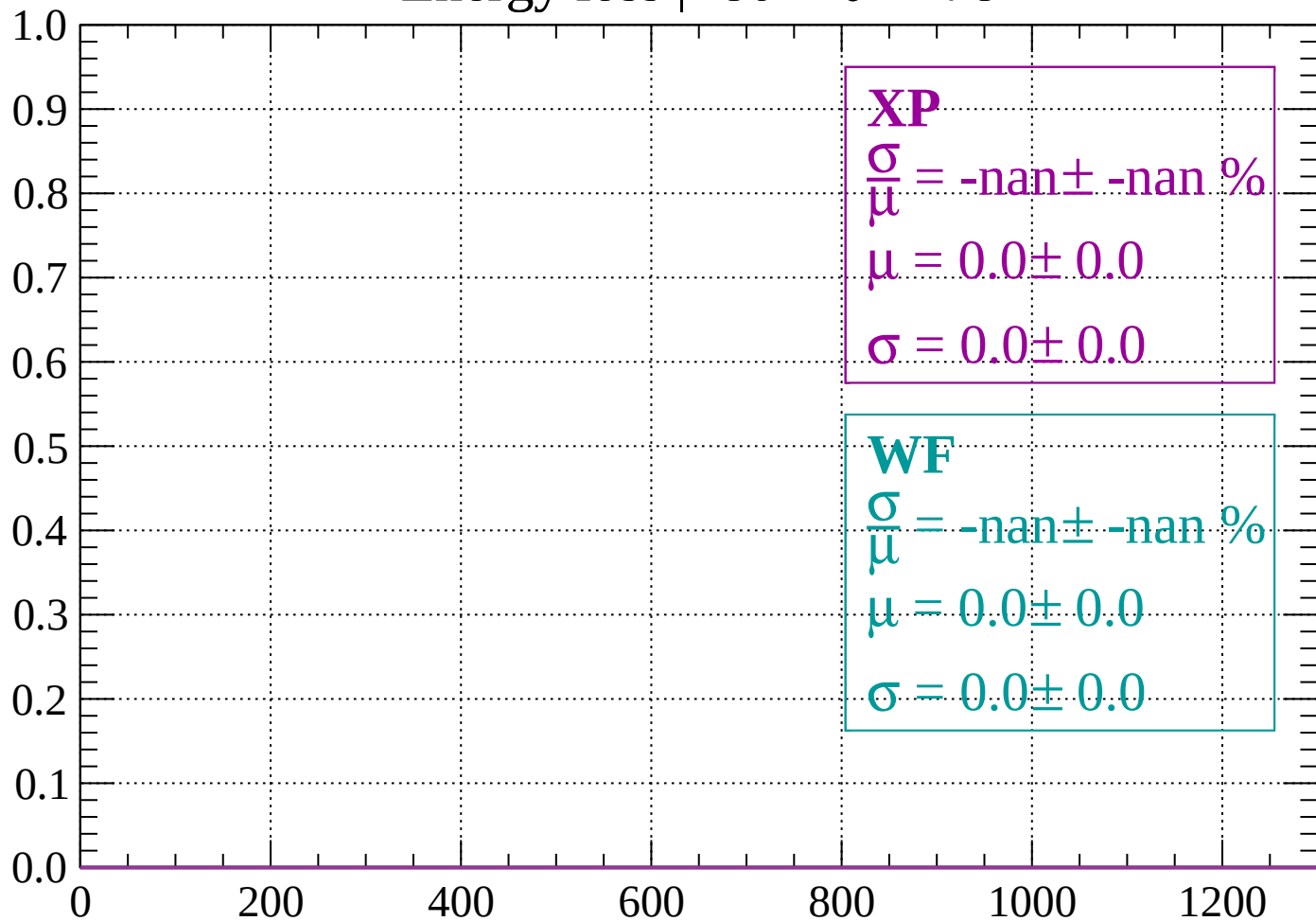
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

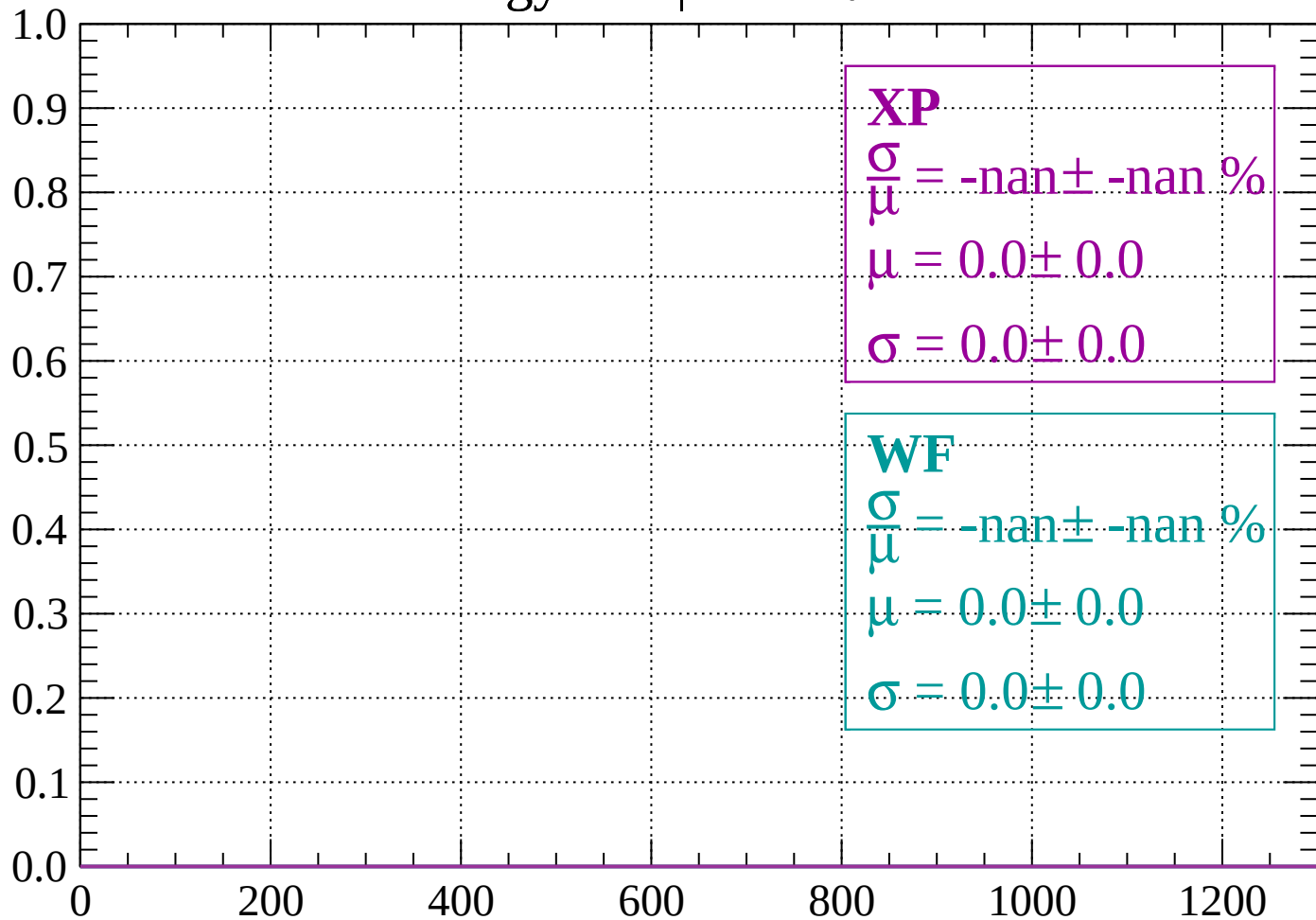
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

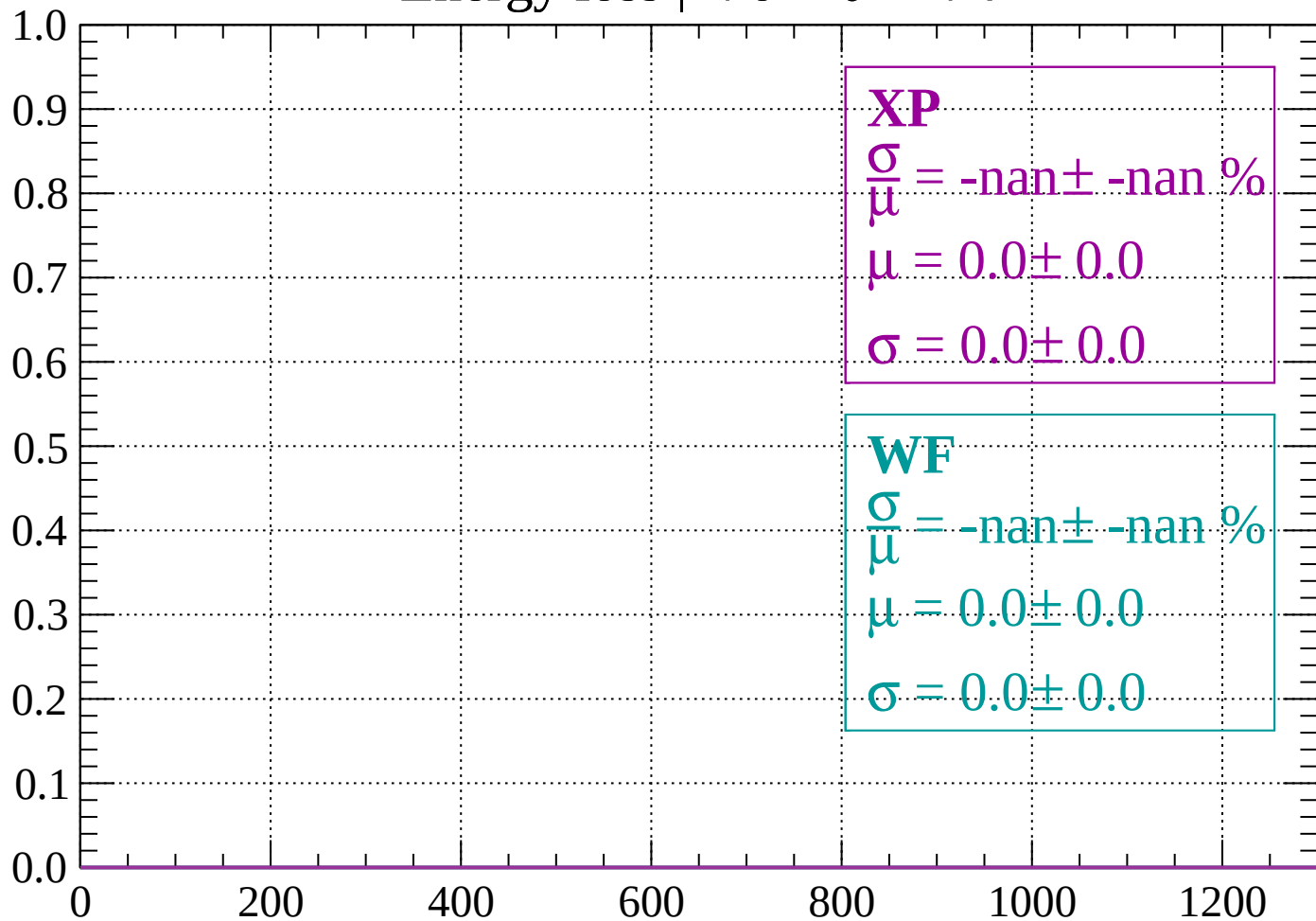
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

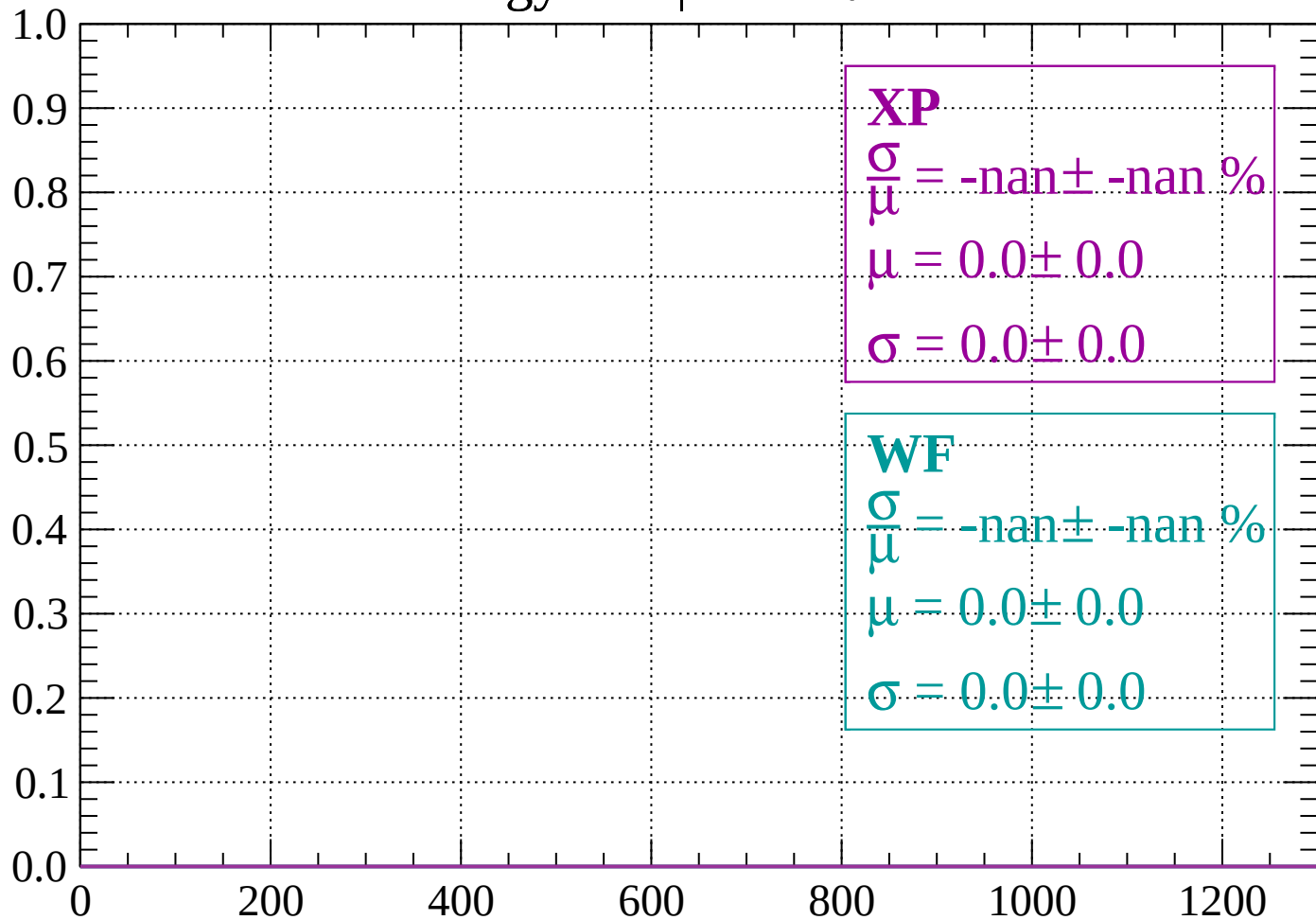
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

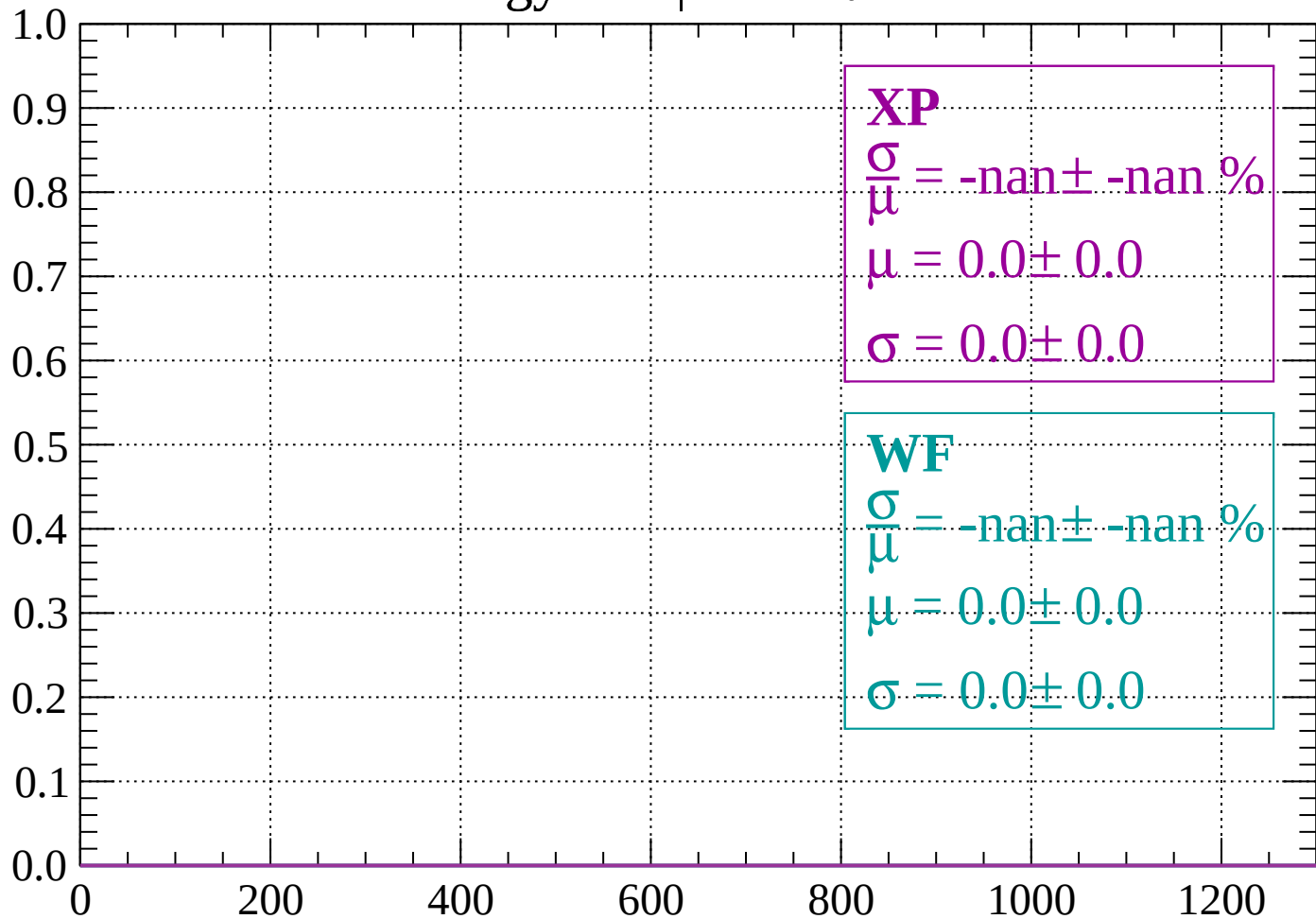
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

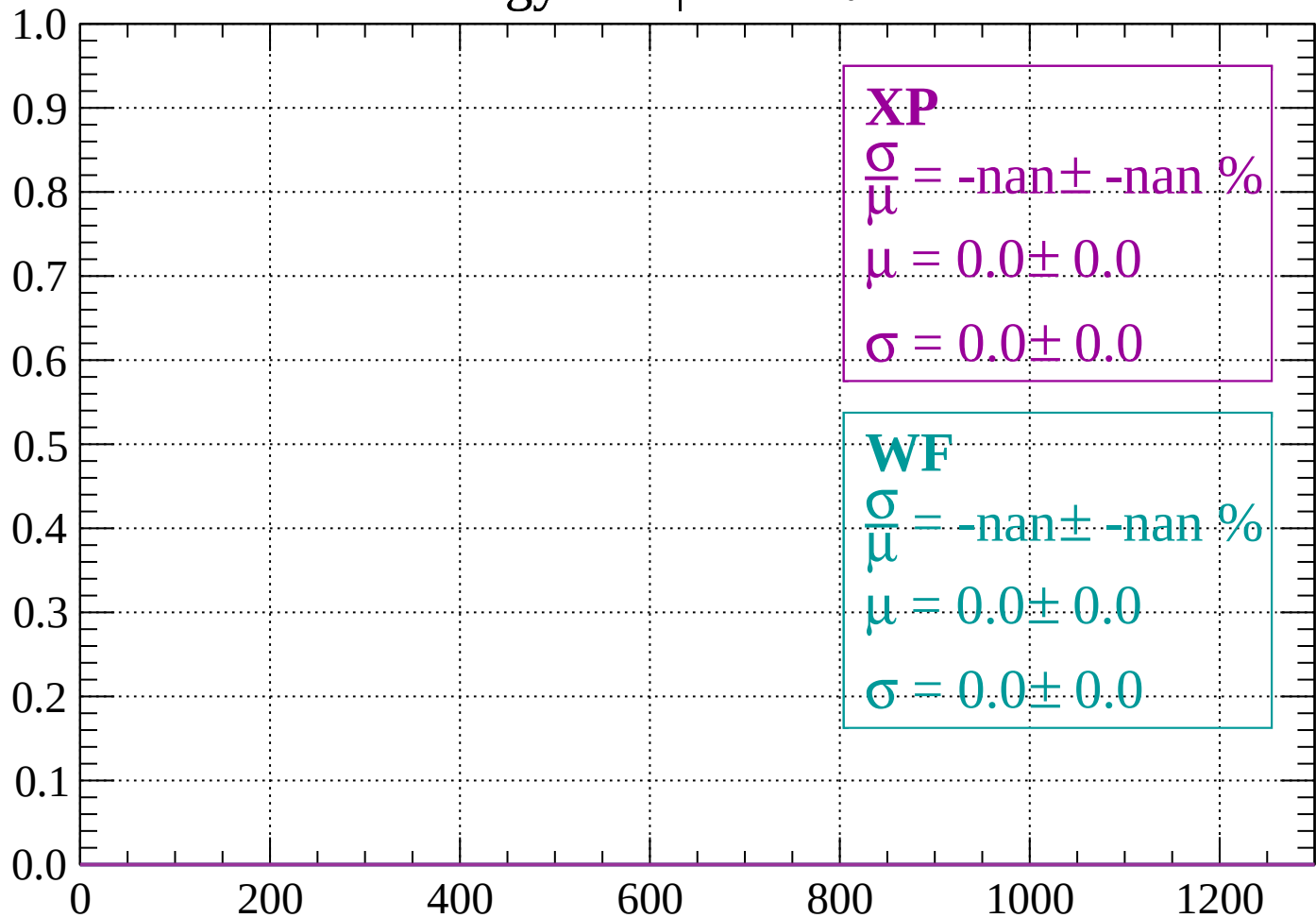
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

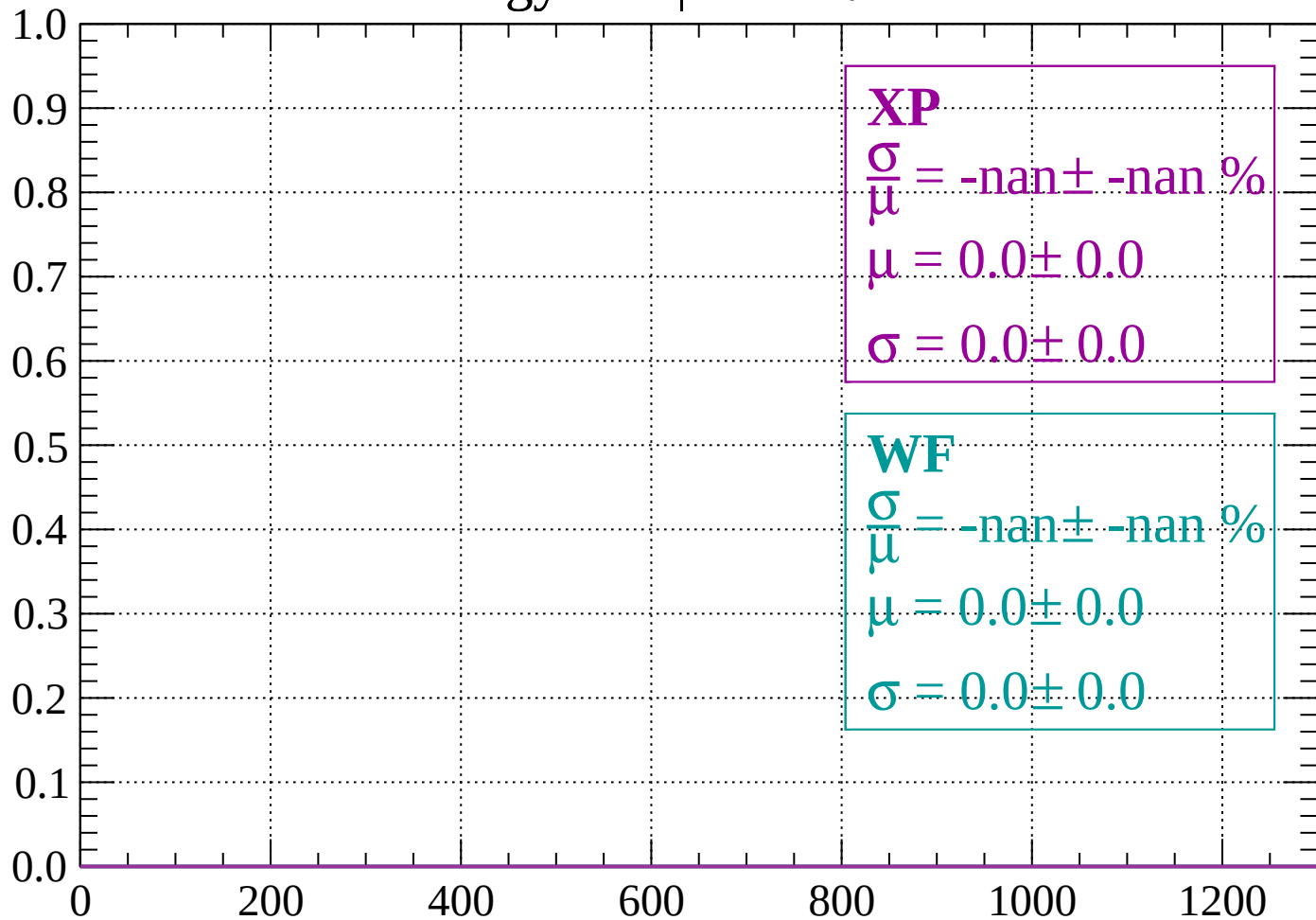
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

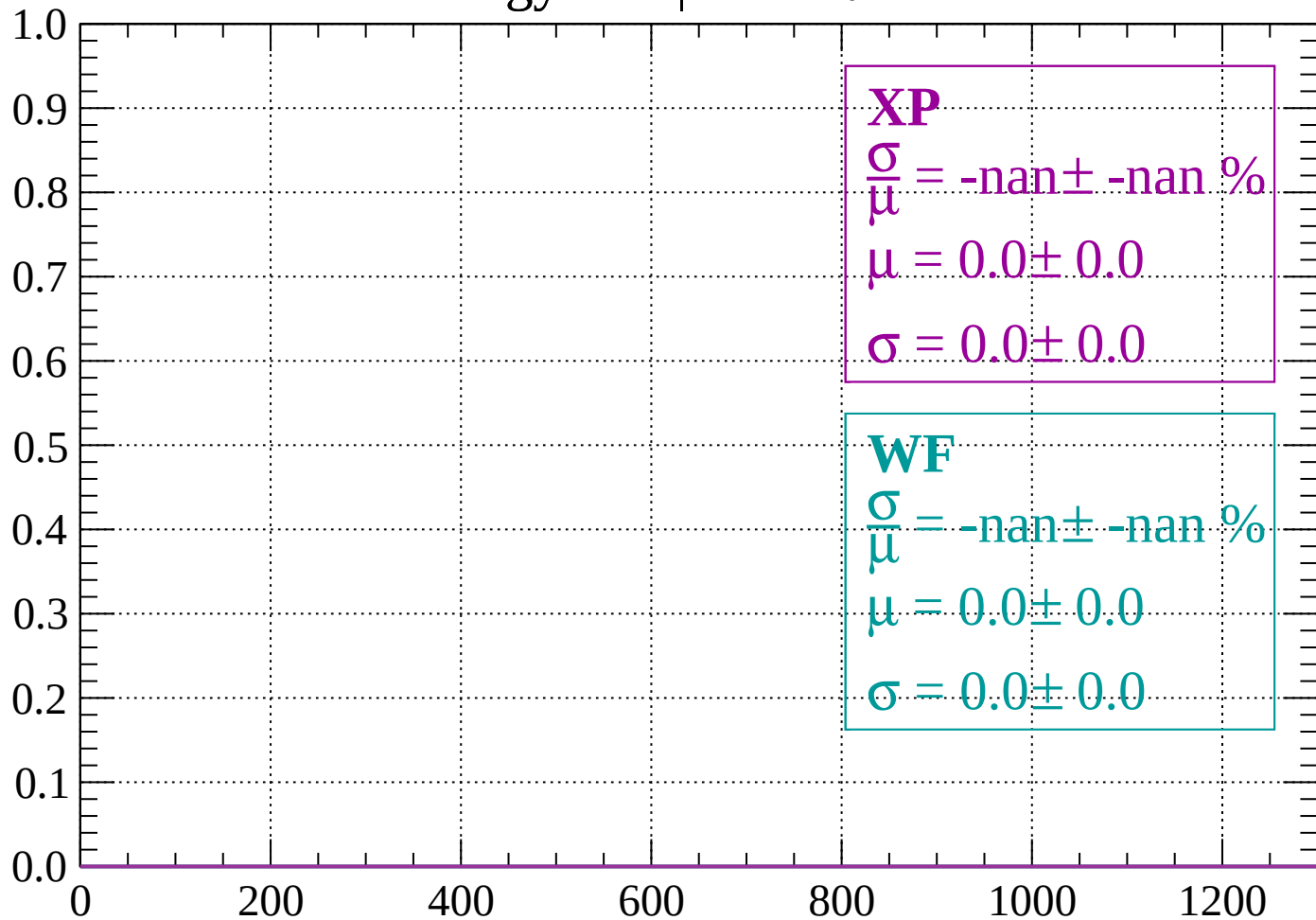
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

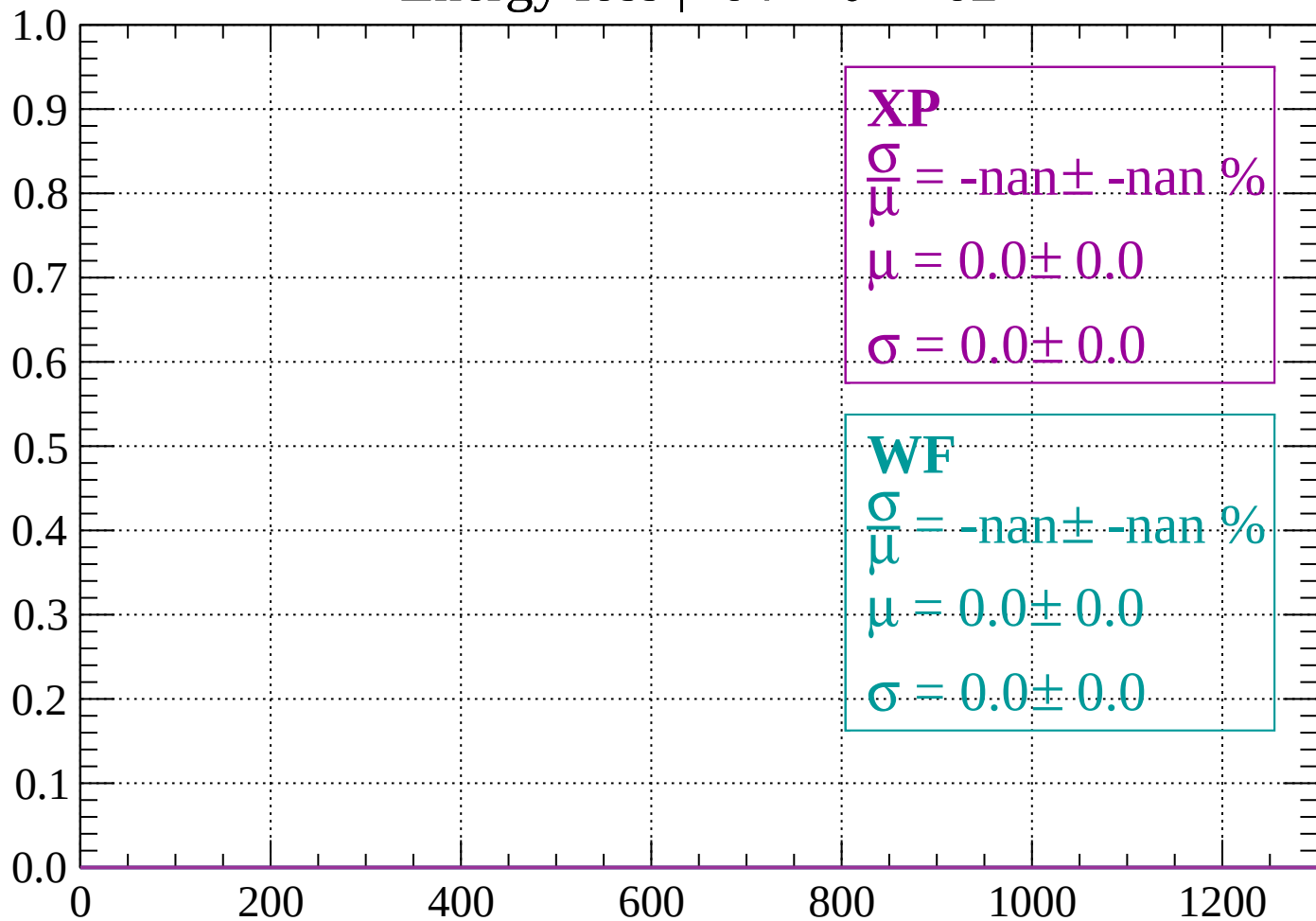
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

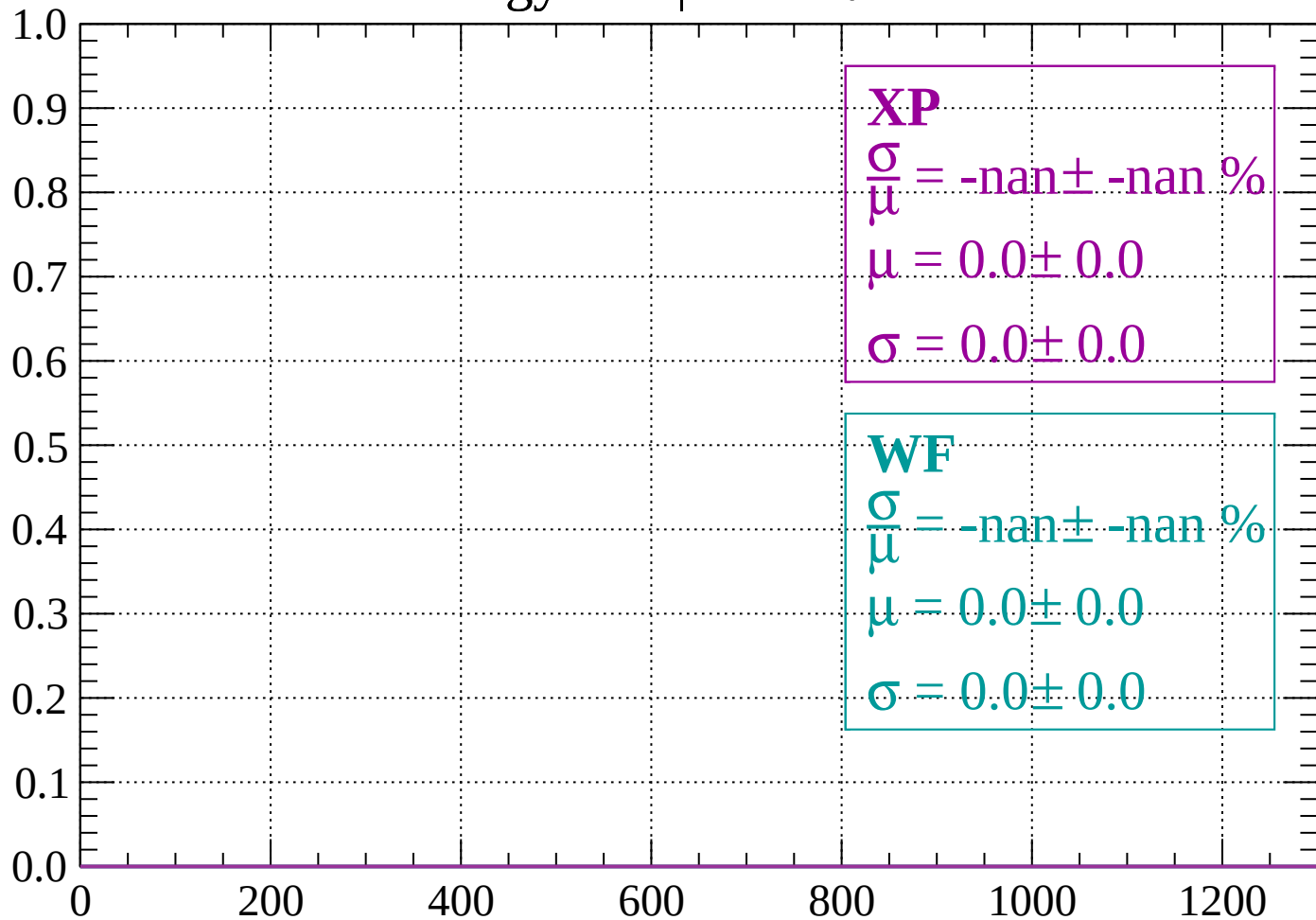
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

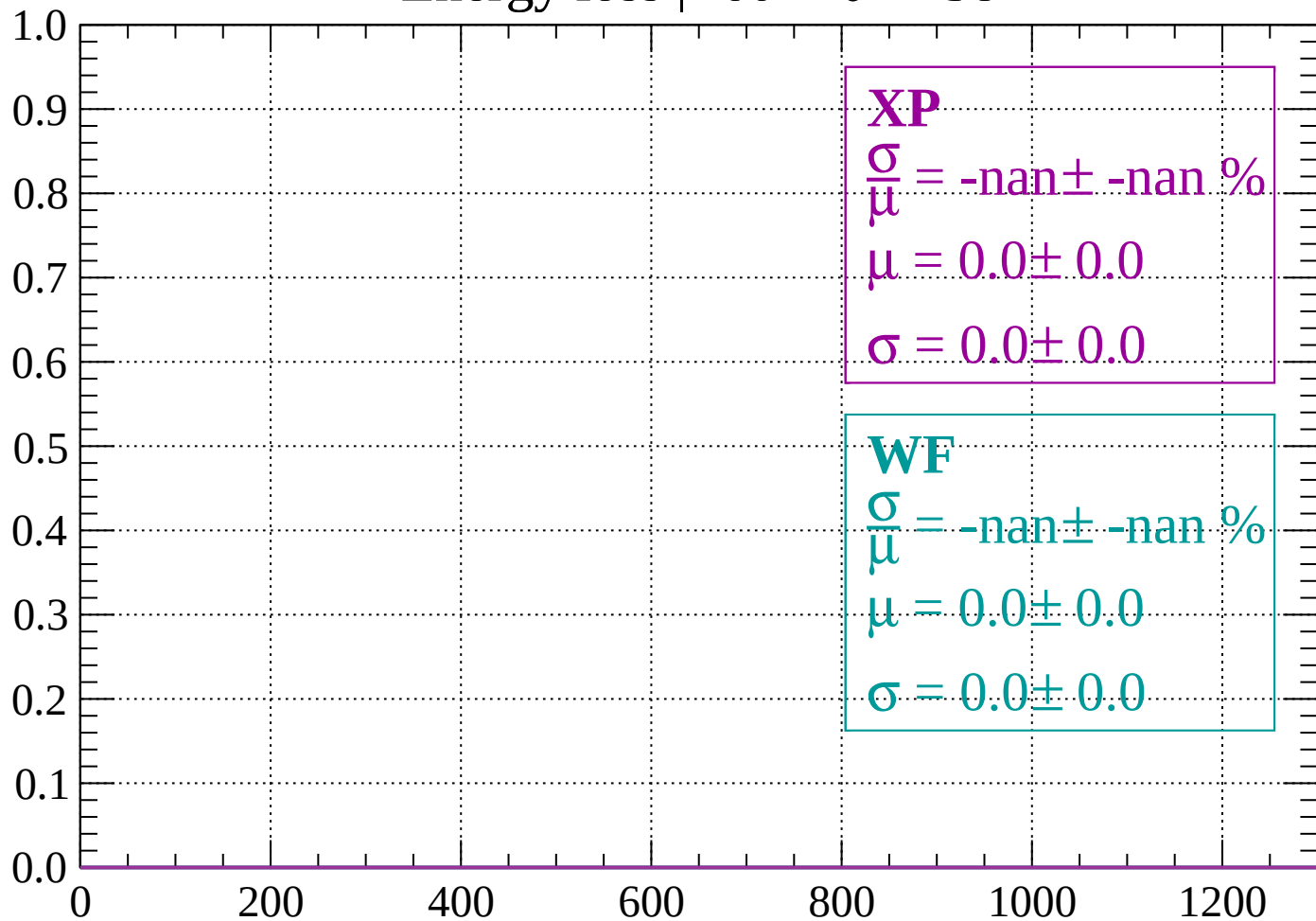
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

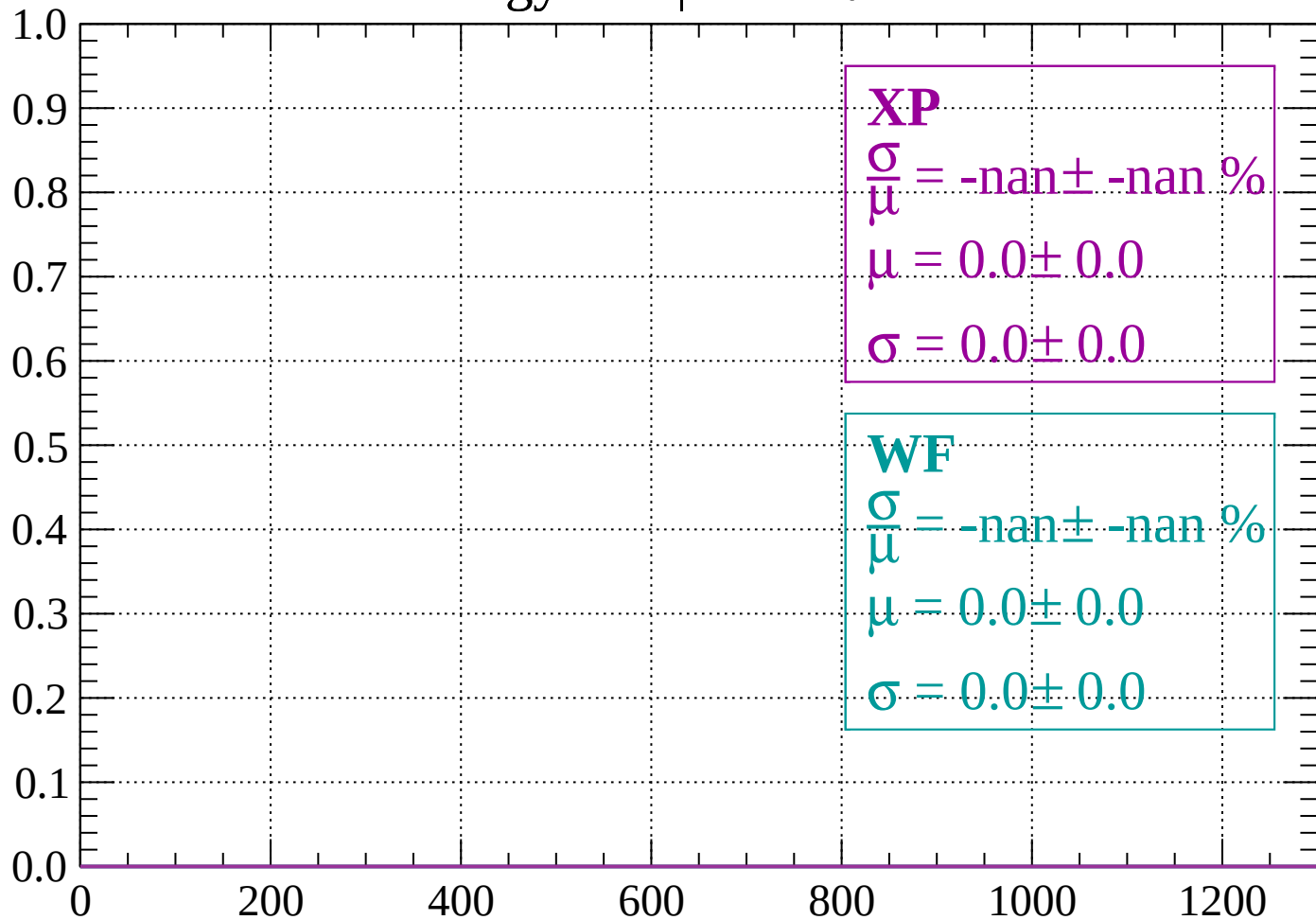
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

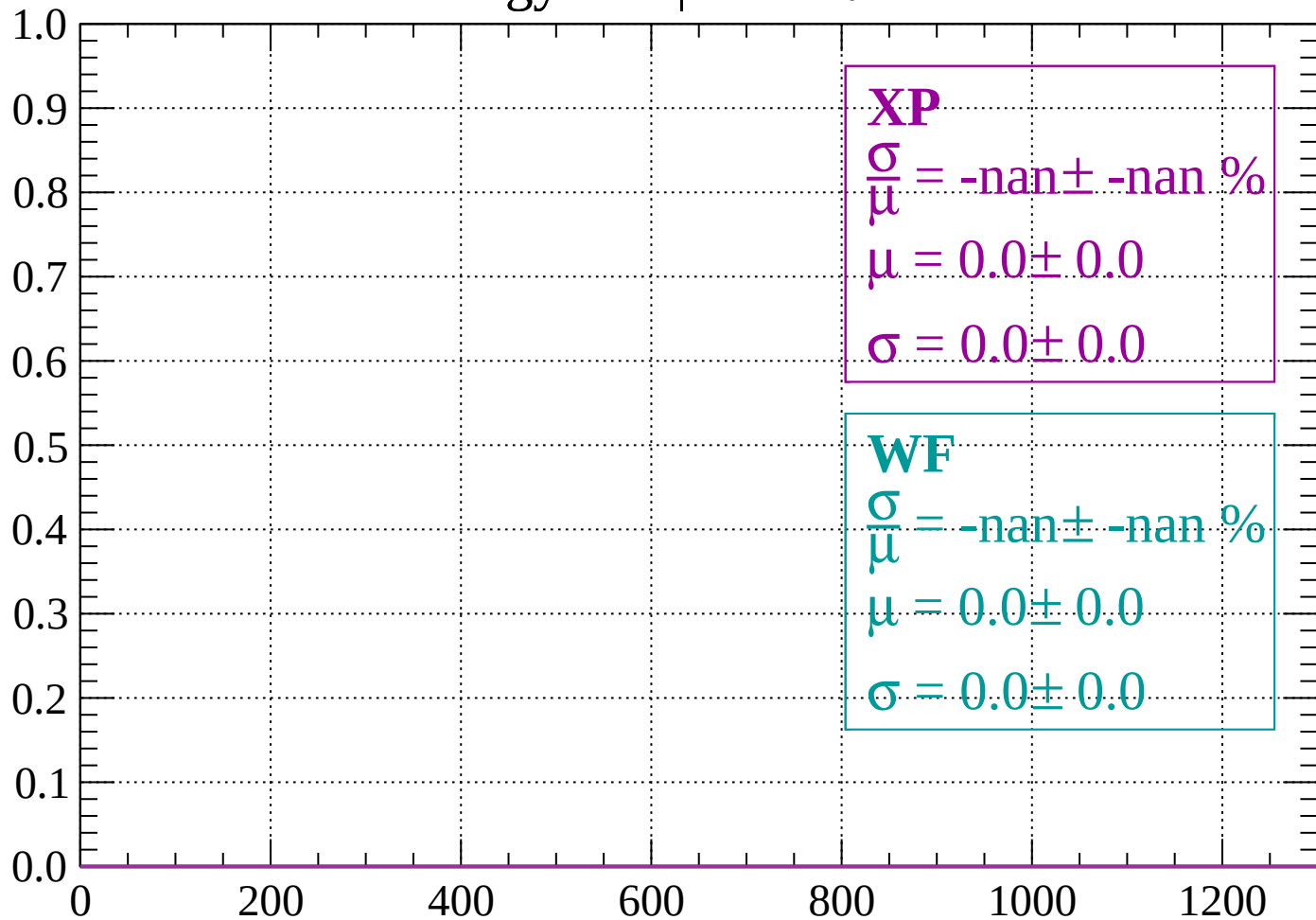
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

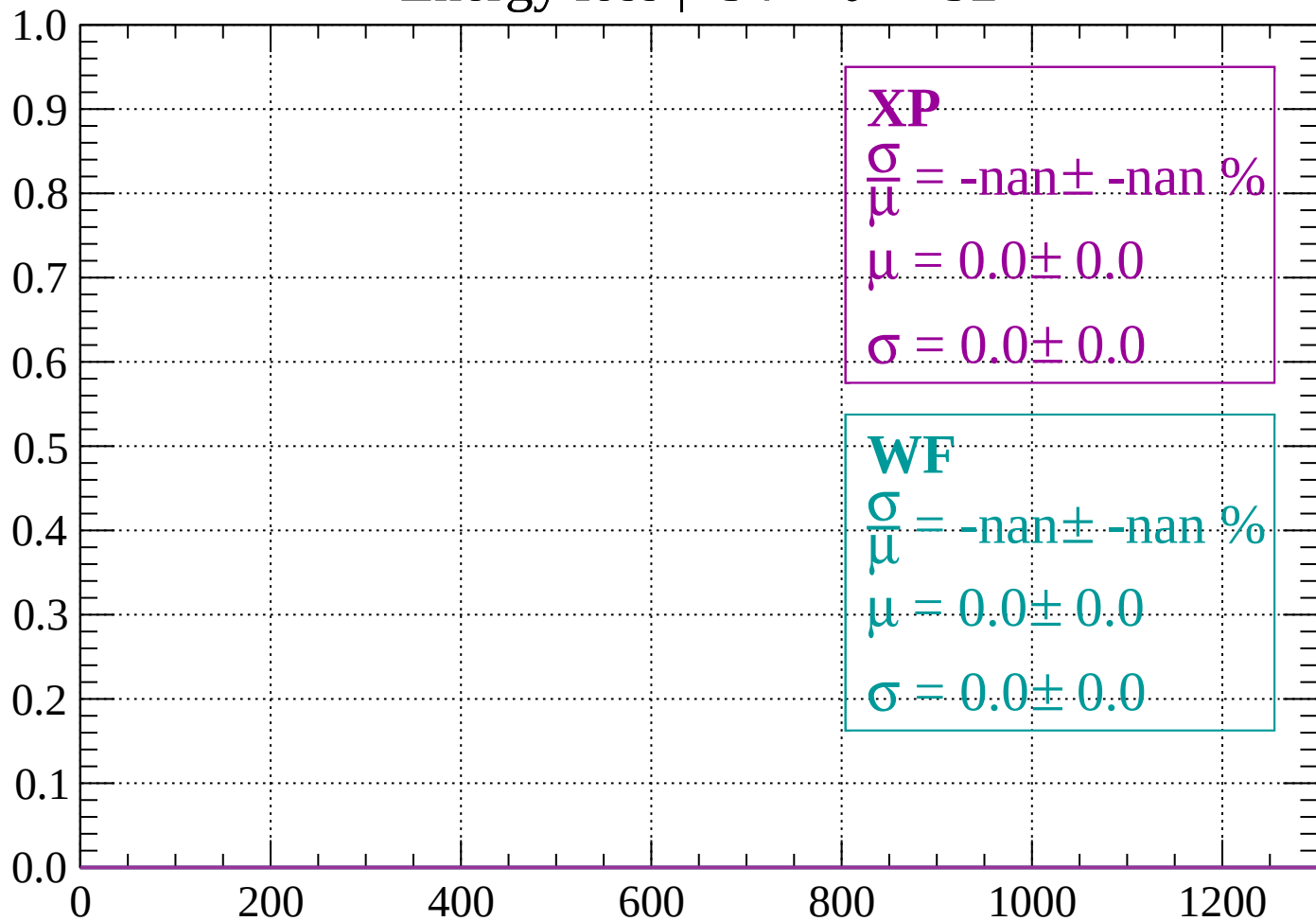
Count



dE/dx (ADC counts/cm)

Energy loss | $-54 < \theta < -52$

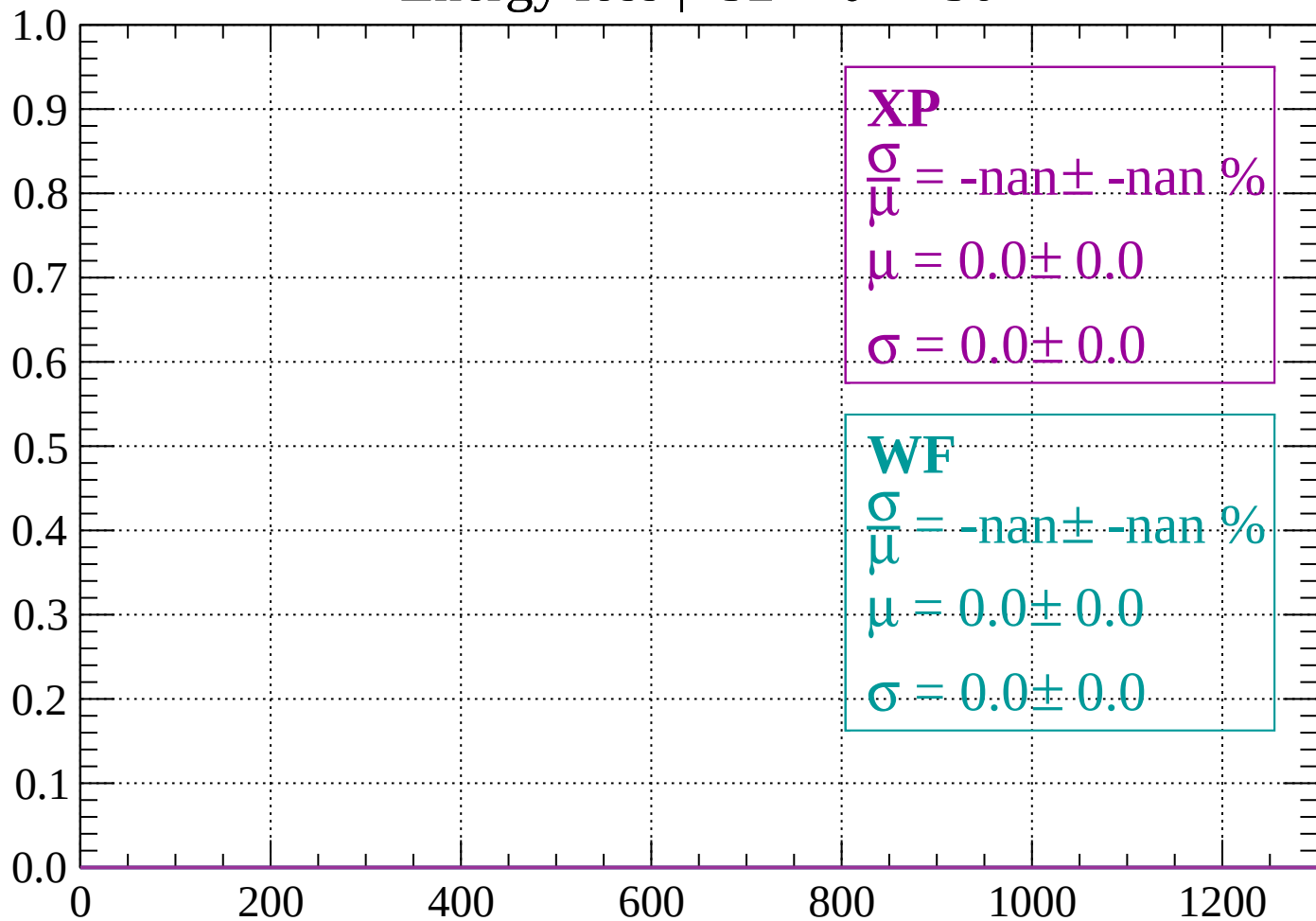
Count



dE/dx (ADC counts/cm)

Energy loss | $-52 < \theta < -50$

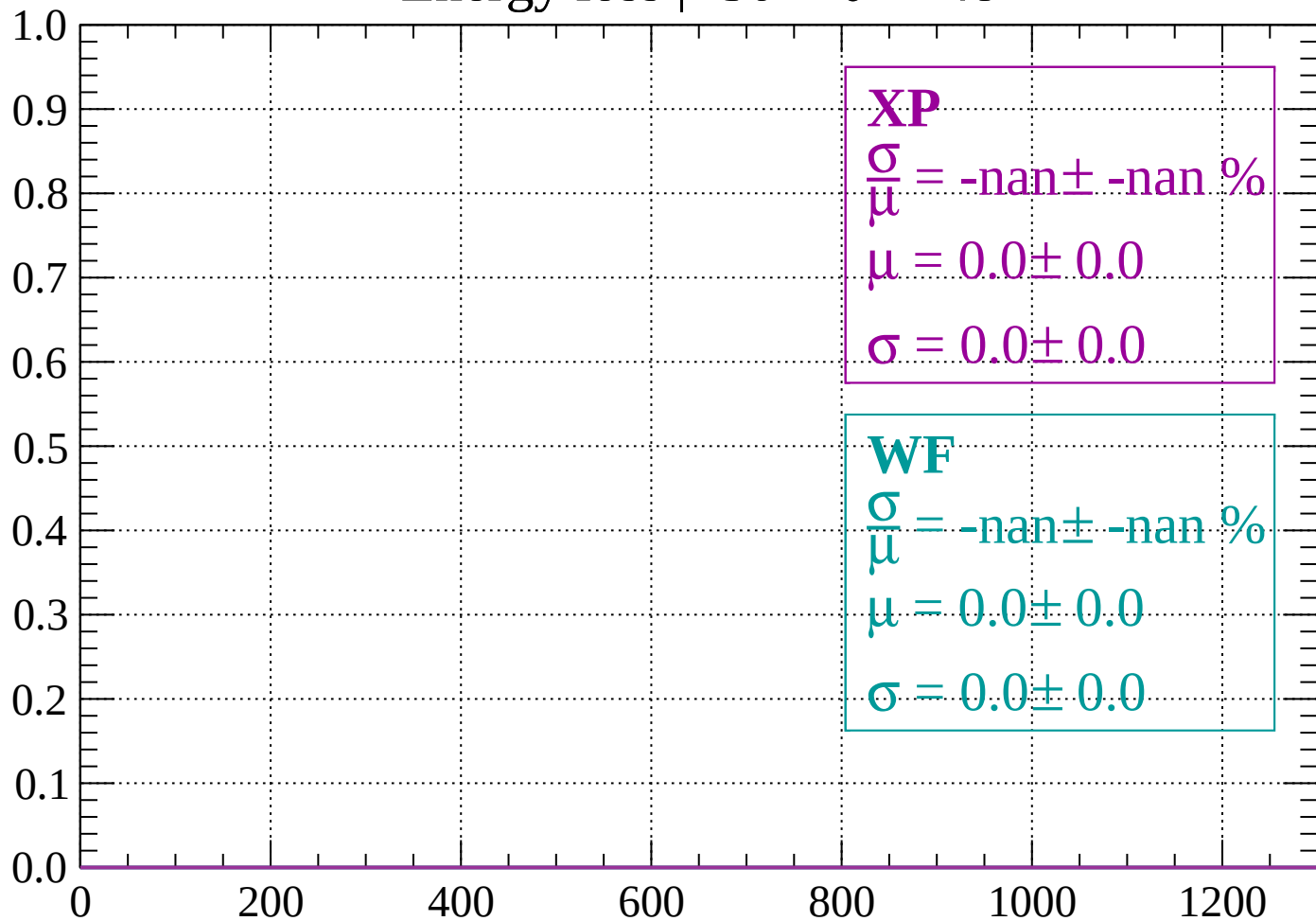
Count



dE/dx (ADC counts/cm)

Energy loss | $-50 < \theta < -48$

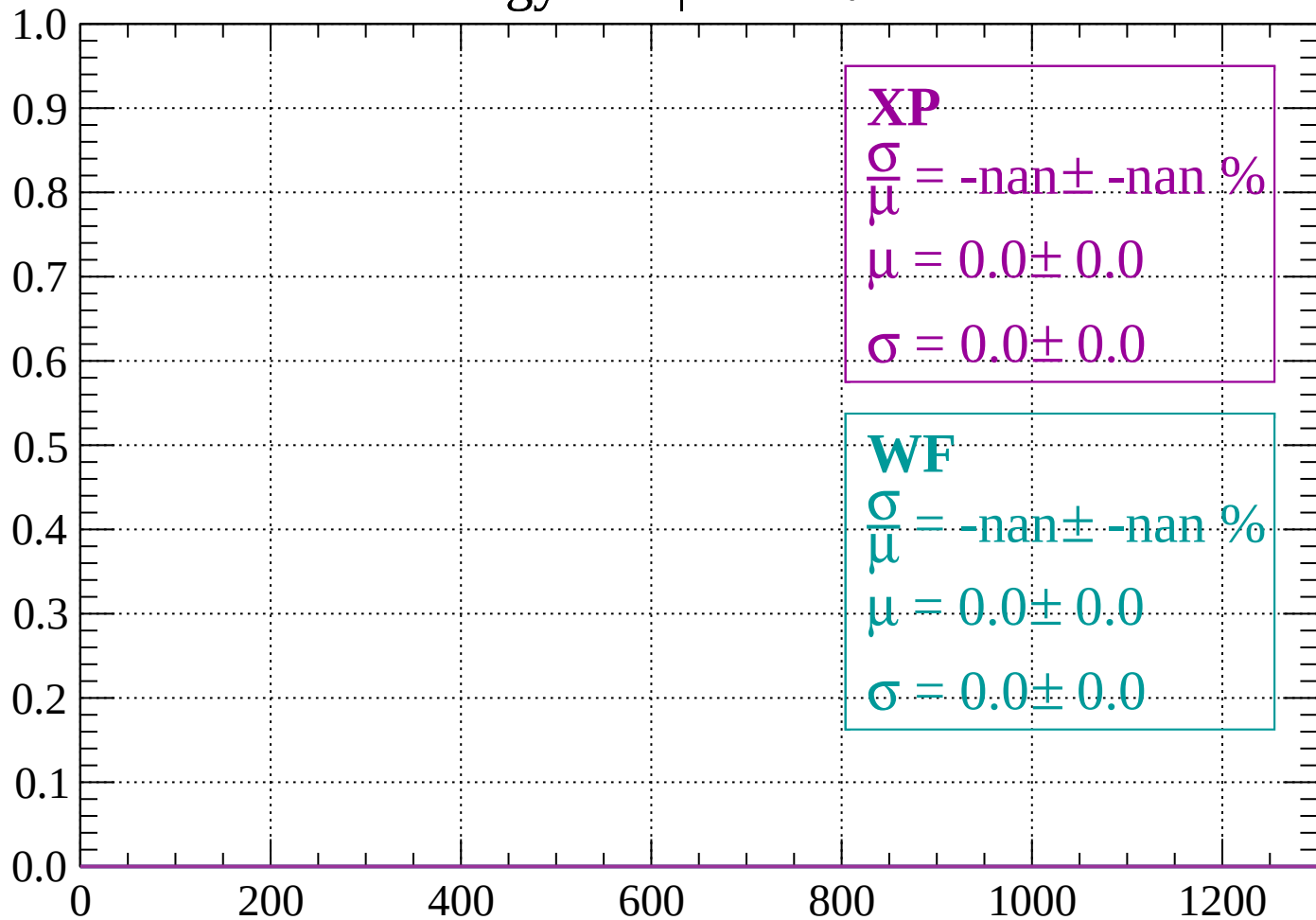
Count



dE/dx (ADC counts/cm)

Energy loss | $-48 < \theta < -46$

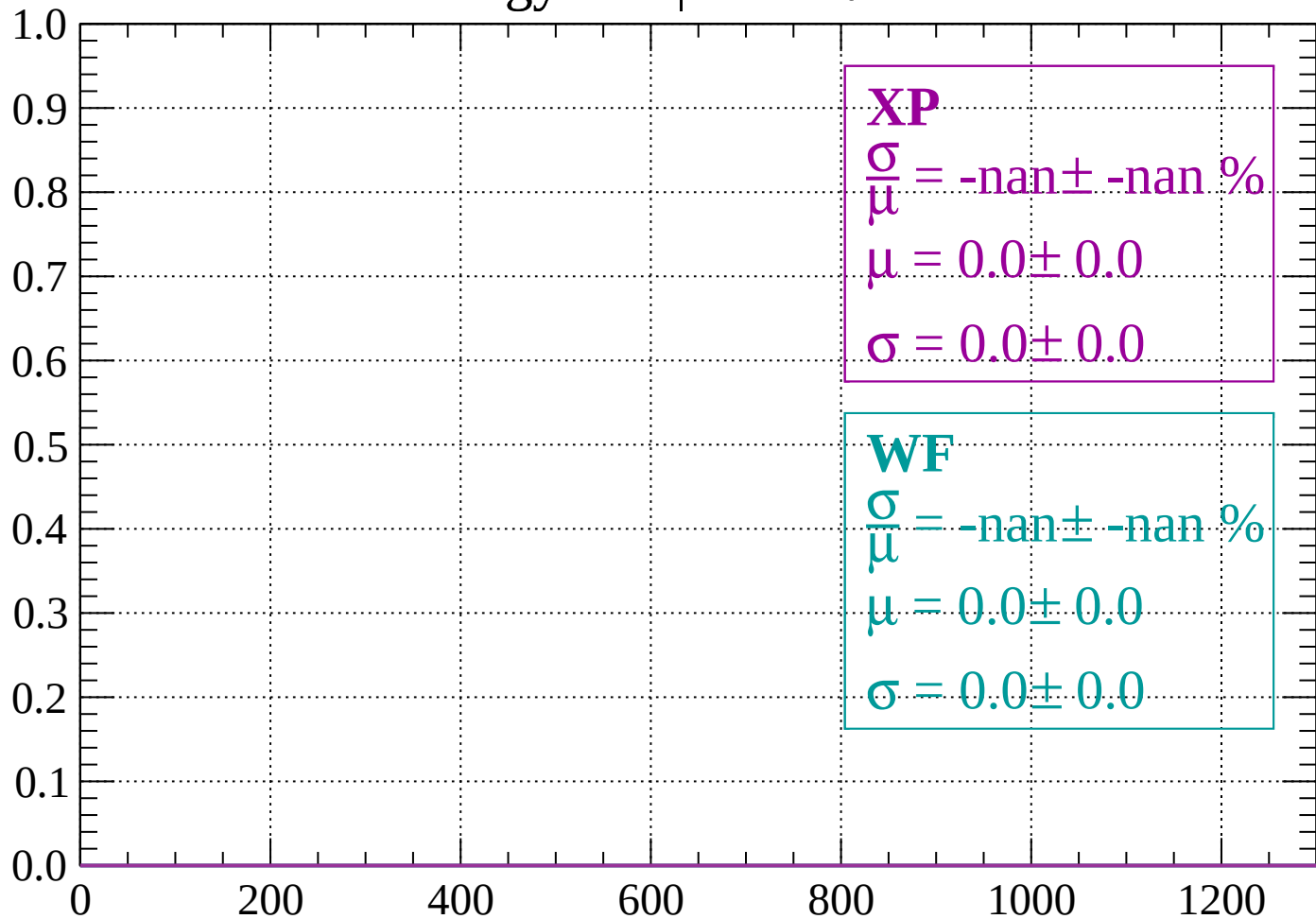
Count



dE/dx (ADC counts/cm)

Energy loss | $-46 < \theta < -44$

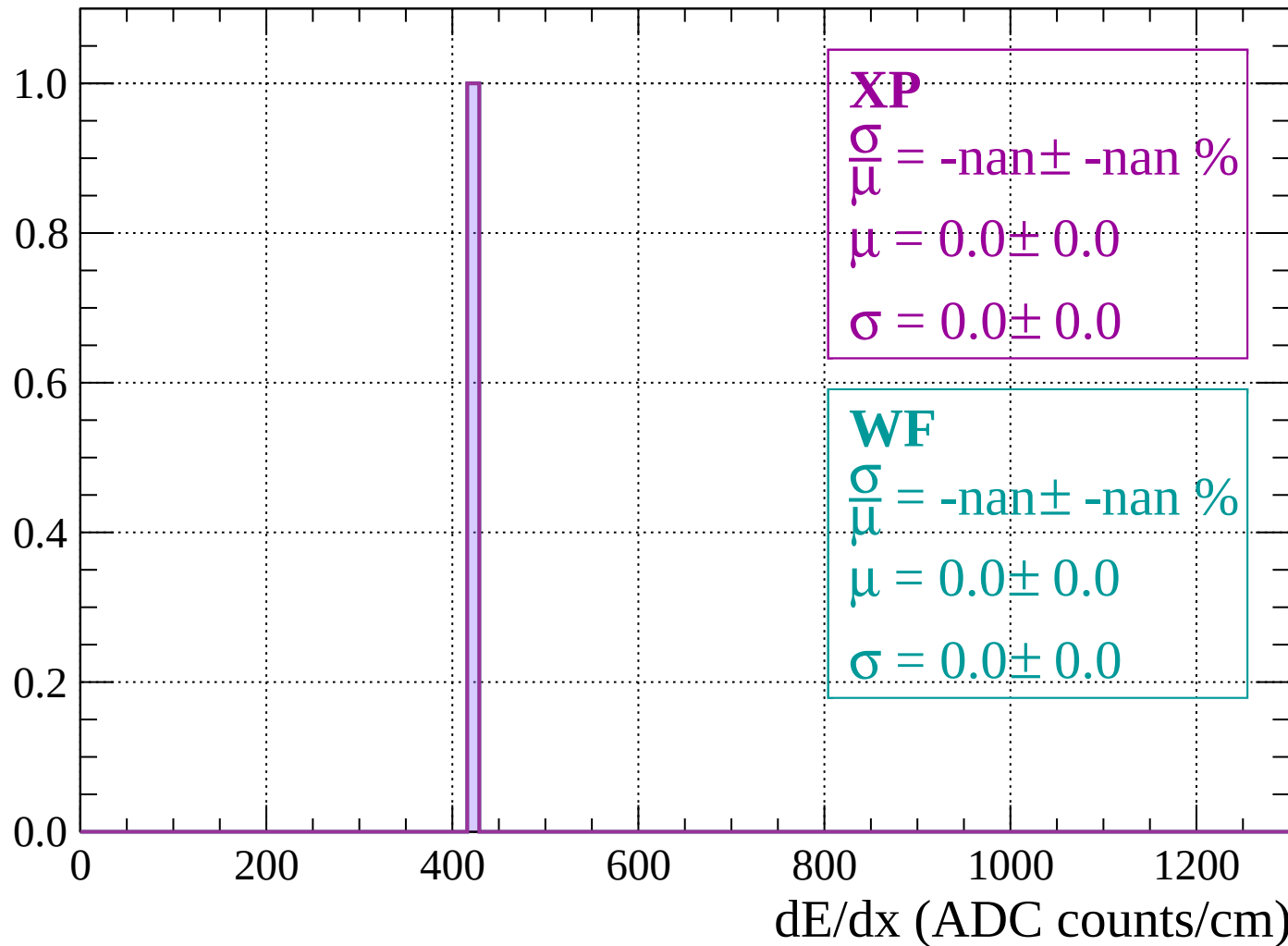
Count



dE/dx (ADC counts/cm)

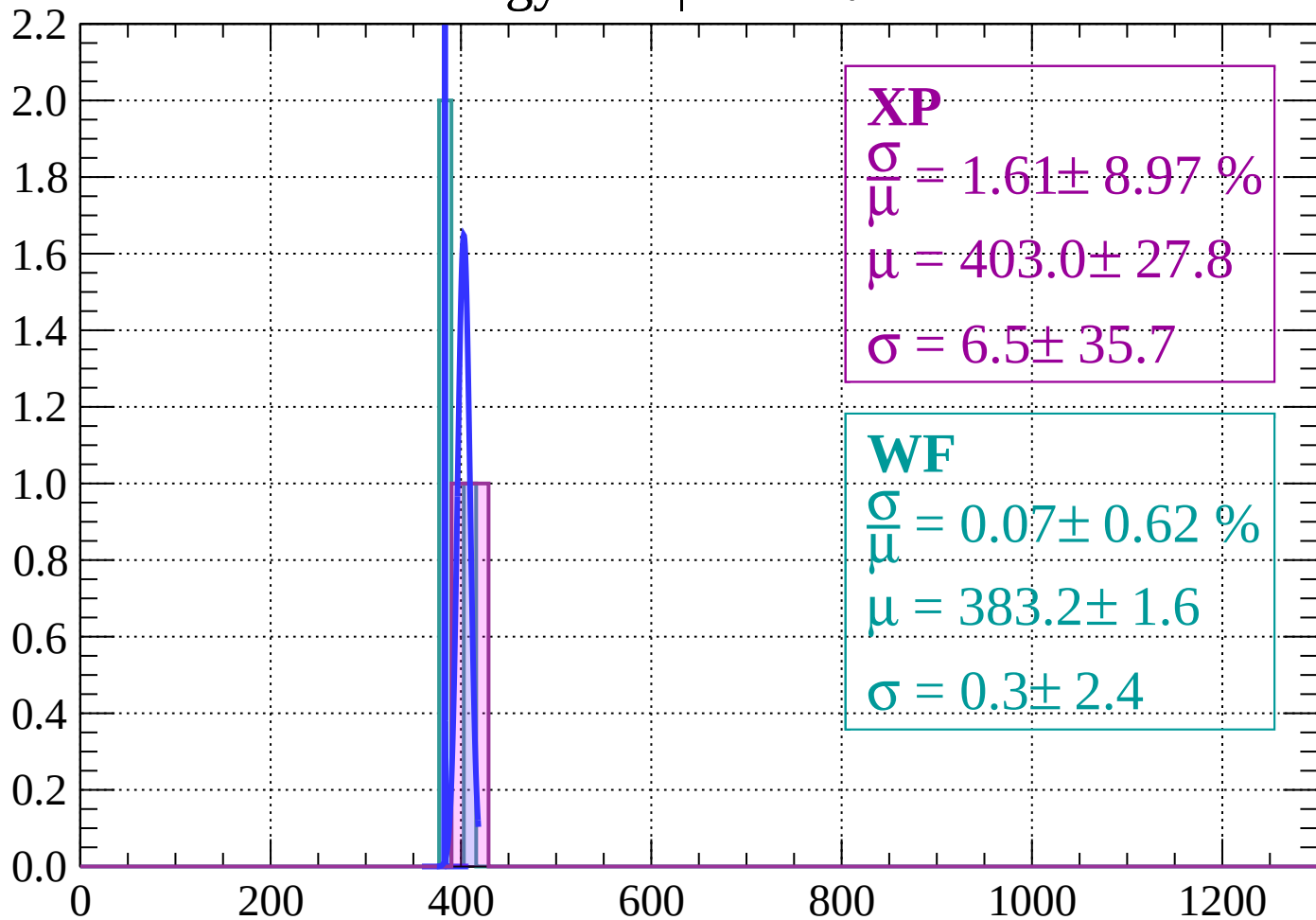
Energy loss | $-44 < \theta < -42$

Count



Energy loss | $-42 < \theta < -40$

Count



XP

$$\frac{\sigma}{\mu} = 1.61 \pm 8.97 \%$$

$$\mu = 403.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

WF

$$\frac{\sigma}{\mu} = 0.07 \pm 0.62 \%$$

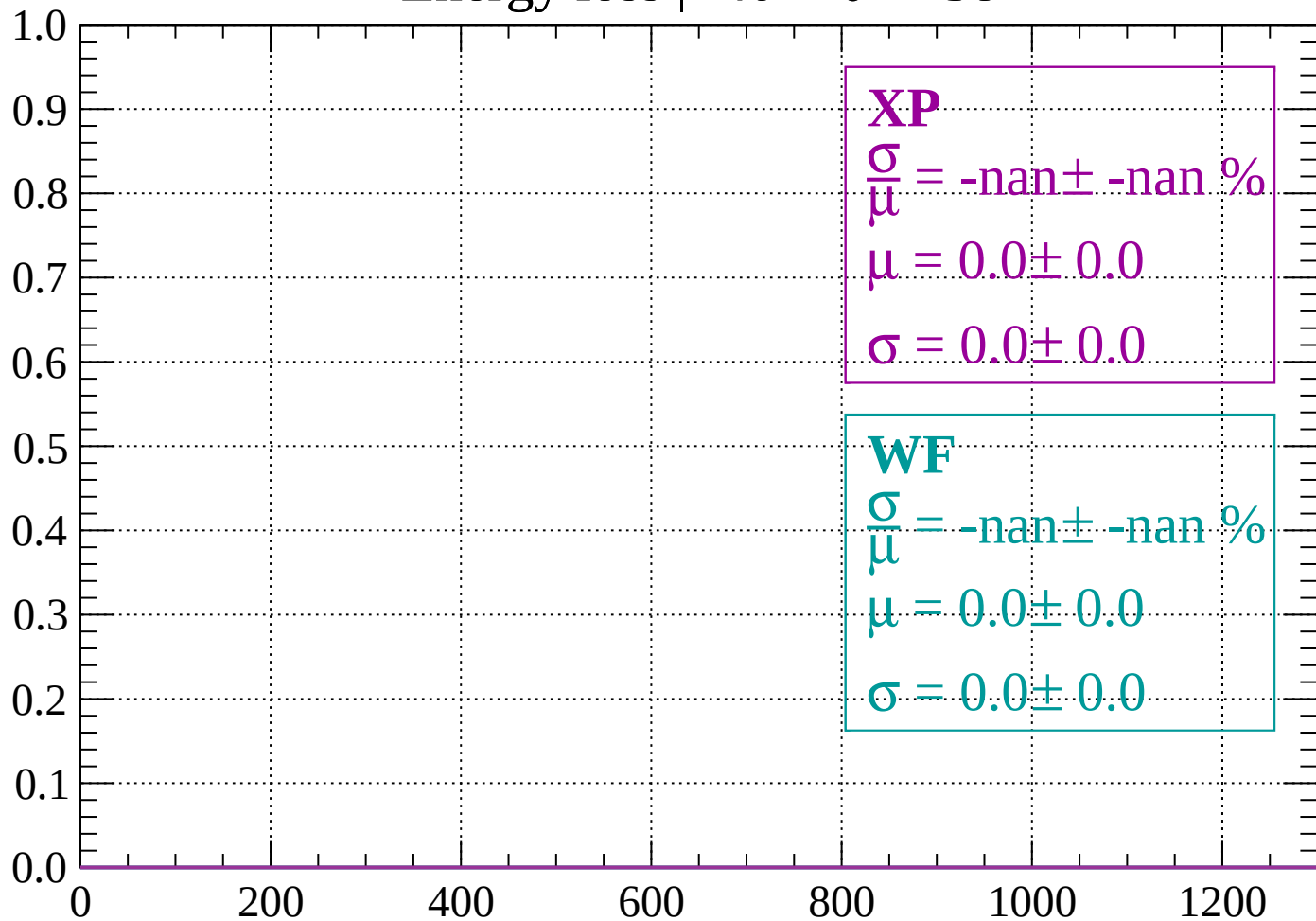
$$\mu = 383.2 \pm 1.6$$

$$\sigma = 0.3 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$

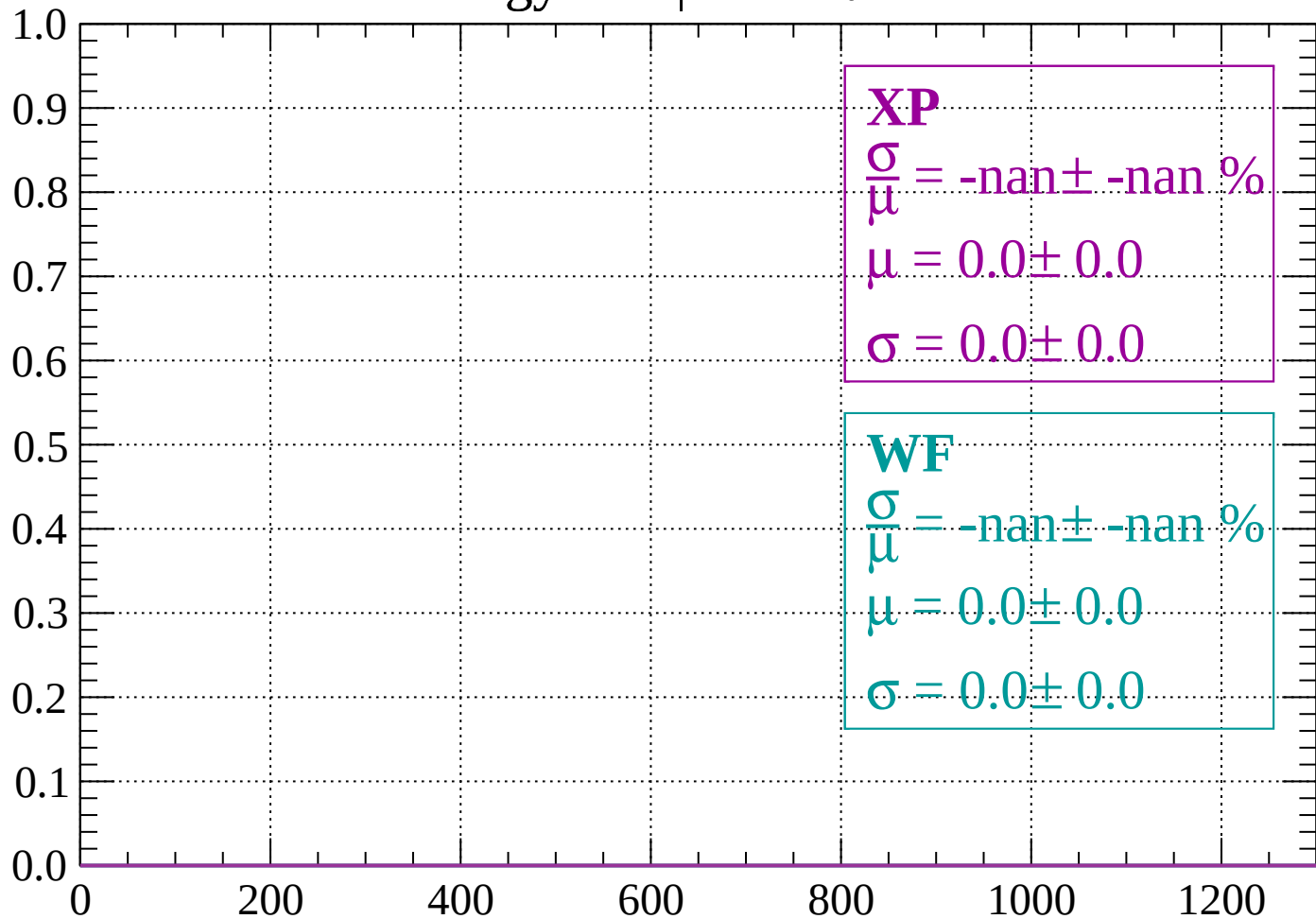
Count



dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$

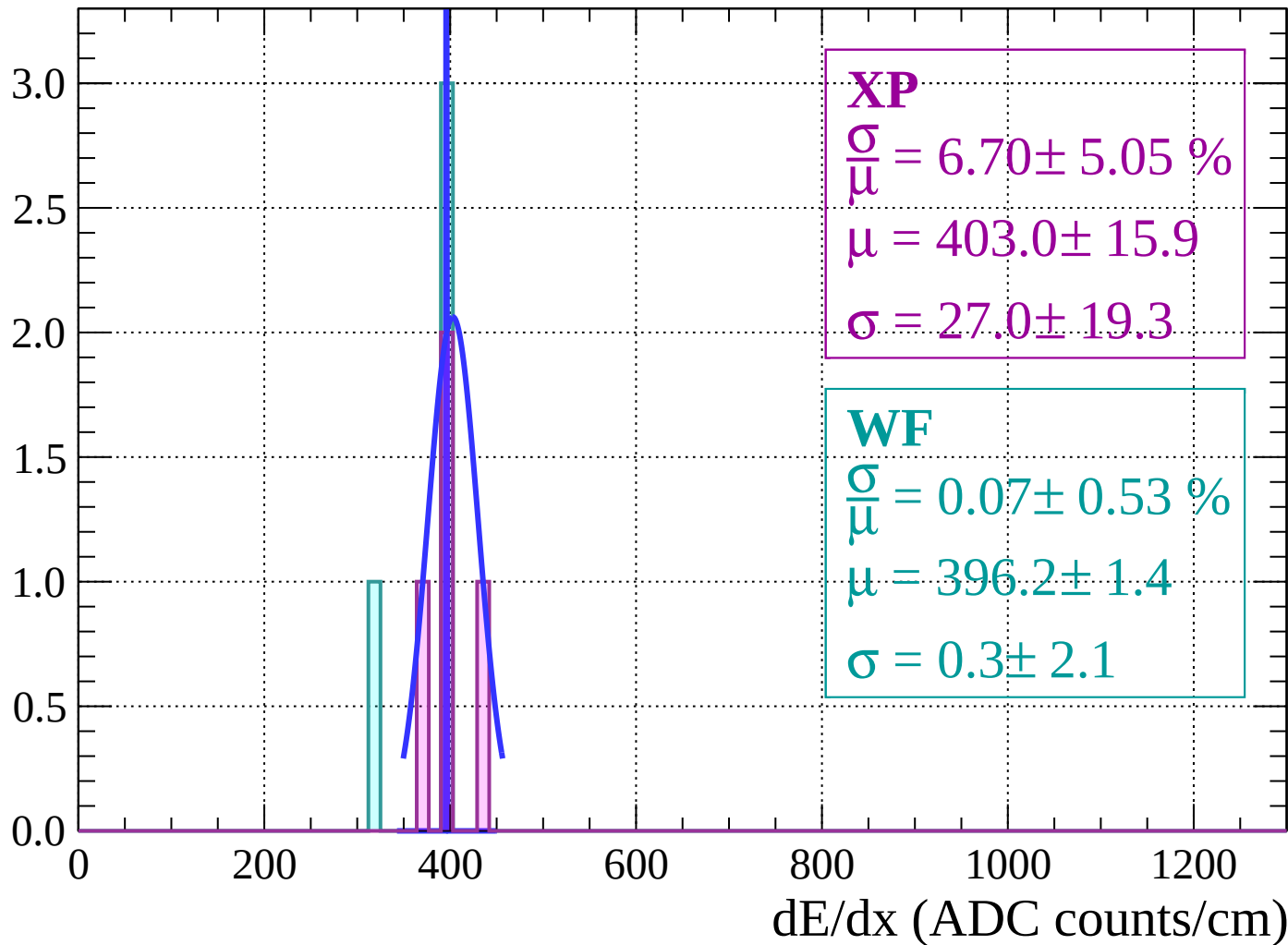
Count



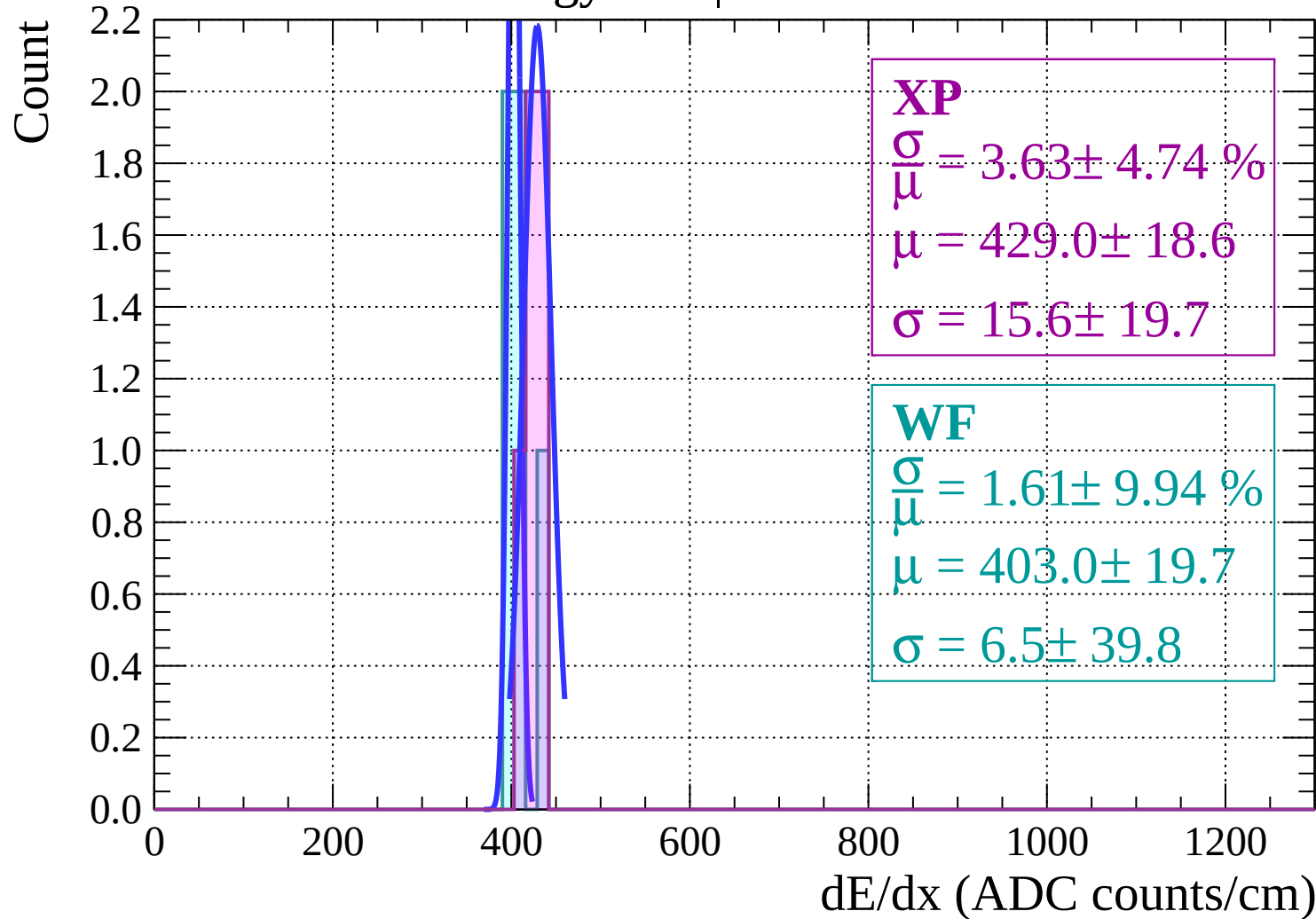
dE/dx (ADC counts/cm)

Energy loss | $-36 < \theta < -34$

Count

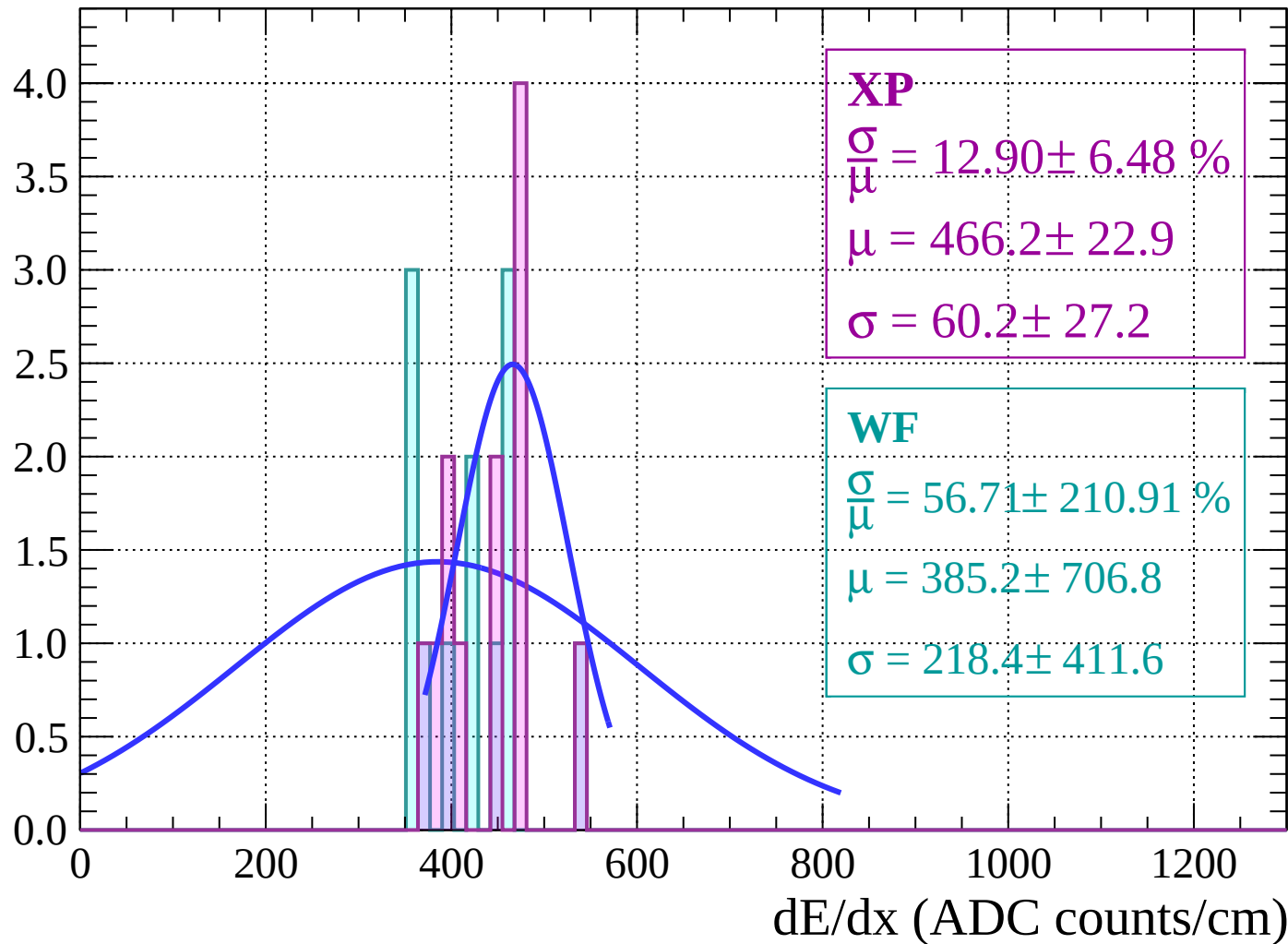


Energy loss | $-34 < \theta < -32$



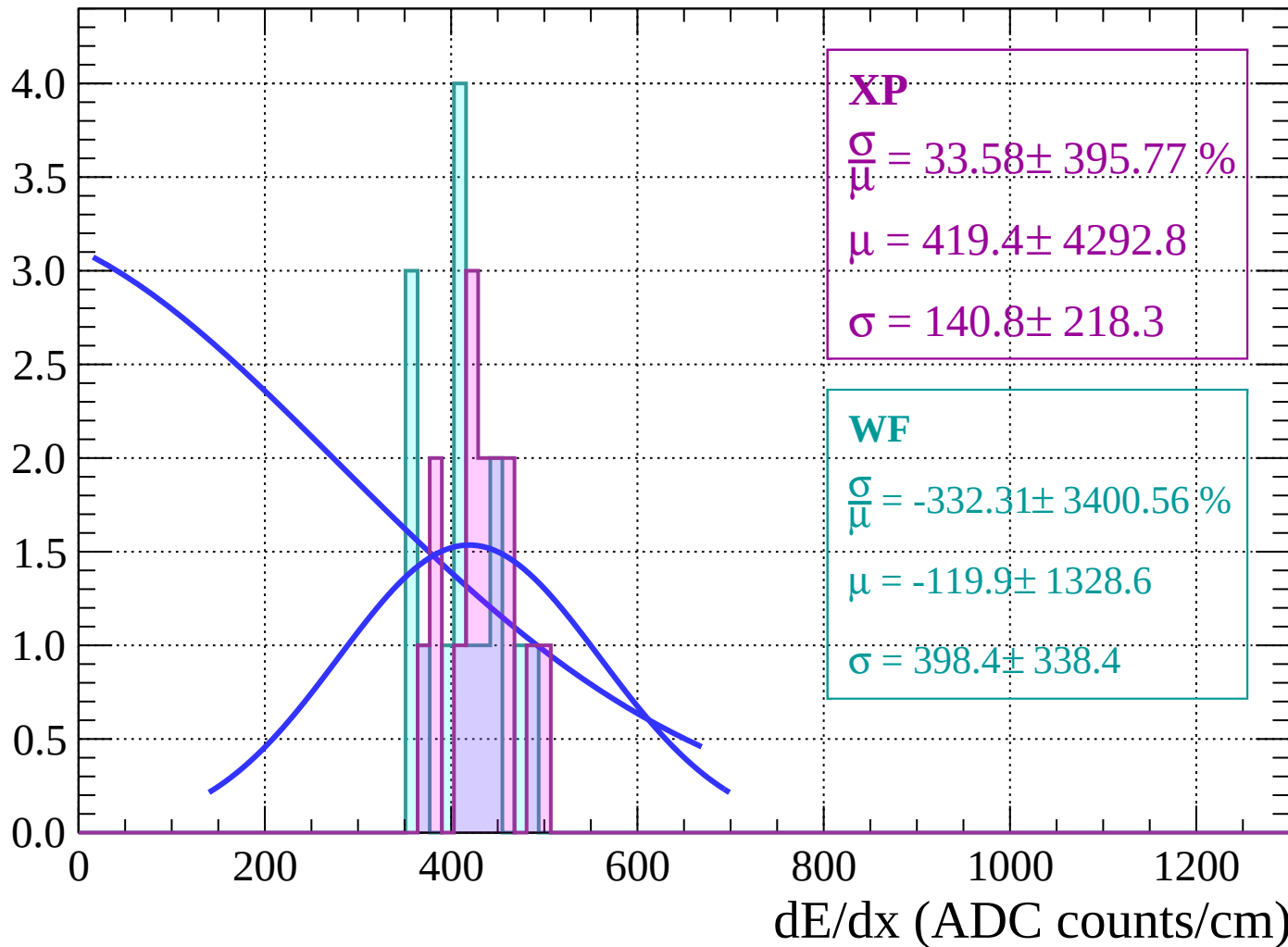
Energy loss | $-32 < \theta < -30$

Count



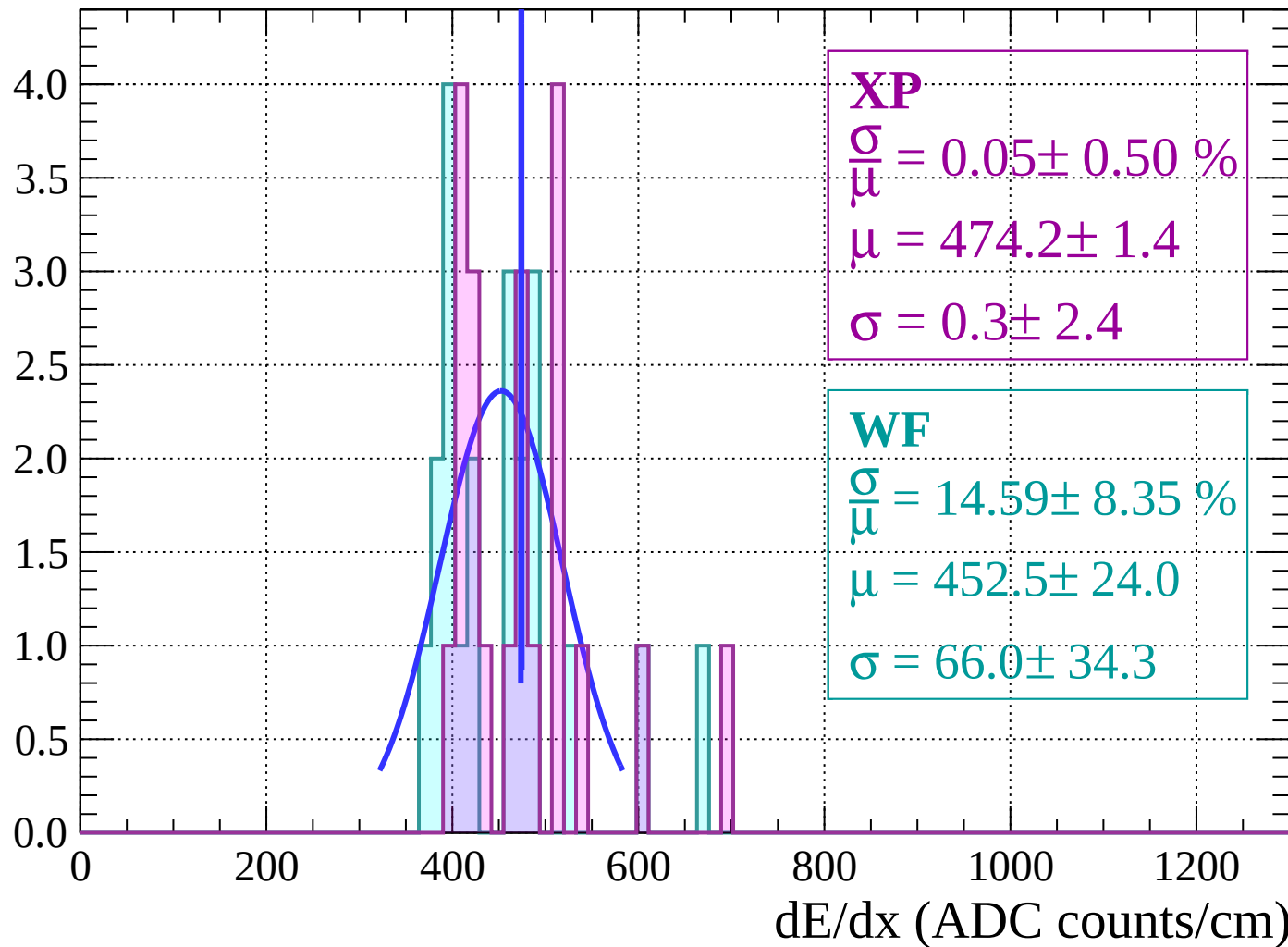
Energy loss | $-30 < \theta < -28$

Count



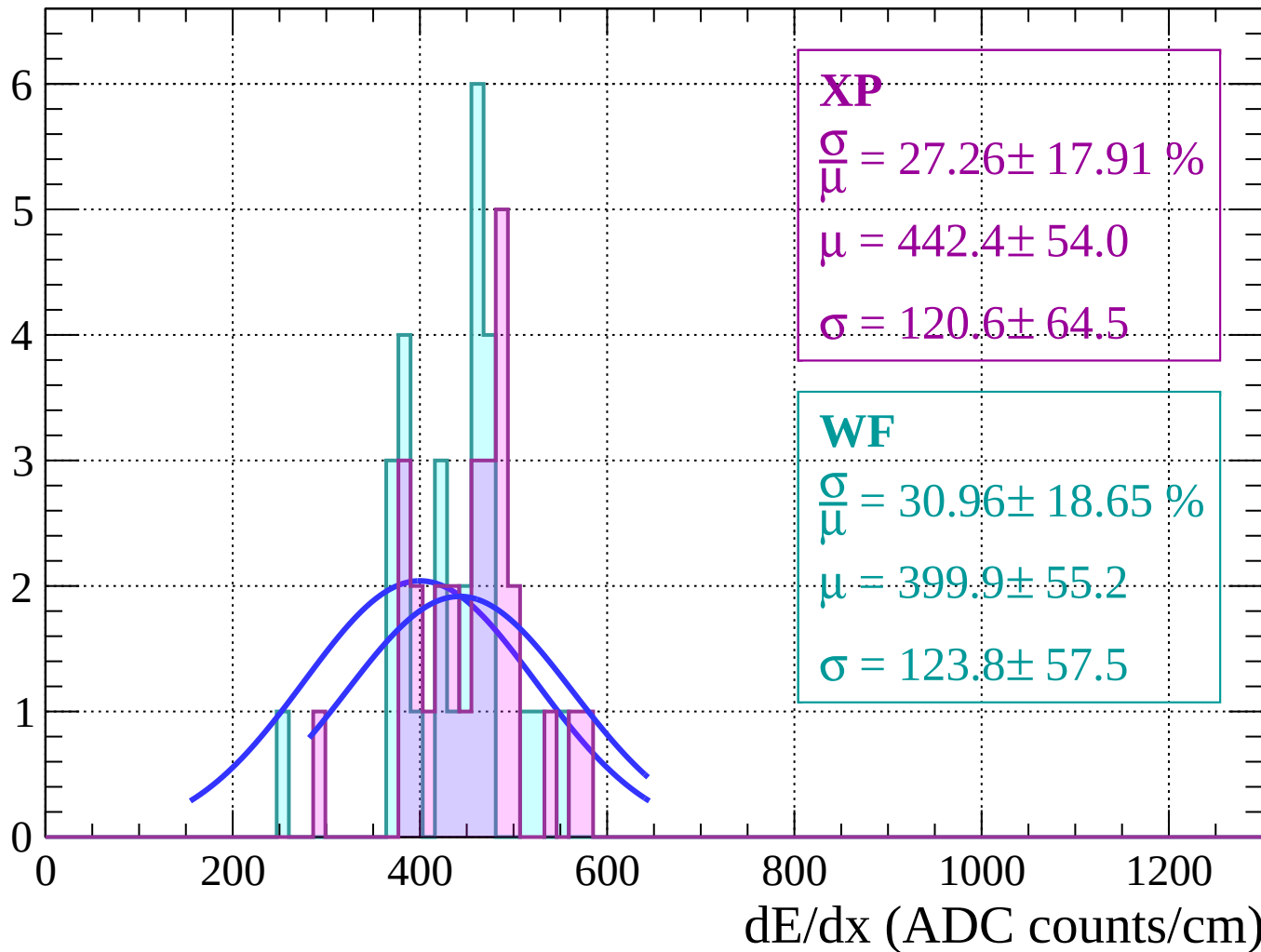
Energy loss $|-28 < \theta < -26$

Count



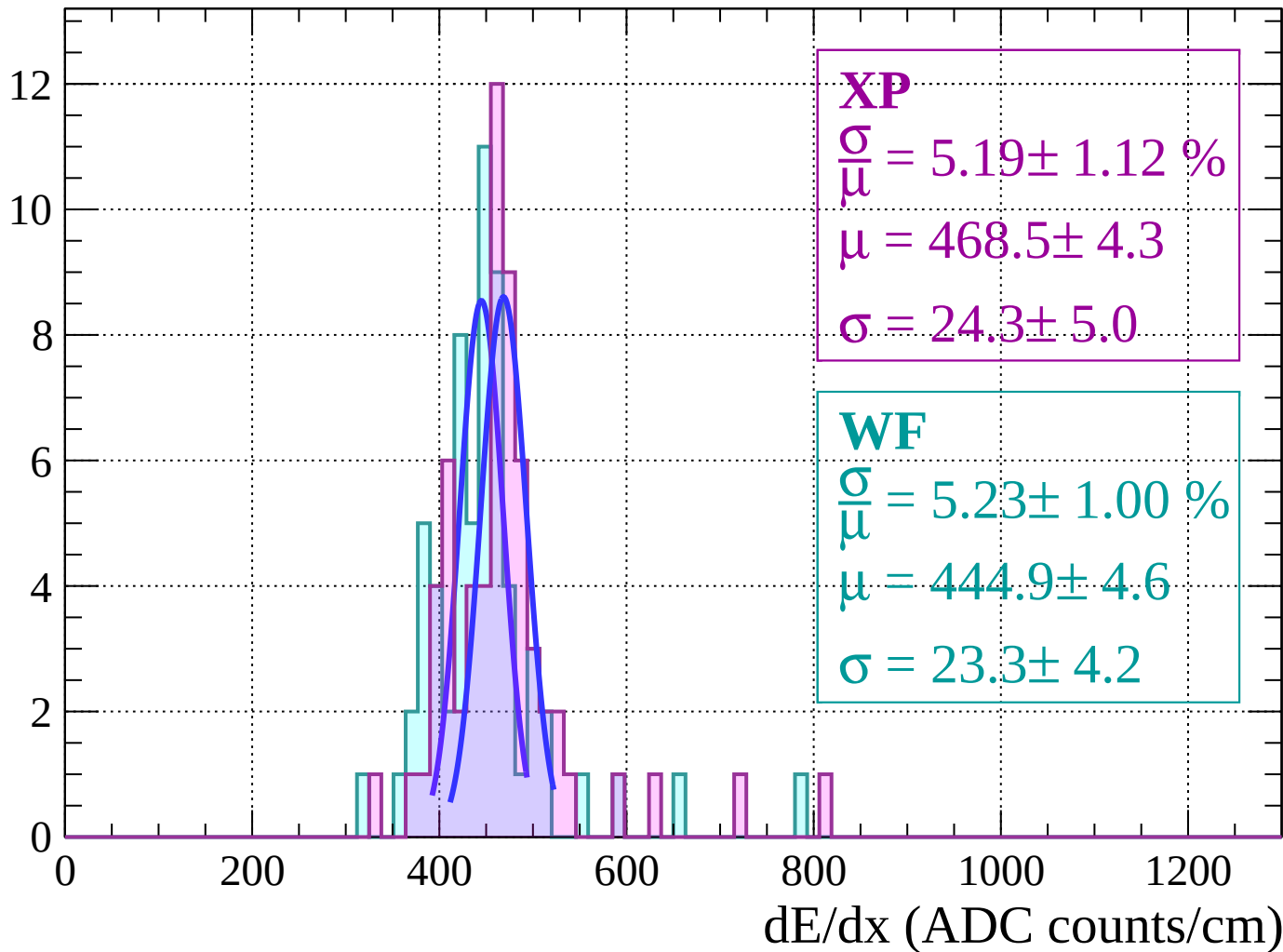
Energy loss | $-26 < \theta < -24$

Count



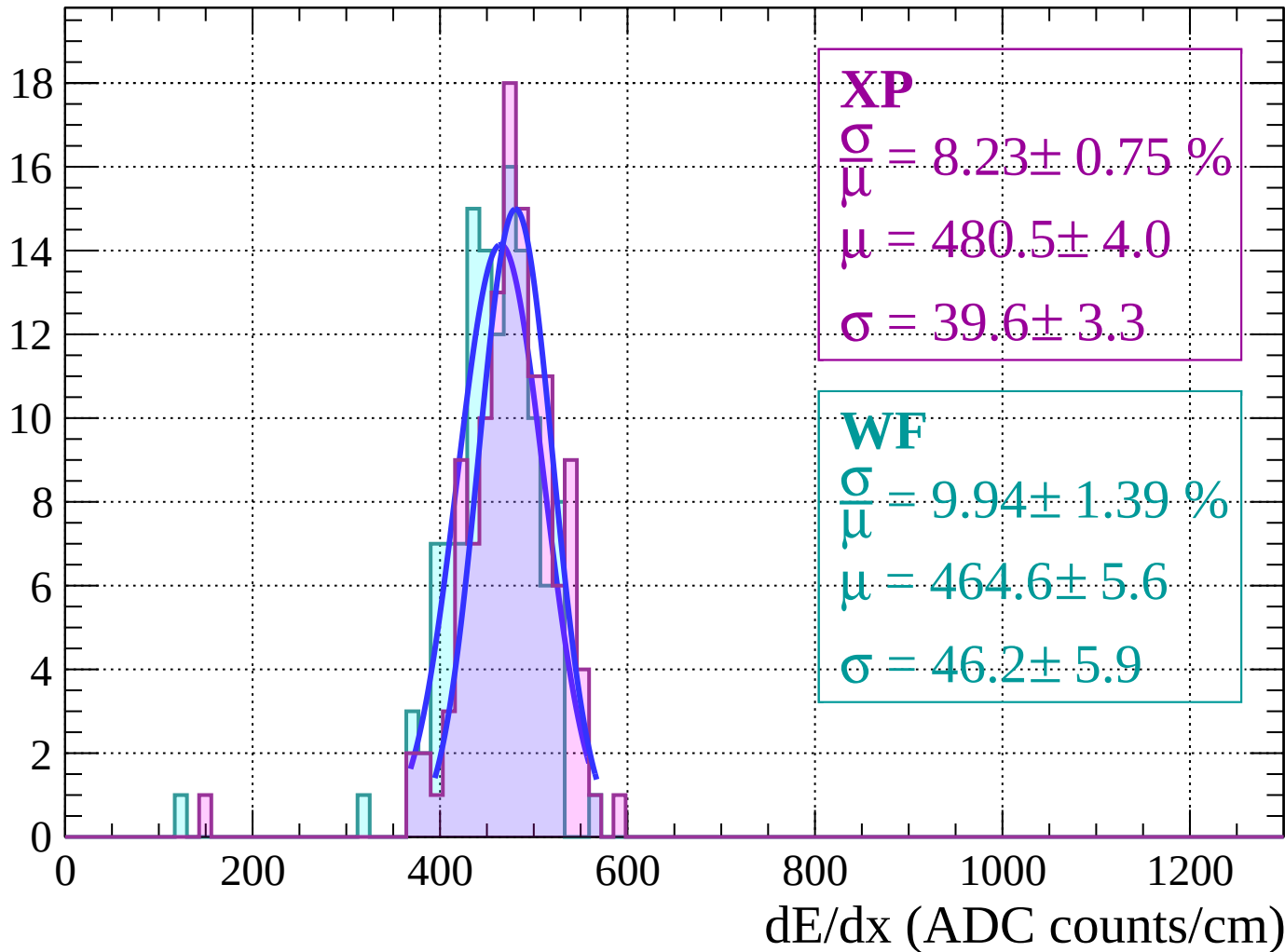
Energy loss | $-24 < \theta < -22$

Count



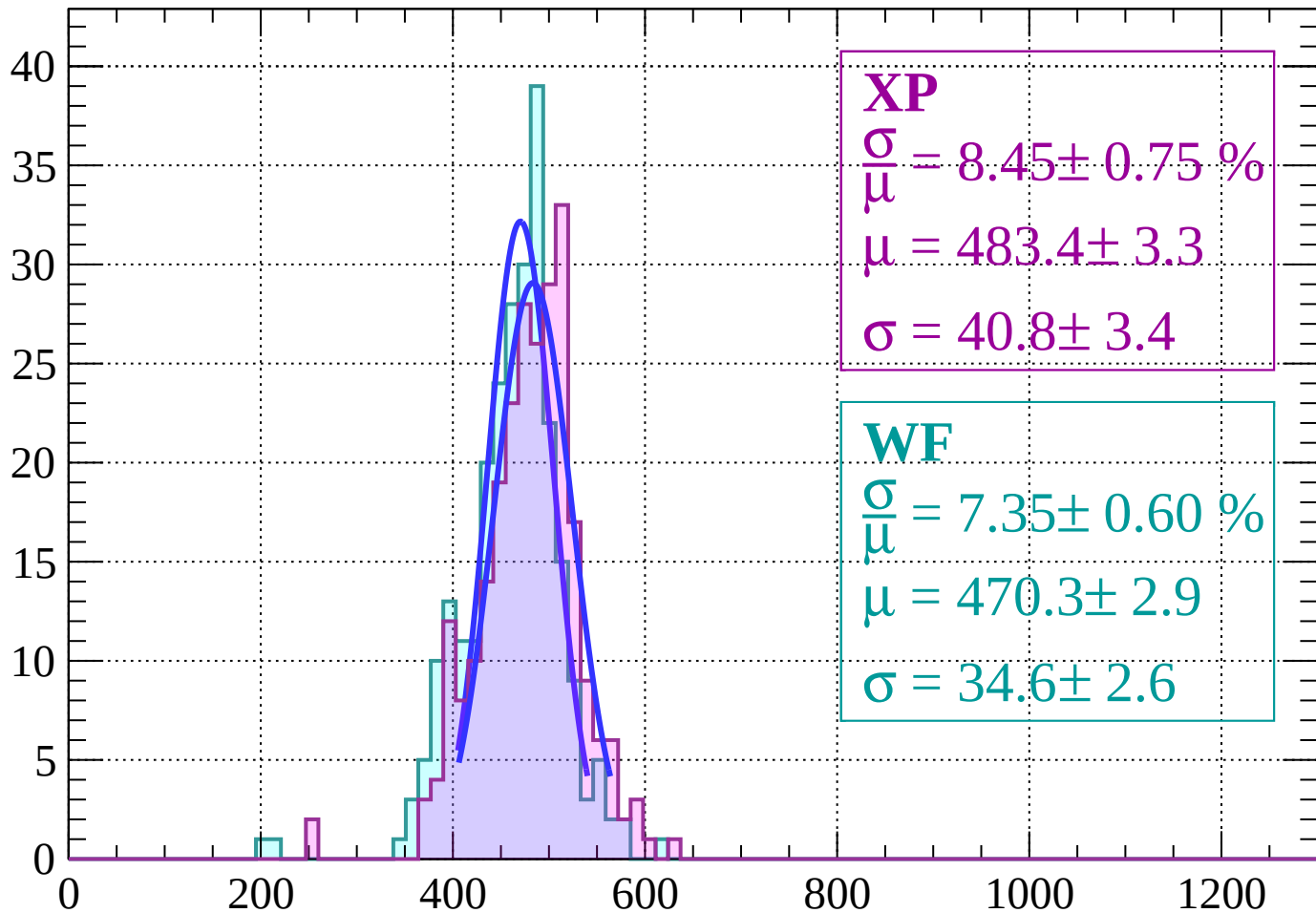
Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.75 \%$$

$$\mu = 483.4 \pm 3.3$$

$$\sigma = 40.8 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 7.35 \pm 0.60 \%$$

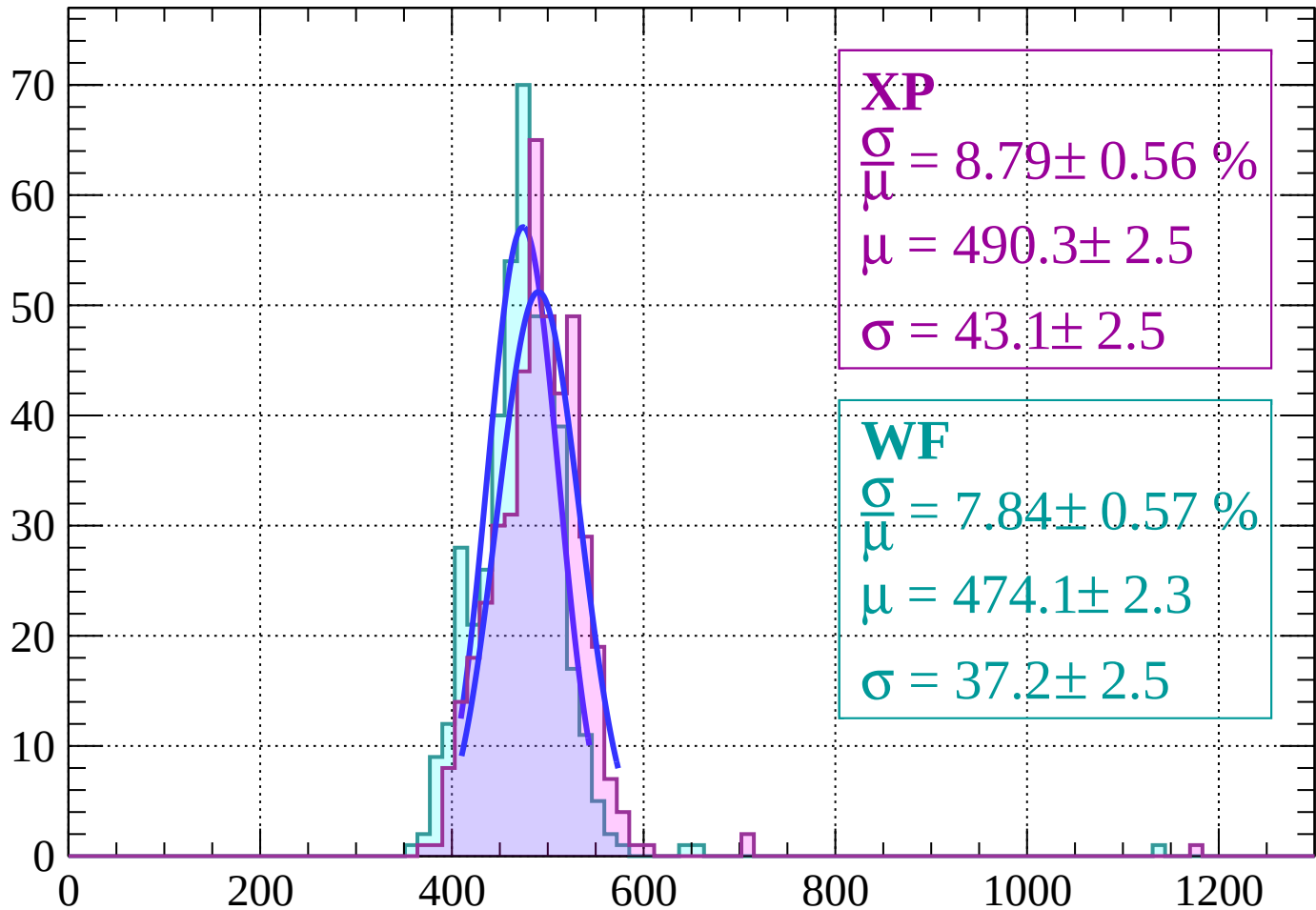
$$\mu = 470.3 \pm 2.9$$

$$\sigma = 34.6 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.56 \%$$

$$\mu = 490.3 \pm 2.5$$

$$\sigma = 43.1 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.84 \pm 0.57 \%$$

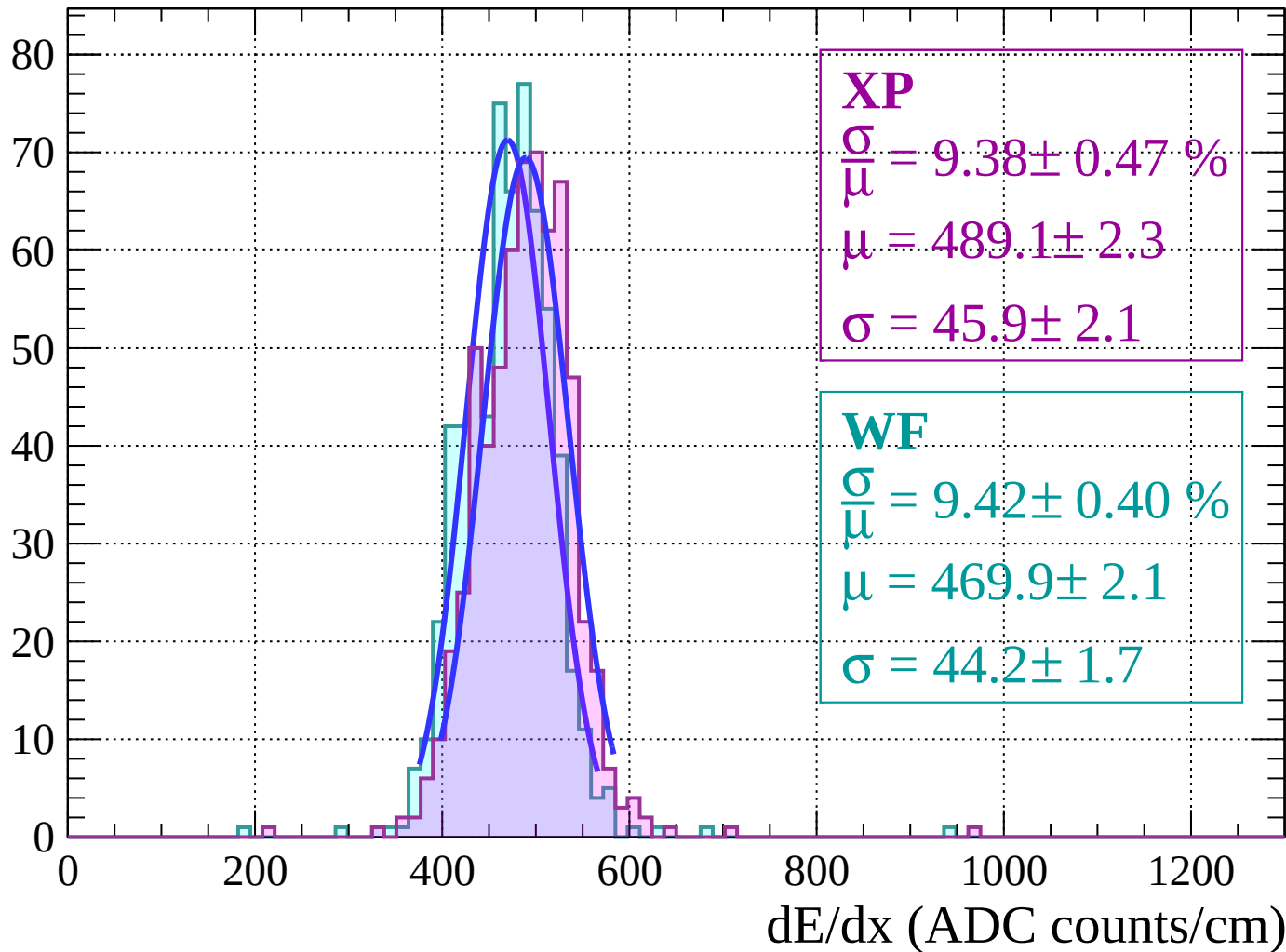
$$\mu = 474.1 \pm 2.3$$

$$\sigma = 37.2 \pm 2.5$$

dE/dx (ADC counts/cm)

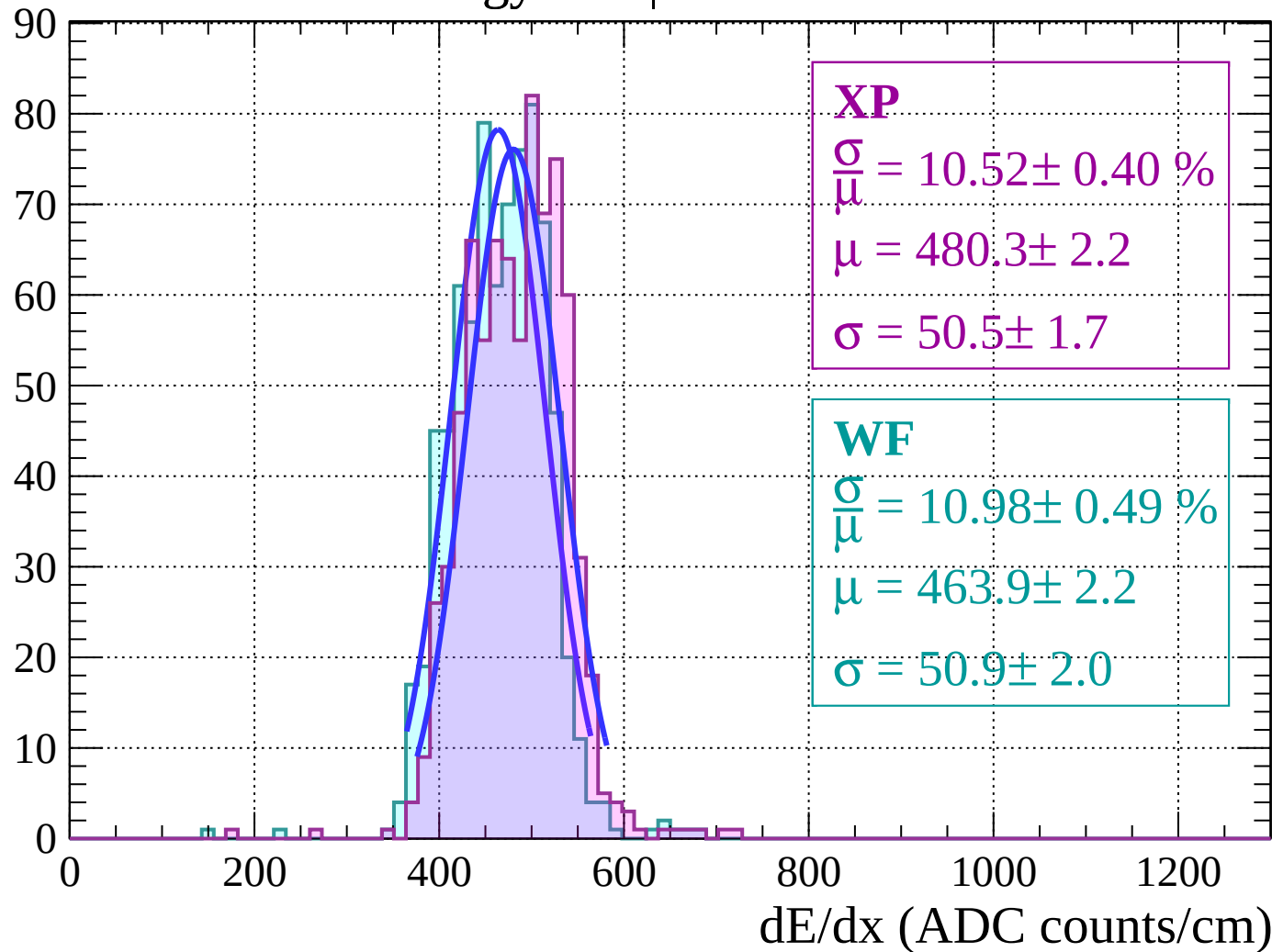
Energy loss $|-16 < \theta < -14$

Count

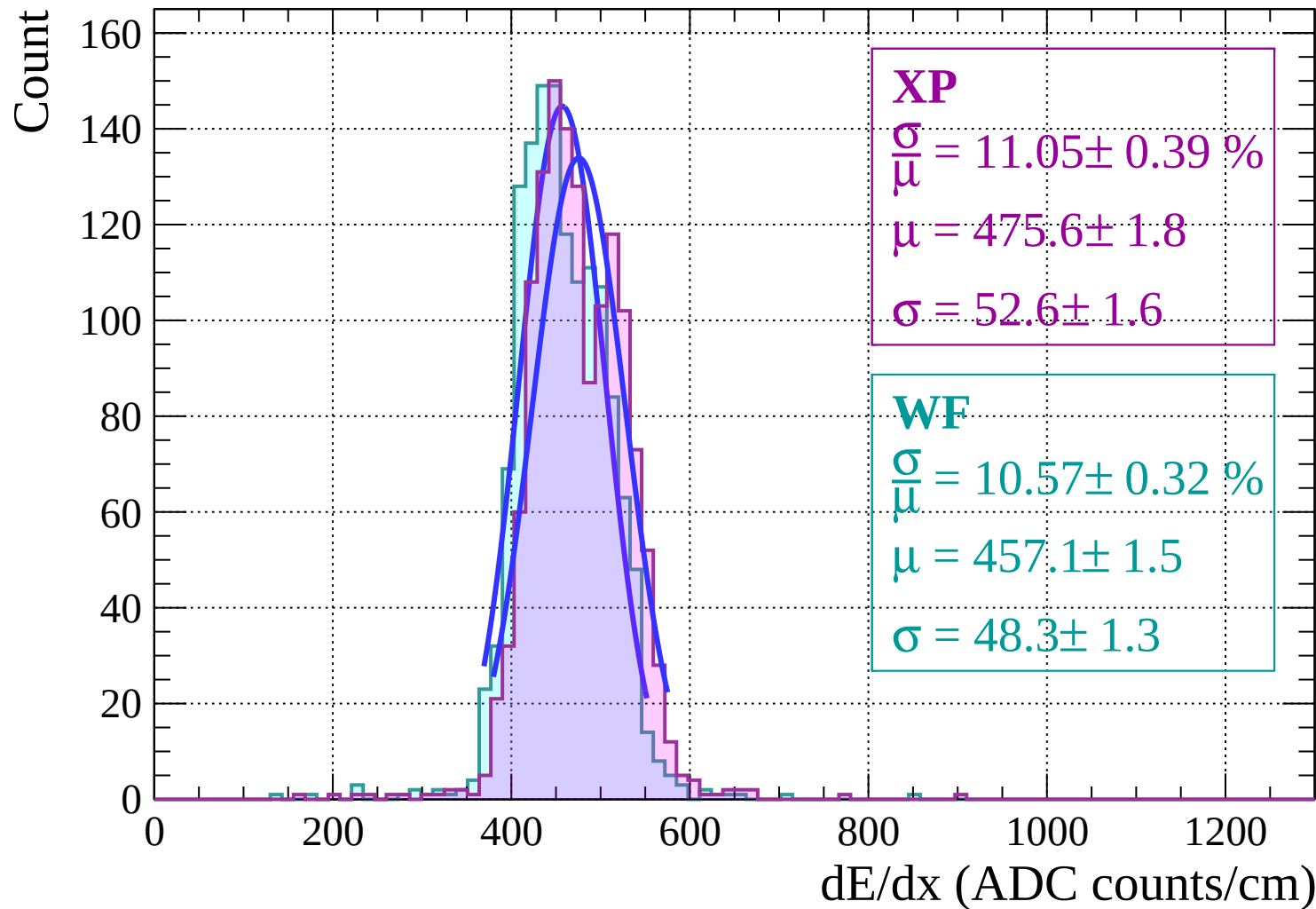


Energy loss | $-14 < \theta < -12$

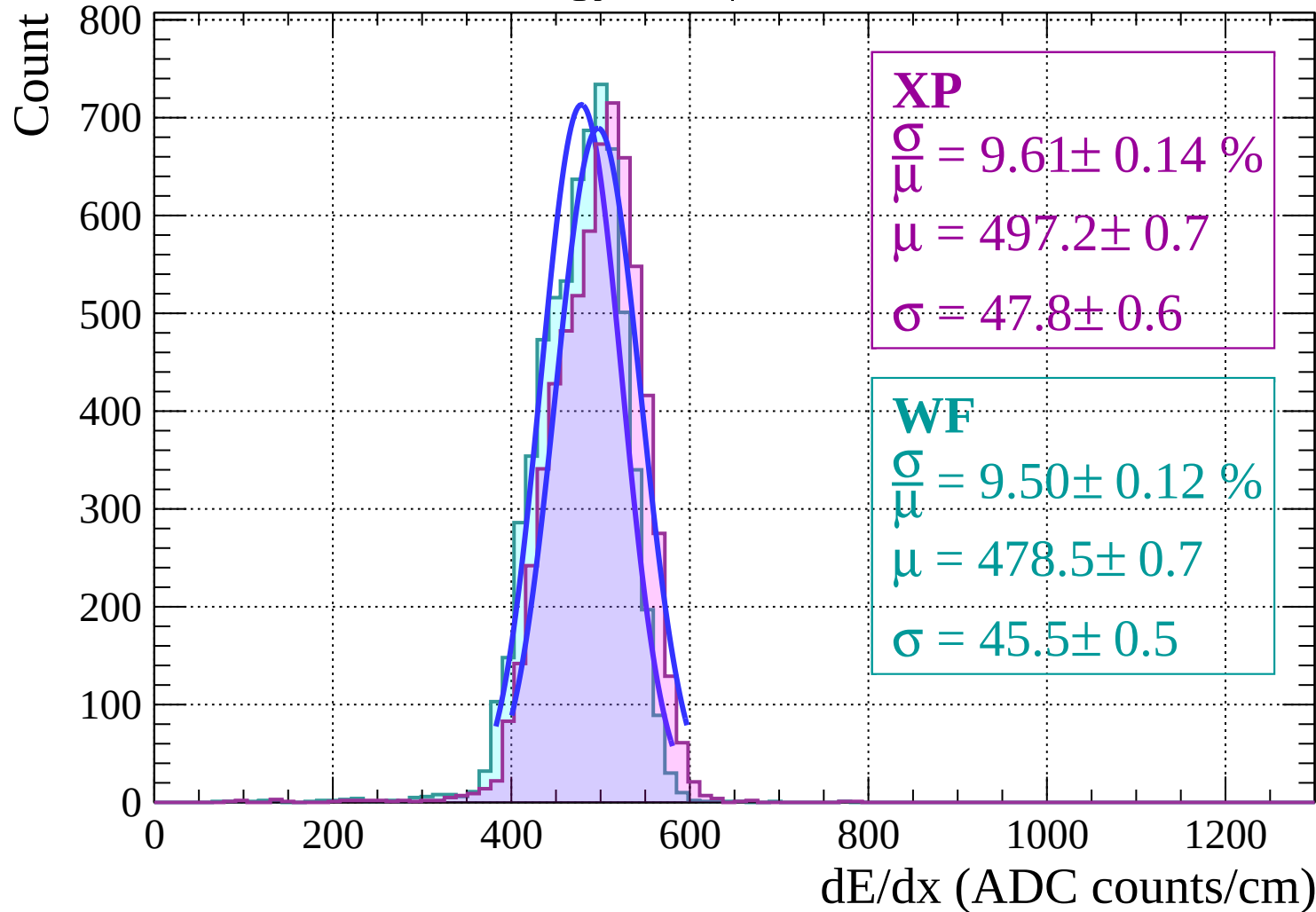
Count



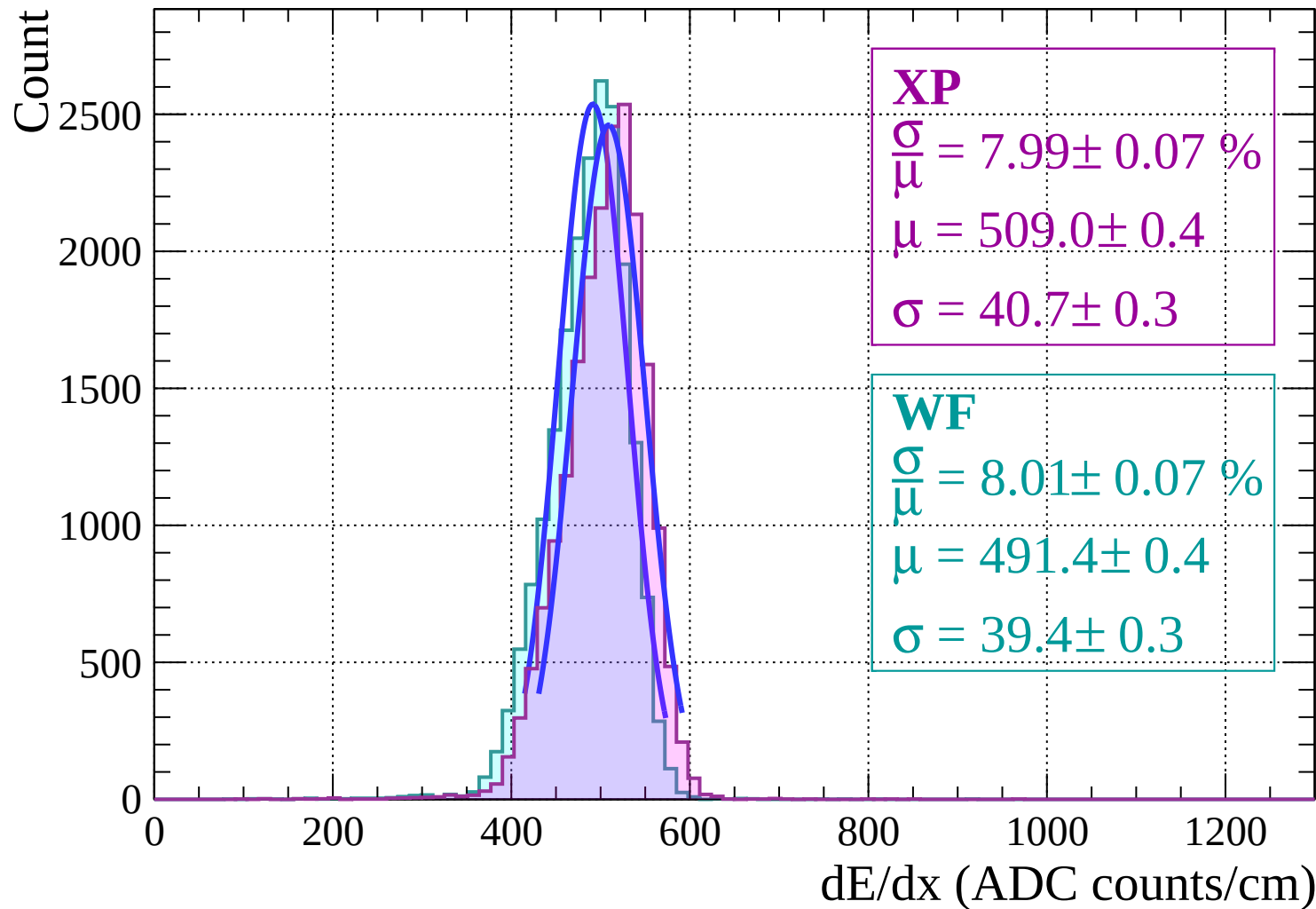
Energy loss | $-12 < \theta < -10$



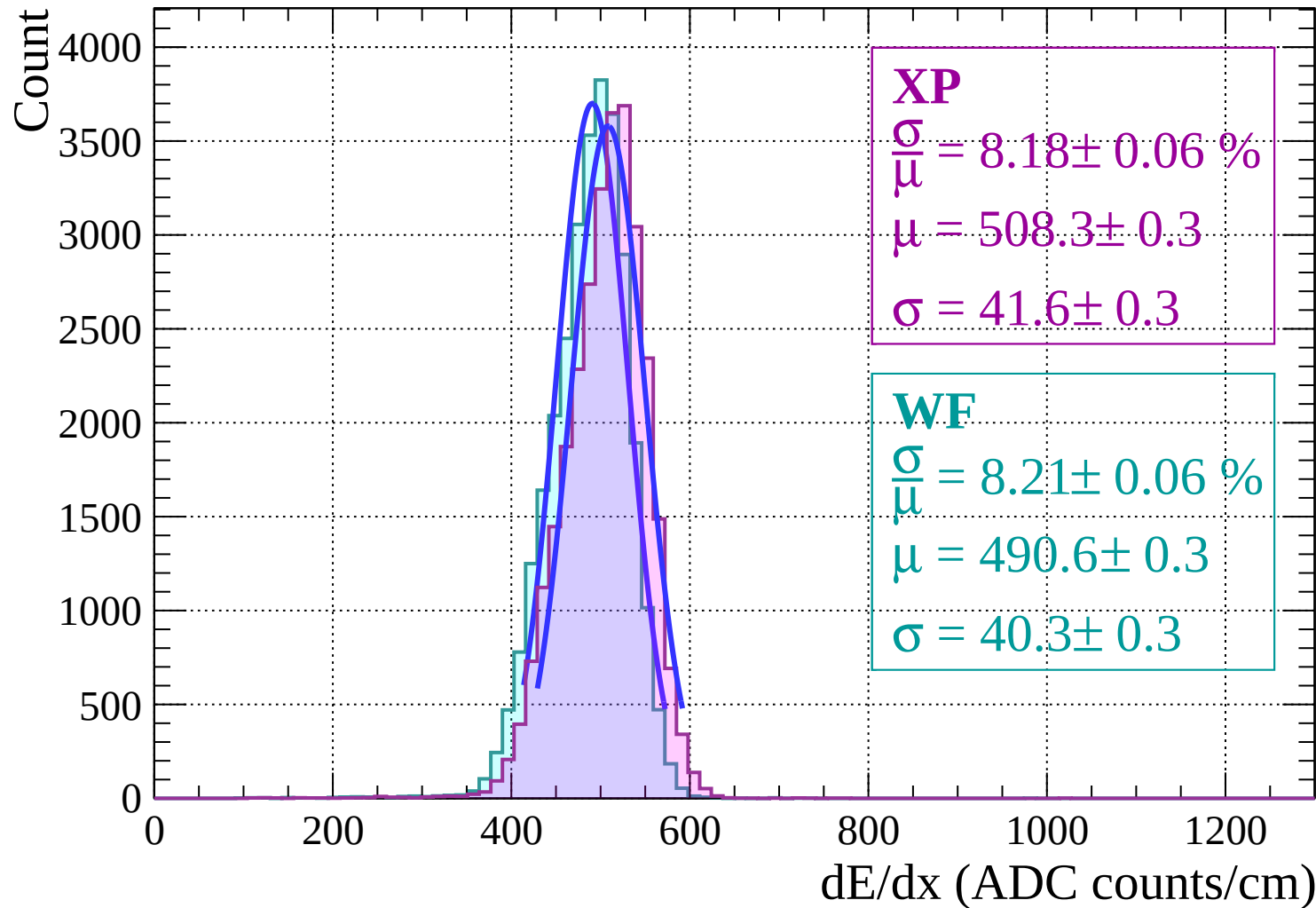
Energy loss | $-10 < \theta < -8$



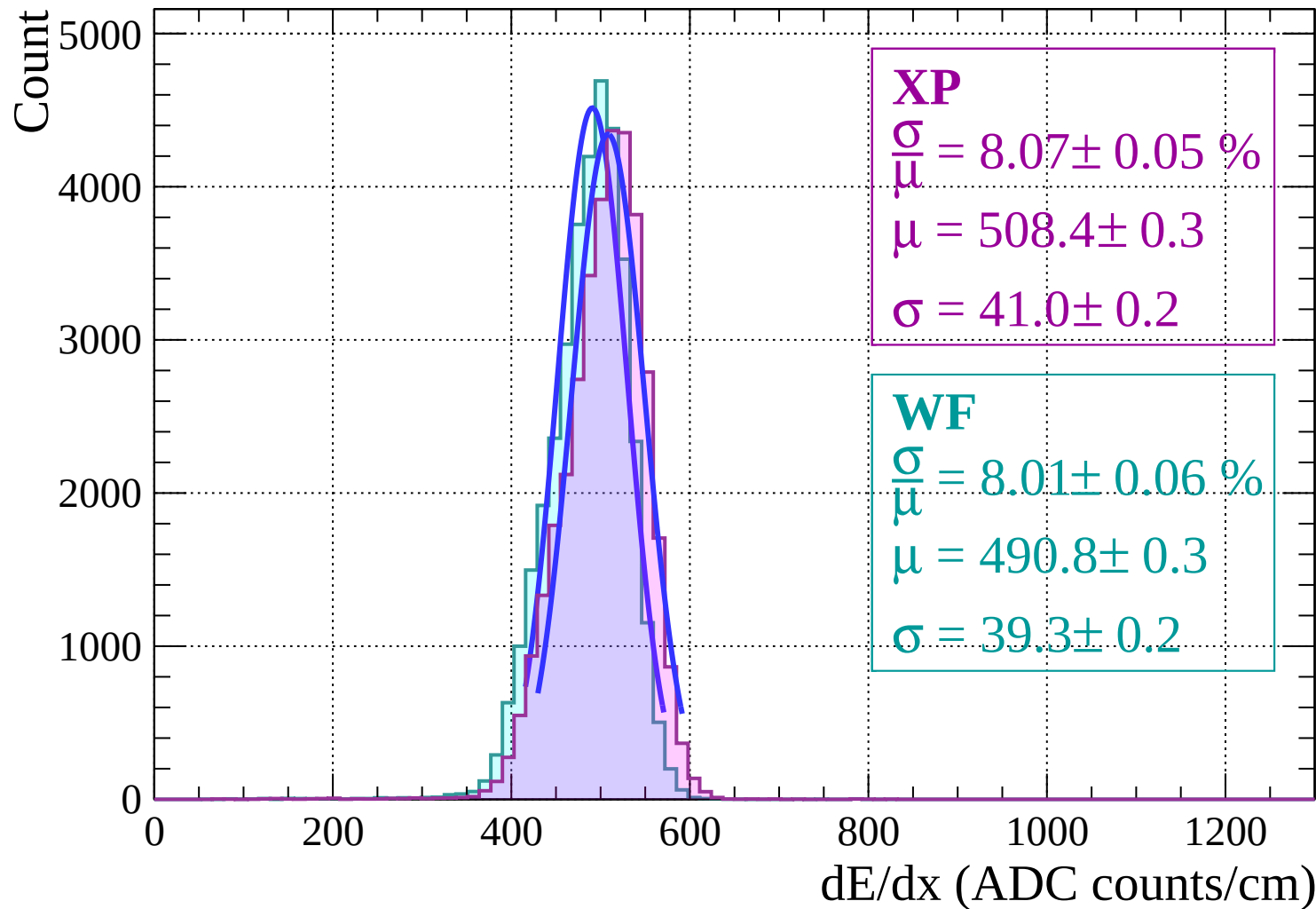
Energy loss $|-8 < \theta < -6$



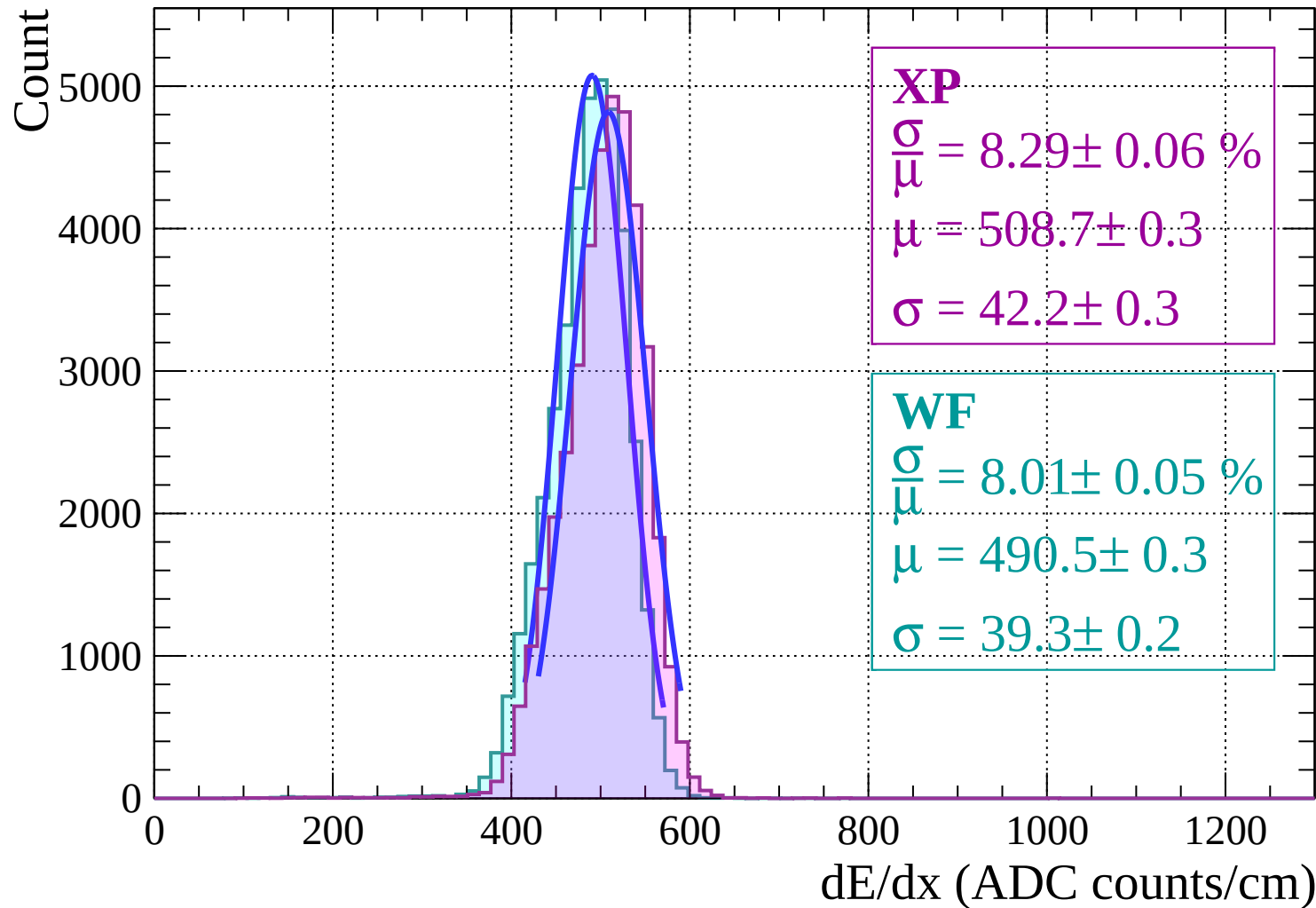
Energy loss | $-6 < \theta < -4$



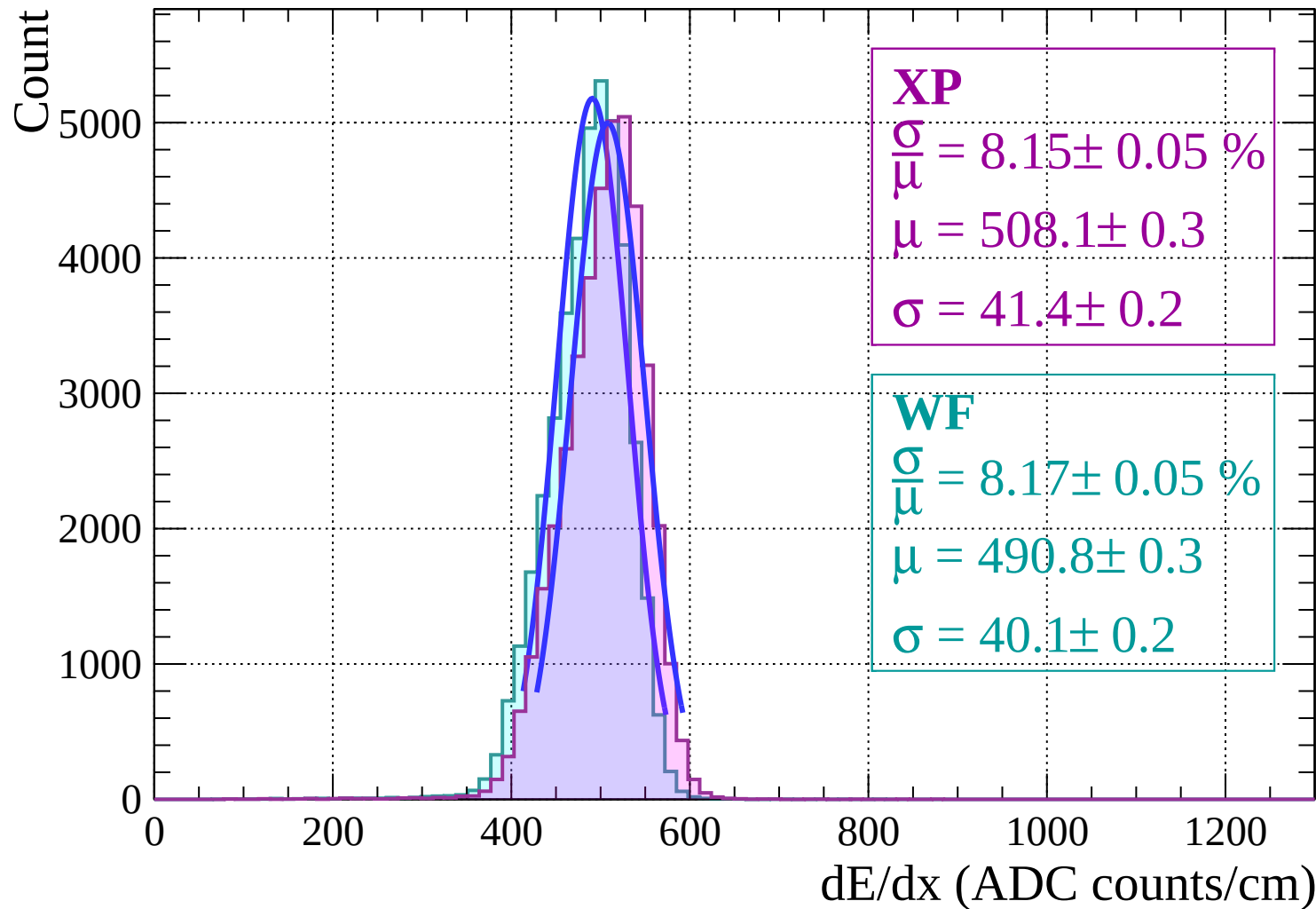
Energy loss | $-4 < \theta < -2$



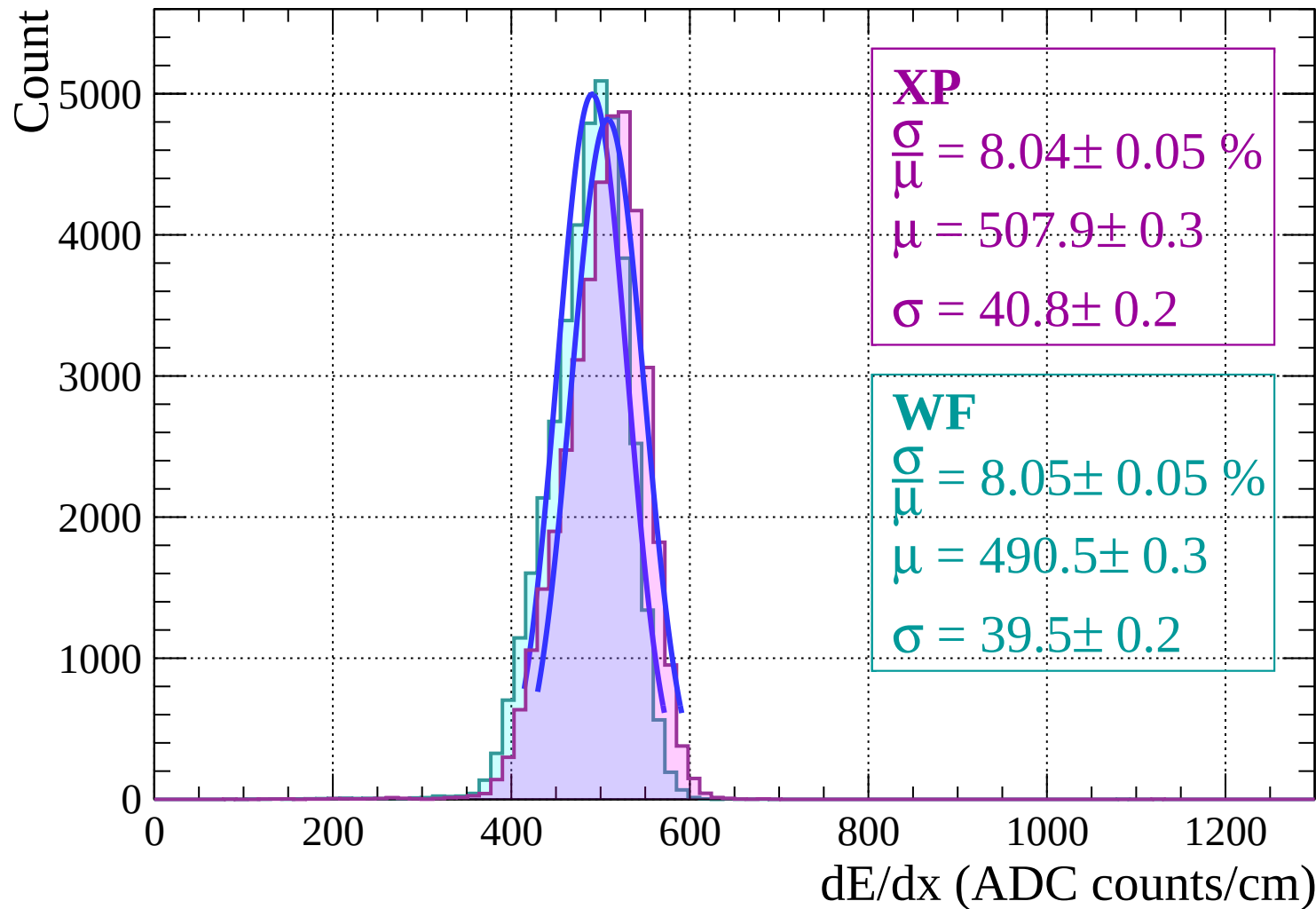
Energy loss | $-2 < \theta < 0$



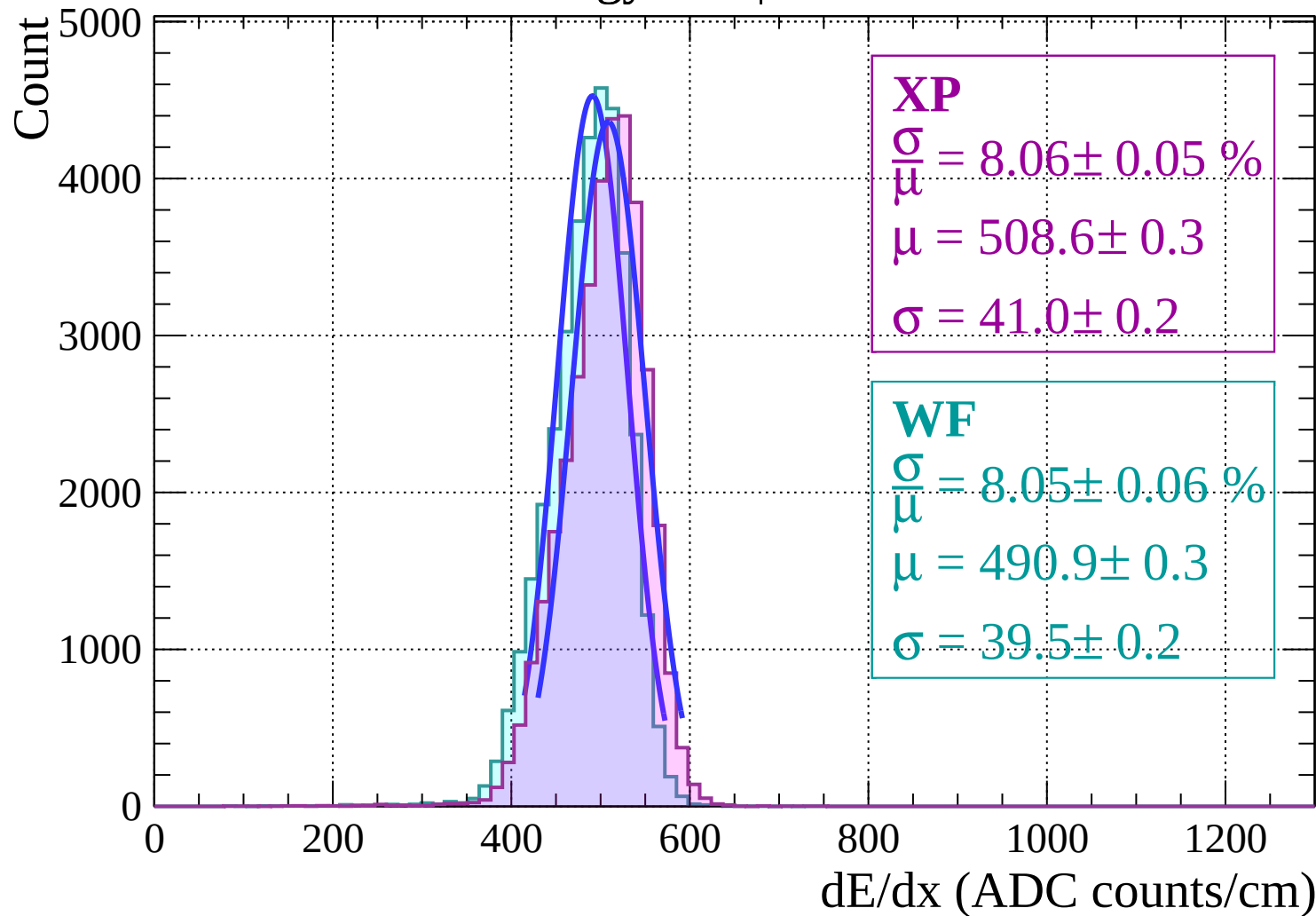
Energy loss | $0 < \theta < 2$



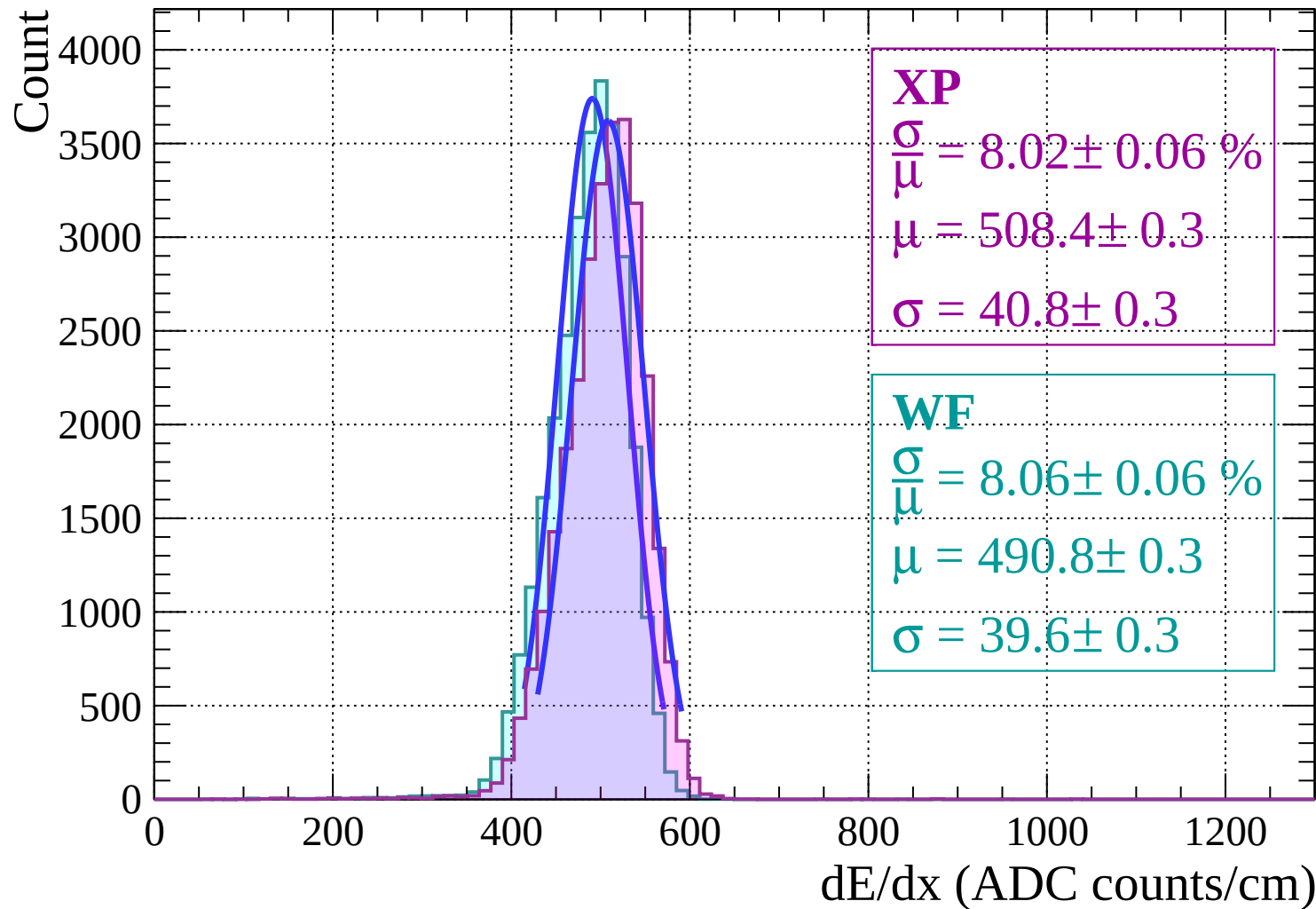
Energy loss | $2 < \theta < 4$



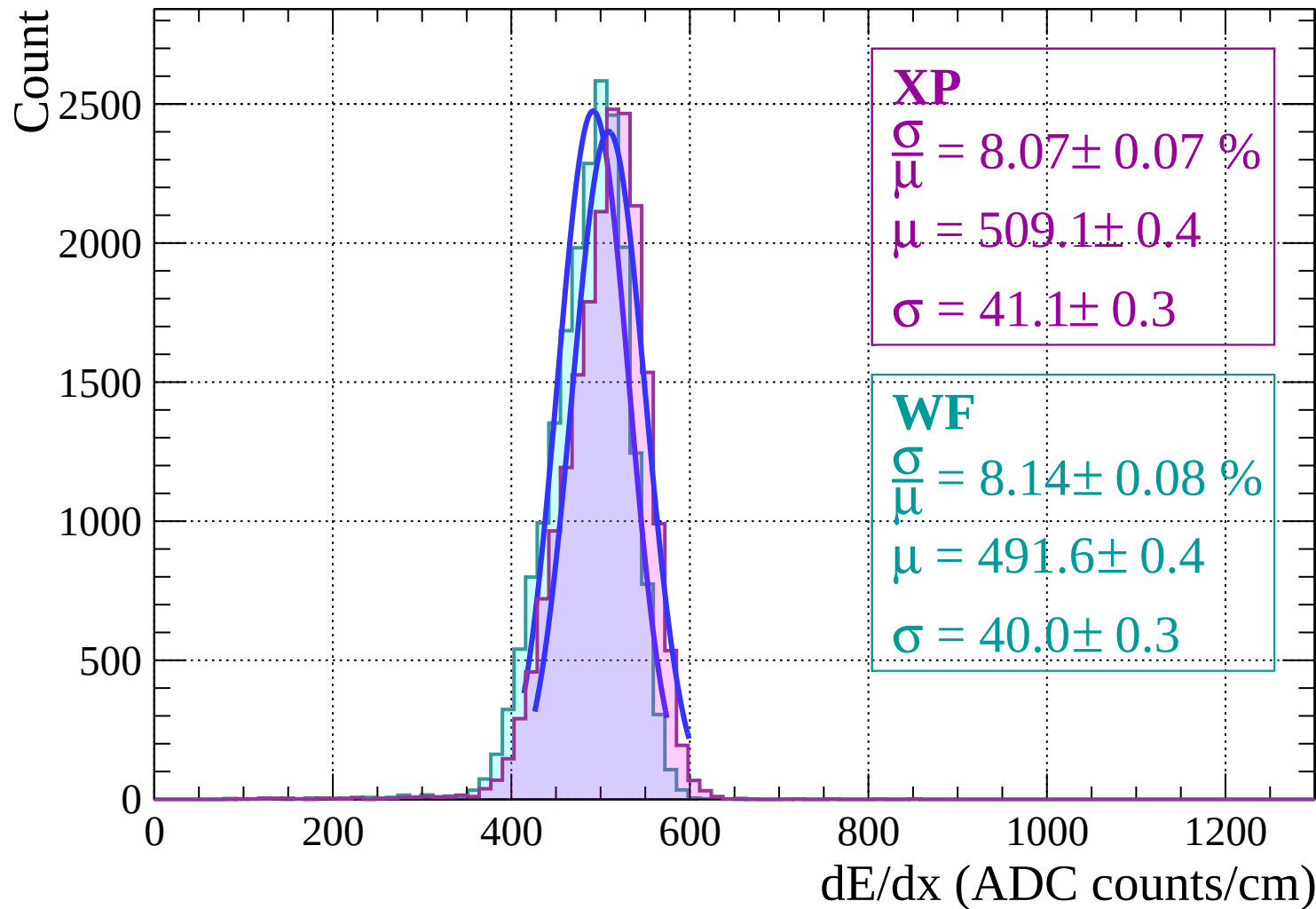
Energy loss | $4 < \theta < 6$



Energy loss | $6 < \theta < 8$

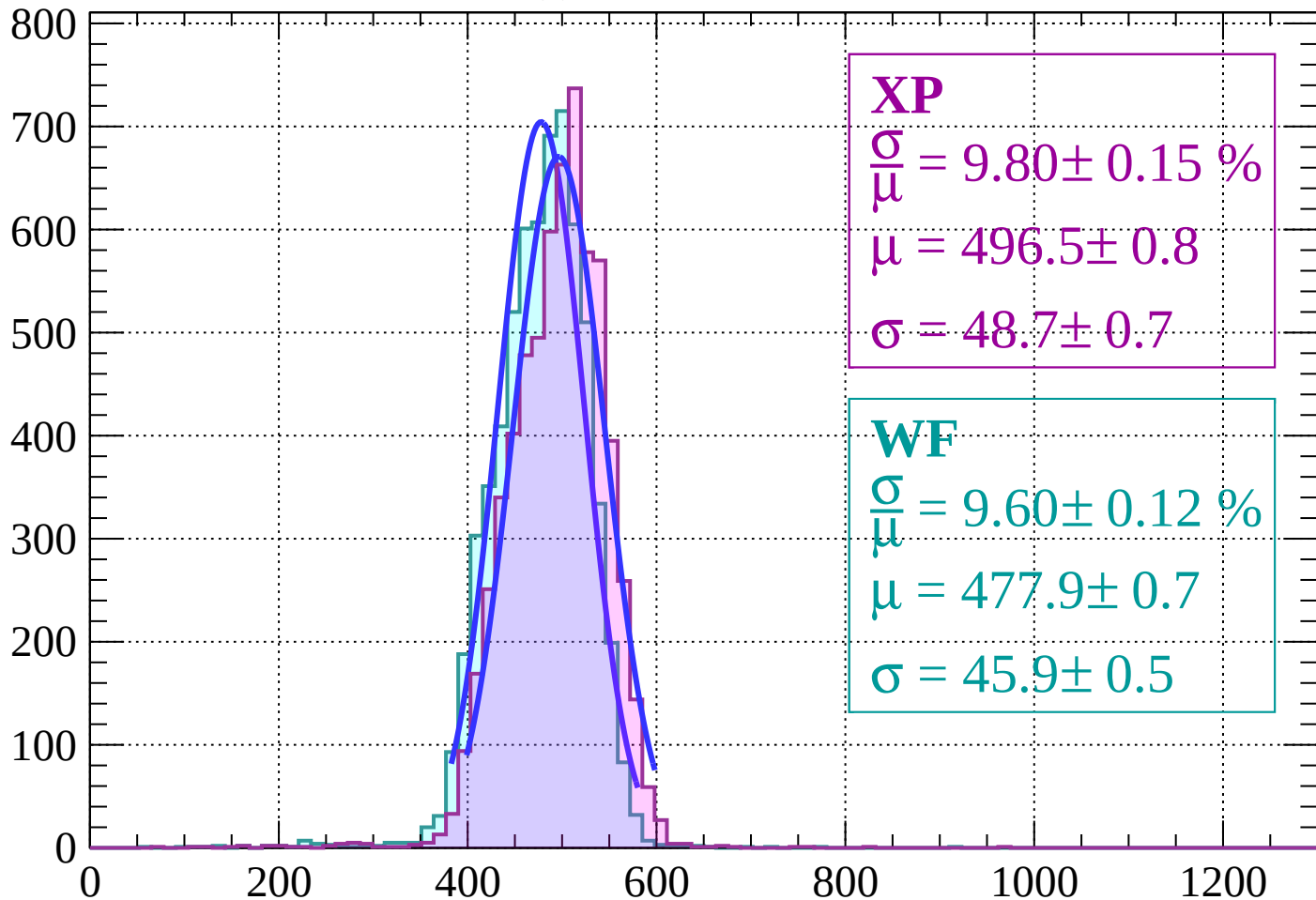


Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.15 \%$$

$$\mu = 496.5 \pm 0.8$$

$$\sigma = 48.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.12 \%$$

$$\mu = 477.9 \pm 0.7$$

$$\sigma = 45.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.41 \pm 0.33 \%$$

$$\mu = 477.9 \pm 1.7$$

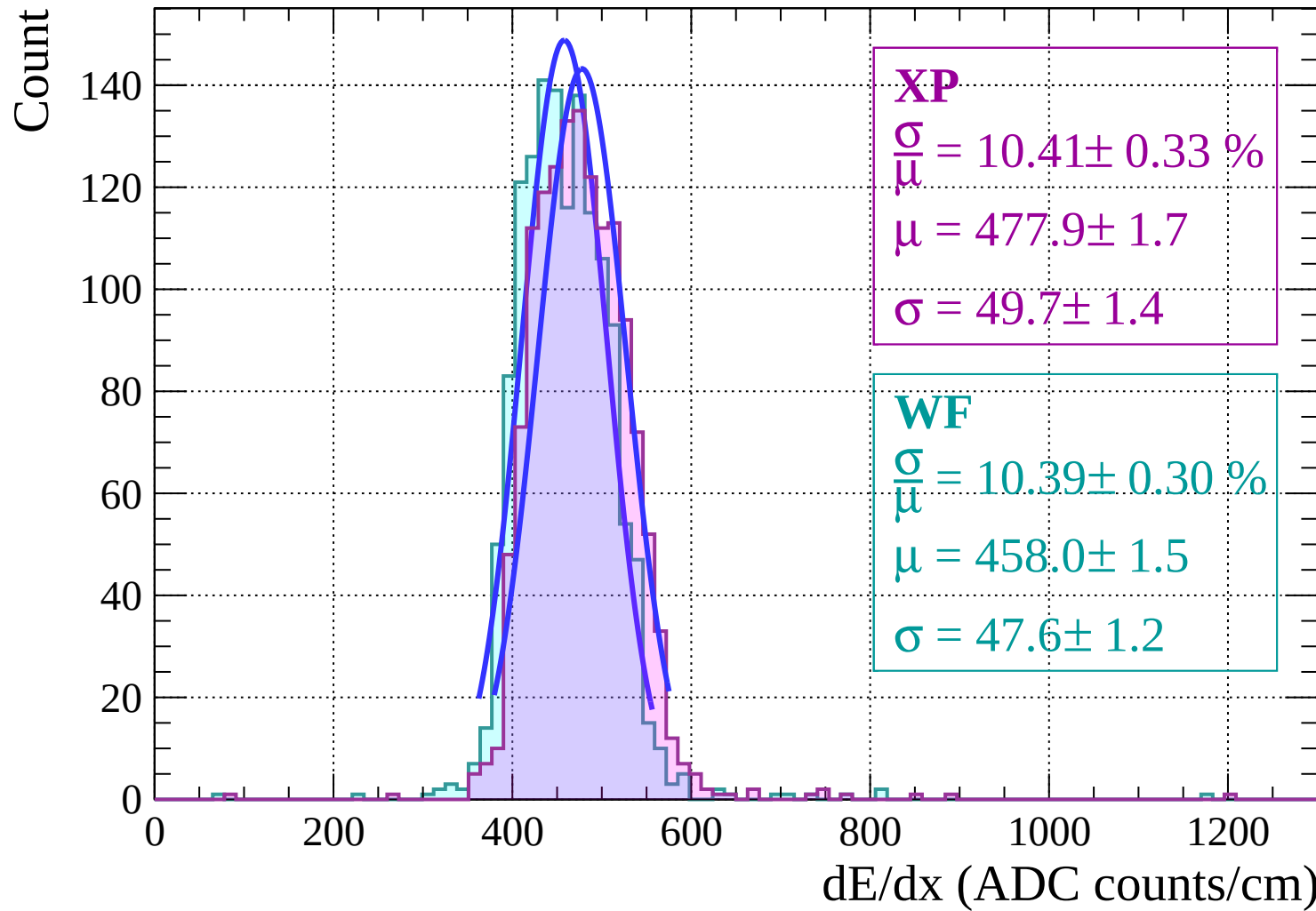
$$\sigma = 49.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.39 \pm 0.30 \%$$

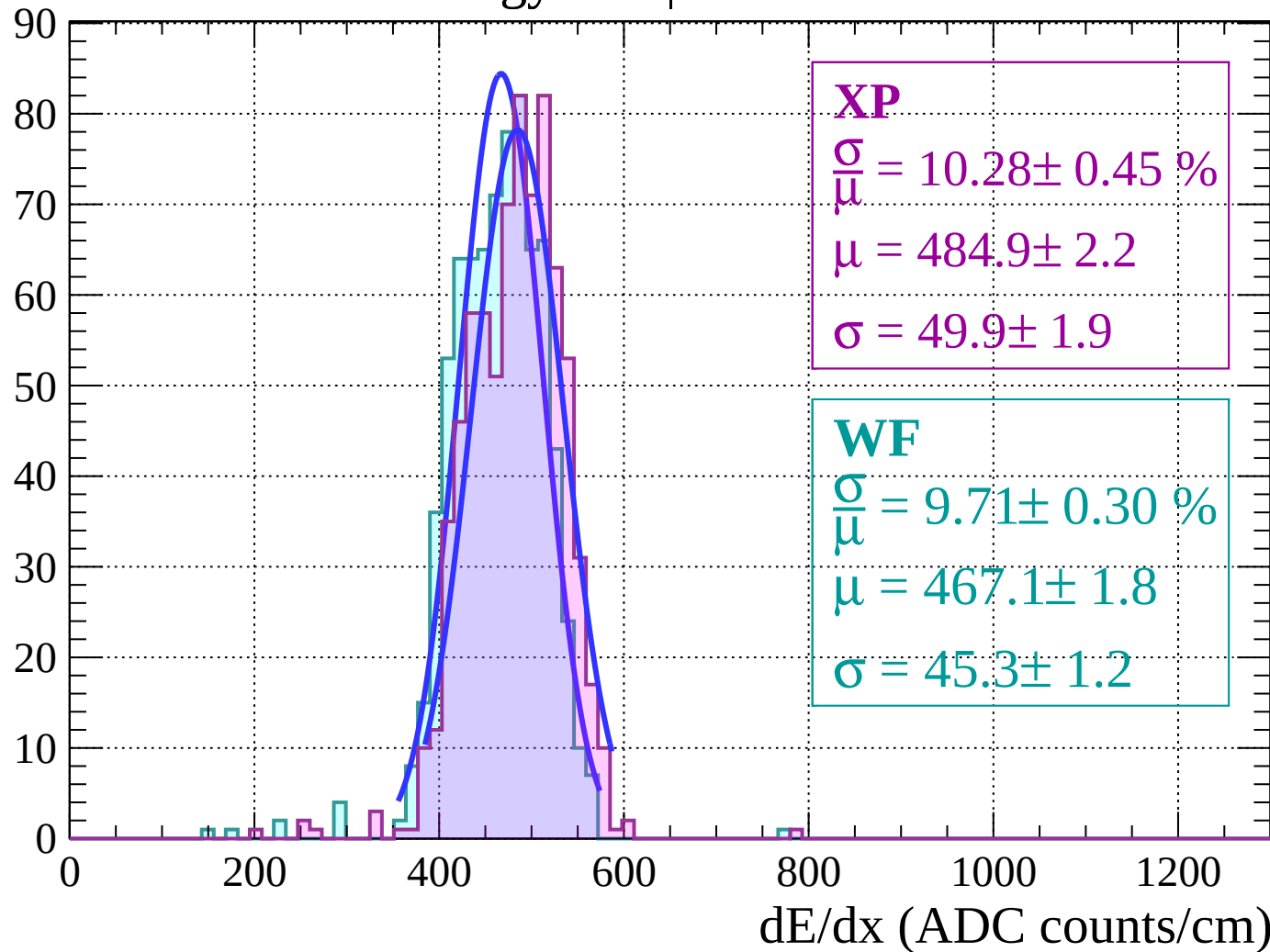
$$\mu = 458.0 \pm 1.5$$

$$\sigma = 47.6 \pm 1.2$$



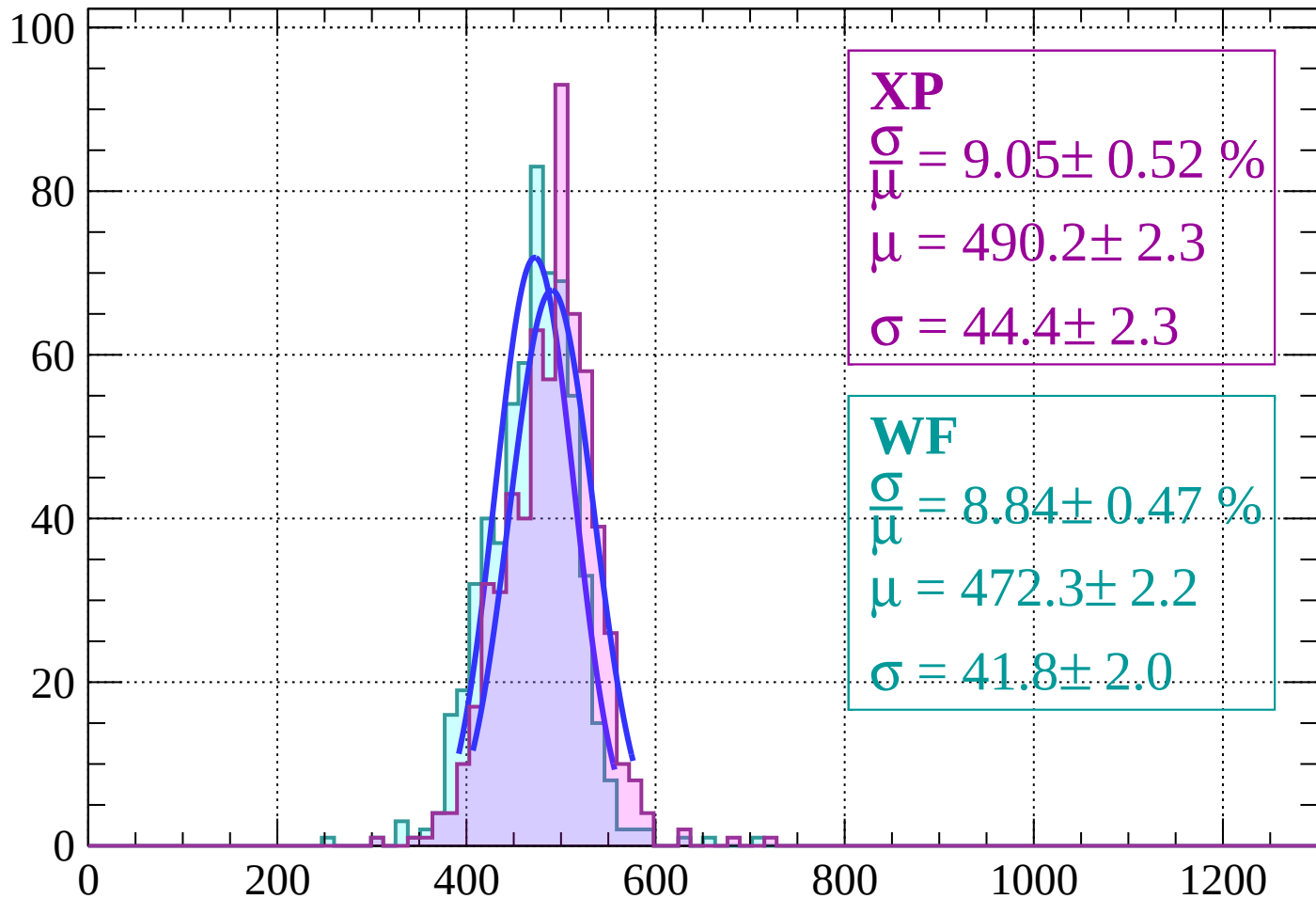
Energy loss | $14 < \theta < 16$

Count



Energy loss | $16 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 9.05 \pm 0.52 \%$$

$$\mu = 490.2 \pm 2.3$$

$$\sigma = 44.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.47 \%$$

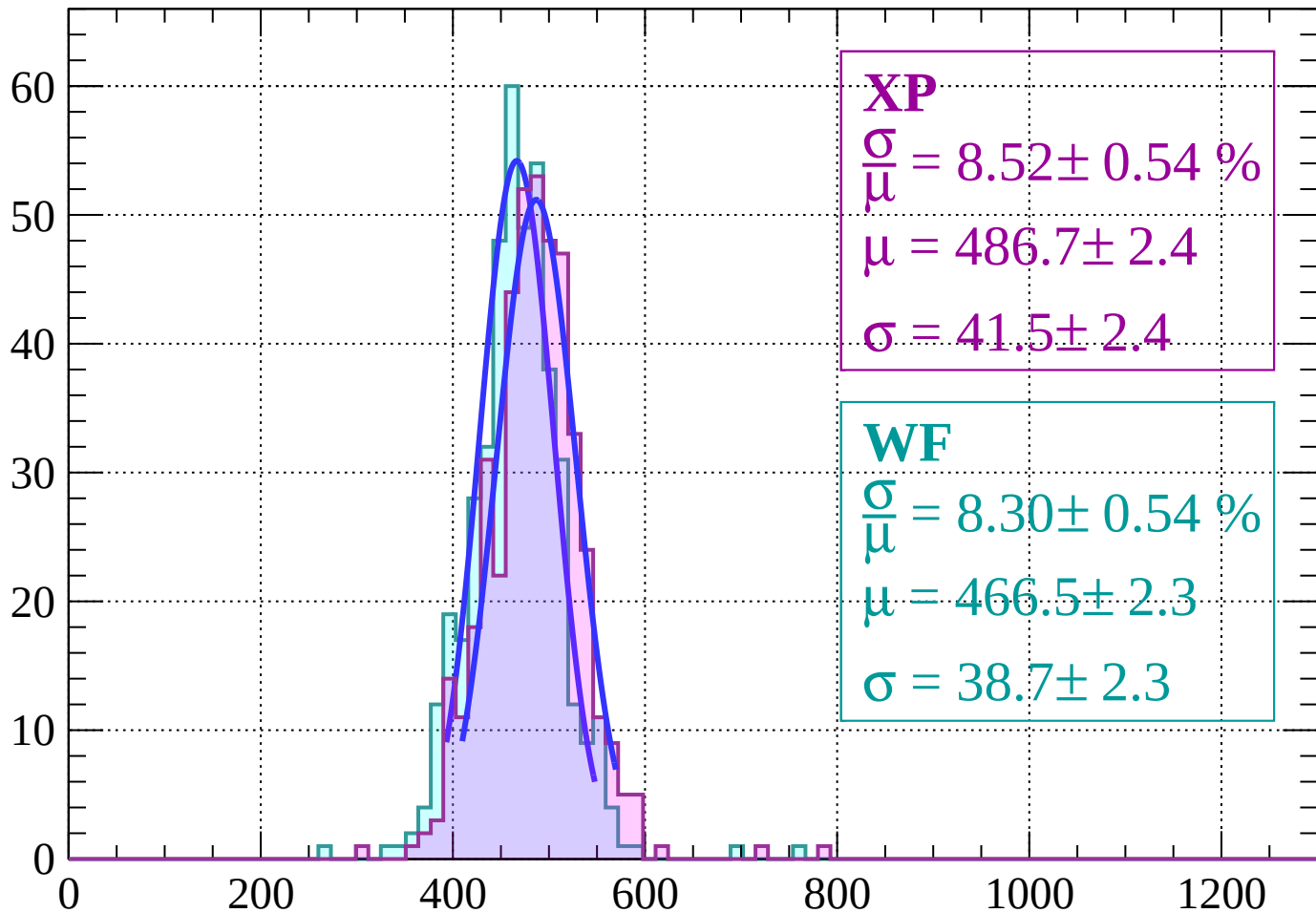
$$\mu = 472.3 \pm 2.2$$

$$\sigma = 41.8 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

Count



XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.54 \%$$

$$\mu = 486.7 \pm 2.4$$

$$\sigma = 41.5 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.54 \%$$

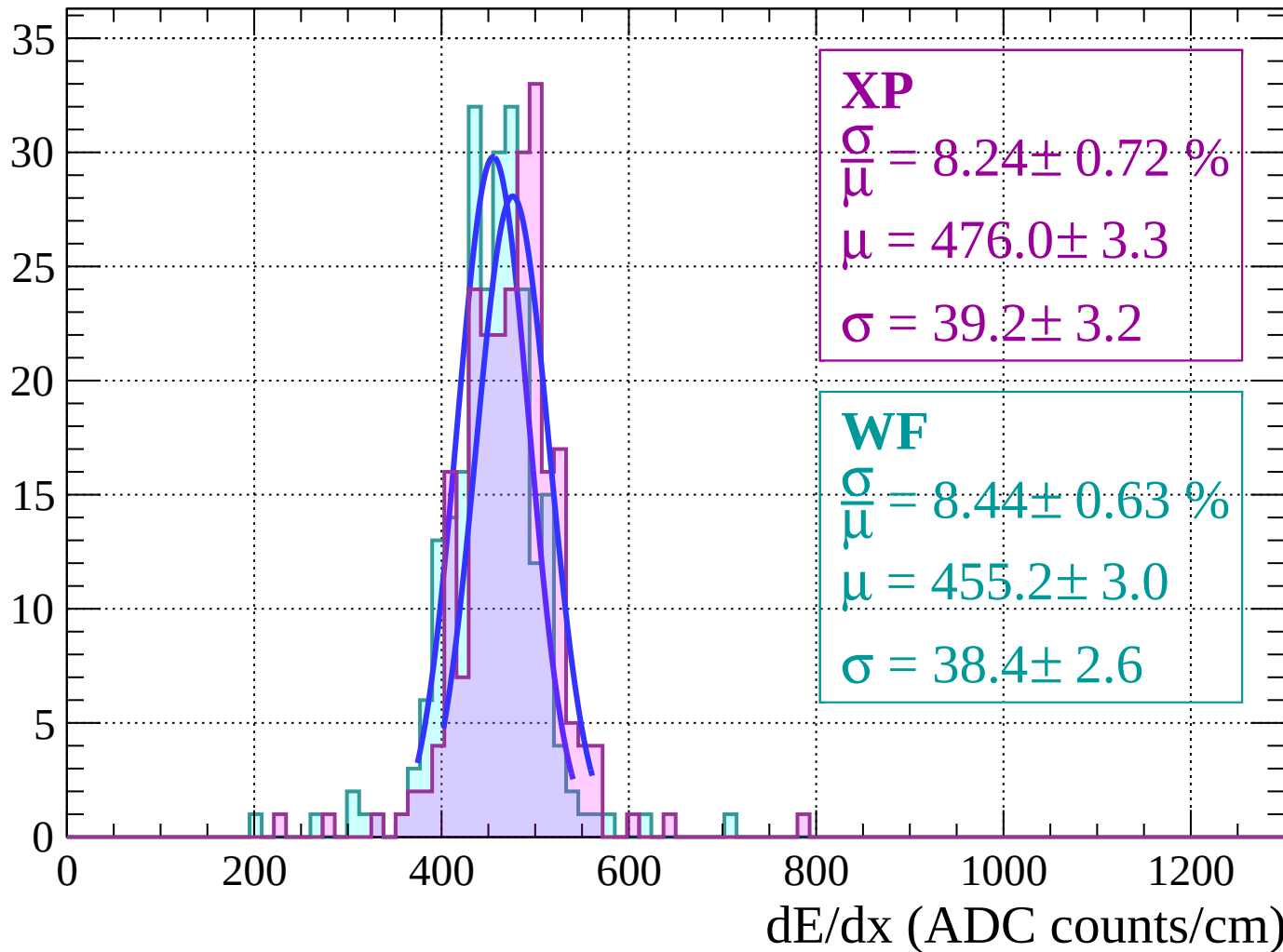
$$\mu = 466.5 \pm 2.3$$

$$\sigma = 38.7 \pm 2.3$$

dE/dx (ADC counts/cm)

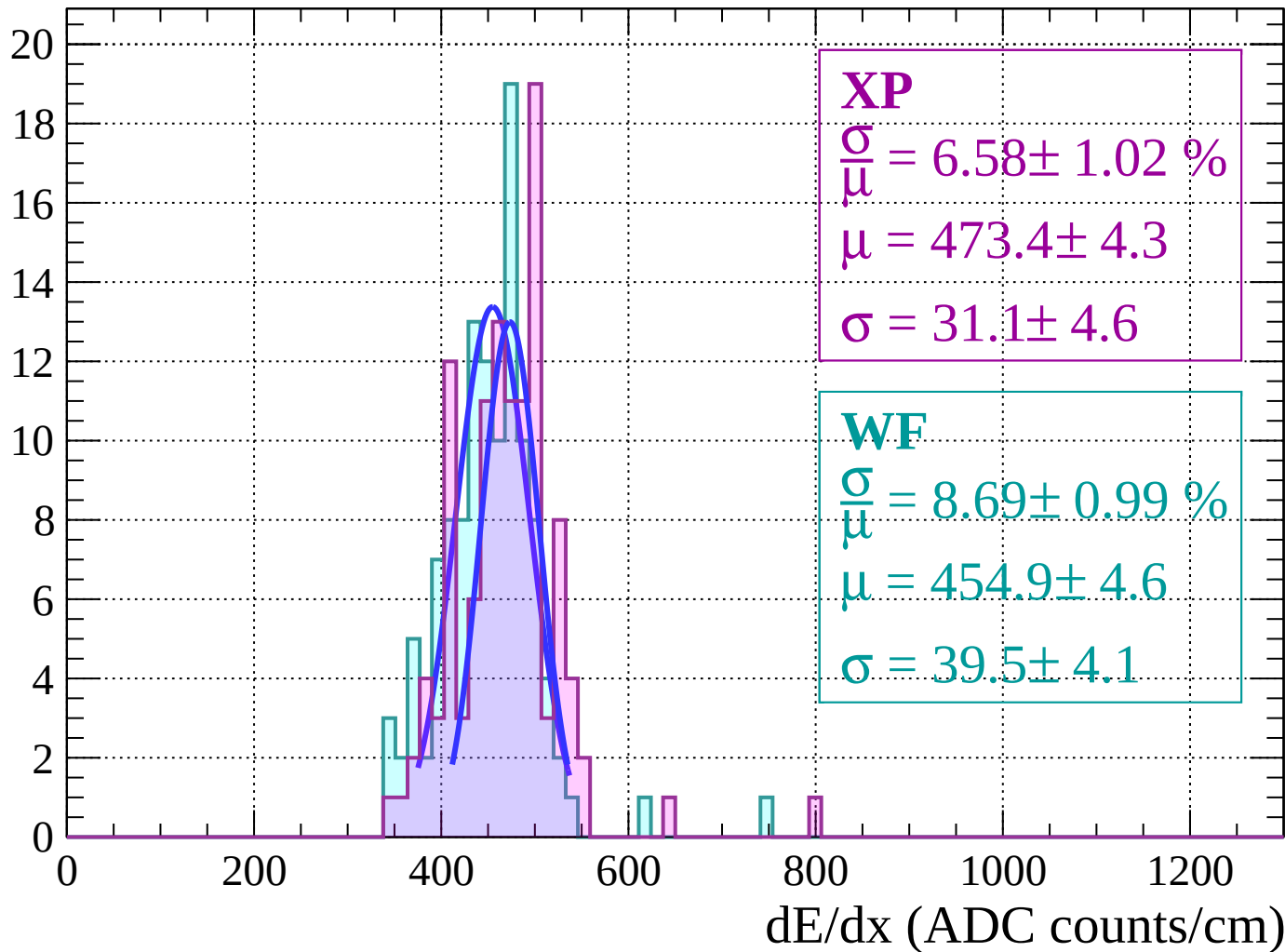
Energy loss | $20 < \theta < 22$

Count



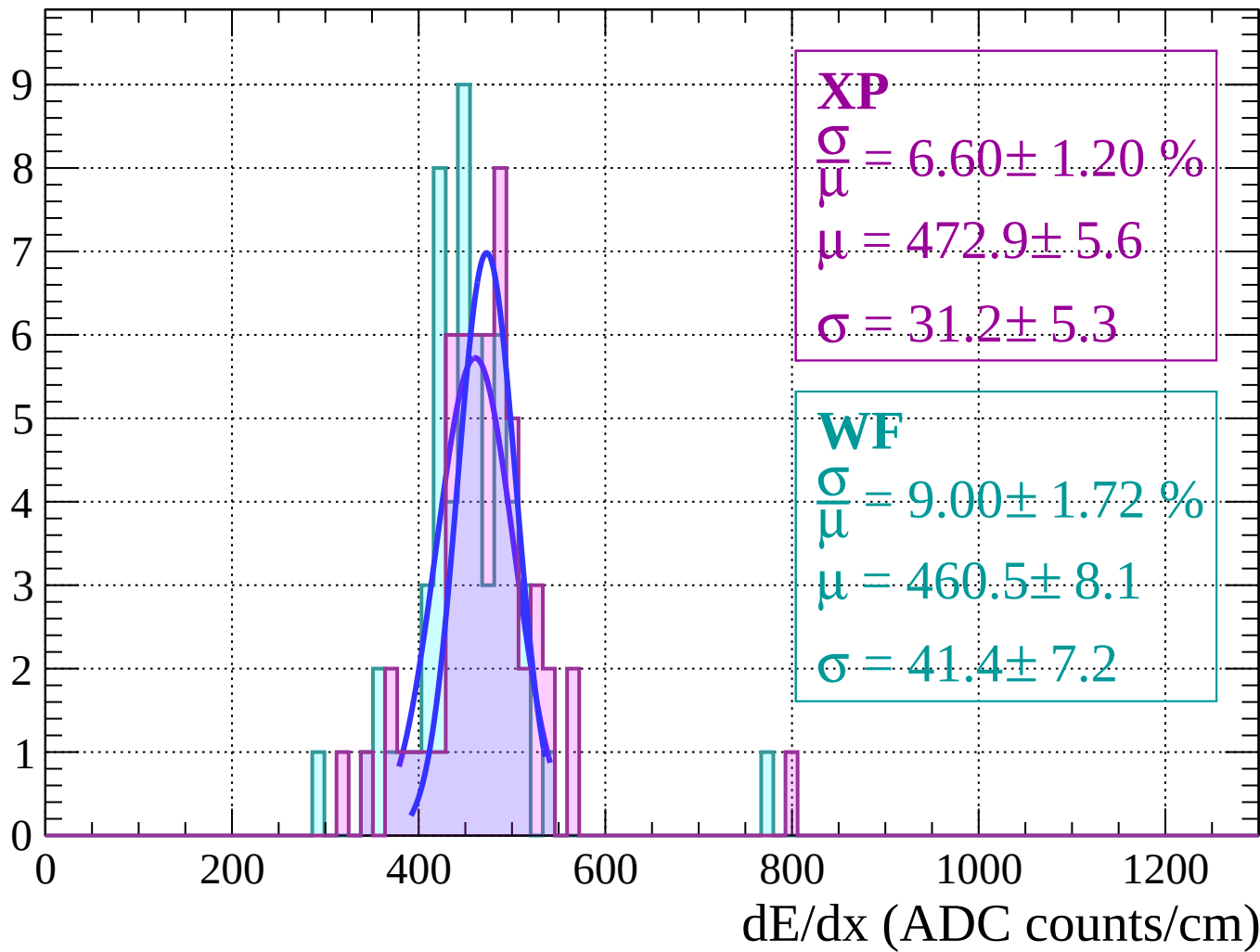
Energy loss | $22 < \theta < 24$

Count



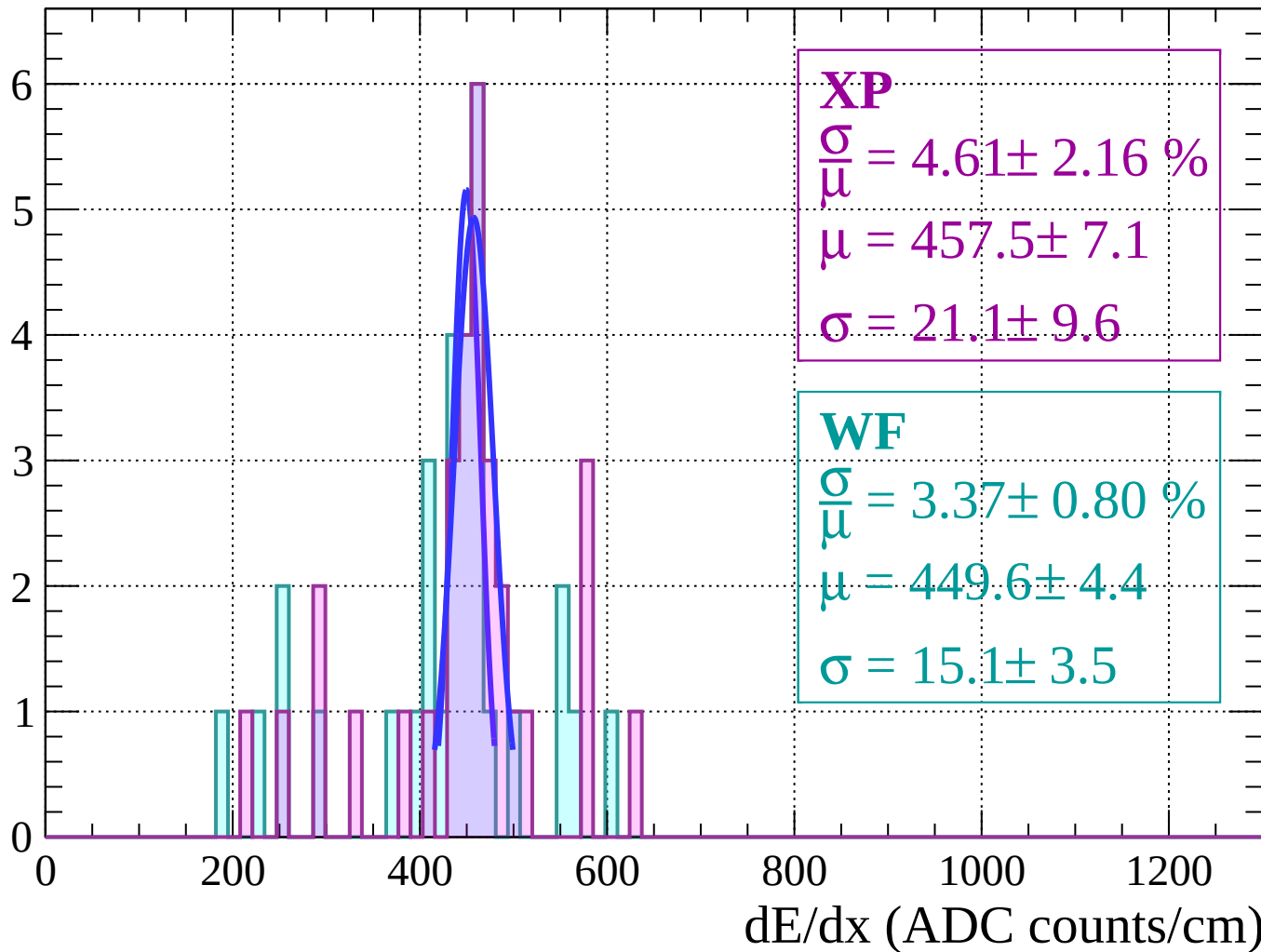
Energy loss | $24 < \theta < 26$

Count



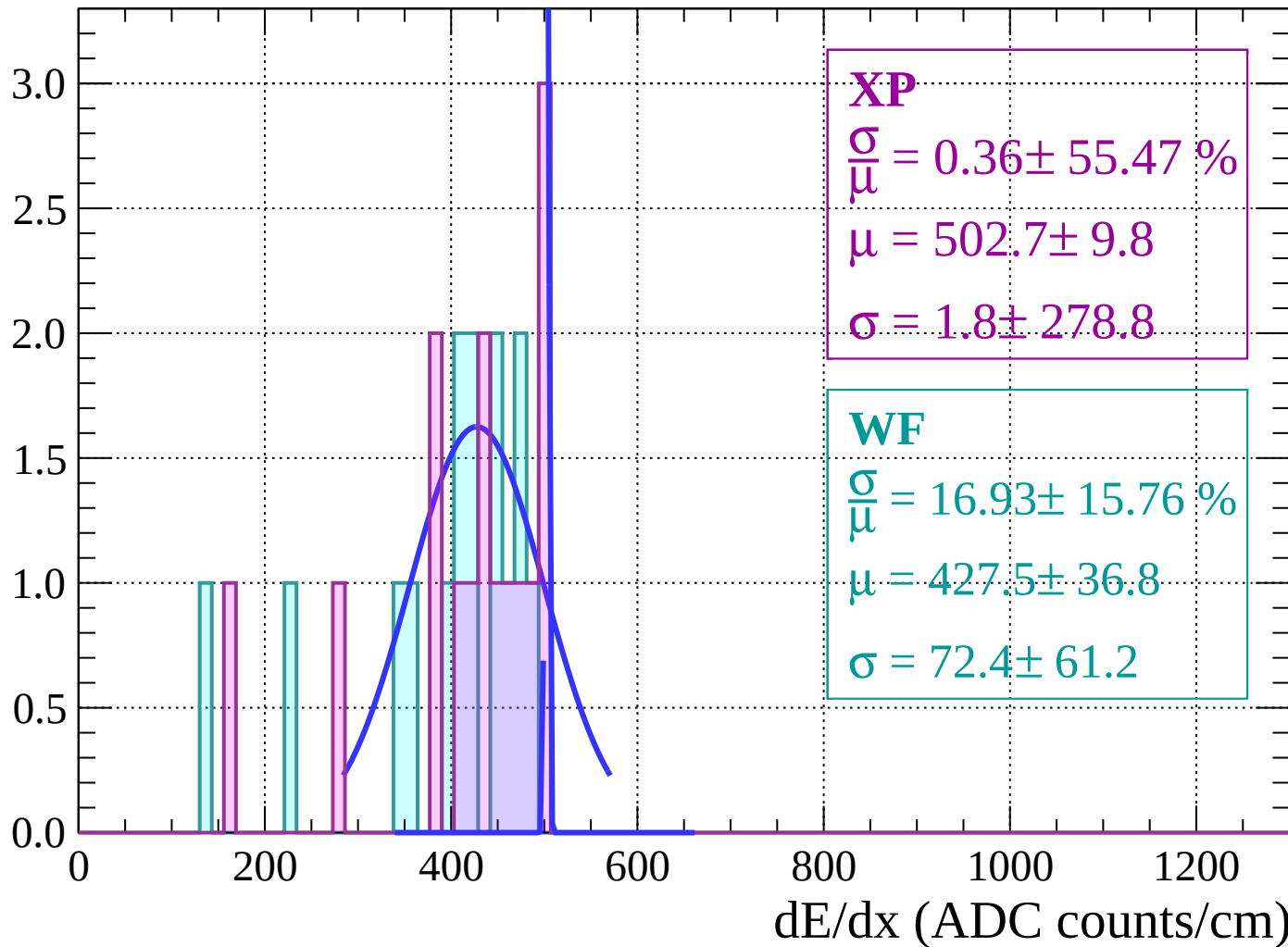
Energy loss | $26 < \theta < 28$

Count



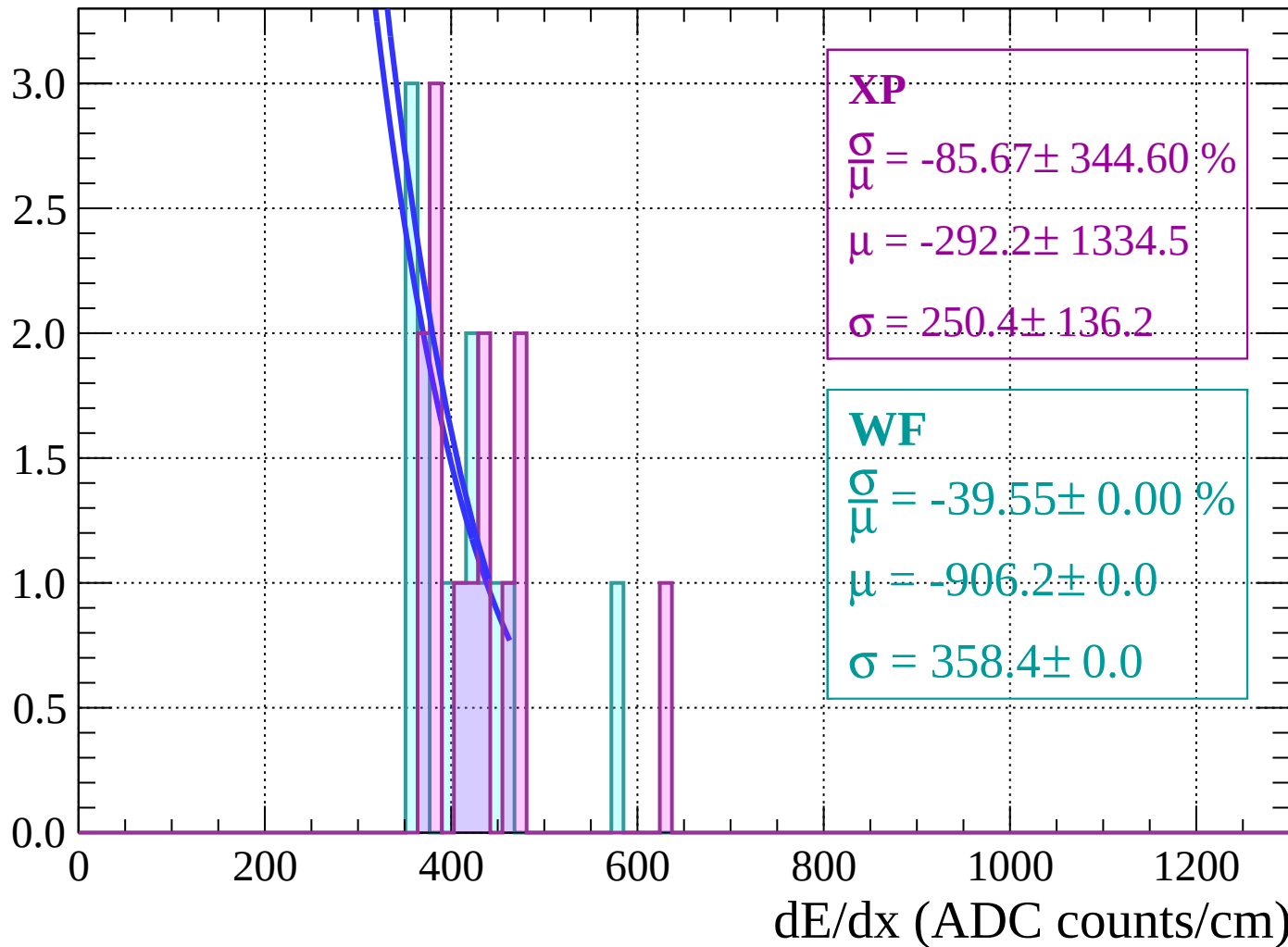
Energy loss | $28 < \theta < 30$

Count



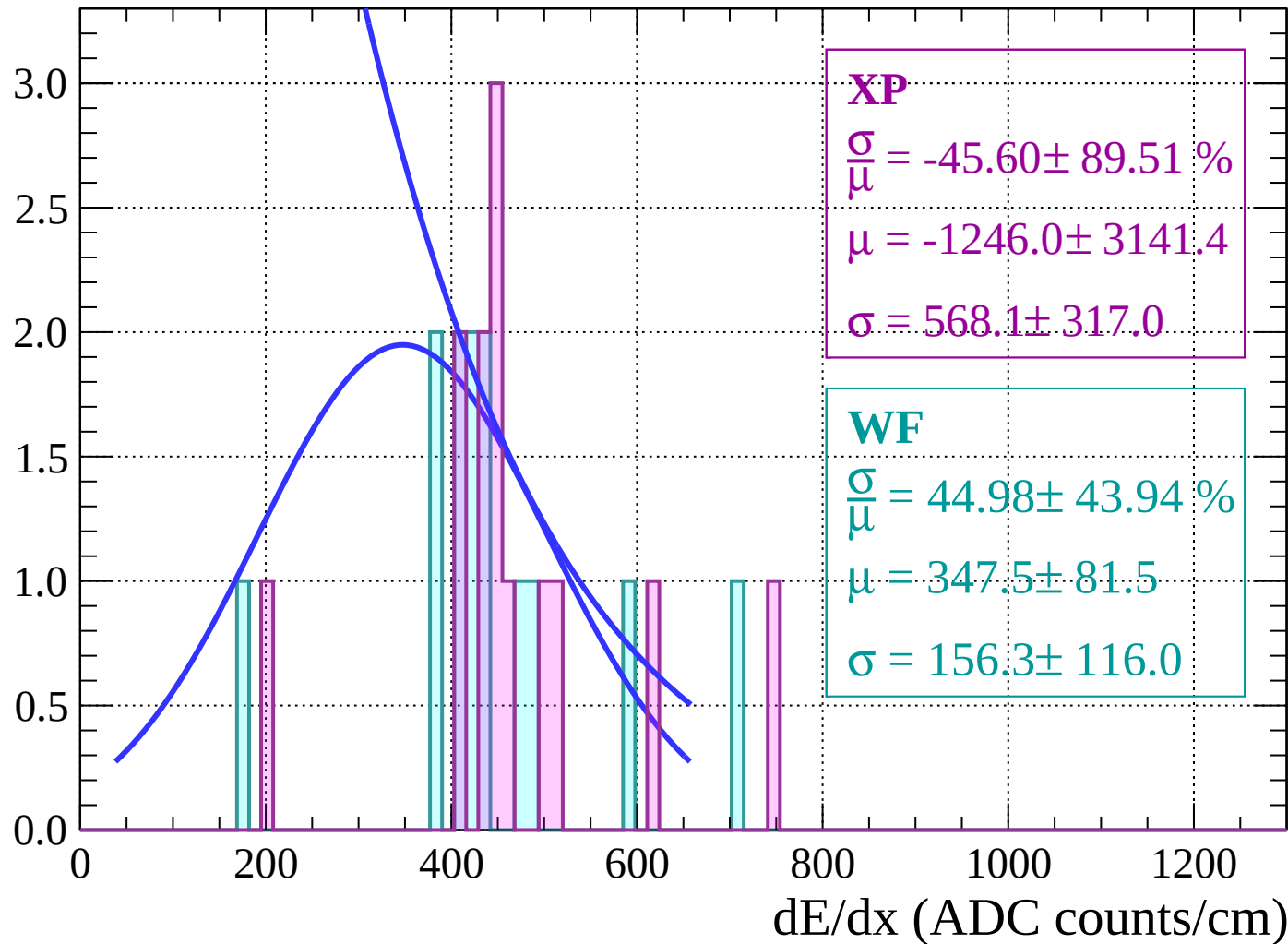
Energy loss | $30 < \theta < 32$

Count



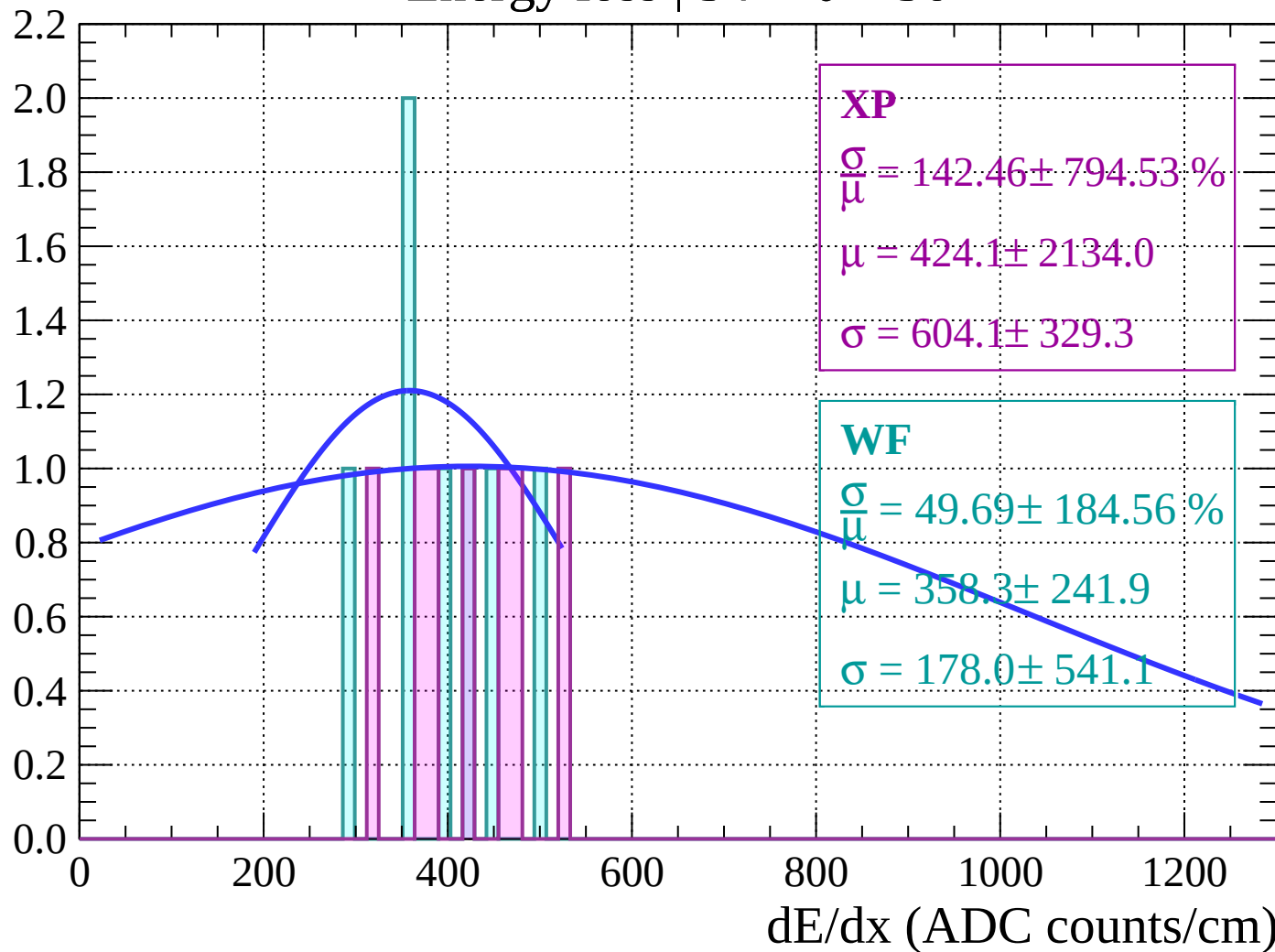
Energy loss | $32 < \theta < 34$

Count



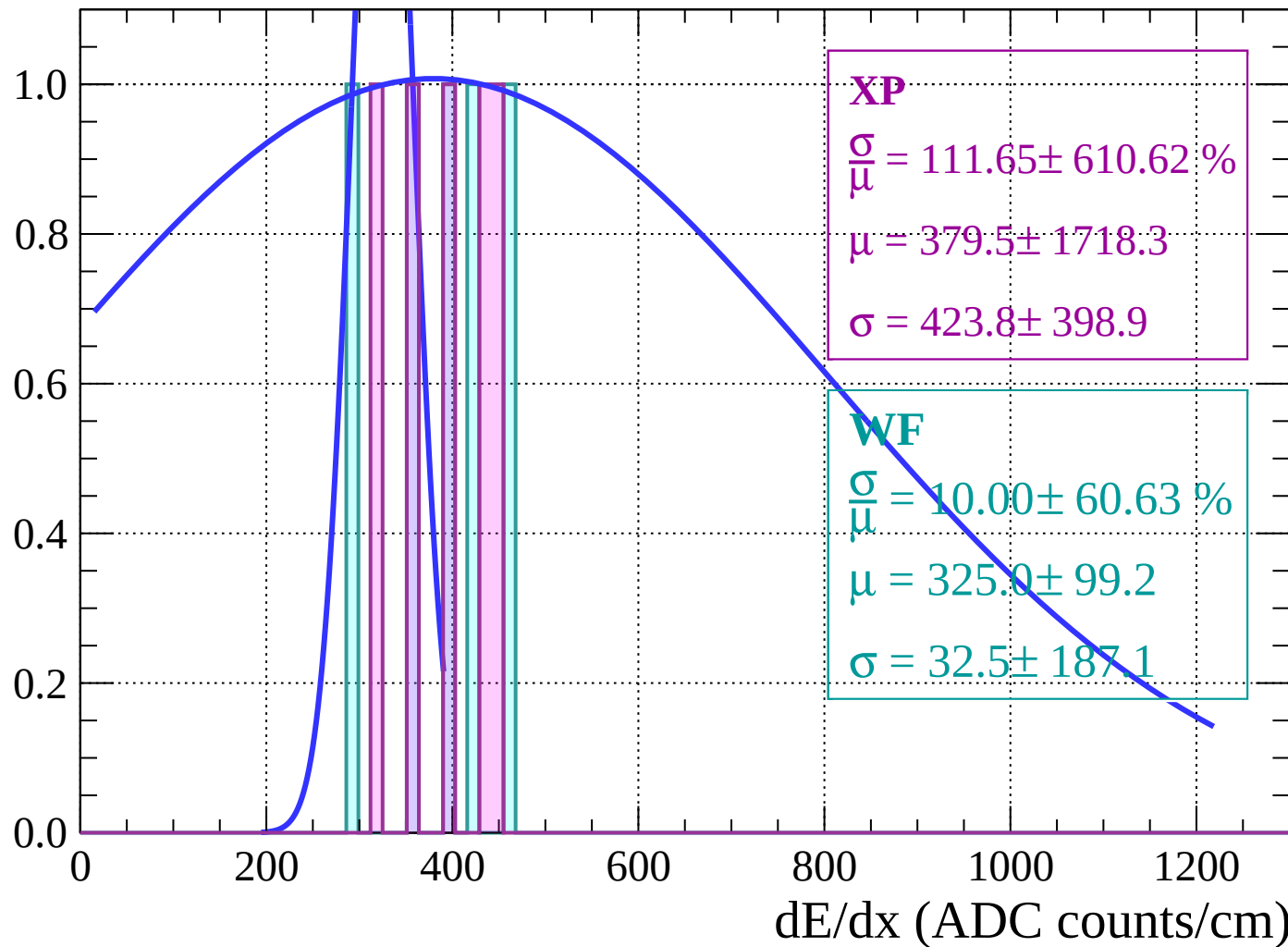
Energy loss | $34 < \theta < 36$

Count



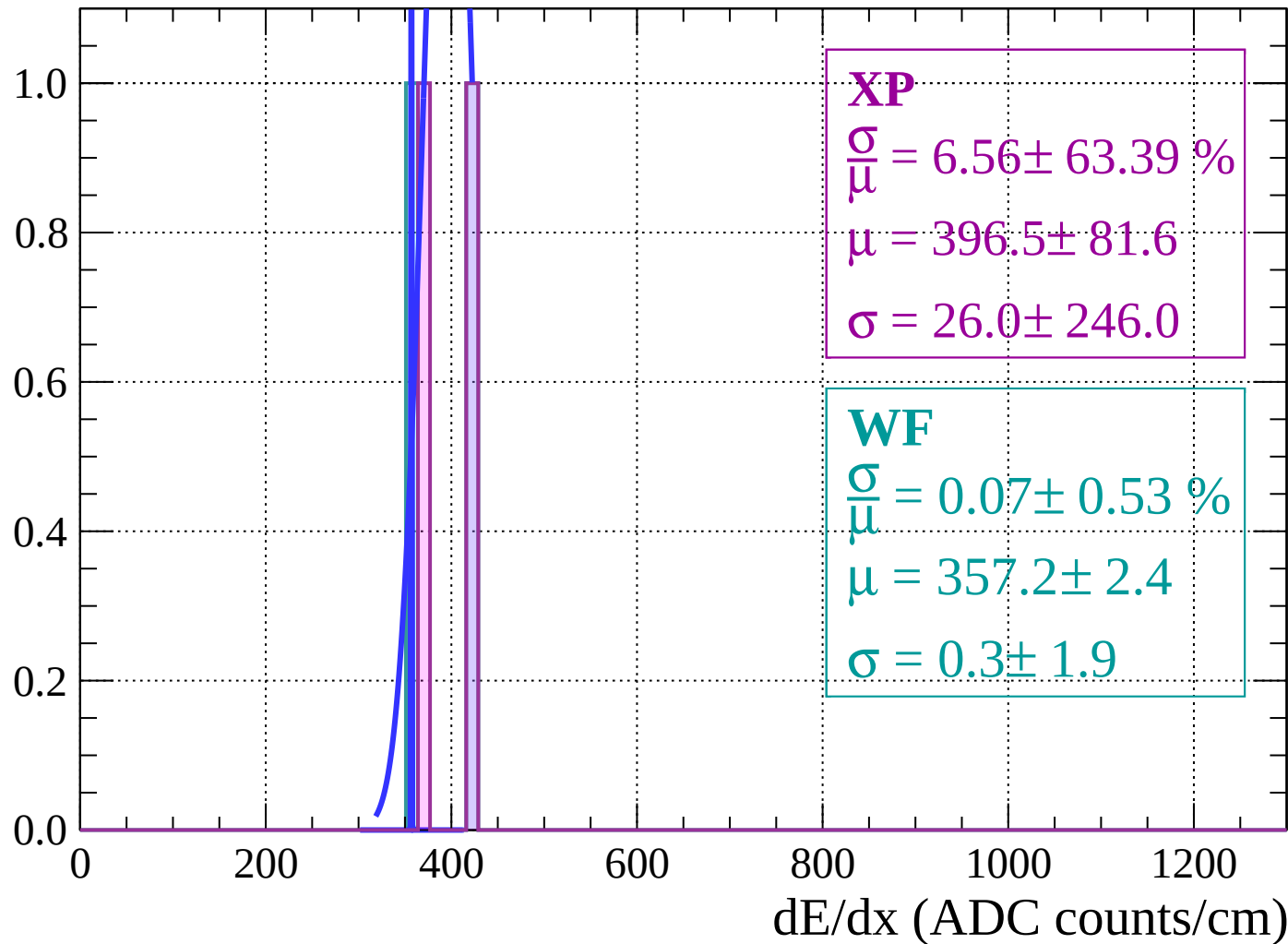
Energy loss | $36 < \theta < 38$

Count



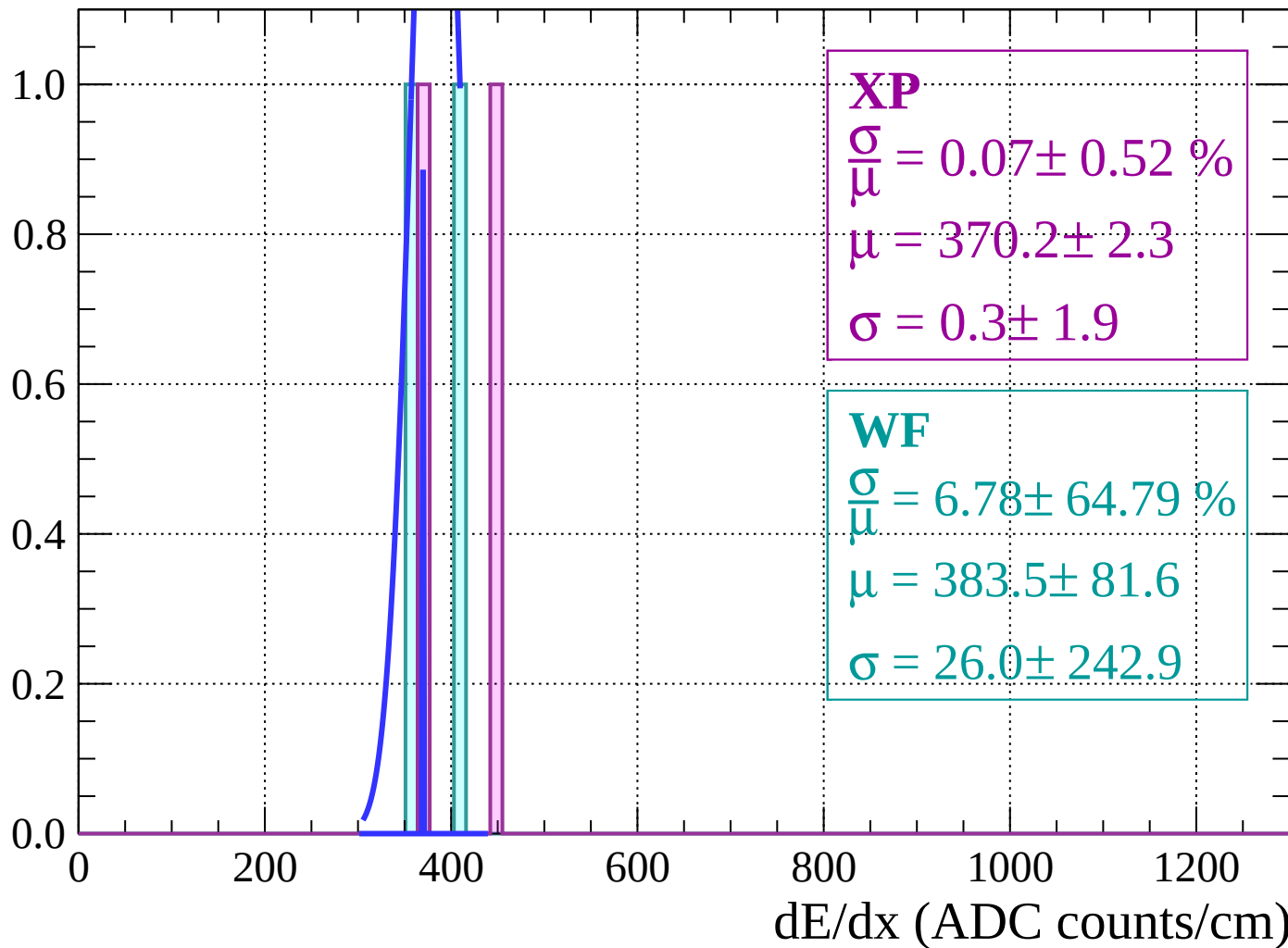
Energy loss | $38 < \theta < 40$

Count



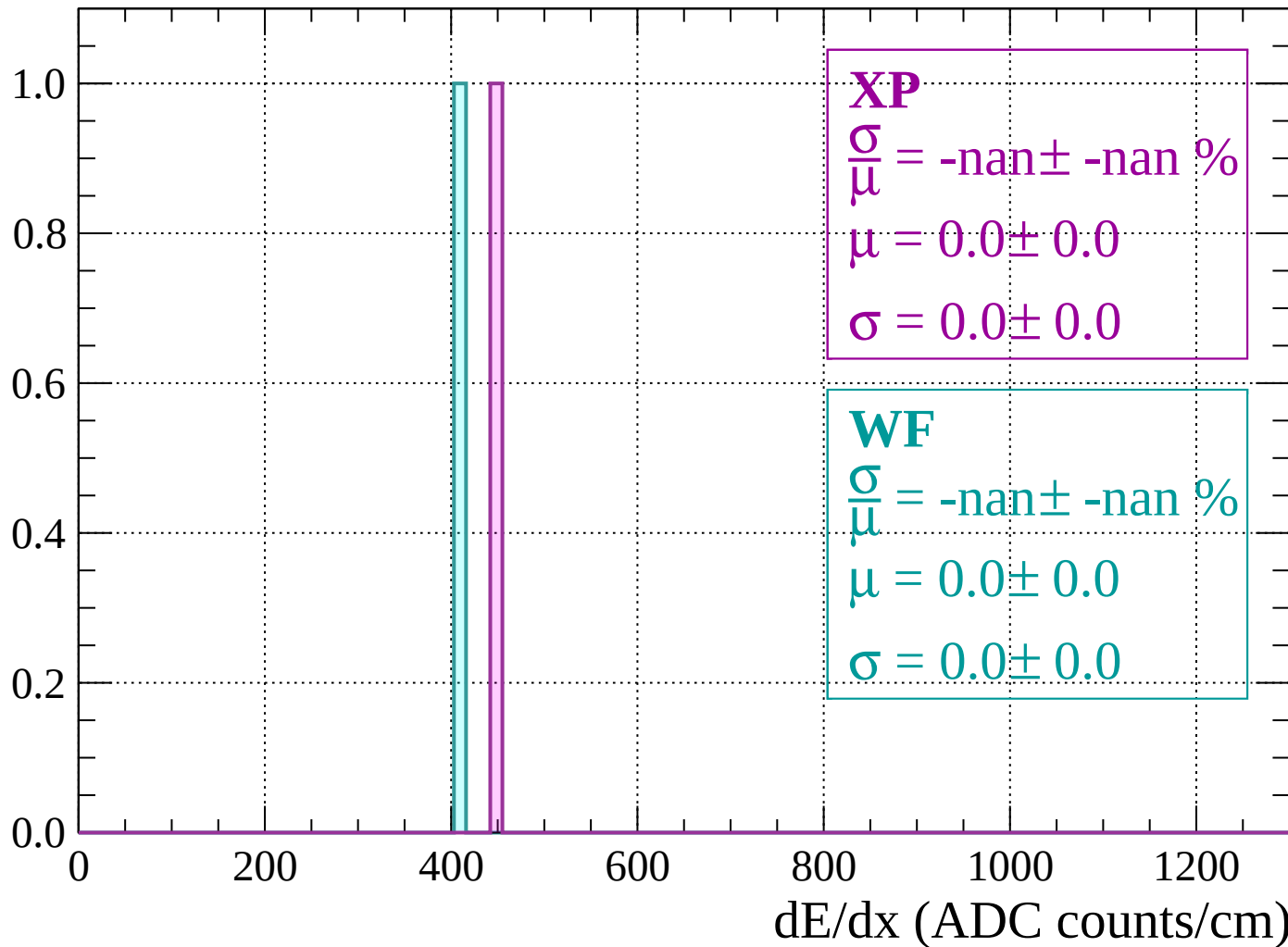
Energy loss | $40 < \theta < 42$

Count



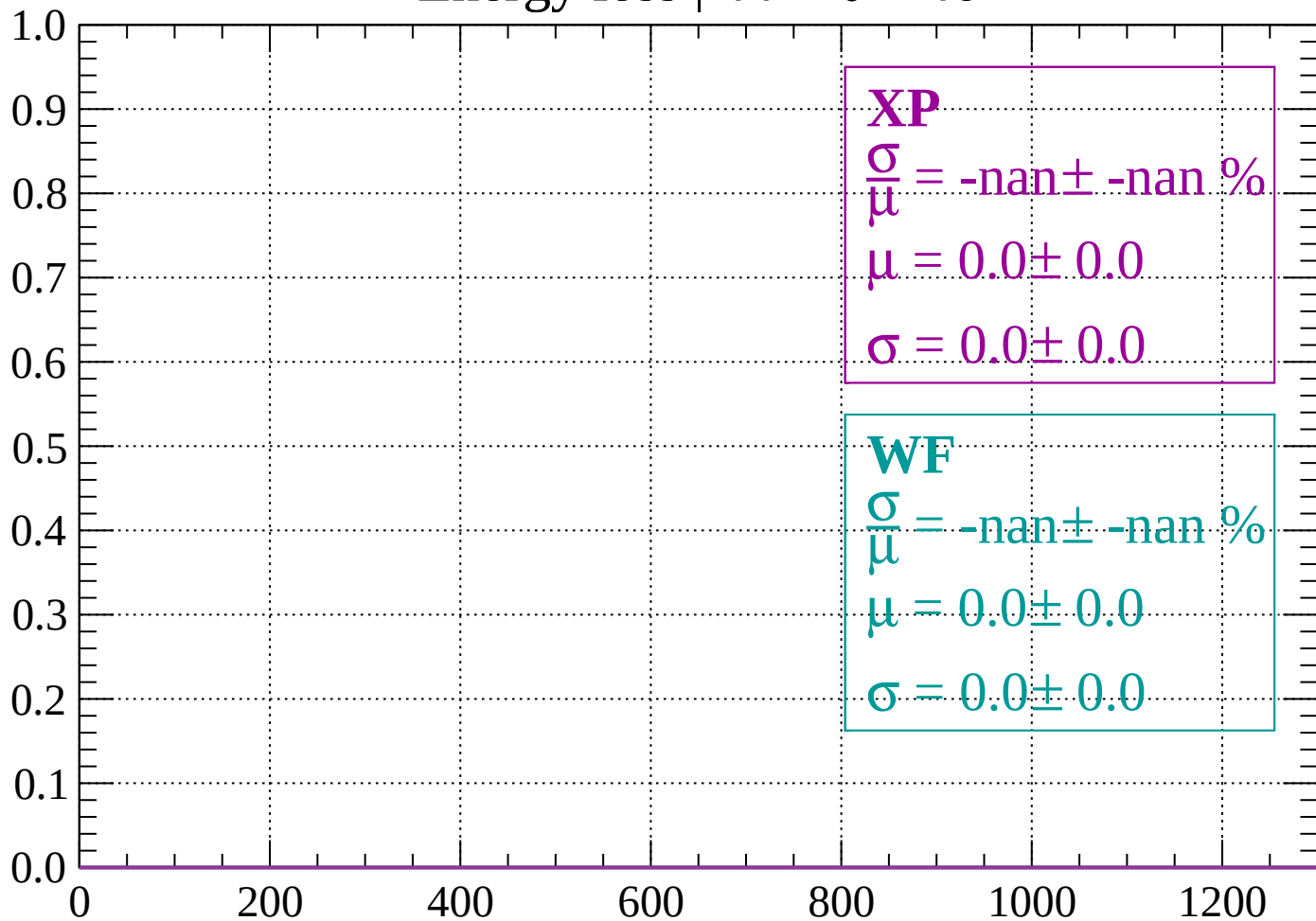
Energy loss | $42 < \theta < 44$

Count



Energy loss | $44 < \theta < 46$

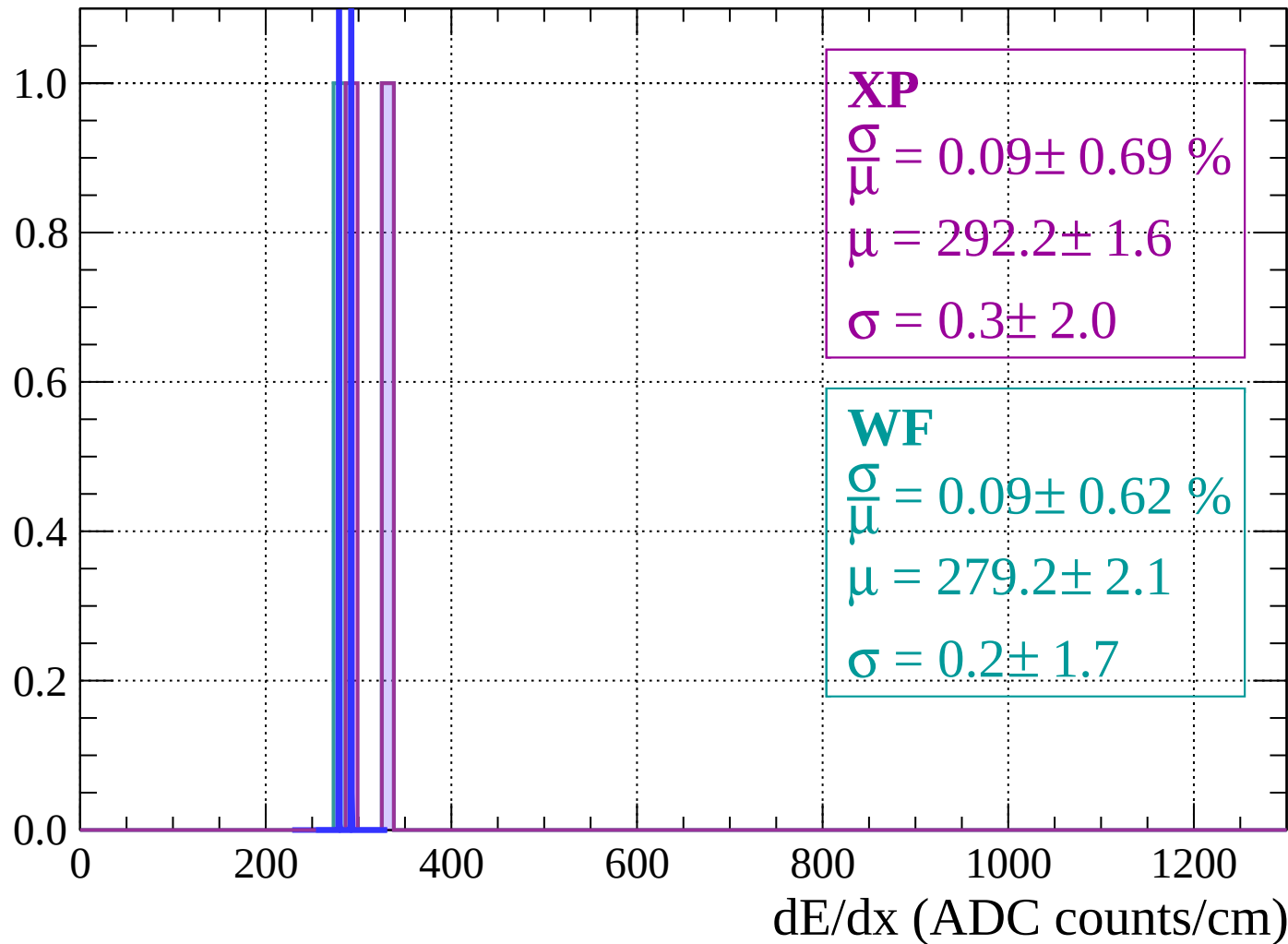
Count



dE/dx (ADC counts/cm)

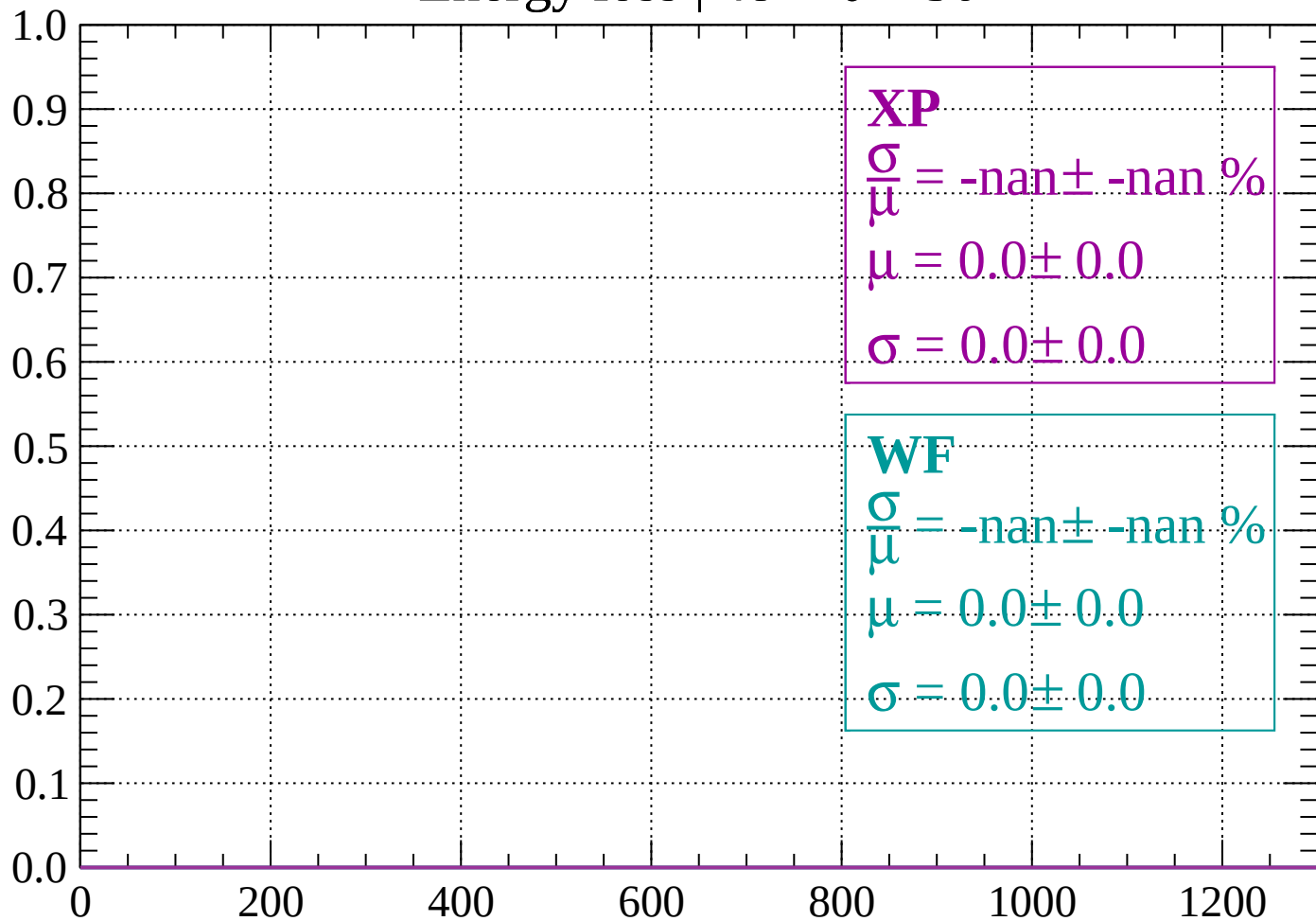
Energy loss | $46 < \theta < 48$

Count



Energy loss | $48 < \theta < 50$

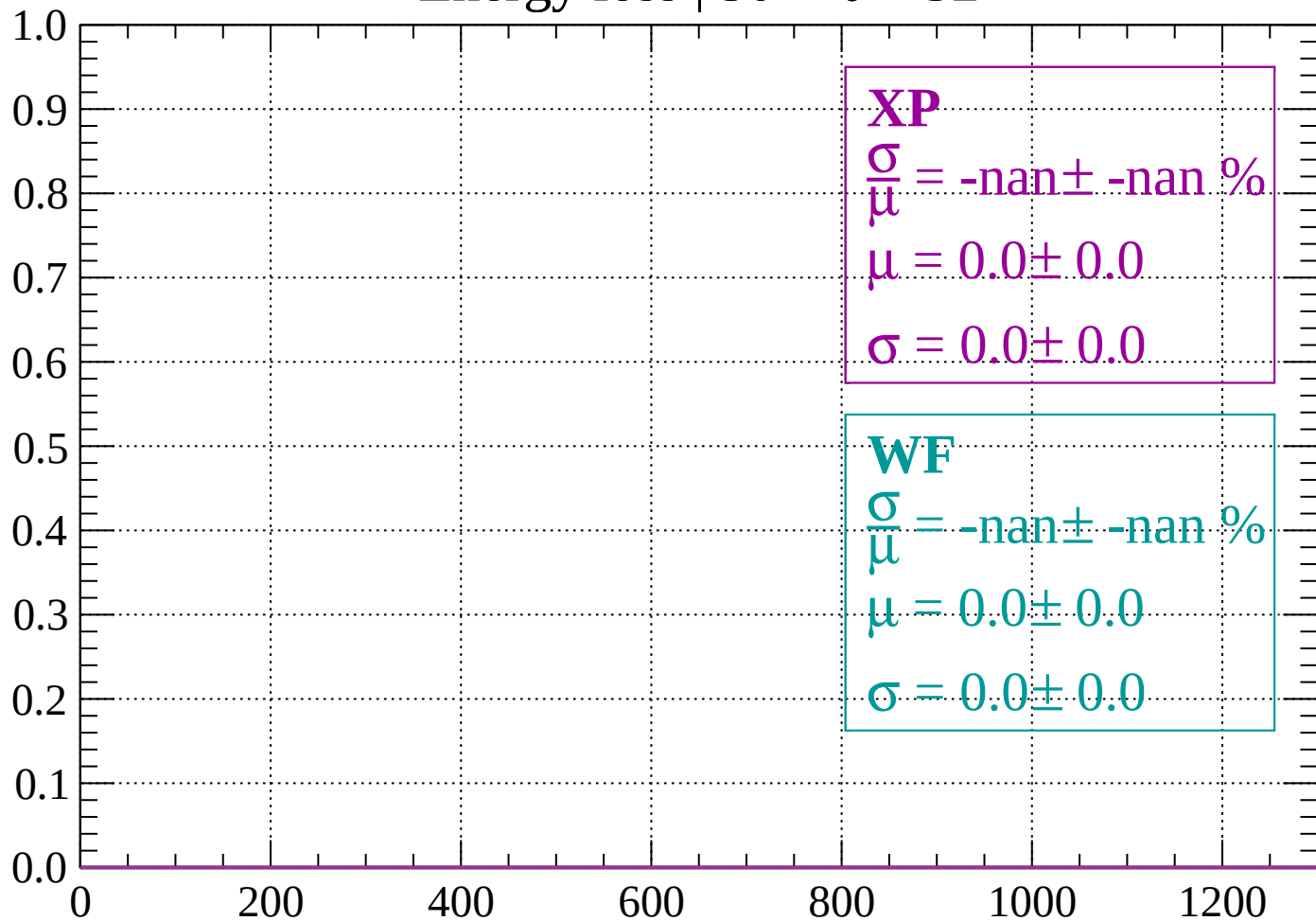
Count



dE/dx (ADC counts/cm)

Energy loss | $50 < \theta < 52$

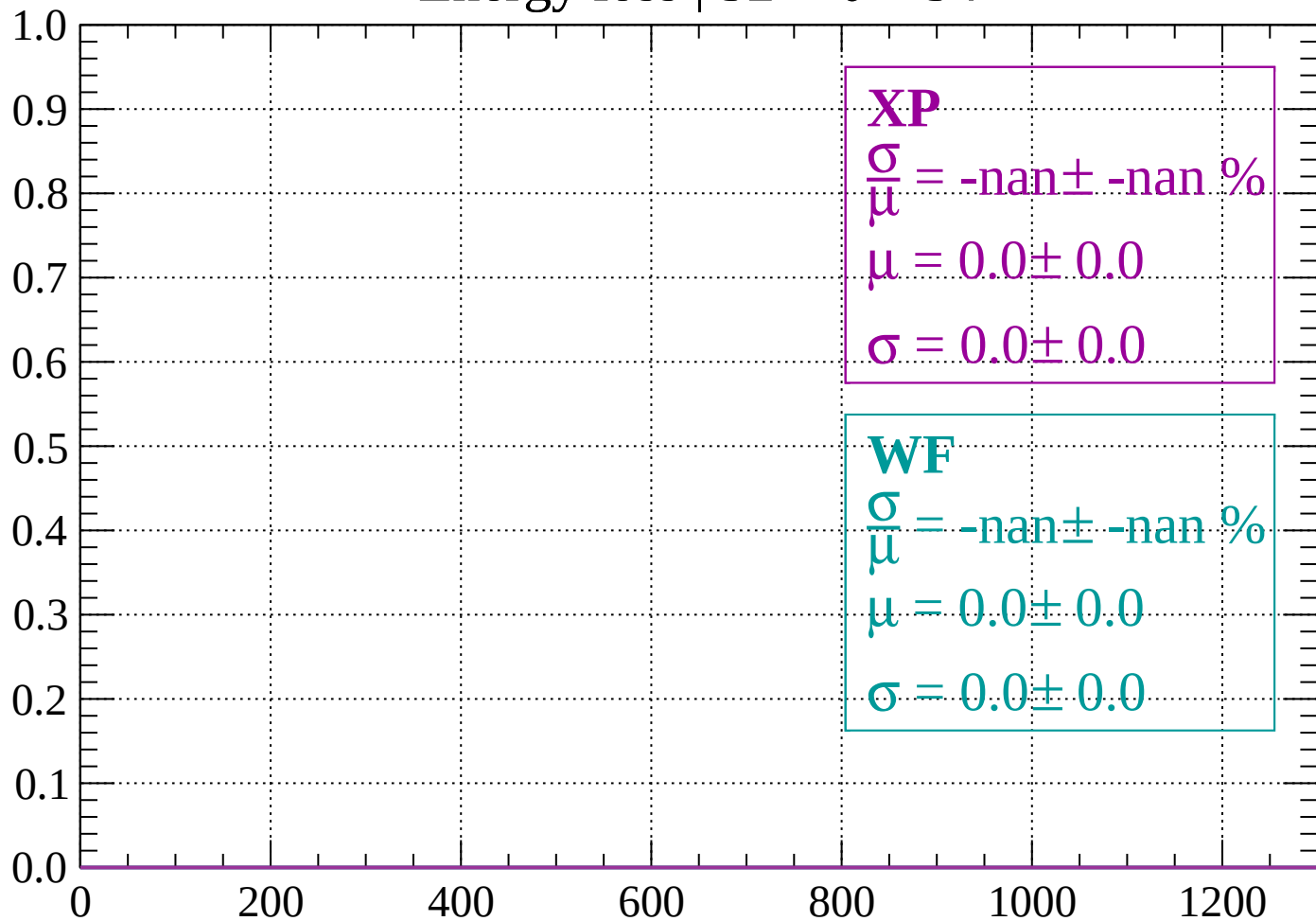
Count



dE/dx (ADC counts/cm)

Energy loss | $52 < \theta < 54$

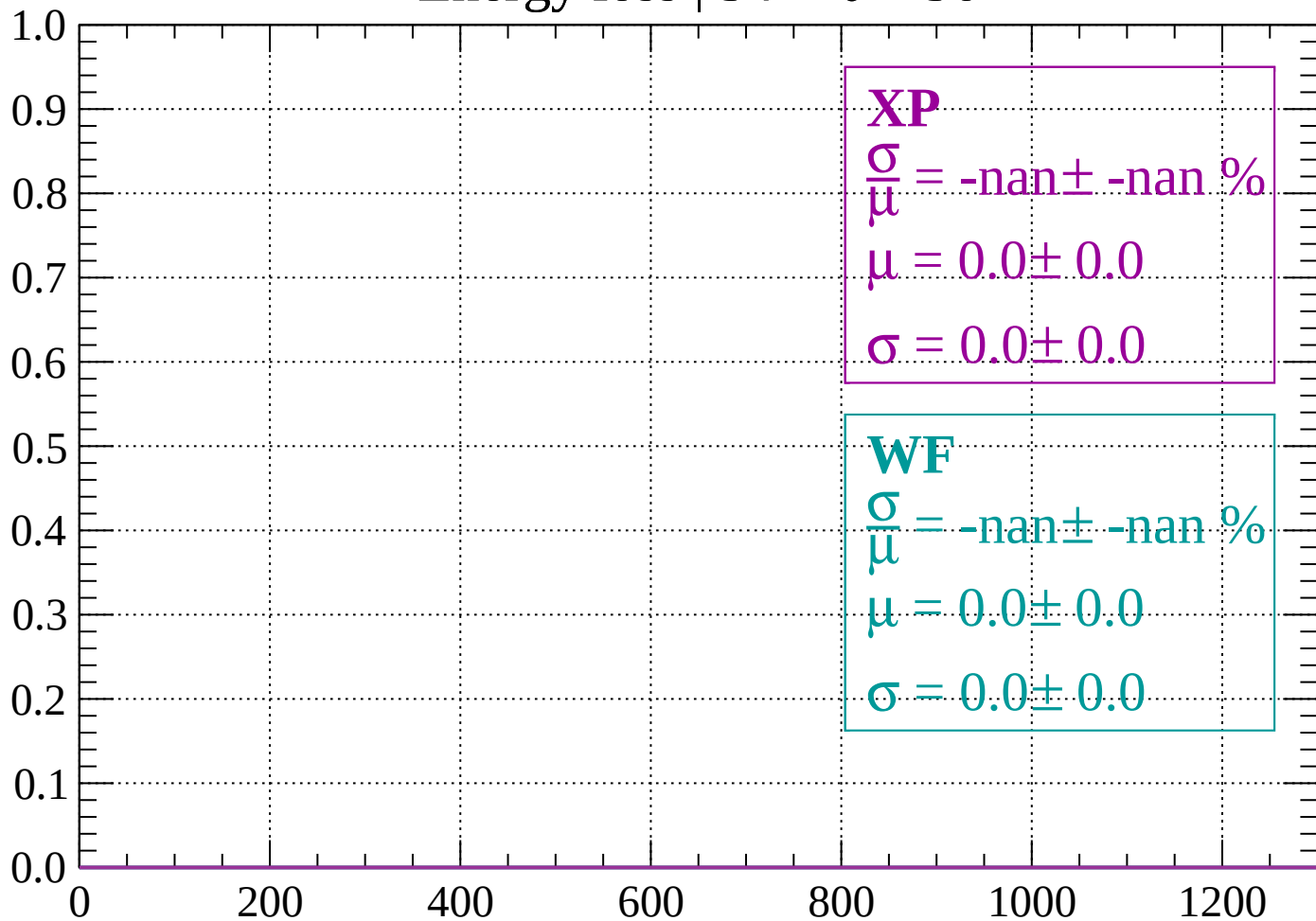
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

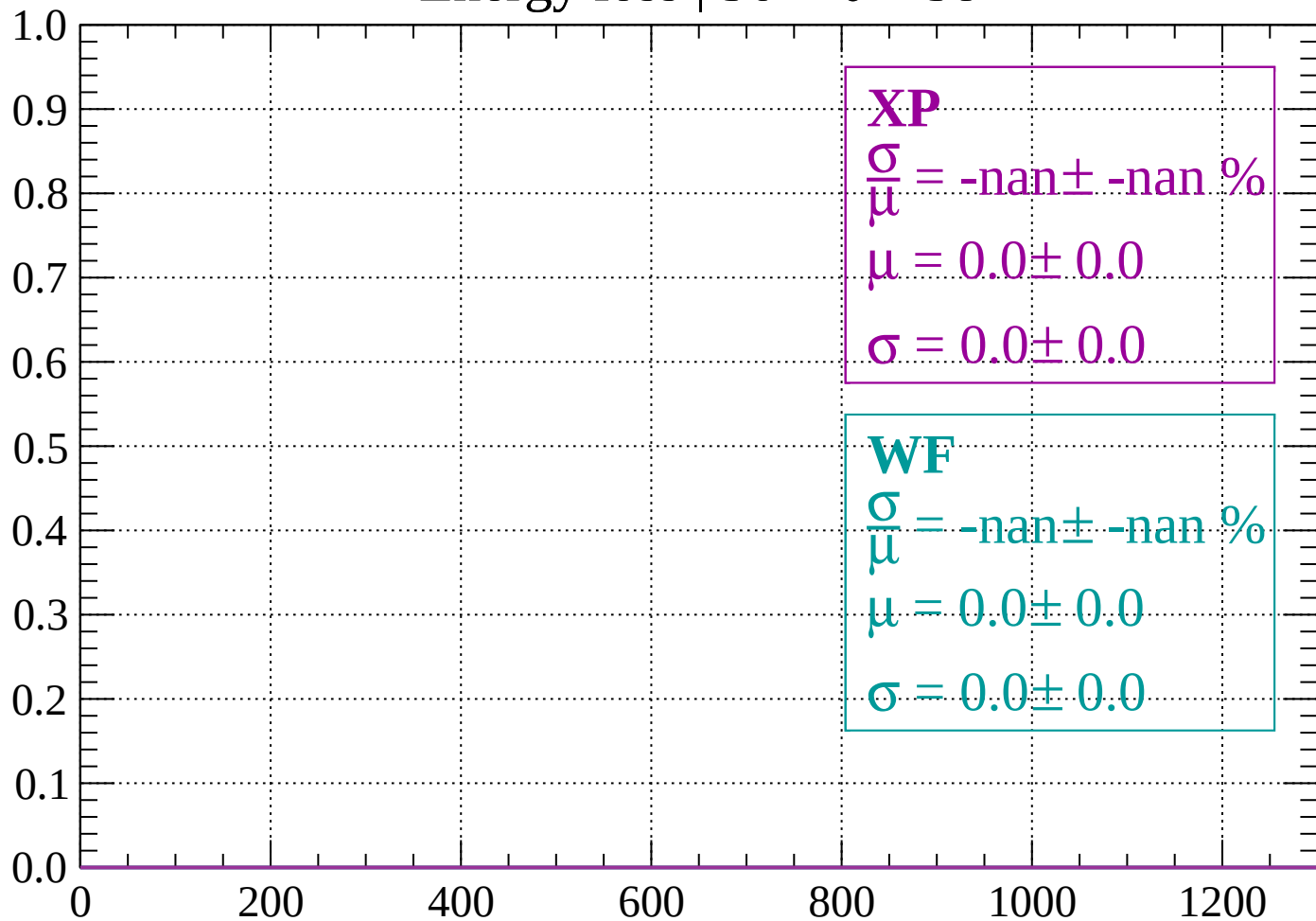
Count



dE/dx (ADC counts/cm)

Energy loss | $56 < \theta < 58$

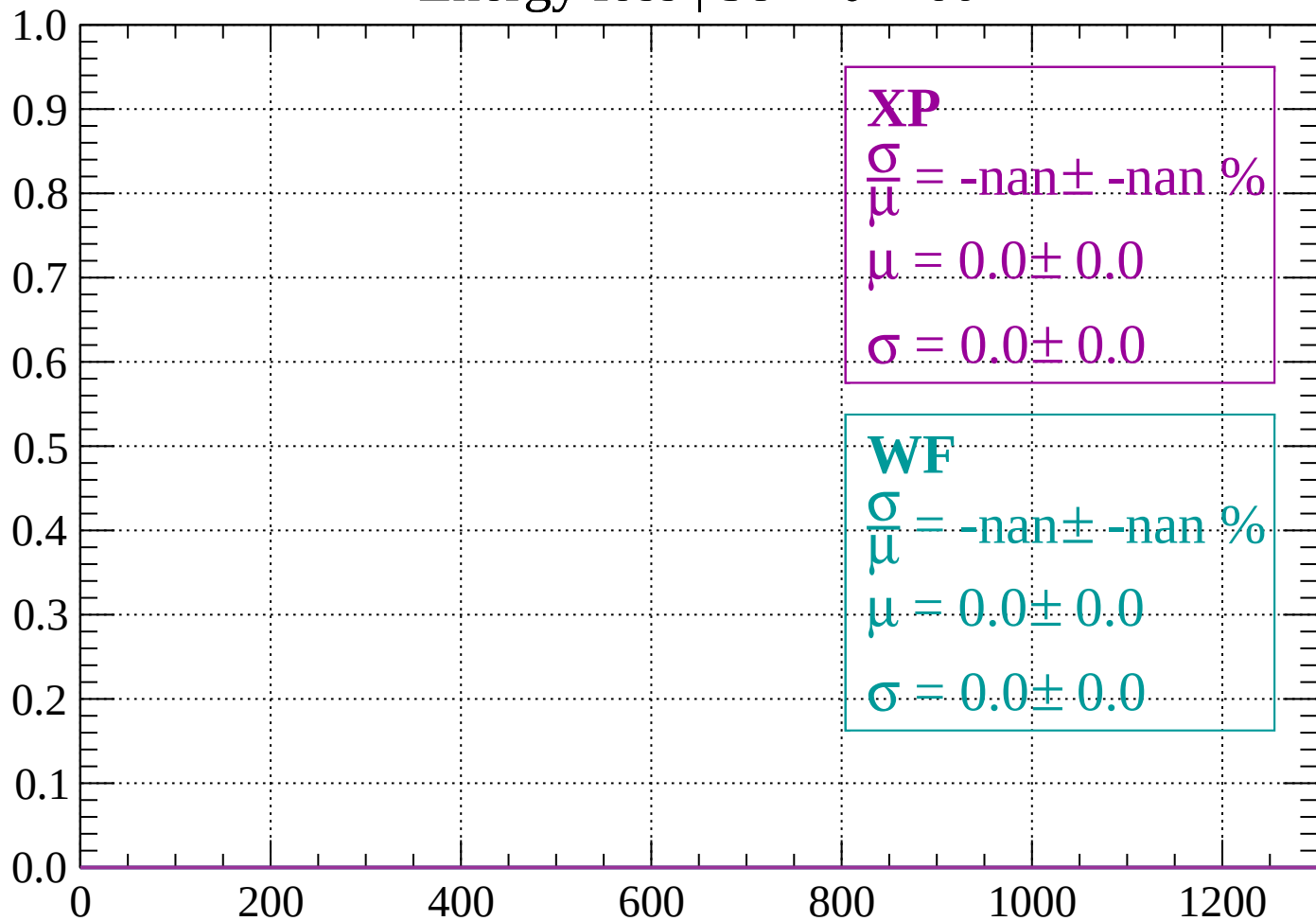
Count



dE/dx (ADC counts/cm)

Energy loss | $58 < \theta < 60$

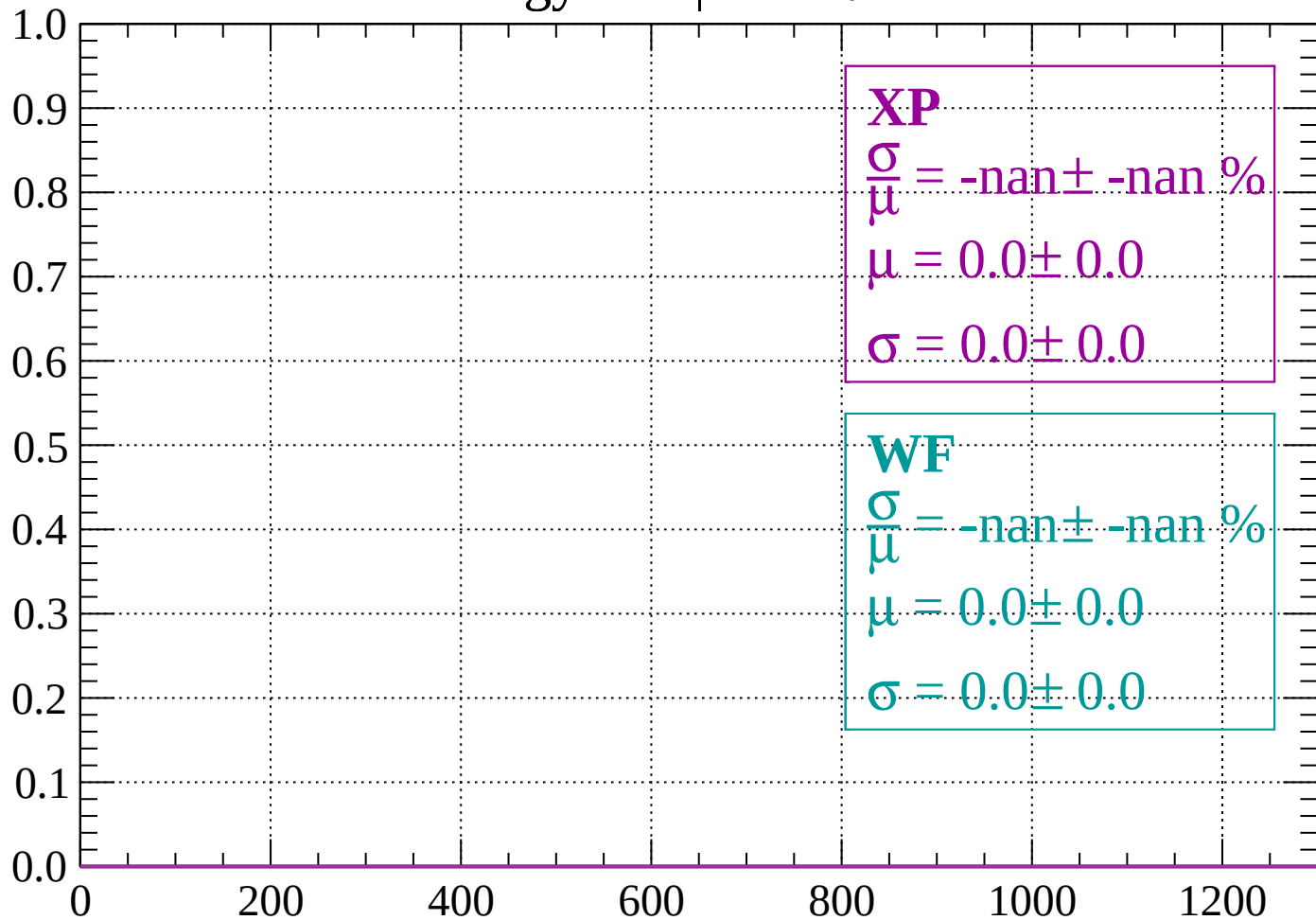
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 62$

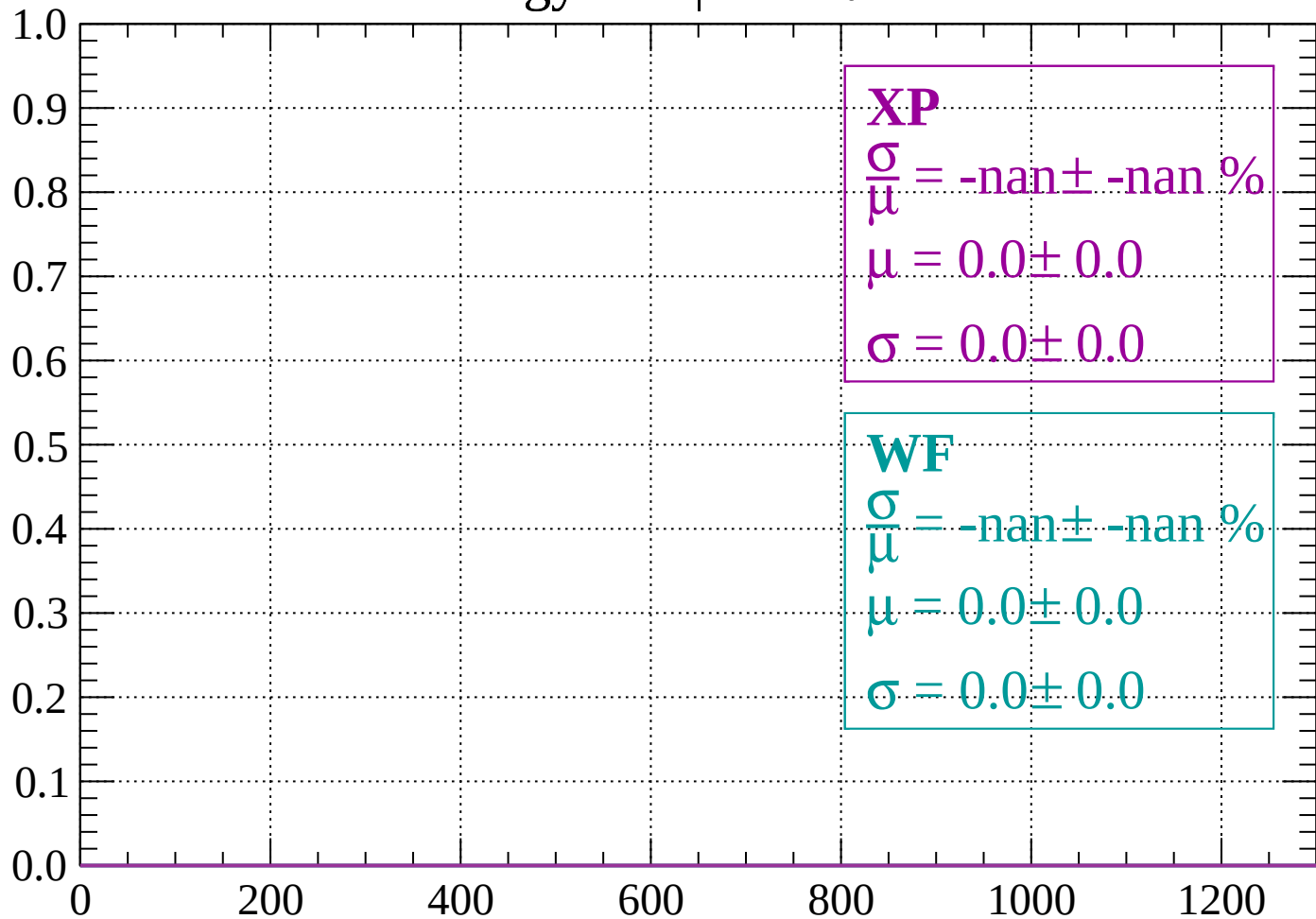
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

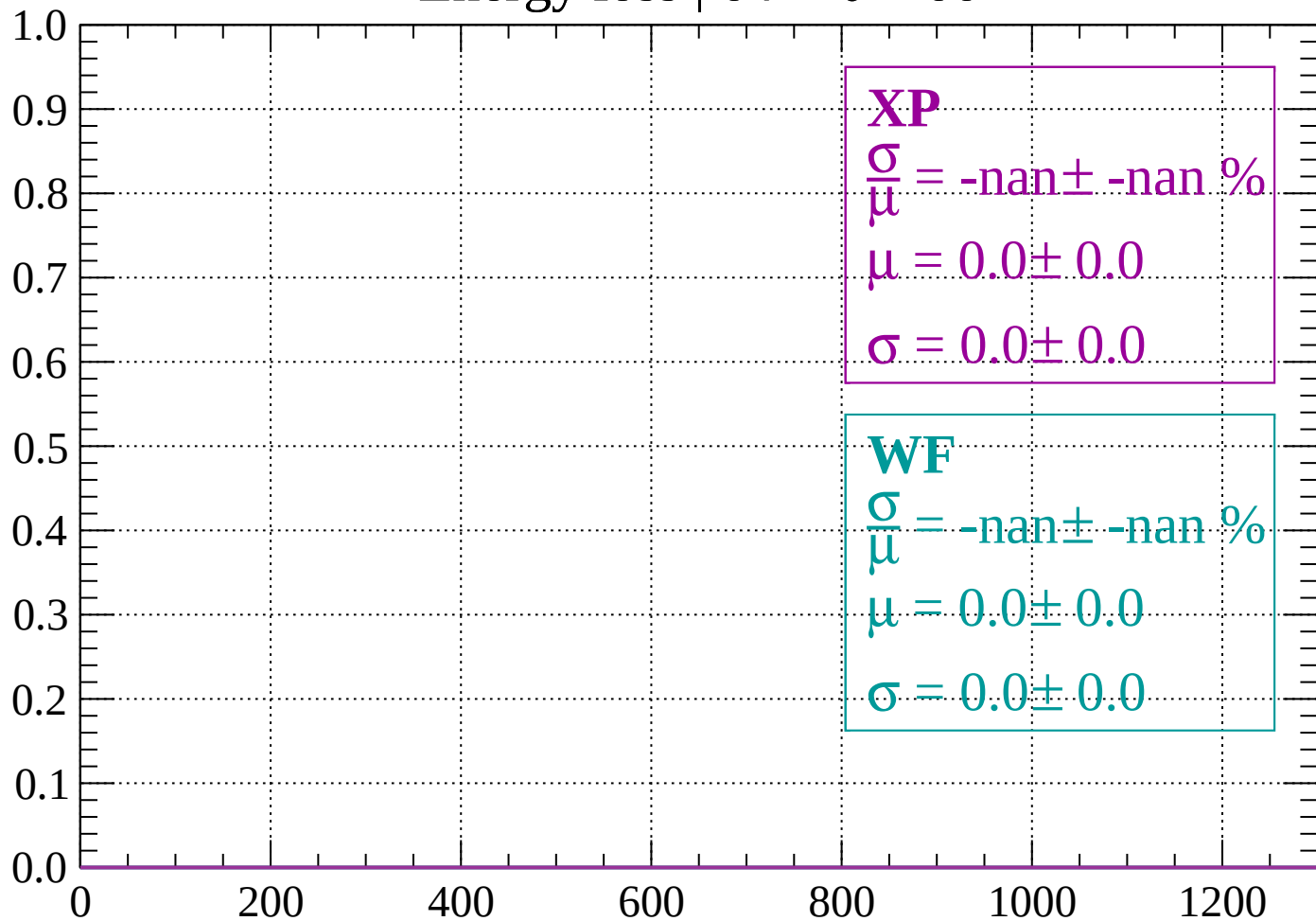
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

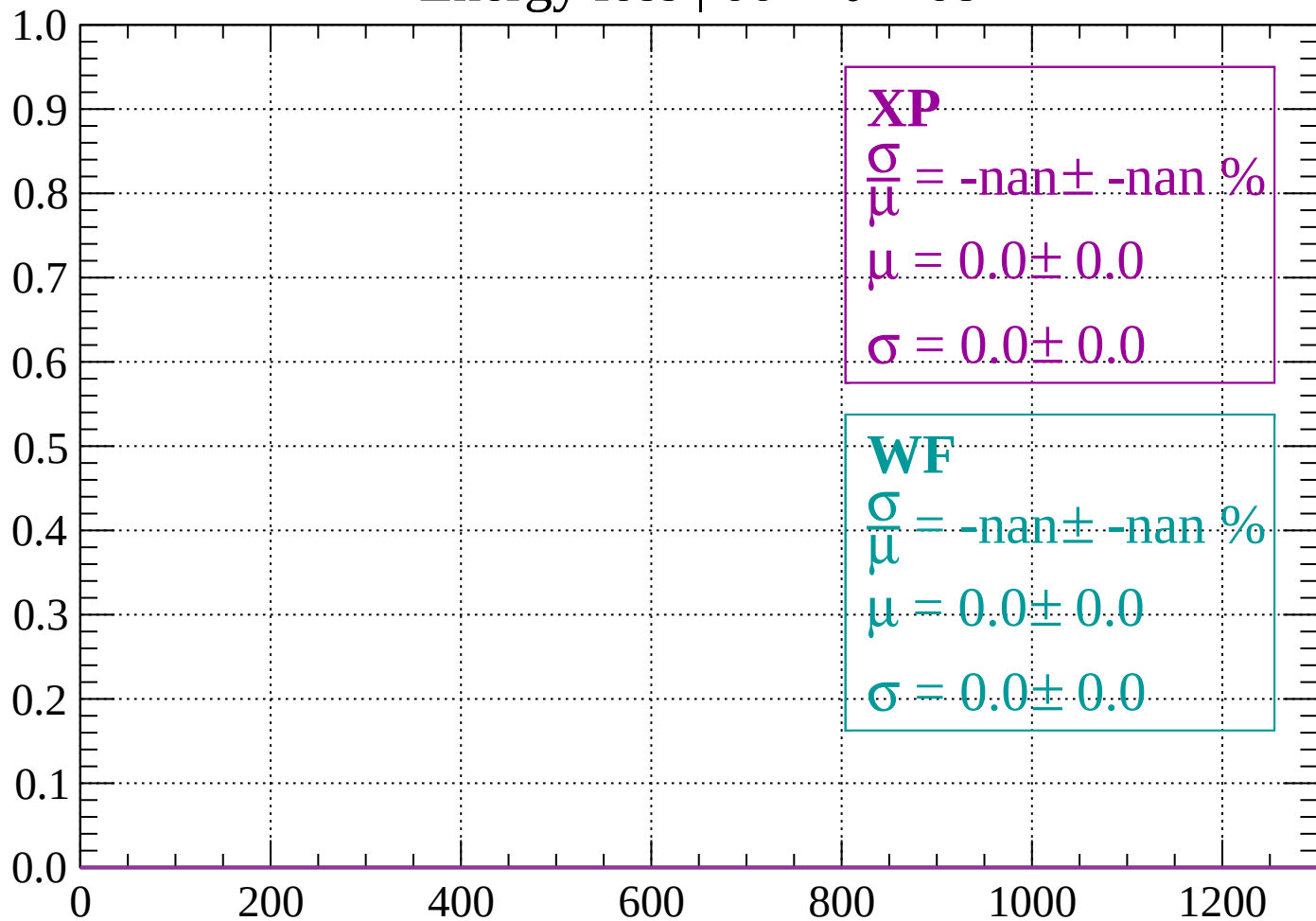
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

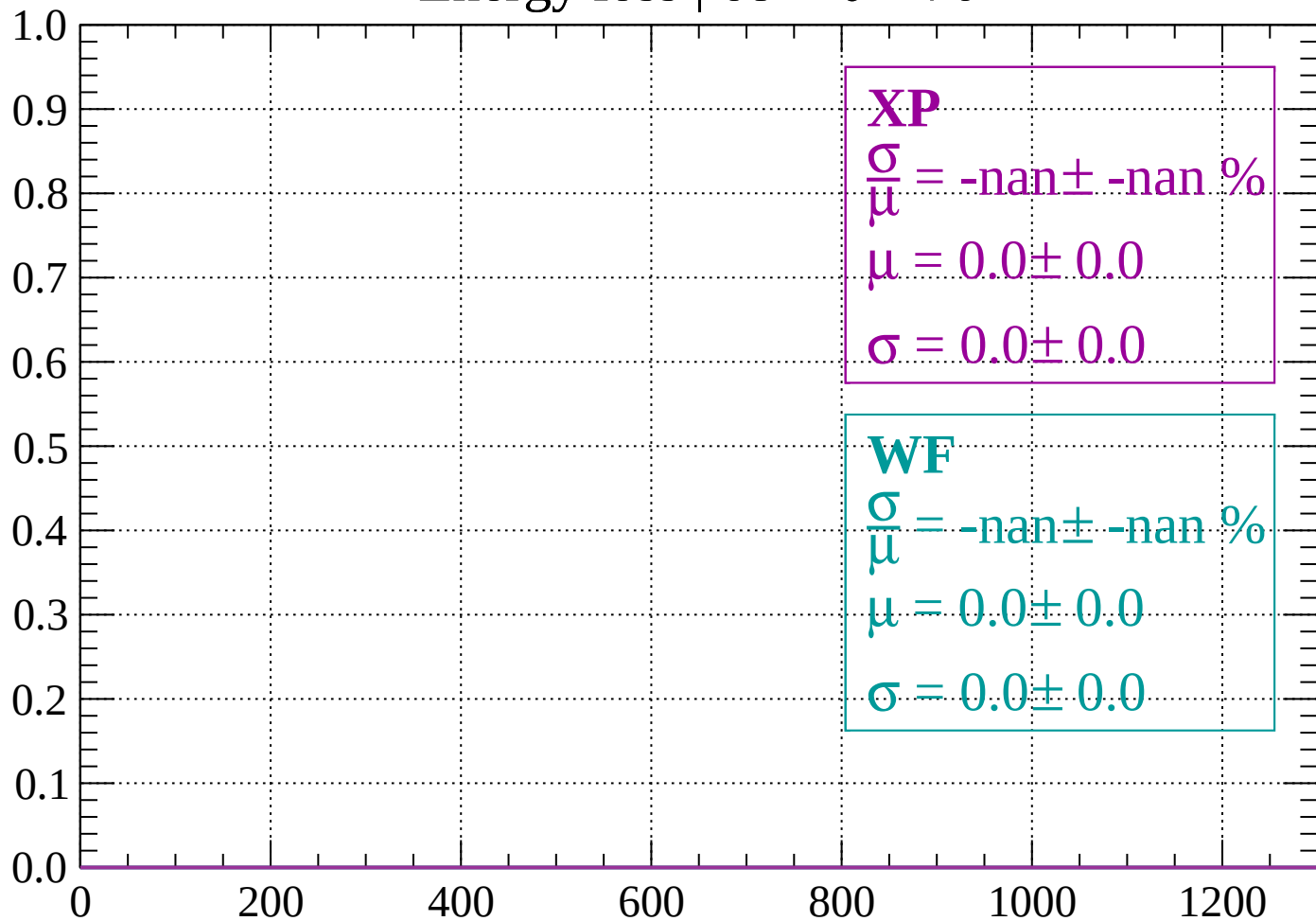
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

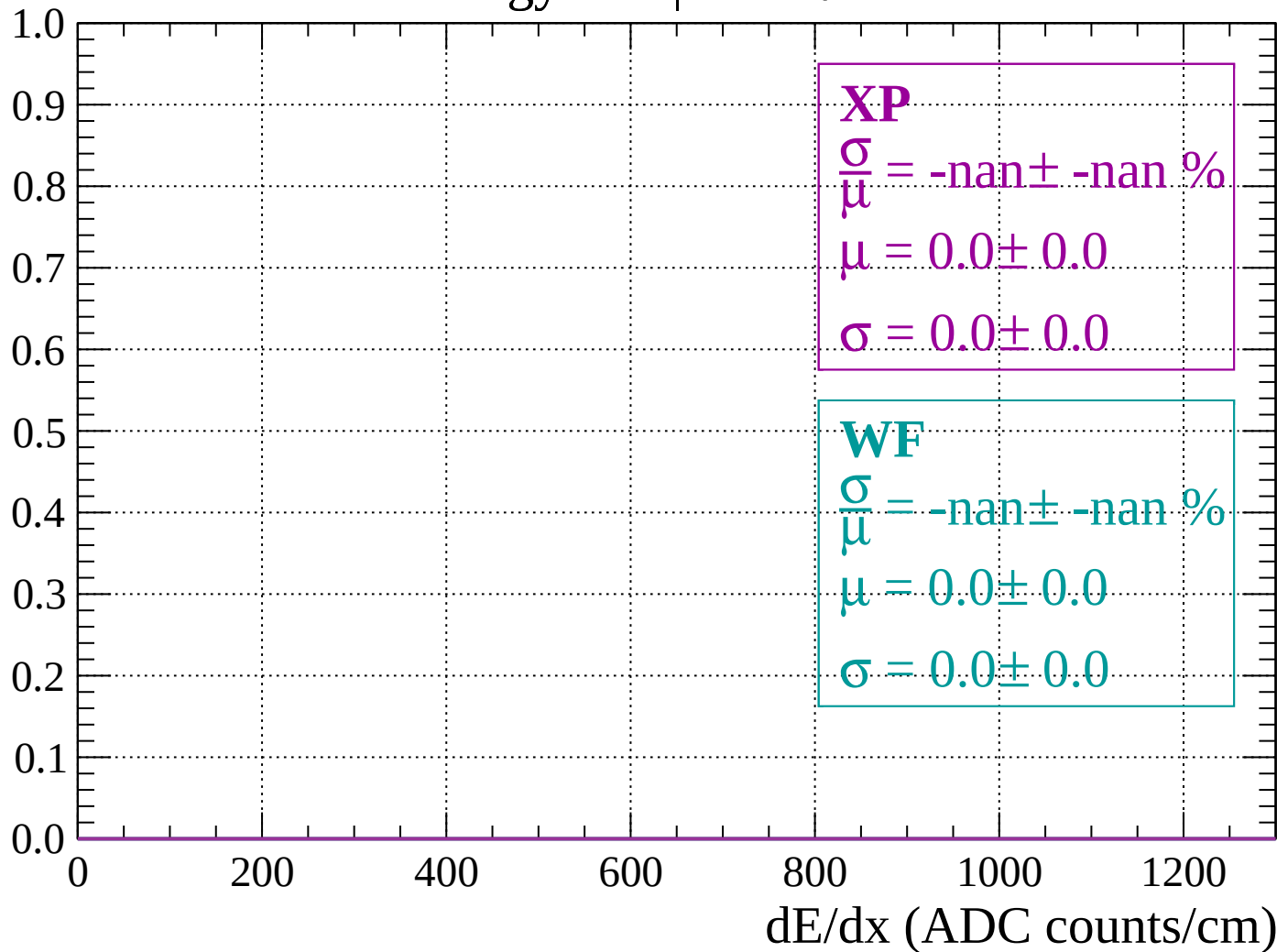
Count



dE/dx (ADC counts/cm)

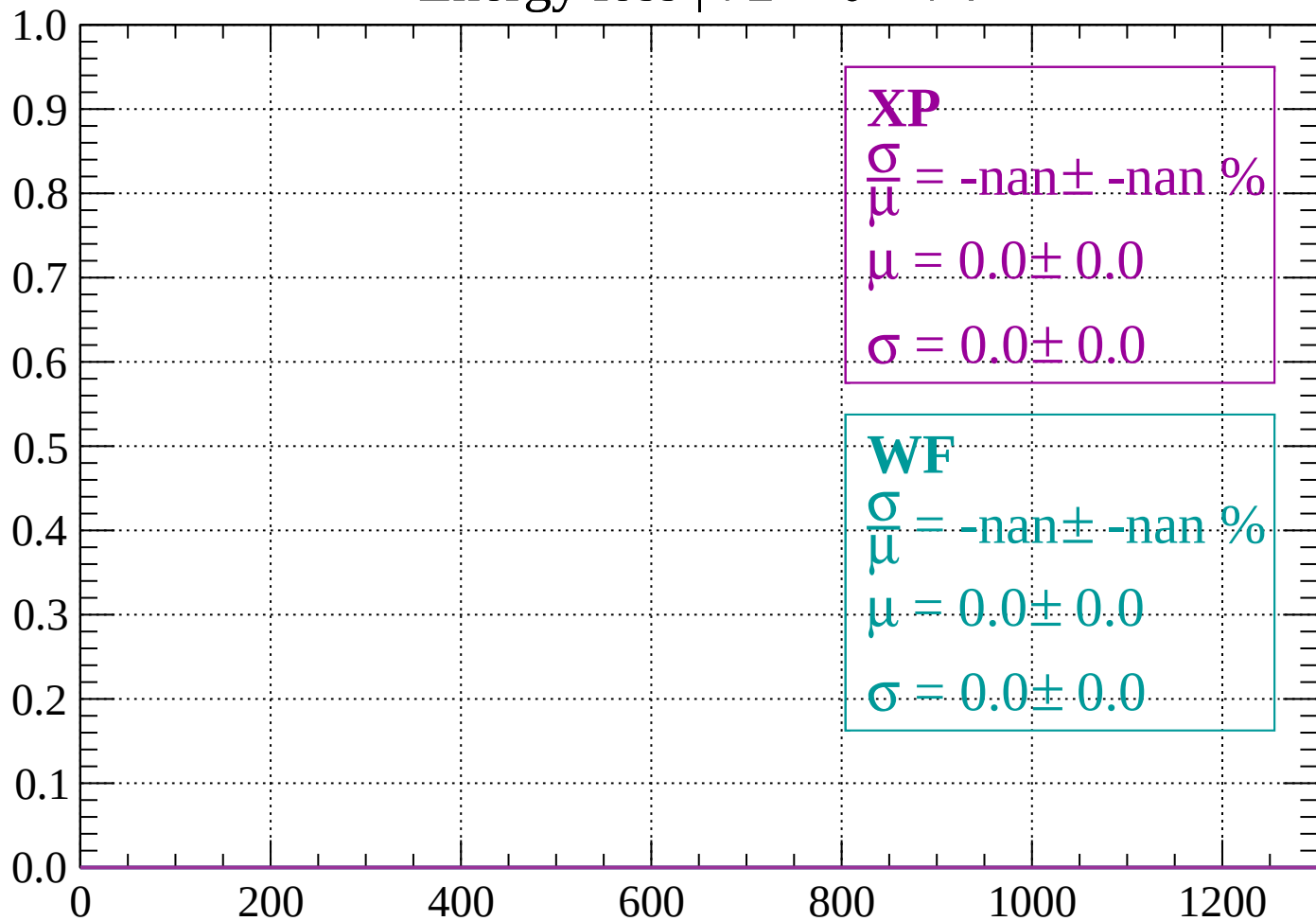
Energy loss | $70 < \theta < 72$

Count



Energy loss | $72 < \theta < 74$

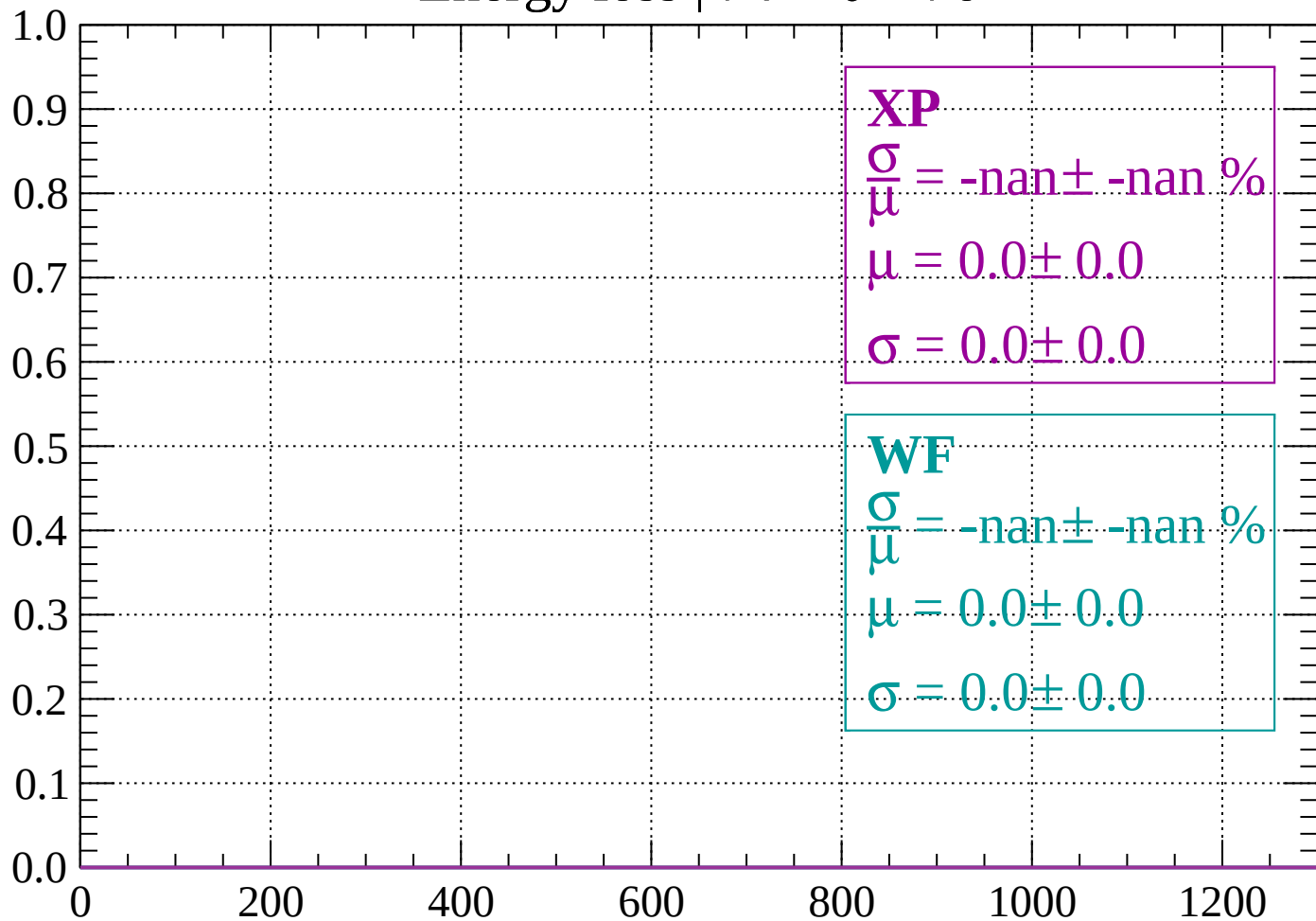
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

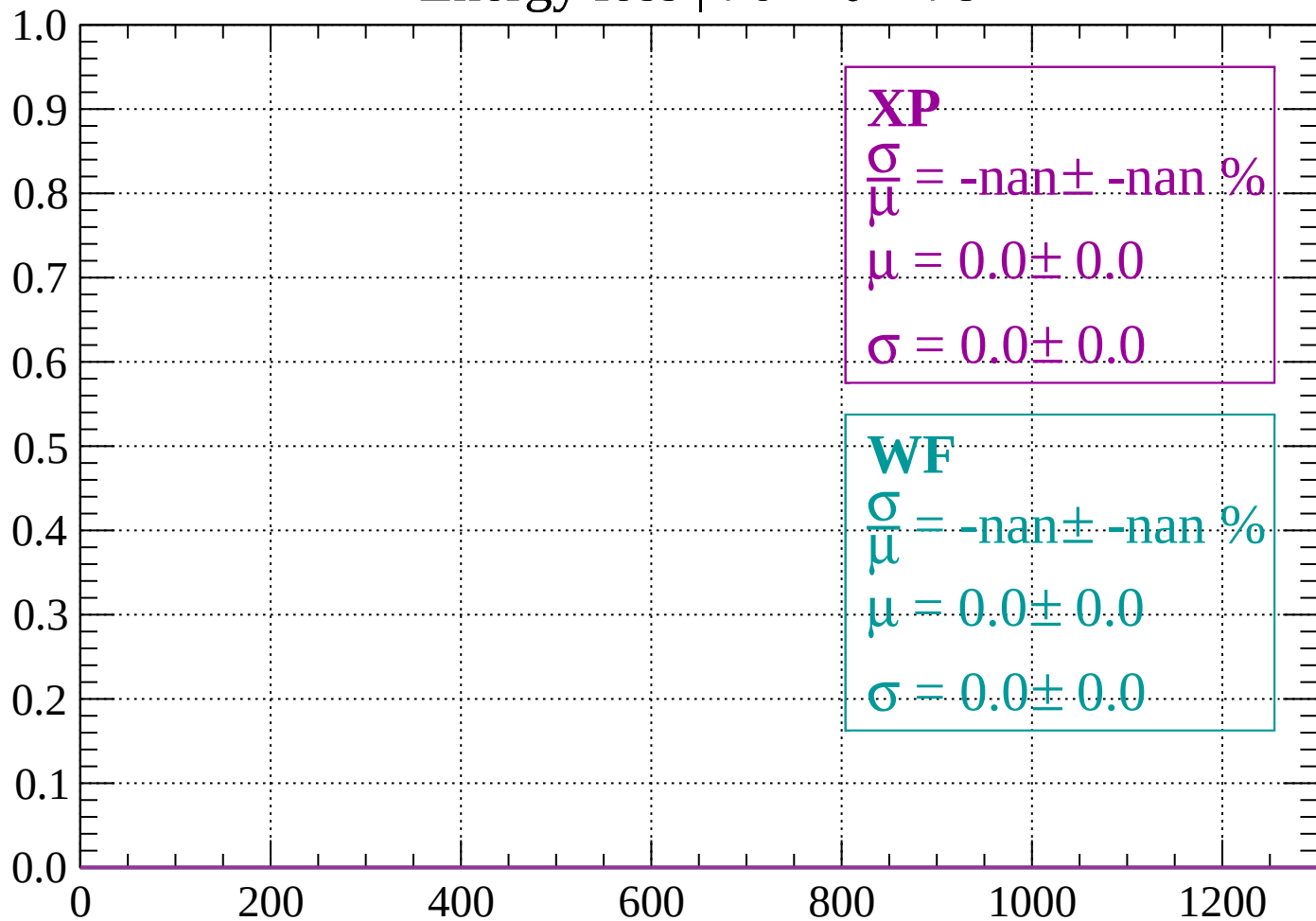
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

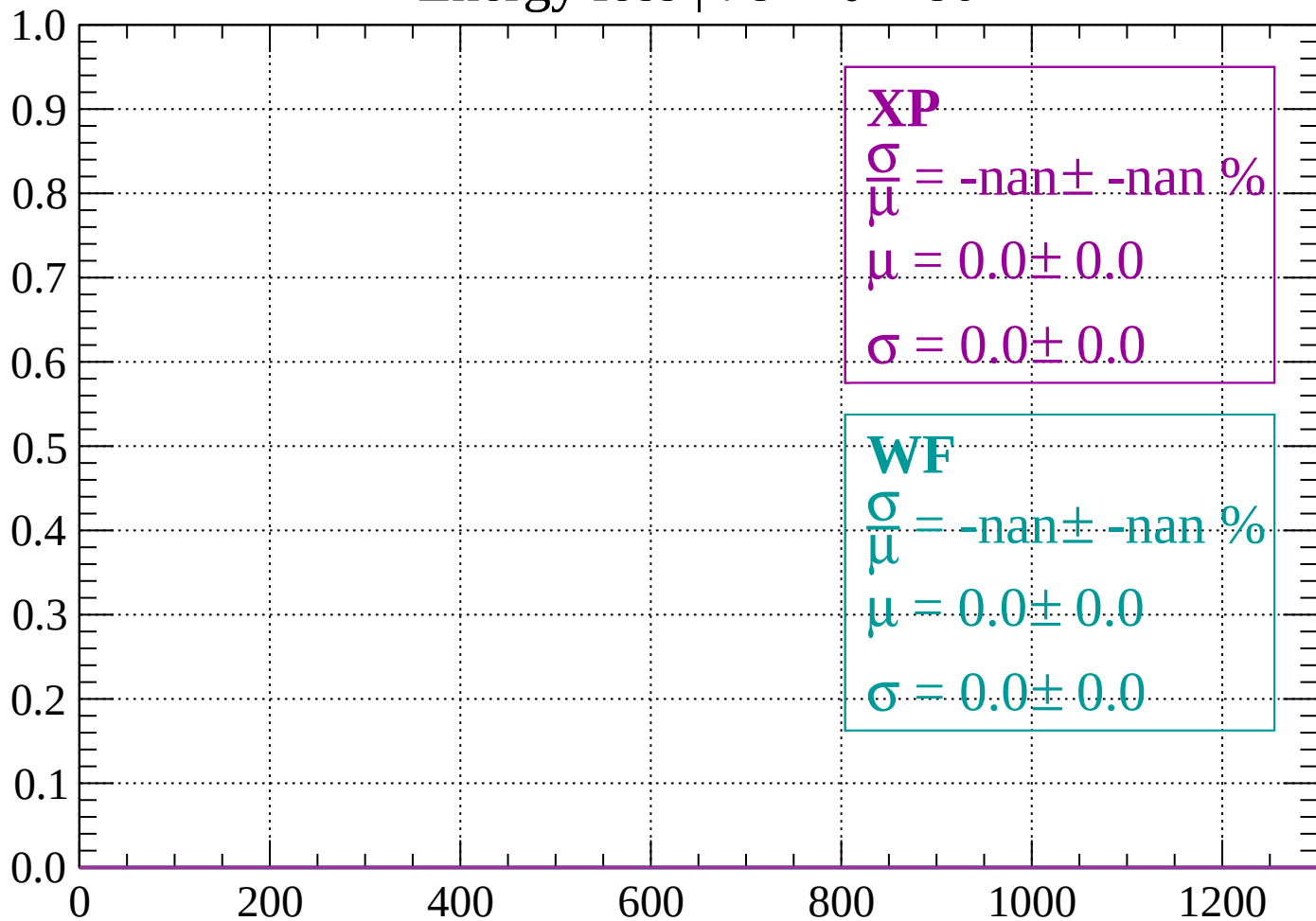
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

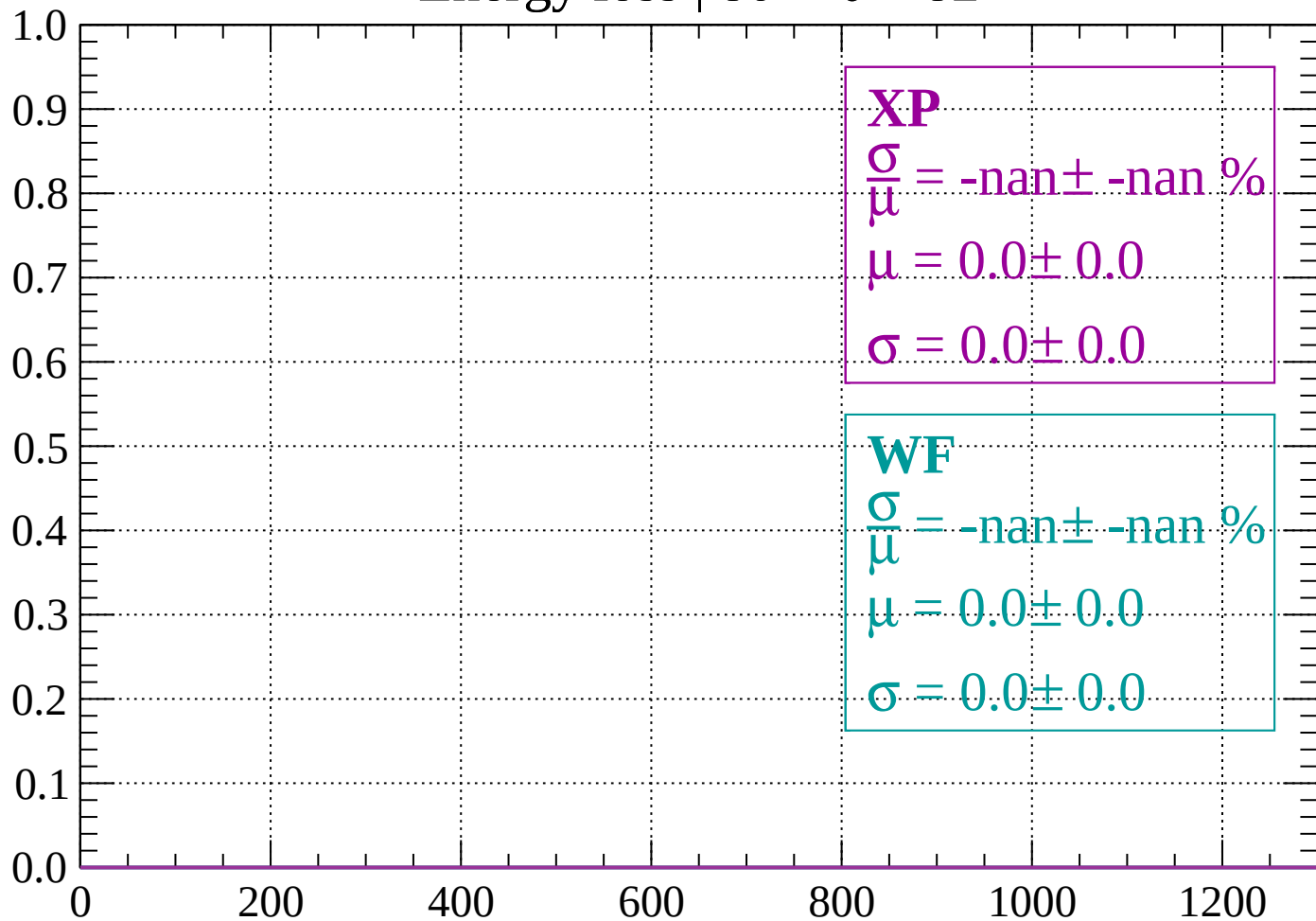
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

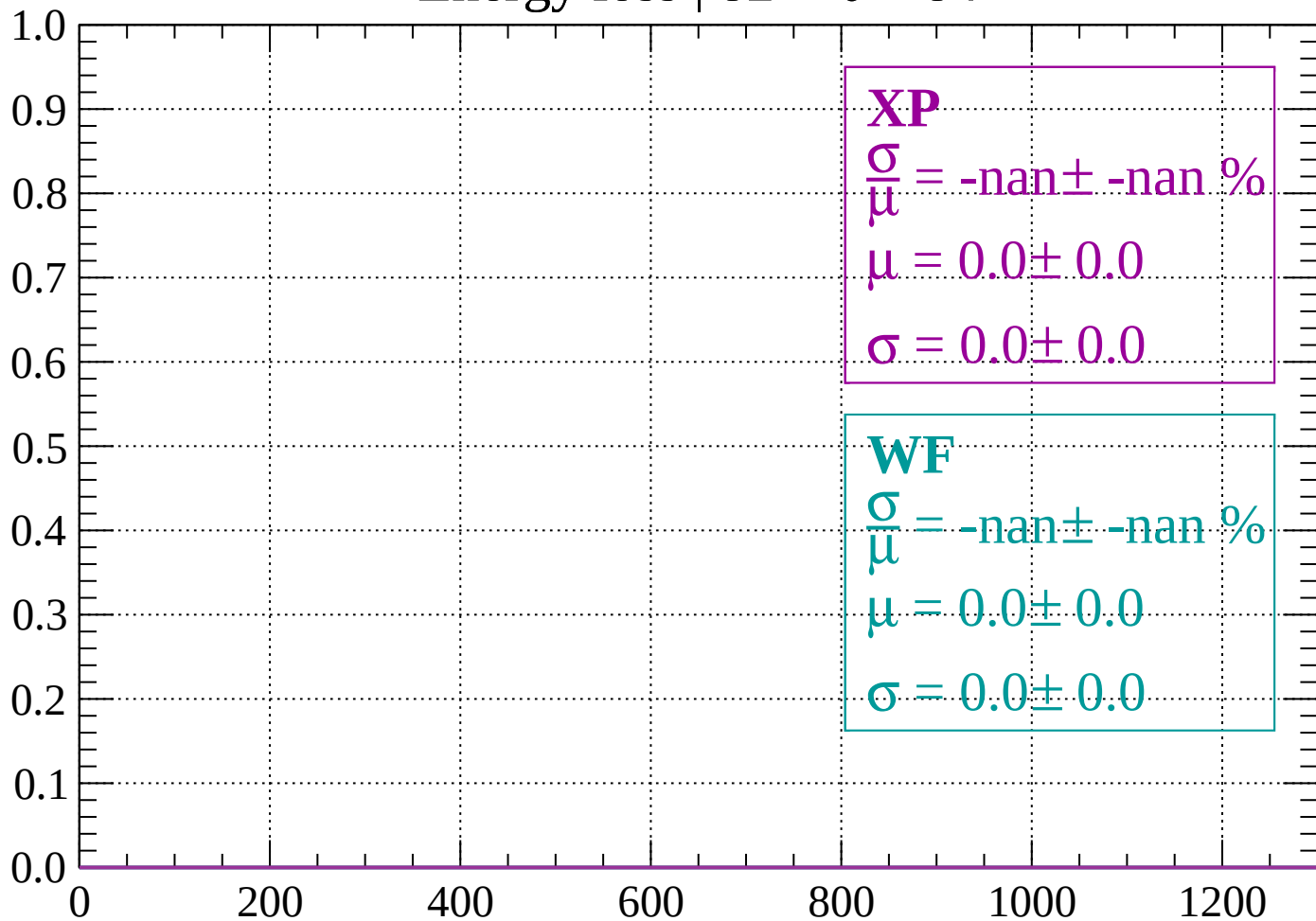
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

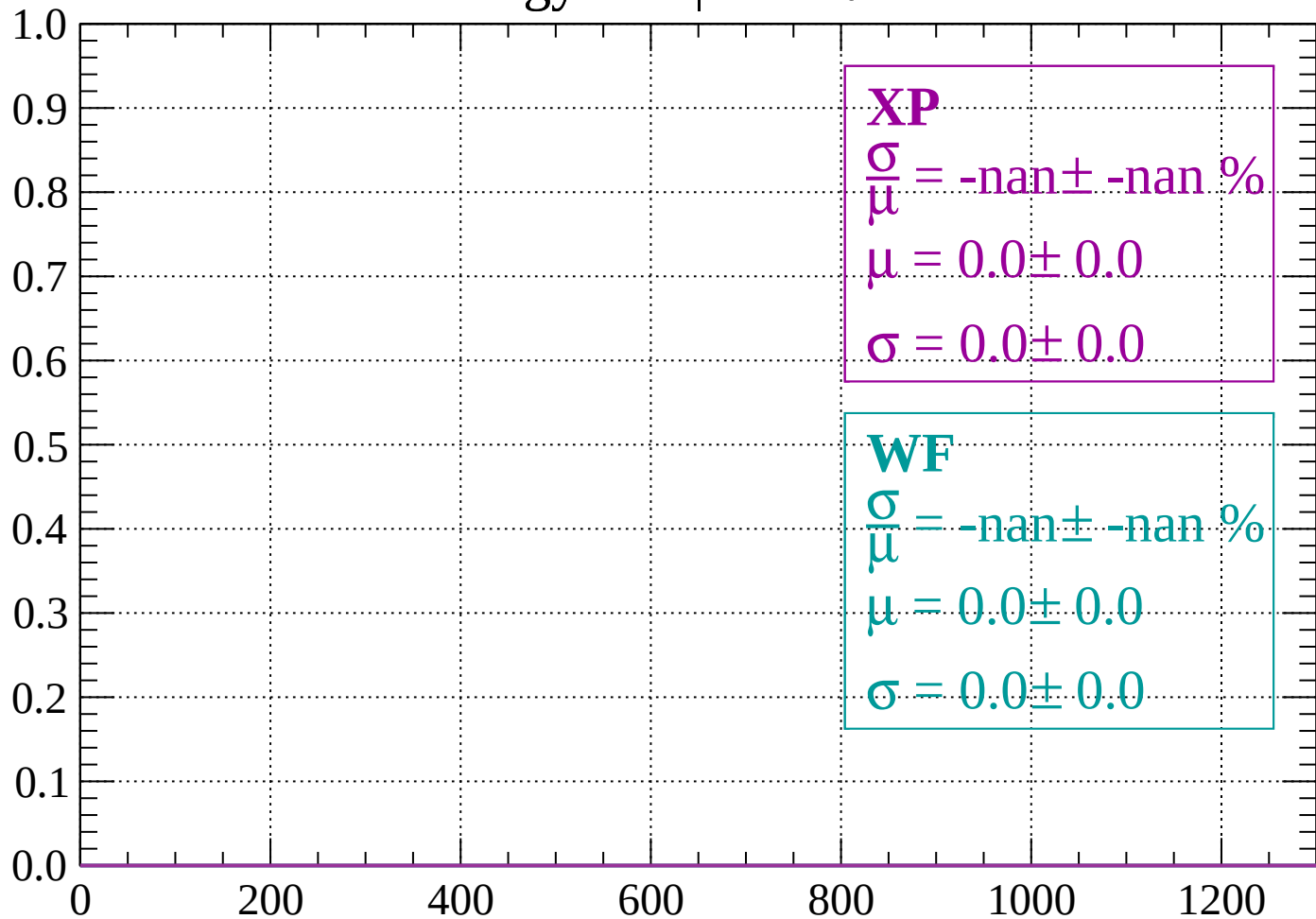
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

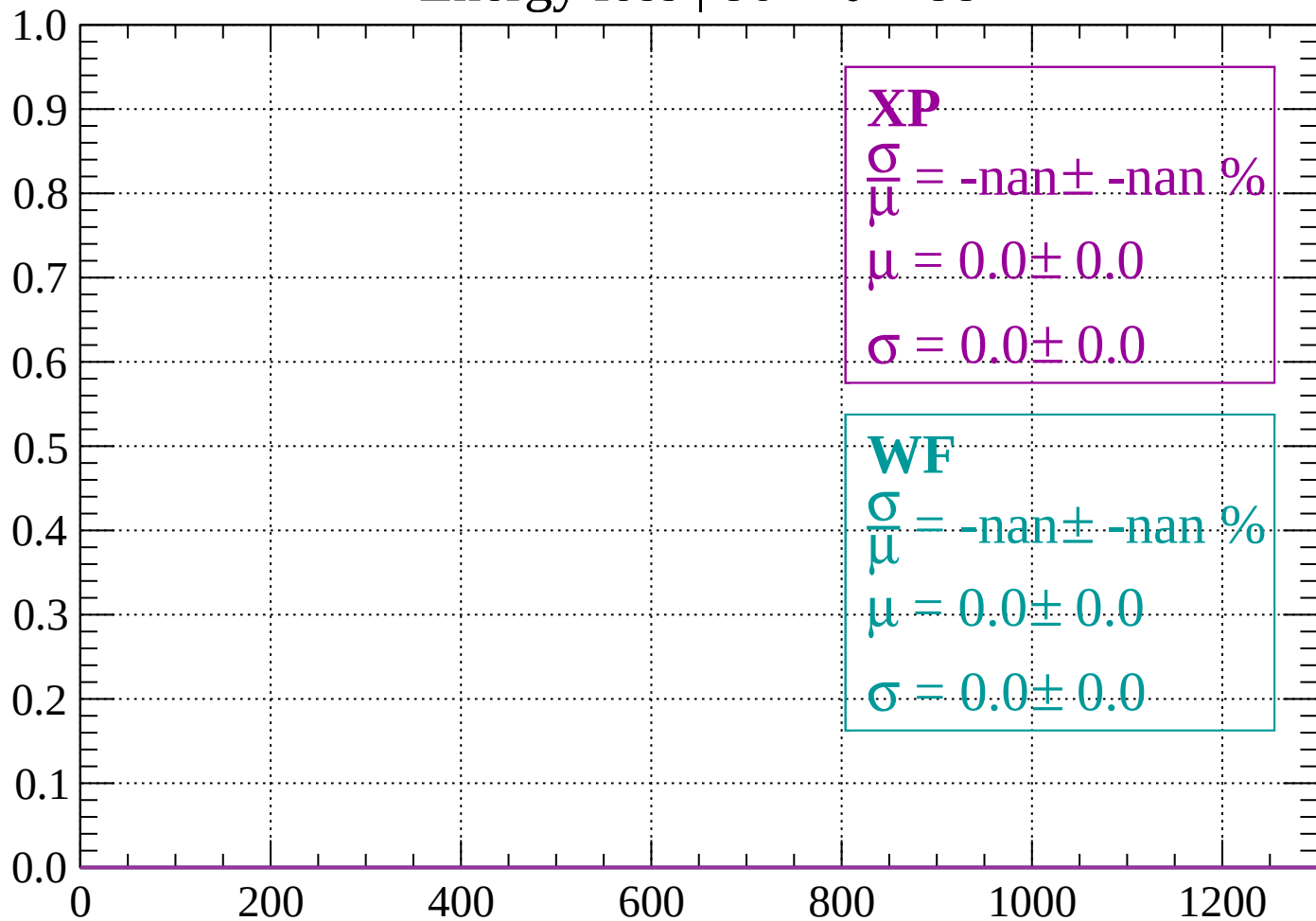
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

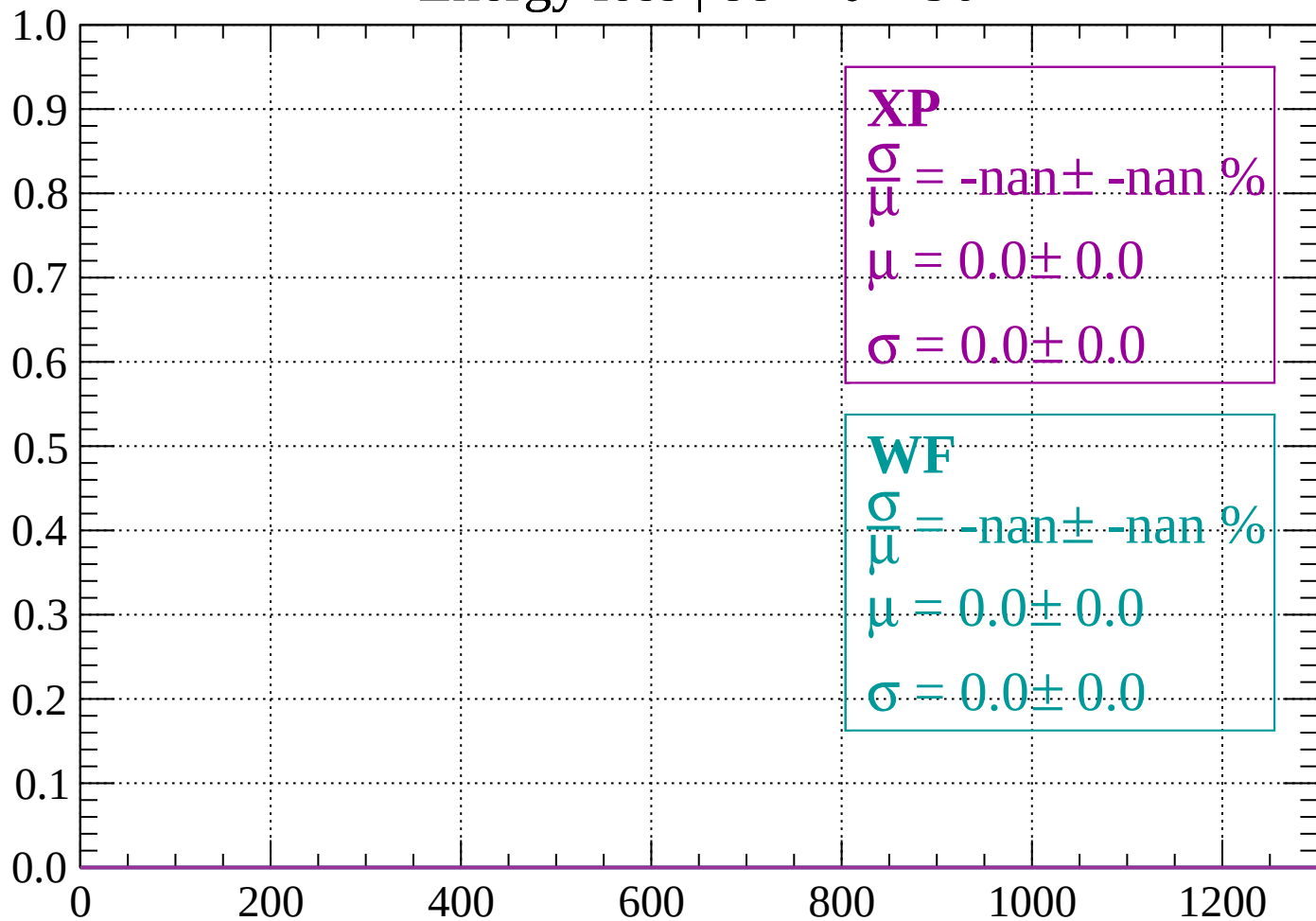
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

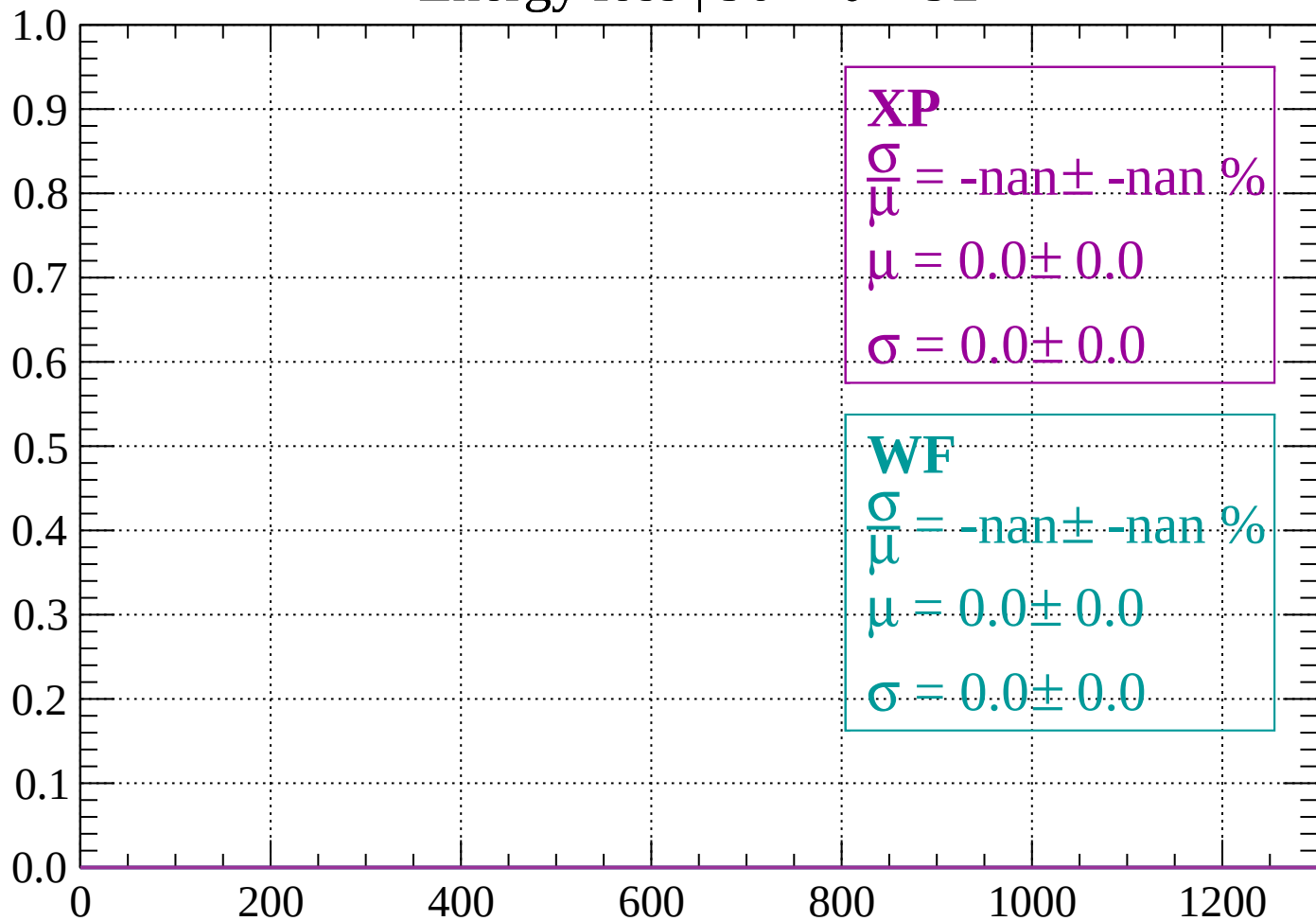
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

